Andrés Armando Mendiburu Zevallos

Flame Propagation Fundamentals

Applications to Sustainable Fuels

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To my wife Sofia

Preface

Combustion is one of the most important means for energy conversion. Since the Industrial Revolution, combustion has become more relevant worldwide. The study of combustion from a scientific point of view has received attention since the late eighteenth century and became a well-established field of research since the nineteenth century. Considering the total energy consumption worldwide in 2023, gas, coal and oil represent 23.30%, 26.47% and 31.70%, respectively, while biofuels represent 0.77%. In Brazil, these contributions are 7.70%, 4.13%, 36.82 and 6.90%, respectively [1]. Therefore, combustion will continue to be an important field of research in the current century. The increasing knowledge of combustion phenomena has led to more efficient and less polluting combustion processes. However, in view of the dangerous scenarios related to the emission of greenhouse gases, energy transition has gained a critical role in mitigating climate change. In the context of combustion science, the study of less polluting fuels is a way to contribute to the mitigation of climate change. Therefore, biofuels, such as ethanol, hydrogen and ammonia, have been receiving increasing attention from the scientific community. Important characteristics of these fuels are related to their combustion in premixed mixtures with air. For example, the laminar flame speed, the flammability limits, and the flame acceleration behavior, among others. Therefore, this book aims to present the fundamentals of premixed flame propagation with emphasis on ethanol, hydrogen and ammonia. The emphasis is given through the examples which focus on those fuels. To provide a clear picture, the book is divided into eight chapters. Chapter 1 deals with the thermodynamics of combustion which represents the basic knowledge to successfully understand this phenomenon. Concepts such as the adiabatic flame temperature, the equilibrium constant and the minimization of the Gibbs free energy are introduced. Chapter 2 discusses the chemical kinetics of combustion. In this case, the interest is focused on understanding the evolution of a combustion reaction through detailed mechanisms formed by elementary reactions. These mechanisms are postulated and must be validated with fundamental experimental data. Chapter 3 presents the derivation of the conservation equations in spherical coordinates. The derivation process provides the reader with the necessary fundamental knowledge regarding the widely known total mass conservation equation, species mass conservation equation, Navier–Stokes equation, and energy conservation equation. More-

over, this coordinate system makes evident the complexities related to the orthogonal unit vectors, which are not observable when the Cartesian (or rectangular) coordinate system is chosen to derive these equations. Chapter 4 discusses the one-dimensional theory of premixed flames. The focus of this chapter is to present a phenomenological interpretation of these flames, introducing relevant concepts as the laminar flame speed. The governing conservation equations are derived and an expression for the laminar flame speed is obtained. This chapter also presents the dimensionless numbers of Lewis, Zeldovich and Damköhler. Chapter 5 is focused on the dynamics of a propagating premixed flame. The analysis begins by treating the flame as a discontinuity and characterizing relevant vectorial quantities. Later, a discussion of flame stretch theory is presented with a derivation of the flame stretch equation. The chapter is closed by discussing the experimental and theoretical observations of flame propagation in closed ducts. Chapter 6 presents selected aspects of compressible flow theory which are necessary to understand the topic of detonations. The isentropic flow theory is first presented, followed by the discussion of normal shock waves theory and oblique shock waves theory. The chapter is closed with an overview of shock reflections. Chapter 7 deals with detonation theory. The chapter starts by presenting the thermodynamics of detonation, with emphasis on the Rayleigh line, the Hugoniot curve and the Chapman–Jouguet theory. Afterwards, the ZND theory, named after Yakov Zeldovich, John von Newman and Werner Döring, is discussed, introducing the concepts of induction zone length and thermicity. The chapter is closed with a discussion on the application of ZND theory together with detailed chemical kinetics mechanisms. Finally, Chap. 8 presents a discussion about the flammability limits. This topic is of great importance to the safety of industrial applications. The simple one-dimensional theory with one-step reaction is first discussed, followed by a presentation of the more complete one-dimensional theory coupled with detailed chemical kinetics mechanisms. The effects of initial temperature, initial pressure and diluent addition are analyzed. Also, the flammability limits of mixtures involving two or more fuels are also discussed. The last section of the chapter is dedicated to presenting some estimation methods of the flammability limits.

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Thermodynamics of Combustion

1

Nomenclature

Symbols

AF_s	Stoichiometric air-to-fuel ratio, kg_{air}/kg_{fuel}
AF	Air-to-fuel ratio, kg_{air}/kg_{fuel}
E_j	Energy of a thermodynamic system in the state j, J
$c_{p,i}$	Heat capacity of species i at constant pressure in mass basis, J/(kg K)
$\overline{c}_{p,i}$	Heat capacity of species i at constant pressure in molar basis, J/(mol K)
$c_{v,i}$	Heat capacity of species i at constant volume in mass basis, J/(kg K)
$\overline{c}_{v,i}$	Heat capacity of species i at constant volume in molar basis, J/(mol K)
F_i	Helmholtz free energy of species i in molar basis, kJ/mol
G_i	Gibbs free energy of species i in molar basis, kJ/mol
ΔG_T°	Variation of the Gibbs free energy over a simple reaction
$\overline{g}_{f,T,i}^\circ$	Gibbs free energy of formation of species i in molar basis, kJ/mol
g	Acceleration of gravity, m/s^2
H_i	Enthalpy of species i, kJ
h_i	Total specific enthalpy of species i in mass basis, kJ/kg
$\overline{h}_i$	Total specific enthalpy of species i in molar basis, kJ/mol
$\overline{h}_{f,i}^\circ$	Enthalpy of formation of species i in molar basis, kJ/mol
Δh_i	Sensible specific enthalpy of species i in mass basis, kJ/kg
$\Delta\overline{h}_i$	Sensible specific enthalpy of species i in molar basis, kJ/mol
K_P	Equilibrium constant
m_i	Mass of species i, kg
$\dot{m}_i$	Mass flow rate of species i, kg/s
n_i	Number of moles of species i, mol

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p	Pressure
p°	Reference pressure of 1 atm or 101.325 kPa
Q_{1-2}	Heat exchanged along a thermodynamic process, W
$\dot{Q}$	Rate of heat exchanged along a thermodynamic process, J
R	Gas constant, kJ/(kg K)
R_u	Universal gas constant, kJ/(mol K)
S_i	Entropy of species i, J/K
s_i	Specific entropy of species i in molar basis, J/(kg K)
$\overline{s}^\circ_{f,T,i}$	Entropy of formation of species i in molar basis, J/(mol K)
T	Temperature, K
U_i	Internal energy of species i, kJ
u_i	Specific internal energy of species i in mass basis, kJ/kg
$\overline{u}_i$	Specific internal energy of species i in molar basis, kJ/mol
v^s_{ar}	Stoichiometric coefficient of air in a global reaction
v_{air}	Coefficient of air in global reaction
v_i	Stoichiometric coefficient of species i in a simple reaction
v	Specific volume, m^3/kg
V	Velocity, m/s
W_i	Molecular weight of species i, kg/kmol
W_{1-2}	Work exchanged along a thermodynamic process, J
$\dot{W}$	Rate of work exchanged along a thermodynamic process, W
x_i	Number of molecules of element i in the species j
z	Height measured from a reference level, m

Greek Letters

ρ_i	Density of species i, kg/m^3
λ_i	Lagrange multiplier corresponding to element i
ϕ	Equivalence ratio
μ_i	Chemical potential of species i, kJ/mol

Subscripts

f	Fuel
R	Reactants
P	Products

1.1 Introduction

Understanding the thermodynamics of combustion is the first step to developing the necessary background in combustion science. This chapter begins with the presentation of the application of the mass conservation law and its use in global combustion reactions in order to determine the number of moles of each species in the products mixture. In a first moment, the analysis is restricted to reactants mixtures which are in stoichiometric or lean proportions. That is, there is enough oxidant to invoke the complete combustion assumption. Subsequently, the energy conservation law is introduced, which allows for the determination of the adiabatic flame temperature of constant pressure and constant volume combustion processes. Finally, the thermodynamic equilibrium assumption is introduced through the use of the equilibrium constant and also through the minimization of the Gibbs free energy and of the Helmholtz free energy for constant pressure and constant volume combustion processes, respectively.

1.2 Mass Conservation

1.2.1 The Global Reaction and Stoichiometric Combustion

Since classical thermodynamics considers only equilibrium states, the combustion process is generally represented in terms of a global combustion reaction which relates the fuel, oxidant and products as shown in Eq. (1.1).

$$\text{Fuel} + \text{Oxidant} \rightarrow \text{Produtcs} \tag{1.1}$$

According to the global reaction, the fuel and oxidant react to form products. This is equivalent to looking at the system at two equilibrium states. One of these states considers the system before the chemical reaction, and the other considers it after the reaction. The most common fuels include hydrocarbons. However, biofuels, such as ethanol, and alternative fuels, such as hydrogen and ammonia, are currently of interest. The most commonly used oxidant is air. Nevertheless, other chemical species, such as pure oxygen, can also be used. The products may contain carbon dioxide (CO_2), carbon monoxide (CO), water vapor (H_2O), hydrogen (H_2), nitrogen oxides (NO_x), and nitrogen (N_2), among others.

The air is considered as an oxidant for simplicity. However, the actual oxidant is the oxygen contained in it. The air used in combustion calculations has a volumetric concentration of 21% O_2 and 79% N_2. This means that for 1 mol of O_2, there are 3.76 mol of N_2 in the air. The actual air composition includes some water vapor (humidity) and small amounts of CO_2 and argon (Ar). Consider, for instance, the reaction of hydrogen (H_2) with air and ethanol (C_2H_5OH) with air shown below.

$$H_2 + v_{air}(O_2 + 3.76\ N_2) \rightarrow n_{H_2O}H_2O + n_{H_2}H_2 + n_{O_2}O_2 + n_{N_2}N_2 + n_{NO}NO + \text{others} \tag{1.2}$$

$$C_2H_5OH + v_{air}(O_2 + 3.76\ N_2) \rightarrow \begin{array}{c} n_{CO_2}CO_2 + n_{CO}CO + n_{H_2O}H_2O \\ + n_{H_2}H_2 + n_{O_2}O_2 + n_{N_2}N_2 + n_{NO}NO + \text{others} \end{array} \tag{1.3}$$

In both reactions above, there are 1 mol of fuel and 4.76 υ_{air} mol of air. The air coefficient is represented by υ_{air} in the reactions. CO_2 and H_2O are considered species of complete combustion, while CO and H_2 are considered species of incomplete combustion. Nitrogen can react to form nitric oxide (NO), and some other products can also exist in equilibrium. The law of mass conservation establishes that the total mass of each element should be conserved. Therefore, it can be stated that the total number of moles of an element should be conserved, even if this element is present in different molecules when looking at the reactants and products. This can be stated for an element "i" as shown in Eq. (1.4), where the subscripts *R* and *P* refer to reactants and products respectively.

$$\sum n_{i,R} = \sum n_{i,P} \tag{1.4}$$

1.2.2 The Equivalence Ratio

A combustion process is considered stoichiometric when the number of moles of oxygen available in the reactants is the strictly necessary amount to achieve complete combustion of carbon, hydrogen, and sulfur in the fuel. Moreover, nitrogen is considered inert to simplify the calculations. The stoichiometric combustion of ethanol with air is represented by the reaction shown in Eq. (1.5). It is important to realize that a stoichiometric air coefficient was represented by v^s_{air}.

$$C_2H_5OH + v^s_{ar}(O_2 + 3.76\ N_2) \rightarrow n_{CO_2}CO_2 + n_{H_2O}H_2O + n_{N_2}N_2 \tag{1.5}$$

For a general fuel molecule of the form $C_{x_C}H_{x_H}O_{x_O}N_{x_N}S_{x_S}$, the global reaction for stoichiometric combustion is given in Eq. (1.6). Note that the reaction has been balanced by applying the mass conservation law to each element.

$$C_{x_C}H_{x_H}O_{x_O}N_{x_N}S_{x_S} + v^s_{air}(O_2 + 3.76\ N_2) \rightarrow x_CCO_2 + \frac{x_H}{2}H_2O + x_SSO_2 + \left(3.76v^s_{air} + \frac{x_N}{2}\right)N_2 \tag{1.6}$$

It is easy to show that the stoichiometric coefficient of air, based on the complete combustion of carbon, hydrogen and sulfur, is given by Eq. (1.7).

$$v_{\text{air}}^{s} = x_{\text{C}} + \frac{x_{\text{H}}}{4} + x_{\text{S}} - \frac{x_{\text{O}}}{2} \tag{1.7}$$

In the case of ethanol, Eq. (1.7) gives $v_{\text{air}}^{s} = 3$, while for hydrogen it gives $v_{\text{air}}^{s} = 0.5$. Another fuel of interest is ammonia (NH_3), for which $v_{\text{air}}^{s} = 0.75$. Note that in this ideal representation of combustion, nitrogen is not forming any pollutant, which is not the case in a real process. However, the formation of NO_x needs the application of chemical kinetics theory and will be considered in Chap. 2.

The air-to-fuel ratio (AF) is defined as the ratio between the mass of air and the mass of fuel present in the reactants. For a reaction involving 1 mol of fuel, the stoichiometric air-to-fuel ratio (AF_s) is defined as shown in Eq. (1.8). The AF_s is considered as a representative parameter of a particular fuel.

$$AF_s = \frac{4.76 v_{\text{air}}^{s} W_{\text{air}}}{W_f} \tag{1.8}$$

In the above expression, W_{air} and W_f are the molecular weights of air and fuel, respectively. The number of moles of air in the reactants can be greater or lesser than the stoichiometric number of moles. This is represented by an air coefficient such that $v_{\text{air}} > v_{\text{air}}^{s}$ or $v_{\text{air}} < v_{\text{air}}^{s}$. In general, the AF can be written as shown in Eq. (1.9).

$$AF = \frac{4.76 \ v_{\text{air}} W_{\text{air}}}{W_f} \tag{1.9}$$

The equivalence ratio (ϕ) is defined as the quotient of the stoichiometric air-to-fuel ratio and the actual air-to-fuel ratio of the reaction, as shown in Eq. (1.10)

$$\phi = \frac{AF_s}{AF} \tag{1.10}$$

Substitution of Eqs. (1.8) and (1.9) into Eq. (1.10) gives

$$\phi = \frac{v_{\text{air}}^{s}}{v_{\text{air}}} \tag{1.11}$$

The equivalence ratio is used to classify mixtures into three classes. A stoichiometric mixture has the exact number of moles of air necessary to achieve complete combustion; therefore, $\phi = 1$. A lean mixture has greater number of moles of air than are necessary to achieve complete combustion; therefore, $\phi < 1$. A rich mixture has less number of moles of air than are necessary to achieve complete combustion; thus, $\phi > 1$. It is clear that complete combustion can only be achieved if $\phi \leq 1$. Moreover, in several real combustion processes, the amount of air is greater than the stoichiometric quantity to reduce the concentration of incomplete combustion species.

In the case of a lean mixture, it can be assumed that the combustion reaction is complete and that the excess oxygen and nitrogen are inert species. This consideration can be applied to obtain the following general global reaction for lean mixtures.

$$\mathrm{C}_{x_\mathrm{C}}\mathrm{H}_{x_\mathrm{H}}\mathrm{O}_{x_\mathrm{O}}\mathrm{N}_{x_\mathrm{N}}\mathrm{S}_{x_\mathrm{S}} + \frac{v_{\mathrm{air}}^{s}}{\phi}(\mathrm{O}_2 + 3.76\ \mathrm{N}_2)$$
$$\rightarrow x_\mathrm{C}\mathrm{CO}_2 + \frac{x_\mathrm{H}}{2}\mathrm{H}_2\mathrm{O} + \left(\frac{v_{\mathrm{air}}^{s}}{\phi} - v_{\mathrm{air}}^{s}\right)\mathrm{O}_2 + x_\mathrm{S}\mathrm{SO}_2 + \left(3.76\frac{v_{\mathrm{air}}^{s}}{\phi} + \frac{x_\mathrm{N}}{2}\right)\mathrm{N}_2 \tag{1.12}$$

Example 1.1
Consider the global combustion reactions of ethanol (C_2H_5OH), hydrogen (H_2), ammonia (NH_3), and octane (C_8H_{18}) with equivalence $\phi = 0.80$. Determine the air-to-fuel ratios and the concentrations of the species in the products.

Solution
The global reactions for these fuels can be written by assuming that complete reaction is achieved, with the excess oxygen and nitrogen treated as inert species. The complete combustion assumption is reasonable because the mixture is lean. The stoichiometric air coefficient is first determined from Eq. (1.7). For ethanol it is found that $v_{\mathrm{air}}^{s} = 3.00$; for hydrogen, $v_{\mathrm{air}}^{s} = 0.50$; for ammonia, $v_{\mathrm{air}}^{s} = 0.75$; and for octane, $v_{\mathrm{air}}^{s} = 12.50$. Therefore, the global reactions are written as shown below by using Eq. (1.12).

Ethanol

$$\mathrm{C_2H_5OH} + 3.75(\mathrm{O_2} + 3.76\ \mathrm{N_2}) \rightarrow 2\ \mathrm{CO_2} + 3\ \mathrm{H_2O} + 0.75\ \mathrm{O_2} + 14.1\ \mathrm{N_2}$$

Hydrogen

$$\mathrm{H_2} + 0.625(\mathrm{O_2} + 3.76\ \mathrm{N_2}) \rightarrow \mathrm{H_2O} + 0.125\ \mathrm{O_2} + 2.35\ \mathrm{N_2}$$

Ammonia

$$\mathrm{NH_3} + 0.938(\mathrm{O_2} + 3.76\ \mathrm{N_2}) \rightarrow 1.5\ \mathrm{H_2O} + 0.188\ \mathrm{O_2} + 4.027\ \mathrm{N_2}$$

Octane

$$\mathrm{C_8H_{18}} + 15.625(\mathrm{O_2} + 3.76\ \mathrm{N_2}) \rightarrow 8\ \mathrm{CO_2} + 9\ \mathrm{H_2O} + 3.125\ \mathrm{O_2} + 58.75\ \mathrm{N_2}$$

The air-to-fuel ratios can be determined directly by using Eqs. (1.8) and (1.10). The molecular weight of air is calculated by considering that for 1 mol of O_2, there are 3.76 mol of N_2. Therefore, $W_{\mathrm{air}} = 28.84$ g/mol. In the case of ethanol, hydrogen, ammonia, and octane, the molecular weights are 46 g/mol, 2 g/mol, 17 g/mol, and 114 g/mol, respectively.

Table 1.1 Concentrations of product species for lean mixtures of ethanol, hydrogen, and ammonia with air for $\phi = 0.80$

Product species		Ethanol/air	Hydrogen/air	Ammonia/air	Octane/air
CO_2	(%)	10.08	0.00	0.00	10.14
H_2O	(%)	15.11	28.78	26.24	11.41
O_2	(%)	3.78	3.60	3.29	3.97
N_2	(%)	71.03	67.62	70.47	74.48
Total	(%)	100.00	100.00	100.00	100.00
n_P	Mol	19.85	3.475	5.715	78.875
m_{CO_2}/m_F	kg_{CO2}/kg_{fuel}	1.91	0.00	0.00	3.09

Ethanol

$$AF_s = \frac{4.76 \times 3 \times 28.84}{46} = 8.96\ \mathrm{kg_{air}/kg_{fuel}}; \quad AF = \frac{8.96}{0.8} = 11.20\ \mathrm{kg_{air}/kg_{fuel}}.$$

Hydrogen

$$AF_s = \frac{4.76 \times 0.5 \times 28.84}{2} = 34.32\ \mathrm{kg_{air}/kg_{fuel}}; \quad AF = \frac{34.32}{0.8} = 42.90\ \mathrm{kg_{air}/kg_{fuel}}.$$

Ammonia

$$AF_s = \frac{4.76 \times 0.75 \times 28.84}{17} = 6.06\ \mathrm{kg_{air}/kg_{fuel}}; \quad AF = \frac{6.06}{0.8} = 7.58\ \mathrm{kg_{air}/kg_{fuel}}.$$

Octane

$$AF_s = \frac{4.76 \times 12.50 \times 28.84}{114} = 15.06\ \mathrm{kg_{air}/kg_{fuel}}; \quad AF = \frac{15.06}{0.8} = 18.82\ \mathrm{kg_{air}/kg_{fuel}}.$$

The concentrations of the species can now be determined from the global reactions. The results of these calculations are presented in Table 1.1. Another result presented in Table 1.1 is the ratio of mass of CO_2 produced by kilogram of fuel.

1.3 Energy Conservation

The first law of thermodynamics for a closed system in steady state establishes that the change in the energy of a system is equal to the difference between the heat exchanged through the system boundaries and the work done on the system boundaries. This relationship is given by Eq. (1.13), where subscripts 1 and 2 refer to the initial and final states, respectively.

$$E_2 - E_1 = Q_{1-2} - W_{1-2} \tag{1.13}$$

The energy of the system includes internal, kinetic, and potential energies. Considering a closed system for which the kinetic and potential energy variations are negligible in comparison to the internal energy variation, the energy conservation is given as shown in Eq. (1.14).

$$U_2 - U_1 = Q_{1-2} - W_{1-2} \tag{1.14}$$

In the case of an open system, or control volume, the energy conservation is written in terms of enthalpy, kinetic, and potential energies as shown in Eq. (1.15).

$$\sum_{\text{out}} \dot{m}\left(h + \frac{1}{2}V^2 + gz\right) - \sum_{\text{in}} \dot{m}\left(h + \frac{1}{2}V^2 + gz\right) = \dot{Q} - \dot{W} \tag{1.15}$$

It is common practice in thermodynamics calculations of combustion to neglect the kinetic and potential energy variations in comparison to the enthalpy change.

$$\sum_{\text{out}} \dot{m}_j h_j - \sum_{\text{in}} \dot{m}_i h_i = \dot{Q} - \dot{W} \tag{1.16}$$

The subscripts i and j were adopted to differentiate between reactants and products. In the context of combustion, there are chemical reactions, and the species present in the products and reactants are not the same. Finally, integrating Eq. (1.16) for a time interval Δt, the expression shown in Eq. (1.17) is obtained.

$$\sum_{\text{out}} m_j h_j - \sum_{\text{in}} m_i h_i = Q_{1-2} - W_{1-2} \tag{1.17}$$

1.3.1 Thermodynamic Properties of Ideal Gases

The total enthalpy (h) of a substance is given in terms of the sensible enthalpy (Δh) and the enthalpy of formation (h_f°), as shown in Eq. (1.18). It is known from thermodynamics that the sensible enthalpy is measuring the change in enthalpy related to temperature and pressure, while the enthalpy of formation is a measure of the energy necessary to form a chemical species at given temperature and pressure. The convention is that species in their natural state have enthalpies of formation equal to zero, for example, hydrogen in gas state.

$$h_i = h_{f,i}^0 + \Delta h_i \tag{1.18}$$

The sensible enthalpy of an ideal gas depends only on the temperature and can be written in polynomial form, as shown in Eq. (1.19). The enthalpy of formation of an ideal gas can be obtained from the NIST–JANAF thermochemical tables [2]. Moreover, the total enthalpy can be directly calculated using the NASA Glenn Polynomials [3], as shown in Eq. (1.20).

$$\Delta h = a_0 + a_1 T + a_2 T^2 + a_3 T^3 + a_4 T^4 + a_5 T^5 \tag{1.19}$$

$$\frac{h(T)}{RT} = -a_1 T^{-2} + a_2 \frac{\ln T}{T} + a_3 + a_4 \frac{T}{2} + a_5 \frac{T^2}{3} + a_6 \frac{T^3}{4} + a_7 \frac{T^4}{5} + \frac{b_1}{T} \tag{1.20}$$

Note that in Eq. (1.20), the units are dictated by the gas constant (R). In the form written above, R is in J/kg K and the enthalpy in J/kg. The gas constant is obtained from the universal gas constant divided by the species molecular weight ($R = R_u/W$). From the definition of enthalpy and the equation of state for ideal gases ($pv = RT$), it is possible to relate the internal energy to the total enthalpy, as shown in Eq. (1.21).

$$h = u + pv = u + RT \tag{1.21}$$

The entropy (S), Helmholtz free energy (F) and Gibbs free energy (G) are defined by the widely known relationships shown in Eq. (1.22).

$$TdS = dH - Vdp; \quad F = U - TS; \quad G = H - TS \tag{1.22}$$

The NASA Glenn Polynomials [3] for the determination of the entropy and specific heat capacity at constant pressure of ideal gases are given in Eqs. (1.23) and (1.24).

$$\frac{s(T)}{RT} = -a_1 \frac{T^{-2}}{2} + a_2 T^{-1} + a_3 \ln T + a_4 T + a_5 \frac{T^2}{2} + a_6 \frac{T^3}{3} + a_7 \frac{T^4}{4} + b_2 \tag{1.23}$$

$$\frac{c_p(T)}{R} = a_1 T^{-2} + a_2 T^{-1} + a_3 + a_4 T + a_5 T^2 + a_6 T^3 + a_7 T^4 \tag{1.24}$$

The values of the coefficients to be used in Eqs. (1.20), (1.23), and (1.24) can be obtained from the original source [3]. Also, the specific heat capacity at constant volume for an ideal gas can be easily determined by $R = c_p - c_v$.

1.3.2 Constant Pressure Combustion

A combustion process at constant pressure can be analyzed by applying the first law of thermodynamics in the form given by Eq. (1.16). It is normally assumed that the combustion process is adiabatic and that there is not work being done. Therefore, the following is obtained.

$$\sum_{\text{products}} m_j h_j - \sum_{\text{reactants}} m_i h_i = 0 \tag{1.25}$$

The above expression is written in mass basis. However, it can also be written in molar basis as shown in Eq. (1.26).

$$\sum_{\text{products}} n_j \bar{h}_j - \sum_{\text{reactants}} n_i \bar{h}_i = 0 \tag{1.26}$$

Depending on the problem at hand, it could be easier to work with Eq. (1.26) because it is directly related to the global combustion reaction. For instance, consider the reaction of a lean mixture shown in Eq. (1.12). The application of Eq. (1.26) to that reaction gives the expression shown below.

$$\begin{aligned} \bar{h}_{\mathrm{F}} + \frac{v_{air}^s}{\phi}\bar{h}_{\mathrm{O_2}} + 3.76\frac{v_{air}^s}{\phi}\bar{h}_{\mathrm{N_2}} = {} & x_{\mathrm{C}}\bar{h}_{\mathrm{CO_2}} + \frac{x_{\mathrm{H}}}{2}\bar{h}_{\mathrm{H_2O}} + \left(\frac{v_{air}^s}{\phi} - v_{ar}^s\right)\bar{h}_{\mathrm{O_2}} \\ & + x_{\mathrm{S}}\bar{h}_{\mathrm{SO_2}} + \left(3.76\frac{v_{air}^s}{\phi} + \frac{x_{\mathrm{N}}}{2}\right)\bar{h}_{\mathrm{N_2}} \end{aligned} \tag{1.27}$$

The above expression corresponds to the reaction of 1 mol of fuel. The reactants are at a known unburned temperature (T_u), while the products are at the adiabatic flame temperature (T_b). The value of T_b can be obtained by solving Eq. (1.27). Note that for rich mixtures, the determination of T_b involves the necessity of determining the number of moles of each species in the products.

1.3.3 Constant Volume Combustion

The first law of thermodynamics given in Eq. (1.14) can be applied to a combustion process at constant volume. When the process is adiabatic and there is no work being done, the process can be represented by any of the two expressions shown below.

$$\sum_{\text{products}} m_j u_j - \sum_{\text{reactants}} m_i u_i = 0 \tag{1.28}$$

$$\sum_{\text{products}} n_j \bar{u}_j - \sum_{\text{reactants}} n_i \bar{u}_i = 0 \tag{1.29}$$

Considering a lean global reaction, as shown in Eq. (1.12), the adiabatic flame temperature for this process can be determined by solving Eq. (1.30).

$$\begin{aligned} \bar{u}_{\mathrm{F}} + \frac{v_{air}^s}{\phi}\bar{u}_{\mathrm{O_2}} + 3.76\frac{v_{air}^s}{\phi}\bar{u}_{\mathrm{N_2}} = {} & x_{\mathrm{C}}\bar{u}_{\mathrm{CO_2}} + \frac{x_{\mathrm{H}}}{2}\bar{u}_{\mathrm{H_2O}} + \left(\frac{v_{air}^s}{\phi} - v_{air}^s\right)\bar{u}_{\mathrm{O_2}} \\ & + x_{\mathrm{S}}\bar{u}_{\mathrm{SO_2}} + \left(3.76\frac{v_{air}^s}{\phi} + \frac{x_{\mathrm{N}}}{2}\right)\bar{u}_{\mathrm{N_2}} \end{aligned} \tag{1.30}$$

Example 1.2
Determine the adiabatic flame temperature of a stoichiometric mixture of ethanol and air with initial temperature and pressure of 298.15 K and 1 atm, respectively.

Consider the processes at constant pressure and constant volume. Use average specific heat values to obtain a first approximation.

Solution

The first step is to write the overall combustion reaction considering complete combustion.

$$C_2H_5OH + 3(O_2 + 3.76\ N_2) \rightarrow 2CO_2 + 3H_2O + 11.28N_2$$

(a) **Adiabatic flame temperature at constant pressure**

The first law of thermodynamics is written by using Eq. (1.26). The total enthalpy is divided into two terms, the formation enthalpy and the sensible enthalpy. In turn, the sensible enthalpy is obtained by assuming average heat capacities at constant pressure.

$$\begin{aligned} \overline{h}^0_{f,f} + \overline{c}_{p,f}(T_i - T_r) & \qquad 2\left[\overline{h}^0_{f,CO_2} + \overline{c}_{p,CO_2}(T_b - T_r)\right] \\ +3\left[\overline{h}^0_{f,O_2} + \overline{c}_{p,O_2}(T_i - T_r)\right] & = +3\left[\overline{h}^0_{f,H_2O} + \overline{c}_{p,H_2O}(T_b - T_r)\right] \\ +11.28\left[\overline{h}^0_{f,N_2} + \overline{c}_{p,N_2}(T_i - T_r)\right] & \quad +11.28\left[\overline{h}^0_{f,N_2} + \overline{c}_{p,N_2}(T_b - T_r)\right] \end{aligned}$$

The reference temperature is generally $T_r = 298$ K; thus, the sensible enthalpies of the reactants are all zero. Moreover, the formation enthalpies of elements in their natural states, like O_2 and N_2, are considered equal to zero. Applying these simplifications and isolating T_b, the following expression is obtained.

$$T_b = \frac{\overline{h}^0_{f,f} - 2\overline{h}^0_{f,CO_2} - \overline{h}^0_{f,H_2O}}{2\overline{c}_{p,CO_2} + 3\overline{c}_{p,H_2O} + 11.28\overline{c}_{p,N_2}} + T_r$$

Equivalently, it is possible to define an average specific heat capacity at constant pressure for the products mixture by calculating the total number of moles of the products and the mole fractions of each species. This information can be easily obtained by following the procedures shown in Example 1.1.

$$n_T\left[y_{CO_2}\overline{c}_{p,CO_2} + y_{H_2O}\overline{c}_{p,H_2O} + y_{N_2}\overline{c}_{p,N_2}\right] = n_T\overline{c}_{p,pr}$$

Then, the adiabatic flame temperature will be determined as

$$T_b = \frac{\overline{h}^0_{f,f} - 2\overline{h}^0_{f,CO_2} - 3\overline{h}^0_{f,H_2O}}{n_T\overline{c}_{p,pr}} + T_r$$

The formation enthalpies at 298 K and 1 atm are obtained from the NIST–JANAF thermochemical tables [2].

$$\overline{h}^0_{f,C_2H_5OH} = -234.95 \text{ kJ/mol}$$

$$\overline{h}^0_{f,CO_2} = -393.522 \text{ kJ/mol}$$

$$\overline{h}^0_{f,H_2O(g)} = -241.826 \text{ kJ/mol}$$

The heat capacities at constant pressure will be determined considering an average temperature of 1200 K; this is a simplification adopted as an example. From the tables it is obtained

$$\overline{c}_{p,CO_2} = 56,342 \text{ kJ/kmol K}$$

$$\overline{c}_{p,H_2O} = 43,768 \text{ kJ/kmol K}$$

$$\overline{c}_{p,N_2} = 33,723 \text{ kJ/kmol K}$$

The average heat capacity of the mixture of products is now calculated.

$$\overline{c}_{p,pr} = 0.1229(56.342) + 0.1842(43.768) + 0.6929(33.723) = 38.353 \text{ kJ/kmol K}$$

Substitution on the expression to determine the adiabatic flame temperature yields

$$T_b = \frac{-234.95 - 2(-393.522) - 3(-241.826)}{16.28\left(38.353 \times 10^{-3}\right)} + 298.15 = 2344.3 \text{ K}$$

(b) **Adiabatic flame temperature at constant volume**
In the case of the process at constant volume, it is necessary to apply Eq. (1.28). However, the internal energy should be obtained using Eq. (1.21). Therefore, considering $T_r = T_i$, the following is obtained.

$$\begin{aligned}\overline{h}^0_{f,f} - 15.28R_uT_i = &\ 2\left[\overline{h}^0_{f,CO_2} + \overline{c}_{p,CO_2}(T_b - T_r) - R_uT_b\right] \\ &+3\left[\overline{h}^0_{f,H_2O} + \overline{c}_{p,H_2O}(T_b - T_r) - R_uT_b\right] \\ &+11.28\left[\overline{c}_{p,N_2}(T_b - T_r) - R_uT_b\right]\end{aligned}$$

Note that the number of moles of the reactants is 15.28, and the number of moles of the products will be 16.28. The adiabatic flame temperature for the process at constant volume will be

$$T_b = \frac{\overline{h}^0_{f,f} - 15.28R_uT_i - 2\overline{h}^0_{f,\mathrm{CO_2}} - 3\overline{h}^0_{f,\mathrm{H_2O}}}{2\overline{c}_{p,\mathrm{CO_2}} + 3\overline{c}_{p,\mathrm{H_2O}} + 11.28\overline{c}_{p,\mathrm{N_2}} - 16.28R_u} + \frac{2\overline{c}_{p,\mathrm{CO_2}} + 3\overline{c}_{p,\mathrm{H_2O}} + 11.28\overline{c}_{p,\mathrm{N_2}}}{2\overline{c}_{p,\mathrm{CO_2}} + 3\overline{c}_{p,\mathrm{H_2O}} + 11.28\overline{c}_{p,\mathrm{N_2}} - 16.28R_u} T_r$$

The denominator of the above expression can be simplified as

$$n_T\overline{c}_{v,pr} = n_T\left(y_{\mathrm{CO_2}}\overline{c}_{p,\mathrm{CO_2}} + y_{\mathrm{H_2O}}\overline{c}_{p,\mathrm{H_2O}} + y_{\mathrm{N_2}}\overline{c}_{p,\mathrm{N_2}} - R_u\right) = n_T\left(\overline{c}_{p,pr} - R_u\right)$$

Therefore,

$$T_b = \frac{\overline{h}^0_{f,f} - 15.28R_uT_i - 2\overline{h}^0_{f,\mathrm{CO_2}} - 3\overline{h}^0_{f,\mathrm{H_2O}}}{n_T\overline{c}_{v,pr}} + \frac{\overline{c}_{p,pr}}{\overline{c}_{v,pr}} T_r$$

The specific heat capacities at constant pressure could be obtained at a higher temperature this time. However, the same values will be used for simplicity. Then,

$$\overline{c}_{v,pr} = 38.353 - 8.3144 = 30.039 \text{ kJ/kmol K}$$

Also, for simplicity first we calculate the term 15.28 R_uT_i.

$$15.28\ \ cR_uT_i = 15.28 \times 8.3144 \times 10^{-3} = 37.859$$

Substitution of the numerical values gives

$$T_b = \frac{-234.95 - 37.859 - 2(-393.522) - 3(-241.826)}{16.28\left(30.039 \times 10^{-3}\right)} + \frac{38.353}{30.039} 298 = 2915.5 \text{ K}$$

Therefore, it is observed that the adiabatic flame temperature at constant volume is higher than the adiabatic flame temperature at constant pressure. Since there is no flow work (pv), more energy is stored in the products. Note that the values of T_b could be more accurate by changing the selection of the average specific heat at constant pressure.

1.3.4 The Low Heating Value and the High Heating Value

The discussion about energy conservation is closed by introducing the low and high heating values. The low heating value (LHV) is defined as the energy that can be released by the complete combustion of a fuel at stoichiometric proportions in air, at reference conditions of temperature and pressure, given that the water in the products is in a vapor state. The high heating value (HHV) is defined as the energy that can be released by the complete combustion of a fuel at stoichiometric proportions in air, at reference conditions of temperature and pressure, given that the water in the products

is in a liquid state. Following the global combustion reaction given in Eq. (1.12), the LHV and HHV are given by Eqs. (1.31) and (1.32).

$$\text{LHV} = \frac{\overline{h}^{\circ}_{f,f} - x_{\text{C}}\overline{h}^{\circ}_{f,\text{CO}_2} - \frac{x_{\text{H}}}{2}\overline{h}^{\circ}_{f,\text{H}_2\text{O}(g)} - x_{\text{S}}\text{SO}_2}{W_f} \tag{1.31}$$

$$\text{HHV} = \frac{\overline{h}^{\circ}_{f,f} - x_{\text{C}}\overline{h}^{\circ}_{f,\text{CO}_2} - \frac{x_{\text{H}}}{2}\overline{h}^{\circ}_{f,\text{H}_2\text{O}(l)} - x_{\text{S}}\text{SO}_2}{W_f} \tag{1.32}$$

Example 1.3
Determine the low heating values of hydrogen, ammonia, ethanol, and methane. Also, calculate the low heating value in volume basis.

Solution
The adopted temperature and pressure conditions are 25 °C and 1 atm. The enthalpy of formation of the product gases, and also of ammonia and methane, is obtained from the NIST–JANAF thermochemical tables [2]. In the case of ethanol, the enthalpy of formation was obtained from the NIST Chemistry WebBook [4]. The following values are obtained for the enthalpies of formation.

$$\overline{h}^{\circ}_{f,\text{CO}_2} = -393.522\ \text{kJ/mol}; \quad \overline{h}^{\circ}_{f,\text{H}_2\text{O}(g)} = -241.826\ \text{kJ/mol};$$

$$\overline{h}^{\circ}_{f,\text{H}_2\text{O}(l)} = -285.830\ \text{kJ/mol}; \quad \overline{h}^{\circ}_{f,\text{H}_2} = 0.0\ \text{kJ/mol};$$

$$\overline{h}^{\circ}_{f,\text{NH}_3} = -45.898\ \text{kJ/mol}; \quad \overline{h}^{\circ}_{f,\text{C}_2\text{H}_5\text{OH}(g)} = -234.95\text{kJ/mol}$$

$$\overline{h}^{\circ}_{f,\text{CH}_4} = -74.873\ \text{kJ/mol}$$

Equation (1.30) is used to obtain the LHV.

$$\text{LHV}_{\text{H}_2} = \frac{0 - (-241.826)}{2} = 120.913\ \text{MJ/kg}$$

$$\text{LHV}_{\text{NH}_3} = \frac{-45.898 - 1.5(-241.826)}{17} = 18.63\ \text{MJ/kg}$$

$$\text{LHV}_{\text{C}_2\text{H}_5\text{OH}} = \frac{-234.95 - 2(-393.522) - 3(-241.826)}{46} = 27.78\ \text{MJ/kg}$$

$$\text{LHV}_{\text{CH}_4} = \frac{-74.873 - (-393.522) - 2(-241.826)}{16} = 50.14\ \text{MJ/kg}$$

So far, we have assumed that ethanol is in the gas phase to perform the calculation. However, it should be observed that ethanol would be in the liquid phase under

the reference conditions cited in this example. Therefore, the LHV should be calculated for its liquid phase. To do so, the enthalpy of vaporization (h_{fg}) is subtracted from the enthalpy of formation of ethanol in the gas phase. The value of h_{fg} was obtained from the NIST Chemistry WebBook [4].

$$\overline{h}^{0}_{f,C_2H_5OH(l)} = \overline{h}^{0}_{f,C_2H_5OH(g)} - h_{fg,C_2H_5OH} = -234.95 - 42.3 = -277.25 \text{ kJ/mol}$$

Therefore, for ethanol in liquid phase the LHV becomes

$$\text{LHV}_{C_2H_5OH(l)} = \frac{-277.25 - 2(-393.522) - 3(-241.826)}{46} = 26.86 \text{ MJ/kg}$$

To convert the LHV to a volume basis, we shall transform MJ/kg to MJ/Nm^3, in the case of gaseous fuels, and to MJ/L, in the case of liquid fuels. Therefore, the obtained values will be multiplied by the fuels' densities at 273.15 K and 101.325 kPa for gaseous fuels. From the equation of state of a perfect gas, the densities of hydrogen, ammonia, and methane will be given by the expression below.

$$\rho_f = \frac{p}{(R_u/W_f)T}$$

Using $R_u = 8.3144$ kJ/kmol K, the following values of fuel densities are obtained.

$$\rho_{f,H_2} = 0.089 \text{ kg/Nm}^3; \quad \rho_{f,NH_3} = 0.759 \text{ kg/Nm}^3; \quad \rho_{f,CH_4} = 0.713 \text{ kg/Nm}^3;$$

On the other hand, the density of ethanol at normal conditions (liquid state) can be adopted as shown below in kg/L, which is more adequate for liquid fuels.

$$\rho_{f,C_2H_5OH(l)} = 0.789 \text{ kg/L}$$

Then, the LHV in MJ/Nm^3 and MJ/L are

$$\text{LHV}_{H_2} = 120.913 \times 0.089 = 10.77 \text{ MJ/Nm}^3$$

$$\text{LHV}_{NH_3} = 18.63 \times 0.759 = 14.14 \text{ MJ/Nm}^3$$

$$\text{LHV}_{C_2H_5OH(l)} = 26.86 \times 0.789 = 21.20 \text{ MJ/L}$$

$$\text{LHV}_{CH_4} = 50.14 \times 0.713 = 35.74 \text{ MJ/Nm}^3$$

The low heating values obtained in this example show that, when compared on a mass basis, hydrogen presents the greatest LHV among the considered fuels. However, when compared on a volume basis, hydrogen presents the smallest LHV among the considered fuels. This apparent paradox is related to the molecular weight of the fuels.

1.4 Chemical Equilibrium

It is demonstrated in classical thermodynamics textbooks that, at equilibrium, the thermodynamic potentials are at a minimum. Therefore, the enthalpy, internal energy, Gibbs free energy, and Helmholtz free energy are all at a minimum at equilibrium [5, 6]. However, the natural variables of these potentials are different. The most convenient thermodynamic potential for equilibrium calculations at constant pressure is the Gibbs free energy, because its natural variables are temperature, pressure, and the number of moles of the species. On the other hand, the Helmholtz free energy has, as natural variables, the temperature, volume, and number of moles of the species, which makes it adequate for evaluating constant volume processes.

Starting from these observations, it is interesting to formulate the constant pressure combustion process in terms of the Gibbs free energy of formation (G) and the constant volume combustion process in terms of the Helmholtz free energy of formation (F).

1.4.1 The Gibbs Free Energy of Formation

The Gibbs free energy of formation of a chemical species ($\overline{g}^{\circ}_{f,T,i}$) is determined from the enthalpy of formation ($\overline{h}^{\circ}_{f,T,i}$), the entropy of formation ($\overline{s}^{\circ}_{f,T,i}$), and the temperature, as shown in Eq. (1.33).

$$\overline{g}^{\circ}_{f,T,i} = \overline{h}^{\circ}_{f,T,i} - T\left(\overline{s}^{\circ}_{f,T,i}\right) \tag{1.33}$$

The enthalpy of formation is already defined, and the entropy of formation is defined from the elements that form the species. For example, if the chemical species i is obtained from the elements j and k, at temperature T, the entropy of formation would be obtained by Eq. (1.34).

$$\overline{s}^{\circ}_{f,T,i} = \overline{s}^{\circ}_{T,i} - \left(\overline{s}^{\circ}_{T,j} + \overline{s}^{\circ}_{T,k}\right) \tag{1.34}$$

The NIST–JANAF thermochemical tables [2] present the values of the Gibbs free energy of formation for several species.

Example 1.4
Determine the Gibbs free energy of formation of carbon dioxide at 1500 K.

Solution
Consider the formation reaction of carbon dioxide (CO_2) from carbon (C) and oxygen (O_2). Note that the natural state of carbon is graphite.

$$C_{(S)} + O_{2(g)} \rightarrow CO_{2(g)}$$

The entropies of formation and the enthalpy of formation are obtained from the NIST–JANAF thermochemical tables [2].

$$\overline{s}^{\circ}_{T,CO_2} = 292.199 \text{ J/mol K}; \quad \overline{s}^{\circ}_{T,C_{(s)}} = 33.718 \text{ J/mol K}; \quad \overline{s}^{\circ}_{T,O_2} = 258.068 \text{ J/mol K}$$

$$\overline{h}^{\circ}_{f,T,CO_2} = -395.668 \text{ kJ/mol K}$$

Applying Eqs. (1.32) and (1.33), the following is obtained.

$$\overline{s}^{\circ}_{f,T,CO_2} = 292.199 - (33.718 + 258.068) = 0.413 \text{ J/mol K}$$

$$\overline{g}^{\circ}_{f,T,i} = -395.668 - 1500\frac{0.413}{1000} = -396.288 \text{ kJ/mol K}$$

Checking the NIST–JANAF tables, we can see that the tabulated value is the same as that obtained in this example.

Similarly to what has been done with the absolute enthalpy, it is possible to divide the Gibbs free energy of a chemical species into two parts, the first one representing the Gibbs free energy at a given temperature and constant pressure and the second one representing the variation of the Gibbs free energy due to pressure [7]. This is shown in Eq. (1.35).

$$\overline{g}_{T,i} = \overline{g}^{\circ}_{T,i} + RT \ln\left(\frac{p_i}{p^{\circ}}\right) \tag{1.35}$$

Again, by analogy with the enthalpy of formation, the first term on Eq. (1.35) can be understood as having two contributions: one from the Gibbs free energy of formation and the other from the variation of the Gibbs free energy when the temperature is different than that of formation. This is shown in Eq. (1.36).

$$\overline{g}^{\circ}_{T,i} = \overline{g}^{\circ}_{f,T,i} + \Delta\overline{g}^{\circ}_{T,i} \tag{1.36}$$

However, equilibrium calculations in combustion consider that the Gibbs free energy is taken at the final temperature of the process; therefore, $\Delta\overline{g}^{\circ}_{T,i} = 0$ and $\overline{g}^{\circ}_{T,i} = \overline{g}^{\circ}_{f,T,i}$.

1.4.2 The Variation of the Gibbs Free Energy in a Reaction

The variation of the Gibbs free energy in a reaction is also important for the thermodynamics of combustion. Consider the arbitrary reaction presented in Eq. (1.37) where chemical species A and B react to form chemical species C and D.

$$v_A A + v_B B \rightarrow v_C C + v_D D \tag{1.37}$$

The variation of the Gibbs free energy for this reaction is given by Eq. (1.38).

$$\Delta G_T^\circ = \left(v_C \overline{g}_{f,T,C}^\circ + v_D \overline{g}_{f,T,D}^\circ\right) - \left(v_A \overline{g}_{f,T,A}^\circ + v_B \overline{g}_{f,T,B}^\circ\right) \tag{1.38}$$

In general terms, the variation is given by Eq. (1.39).

$$\Delta G_T^\circ = \sum_{i=1}^{P} v_i \overline{g}_{f,T,i}^\circ - \sum_{j=1}^{R} v_j \overline{g}_{f,T,j}^\circ \tag{1.39}$$

1.4.3 The Equilibrium Constant

The context of application of the equilibrium condition in combustion processes at constant pressure can be understood more clearly if a reaction with species of incomplete combustion, such as CO and H_2, is considered. The combustion reaction of a rich mixture of an arbitrary fuel and air is written as shown in Eq. (1.40), where an arbitrary number of chemical species was considered in the products.

$$\begin{aligned} C_{x_C} H_{x_H} O_{x_O} + v_{ar}(O_2 + 3.76\ N_2) \rightarrow n_{CO_2} CO_2 + n_{CO} CO + n_{H_2O} H_2O \\ + n_{H_2} H_2 + n_{N_2} N_2 + \cdots \end{aligned} \tag{1.40}$$

If the species in the products are at thermodynamic equilibrium, the temperature, pressure, and composition of the mixture no longer change. However, from the point of view of chemical reactions, this means that the reactions proceed at the same rate in both directions. For example, within the mixture of gases produced by combustion, carbon monoxide could react with water vapor to form carbon dioxide and hydrogen and vice versa. This reaction is known as the water–gas homogeneous reaction and is given in Eq. (1.41).

Water–gas homogeneous reaction

$$CO + H_2O \leftrightarrow CO_2 + H_2 \tag{1.41}$$

If the species in the products were not at equilibrium, the concentrations of CO_2 and H_2 could still increase at the expense of the CO and H_2O content. However, at equilibrium the concentrations do not change since CO_2 and H_2 are produced at the same rate at which they are consumed by the water–gas homogeneous reaction. The rate of reactions will be studied in more detail in Chap. 2. In order to take advantage of the equilibrium condition, the equilibrium constant of a reaction will be obtained. The reaction is rewritten for 1 mol of species A.

$$A + \frac{v_B}{v_A} B \rightarrow \frac{v_C}{v_A} C + \frac{v_D}{v_A} D \tag{1.42}$$

The equilibrium condition states that $dG = 0$. Under conditions of constant pressure and temperature, the variation in Gibbs free energy is related to the variation in the number of moles of chemical species in the reaction.

$$dG_{T,p} = \sum \overline{g}_{T,i} dn_i = 0 \tag{1.43}$$

In particular, for the reaction shown in Eq. (1.42)

$$dG_{T,p} = \overline{g}_{T,C} dn_C + \overline{g}_{T,D} dn_D + \overline{g}_{T,A} dn_A + \overline{g}_{T,B} dn_B = 0 \tag{1.44}$$

When the number of moles of reactant A presents a variation, then, the variation of the number of moles of the other species is determined by the stoichiometry of the reaction given in Eq. (1.41). In this case, reactant A is being consumed by the reaction; thus, the following can be written.

$$dn_A = -dn; \quad dn_B = -\frac{v_B}{v_A} dn; \quad dn_C = \frac{v_C}{v_A} dn; \quad dn_D = \frac{v_D}{v_A} dn$$

Using the above into Eq. (1.44) yields

$$dG_{T,p} = \left(\frac{v_C}{v_A} \overline{g}_{T,C} + \frac{v_D}{v_A} \overline{g}_{T,D} - \overline{g}_{T,A} - \frac{v_B}{v_A} \overline{g}_{T,B} \right) dn = 0 \tag{1.45}$$

Considering the species to be perfect gases, the expression given in Eq. (1.35) is used to express their Gibbs free energies. Substituting these expressions, multiplying by v_A/dn, and rearranging gives the expression in Eq. (1.46).

$$\begin{aligned} &- \frac{1}{RT} \left(v_C \overline{g}^{\circ}_{f,T,C} + v_D \overline{g}^{\circ}_{f,T,D} - v_A \overline{g}^{\circ}_{f,T,A} - v_B \overline{g}^{\circ}_{f,T,B} \right) \\ &= \left[v_C \ln \left(\frac{p_C}{p^{\circ}} \right) + v_D \ln \left(\frac{p_D}{p^{\circ}} \right) - v_A \ln \left(\frac{p_A}{p^{\circ}} \right) - v_B \ln \left(\frac{p_B}{p^{\circ}} \right) \right] \end{aligned} \tag{1.46}$$

It can be seen that the left-hand side of the equation above is the variation of the Gibbs free energy of the reaction. By performing some algebraic operations, the following expression is obtained.

$$K_p = \exp\left(- \frac{\Delta G_T}{RT} \right) \tag{1.47}$$

The equilibrium constant (K_p) is also given by Eq. (1.48).

$$K_p = \frac{\left(\frac{p_C}{p^{\circ}} \right)^{v_C} \left(\frac{p_D}{p^{\circ}} \right)^{v_D}}{\left(\frac{p_A}{p^{\circ}} \right)^{v_A} \left(\frac{p_B}{p^{\circ}} \right)^{v_B}} \tag{1.48}$$

It is observed that the equilibrium constant depends on the variation of the Gibbs free energy of the reaction. The value of the equilibrium constant can also be related to the number of moles of each species in the combustion products by multiplying by p/p, where p is the total pressure of the combustion gas mixture. This procedure yields

$$\frac{p_i}{p^\circ} = \frac{p_i}{p}\frac{p}{p^\circ} = \frac{n_i}{n_P}\frac{p}{p^\circ}$$

Therefore, Eq. (1.48) can also be written as shown in Eq. (1.49).

$$K_p = \frac{(n_C)^{\nu_C}(n_D)^{\nu_D}}{(n_A)^{\nu_A}(n_B)^{\nu_B}}\left(\frac{1}{n_P}\frac{p}{p^\circ}\right)^{\nu_C+\nu_D-\nu_A-\nu_B} \tag{1.49}$$

In general terms, the equilibrium constant can be represented as shown below.

$$K_p = \frac{\prod_{i=1}^{P}(n_i)^{\nu_i}}{\prod_{j=1}^{R}(n_j)^{\nu_j}}\left(\frac{1}{n_P}\frac{p}{p^\circ}\right)^{\sum\nu_i-\sum\nu_j} \tag{1.50}$$

It should be noted that the products and reactants in Eq. (1.50) are those directly related to the simple reaction that is taking place among some of the species in the products. An example of such is the water–gas homogeneous reaction, where the reactants are CO and H_2O and the products are CO_2 and H_2.

Example 1.5

Determine the equilibrium constant for the reactions shown below at 2000 K.

Water–gas homogeneous reaction

$$CO + H_2O \leftrightarrow CO_2 + H_2$$

Carbon monoxide dissociation reaction

$$CO_2 \leftrightarrow CO + \frac{1}{2}O_2$$

Methane reforming reaction

$$CH_4 + H_2O = CO + 3H_2$$

Solution

First, the variations in the Gibbs free energy of the reactions are determined, and then the equilibrium constant is calculated using this information. The Gibbs free energies of formation were obtained from the NIST–JANAF thermochemical tables [2].

Water–gas homogenous reaction

$$\Delta G_T^0 = (-396.333 + 0) - (-286.034 - 135.528) = 25.229 \text{ kJ/mol}$$

$$K_P = \exp\left(-\frac{25.229}{8.3144 \times 10^{-3} \times 2000}\right) = 0.219$$

Carbon monoxide dissociation reaction

$$\Delta G_T^0 = (-286.034 + 0.5 \times 0) - (-396.333) = 110.299 \text{ kJ/mol}$$

$$K_P = \exp\left(-\frac{110.229}{8.3144 \times 10^{-3} \times 2000}\right) = 1.322 \times 10^{-3}$$

Methane reforming reaction

$$\Delta G_T^0 = (-286.034 + 3 \times 0) - (130.802 - 135.528) = -281.614 \text{ kJ/mol}$$

$$K_P = \exp\left(-\frac{-281.308}{8.3144 \times 10^{-3} \times 2000}\right) = 2.222 \times 10^7$$

Example 1.6
A mixture of ethanol and air with an equivalence ratio of 1.4 undergoes a combustion process at constant pressure, with the initial temperature of ethanol being 298.15 K and that of air being 400 K. Determine the adiabatic flame temperature and the composition of the combustion products.

Solution
Although this problem could be easily solved using a computational routine, it is interesting to understand the use of the thermodynamic equilibrium condition in a constant pressure combustion process. Therefore, the solution will be obtained using tables. We will first determine the air coefficient.

$$v_{air} = \frac{v_{ar}^s}{\phi} = \frac{3}{1.4} = 2.142$$

The overall combustion reaction can now be written. In order to keep the problem simple, it is assumed that nitrogen is inert and that the incomplete combustion species will be CO and H_2.

$$C_2H_5OH + 2.142(O_2 + 3.76\ N_2) \rightarrow n_{CO_2}CO_2 + n_{CO}CO + n_{H_2O}H_2O + n_{H_2}H_2 + 8.054\ N_2$$

A mass balance is performed for the elements.

$$n_{CO_2} + n_{CO} = 2 \qquad \text{(a)}$$

$$n_{H_2O} + n_{H_2} = 3 \tag{b}$$

$$2n_{CO_2} + n_{CO} + n_{H_2O} = 5.284 \tag{c}$$

In this case, the mass conservation of the elements is not enough to balance the global combustion reaction. Conversely, for stoichiometric and lean combustion, the mass conservation of the elements would have been enough. There are five unknowns in this problem: the adiabatic flame temperature and the number of moles of CO_2, CO, H_2O, and H_2.

An energy balance is performed between the reactants and products. The sensible enthalpy of ethanol will be zero at 298.15 K, and the enthalpies of formation of the elements in their natural states are zero. Using the absolute enthalpy notation, the following is obtained.

$$H_R = H_P \tag{d}$$

$$H_p = n_{CO_2}\overline{h}_{CO_2} + n_{CO}\overline{h}_{CO} + n_{H_2O}\overline{h}_{H_2O} + n_{H_2}\Delta\overline{h}_{H_2} + 8.054\Delta\overline{h}_{N_2}$$

$$H_R = \overline{h}^0_{f,f} + 2.142\left(\Delta\overline{h}_{O_2} + 3.76\Delta\overline{h}_{N_2}\right)$$

At the moment we have four equations and five unknowns. To obtain a new equation, we consider the water–gas homogenous reaction.

$$CO + H_2O \leftrightarrow CO_2 + H_2$$

The equilibrium constant equation, given in Eq. (1.49), provides the fifth equation.

$$K_P = \frac{n_{CO_2} n_{H_2}}{n_{CO} n_{H_2O}} \tag{e}$$

Where, from Eqs. (1.47) and (1.38),

$$K_p = \exp\left(-\frac{\Delta G_T}{RT}\right)$$

$$\Delta G^\circ_T = \left(\overline{g}^\circ_{f,T,CO_2} + \overline{g}^\circ_{f,T,H_2}\right) - \left(\overline{g}^\circ_{f,T,CO} + \overline{g}^\circ_{f,T,H_2O}\right)$$

Since Eqs. (a), (b), (c), and (d) are linear, we can express all the numbers of moles of the species in the products as a function of the number of moles of a single species. We choose the number of moles of CO_2 to obtain the solution.

$$n_{CO} = 2 - n_{CO_2} \tag{f}$$

$$n_{H_2O} = 3.284 - n_{CO_2} \tag{g}$$

$$n_{H_2} = n_{CO_2} - 0.284 \tag{h}$$

Substitution of Eqs. (f)–(h) into Eq. (e) yields

$$(1-K_P)(n_{CO_2})^2+(5.284K_P-0.284)n_{CO_2}-6.568K_P=0 \tag{i}$$

Substitution of Eqs. (f)–(h) into Eq. (d) yields

$$H_R=H_P=\begin{array}{c} n_{CO_2}\overline{h}_{CO_2}+(2-n_{CO_2})\overline{h}_{CO}+(3.284-n_{CO_2})\overline{h}_{H_2O} \\ +(n_{CO_2}-0.284)\Delta\overline{h}_{H_2}+8.054\Delta\overline{h}_{N_2} \end{array} \tag{j}$$

Equation (i) gives the number of moles of CO_2 if the equilibrium constant is known. Since the equilibrium constant is temperature dependent, we must assume values for the adiabatic flame temperature to determine the number of moles of CO_2. Then, the enthalpy of the products, given in Eq. (j), is calculated to verify that the number of moles of CO_2 is the correct value.

The enthalpy of the reactants is a known value. For this we obtain the enthalpy values from the tables.

$$H_R=-234.95+2.142\times(3.025+3.76\times2.971)=-204.542\text{ kJ}$$

First iteration: adopting a temperature of 2000 K and reference of 298.15 K

Species	$\overline{h}^0_{f,i}$ (kJ/mol)	$\Delta\overline{h}_i$ (kJ/mol)	$\overline{h}_i$ (kJ/mol)	$\overline{g}^\circ_{f,T,i}$
CO_2	−393.522	91.439	−302.083	−396.333
CO	−110.527	56.744	−53.783	−286.034
H_2O	−241.826	72.790	−169.036	−135.528
H_2	0.0	52.951	52.951	0.0
N_2	0.0	56.137	56.137	0.0

The equilibrium constant is calculated.

$$\Delta G^\circ_T=(-396.333+0)-(-286.034-135.528)=25.229\text{ kJ/mol}$$

$$K_p=\exp\left(-\frac{25.299}{8.3144\times10^{-3}\times2000}\right)=0.22$$

Solving Eqs. (i) and (j)

$$n_{CO_2}=0.91\text{ mol}$$

$$H_P=-249.536\text{ kJ}$$

$$H_R-H_P=-204.542-(-249.536)=44.994\text{ kJ}$$

Since the enthalpy of the reactants is greater than the enthalpy of the products, we must increase the temperature value.

Second iteration: adopting a temperature of 2200 K and reference of 298.15 K

Species	$\overline{h}^0_{f,i}$ (kJ/mol)	$\Delta\overline{h}_i$ (kJ/mol)	$\overline{h}_i$ (kJ/mol)	$\overline{g}^\circ_{f,T,i}$
CO_2	−393.522	103.562	−289.960	−396.262
CO	−110.527	64.021	−46.506	−302.672
H_2O	−241.826	83.153	−158.673	−123.905
H_2	0.0	59.876	59.876	0.0
N_2	0.0	63.361	63.361	0.0

The equilibrium constant is calculated.

$$\Delta G^\circ_T = 30.315 \text{ kJ/mol}$$

$$K_p = \exp\left(-\frac{30.315}{8.3144 \times 10^{-3} \times 2200}\right) = 0.19$$

Solving Eqs. (i) and (j)

$$n_{CO_2} = 0.874 \text{ mol}$$

$$H_P = -142.557 \text{ kJ}$$

$$H_R - H_P = -204.542 - (-142.557) = -61.985 \text{ kJ}$$

For this temperature we have that the enthalpy of the products is greater than that of the reactants. We perform an iteration to determine the temperature between 2000 and 2200 K.

$$T_b = 2200 - (-142.557 - [-204.542])\left(\frac{2200 - 2000}{-142.557 - [-249.536]}\right)$$

$$T_b = 2084 \text{ K}$$

Third iteration: adopting a temperature of 2084 K and reference of 298.15 K

Species	$\overline{h}^0_{f,i}$ (kJ/mol)	$\Delta\overline{h}_i$ (kJ/mol)	$\overline{h}_i$ (kJ/mol)	$\overline{g}^\circ_{f,T,i}$
CO_2	−393.522	96.520	−297.002	−396.308
CO	−110.527	59.794	−50.733	−293.038
H_2O	−241.826	77.117	−164.709	−130.650
H_2	0.0	55.846	55.846	0.0
N_2	0.0	59.166	59.166	0.0

The equilibrium constant is calculated.

$$\Delta G^\circ_T = 27.38$$

$$K_p = \exp\left(-\frac{27.38}{8.3144 \times 10^{-3} \times 2084}\right) = 0.206$$

Solving Eqs. (i) and (j)

$$n_{CO_2} = 0.893$$

$$H_P = -204.670$$

$$H_R - H_P = -204.542 - (-204.670) = 0.128$$

The difference between H_R and H_P is now small. Therefore, a temperature of 2084 K can be adopted as the adiabatic flame temperature. In fact, another iteration on the temperature gives a new value of 2084.24 K which for practical purposes does not represent a large difference.

Finally, solving Eqs. (f)–(h) we obtain the concentration of the products.

Species	n_i (mol)	%Vol
CO_2	0.893	6.84
CO	1.107	8.49
H_2O	2.391	18.31
H_2	0.609	4.66
N_2	8.054	61.70
Total	13.054	100.00

1.4.4 The Minimization of Thermodynamic Potentials

The product gas composition and adiabatic flame temperature of constant pressure and constant volume combustion processes can also be studied by applying the method of minimization of thermodynamic potentials using the Lagrange multipliers [8]. The following global combustion reaction is adopted.

$$\begin{aligned} & C_{x_C} H_{x_H} O_{x_O} + v_{air}(O_2 + 3.76\ N_2) \\ & \quad n_{CO_2} CO_2 + n_{CO} CO + n_{H_2O} H_2O + n_{H_2} H_2 + n_H H \\ \rightarrow & \quad + n_{O_2} O_2 + n_O O + n_{OH} OH + n_{CH_2O} CH_2O \\ & \quad + n_{CH_4} CH_4 + n_{NO} NO + n_{N_2} N_2 \end{aligned} \tag{1.51}$$

Applying the conservation of mass of the elements, the following equations are obtained.

$$n_{CO_2} + n_{CO} + n_{CH_2O} + n_{CH_4} - x_C = 0 \tag{1.52}$$

$$n_{H_2O} + n_{H_2} + n_{CH_2O} + 2n_{CH_4} + \frac{1}{2}(n_H + n_{OH}) - \frac{x_H}{2} = 0 \tag{1.53}$$

$$n_{CO_2} + n_{O_2} + \frac{1}{2}(n_{CO} + n_{H_2O} + n_O + n_{OH} + n_{CH_2O} + n_{NO}) - \frac{x_O}{2} - v_{air} = 0 \tag{1.54}$$

$$\frac{1}{2}n_{NO} + n_{N_2} - 3.76v_{air} = 0 \tag{1.55}$$

A general form of the conservation of mass of the elements can be proposed, as shown in Eq. (1.56). In that equation the b_j represent the number of moles of the element j, while a_{ij} represent the number of molecules of element j in the molecule of species i. For example, in the case of CO_2, there would be 1 molecule of carbon and 2 molecules of oxygen.

$$\sum_{i=1}^{N} a_{ji}n_i - b_j = 0 \tag{1.56}$$

The first law of thermodynamics applied to the constant pressure process is written as shown in Eq. (1.26), while for the constant volume process, it is written as shown in Eq. (1.29). The minimization of the thermodynamic potential would produce the necessary equations to close the system. The constant pressure process is represented by the Gibbs free energy, while the constant volume process is represented by the Helmholtz free energy. At constant temperature, considering ideal gas and taking the summation for the N species in the mixture, the following expressions are obtained.

$$G = \sum_{i=1}^{N}\left(G^{\circ}_{T,i} + n_i RT \ln \frac{p_i}{p^{\circ}}\right) \tag{1.57}$$

$$F = \sum_{i=1}^{N}\left(F^{\circ}_{T,i} - n_i RT \ln \frac{V}{V^{\circ}}\right) \tag{1.58}$$

Each element in the system will have a Lagrange multiplier, λ_i. Then, taking the summation over the K elements in Eq. (1.56), the function Eq. (1.59) is obtained. Moreover, two functions are defined, as shown in Eq. (1.60).

$$L = \sum_{j=1}^{K}\lambda_j\left(\sum_{i=1}^{N} a_{ji}n_i - b_j\right) = 0 \tag{1.59}$$

$$\psi_1 = \frac{G}{RT} - L; \quad \psi_2 = \frac{F}{RT} - L \tag{1.60}$$

The partial derivatives of the functions ψ_1 and ψ_2, with respect to n_i and λ_i, will be zero at the minimum. It is important to notice that the partial derivatives with respect to λ_i will produce the mass conservation relations shown in Eqs. (1.52)–(1.55). On the other hand, the expressions for the partial derivatives with respect to n_i will have the form shown below.

Constant pressure combustion

$$\frac{\partial \psi_1}{\partial n_i} = \left(\frac{\overset{\circ}{\mu}_{T,i}}{RT} + \ln \frac{n_i}{n_T} - \frac{n_i}{n_T} + \ln \frac{p}{p^\circ} + 1 \right) - \frac{\partial}{\partial n_i} \left[\sum_{j=1}^{K} \lambda_j \left(\sum_{i=1}^{N} a_{ji} n_i - b_j \right) \right] = 0 \tag{1.61}$$

Constant volume combustion

$$\frac{\partial \psi_2}{\partial n_i} = \left(\frac{\overset{\circ}{\mu}_{T,i}}{RT} + \ln \frac{n_i RT}{Vp^\circ} + 1 \right) - \frac{\partial}{\partial n_i} \left[\sum_{j=1}^{K} \lambda_j \left(\sum_{i=1}^{N} a_{ji} n_i - b_j \right) \right] = 0 \tag{1.62}$$

The chemical potential μ_i is defined as shown in Eq. (1.63).

$$\left(\frac{\partial G}{\partial n_i} \right)_{T,p,n_{j \neq i}} = \left(\frac{\partial A}{\partial n_i} \right)_{T,V,n_{j \neq i}} = \mu_i \tag{1.63}$$

In both cases, the variables are the number of moles of the species, the Lagrange multipliers, and the adiabatic flame temperature. Therefore, a total of $N + K + 1$ equations are necessary to close the system. This means that there are N equations (1 for each species in the products) of the form shown in Eqs. (1.61) and (1.62). As an example, the equations for CO_2 are shown below.

Constant pressure combustion

$$\frac{\overset{\circ}{\mu}_{T,CO_2}}{RT} + \ln \frac{n_{CO_2}}{n_T} + \ln \frac{p}{p^\circ} + 1 - \lambda_C - \lambda_{O_2} = 0 \tag{1.64}$$

Constant volume combustion

$$\frac{\overset{\circ}{\mu}_{T,CO_2}}{RT} + \ln \left(n_{CO_2} \frac{RT}{Vp^\circ} \right) + 1 - \lambda_C - \lambda_{O_2} = 0 \tag{1.65}$$

Example 1.7

Analyze the adiabatic flame temperature for combustion processes at constant pressure and constant volume of hydrogen, ammonia, and ethanol with air at an initial temperature of 298 K. For constant pressure combustion, consider a pressure of 101.325 kPa, and for constant volume combustion, consider a volume of 0.0105 m^3.

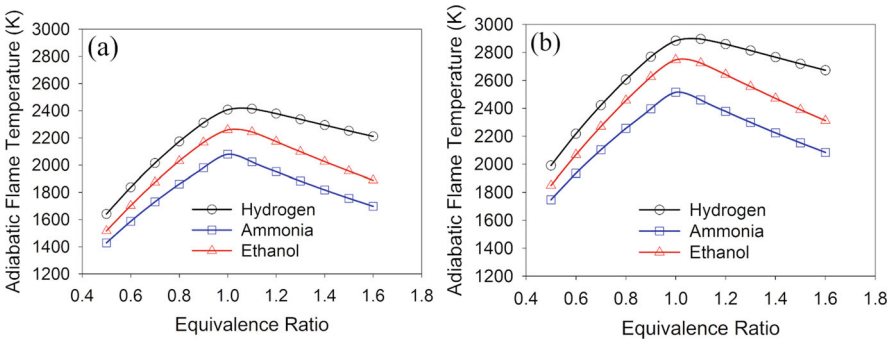


Fig. 1.1 Comparison of adiabatic flame temperatures for (**a**) constant pressure and (**b**) constant volume processes

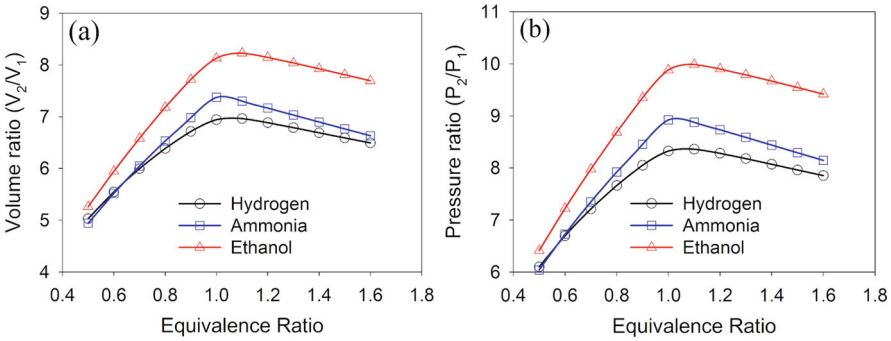


Fig. 1.2 Comparison of (**a**) volume ratio for a constant pressure process and (**b**) pressure ratio for a constant volume process

Solution

To solve this problem, it is necessary to implement a computational routine which is capable of solving the system of equations described in Sect. 1.4.4.

The variation of the adiabatic flame temperature will be analyzed by considering the equivalence ratio ranging from 0.5 to 1.6. The results of these calculations are depicted in Figs. 1.1 and 1.2. It is clearly observed that the T_b for the processes at constant volume is higher than those obtained at constant pressure. Moreover, the T_b of hydrogen is greater than the T_b of ethanol and ammonia. For instance, at $\phi = 1$ the T_b for H_2 is 2407 K, for CH_5OH is 2259 K and for NH_3 is 2080 K. As will be seen in Chap. 2, the formation of NO strongly depends on temperature, its production rate being enhanced at higher temperatures.

Other important parameters to be determined are the volume ratio for a constant pressure process and the pressure ratio for a constant volume process. These ratios compare the value of the property at the product's thermodynamic state with the value of the property at the reactant's thermodynamic state. Note that the volume and pressure ratios are greater for ethanol, followed by ammonia and hydrogen.

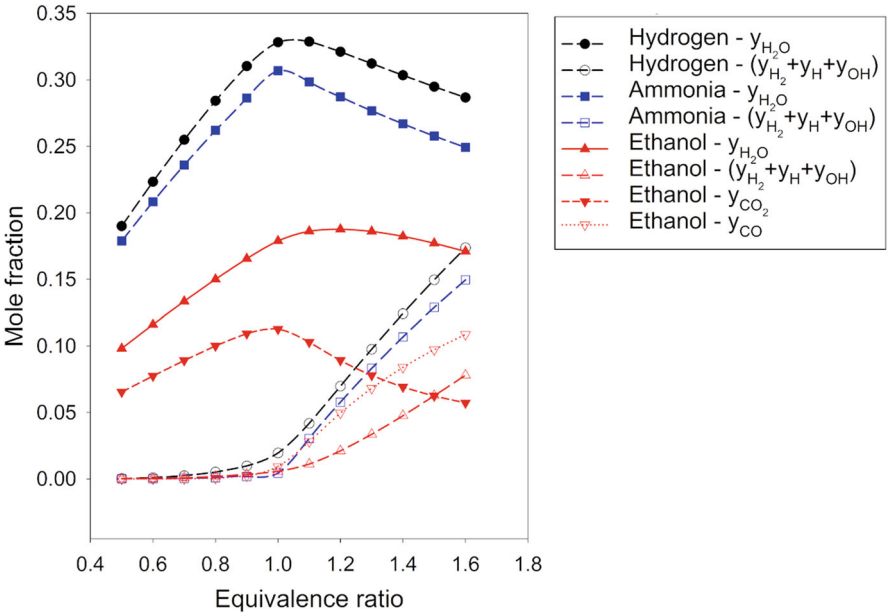


Fig. 1.3 Variation of species mole fractions for the constant pressure combustion process of hydrogen, ammonia, and ethanol

The variation of the mole fractions of different species in the products, for the combustion process at constant pressure, is shown in Fig. 1.3.

Hydrogen and ammonia do not produce CO_2 or CO; therefore, these species are only shown for ethanol combustion. The minor species in the products, H_2, H, and OH were summed over and represent the extent of incomplete combustion. Note that the mole fractions of species of incomplete combustion increase with increasing equivalence ratio, as expected, because there is not enough oxygen to convert these species into CO_2 or H_2O. There are other species, like O_2, O, N_2, and NO, that were not shown in Fig. 1.3.

1.5 Closure

This chapter presented the thermodynamics of combustion. The mass conservation of the elements was applied to balance the global combustion reaction of stoichiometric and lean mixtures. In this context, the concepts of stoichiometric, lean, and rich mixtures were introduced together with the definition of the equivalence ratio. Also, energy conservation was applied to determine the adiabatic flame temperature of global combustion reactions, considering constant pressure and constant volume processes. The approach was demonstrated for constant heat capacities and for variable enthalpies. The equilibrium condition was applied to study incomplete combustion reactions, which are characteristic of rich mixtures. The equilibrium

condition was applied to the products by following two methods, namely, the method of the equilibrium constant and the Gibbs free energy minimization method. The latter method is more adequate to analyze global reactions with many species in the products. The examples presented in this chapter showed the application of thermodynamic theory to the combustion processes of hydrogen, ammonia, and ethanol.

Chemical Kinetics

2

Nomenclature

Symbols

C_i	Concentration of species i, kmol/m^3
E_a	Activation energy, kJ/kmol
k_f	Specific rate constant in the forward direction, m^3/(kmol s)
k_b	Specific rate constant in the reverse direction, m^3/(kmol s)
K_P	Equilibrium constant
$\dot{m}_{R,i}$	Reaction rate of species i in mass basis, kg/(m^3 s)
M	Representation of an arbitrary species in an elementary reaction
n_i	Number of moles of species i, mol
R	Universal ideal gas constant, kJ/(kmol K)
t	Time, s
T	Temperature, K
T_a	Activation temperature, K
v	Flow velocity, m/s
$v_{i,\text{diff}}$	Diffusion velocity of species i, m/s
υ_i	Stoichiometric coefficient of species i in an elementary reaction
υ^s_{air}	Stoichiometric coefficient of air (or oxygen) in a one-step reaction
W_i	Molecular weight of species i, kg/kmol
Y_i	Mass fraction of species i
z_i	Catalytic efficiency of species i

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Greek Letters

ρ Density, kg/m^3
ω_k Reaction rate of the elementary reaction k, kmol/(m^3 s)
$\widehat{\omega}_i$ Reaction rate of species i, kmol/(m^3 s)

Subscripts

P Products
R Reactants
prod Produced
cons Consumed

2.1 Introduction

This chapter is aimed to present the fundamental aspects of chemical kinetics which are applied to deflagration and detonation phenomena. In this regard, the chapter will start by giving brief descriptions of hydrocarbons, alcohols, and free radicals. This chapter also introduces the definitions necessary to understand the chemical kinetics of combustion, that is, elementary chemical reactions, their types, and the representation of the reaction rates of the chemical species. The mathematical relations will be formed by introducing the law of mass action and Arrhenius Law. The time evolution of chemical species concentrations is also presented, and explicit expressions for species concentrations as a function of time will be obtained for unimolecular, bimolecular, and consecutive reactions. The relation between the forward and reverse specific reaction rate constants is also explained. Since the reaction rates of elementary reactions can also depend on the pressure, Lindeman's theory is also discussed. Finally, the reaction mechanisms, their postulation, particularities, and reduction techniques are introduced. For further discussion the classical textbooks by Kuo [9], Glassman et al. [10], Law [11], Lewis and von Elbe [12], Coelho and Costa [13], and Turns [7] are recommended.

2.2 Fuels and Chemical Species

2.2.1 Hydrocarbons

The hydrocarbons can be categorized in alkanes (C_nH_{2n+n}), alkenes (C_2H_{2n}), alkynes (C_2H_{2n-2}), and aromatics. The alkanes do not present multiple bonds between the carbon atoms. The alkanes present at least one double bond between carbon atoms and the alkynes have at least one triple bond between carbon atoms. Some examples are presented in Fig. 2.1.

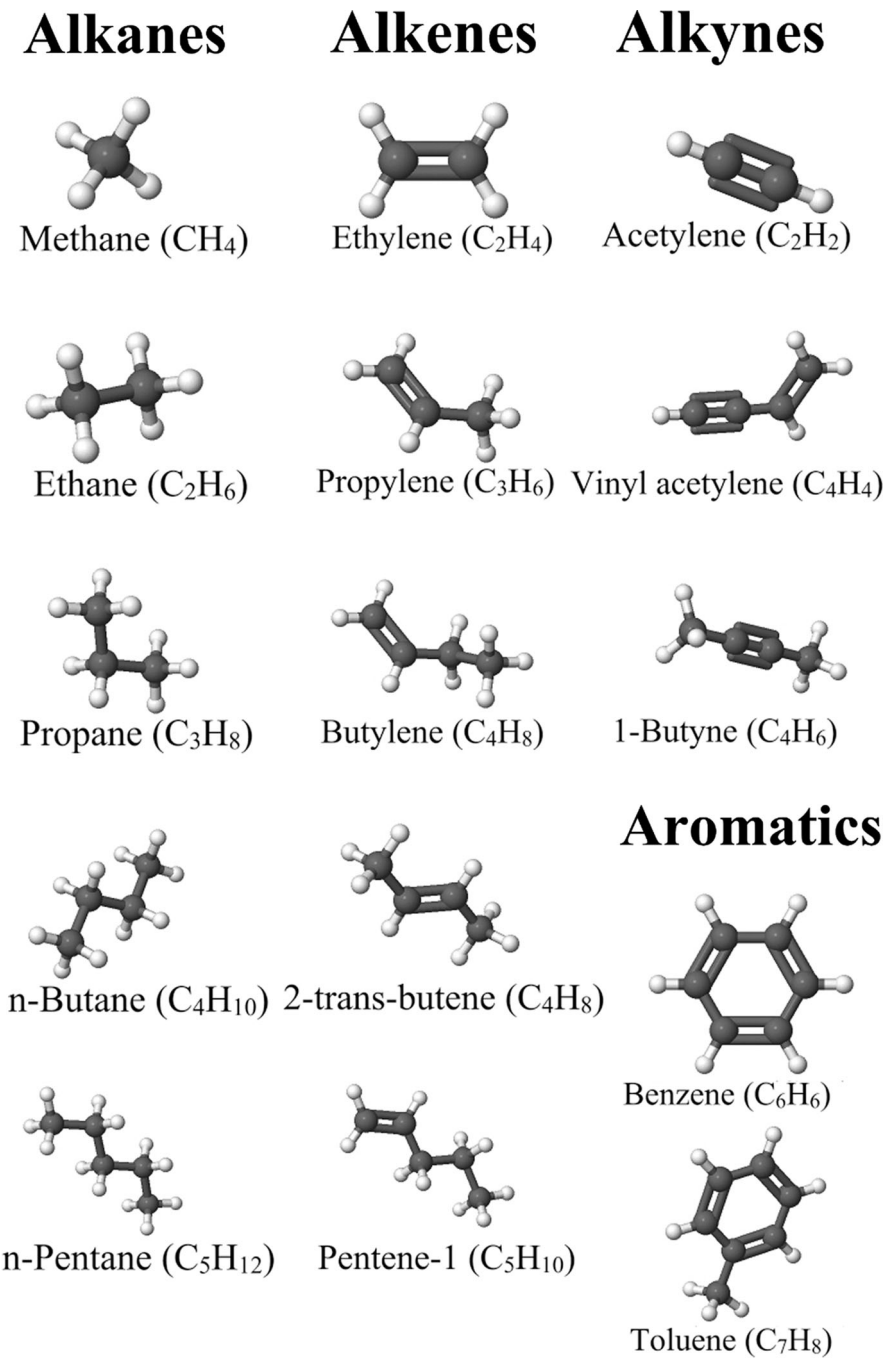


Fig. 2.1 Examples of hydrocarbon molecules. Molecule graphs generated by ChemSpider website [14]

The alkyl radicals, represented by the letter R, are obtained when a hydrogen atom is removed from an alkane molecule. Some examples of these radicals are the methyl radical (CH_3), the ethyl radical (C_2H_5), the propyl radical (C_3H_7), and the iso-propyl radical (C_3H_7). Therefore, an alkane can be represented in general as R–H. The phenyl group is a benzene ring bonded to another group of atoms in a molecule. The phenyl radical (C_6H_5) appears when a hydrogen atom is removed from a benzene ring. This group is generally represented as Ph. The benzyl group ($C_6H_5CH_2$) is formed by the combination of a phenyl group and a methylene group (CH_2). In general, the benzyl group is represented by Bn.

2.2.2 Alcohols

An alcohol is a species which contains the functional group hydroxyl (OH). Examples of alcohols are methanol (CH_3OH), ethanol (C_2H_5OH), and propanol (C_3H_7OH). These are shown in Fig. 2.2. A primary alcohol is one in which the OH group is bonded to a carbon atom that is also bonded to another carbon atom. A secondary alcohol has the OH group bonded to a carbon atom that is in turn bonded to another two carbon atoms. Finally, a tertiary alcohol has the OH group bonded to a carbon atom that is also bonded to three other carbon atoms.

2.2.3 Free Radicals

The radicals are very important for the combustion process to take place because they are (mostly) highly reactive species that appear and disappear very fast. Moreover, a radical is a chemical species with free electrons [15]. Consider the example shown in Fig. 2.3. In that figure, species A and B are bonded through a covalent bond, and upon a homolysis reaction, they form two free radicals.

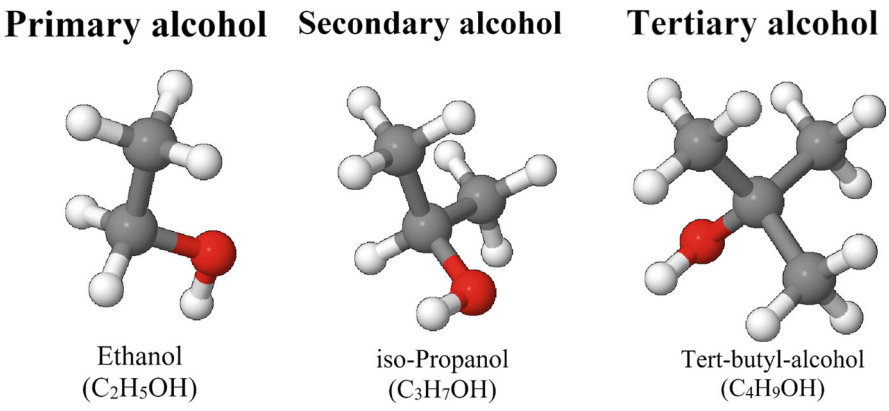


Fig. 2.2 Examples of alcohol molecules. Molecule graphs generated by ChemSpider website [14]



Fig. 2.3 Examples of free radicals that are formed by homolysis

2.3 Definitions and Fundamental Concepts

2.3.1 Elementary Chemical Reactions

The global effect of the chemical reaction between a fuel and an oxidant can be expressed in terms of a global reaction, like those shown in Eqs. (2.1) and (2.2), for hydrogen and ethanol, respectively, reacting with oxygen.

$$H_2 + \frac{1}{2}O_2 \rightarrow H_2O \tag{2.1}$$

$$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O \tag{2.2}$$

The above reactions describe only the initial and final compositions of the combustion process. However, they do not give any information regarding the path followed to arrive from the initial to the final composition. The path of a combustion reaction is determined by a number of elementary chemical reactions, like that shown in Eq. (2.3). In this case, a hydrogen molecule (H_2) reacts with an oxygen molecule (O_2) to form a hydrogen atom (H) and a hydroperoxyl radical (HO_2).

$$H_2 + O_2 \leftrightarrow HO_2 + H \tag{2.3}$$

In the above reaction, the bond between the hydrogen atoms in the H_2 molecule is broken by the collision between H_2 and O_2 molecules. This liberates the H atom and allows the formation of a new bond between O_2 and H, thus forming the HO_2 radical. A chemical kinetics description of the combustion process involves the adoption of a mechanism comprising dozens, hundreds, or even thousands of elementary reactions.

The elementary reaction can be understood by considering a gas mixture. The molecules in this mixture are in constant motion and collide with each other with a certain frequency. The opportunity for a chemical reaction to take place is the instant when two or three molecules collide. This opportunity is a probability that a chemical reaction might occur under given conditions. Note that collisions of more than three molecules are infrequent.

2.3.2 Types of Elementary Chemical Reactions

In view of the number of molecules involved in an elementary chemical reaction, these can be classified as unimolecular, bimolecular, or three-molecular elementary chemical reactions. As aforementioned, elementary reactions involving more than three molecules are infrequent. As an example, consider the reactions below, which involve hydrogen.

Unimolecular reaction

$$H_2 \rightarrow H + H \tag{2.4}$$

Bimolecular reaction

$$H_2 + O \rightarrow OH + H \tag{2.5}$$

Three-molecular reaction

$$H + O_2 + M \rightarrow HO_2 + M \tag{2.6}$$

Note that M is a third body participating in the reaction. This molecule could be any species present in the mixture. The elementary chemical reactions take place in time, and they consume one or more species to form other species. Therefore, if a species is quantified through its concentration (C_i), the rate of change of its concentration is linked to the chemical reaction and is usually called reaction rate (dC_i/dt). Considering the reaction given in Eq. (2.5), the necessary reaction rates are dC_{H2}/dt, $dC_O/_{dt}$, dC_{OH}/dt, and dC_H/dt. In the next subsections, these reaction rates will be further studied. In the SI system, the reaction rates are given in mol/(cm^3 s) or kmol/(m^3 s).

2.3.3 Relation Between Reaction Rates

The bimolecular reaction presented in Eq. (2.5) can be analyzed by noting the proportion between reactants and products. For instance, for 1 mol of H_2 consumed by the reaction, there is 1 mol of O_2 also consumed, and 1 mol of HO_2 and H will be produced, respectively. Therefore, the following relation can be written.

$$-\frac{dn_{H_2}}{dt} = -\frac{dn_{O_2}}{dt} = \frac{dn_{HO_2}}{dt} = \frac{dn_H}{dt} \tag{2.7}$$

The negative signs in Eq. (2.7) were attributed to the species which are being consumed by the reaction. Assuming that the reactions take place in the same volume, then, the number of moles can be converted into concentrations by simply dividing through by the volume.

$$-\frac{dC_{H_2}}{dt} = -\frac{dC_{O_2}}{dt} = \frac{dC_{HO_2}}{dt} = \frac{dC_H}{dt} \tag{2.8}$$

A three-molecular reaction is presented in Eq. (2.9). In that reaction two hydrogen atoms react to form a hydrogen molecule. Note that a third hydrogen atom is participating as a third body; it participates in the collision, but it does not react.

$$\mathrm{H} + \mathrm{H} + \mathrm{H} \rightarrow \mathrm{H_2} + \mathrm{H} \tag{2.9}$$

Analyzing the above reaction, it is observed that 2 mol of H react to form 1 mol of H_2. Therefore, the following proportionality relation can be written.

$$-\frac{dC_{\mathrm{H}}}{dt} \propto \frac{dC_{\mathrm{H_2}}}{dt} \tag{2.10}$$

The proportionality between these reaction rates is defined in terms of the number of moles of each species participating in the elementary reaction.

$$-\frac{1}{2}\frac{dC_{\mathrm{H}}}{dt} = \frac{dC_{\mathrm{H_2}}}{dt} \tag{2.11}$$

Equation (2.11) states that the reaction rate of consumption of hydrogen atoms is twice faster than the reaction rate of production of hydrogen molecules. A generalization of the relation given in Eq. (2.11) can be obtained by introducing the elementary reaction shown in Eq. (2.12) and using the stoichiometric coefficients (υ_i) instead of the mole numbers in order to define the proportionality.

$$v_{a,R}\mathrm{A} + v_{b,R}\mathrm{B} + v_{c,R}\mathrm{C} + v_{d,R}\mathrm{D} \leftrightarrow v_{a,P}\mathrm{A} + v_{b,P}\mathrm{B} + v_{c,P}\mathrm{C} + v_{d,P}\mathrm{D} \tag{2.12}$$

The subscripts R and P refer to reactants and products, respectively. Then, the reaction rates are given by

$$\frac{1}{v_{a,P} - v_{a,R}}\frac{dC_{\mathrm{A}}}{dt} = \frac{1}{v_{b,P} - v_{b,R}}\frac{dC_{\mathrm{B}}}{dt} = \frac{1}{v_{c,P} - v_{c,R}}\frac{dC_{\mathrm{C}}}{dt} = \frac{1}{v_{d,P} - v_{d,R}}\frac{dC_{\mathrm{D}}}{dt} \tag{2.13}$$

An alternative way to represent the reaction rate is shown in Eq. (2.14). Then, the reaction rate of an elementary reaction can be given by Eq. (2.15).

$$\widehat{\omega}_{\mathrm{A}} = \frac{dC_{\mathrm{A}}}{dt} \tag{2.14}$$

$$\omega_k = \frac{\widehat{\omega}_{\mathrm{A}}}{v_{a,P} - v_{a,R}} = \frac{\widehat{\omega}_{\mathrm{B}}}{v_{b,P} - v_{b,R}} = \frac{\widehat{\omega}_{\mathrm{C}}}{v_{c,P} - v_{c,R}} = \frac{\widehat{\omega}_{\mathrm{D}}}{v_{d,P} - v_{d,R}} \tag{2.15}$$

The subscript k refers to an elementary reaction within a reaction mechanism. The reaction rate of a species represents the number of moles of the species, per unit volume, which are consumed (or produced) in a unit time by the reaction. On the other hand, the reaction rate of the reaction represents the number of moles, per unit volume, which would be consumed by unit time, if all the species had unitary stoichiometric coefficients.

Example 2.1
Consider the reactions shown below, and write the reaction rate of the reaction and the relation between the reaction rates of the species.

$$\mathrm{M} + \mathrm{Br_2} \rightarrow \mathrm{Br} + \mathrm{Br} + \mathrm{M}$$

$$\mathrm{Br} + \mathrm{H_2} \rightarrow \mathrm{HBr} + \mathrm{H}$$

$$\mathrm{H} + \mathrm{Br_2} \rightarrow \mathrm{HBr} + \mathrm{Br}$$

$$\mathrm{H} + \mathrm{HBr} \rightarrow \mathrm{H_2} + \mathrm{Br}$$

$$\mathrm{M} + 2\mathrm{Br} \rightarrow \mathrm{Br_2} + \mathrm{M}$$

Solution
The reactions (R1)–(R5) represent a chemical kinetics mechanism in which the following species participate: bromine (Br_2), bromine atom (Br), hydrogen (H_2), hydrogen atom (H), and hydrogen bromide (HBr). The third body M is not reacting, but it collides with the other molecules. Therefore, the following is obtained by applying the developments of this section.

$$\omega_{\mathrm{R1}} = -\widehat{\omega}_{\mathrm{Br_2}} = \frac{1}{2}\widehat{\omega}_{\mathrm{Br}} \tag{a}$$

$$\omega_{\mathrm{R2}} = -\widehat{\omega}_{\mathrm{Br}} = -\widehat{\omega}_{\mathrm{H_2}} = \widehat{\omega}_{\mathrm{HBr}} = \widehat{\omega}_{\mathrm{H}} \tag{b}$$

$$\omega_{\mathrm{R3}} = -\widehat{\omega}_{\mathrm{H}} = -\widehat{\omega}_{\mathrm{Br_2}} = \widehat{\omega}_{\mathrm{HBr}} = \widehat{\omega}_{\mathrm{Br}} \tag{c}$$

$$\omega_{\mathrm{R4}} = -\widehat{\omega}_{\mathrm{H}} = -\widehat{\omega}_{\mathrm{HBr}} = \widehat{\omega}_{\mathrm{H_2}} = \widehat{\omega}_{\mathrm{Br}} \tag{d}$$

$$\omega_{\mathrm{R5}} = -\frac{1}{2}\widehat{\omega}_{\mathrm{Br}} = \widehat{\omega}_{\mathrm{Br_2}} \tag{e}$$

2.3.4 The Law of Mass Action

The law of mass action establishes phenomenologically that the reaction rate of a reaction is proportional to the product of the reactants' concentrations elevated to a power equal to their stoichiometric coefficients in the elementary reaction.

$$\omega \propto \prod_{i}^{N} C_i^{\nu_i} \tag{2.16}$$

The proportionality coefficient is known as the specific reaction rate constant (k_f) which is fundamentally a function of temperature. The subscript f denotes the forward direction of the reaction (from left to right).

$$\omega = k_f \prod_{i}^{N} C_i^{\nu_i} \tag{2.17}$$

Consider again the reactions shown in Eqs. (2.4)–(2.6), which are going to be denoted by the subscripts I, II, and III, respectively. Applying the law of mass action, the following relations are obtained.

$$\frac{dC_{H_2}}{dt} = -\omega_I = -k_{f,I} C_{H_2} \tag{2.18}$$

$$\frac{dC_{H_2}}{dt} = \frac{dC_O}{dt} = -\omega_{II} = -k_{f,II} C_{H_2} C_O \tag{2.19}$$

$$\frac{dC_H}{dt} = \frac{dC_{O_2}}{dt} = -\omega_{III} = -k_{f,III} C_H C_{O_2} C_M \tag{2.20}$$

Since the third body, M, can be any species present in the reaction, the concentration of the third body is usually calculated by the expression given in Eq. (2.21).

$$C_M = \sum_i^N z_i C_i \tag{2.21}$$

In the above expression, the catalytic efficiencies (z_i) must be known from empirical studies. The reaction order is defined as the sum of the exponents of the concentrations when the law of mass action is applied to a reaction. For example, Eq. (2.18) defines a first-order reaction, while Eq. (2.19) defines a second-order reaction.

2.3.5 Arrhenius Law

Arrhenius Law gives the dependence between the specific reaction rate constant and the temperature. Moreover, k_f also depends on a pre-exponential factor (A) and the activation energy (E_a), which is the energy necessary for the chemical reaction to occur.

$$k_f = A \exp\left(-\frac{E_a}{RT}\right) \tag{2.22}$$

The activation energy is determined experimentally by plotting the function $\ln k_f$ vs. $1/T$. For constant values of A and E_a, a linear relationship, as shown in Fig. 2.4, is obtained.

As observed in Fig. 2.4, the pre-exponential factor is found as the interception with the ordinate, while the activation energy, or more specifically $-E_a/R$, is found as the slope of the line. Equation (2.22) can be rewritten as shown below.

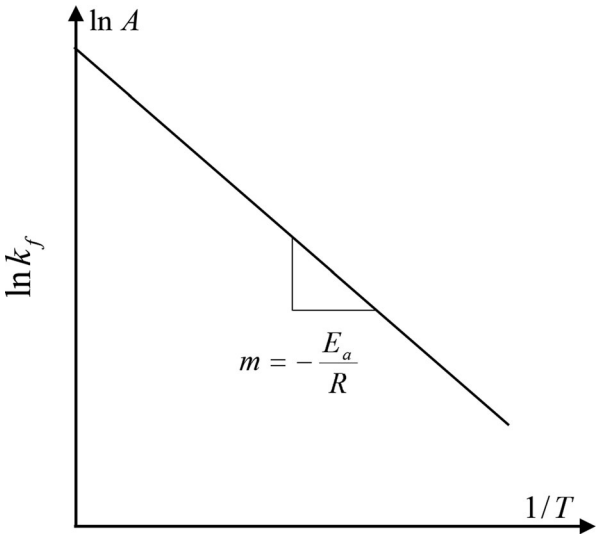


Fig. 2.4 Functional dependence between the activation energy and $\ln k_f$ and $1/T$

$$\frac{1}{k_f}\frac{dk_f}{dT} = \frac{E_a}{R}\frac{1}{T^2} \tag{2.23}$$

The quotient between the activation energy and the universal gas constant has temperature units (K). Therefore, the activation temperature (T_a) is defined as shown in Eq. (2.24).

$$T_a = \frac{E_a}{R} \tag{2.24}$$

Since in most cases the pre-exponential factor depends on temperature, Eq. (2.22) is modified by introducing a temperature dependence before the exponential.

$$k_f = BT^n \exp\left(-\frac{E_a}{RT}\right) \tag{2.25}$$

Example 2.2

Determine the activation energy (E_a) and the pre-exponential factor (A) of the reaction between species B and C to form species D using the data shown in the table. This problem was inspired by an example presented in the book by Annamalai and Puri [16]. The chemical reaction is shown below.

$$B + C \rightarrow 2D$$

T (K)	$\frac{dC_D}{dt}$ [kmol/(m^3 s)]	$\frac{1}{2}\frac{dC_D/dt}{C_B C_C}$ [m^3/(kmol s]
800	1.075×10^{-4}	6.456×10^{-1}
1000	4.720×10^{-1}	3.068×10^{3}
1200	7.240×10^{1}	5.068×10^{5}
1400	2.074×10^{3}	1.556×10^{7}

Solution

Arrhenius Law applied to the reaction establishes the following relationship.

$$\ln k_f = \ln A + \left(-\frac{E_a}{R}\right)\frac{1}{T} \tag{a}$$

The relationship between the reaction rates of reactants and products is also written as

$$-\frac{dC_B}{dt} = -\frac{dC_C}{dt} = \frac{1}{2}\frac{dC_D}{dt} \tag{b}$$

Also, applying the Arrhenius Law to determine the reaction rate of species D,

$$\frac{dC_D}{dt} = 2k_f C_B C_C \tag{c}$$

Isolating k_f from Eq. (c) and taking natural logarithm in both sides,

$$\ln k_f = \ln\left(\frac{1}{2}\frac{dC_D/dt}{C_B C_C}\right) \tag{d}$$

Then, in order to determine the interception and the slope, the method of minimum squares is applied. In this case, $x = 1/T$ e $y = \ln k_f$.

$$\ln A = \frac{\sum x_i^2 \sum y_i - \sum x_i \sum x_i y_i}{N\sum x_i^2 - \left(\sum x_i\right)^2} \tag{e}$$

$$\frac{E_a}{R} = -\frac{N\sum x_i y_i - \sum x_i \sum y_i}{N\sum x_i^2 - \left(\sum x_i\right)^2} \tag{f}$$

The following table can be obtained using the experimental data provided in the example.

N	x_i	y_i	$x_i y_i$	x_i^2

(continued)

N	x_i	y_i	$x_i y_i$	x_i^2
1	1.250×10^{-3}	-4.376×10^{-1}	-5.470×10^{-4}	1.563×10^{-6}
2	1.000×10^{-3}	8.029	8.029×10^{-3}	1.000×10^{-6}
3	8.333×10^{-4}	1.314×10^{1}	1.095×10^{-2}	6.944×10^{-7}
4	7.143×10^{-4}	1.656×10^{1}	1.183×10^{-2}	5.102×10^{-7}
$\sum$	3.798×10^{-3}	3.729×10^{1}	3.026×10^{-2}	3.767×10^{-6}

Therefore, substitution of the numerical values gives

$$\ln A = \frac{\left(3.767 \times 10^{-6}\right)(37.287) - \left(3.798 \times 10^{-3}\right)\left(3.026 \times 10^{-3}\right)}{4\left(3.767 \times 10^{-6}\right) - \left(3.798 \times 10^{-3}\right)^2}$$

The above equation was actually solved using spread sheet software; therefore, small differences may appear when compared to hand calculations.

$$A = \exp(39.527) = 1.1467 \times 10^{17}\ \mathrm{m^3/(kmol\ s)}$$

$$\frac{E_a}{R} = -\frac{4\left(3.026 \times 10^{-2}\right) - \left(3.798 \times 10^{-3}\right)(37.287)}{4\left(3.767 \times 10^{-6}\right) - \left(3.798 \times 10^{-3}\right)^2}$$

$$T_a = \frac{E_a}{R} = 31,815\ \mathrm{K}$$

$$E_a = 264.25\ \mathrm{kJ/mol}$$

The order of magnitude of the pre-exponential factor and of the activation energy is reasonable. For example, for the dissociation reaction $H_2 + M \rightarrow H + H + M$, the values are $A = 4.58 \times 10^{19}$ and $E_a = 435.16$ kJ/mol with an exponent $n = -1.40$ [10]. In the current example, the exponent n was not considered, since the unmodified Arrhenius expression was used.

2.3.6 Time Evolution of Concentration

The time evolution of species concentration of some simple reactions can be obtained. These are classical developments also available in other textbooks [9, 13]. However, in this case the analysis will be applied to specific reactions in order to show the application.

2.3.6.1 Unimolecular Reaction

Consider the unimolecular reaction shown in Eq. (2.4). In this case the integration is performed by first separating the variables in Eq. (2.18).

$$\frac{dC_{H_2}}{C_{H_2}} = -k_{f,I}dt \tag{2.26}$$

The above equation is integrated from the initial time $t_0 = 0$, at which the concentration is $C_{H2,0}$ to an arbitrary time and concentration. Therefore, integrating and rearranging,

$$C_{H_2} = C_{H_2,0}e^{-kt} \tag{2.27}$$

2.3.6.2 Bimolecular Reaction

Consider the bimolecular reaction given in Eq. (2.5). The integration in this case is not performed so easily as before. First, the relationship between the reaction rates is considered (Eq. 2.8). For the sake of simplicity in the analysis, only HO_2 is considered. The H atom in the products could be analyzed in the same manner.

At the beginning of the reaction, the concentrations of H_2 and O_2 are known and different than zero, that is, $C_{H2,0}$ and $C_{O2,0}$ are known. The concentration of HO_2 is also known and equal to zero, that is, $C_{HO2,0} = 0$. The following can be written.

$$-\frac{dC_{H_2}}{dt} = -\frac{dC_{O_2}}{dt} = \frac{dC_{HO_2}}{dt} \tag{2.28}$$

Integrating the above equations between the initial concentrations and any arbitrary concentration,

$$C_{H_2} - C_{H_2,0} = C_{O_2} - C_{O_2,0} = -C_{HO_2} \tag{2.29}$$

The reaction rate of HO_2 is given by a combination of Eqs. (2.8) and (2.19).

$$\frac{dC_{HO_2}}{dt} = k_{f,II}C_{H_2}C_O \tag{2.30}$$

It is noticed from Eq. (2.29) that $C_{H2} = C_{H2,0} - C_{HO2}$ and $C_{O2} = C_{O2,0} - C_{HO2}$. Then, differentiating these relationships gives

$$d(C_{H_2,0} - C_{HO_2}) = d(C_{O_2,0} - C_{HO_2}) = -dC_{HO_2} \tag{2.31}$$

The substitution of Eq. (2.31) into Eq. (2.30) allows the reaction rate of HO_2 to be expressed in terms of its own concentration.

$$\frac{dC_{HO_2}}{dt} = k_{f,II}(C_{H_2,0} - C_{HO_2})(C_{O_2,0} - C_{HO_2}) \tag{2.32}$$

Considering that $C_{H2,0} \neq C_{O2,0}$, multiplying both sides by $C_{H2,0} - C_{O2,0}$, and rearranging,

$$\frac{dC_{HO_2}}{dt}\frac{(C_{O_2,0}-C_{H_2,0})}{(C_{H_2,0}-C_{HO_2})(C_{O_2,0}-C_{HO_2})}=k_f(C_{O_2,0}-C_{H_2,0})$$

The following is observed.

$$\frac{(C_{O_2,0}-C_{H_2,0})}{(C_{H_2,0}-C_{HO_2})(C_{O_2,0}-C_{HO_2})}=\frac{1}{C_{H_2,0}-C_{HO_2}}-\frac{1}{C_{O_2,0}-C_{HO_2}}$$

Therefore,

$$\frac{dC_{HO_2}}{dt}\left(\frac{1}{C_{H_2,0}-C_{HO_2}}-\frac{1}{C_{O_2,0}-C_{HO_2}}\right)=k_f(C_{O_2,0}-C_{H_2,0}) \tag{2.33}$$

Substitution of Eq. (2.31) into Eq. (2.33) gives

$$-\frac{d(C_{H_2,0}-C_{HO_2})}{C_{H_2,0}-C_{HO_2}}+\frac{d(C_{O_2,0}-C_{HO_2})}{C_{O_2,0}-C_{HO_2}}=k_{f,II}(C_{O_2,0}-C_{H_2,0})dt \tag{2.34}$$

The integration is performed from $t_0 = 0$, when $C_{HO2,0} = 0$ to the arbitrary conditions of time and concentration of HO_2.

$$-[\ln(C_{H_2,0}-C_{HO_2})-\ln C_{H_2,0}]+[\ln(C_{O_2,0}-C_{HO_2})-\ln C_{O_2,0}]=k_{f,II}t(C_{O_2,0}-C_{H_2,0})$$

Rearranging,

$$\ln\left(\frac{C_{O_2,0}-C_{HO_2}}{C_{H_2,0}-C_{HO_2}}\right)-\ln\left(\frac{C_{O_2,0}}{C_{H_2,0}}\right)=k_{f,II}t(C_{O_2,0}-C_{H_2,0}) \tag{2.35}$$

Isolating $k_{f,II}$ and notting that $C_{H2} = C_{H2,0} - C_{HO2}$ and $C_{O2} = C_{O2,0} - C_{HO2}$, the following is obtained.

$$k_{f,II}=\frac{1}{t(C_{O_2,0}-C_{H_2,0})}\ln\left(\frac{C_{H_2,0}}{C_{O_2,0}}\frac{C_{O_2}}{C_{H_2}}\right) \tag{2.36}$$

Finally, from Eq. (2.29), it is possible to write

$$C_{O_2}=C_{H_2}+\left(C_{O_2,0}-C_{H_2,0}\right) \tag{2.37}$$

The substitution into Eq. (2.36) and some rearrangement give

$$C_{H_2}=C_{H_2,0}\left(\frac{C_{H_2,0}-C_{O_2,0}}{C_{H_2,0}-C_{O_2,0}\exp\left[k_{f,II}t(C_{O_2,0}-C_{H_2,0})\right]}\right) \tag{2.38}$$

The procedure shown here can be applied to any other bimolecular reaction. Also, note that H could have been used instead of HO_2. Representing H_2 by A, O_2 by B, and HO_2 by C, the general forms of Eqs. (2.36)–(2.38) can be obtained.

$$k_f = \frac{1}{t(C_{B,0} - C_{A,0})} \ln\left(\frac{C_{A,0}}{C_{B,0}} \frac{C_B}{C_A}\right) \tag{2.39}$$

$$C_A = C_{A,0}\left(\frac{C_{A,0} - C_{B,0}}{C_{A,0} - C_{B,0} \exp\left[k_f t(C_{B,0} - C_{A,0})\right]}\right) \tag{2.40}$$

$$C_B = C_A + (C_{B,0} - C_{A,0}) \tag{2.41}$$

Example 2.3
Initially 4 mol of hydroxyl radical (OH) and 6 mol of H_2 are present in a mixture. The bimolecular reaction between these species takes place to form H_2O and H. The experimentally determined parameters for the modified Arrhenius Law are given below [7]: $B = 1.5 \times 10^8$ $(cm^3/mol)^{n-1}/(s\ K^{\alpha})$, $n = 1.6$ e $E_a = 13.8$ kJ/mol. Determine the time necessary to consume 95% of the initial concentration of OH at constant temperature and pressure conditions of 2200 K and 101,325 kPa, respectively.

Solution
The elementary bimolecular reaction is shown below.

$$OH + H_2 \rightarrow H_2O + H$$

Since 95% of the OH radical will be consumed, it is possible to write $C_{OH} = 0.05\ C_{OH,\ 0}$. From Eq. (2.29), in general $C_B = C_A - C_{A,\ 0} + C_{B,\ 0}$. Therefore,

$$C_{H_2} = C_{H_2,0} - 0.95 C_{OH,0}$$

Rearranging Eq. (2.37), with A = OH and B = H_2.

$$t_{95\%} = \frac{1}{k_f(C_{H_2,0} - C_{OH,0})} \ln\left(\frac{C_{OH,0}}{C_{H_2,0}} \frac{C_{H_2}}{C_{OH}}\right)$$

The initial concentrations are now determined.

$$C_{OH,0} = \frac{y_{OH,0}p}{RT} = \frac{0.4 \times 101.325}{8.3144 \times 10^{-3} \times 2200} = 2.2158\ \text{mol/m}^3$$

$$C_{H_2,0} = \frac{y_{H_2,0}p}{RT} = \frac{0.6 \times 101.325}{8.3144 \times 10^{-3} \times 2200} = 3.3237\ \text{mol/m}^3$$

Applying the modified Arrhenius Law,

$$k_f = 1.5 \times 10^8 \times (2200)^{1.6} \exp\left(-\frac{13.8}{8.3144 \times 10^{-3} \times 2200}\right)$$
$$= 1.5716 \times 10^{13}\ \mathrm{cm}^3/(\mathrm{mol\ s})$$

Substitution into the expression for time gives

$$t_{95\%} = \frac{1}{1.5716 \times 10^{13}(3.3237 - 2.2158) \times 10^{-6}} \ln\left(\frac{2.2158}{3.3237} \times \frac{3.3237 - 0.95 \times 2.2158}{0.05 \times 2.2158}\right)$$

$$t_{95\%} = 1.1443 \times 10^{-7}\ \mathrm{s}.$$

2.3.6.3 Consecutive Reactions

Consecutive reactions take place when a set of reactants form products, and these go through another reaction in the sequence to form new products. An example of a reaction whose behavior cannot be explained by a direct elementary reaction is the formation of hydrogen peroxide (H_2O_2) and O_2 from two molecules of HO_2. According to Compton and Hancock [17], there is an intermediary reaction taking place, as shown below.

$$HO_2 + HO_2 \leftrightarrow \begin{matrix} H_2O_4{}^* \\ M\updownarrow \\ H_2O_4 \end{matrix} \rightarrow H_2O_2 + O_2 \tag{2.42}$$

Note that the tetraoxidane (H_2O_4) molecule interacts with a third body in order to activate the next step. A simpler reaction emulating that behavior is represented in general form in Eq. (2.43).

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C + D \tag{2.43}$$

Since the first reaction is a unimolecular reaction, the result shown in Eq. (2.27) can be adapted to obtain the concentration of A.

$$C_A = C_{A,0} e^{-k_1 t} \tag{2.44}$$

Since species B is being produced by the first reaction and consumed by the second reaction, its reaction rate is a combination of two terms, as shown in Eq. (2.45).

$$\frac{dC_B}{dt} = k_1 C_A - k_2 C_B = k_1 C_{A,0} e^{-k_1 t} - k_2 C_B \tag{2.45}$$

In order to integrate Eq. (2.45), it is interesting to use Laplace transforms. For instance,

$$L\left\{\frac{dC_B}{dt}\right\} + k_2 L\{C_B\} = k_1 C_{A,0} L\left\{e^{-k_1 t}\right\}$$

where the term-by-term transforms are

$$L\{C_B\} = Y(s); \quad L\left\{\frac{dC_B}{dt}\right\} = sY(s) - C_{B,0} = sY(s); \quad L\left\{e^{-k_1 t}\right\} = \frac{1}{s+k_1}$$

Then, the differential equation is reduced to the following expression.

$$sY(s) + k_2 Y(s) = k_1 C_{A,0} \frac{1}{s+k_1} \tag{2.46}$$

Solving the above equation for $Y(s)$ and applying partial fractions,

$$Y(s) = C_{A,0} k_1 \frac{1}{(s+k_1)(s+k_2)} = C_{A,0} \frac{k_1}{k_2 - k_1}\left(\frac{1}{s+k_1} - \frac{1}{s+k_2}\right) \tag{2.47}$$

Thus, taking the inverse Laplace transform, the following is observed.

$$L^{-1}\{Y(s)\} = C_B; \quad L^{-1}\left\{\frac{1}{s+k}\right\} = e^{-kt}$$

The expression of C_B is then obtained by substitution of the above expressions into Eq. (2.47).

$$C_B = C_{A,0} \frac{k_1}{k_2 - k_1}\left(e^{-k_1 t} - e^{-k_2 t}\right) \tag{2.48}$$

Moreover, the expression of C_C can also be obtained by noticing that $dC_C/dt = k_2 C_B$ and integrating this using the results shown in Eq. (2.48).

$$C_C = \frac{C_{A,0}}{k_2 - k_1}\left[k_2\left(1 - e^{-k_1 t}\right) - k_1\left(1 - e^{-k_2 t}\right)\right] \tag{2.49}$$

Although this reaction was presented as an example by Kuo [9], in this case the complete integration process has been developed. Another more complex example of a consecutive reaction would be the one presented in Eq. (2.50).

$$A + B \xrightarrow{k_1} AB \xrightarrow{k_2} C + D \tag{2.50}$$

Again, species AB is being produced by the first reaction and consumed by the second one. Therefore, the net reaction rate of AB is given by Eq. (2.51).

$$\frac{dC_{AB}}{dt} = k_1 C_A C_B - k_2 C_{AB} \tag{2.51}$$

Making $\Delta C_0 = C_{A,0} - C_{B,0}$, and using Eqs. (2.40) and (2.41), it is possible to write

$$C_A C_B = \frac{C_{A,0}^2 \Delta C_0^2}{(C_{A,0} - C_{B,0} e^{-k_1 \Delta C_o t})^2} - \frac{C_{A,0} \Delta C_0^2}{C_{A,0} - C_{B,0} e^{-k_1 \Delta C_0 t}} \tag{2.52}$$

The substitution into Eq. (2.51) and some rearrangements yield

$$\begin{aligned} \frac{dC_{AB}}{dt} + k_2 C_{AB} &= \frac{C_{A,0} \Delta C_0 (k_1 \Delta C_0 C_{B,0}) e^{-k_1 \Delta C_o t}}{(C_{A,0} - C_{B,0} e^{-k_1 \Delta C_o t})^2} \\ &= -C_{A,0} \Delta C_0 \frac{d}{dt} \left(\frac{1}{C_{A,0} - C_{B,0} e^{-k_1 \Delta C_0 t}} \right) \end{aligned} \tag{2.53}$$

The form chosen in the right-hand side of Eq. (2.53) is convenient for the application of the Laplace transform

$$\frac{1}{C_{A,0}} L\left\{ \frac{d}{dt} \left(\frac{1}{1 - (C_{B,0}/C_{A,0}) e^{-k_1 \Delta C_0 t}} \right) \right\} = \frac{1}{C_{A,0}} L\left\{ \frac{d}{dt} f(t) \right\} = \frac{1}{C_{A,0}} [sF(s) - f(0)]$$

Note that the function on the right-hand side of Eq. (2.53) is not easily transformed. However, it can be expressed as an expansion in power series up to the second power, as shown below.

$$\frac{1}{1 - (C_{B,0}/C_{A,0}) e^{-k_1 \Delta C_0 t}} = 1 + \left(\frac{C_{B,0}}{C_{A,0}} \right) e^{-k_1 \Delta C_0 t} + \left(\frac{C_{B,0}}{C_{A,0}} \right)^2 e^{-2k_1 \Delta C_0 t}$$

Applying the Laplace transform in Eq. (2.53), and noting that $C_{AB,0} = 0$, then,

$$\frac{Y(s)}{\Delta C_0} = -\left(\frac{C_{B,0}}{C_{A,0}} \right) \frac{s}{(s + k_1 \Delta C_0)(s + k_2)} - \left(\frac{C_{B,0}}{C_{A,0}} \right)^2 \frac{s}{(s + 2k_1 \Delta C_0)(s + k_2)} \tag{2.54}$$

The terms on the right-hand side of Eq. (2.53) can be separated by partial fraction, and then the inverse Laplace transform can be obtained.

$$\begin{aligned} Y(s) = &- \frac{(C_{B,0}/C_{A,0}) \Delta C_0}{1 - (k_1/k_2) \Delta C_0} \left(\frac{1}{s + k_2} - \frac{(k_1/k_2) \Delta C_0}{s + k_1 \Delta C_0} \right) \\ &- \frac{(C_{B,0}/C_{A,0})^2 \Delta C_0}{1 - (2k_1/k_2) \Delta C_0} \left(\frac{1}{s + k_2} - \frac{(2k_1/k_2) \Delta C_0}{s + 2k_1 \Delta C_0} \right) \end{aligned}$$

Applying Laplace transform and rearranging,

$$C_{AB} = \frac{k_1}{k_2} (C_{A,0} - C_{B,0}) (a_1 e^{-k_1 \Delta C_0 t} + 2a_2 e^{-2k_1 \Delta C_0 t}) - (a_1 + a_2) e^{-k_2 t} \tag{2.55}$$

where

$$a_1 = \frac{(C_{B,0}/C_{A,0})(C_{A,0} - C_{B,0})}{1 - (k_1/k_2)(C_{A,0} - C_{B,0})}; \quad a_2 = \frac{(C_{B,0}/C_{A,0})^2(C_{A,0} - C_{B,0})}{1 - (2k_1/k_2)(C_{A,0} - C_{B,0})} \tag{2.56}$$

This section is concluded with the above derivation. Another important aspect of chemical kinetics is that forward and backward reactions compete with each other. This will be the topic of the next section.

2.3.7 Reverse Reaction Rate

Considering the bimolecular reaction shown in Eq. (2.57), in general a reaction can take place in both directions. That is, the products C and D are formed when the reaction is taking place in the forward direction, and the products A and B are formed when the reaction is taking place in the reverse direction.

$$\nu_A A + \nu_B B \underset{k_b}{\overset{k_f}{\rightleftarrows}} \nu_C C + \nu_D D \tag{2.57}$$

Arrhenius Law allows for the determination of the forward specific rate constant, detonated by k_f. However, it is still necessary to determine the reverse reaction (right to left) which has a specific rate constant denoted by k_b. There is a direct relation between k_f and k_b that will be obtained in this section. Applying the law of mass action for the reaction rate of species A, the following is obtained.

$$\frac{dC_A}{dt} = -k_f C_A^{\nu_A} C_B^{\nu_B} + k_b C_C^{\nu_C} C_D^{\nu_D} \tag{2.58}$$

The first term on the right-hand side of Eq. (2.58) represents the rate of consumption of species A, while the second term is the rate of production of the species. In order to determine the relationship between k_f and k_b, the equilibrium condition is assumed, that is, $dC_A/dt = 0$. After some rearrangements,

$$\frac{k_f}{k_b} = \frac{C_C^{\nu_C} D_D^{\nu_D}}{C_A^{\nu_A} C_B^{\nu_B}} = K_C \tag{2.59}$$

A general form of writing K_C is shown below.

$$K_C = \frac{\prod_{i=1}^{P} (C_i)^{\nu_i}}{\prod_{j=1}^{R} (C_j)^{\nu_j}} \tag{2.60}$$

Equation (2.59) shows the relationship between k_f and k_b, which is a function of the species concentrations and the stoichiometric coefficients of the reaction. The value of K_C can be obtained from the equilibrium constant, defined in Chap. 1. The

equilibrium constant in terms of the number of moles and stoichiometric coefficients is given below.

$$K_p = \frac{\prod_{i=1}^{P} (n_i)^{\nu_i}}{\prod_{j=1}^{R} (n_j)^{\nu_j}} \left(\frac{1}{n_T}\frac{p}{p^\circ}\right)^{\sum \nu_i - \sum \nu_j} \tag{2.61}$$

On the right-hand side of Eq. (2.61), the n_i in the numerator is multiplied by $V^{\nu i}/V^{\nu i}$, and the n_j in the denominator are multiplied by $V^{\nu j}/V^{\nu j}$. In that manner, the concentrations of each species will appear in the equations. In addition, recognizing that $pV = n_T RT$, the expression shown in Eq. (2.62) is obtained.

$$K_p = \frac{\prod_{i=1}^{P} (C_i)^{\nu_i}}{\prod_{j=1}^{R} (C_j)^{\nu_j}} \left(\frac{RT}{p^\circ}\right)^{\sum \nu_i - \sum \nu_j} = K_C (RT/p^o)^{\sum \nu_i - \sum \nu_j} \tag{2.62}$$

Note that for a bimolecular reaction with unitary stoichiometric coefficients, as the one shown in Eq. (2.5), $K_p = K_C$.

Example 2.4
Obtain the reverse specific rate constant for the reaction shown below. Assume constant temperature and pressure of 2000 K and 1 atm. The empirical parameters for determining the forward specific rate constant are $B = 3.34 \times 10^4$ $(\text{cm}^3/\text{mol})^{\text{n-1}}/\text{s K}^{\alpha}$, $n = 2.42$ e $E_a = -80.76$ kJ/mol, as presented by Glassman et al. [10].

$$\text{OH} + \text{OH} \leftrightarrow \text{O} + \text{H}_2\text{O}$$

Solution
First the forward specific rate constant is obtained.

$$k_f = \left(3.34 \times 10^4\right)(2000)^{2.42} \exp\left[-\frac{-80.76}{8.3144 \times 10^{-3} \times 2000}\right]$$

$$k_f = 3.592 \times 10^{14}\ \text{cm}^3/(\text{mol s})$$

The next step is to determine the equilibrium constant. Therefore, the molar Gibbs free energies are obtained from thermodynamic tables.

$$\overline{g}^{\circ}_{f,\text{OH},T} = 9.197\ \text{kJ/mol}; \quad \overline{g}^{\circ}_{f,\text{O},T} = 121.552\ \text{kJ/mol}; \quad \overline{g}^{\circ}_{f,\text{H}_2\text{O},T} = -135.528\ \text{kJ/mol}$$

Then, from Chap. 1,

$$K_p = \exp\left(-\frac{\Delta G_T}{RT}\right) = \exp\left(-\frac{-135.528 + 121.552 - 2 \times 9.197}{8.3144 \times 10^{-3} \times 2000}\right) = 7.005$$

Since the stoichiometric coefficients are all unitary,

$$K_C = K_p = 7.005.$$

Finally, applying Eq. (2.58), it is observed that the k_f is seven times higher than k_b.

$$k_b = \frac{k_f}{K_C} = \frac{3.592 \times 10^{14}}{7.005} = 5129 \times 10^{13} \text{ cm}^3/(\text{mol s})$$

2.3.8 Pressure Dependence of Elementary Chemical Reactions

The dependence of the reaction rate with pressure can be explained by Lindeman's theory. The following two-step reaction is considered.

$$\text{A} + \text{M} \underset{k_{b1}}{\overset{k_{f1}}{\leftrightarrow}} \text{A}^* + \text{M} \tag{2.63}$$

$$\text{A}^* \xrightarrow{k_{f2}} \text{B} \tag{2.64}$$

In the above system, the reactant is molecule A, which is energized through collisions with molecules M. The energized molecule is denoted by A*. Afterwards, A* reacts to form the product, represented by molecule B. Applying the law of mass action to each one of the reactions in which A* participates and summing up all these rates, the net rate of reaction of A* is obtained.

$$\frac{dC_{\text{A}^*}}{dt} = k_{f1} C_{\text{A}} C_{\text{M}} - k_{b1} C_{\text{A}^*} C_{\text{M}} - k_{f2} C_{\text{A}^*} \tag{2.65}$$

It is possible to assume that the energized molecule A* is produced and consumed very fast. Therefore, the net reaction rate for this molecule could be assumed as equal to zero, that is, $dC_{\text{A}*}/dt = 0$.

$$C_{\text{A}^*} = \frac{k_{f1} C_{\text{A}} C_{\text{M}}}{k_{f2} + k_{b1} C_{\text{M}}} \tag{2.66}$$

Now consider the reaction rate of product B. The law of mass action is applied together with the expression for $C_{\text{A}*}$ given in Eq. (2.66).

$$\frac{dC_{\text{B}}}{dt} = k_{f2} C_{\text{A}^*} = \frac{k_{f2} k_{f1} C_{\text{M}}}{k_{f2} + k_{b1} C_{\text{M}}} C_{\text{A}} \tag{2.67}$$

The effect of pressure can be evaluated from Eq. (2.67). For considerably high pressures, the concentrations of all the species are increased. Particularly, the

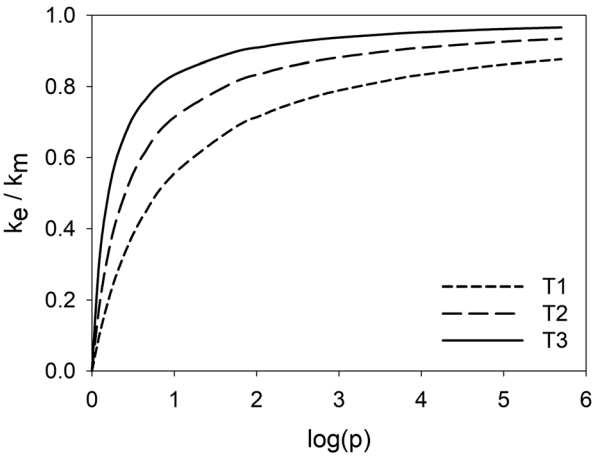


Fig. 2.5 Schematic representation of the behavior of fall-off curves

concentration C_M can be high enough as to make the product $k_{b1}C_M$ dominant over k_{f2} in the denominator on the right-hand side of Eq. (2.67). Therefore,

$$\frac{dC_B}{dt} = \frac{k_{f2}k_{f1}}{k_{b1}}C_A = k_m C_A \tag{2.68}$$

On the other hand, for considerably low pressures, the concentration C_M can be low enough as to make k_{f2} dominant over the product $k_{b1}C_M$ in the denominator of Eq. (2.67). Then,

$$\frac{dC_B}{dt} = k_{f1}C_A C_M \tag{2.69}$$

It is evident that at high pressures, the reaction behaves as first-order [Eq. (2.68)], and at low pressures, it behaves as second-order (Eq. 2.69). As a final observation, the right-hand side term of Eq. (2.67), which is multiplying the concentration C_A, can be considered as an equivalent specific rate constant (k_e). Dividing this by the k_m defined in Eq. (2.68), the following is obtained.

$$\frac{k_e}{k_m} = \frac{k_{b1}C_M}{k_{f2} + k_{b1}C_M} \tag{2.70}$$

Analyzing Eq. (2.70), it is clear that at considerably high pressure, $k_{b1}C_M >> k_{f2}$, and $k_e/k_m \rightarrow 1$. As the pressure is reduced, the quotient k_e/k_m decreases. The curves obtained for k_e/k_m vs. $\log(p)$ at constant temperatures are called fall-off curves. In order to observe this behavior, it was assumed that k_{b1} and k_{f2} are constant for a particular temperature. Then, since C_M is directly proportional to pressure, the curves shown in Fig. 2.5 are obtained.

2.4 Reactions and Mechanisms

The chemical kinetic mechanisms are formed by a set of elementary reactions which represent the many steps for the combustion process of a fuel. It is important to notice that these mechanisms are postulated based on theoretical and empirical observations and must be validated against fundamental experimental data. The fundamental experiments for validation of chemical kinetic mechanisms are listed below.

- Laminar flame speed measurements in spherical bombs, heat flux burners, or counterflow burners
- Ignition delay time measurements in shock tubes and rapid compression machines
- Species profile measurements in plug flow reactors
- Fuel consumption and product formation in perfectly stirred reactors

2.4.1 Reaction Mechanism of Hydrogen

Since mechanisms are postulated, it is possible to carry on the exercise of elaborating on a mechanism for hydrogen combustion with oxygen. H_2 and O_2 will ultimately produce H_2O; nevertheless, radicals like H, O, and OH (at least) must appear in the reaction path. Therefore, the first simple mechanism proposed here is shown in Fig. 2.6a. However, such a mechanism cannot represent with sufficient accuracy the combustion of hydrogen since it does not consider intermediary species. Thus, HO_2 is added as an intermediate species, and the mechanism is modified as shown in Fig. 2.6b. These particular mechanisms are not realistic since the H and O atoms are likely to be produced along the way and they do not appear in any of the simple mechanisms postulated to this point.

A mechanism for the combustion of hydrogen with oxygen was postulated by Sánchez and Williams [18], and it is shown in Table 2.1. A line of reasoning will be developed in order to imagine how this mechanism could have been postulated. Since H and O should be formed at the initial steps, the reactions (R7), (R12), and

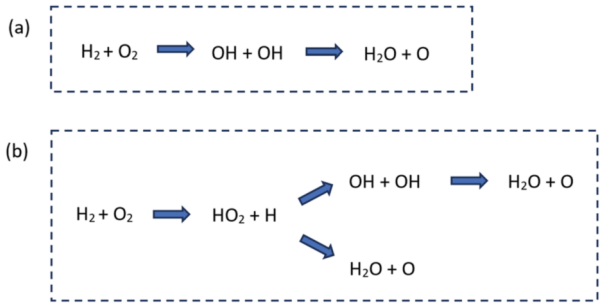


Fig. 2.6 Mechanisms for hydrogen oxidation that would fail to represent the process

Table 2.1 Hydrogen oxidation mechanism [18]

Reactions		B	n	T_a	
Type 1					
$H + O_2 \leftrightarrow OH + O$		3.52×10^{16}	−0.70	8590	(R1)
$H_2 + O \leftrightarrow OH + H$		5.06×10^{4}	2.67	3165	(R2)
$H_2 + OH \leftrightarrow H_2O + H$		1.17×10^{9}	1.30	1825	(R3)
$H_2O + O \leftrightarrow OH + OH$		7.00×10^{5}	2.33	7321	(R4)
Type 2					
$H + O_2 + M \leftrightarrow HO_2 + M$[b]	k_e	5.75×10^{19}	−1.40	0.0	(R5)
	k_m	4.65×10^{12}	0.44	0.0	
$HO_2 + H \leftrightarrow OH + OH$		7.08×10^{13}	0.0	148	(R6)
$HO_2 + H \leftrightarrow H_2 + O_2$		1.66×10^{13}	0.0	414	(R7)
$HO_2 + H \leftrightarrow H_2O + O$		3.10×10^{13}	0.0	866	(R8)
$HO_2 + O \leftrightarrow OH + O_2$		2.00×10^{13}	0.0	0.0	(R9)
$HO_2 + OH \leftrightarrow H_2O + O_2$		2.89×10^{13}	0.0	−250	(R10)
		4.50×10^{14}	0.0	5500	
Type 3					
$H + OH + M \leftrightarrow H_2O + M$[c]		4.00×10^{22}	−2.00	0.0	(R11)
$H + H + M \leftrightarrow H_2 + M$[d]		1.30×10^{18}	−1.00	0.0	(R12)
$O + O + M \leftrightarrow O_2 + M$[e]		6.17×10^{15}	−0.50	0.0	(R13)
$H + O + M \leftrightarrow OH + M$[f]		4.71×10^{18}	−1.00	0.0	(R14)
Type 4					
$OH + OH + M \leftrightarrow H_2O_2 + M$[g]	k_e	2.76×10^{25}	−3.20	0.0	(R15)
	k_m	9.55×10^{13}	−0.27	0.0	
$HO_2 + HO_2 \leftrightarrow H_2O_2 + O_2$		1.03×10^{14}	0.0	5556	(R16)
		1.94×10^{11}	0.0	−709	
$H_2O_2 + H \leftrightarrow HO_2 + H_2$		2.30×10^{13}	0.0	4000	(R17)
$H_2O_2 + H \leftrightarrow H_2O + OH$		1.00×10^{13}	0.0	1804	(R18)
$H_2O_2 + OH \leftrightarrow H_2O + HO_2$		1.74×10^{12}	0.0	160	(R19)
		7.59×10^{13}	0.0	3660	
$H_2O_2 + O \leftrightarrow HO_2 + OH$		9.63×10^{6}	2.0	2009	(R20)

[a] Units are mol, s, cm^3 e K

Third-body efficiencies: value of 1.0 for not specified species

[b] H_2(2.5). H_2O(16.0). CO(1.2). CO_2(2.4). Ar(0.7). He(0.7). Troe decay with $F_c = 0.5$

[c] H_2(2.5). H_2O(12.0). CO(1.9). CO_2(3.8). Ar(0.4). He(0.4)

[d] H_2(2.5). H_2O(12.0). CO(1.9). CO_2(3.8). Ar(0.5). He(0.5)

[e] H_2(2.5). H_2O(12.0). CO(1.9). CO_2(3.8). Ar(0.2). He(0.2)

[f] H_2(2.5). H_2O(12.0). CO(1.9). CO_2(3.8). Ar(0.7). He(0.7)

[g] H_2(2.5). H_2O(6.0). H_2O_2(6.0). CO(1.5). CO_2(2.0). Ar(0.7). He(0.4). Troe decay with $F_c = 0.43$

(R13) can be considered. The last two reactions involve a third body, M, which can be any species present in the combustion process.

$$H_2 + M \leftrightarrow H + H + M \quad (R12)$$

$$O_2 + M \leftrightarrow O + O + M \quad \text{(R13)}$$

$$H_2 + O_2 \leftrightarrow HO_2 + H \quad \text{(R7)}$$

The above three reactions will create the starting radical pool. The following species are now present in the system: H_2, O_2, H, O, and HO_2. It is possible to continue by thinking about all the experimentally feasible elementary reactions that involve these five species. However, an interesting approach would be to consider first the possible combinations of binary elementary reactions. That is equivalent to calculate the combinations of these $n = 5$ species taken $k = 2$ at a time. Since the number of combinations can be calculated by $n!/[k!(n-k)!]$, in this case a number of ten binary elementary reactions is possible. Nevertheless, this does not mean that all these ten reactions are actually feasible. With this said, it is now interesting to note that species such as H_2O and OH are still not present. Therefore, more options should be analyzed. Figure 2.7 shows the possible reactions involving HO_2 and the other species available to this point.

The analysis shown in Fig. 2.7 brings three more species, namely, H_2O, OH, and H_2O_2. Therefore, up to this point, there are eight species, which means that 28 binary elementary reactions are possible. However, as shown for the case of HO_2 and O_2 (see Fig. 2.7), some of these combinations will not be realistic, since they will not produce known chemical species. The same procedure can be applied to reactions involving H_2, O_2, H, and O. The feasible reactions are shown below.

$$O_2 + H \leftrightarrow OH + O \quad \text{(R1)}$$

$$H_2 + O \leftrightarrow OH + H \quad \text{(R2)}$$

Since the species OH, H_2O_2, and H_2O have also appeared, their binary reactions with the available species must be included. A reaction is considered repeated if it can be produced by considering any direction (forward or reverse) of an already

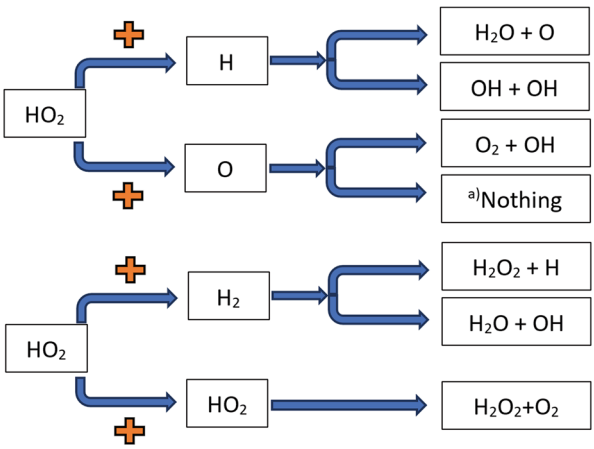


Fig. 2.7 Possible reactions involving HO_2, H, O, H_2, and O_2. To the author's knowledge, there are no probable reactions between HO_2 and O_2

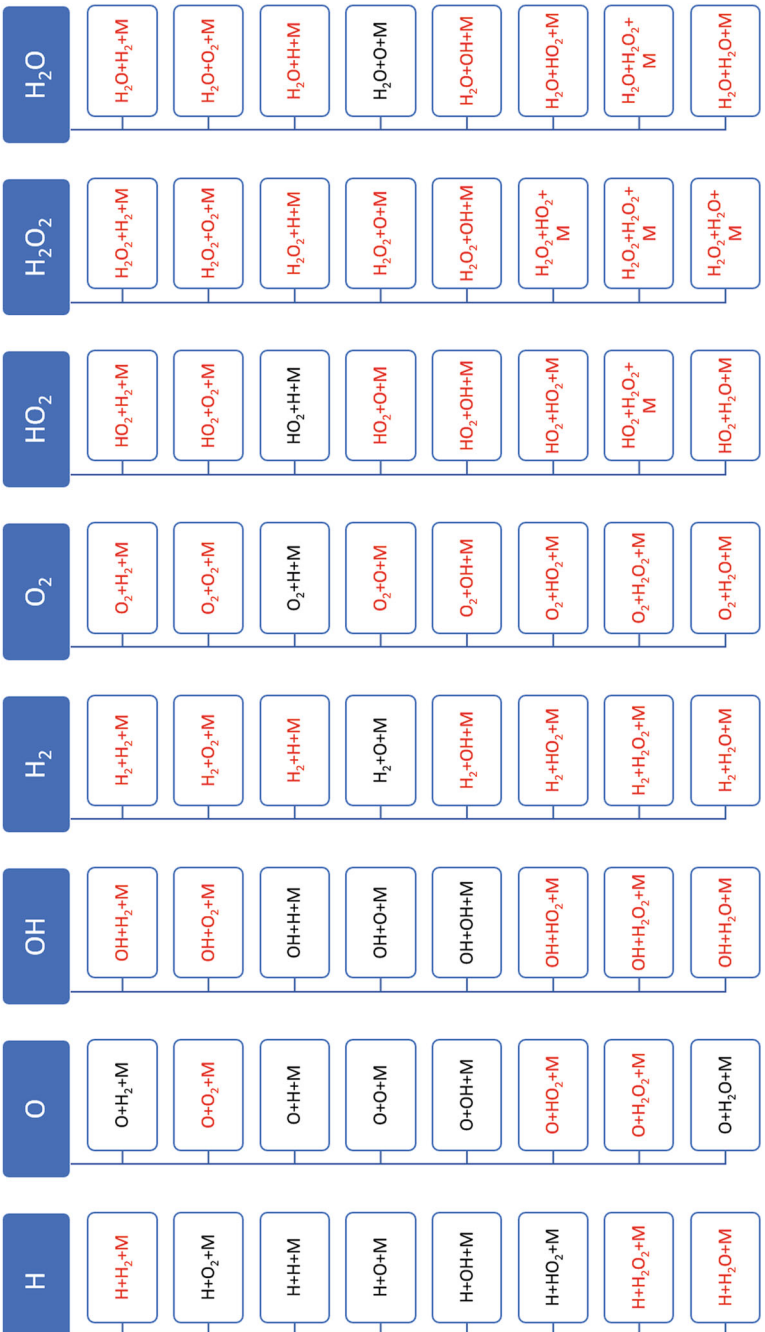


Fig. 2.8 Possible combinations for third-body reactions

identified reaction. Thus, dropping the repeated reactions, the following are obtained.

$$OH + H_2 \leftrightarrow H_2O + H \quad (R3)$$

$$OH + O_2 \leftrightarrow HO_2 + O \quad (R9)$$

$$OH + OH \leftrightarrow H_2O + O \quad (R4)$$

$$OH + OH \leftrightarrow HO_2 + H \quad (R6)$$

Looking now into the reactions involving H_2O and dropping repeated reactions,

$$H_2O + O \leftrightarrow HO_2 + H \quad (R8)$$

$$H_2O + OH \leftrightarrow HO_2 + H_2 \quad (R17)$$

$$H_2O + OH \leftrightarrow H_2O_2 + H \quad (R18)$$

Analyzing the information given in Fig. 2.7 and the binary elementary reactions identified to this point, there are 12 reactions, namely, R1, R2, R3, R4, R6, R7, R8, R9, R12, R13, R16, R17 and R18. It is now interesting to consider some other third-body reactions; among these, only R12 and R13 have been identified. A strategy to form these third-body reactions is to combine pairs of species and add a third body to these pairs. Considering the eight species involved in the reaction mechanism, Fig. 2.8 was elaborated. The third-body reactions in red are considered less probable or unfeasible within the context of this combustion process.

The reactions of two species and a third body that are possible are shown below.

$$H + O_2 + M \leftrightarrow HO_2 + M \quad (R5)$$

$$H + OH + M \leftrightarrow H_2O + M \quad (R11)$$

$$H + H + M \leftrightarrow H_2 + M \quad (R12)$$

$$O + O + M \leftrightarrow O_2 + M \quad (R13)$$

$$H + O + M \leftrightarrow OH + M \quad (R14)$$

$$OH + OH + M \leftrightarrow H_2O_2 + M \quad (R15)$$

$$H + HO_2 + M \leftrightarrow H_2O_2 + M \quad (R21)$$

$$H_2 + O + M \leftrightarrow H_2O + M \quad (R22)$$

$$O + OH + M \leftrightarrow HO_2 + M \quad (R23)$$

$$O + H_2O + M \leftrightarrow H_2O_2 + M \quad (R24)$$

It is at this stage that the empirical evidence starts to become more necessary to move forward. For example, the reactions R21–R24 have not been considered by

Sánchez and Williams in the mechanism presented in Table 2.1. Finally, binary elementary reactions involving H_2O_2 and radicals can also be included. The reactions of H_2O_2 with O_2 and H have already been included as R16, R17, and R18. On the other hand, the reactions of H_2O_2 with O and OH are still not included.

$$H_2O_2 + OH \leftrightarrow H_2O + HO_2 \tag{R19}$$

$$H_2O_2 + O \leftrightarrow HO_2 + OH \tag{R20}$$

The reactions of H_2O_2 with H_2, HO_2, and H_2O are not considered in the mechanism. However, the interested reader can evaluate their feasibility. Therefore, a reaction mechanism for hydrogen combustion with oxygen, formed by the reactions R1–R20, is postulated and must be validated with experimental data. This validation has already been carried out [18].

As a final concept, consider the net reaction rate of a species in a mechanism. This rate must take into account all the consumption and reaction rates of the particular species in the different elementary reactions which constitute the mechanism. For instance, consider the reaction rate of H_2 in the mechanism presented in Table 2.1. H_2 participates in reactions R2, R3, R7, R12, and R17. Therefore, its reaction rate is given by the expression below.

$$\begin{aligned}\frac{dC_{H_2}}{dt} = & -k_{f2}C_{H_2}C_O + k_{b2}C_{OH}C_H - k_{f3}C_{H_2}C_{OH} + k_{b3}C_{H_2O}C_H \\ & +k_{f7}C_{HO_2}C_H - k_{b7}C_{H_2}C_{O_2} + k_{f12}C_H^2C_M - k_{b12}C_{H_2}C_M \\ & +k_{f17}C_{H_2O_2}C_H - k_{b17}C_{HO_2}C_{H_2}\end{aligned} \tag{2.71}$$

The net reaction rate of any species k (kmol/m^3 s) in a chemical kinetic mechanism can be written in general terms by analyzing Eq. (2.71), and taking into account Eq. (2.15), the result is as shown in Eq. (2.72).

$$\widehat{\omega}_k = \sum_{i=1}^{L}\left(v_{ji,P} - v_{ji,R}\right)\left[k_{fi}\prod_{j=1}^{N}\left(C_j\right)^{v_{ji,R}} - k_{bi}\prod_{j=1}^{N}\left(C_j\right)^{v_{ji,P}}\right] \tag{2.72}$$

In the above expression, the net reaction rate of the species k depends on its interaction with the N chemical species in the mechanism, represented by the subscript j. In addition, there are L chemical reactions in the mechanism, represented by the subscript i. The stoichiometric coefficient of the jth species in the ith reaction is given by $v_{ji,R}$ for the reactants' side and by $v_{ji,P}$ for the products' side. The k_{fi} and k_{bi} are the forward and backward reaction rate constants for the ith reaction. Note that in cases where a particular species is not participating in an elementary reaction, its stoichiometric coefficient is simply equal to zero. Finally, the net production rate of species k (kg/m^3 s) can be obtained by multiplying Eq. (2.72) by the molecular weight of the species.

$$\dot{m}_{k,R} = W_k \widehat{\omega}_k = W_k \sum_{i=1}^{L} \left(\nu_{ji,P} - \nu_{ji,R}\right) \left[k_{fi} \prod_{j=1}^{N} \left(C_j\right)^{\nu_{ji,R}} - k_{bi} \prod_{j=1}^{N} \left(C_j\right)^{\nu_{ji,P}} \right] \tag{2.73}$$

The formulations provided in Eqs. (2.72) and (2.73) are useful for models accounting for detailed chemical kinetics.

2.4.2 Types of Elementary Reactions

In this section some types of reactions are defined in order to better understand the construction of chemical kinetics mechanisms.

2.4.2.1 Chain Initiation Reactions

These reactions are responsible for the formation of free radicals from stable species. The reverse reaction R12 on Table 2.1 is an example.

$$H_2 + M \rightarrow H + H + M \tag{R12}$$

2.4.2.2 Chain Propagation Reactions

This type of reaction maintains the same number of radical species in the reaction. For example, in the reaction R3, a radical OH reacts with H_2 and forms a water molecule and a radical H.

$$H_2 + OH \rightarrow H_2O + H \tag{R3}$$

2.4.2.3 Chain Branching Reactions

These reactions have the characteristic of increasing the radical pool. That is, they produce more radicals than those they consume. For example, H reacts with HO_2 in reaction R6, and it produces two molecules of OH. Thus, the radical pool is increased. Note that in this case HO_2 is more stable than OH.

$$HO_2 + H \leftrightarrow OH + OH \tag{R6}$$

2.4.2.4 Chain Termination Reactions

The reaction of one or more radicals that produce a stable species is called a chain termination reaction. An example is reaction R11 where H and OH react to form H_2O. This type of reaction often includes a third body.

$$H + OH + M \rightarrow H_2O + M \tag{R11}$$

2.4.3 Reaction Mechanism of Hydrocarbons

Classical articles studying the chemical kinetics of hydrocarbon combustion are those by Westbrook and Dryer [19], Simmie [20], Metcalfe et al. [21], and Pollard [22]. The chemical kinetics of the combustion of aromatic hydrocarbons has been studied in a review article by Jin et al. [23]. It is important to remember that, a reaction mechanism is always an approximation. Also, note that ionized species, or even excited states, must be considered in some applications. Following the discussion of the H_2–O_2 mechanism, presented in the preceding section, it could be noticed that hydrocarbon combustion would include a larger number of species and elementary reactions. Although this is true, the mechanisms are constructed in a hierarchical manner; thus, the H_2–O_2 mechanism can be treated as a sub-mechanism for any hydrocarbon reaction mechanism. An example of this hierarchical treatment is presented in Fig. 2.9.

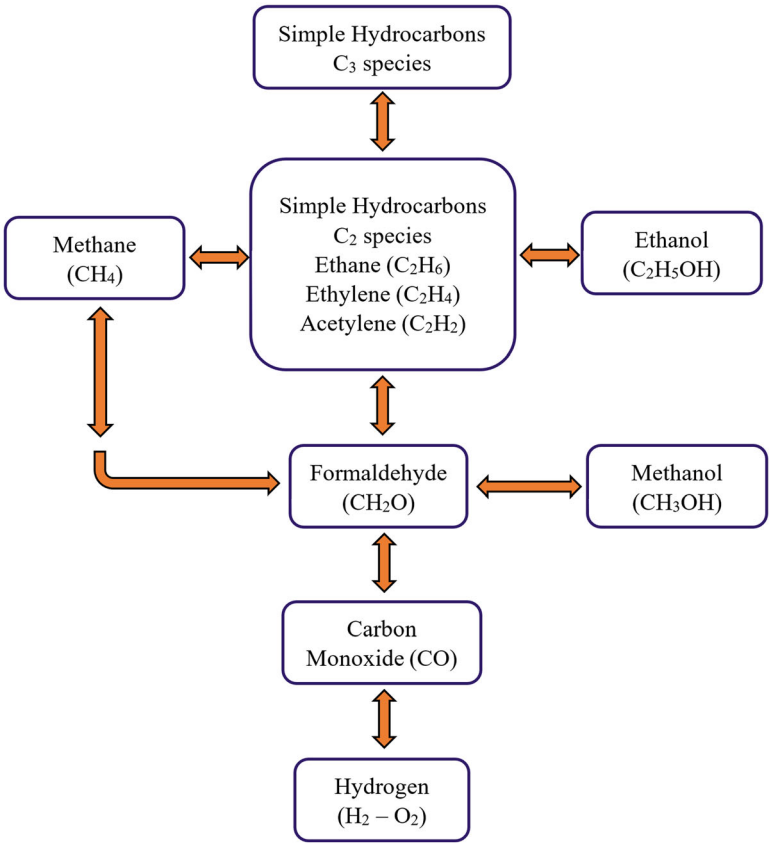


Fig. 2.9 Hierarchical structure for simple hydrocarbon combustion. Adapted from Westbrook and Dryer [19]. (Adapted from Progress in Energy and Combustion Science, Vol 10/1, Charles K. Westbrook, Frederick L. Dryer, Chemical kinetic modeling of hydrocarbon combustion, 1–57, Copyright (1984), with permission from Elsevier)

As observed in Fig. 2.9, a C_3 hydrocarbon, such as propane, will first dissociate to form C_2 hydrocarbons. These may include ethane, ethylene, and/or acetylene. Afterwards, methane and ethanol would be formed. Eventually, methanol will also be produced. Once formaldehyde has been formed, the carbon monoxide and hydrogen sub-mechanisms take place; therefore, CO_2 and H_2O are formed. It is out of the scope of this book to present a detailed discussion of each sub-mechanism that can be part of a hydrocarbon reaction mechanism. However, some important sub-mechanisms will be presented.

2.4.3.1 Ozone (O_3) Sub-mechanism

Ozone (O_3) is a species that could have been considered in the H_2–O_2 mechanism. Actually, some reactions that suggested the formation of O_3 were discarded because they are considered not relevant for H_2 oxidation. However, recent articles have shown that a small amount of O_3 has the potential to promote detonations or enhance chemical reactions [24, 25]. The following reactions were suggested y Westbrook and Dryer [19].

$$O_3 + M \leftrightarrow O_2 + O + M \quad \text{(R25)}$$

$$O_3 + O \leftrightarrow O_2 + O_2 \quad \text{(R26)}$$

$$O_3 + H \leftrightarrow OH + O_2 \quad \text{(R27)}$$

$$O_3 + OH \leftrightarrow HO_2 + O_2 \quad \text{(R28)}$$

2.4.3.2 Formaldehyde (CH_2O) Sub-mechanism

Formaldehyde (CH_2O) is an intermediate product that leads to formyl radical (HCO), which in turns decomposes to form CO. As stated by Westbrook and Dryer [19], there are two possible reactions for CH_2O thermal dissociation.

$$CH_2O + M \leftrightarrow HCO + H + M \quad \text{(R29)}$$

$$CH_2O + M \leftrightarrow CO + H_2 + M \quad \text{(R30)}$$

Also, the reactions of CH_2O with radicals like H, O, OH, and HO_2 are as follows.

$$CH_2O + H \leftrightarrow HCO + H_2 \quad \text{(R31)}$$

$$CH_2O + O \leftrightarrow HCO + OH \quad \text{(R32)}$$

$$CH_2O + OH \leftrightarrow HCO + H_2O \quad \text{(R33)}$$

$$CH_2O + HO_2 \leftrightarrow HCO + H_2O_2 \quad \text{(R34)}$$

The reaction involving HO_2 is comparatively less important than those involving H, O, and OH radicals. The reactions in which CO is formed from HCO are shown below.

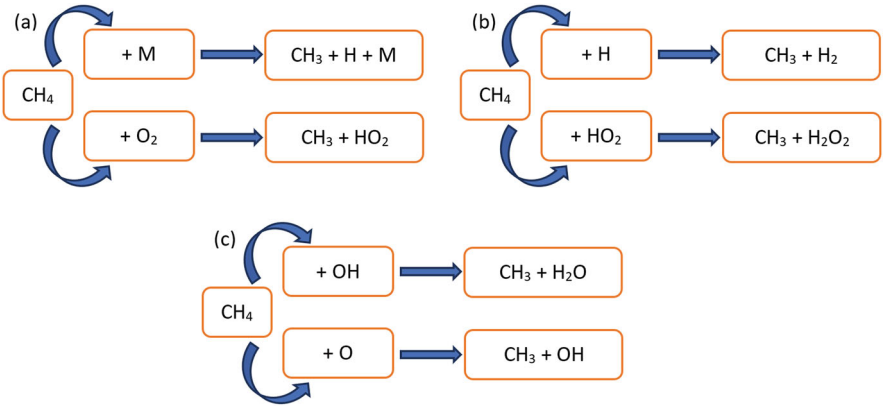


Fig. 2.10 Initiation and hydrogen abstraction reactions for CH_4 oxidation

$$HCO + M \leftrightarrow CO + H + M \quad \text{(R35)}$$

$$HCO + O_2 \leftrightarrow CO + HO_2 \quad \text{(R36)}$$

2.4.3.3 Carbon Monoxide (CO) Sub-mechanism

Since the combustion products must include CO_2 for a complete combustion process, there should be a mechanism which brings the CO formed during the hydrocarbon oxidation to the final product, CO_2. The reaction mechanism proposed by Westbrook and Dryer [19] is shown below.

$$CO + O + M \leftrightarrow CO_2 + M \quad \text{(R37)}$$

$$CO + O_2 \leftrightarrow CO_2 + O \quad \text{(R38)}$$

$$CO + OH \leftrightarrow CO_2 + H \quad \text{(R39)}$$

$$CO + HO_2 \leftrightarrow CO_2 + OH \quad \text{(R40)}$$

2.4.3.4 The Methane Combustion Mechanism

The combustion of methane is typically explained in combustion textbooks [7, 9] due to its importance in the hierarchical construction of chemical kinetic mechanisms. In this case, the source used for the mechanism will be the article by Westbrook and Dryer [19]. The initiation can occur through the thermal decomposition of CH_4, producing an alkyl radical (CH_3) and a H radical. Also, CH_4 can react with O_2, producing CH_3 and HO_2 radicals (Fig. 2.10a). Afterwards, hydrogen abstraction from CH_4 can take place by reactions with the H and HO_2 radicals to form CH_3, H_2, and H_2O_2, as shown in Fig. 2.10b. In addition, since H_2 has been formed, from the H_2–O_2 sub-mechanism, new radicals can be formed, in this case, O

and OH. Therefore, hydrogen abstraction can also occur by reactions of CH_4 with O and OH to form CH_3, OH, and H_2O (Fig. 2.10c).

The information in Fig. 2.10 can also be presented in terms of elementary reactions as shown below.

$$CH_4 + M \leftrightarrow CH_3 + H + M \quad \text{(R41)}$$

$$CH_4 + O_2 \leftrightarrow CH_3 + HO_2 \quad \text{(R42)}$$

$$CH_4 + H \leftrightarrow CH_3 + H_2 \quad \text{(R43)}$$

$$CH_4 + HO_2 \leftrightarrow CH_3 + H_2O_2 \quad \text{(R44)}$$

$$CH_4 + OH \leftrightarrow CH_3 + H_2O \quad \text{(R45)}$$

$$CH_4 + O \leftrightarrow CH_3 + OH \quad \text{(R46)}$$

The methyl radical formed up to this point will have the possibility to recombine, thus forming C_2 species as shown by the connection between the CH_4 box and the C_2 species box in Fig. 2.9. The reactions shown in this process are R47–R49.

$$CH_3 + CH_3 \leftrightarrow C_2H_6 \quad \text{(R47)}$$

$$CH_3 + CH_3 \leftrightarrow C_2H_5 + H \quad \text{(R48)}$$

$$CH_3 + CH_3 \leftrightarrow C_2H_4 + H_2 \quad \text{(R49)}$$

The path of CH_3 includes oxidation to generate an O radical and a methoxide radical (CH_3O) through R50. In addition, at high temperatures, CH_3 can dissociate as shown in R51.

$$CH_3 + O_2 \leftrightarrow CH_3O + O \quad \text{(R50)}$$

$$CH_3 + M \leftrightarrow CH_2 + H + M \quad \text{(R51)}$$

Moreover, the methyl radical can react with other radicals according to the reactions R52–R57.

$$CH_3 + O \leftrightarrow CH_2O + H \quad \text{(R52)}$$

$$CH_3 + OH \leftrightarrow CH_2O + H_2 \quad \text{(R53)}$$

$$CH_3 + OH \leftrightarrow CH_3O + H \quad \text{(R54)}$$

$$CH_3 + CH_2O \leftrightarrow CH_4 + HCO \quad \text{(R55)}$$

$$CH_3 + HCO \leftrightarrow CH_4 + CO \quad \text{(R56)}$$

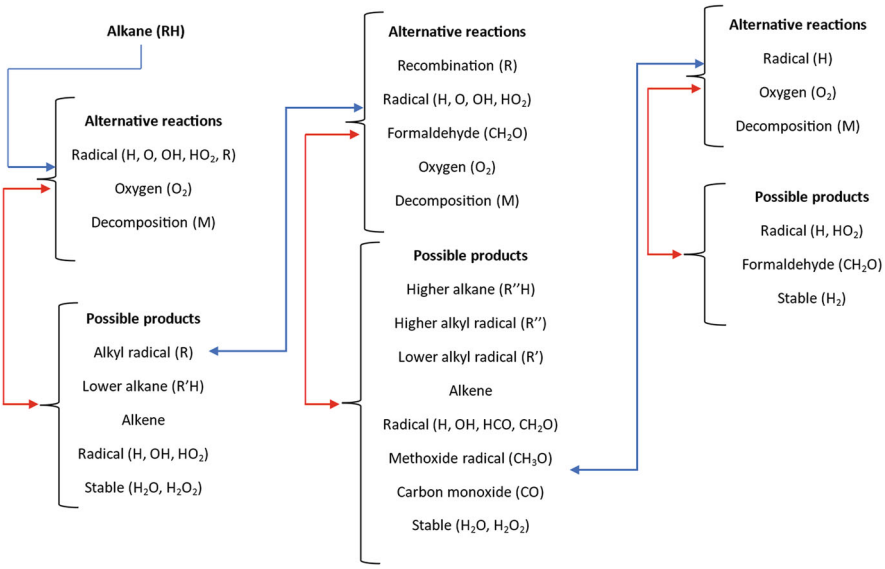


Fig. 2.11 Schematic representation of hydrocarbon oxidation. Based on the work by Pollard [22]

$$CH_3 + HO_2 \leftrightarrow CH_3O + OH \tag{R57}$$

The methoxide radical (CH_3O) reacts to form CH_2O, which forms CO through its own sub-mechanisms.

$$CH_3O + M \leftrightarrow CH_2O + H + M \tag{R58}$$

$$CH_3O + O_2 \leftrightarrow CH_2O + HO_2 \tag{R59}$$

$$CH_3O + H \leftrightarrow CH_2O + H_2 \tag{R60}$$

Once the methane reaction path has led to formaldehyde, then, the reaction sub-mechanism of CH_2O takes places, leading to CO, which will eventually form CO_2. Note that the H_2–O_2 sub-mechanism is also taking place to produce H_2O. A widely accepted mechanism for CH_4 oxidation is the GRI-Mech 3.0 mechanism [26]. More details about reaction rates and preferred paths can be found in the classical articles [19–22].

2.4.3.5 A Simplified Scheme for Alkane Oxidation

This section is closed by presenting a simplified view of the oxidation process of an alkane, as presented in Fig. 2.11. The simplified scheme will be taken up to the formation of formaldehyde, because from there, the sub-mechanisms have already been discussed in more detail in the preceding subsections. An alkane (RH) will usually react with a radical, an oxygen molecule, or by thermal decomposition, to produce an alkyl radical (R). In case of higher hydrocarbons, this first step can also

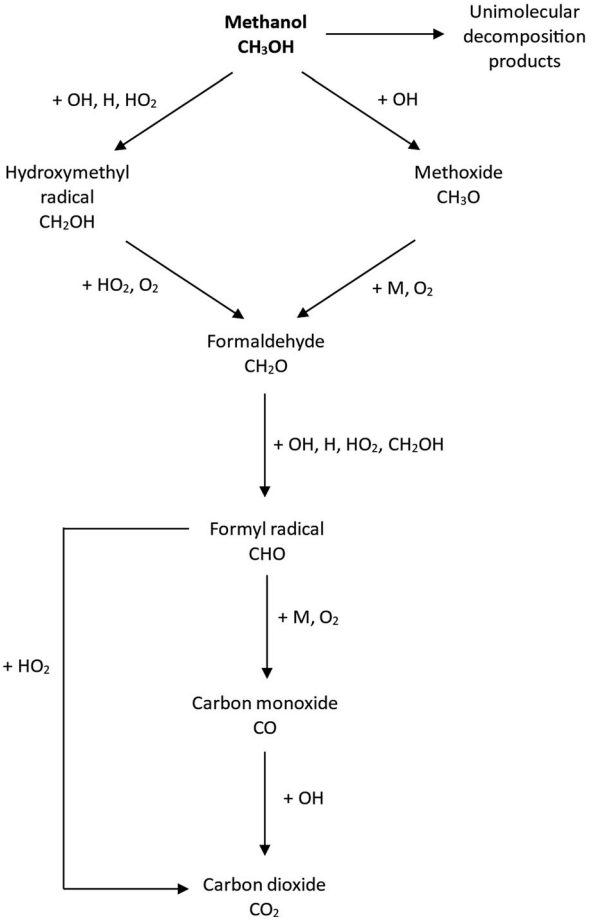


Fig. 2.12 Reaction path diagram for methanol oxidation. (Adapted from Sarathy et al. [27]. Adapted from Progress in Energy and Combustion Science, Vol 44, S. Mani Sarathy, Patrick Oßwald, Nils Hansen, Katharina Kohse-Höinghaus, Alcohol combustion chemistry, 10–102, Copyright (2014), with permission from Elsevier)

produce lower hydrocarbons (R'H) or alkenes. For example, methane could be produced from ethane reacting with a methyl radical.

The second main steps involve reactions of alkyl radicals with other species such as radicals, oxygen, and formaldehyde and even recombination. The products in this case may be higher alkanes, higher and lower alkyl radicals, alkenes, other radicals, and methoxide radical, among others (see Fig. 2.11). The methoxide radical will be the protagonist of the third main step, in which it reacts with other radicals or oxygen or decomposes to produce formaldehyde, other radicals, and some stable species. At this point, formaldehyde will take the leading role (in this simplified scheme) and produce the formyl radicals that will react to form CO. The sub-mechanisms of CH_2O, H_2, and CO have already been discussed in the previous subsections.

2.4.4 Reaction Mechanism of Alcohols

Having introduced in enough detail the mechanism of methane and some sub-mechanism, in this section, the mechanism of methanol and ethanol will be presented as examples. The article by Sarathy et al. [27] presents an extensive discussion about alcohol oxidation. A widely adopted mechanism for ethanol oxidation is that proposed by Marinov [28]. In a review article about ethanol combustion, Mendiburu et al. [8] compared several mechanisms for ethanol combustion under different conditions.

2.4.4.1 Methanol Oxidation Mechanism

As described by Sarathy et al. [27], the oxidation of methanol will be initiated by its reactions with radicals such as OH, HO_2, and H, which produce the abstraction of H atoms. Two possible outcomes are the formation of hydroxymethyl radical (CH_2OH) and methoxide (CH_3O), respectively, with the first being more important than the second. Nevertheless, CH_2OH and CH_3O will produce CH_2O by reacting with HO_2 and O_2 in the first case and by reaction in third-body reaction and with O_2 in the second case. Once formaldehyde is formed, it will react with radicals like OH, O, HO_2, and CH_2OH to produce the formyl radical. The CHO will react with O_2 and through third-body reactions to form CO. Also, CHO will react with HO_2 to form CO_2. The sub-mechanism of H_2 and CO oxidation have already been discussed. The oxidation mechanism of methanol is as shown in Fig. 2.12. The thermal decomposition reactions of methanol are presented as R61–R63.

$$CH_3OH + M \leftrightarrow CH_3 + OH \quad (R61)$$

$$CH_3OH + M \leftrightarrow CH_2 + H_2O \quad (R62)$$

$$CH_3OH + M \leftrightarrow CH_2OH + H \quad (R63)$$

Hydrogen abstraction reactions producing CH_2OH and CH_3O are presented in R64–R69.

$$CH_3OH + H \leftrightarrow CH_2OH + H_2 \quad (R64)$$

$$CH_3OH + H \leftrightarrow CH_3O + H_2 \quad (R65)$$

$$CH_3OH + OH \leftrightarrow CH_2OH + H_2O \quad (R66)$$

$$CH_3OH + OH \leftrightarrow CH_3O + H_2O \quad (R67)$$

$$CH_3OH + HO_2 \leftrightarrow CH_2OH + H_2O_2 \quad (R68)$$

$$CH_3OH + HO_2 \leftrightarrow CH_3O + H_2O_2 \quad (R69)$$

Note that methanol reacting with the radicals H, OH, and HO_2 can produce CH_2OH or CH_3O. However, the formation of CH_2OH seems to be the preferred path [27]. In addition, note that Westbrook and Dryer [19] also provided some steps of methanol's oxidation.

2.4.4.2 Ethanol Oxidation Mechanism

Some important features of the ethanol oxidation mechanism are depicted in Fig. 2.13. The decomposition reactions involving ethanol produce species such as ethylene, methyl radical, ethyl radical (C_2H_5), H_2O, CH_2OH, and OH, as shown in R70–R72.

$$C_2H_5OH + M \leftrightarrow C_2H_4 + H_2O + M \quad (R70)$$

$$C_2H_5OH + M \leftrightarrow CH_3 + CH_2OH + M \quad (R71)$$

$$C_2H_5OH + M \leftrightarrow C_2H_5 + OH + M \quad (R72)$$

The hydrogen abstraction reactions, involving ethanol and radical species, lead to the formation of ethyl hydroxy (C_2H_5O), α-hydroxyethyl (CH_3CHOH), and β-hydroxyethyl (CH_2CH_2OH). The reactions of ethanol and H atom are presented in reactions R73–R75.

$$C_2H_5OH + H \leftrightarrow C_2H_5O + H_2 \quad (R73)$$

$$C_2H_5OH + H \leftrightarrow CH_3CHOH + H_2 \quad (R74)$$

$$C_2H_5OH + H \leftrightarrow CH_2CH_2OH + H_2 \quad (R75)$$

Ethanol reacting with OH produces the elementary reactions presented as R76–R78.

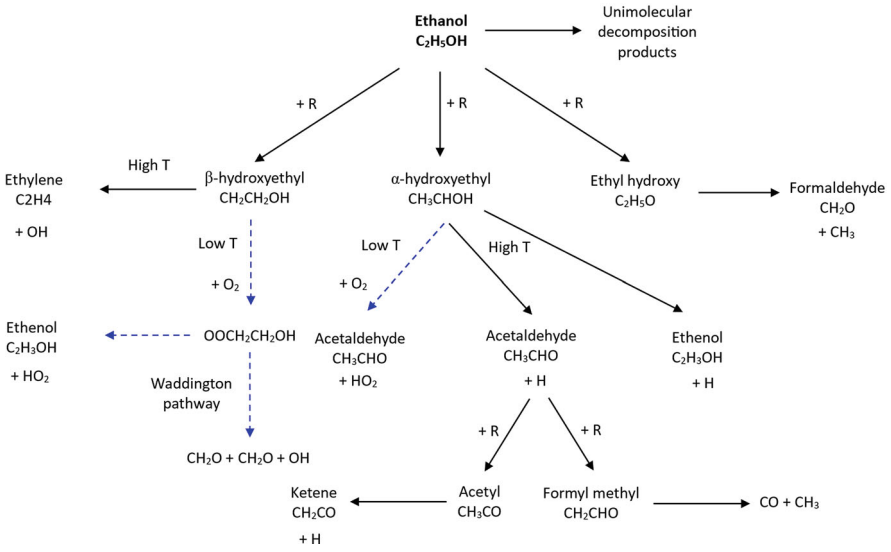


Fig. 2.13 Reaction path diagram for ethanol oxidation. (Adapted from Sarathy et al. [27]. Adapted from Progress in Energy and Combustion Science, Vol 44, S. Mani Sarathy, Patrick Oßwald, Nils Hansen, Katharina Kohse-Höinghaus, Alcohol combustion chemistry, 10–102, Copyright (2014), with permission from Elsevier)

$$C_2H_5OH + OH \leftrightarrow C_2H_5O + H_2O \quad (R76)$$

$$C_2H_5OH + OH \leftrightarrow CH_3CHOH + H_2O \quad (R77)$$

$$C_2H_5OH + OH \leftrightarrow CH_2CH_2OH + H_2O \quad (R78)$$

The reactions involving ethanol and HO_2 are presented in R79–R81.

$$C_2H_5OH + HO_2 \leftrightarrow C_2H_5O + H_2O_2 \quad (R79)$$

$$C_2H_5OH + HO_2 \leftrightarrow CH_3CHOH + H_2O_2 \quad (R80)$$

$$C_2H_5OH + HO_2 \leftrightarrow CH_2CH_2OH + H_2O_2 \quad (R81)$$

As shown in Fig. 2.13, the ethyl hydroxy may react to form formaldehyde, which leads to the already known path discussed in the previous subsections.

β-Hydroxyethyl reacts to form ethylene; however, it can also react through a low-temperature path as shown in Fig. 2.13 by the traced lines. The low-temperature path leads to the formation of ethenol (C_2H_3OH) and formaldehyde. α-Hydroxyethyl can also react through high- and low-temperature paths to form acetaldehyde in both cases. Following the high-temperature path in Fig. 2.13, acetaldehyde will further react to form acetyl (CH_3CO) and formyl methyl (CH_2CHO). The acetyl will continue the reaction by forming ketene (CH_2CO), while formyl methyl produces CO and CH_3. More details about the oxidation mechanism of ethanol can be found in the work by Sarathy et al. [27]. However, it is also important to note that the reaction mechanism developed by Marinov et al. [28] has been found to be more accurate at high pressures [8].

2.5 Reduction of Chemical Kinetic Mechanisms

Some assumptions can be applied to reduce chemical kinetics mechanism. These assumptions are the steady-state approximation and the partial equilibrium approximation. These assumptions and two reduced mechanisms will be presented in this section.

2.5.1 Steady-State Approximation

The steady-state approximation is applied to the free radicals because these species have the characteristic of increasing their concentration very fast, while, at the same time, these radicals are also consumed very fast. Therefore, the production and consumption rates can be considered equal. In order to understand the application of this approximation, the thermal mechanism of NO formation, developed by Zeldovich, will be considered. The mechanism is shown in reactions R82–R84.

$$O + N_2 \leftrightarrow NO + N \tag{R82}$$

$$N + O_2 \leftrightarrow NO + O \tag{R83}$$

$$N + OH \leftrightarrow NO + H \tag{R84}$$

The reaction rate of the nitrogen atom (N) is presented in Eq. (2.74).

$$\frac{dC_N}{dt} = \begin{array}{c} k_{f1}C_O C_{N_2} - k_{b1}C_{NO}C_N - k_{f2}C_N C_{O_2} \\ +k_{b2}C_{NO}C_O - k_{f3}C_N C_{OH} + k_{b3}C_{NO}C_H \end{array} \tag{2.74}$$

The steady-state approximation, applied to the radical N, establishes that its net reaction rate is approximately equal to zero ($dC_N/dt \sim 0$). Applying that condition to Eq. (2.74) and rearranging,

$$C_N = \frac{k_{f1}C_O C_{N_2} + k_{b2}C_{NO}C_O + k_{b3}C_{NO}C_H}{k_{b1}C_{NO} + k_{f2}C_{O_2} + k_{f3}C_{OH}} \tag{2.75}$$

Therefore, the ordinary differential equation (ODE) given in Eq. (2.74) can be replaced by Eq. (2.75), because both give information about the concentration of the radical N. Since in a reaction mechanism each species concentration is given through an ordinary differential equation, the steady-state approximation can be used to replace one or more ODEs with an algebraic equation. The application of this approximation is an important step in the reduction of chemical kinetic mechanisms.

A more fundamental interpretation of the steady-state approximation is given in terms of the species conservation equation [18], as shown in Eq. (2.76). In that equation, the units are 1/s.

$$\frac{\partial Y_i}{\partial t} + v \cdot \nabla Y_i + \frac{1}{\rho}\nabla \cdot (\rho Y_i v_{i,\text{diff}}) = \frac{W_i}{\rho}\frac{dC_i}{dt} = \left(\frac{\dot{m}_{R,i}}{\rho}\right)_{\text{prod}} - \left(\frac{\dot{m}_{R,i}}{\rho}\right)_{\text{cons}} \tag{2.76}$$

The flow velocity is denoted by v, while the diffusion velocity is denoted by $v_{i,\text{diff}}$. The net rate of reaction of the species has been represented as the difference of the contribution of the reactions which produce the species minus the contribution of the reactions consuming the species, represented in the right-hand side of Eq. (2.76). The characteristic production and consumption times are then defined in terms of the corresponding rates.

$$\left(\frac{\dot{m}_{R,i}}{\rho}\right)_{\text{prod}} \rightarrow t_{\text{prod}}; \quad \left(\frac{\dot{m}_{R,i}}{\rho}\right)_{\text{cons}} \rightarrow t_{\text{cons}}$$

On the other hand, the fluid mechanical times are defined in terms of the characteristic times of accumulation, flow, and diffusion, all of which are given on the left-hand side of Eq. (2.76).

$$\frac{\partial Y_i}{\partial t} \rightarrow t_{M,1}; \quad v \cdot \nabla Y_i \rightarrow t_{M,2}; \quad \frac{1}{\rho} \nabla \cdot (\rho Y_i v_{i,\text{diff}}) \rightarrow t_{M,3}$$

The smallest characteristic time is taken as a measure of the fluid mechanics time scale (t_M).

$$t_M = \min(t_{M,1}, t_{M,2}, t_{M,3}) \tag{2.77}$$

Therefore, the steady-state approximation, as presented at the beginning of this section, should also require that t_M is significantly larger than t_{prod} and t_{cons}. That is, the accumulation, flow, and diffusion times should all be considerably larger than the chemical times for the production and consumption of the species.

2.5.2 Partial Equilibrium Approximation

The partial equilibrium approximation is another possible simplification. The process involves fast and slow reactions, with the fast reactions being formed in both directions. Generally, the fast reactions are chain propagation or chain branching reactions. On the other hand, the slow reactions are chain termination reactions. Therefore, some fast reactions may be considered in equilibrium, that is, the forward and reverse reaction rates are considered equal for a particular chain branching or chain propagation reaction.

Following Coelho and Costa [13], this approximation can be exemplified by considering the chain branching reactions (R1) and (R4) and the chain propagation reaction (R3), all of which are given in Table 2.1. Again, the analysis will be developed by providing more detail.

$$\mathrm{H} + \mathrm{O_2} \leftrightarrow \mathrm{OH} + \mathrm{O} \tag{R1}$$

$$\mathrm{H_2} + \mathrm{OH} \leftrightarrow \mathrm{H_2O} + \mathrm{H} \tag{R3}$$

$$\mathrm{H_2O} + \mathrm{O} \leftrightarrow \mathrm{OH} + \mathrm{OH} \tag{R4}$$

Applying the partial equilibrium consideration, the forward and reverse reaction rates should be equal.

$$k_{f1} C_{\mathrm{H}} C_{\mathrm{O_2}} = k_{b1} C_{\mathrm{OH}} C_{\mathrm{O}} \tag{2.78}$$

$$k_{f3} C_{\mathrm{H_2}} C_{\mathrm{OH}} = k_{b3} C_{\mathrm{H_2O}} C_{\mathrm{H}} \tag{2.79}$$

$$k_{f4} C_{\mathrm{H_2O}} C_{\mathrm{O}} = k_{b4} C_{\mathrm{OH}}^2 \tag{2.80}$$

Note that there are three reactions, involving six chemical species. Since there are three equations, only three species concentrations will be independent. Thus, it is possible to express the concentrations of three species in terms of the other three. The concentrations chosen to be independent are those corresponding to more stable

species, which would be easier to measure. Therefore, C_{H2}, C_{O2}, and C_{H2O} are chosen as independent concentrations. Rearranging Eqs. (2.78)–(2.80).

$$C_H = \frac{k_{b1}}{k_{f1}} \frac{C_{OH} C_O}{C_{O_2}} \tag{2.81}$$

$$C_{OH} = \frac{k_{b3}}{k_{f3}} \frac{C_{H_2O} C_H}{C_{H_2}} \tag{2.82}$$

$$C_{OH}^2 = \frac{k_{f4}}{k_{b4}} C_{H_2O} C_O \tag{2.83}$$

Equations (2.81) and (2.83) can be combined to obtain an expression for C_O in terms of the independent species concentrations.

$$C_O = \frac{k_{f1} k_{f3}}{k_{b1} k_{b3}} \frac{C_{H_2} C_{O_2}}{C_{H_2O}} \tag{2.84}$$

Substitution of Eq. (2.84) into Eq. (2.83) gives an expression for C_{OH} in terms of the C_{H2} and C_{O2} concentrations.

$$C_{OH} = \left(\frac{k_{f1} k_{f3} k_{f4}}{k_{b1} k_{b3} k_{b4}} C_{H_2} C_{O_2} \right)^{1/2} \tag{2.85}$$

Finally, substitution of Eqs. (2.84) and (2.85) into Eq. (2.81) provides an expression for C_H in terms of the chosen independent concentrations.

$$C_H = \frac{k_{b1} k_{f1} k_{f3}}{k_{f1} k_{b1} k_{b3}} \frac{k_{f1}^{1/2} k_{f3}^{1/2} k_{f4}^{1/2}}{k_{b1}^{1/2} k_{b2}^{1/2} k_{b4}^{1/2}} \frac{C_{H_2} C_{O_2} C_{H_2}^{1/2} C_{O_2}^{1/2}}{C_{H_2O} C_{O_2}}$$

Then, rearranging,

$$C_H = \left(\frac{k_{f1} k_{f3}^3 k_{f4}}{k_{b1} k_{b3}^3 k_{b4}} \frac{C_{H_2}^3 C_{O_2}}{C_{H_2O}^2} \right)^{1/2} \tag{2.86}$$

The desired expressions that relate the concentrations of radical species, namely, C_O, C_{OH}, and C_H, in terms of the concentrations of stable species (C_{H2}, C_{O2}, and C_{H2O}) are given in Eqs. (2.84)–(2.86). This procedure can be applied many times to reduce more complex chemical kinetic mechanisms.

2.5.3 Reduced Mechanism for Hydrogen

The chemical kinetic mechanism for hydrogen oxidation is accurate for several processes. However, as mentioned by Sánchez et al. [18], this mechanism is not accurate for many detonative processes in which extreme conditions of temperature and pressure dominate. The reduction of the hydrogen mechanism consists in first

Table 2.2 Skeletal mechanism for hydrogen oxidation [30]

Reactions		B	n	T_a	
Type 1					
$H + O_2 \leftrightarrow OH + O$	k_{f1}	3.52×10^{16}	−0.70	8590	(R1)
	k_{b1}	3.52×10^{4}	−0.26	72	
$H_2 + O \leftrightarrow OH + H$	k_{f2}	5.06×10^{4}	2.67	3165	(R2)
	k_{b2}	3.03×10^{4}	2.63	2433	
$H_2 + OH \leftrightarrow H_2O + H$	k_{f3}	1.17×10^{9}	1.30	1825	(R3)
	k_{b3}	1.28×10^{10}	1.19	9412	
Type 2					
$H + O_2 + M \rightarrow HO_2 + M^b$	k_e	5.75×10^{19}	−1.40	0.0	(R5f)
	k_m	4.65×10^{12}	0.44	0.0	
$HO_2 + H \rightarrow OH + OH$	k_{f6}	7.08×10^{13}	0.00	148	(R6f)
$HO_2 + H \leftrightarrow H_2 + O_2$	k_{f7}	1.66×10^{13}	0.00	414	(R7)
	k_{b7}	2.69×10^{12}	0.36	27,888	
$HO_2 + OH \rightarrow H_2O + O_2$	k_{f10a}	2.89×10^{13}	0.00	−250	(R10f)
	k_{f10b}	4.50×10^{14}	0.00	5500	
Type 3					
$H + OH + M \leftrightarrow H_2O + M^c$	k_{f11}	4.00×10^{22}	−2.00	0.0	(R11)
	k_{b11}	1.03×10^{23}	−1.75	59,675	
$H + H + M \leftrightarrow H_2 + M^d$	k_{f12}	1.30×10^{18}	−1.00	0.0	(R12)
	k_{b12}	3.04×10^{17}	−0.65	52,092	
Type 4					
$H_2O_2 + M \rightarrow OH + OH + M^g$	k_e	7.60×10^{30}	−4.20	25,703	(R15b)
	k_m	2.63×10^{19}	−1.27	25,703	
$HO_2 + HO_2 \rightarrow H_2O_2 + O_2$	k_{f16a}	1.03×10^{14}	0.00	5556	(R16f)
	k_{f16b}	1.94×10^{11}	0.00	−709	
$HO_2 + H_2 \rightarrow H_2O_2 + H$	k_{17b}	7.80×10^{10}	0.61	12,045	(R17b)

Adapted from Proceedings of the Combustion Institute, Vol 33, P. Boivin, C. Jiménez, A.L. Sánchez, F.A. Williams, An explicit reduced mechanism for H2–air combustion, 517–523, Copyright (2011), with permission from Elsevier

[a] Units are mol, s, cm^3 e K

Third-body efficiencies: value of 1.0 for not specified species

[b] H_2(2.5). H_2O(16.0). CO(1.2). CO_2(2.4). Ar(0.7). He(0.7). Troe decay with $F_c = 0.5$

[c] H_2(2.5). H_2O(12.0). CO(1.9). CO_2(3.8). Ar(0.4). He(0.4)

[d] H_2(2.5). H_2O(12.0). CO(1.9). CO_2(3.8). Ar(0.5). He(0.5)

[e] H_2(2.5). H_2O(12.0). CO(1.9). CO_2(3.8). Ar(0.2). He(0.2)

[f] H_2(2.5). H_2O(12.0). CO(1.9). CO_2(3.8). Ar(0.7). He(0.7)

[g] H_2(2.5). H_2O(6.0). H_2O_2(6.0). CO(1.5). CO_2(2.0). Ar(0.7). He(0.4). Troe decay with $F_c = 0.43$

obtaining a skeletal mechanism for its oxidation from low to high pressures, for premixed and non-premixed flames, and for autoignition. The result of this process is shown in Table 2.2.

A skeletal mechanism should retain the most important reactions that are able to accurately represent the combustion processes of interest. The obtention of such a mechanism is out of the scope of this book; however, the interested reader should

refer to the article by Xue et al. [29]. The skeletal mechanism presented in Table 2.2 represents a reduction of the original mechanism presented in Table 2.1. In this case, only 12 reactions are needed, and among them, only 6 reactions are considered in both directions.

The reaction rates for the species shown in Table 2.2 are written by using the nomenclature shown in Eqs. (2.18)–(2.20); however, for simplicity, $\omega_k = \omega_{kf} - \omega_{kb}$.

$$\frac{dC_{H_2}}{dt} = -\omega_2 - \omega_3 + \omega_7 + \omega_{12} - \omega_{17b} \tag{2.87}$$

$$\frac{dC_{O_2}}{dt} = -\omega_1 - \omega_{5f} + \omega_7 + \omega_{10f} + \omega_{16f} \tag{2.88}$$

$$\frac{dC_{H_2O}}{dt} = \omega_3 + \omega_{10f} + \omega_{11} \tag{2.89}$$

$$\frac{dC_{H}}{dt} = -\omega_1 + \omega_2 + \omega_3 - \omega_{5f} - \omega_{6f} - \omega_7 - \omega_{11} - 2\omega_{12} + \omega_{17b} \tag{2.90}$$

$$\frac{dC_{HO_2}}{dt} = \omega_{5f} - \omega_{6f} - \omega_7 - \omega_{10f} - 2\omega_{16f} - \omega_{17b} \tag{2.91}$$

$$\frac{dC_{H_2O_2}}{dt} = -\omega_{15b} + \omega_{16f} + \omega_{17b} \tag{2.92}$$

$$\frac{dC_{OH}}{dt} = \omega_1 + \omega_2 - \omega_3 + 2\omega_{6f} - \omega_{10f} - \omega_{11} + 2\omega_{15b} \tag{2.93}$$

$$\frac{dC_{O}}{dt} = \omega_1 - \omega_2 \tag{2.94}$$

The reaction rates to be considered equal to zero, through the steady-state approximation, correspond to the radicals OH and O. Therefore, the first step is to rewrite the reaction rates by adding or subtracting the reaction rates of OH and O to the reaction rates of H_2, H_2O, and H.

$$\frac{dC_{H_2}}{dt} - \frac{dC_{OH}}{dt} - 2\frac{dC_{O}}{dt} = -3\omega_1 - 2\omega_{6f} + \omega_7 + \omega_{10f} + \omega_{11} + \omega_{12} - 2\omega_{15b} - \omega_{17b} \tag{2.95}$$

$$\frac{dC_{H_2O}}{dt} + \frac{dC_{OH}}{dt} + \frac{dC_{O}}{dt} = 2\omega_1 + 2\omega_{6f} + 2\omega_{15b} \tag{2.96}$$

$$\frac{dC_{\mathrm{H}}}{dt}+\frac{dC_{\mathrm{OH}}}{dt}+2\frac{dC_{\mathrm{O}}}{dt}=2\omega_1-\omega_{5f}+\omega_{6f}-\omega_7-\omega_{10f}-2\omega_{11}-2\omega_{12}+2\omega_{15b}+\omega_{17b} \tag{2.97}$$

Since the reaction rates of HO_2 and H_2O_2 have not been modified on the right-hand side, then it is possible to define the following combinations of ω_i.

$$\omega_{\mathrm{III}}=\omega_{5f}-\omega_{6f}-\omega_7-\omega_{10f}-2\omega_{16f}-\omega_{17b} \tag{2.98}$$

$$\omega_{\mathrm{IV}}=-\omega_{15b}+\omega_{16f}+\omega_{17b} \tag{2.99}$$

Adding and subtracting the right-hand sides and the left-hand sides of Eqs. (2.98) and (2.99) into the right-hand side of the reaction rate of O_2 yield

$$\frac{dC_{\mathrm{O_2}}}{dt}=-\omega_1-\omega_{6f}-\omega_{15b}-\omega_{\mathrm{III}}-\omega_{\mathrm{IV}}$$

The above combination suggests the following definition.

$$\omega_{\mathrm{I}}=\omega_1+\omega_{6f}+\omega_{15b} \tag{2.100}$$

Therefore, for the reaction rate of H_2O, given in Eq. (2.96), the following is obtained.

$$\frac{dC_{\mathrm{H_2O}}}{dt}+\frac{dC_{\mathrm{OH}}}{dt}+\frac{dC_{\mathrm{O}}}{dt}=2\omega_{\mathrm{I}} \tag{2.101}$$

The reaction rate for H, given in Eq. (2.95), can be rearranged in the same manner shown for the reaction rate of O_2. This procedure suggests the definition of a ω_{II} as shown in Eq. (2.102), and the reaction rate of H is written as shown in Eq. (2.103).

$$\omega_{\mathrm{II}}=\omega_{5f}+\omega_{11}+\omega_{12}-\omega_{16f}-\omega_{17f} \tag{2.102}$$

$$\frac{dC_{\mathrm{H}}}{dt}+\frac{dC_{\mathrm{OH}}}{dt}+2\frac{dC_{\mathrm{O}}}{dt}=2\omega_{\mathrm{I}}-2\omega_{\mathrm{II}}+\omega_{\mathrm{III}} \tag{2.103}$$

Then, the reaction rate for H_2, given in Eq. (2.95), can be expressed as shown below.

$$\frac{dC_{\mathrm{H_2}}}{dt}-\frac{dC_{\mathrm{OH}}}{dt}-2\frac{dC_{\mathrm{O}}}{dt}=-3\omega_{\mathrm{I}}+\omega_{\mathrm{II}}-\omega_{\mathrm{III}}-\omega_{\mathrm{IV}} \tag{2.104}$$

Applying the steady-state approximation, the reactions rates of OH and O in Eqs. (2.93) and (2.94) are equal to zero. Then, the concentrations of OH and O can be obtained by solving the algebraic equations shown in Eqs. (2.105) and (2.106).

$$\omega_1+\omega_2-\omega_3+2\omega_{6f}-\omega_{10f}-\omega_{11}+2\omega_{15b}=0 \tag{2.105}$$

$$\omega_1 - \omega_2 = 0 \tag{2.106}$$

The reaction rate equations for H_2, O_2, H_2O, H, HO_2, and H_2O_2 can be rewritten by making the reaction rates of OH and O zero and by considering the definitions of ω_I, ω_{II}, ω_{III}, and ω_{IV}, as given in Eqs. (2.100), (2.102), (2.98), and (2.99), respectively.

$$\frac{dC_{H_2}}{dt} = -3\omega_I + \omega_{II} - \omega_{III} - \omega_{IV} \tag{2.107}$$

$$\frac{dC_{O_2}}{dt} = -\omega_I - \omega_{III} - \omega_{IV} \tag{2.108}$$

$$\frac{dC_{H_2O}}{dt} = 2\omega_I \tag{2.109}$$

$$\frac{dC_H}{dt} = 2\omega_I - 2\omega_{II} + \omega_{III} \tag{2.110}$$

$$\frac{dC_{HO_2}}{dt} = \omega_{III} \tag{2.111}$$

$$\frac{dC_{H_2O_2}}{dt} = \omega_{IV} \tag{2.112}$$

Since there are four modified rates of reaction and six species, it is possible to propose a four-step mechanism involving the six species. By inspecting Eqs. (2.107)–(2.112), it is noticed that the first reaction involves H_2, O_2, H_2O, and H, as shown in RI. The second reaction involves H_2 and H, and it is shown as RII. The third reaction has the participation of H_2, O_2, H, and HO_2, as shown in RIII. Finally, the fourth reaction involves H_2, O_2, and H_2O_2, as given in RIV. The rates for the reactions RI–RIV are ω_I to ω_{IV}, respectively.

$$3H_2 + O_2 \overset{I}{\leftrightarrow} 2H_2O + 2H \tag{RI}$$

$$H + H + M \overset{II}{\leftrightarrow} H_2 + M \tag{RII}$$

$$H_2 + O_2 \overset{III}{\leftrightarrow} HO_2 + H \tag{RIII}$$

$$H_2 + O_2 \overset{IV}{\leftrightarrow} H_2O_2 \tag{RIV}$$

Since it is easier to solve the system by having the concentrations of OH and O in explicit form, the following can be obtained by isolating those concentrations in Eqs. (2.103) and (2.104), respectively.

$$C_{OH} = \frac{k_{1f}C_H C_{O_2} + k_{2f}C_{H_2}C_O + k_{3b}C_{H_2O}C_H + 2k_{6f}C_{HO_2}C_H + k_{11b}C_{H_2O}C_{M11} + 2k_{15b}C_{H_2O_2}C_{M15}}{k_{1b}C_O + k_{2b}C_H + k_{3f}C_{H_2} + k_{10f}C_{HO_2} + k_{11f}C_H C_{M11}} \tag{2.113}$$

$$C_O = \frac{k_{1f}C_H C_{O_2} + k_{2b}C_{OH}C_H}{k_{1b}C_{OH} + k_{2f}C_{H_2}} \tag{2.114}$$

Therefore, the procedure for obtaining a reduced mechanism has been shown using the hydrogen oxidation mechanism as an example and following the procedure presented by Sánchez et al. [18]. Other examples of reduced mechanisms are available. For instance, the book edited by Peters and Rogg [31] presents reduced mechanisms for hydrogen, carbon monoxide, methane, propane, ethylene, acetylene, and methanol, deduced for unstretched premixed flames and for counterflow diffusion flames. Also, multipurpose reduced mechanisms for methanol and ethanol have been obtained by Fernández–Tarrazo et al. [32] and by Millán–Merino et al. [33], respectively.

2.6 One-Step Chemical Reaction

The use of one-step chemical kinetic reactions for the study of deflagrations or detonations dynamics is widely extended. The reason for this is that lower computational capabilities are required when this kind of reaction is used. Among the various approaches, the one-step mechanisms obtained by Westbrook e Dryer [19] are of key importance. They assumed the reaction shown in Eq. (2.115).

$$C_{x_C}H_{x_H}O_{x_O} + v^s_{ar}O_2 \rightarrow x_C CO_2 + \frac{x_H}{2}H_2O \tag{2.115}$$

The fuel's reaction rate is represented by Eq. (2.116). The activation energy was assumed, and the pre-exponential factor was adjusted together with the exponents using laminar flame speed data. Note that in the original article, four velocity points were considered: one for the stoichiometric composition, one for the maximum laminar flame velocity, and two velocities near the flammability limits.

$$\frac{dC_F}{dt} = -k_f C_F^x C_{O_2}^y \tag{2.116}$$

The specific reaction rate constant is determined by the expression given in Eq. (2.22). The reaction rates of fuel and oxygen are related through Eq. (2.117).

$$\frac{dC_{O_2}}{dt} = v^s_{ar}\frac{dC_F}{dt} = -v^s_{ar}k_f C_F^x C_{O_2}^y \tag{2.117}$$

The activation energies and the exponents x and y obtained by Westbrook and Dryer for several hydrocarbons are presented in Table 2.3. With the advancement of computational capabilities, at present, researchers mostly adjust their own one-step

Table 2.3 Results for one-step mechanism obtained by Westbrook and Dryer [19, 35]

Fuel	Formula	$A_0 \left(\frac{\text{mol}}{\text{cm}^3}\right)^{1-x-y}/\text{s}$	$E_a \left(\frac{\text{kcal}}{\text{mol}}\right)$	x	y
Methane	CH_4[a]	1.3×10^8	48.4	−0.30	1.30
Methane	CH_4	8.3×10^5	30.0	−0.30	1.30
Ethane	C_2H_6	1.1×10^{12}	30.0	0.10	1.65
Propane	C_3H_8	8.6×10^{11}	30.0	0.10	1.65
Butane	C_4H_{10}	7.4×10^{11}	30.0	0.15	1.60
Pentane	C_5H_{12}	6.4×10^{11}	30.0	0.25	1.50
Hexane	C_6H_{14}	5.7×10^{11}	30.0	0.25	1.50
Heptane	C_7H_{16}	5.1×10^{11}	30.0	0.25	1.50
Octane	C_8H_{18}	4.6×10^{11}	30.0	0.25	1.50
Octane	C_8H_{18}[b]	7.2×10^{12}	40.0	0.25	1.50
Nonane	C_9H_{20}	4.2×10^{11}	30.0	0.25	1.50
Decane	$C_{10}H_{22}$	3.8×10^{11}	30.0	0.25	1.50
Methanol	CH_3OH	3.2×10^{12}	30.0	0.25	1.50
Ethanol	C_2H_5OH	1.5×10^{12}	30.0	0.15	1.60
Benzene	C_6H_6	2.0×10^{11}	30.0	−0.10	1.85
Toluene	C_7H_8	1.6×10^{11}	30.0	−0.10	1.85
Ethylene	C_2H_4	2.0×10^{12}	30.0	0.10	1.65
Propylene	C_3H_6	4.2×10^{11}	30.0	−0.10	1.85
Acetylene	C_2H_2	6.5×10^{12}	30.0	0.50	1.25

Adapted from Progress in Energy and Combustion Science, Vol 10, G. Charles K. Westbrook, Frederick L. Dryer, Chemical kinetic modeling of hydrocarbon combustion, 1–57, Copyright (1984), with permission from Elsevier

Adapted from Combustion Science & Technology, Vol 27, G. Charles K. Westbrook, Frederick L. Dryer, Simplified Reaction Mechanisms for the Oxidation of Hydrocarbon Fuels in Flames, Copyright (2007), with permission from Taylor & Francis

chemical kinetic reaction rate before conducting a simulation. An example of such an approximation is presented in the work by Xiao et al. [34].

$$\frac{dY_F}{dt} = A\rho Y_F \exp\left(-\frac{E_a}{RT}\right) \tag{2.118}$$

This concludes the chapter regarding chemical kinetics. As mentioned in the introduction, the chemical kinetics of combustion is a very ample subject, and this chapter was intended to provide the necessary background for the developments presented in next chapters.

2.7 Closure

This chapter presented the fundamental concepts and definitions of chemical kinetics theory applied to combustion science. Chemical kinetics allows for the introduction of time as an independent variable. Therefore, extending the understanding of

combustion processes. The chapter discussed the elementary reactions, and their types, in the context of combustion. The Law of mass action and the Arrhenius Law were introduced to study the time evolution of species concentration in elementary reactions. The specific reaction rate constant is defined through the Arrhenius Law and the specific reaction rate constant of the reverse reaction is obtained by applying chemical equilibrium considerations. The chapter also discusses chemical kinetic mechanisms with emphasis on hydrogen, hydrocarbons, methanol and ethanol. It is emphasized that detailed mechanisms are postulated, which means that they need to be validated with data from fundamental experiments. Finally, methods for reducing chemical kinetic mechanisms and general aspects of one-step chemical kinetic mechanisms were also presented.

Conservation Equations

3

Nomenclature

Symbols

A_{ij}	Area of the face of the control volume in the plane ij, m^2
c_p	Specific heat capacity at constant pressure, J/kg K
$c_{p,\ k}$	Specific heat capacity at constant pressure of the species k, J/kg s
D	Mass diffusion coefficient, m^2/s
$\vec{e}_{\ i}$	Unit vector in coordinate i
e	Total specific energy, J/kg
E	Total energy, J
$\vec{F}$	Resultant force, N
$\vec{f}_{\ vc}$	Surface force per unit volume, N/m^3
h	Specific enthalpy, J/kg
h_k	Specific enthalpy of the species k, J/kg
$h^0_{f,k}$	Formation enthalpy of species k, J/kg
k	Thermal conductivity, W/m K
m_{vc}	Mass of control volume, kg
m	Mass, kg
$\dot{m}_i$	Mass flow rate in coordinate i, kg/s
$\dot{m}_{k,i,\text{diff}}$	Mass flux of the species k, due to the diffusion of the species in the coordinate i, kg/m^2 s
$\vec{\dot{m}}_{k,i,\text{diff}}$	Species mass flux vector k, due to the diffusion of the species in the coordinate i, kg/m^2 s
$\dot{m}_{k,R}$	Net production rate of the species k per unit volume, kg/m^3 s
p	Pressure, Pa

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$\dot{q}_{\text{cond},vc}$	Heat transfer by conduction in the control volume, W/kg
$\dot{q}_{ge,vc}$	Heat generated within the control volume, W/kg
r	Radial distance from origin, m
T	Temperature, K
u_i	Velocity component in the coordinate i, m/s
$u_{k,\ \text{diff},\ i}$	Component of the species diffusion rate k at the coordinate i, m/s
$\vec{V}$	Velocity vector, m/s.
$\vec{V}_k$	Velocity vector of the species k due to the total flow rate of the mixture, m/s
$\vec{V}_M$	Velocity vector of point M of a fluid element, m/s
$\vec{V}_{k,\text{diff}}$	Mass diffusion velocity vector of the species k, m/s
$\dot{w}_{vc}$	Work done on the control volume, W/kg
Y_k	Mass fraction of the species k

Greek Letters

δ_{ij}	Kronecker delta function
∇	Nabla operator
$\dot{\varepsilon}_{ij}$	Deformation rate in coordinates ij
θ	Positive angle between r and the z axis, also known as polar angle, rad
υ_{vc}	Volume do volume de controle, m^3
ρ	Density, kg/m^3
ρ_k	Density of species k, kg/m^3
ρB_{vc}	Body force per unit volume, N/m^3
σ_{ij}	Stress tensor component acting in the plane normal to coordinate i and in the direction of the coordinate j, N/m^2
$\overleftrightarrow{\sigma}$	Second-order tensor, stress tensor, N/m^2
μ	Dynamic viscosity, Pa.s
μ'	Second coefficient of viscosity, Pa.s
φ	Positive angle between the projection of r on the xy plane and the x axis, also known as azimuthal angle, rad
Φ_v	Viscous dissipation function

3.1 Introduction

The analysis of the flame propagation process requires the application of the conservation equations, most of the time in differential form. Therefore, in this chapter, the conservation equations for the total mass, the mass of the chemical species, and the energy will be derived together with the equation of momentum. Generally, these derivations are given for a rectangular geometry. In this case, the conservation equations will be derived for spherical coordinates, which are relevant

for the classical spherical flame propagation. The reader will find it interesting how the derivation performed for spherical coordinates reveals important considerations that are not so obvious when rectangular coordinates are adopted. The conservation equations will also be given in vector form so that their conversion to other coordinate systems will be straightforward. The derivations are based on the classic books by Kuo [9], Williams [36], Turns [7], and Bird et al. [37]. Moreover, the textbooks by Law [11], Poinsot and Veynante [38], and Clavin and Searby [39] were also considered to develop the derivations. Although the classical experiments on the propagation of a spherical flame may justify the use of the spherical coordinate system, carrying out the derivations in this coordinate system is also a good exercise to dominate the principles behind the use of conservation equations in differential form.

3.2 Total Mass Conservation

The total mass of a system is a conserved scalar quantity. Figure 3.1 presents the volume element in spherical coordinates. The figure shows the dimensions, areas, and volume related. The flow entering and leaving the control volume is shown in Fig. 3.2 in three dimensions. Remember that the relations between spherical and rectangular coordinates are given by Eq. (3.1).

$$x = r\sin\theta\cos\varphi; \quad y = r\sin\theta\sin\varphi; \quad z = r\cos\theta \tag{3.1}$$

The physical quantities presented in Fig. 3.2 are listed and described below for completeness.

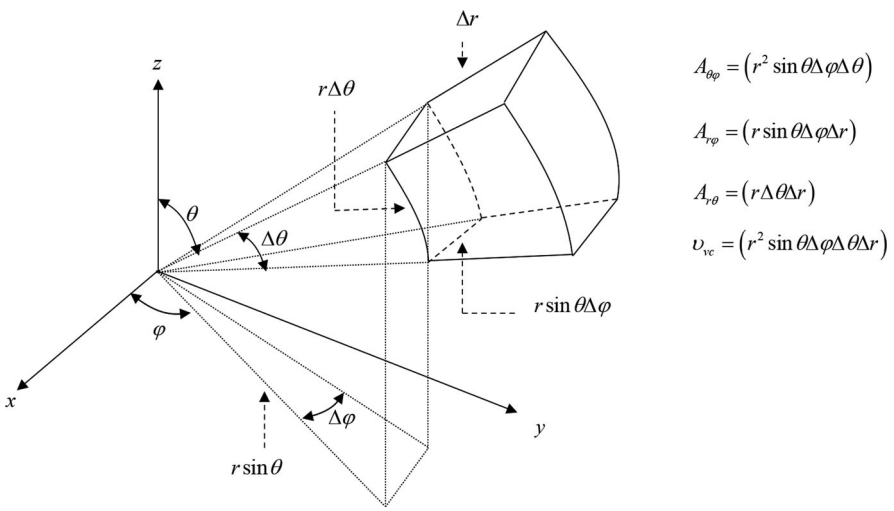


Fig. 3.1 Dimensions, areas, and volume of the control volume in spherical coordinates

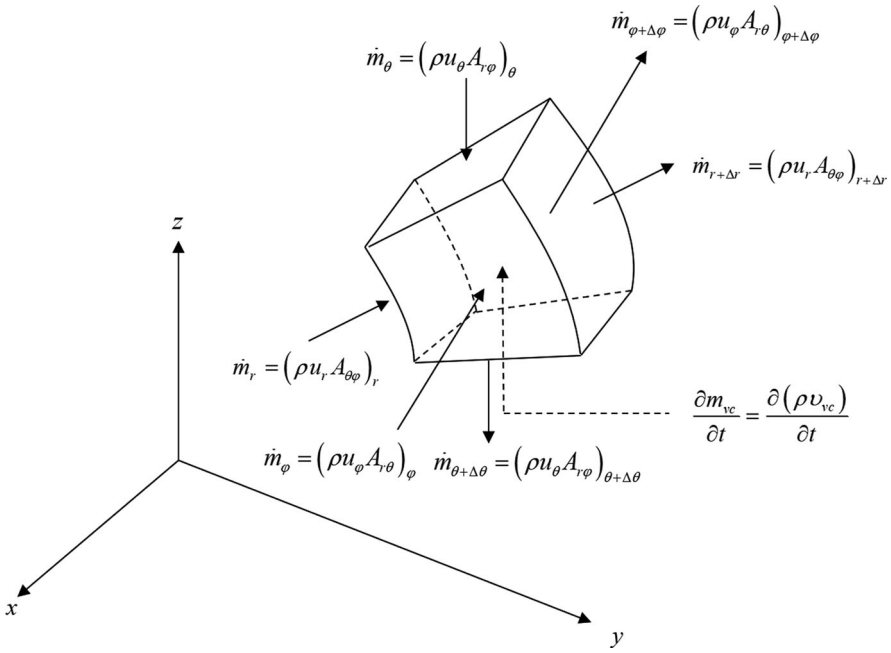


Fig. 3.2 Control volume for determining the total mass conservation equation in spherical coordinates

$\dot{m}_r = \left(\rho u_r A_{\theta\varphi}\right)_r$	:	Mass flow rate entering the control volume in the radial direction (kg/s)
$\dot{m}_{r+\Delta r} = \left(\rho u_r A_{\theta\varphi}\right)_{r+\Delta r}$	:	Mass flow rate leaving the control volume in the radial direction r (kg/s)
$\dot{m}_\theta = \left(\rho u_\theta A_{r\varphi}\right)_\theta$	:	Mass flow rate entering the control volume in the polar angle direction θ (kg/s)
$\dot{m}_{\theta+\Delta\theta} = \left(\rho u_\theta A_{\theta\varphi}\right)_{\theta+\Delta\theta}$	:	Mass flow rate leaving the control volume in the polar angle direction θ (kg/s)
$\dot{m}_\varphi = \left(\rho u_\varphi A_{r\theta}\right)_\varphi$	:	Mass flow rate entering the control volume in the azimuthal angle direction φ (kg/s)
$\dot{m}_{\varphi+\Delta\varphi} = \left(\rho u_\varphi A_{r\theta}\right)_{\varphi+\Delta\varphi}$	:	Mass flow rate leaving the control volume in the azimuthal angle direction φ (kg/s)
$\frac{\partial m_{vc}}{\partial t} = \frac{\partial(\rho v_{vc})}{\partial t}$	:	Mass accumulation rate within the control volume (kg/s)

The conservation of total mass can be expressed in the following terms.

Mass accumulation rate in the control volume	=	Mass flow rate entering the control volume	−	Mass flow rate leaving the control volume

The flow velocity ($\vec{V}$) can be written in terms of its components in spherical coordinates, as shown below.

$$\vec{V} = u_r \vec{e}_r + u_\theta \vec{e}_\theta + u_\varphi \vec{e}_\varphi \tag{3.2}$$

$\vec{e}_r$, $\vec{e}_\theta$, $\vec{e}_\varphi$ represent the orthogonal unit vectors of the spherical coordinates, and u_r, u_θ, and u_φ are the scalar velocity components in each coordinate. Applying the total mass conservation and the values shown in Fig. 3.2, Eq. (3.3) is obtained.

$$\frac{\partial m_{vc}}{\partial t} = (\dot{m}_r)_r - (\dot{m}_r)_{r+\Delta r} + (\dot{m}_\theta)_\theta - (\dot{m}_\theta)_{\theta+\Delta\theta} + (\dot{m}_\varphi)_\varphi - (\dot{m}_\varphi)_{\varphi+\Delta\varphi} \tag{3.3}$$

Substituting into Eq. (3.3) the expressions for the mass flow rates shown in Fig. 3.2, subsequently, substituting the expressions for the areas and volume shown in Fig. 3.1, the Eq. (3.4) is obtained.

$$\begin{aligned} r^2 \sin\theta \Delta\varphi \Delta\theta \Delta r \frac{\partial \rho}{\partial t} = {} & (\rho u_r r^2 \sin\theta \Delta\varphi \Delta\theta)_r - (\rho u_r r^2 \sin\theta \Delta\varphi \Delta\theta)_{r+\Delta r} \\ & + (\rho u_\theta r \sin\theta \Delta\varphi \Delta r)_\theta - (\rho u_\theta r \sin\theta \Delta\varphi \Delta r)_{\theta+\Delta\theta} \\ & + (\rho u_\varphi r \Delta\theta \Delta r)_\varphi - (\rho u_\varphi r \Delta\theta \Delta r)_{\varphi+\Delta\varphi} \end{aligned} \tag{3.4}$$

It is important to note that the control volume is fixed in space; this means that the volume $v_{vc} = r^2 \sin\theta \Delta\varphi \Delta\theta \Delta r$ is constant in time. Dividing Eq. (3.4) by $\Delta\varphi \Delta\theta \Delta r$,

$$\begin{aligned} r^2 \sin\theta \frac{\partial \rho}{\partial t} = {} & -\frac{(\rho u_r r^2 \sin\theta)_{r+\Delta r} - (\rho u_r r^2 \sin\theta)_r}{\Delta r} - \frac{(\rho u_\theta r \sin\theta)_{\theta+\Delta\theta} - (\rho u_\theta r \sin\theta)_\theta}{\Delta\theta} \\ & - \frac{(\rho u_\varphi r)_{\varphi+\Delta\varphi} - (\rho u_\varphi r)_\varphi}{\Delta\varphi} \end{aligned}$$

Taking the limit when Δr, $\Delta\theta$, $\Delta\varphi \to 0$ in the above equation, the partial derivative definition appears. Thus, after taking the limits and rearranging, the total mass conservation equation is obtained.

Total mass conservation equation in spherical coordinates

$$\frac{\partial \rho}{\partial t} = -\frac{1}{r^2} \frac{\partial (\rho u_r r^2)}{\partial r} - \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} (\rho u_\theta \sin\theta) - \frac{1}{r \sin\theta} \frac{\partial (\rho u_\varphi)}{\partial \varphi} \tag{3.5}$$

Equation (3.5) is the total mass conservation equation in spherical coordinates. It is possible to recognize the divergence of $(\rho \vec{V})$.

$$\nabla \cdot \left(\rho \vec{V}\right) = \frac{1}{r^2} \frac{\partial (\rho u_r r^2)}{\partial r} + \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} (\rho u_\theta \sin\theta) + \frac{1}{r \sin\theta} \frac{\partial (\rho u_\varphi)}{\partial \varphi} \tag{3.6}$$

The Nabla operator (∇) is given in spherical coordinates by the expression in Eq. (3.7). The reader should look at the Appendix of the book for more details about how to obtain the expression for $(\nabla \cdot \vec{V})$.

$$\nabla = \vec{e}_r \frac{\partial}{\partial r} + \vec{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \vec{e}_\varphi \frac{1}{r \sin\theta} \frac{\partial}{\partial \varphi} \tag{3.7}$$

Although Eq. (3.5) is useful for a spherical geometry, it is also important to obtain an expression that can be extended to other coordinate systems, such as cylindrical and rectangular systems. This is possible by considering the definition of the divergence given in Eq. (3.6). Therefore, applying that definition to Eq. (3.5) yields the total mass conservation equation in terms of the partial derivative of the density with respect to time and the divergence of the velocity multiplied by the density.

Mass conservation equation in compact form

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho \vec{V} \right) = 0 \tag{3.8}$$

The Eq. (3.8) completes the derivation of the total mass conservation equation. To obtain this equation in other coordinate systems, it will be necessary to consider the corresponding expression for the Nabla operator (∇) and apply the divergence to the velocity vector.

3.3 Species Mass Conservation

The combustion of a mixture of reactants involves several chemical species which are produced and/or consumed by the elemental chemical reactions. The number of species present in the combustion reaction will depend on the chemical kinetic mechanism that is postulated, as was discussed in Chap. 2. Therefore, the law of conservation of mass of the chemical species can be established as shown below for any species A.

$$\begin{array}{c} \text{Mass accumulation} \\ \text{rate for species A in} \\ \text{the control volume} \end{array} = \begin{array}{c} \text{Mass flow rate of} \\ \text{species A entering} \\ \text{the control volume} \end{array} - \begin{array}{c} \text{Mass flow rate of} \\ \text{species A leaving the} \\ \text{control volume} \end{array}$$

$$+ \left(\begin{array}{c} \text{Rate of production} \\ \text{and/or consumption} \\ \text{of species A by} \\ \text{chemical reactions} \end{array} \right)$$

Figure 3.3 shows the control volume in spherical coordinates. The term $\dot{m}_{A,R}$ depends on the chemical kinetic mechanism postulated for the mixture. However, in this chapter it is enough to imagine that the chemical species A is being produced and

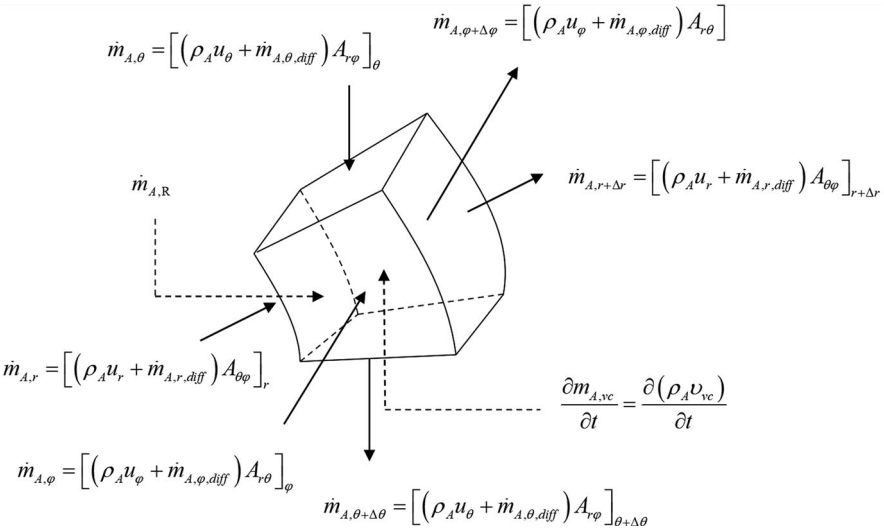


Fig. 3.3 Control volume for the species mass conservation in spherical coordinates

consumed by several elementary chemical reactions which form the mechanism. Then, the net production rate of the chemical species A is denoted by $\dot{m}_{A,R}$. The following physical quantities can be observed on Fig. 3.3.

$(\rho_A u_i)_i$	:	Mass flux of species A, due to the bulk velocity of the mixture at coordinate, entering the control volume (kg/m^2 s)
$(\rho_A u_i)_{i\,+\,\Delta i}$	:	Mass flux of species A, due to the bulk velocity of the mixture at coordinate, leaving the control volume (kg/m^2 s)
$(\dot{m}_{A,i,dif})_i$	:	Mass flux of species A due to diffusion of the species in the coordinate, entering the control volume (kg/m^2 s)
$(\dot{m}_{A,i,diff})_{i+\Delta i}$	:	Mass flux of species A due to diffusion of the species in the coordinate, leaving the control volume (kg/m^2 s)
$\frac{\partial(\rho_A \upsilon_{vc})}{\partial t}$	:	Mass accumulation rate of species A in the control volume (kg/s)
$\dot{m}_{A,R}$	:	Net production rate of species A per unit volume (kg/m^3 s)

The species mass conservation equation for a chemical species A is represented mathematically by writing a balance between the mass flows of species A entering and leaving the control volume in the three directions of the coordinate system. Afterwards, the accumulation and production of the species in the control volume is considered. The result of this balance is shown in Eq. (3.9).

$$r^2 \sin\theta \Delta\varphi \Delta\theta \Delta r \frac{\partial \rho_A}{\partial t} = \left[(\rho_A u_r + \dot{m}_{A,r,\mathrm{diff}}) r^2 \sin\theta \Delta\varphi \Delta\theta\right]_r - \left[(\rho_A u_r + \dot{m}_{A,r,\mathrm{diff}}) r^2 \sin\theta \Delta\varphi \Delta\theta\right]_{r+\Delta r}$$

$$+[(\rho_A u_\theta + \dot{m}_{A,\theta,\text{diff}})r\sin\theta\Delta\varphi\Delta r]_\theta - [(\rho_A u_\theta + \dot{m}_{A,\theta,\text{diff}})r\sin\theta\Delta\varphi\Delta r]_{\theta+\Delta\theta}$$

$$+[(\rho_A u_\varphi + \dot{m}_{A,\varphi,\text{diff}})r\Delta\theta\Delta r]_\varphi - [(\rho_A u_\varphi + \dot{m}_{A,\varphi,\text{diff}})r\Delta\theta\Delta r]_{\varphi+\Delta\varphi}$$

$$+\dot{m}_{A,R} r^2 \sin\theta\Delta\varphi\Delta\theta\Delta r \tag{3.9}$$

Dividing by $\Delta\varphi\Delta\theta\Delta r$, taking the limit for Δr, $\Delta\theta$, $\Delta\varphi \to 0$, and remembering that the control volume is fixed in space,

$$r^2\sin\theta\frac{\partial\rho_A}{\partial t} = -\frac{\partial}{\partial r}[(\rho_A u_r + \dot{m}_{A,r,\text{diff}})r^2\sin\theta] - \frac{\partial}{\partial\theta}[(\rho_A u_\theta + \dot{m}_{A,\theta,\text{diff}})r\sin\theta]$$

$$-\frac{\partial}{\partial\varphi}[(\rho_A u_\varphi + \dot{m}_{A,\varphi,\text{diff}})r] + \dot{m}_{A,R} r^2\sin\theta$$

Rearranging and dividing by $r^2\sin\theta$, the expression for the mass conservation of species A is obtained, as shown in Eq. (3.10).

Mass conservation equation of species A in spherical coordinates

$$\frac{\partial\rho_A}{\partial t} = -\frac{1}{r^2}\frac{\partial(\rho_A u_r r^2)}{\partial r} - \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\rho_A u_\theta \sin\theta) - \frac{1}{r\sin\theta}\frac{\partial(\rho_A u_\varphi)}{\partial\varphi}$$

$$-\frac{1}{r^2}\frac{\partial(\dot{m}_{A,r,\text{diff}} r^2)}{\partial r} - \frac{1}{r\sin\theta}\frac{\partial(\dot{m}_{A,\theta,\text{diff}}\sin\theta)}{\partial\theta} - \frac{1}{r\sin\theta}\frac{\partial(\dot{m}_{A,\varphi,\text{diff}})}{\partial\varphi} + \dot{m}_{A,R} \tag{3.10}$$

Equation (3.10) is the mass conservation equation for species A in spherical coordinates. It is possible to recognize the divergence of the sum of the vectors $\rho_A\vec{V}$ and $\vec{\dot{m}}_{A,\text{dif}}$; therefore, Eq. (3.10) can be rewritten in the compact form given in Eq. (3.11).

Mass conservation equation of chemical species A in compact form

$$\frac{\partial\rho_A}{\partial t} + \nabla\cdot\left(\rho_A\vec{V} + \vec{\dot{m}}_{A,\text{diff}}\right) = \dot{m}_{A,R} \tag{3.11}$$

The two expressions given above can be further developed. The average velocity of species A is introduced as $\vec{V}_A$. Likewise, the other species k present in the fluid will have their own average velocities $\vec{V}_k$. So far, only the velocity of the mixture has been considered. The mass flow rate of the species has been represented by the sum of two flow rates: (a) the mass flow rate due to the bulk mixture velocity and (b) the mass flow rate due to the diffusion of the species. Denoting the mass flux of species A as $\dot{m}_A$ and noting that $\rho_A = \rho Y_A$, the following is obtained.

$$\dot{m}_A = \rho Y_A\vec{V}_A = \rho Y_A\left(u_{A,r}\vec{e}_r + u_{A,\theta}\vec{e}_\theta + u_{A,\varphi}\vec{e}_\varphi\right) \tag{3.12}$$

Naturally, the mass flux of the mixture will be equal to the sum of the mass fluxes of all species, given that the area is constant. Writing each mass flux as shown in

Eq. (3.12) and equating it to the mass flux of the mixture, which is determined using the bulk mixture velocity, the following result will be obtained.

$$\vec{V} = \sum Y_k \vec{V}_k \tag{3.13}$$

Rewriting the mass flux of species A as the sum of the mass fluxes due to the bulk mixture flow and diffusion, the expression given in Eq. (3.14) is obtained for the diffusion velocity of species A.

$$\vec{V}_{A,\text{dif}} = \vec{V}_A - \vec{V} \tag{3.14}$$

Therefore, Eq. (3.11) can be rewritten in terms of the average velocity of species A and the mole fraction of that species. The result of this procedure is another version of the mass conservation equation of species A, given in Eq. (3.15).

Mass conservation equation of species A in compact form and in terms of the average velocity of species A

$$\frac{\partial \rho Y_A}{\partial t} + \nabla \cdot \left(\rho Y_A \vec{V}_A \right) = \dot{m}_{A,R} \tag{3.15}$$

The sum of the mass conservation equations of the N species, written according to Eq. (3.15), is equal to the total mass conservation equation, recognizing that, when some species are produced, others are consumed. That is, $\sum \dot{m}_{k,R} = 0$. The mass flow rate of chemical species A due to diffusion can be expressed by applying Fick's law of diffusion [37], which is given below for spherical coordinates.

$$\vec{\dot{m}}_{A,\text{diff}} = -\rho D \left(\vec{e}_r \frac{\partial Y_A}{\partial r} + \vec{e}_\theta \frac{1}{r} \frac{\partial Y_A}{\partial \theta} + \vec{e}_\varphi \frac{1}{r \sin\theta} \frac{\partial Y_A}{\partial \varphi} \right) = -\rho D \nabla Y_A \tag{3.16}$$

The mass diffusivity (D) appears in Eq. (3.16). The units of the mass diffusivity are m^2/s. Diffusivity is usually defined for a binary mixture (species A and B). However, in multicomponent systems, it is not uncommon for the diffusivities of different pairs of species to be considered equal [40]. Substituting Fick's law into Eq. (3.11), the compact form of the mass conservation equation for chemical species A is obtained as shown in Eq. (3.17).

Mass conservation equation of species A in compact form using Fick's law

$$\frac{\partial (\rho Y_A)}{\partial t} + \vec{\nabla} \cdot \left(\rho \vec{V} Y_A - \rho D \vec{\nabla} Y_A \right) = \dot{m}_{A,R} \tag{3.17}$$

In spherical coordinates the mass fraction gradient is given by Eq. (3.18); therefore, Eq. (3.10) can be expressed as shown in Eq. (3.19).

$$\vec{\nabla} Y_A = \vec{e}_r \frac{\partial Y_A}{\partial r} + \vec{e}_\theta \frac{1}{r} \frac{\partial Y_A}{\partial \theta} + \vec{e}_\varphi \frac{1}{r \sin\theta} \frac{\partial Y_A}{\partial \varphi} \tag{3.18}$$

Mass conservation equation of species A in spherical coordinates using Fick's law

$$\frac{\partial(\rho Y_A)}{\partial t} = -\frac{1}{r^2}\frac{\partial(\rho Y_A u_r r^2)}{\partial r} - \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\rho Y_A u_\theta \sin\theta) - \frac{1}{r\sin\theta}\frac{\partial(\rho Y_A u_\varphi)}{\partial\varphi} + \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\rho D\frac{\partial Y_A}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\rho D\frac{\partial Y_A}{\partial\theta}\sin\theta\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial\varphi}\left(\rho D\frac{\partial Y_A}{\partial\varphi}\right) + \dot{m}_{A,R} \tag{3.19}$$

Equation (3.17) is widely used to express the mass conservation of chemical species. For example, the conservation equations of the total mass and of the mass of chemical species, Eqs. (3.8) and (3.17), were applied by Matalon and Metzener [41] in their study on the propagation of premixed flames in tubes.

3.4 The Momentum Equation

In this section, the Navier–Stokes equations will be obtained. Also, Euler equations will be presented as a particular form of the Navier–Stokes equations. Since momentum is a vector defined as the product of mass and velocity ($m\vec{V}$), it is balanced by forces acting on the control volume.

Accumulation rate of momentum in the control volume	=	Rate of momentum entering the control volume	−	Rate of momentum leaving the control volume
	+	Sum of forces acting on the control volume		

The resultant force ($\vec{F}$) acting on the control volume can be divided into two components: (a) the force corresponding to all surface forces per unit volume ($\vec{f}_{vc}$) and (b) the force corresponding to all the body forces per unit volume ($\rho\vec{B}_{vc}$). Therefore, Eq. (3.20) represents the resultant force acting on the control volume.

$$\vec{F} = \vec{f}_{vc}\upsilon_{vc} + \rho\vec{B}_{vc}\upsilon_{vc} \tag{3.20}$$

The body forces are distributed throughout the control volume without the need for direct contact. Two examples of this kind of force are the gravitational force and the force due to magnetic fields. For the purposes of this derivation, it will suffice to write the resultant body forces as a function of three components, one for each coordinate of the spherical system.

$$\rho \vec{B}_{vc} v_{vc} = \rho B_r v_{vc} \vec{e}_r + \rho B_\theta v_{vc} \vec{e}_\theta + \rho B_\varphi v_{vc} \vec{e}_\varphi \tag{3.21}$$

The body force given in Eq. (3.21) acts on the entire control volume, without differentiating between the species in the mixture. A force per unit mass acting on species k, is denoted by $\vec{f}_k$. The scalar component of this force in the ith coordinate is denoted by $f_{k,i}$. Thus, each scalar component of the body force can be written in terms of forces acting on the species per unit mass.

$$\rho B_i = \rho \sum_{k=1}^{N} Y_k f_{k,i} \tag{3.22}$$

Surface forces are caused by the vicinity and act directly on the surfaces of the control volume. Normal stresses act in the direction normal to the surface plane (like pressure), and shear stresses act on the surface plane. All these stresses are included in the second-order tensor that will be simply called stress tensor. Figure 3.4 shows the stress tensor components acting on the radial direction, and Fig. 3.5 shows all the stress tensor components. When the derivation of the momentum conservation equation is obtained in Cartesian (rectangular) coordinates, each coordinate can be considered separately. However, in the spherical coordinate system, it is necessary to consider all the components at the same time, as will be shown in the following

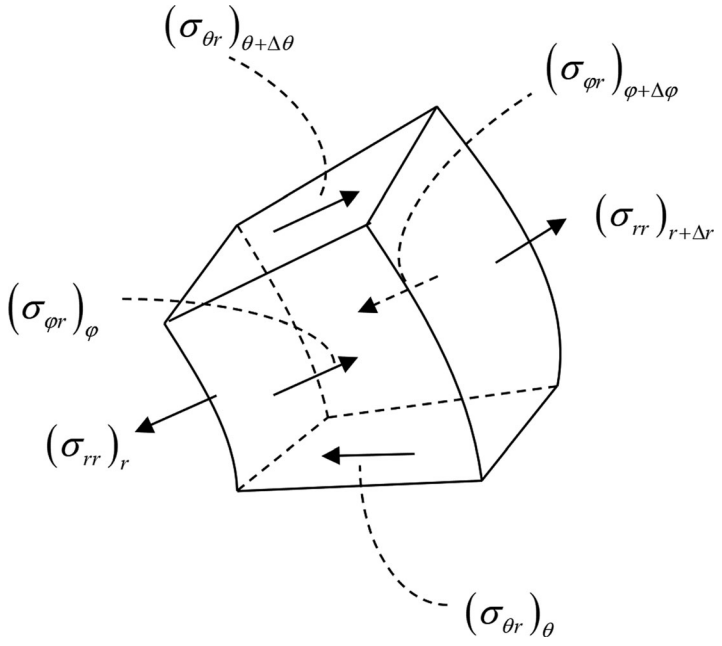


Fig. 3.4 Control volume in spherical coordinates showing stresses acting in the radial direction

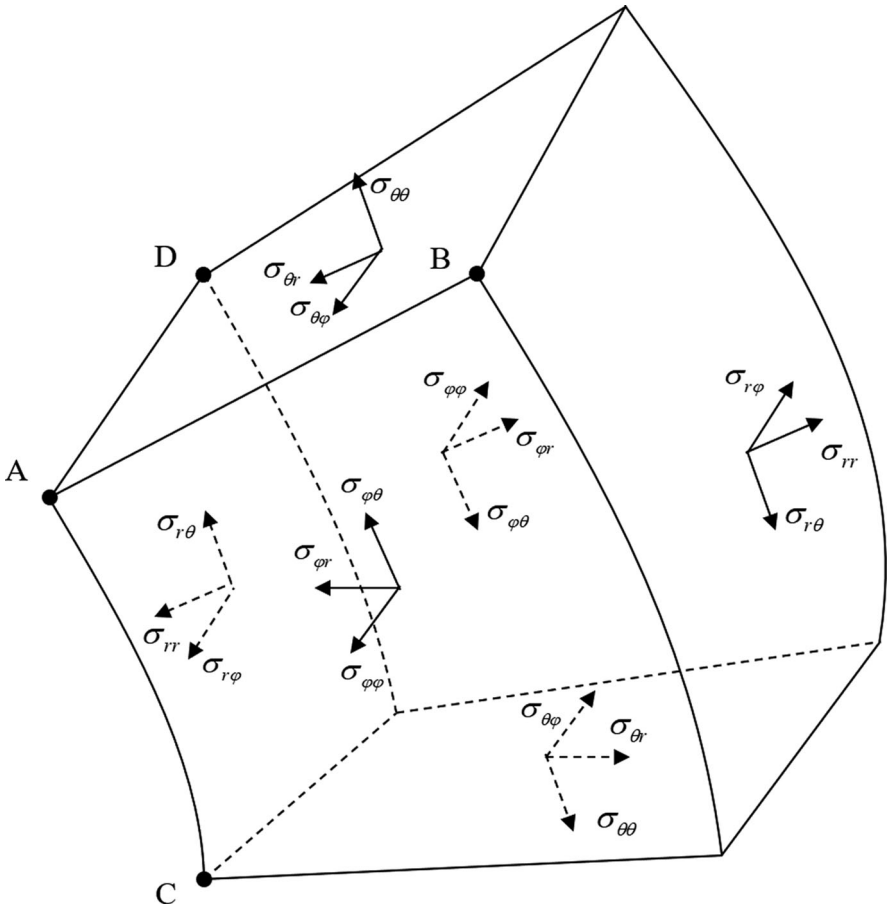


Fig. 3.5 Fluid element showing all stress components

procedure. Figure 3.4 shows the following stresses acting on the surface of the volume element.

σ_{rr}	:	Stress acting on the plane normal to r (plane $\theta\varphi$) and in the radial direction (normal stress)
$\sigma_{\theta r}$	:	Stress acting on the plane normal to θ (plane $r\varphi$) and in the radial direction (shear stress)
$\sigma_{\varphi r}$	:	Stress acting on the plane normal to φ (plane $r\theta$) and in the radial direction (shear stress)

Since the stress tensor is a second-order tensor, it is expressed in full as shown in Eq. (3.23).

$$\overleftrightarrow{\sigma} = \begin{matrix} \sigma_{rr}\vec{e}_r\vec{e}_r + \sigma_{r\theta}\vec{e}_\theta\vec{e}_r + \sigma_{r\varphi}\vec{e}_\varphi\vec{e}_r \\ +\sigma_{\theta r}\vec{e}_r\vec{e}_\theta + \sigma_{\theta\theta}\vec{e}_\theta\vec{e}_\theta + \sigma_{\theta\varphi}\vec{e}_\varphi\vec{e}_\theta \\ +\sigma_{\varphi r}\vec{e}_r\vec{e}_\varphi + \sigma_{\varphi\theta}\vec{e}_\theta\vec{e}_\varphi + \sigma_{\varphi\varphi}\vec{e}_\varphi\vec{e}_\varphi \end{matrix} \tag{3.23}$$

The stresses for which $i = j$ represent normal stresses, and those for which $i \neq j$ represent shear stresses. Each of these stress components is related to a force acting on a given surface in a particular direction. In order to keep the deduction as straightforward as possible, an Appendix with the intermediate mathematical expressions used is presented at the end of this chapter. The resultant surface force is given by the summation presented in Eq. (3.96) in the Appendix at the end of this chapter. It is important to note that the orthogonal unit vectors were considered in writing Eq. (3.96). Dividing by $\Delta\varphi\Delta\theta\Delta r$, taking the limit for Δr, $\Delta\theta$, $\Delta\varphi \to 0$, and rearranging, the following is obtained.

$$\begin{aligned} \vec{f}_{vc} = {} & \frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}\left(\sigma_{rr}r^2\sin\theta\,\vec{e}_r\right) + \frac{\partial}{\partial\theta}\left(\sigma_{\theta r}r\sin\theta\,\vec{e}_r\right) + \frac{\partial}{\partial\varphi}\left(\sigma_{\varphi r}r\,\vec{e}_r\right)\right] \\ & + \frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}\left(\sigma_{r\theta}r^2\sin\theta\,\vec{e}_\theta\right) + \frac{\partial}{\partial\theta}\left(\sigma_{\theta\theta}r\sin\theta\,\vec{e}_\theta\right) + \frac{\partial}{\partial\varphi}\left(\sigma_{\varphi\theta}r\,\vec{e}_\theta\right)\right] \\ & + \frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}\left(\sigma_{r\varphi}r^2\sin\theta\,\vec{e}_\varphi\right) + \frac{\partial}{\partial\theta}\left(\sigma_{\theta\varphi}r\sin\theta\,\vec{e}_\varphi\right) + \frac{\partial}{\partial\varphi}\left(\sigma_{\varphi\varphi}r\,\vec{e}_\varphi\right)\right] \end{aligned} \tag{3.24}$$

The derivatives will be performed separately. It is necessary to remember that in the spherical coordinate system, the orthogonal unit vectors are not always constant with respect to the direction, unlike what would happen in the rectangular system [42]. Afterwards, Eqs. (3.97a)–(3.97i) are obtained, as shown in the Appendix at the end of this chapter. As can be seen, the application of the derivatives produces more terms than would be obtained in the rectangular coordinate system. It is important to realize that the derivation in the rectangular system also includes the orthogonal unit vectors; however, because the derivatives of these vectors, with respect to the direction, are zero, in many textbooks they are commonly omitted in the derivation of the momentum equation. This can sometimes give the wrong impression about the nature of the momentum equation. Substituting the results shown in Eqs. (3.97a)–(3.97i) into Eq. (3.24), a mathematical expression for the resultant of the surface forces is obtained.

The resultant of surface forces in spherical coordinates

$$\vec{f}_{vc} = \vec{e}_r\left[\frac{1}{r^2}\frac{\partial(r^2\sigma_{rr})}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sigma_{\theta r}\sin\theta) + \frac{1}{r\sin\theta}\frac{\partial\sigma_{\varphi r}}{\partial\varphi} - \frac{\sigma_{\theta\theta}+\sigma_{\varphi\varphi}}{r}\right]$$

$$+\vec{e}_{\theta}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\sigma_{r\theta}\right)+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\left(\sigma_{\theta\theta}\sin\theta\right)+\frac{1}{r\sin\theta}\frac{\partial\sigma_{\varphi\theta}}{\partial\varphi}+\frac{\sigma_{\theta r}}{r}-\frac{\sigma_{\varphi\varphi}}{r}\cot\theta\right]$$

$$+\vec{e}_{\varphi}\left[\frac{1}{r^2}\frac{\partial\left(r^2\sigma_{r\varphi}\right)}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\left(\sigma_{\theta\varphi}\sin\theta\right)+\frac{1}{r\sin\theta}\frac{\partial\sigma_{\varphi\varphi}}{\partial\varphi}+\frac{\sigma_{\varphi r}}{r}+\frac{\sigma_{\varphi\theta}}{r}\cot\theta\right] \tag{3.25}$$

With the objective of maintaining the focus on the derivation procedure and not so much on the operations with tensors, reference will be made to Appendix A of the book. Analyzing Eq. (A.30) of Appendix A, it is observed that the surface force per unit volume is equal to the divergence of the stress tensor, which yields a vector. This is shown mathematically in Eq. (3.26).

The resultant of surface forces in the general form

$$\vec{f}_{vc}=\nabla\cdot\overleftrightarrow{\sigma} \tag{3.26}$$

Once the relationship between the resultant of surface forces and stress has been determined, the attention is focused on the momentum, which is defined as the product of mass and velocity ($m\vec{V}$). There is a mass flow entering ($\dot{m}_{in}$) and another mass flow leaving ($\dot{m}_{out}$) the control volume. As a result of the several interactions, the mass within the control volume can change with time ($\partial m_{vc}/\partial t$). Thus, there is a momentum rate entering $(\dot{m}\vec{V})_{in}$ and a momentum rate leaving $(\dot{m}\vec{V})_{out}$ the control volume. The rate at which momentum is accumulated within the control volume is given by $[\partial\left(m_{vc}\vec{V}\right)/\partial t]$. Applying the balance between the resulting force and the momentum, Eq. (3.27) is obtained, where the velocity was expressed as a function of its components. The mass flows given in Eq. (3.27) can be written as shown in Eqs. (3.28a) and (3.28b).

$$\frac{\partial\left(m_{vc}\vec{V}\right)}{\partial t}=\left[\dot{m}\left(u_r\vec{e}_r+u_\theta\vec{e}_\theta+u_\varphi\vec{e}_\varphi\right)\right]_{\text{in}}-\left[\dot{m}\left(u_r\vec{e}_r+u_\theta\vec{e}_\theta+u_\varphi\vec{e}_\varphi\right)\right]_{\text{out}}+\vec{f}_{vc}\upsilon_{vc}+\rho\vec{B}_{vc}\upsilon_{vc} \tag{3.27}$$

$$\dot{m}_{\text{in}}=\left[\rho u_r r^2\sin\theta\Delta\varphi\Delta\theta\right]_r+\left[\rho u_\theta r\sin\theta\Delta\varphi\Delta r\right]_\theta+\left[\rho u_\varphi r\Delta\theta\Delta r\right]_\varphi \tag{3.28a}$$

$$\dot{m}_{\text{out}}=\left[\rho u_r r^2\sin\theta\Delta\varphi\Delta\theta\right]_{r+\Delta r}+\left[\rho u_\theta r\sin\theta\Delta\varphi\Delta r\right]_{\theta+\Delta\theta}+\left[\rho u_\varphi r\Delta\theta\Delta r\right]_{\varphi+\Delta\varphi} \tag{3.28b}$$

Substituting Eqs. (3.28a) and (3.28b) into Eq. (3.27), a rather lengthy expression is obtained, which is reproduced in the Appendix at the end of this chapter as

Eq. (3.98). Dividing by $\Delta\varphi\Delta\theta\Delta r$, taking the limit when Δr, $\Delta\theta$, $\Delta\varphi \to 0$, and rearranging yield the expression in Eq. (3.29).

$$\begin{aligned}
&\frac{\partial\left(\rho\vec{V}\right)}{\partial t} = \\
&-\frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}\left(\rho u_r u_r r^2\sin\theta\,\vec{e}_r\right)+\frac{\partial}{\partial\theta}\left(\rho u_\theta u_r r\sin\theta\,\vec{e}_r\right)+\frac{\partial}{\partial\varphi}\left(\rho u_\varphi u_r r\,\vec{e}_r\right)\right] \\
&-\frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}\left(\rho u_r u_\theta r^2\sin\theta\,\vec{e}_\theta\right)+\frac{\partial}{\partial\theta}\left(\rho u_\theta u_\theta r\sin\theta\,\vec{e}_\theta\right)+\frac{\partial}{\partial\varphi}\left(\rho u_\varphi u_\theta r\,\vec{e}_\theta\right)\right] \\
&-\frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}\left(\rho u_r u_\varphi r^2\sin\theta\,\vec{e}_\varphi\right)+\frac{\partial}{\partial\theta}\left(\rho u_\theta u_\varphi r\sin\theta\,\vec{e}_\varphi\right)+\frac{\partial}{\partial\varphi}\left(\rho u_\varphi u_\varphi r\,\vec{e}_\varphi\right)\right] \\
&\qquad\qquad +\vec{f}_{vc}+\rho\vec{B}_{vc}
\end{aligned} \tag{3.29}$$

The partial derivatives were applied separately, and the result is presented in Eqs. (3.99a)–(3.99i) in the Appendix at the end of this chapter. Likewise, on the left-hand side of Eq. (3.29),

$$\frac{\partial\left(\rho\vec{V}\right)}{\partial t}=\rho\frac{\partial\vec{V}}{\partial t}+\vec{V}\frac{\partial\rho}{\partial t}= \begin{array}{l} \vec{e}_r\rho\dfrac{\partial u_r}{\partial t}+\vec{e}_\theta\rho\dfrac{\partial u_\theta}{\partial t}+\vec{e}_\varphi\rho\dfrac{\partial u_\varphi}{\partial t} \\ +\vec{e}_r u_r\dfrac{\partial\rho}{\partial t}+\vec{e}_\theta u_\theta\dfrac{\partial\rho}{\partial t}+\vec{e}_\varphi u_\varphi\dfrac{\partial\rho}{\partial t} \end{array} \tag{3.30}$$

Using the results obtained in Eqs. (3.99a)–(3.99i) and Eq. (3.30) into Eq. (3.29), rearranging and identifying the divergence of $(\rho\vec{V})$, the expression given in Eq. (3.31) is obtained.

$$\begin{aligned}
&\vec{f}_{vc}+\rho\vec{B}_{vc}=\vec{e}_r\rho\left[\frac{\partial u_r}{\partial t}+u_r\frac{\partial u_r}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_r}{\partial\theta}+\frac{u_\varphi}{r\sin\theta}\frac{\partial u_r}{\partial\varphi}-\frac{u_\theta^2+u_\varphi^2}{r}+\frac{u_r}{\rho}\left(\frac{\partial\rho}{\partial t}+\vec{\nabla}\cdot\left(\rho\vec{V}\right)\right)\right] \\
&+\vec{e}_\theta\rho\left[\frac{\partial u_\theta}{\partial t}+u_r\frac{\partial u_\theta}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_\theta}{\partial\theta}+\frac{u_\varphi}{r\sin\theta}\frac{\partial u_\theta}{\partial\varphi}+\frac{u_r u_\theta}{r}-\frac{u_\varphi^2}{r}\cot\theta+\frac{u_\theta}{\rho}\left(\frac{\partial\rho}{\partial t}+\vec{\nabla}\cdot\left(\rho\vec{V}\right)\right)\right] \\
&+\vec{e}_\varphi\rho\left[\frac{\partial u_\varphi}{\partial t}+u_r\frac{\partial u_\varphi}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_\varphi}{\partial\theta}+\frac{u_\varphi}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}+\frac{u_r u_\varphi}{r}+\frac{u_\theta u_\varphi}{r}\cot\theta+\frac{u_\varphi}{\rho}\left(\frac{\partial\rho}{\partial t}+\vec{\nabla}\cdot\left(\rho\vec{V}\right)\right)\right]
\end{aligned} \tag{3.31}$$

The derivatives were applied in such a way that the equation for conservation of total mass, which is equal to zero, was obtained in each of the coordinates. Therefore, the momentum equation in spherical coordinates was obtained in terms of the stress tensor by substituting Eqs. (3.21) and (3.25) into Eq. (3.31). The result is shown in Eqs. (3.32a)–(3.32c).

Momentum equation in spherical coordinates in terms of the stress tensor
At coordinate r

$$\frac{1}{r^2}\frac{\partial(r^2\sigma_{rr})}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sigma_{\theta r}\sin\theta)+\frac{1}{r\sin\theta}\frac{\partial\sigma_{\varphi r}}{\partial\varphi}-\frac{\sigma_{\theta\theta}+\sigma_{\varphi\varphi}}{r}+\rho B_r$$
$$=\rho\left(\frac{\partial u_r}{\partial t}+u_r\frac{\partial u_r}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_r}{\partial\theta}+\frac{u_\varphi}{r\sin\theta}\frac{\partial u_r}{\partial\varphi}-\frac{u_\theta^2+u_\varphi^2}{r}\right) \tag{3.32a}$$

At coordinate θ

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\sigma_{r\theta}\right)+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sigma_{\theta\theta}\sin\theta)+\frac{1}{r\sin\theta}\frac{\partial\sigma_{\varphi\theta}}{\partial\varphi}+\frac{\sigma_{\theta r}}{r}-\frac{\sigma_{\varphi\varphi}}{r}\cot\theta+\rho B_\theta$$
$$=\rho\left(\frac{\partial u_\theta}{\partial t}+u_r\frac{\partial u_\theta}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_\theta}{\partial\theta}+\frac{u_\varphi}{r\sin\theta}\frac{\partial u_\theta}{\partial\varphi}+\frac{u_r u_\theta}{r}-\frac{u_\varphi^2}{r}\cot\theta\right) \tag{3.32b}$$

At coordinate φ

$$\frac{1}{r^2}\frac{\partial\left(r^2\sigma_{r\varphi}\right)}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\left(\sigma_{\theta\varphi}\sin\theta\right)+\frac{1}{r\sin\theta}\frac{\partial\sigma_{\varphi\varphi}}{\partial\varphi}+\frac{\sigma_{\varphi r}}{r}+\frac{\sigma_{\varphi\theta}}{r}\cot\theta+\rho B_\varphi$$
$$=\rho\left(\frac{\partial u_\varphi}{\partial t}+u_r\frac{\partial u_\varphi}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_\varphi}{\partial\theta}+\frac{u_\varphi}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}+\frac{u_r u_\varphi}{r}+\frac{u_\theta u_\varphi}{r}\cot\theta\right) \tag{3.32c}$$

Analyzing Eqs. (3.32a)–(3.32c), it is possible to recognize the definition of the material derivative of the velocity vector, which together with Eq. (3.26) allows to write the momentum equation in compact form. The procedure for obtaining the material derivative of the velocity vector in spherical coordinates is shown in Appendix A of the book, and the result of this procedure is found in Eq. (A.32).

Momentum equation in compact form

$$\rho\frac{D\vec{V}}{Dt}=\nabla\cdot\overleftrightarrow{\sigma}+\rho\vec{B}_{vc} \tag{3.33}$$

An expression has been adopted to represent the vector of body forces per unit volume ($\vec{B}_{vc}$); it was also established that the surface force per unit volume ($\vec{f}_{vc}$) is equal to the divergence of the stress tensor ($\overleftrightarrow{\sigma}$). Finally, an equation that represents the balance between force and momentum has been determined. It is still necessary to find a way to represent $\overleftrightarrow{\sigma}$ as a function of velocity, viscosity, and pressure. This expression is called a constitutive relation and will be determined in the next section.

3.5 Constitutive Relation for the Stress Tensor

Now it will be necessary to determine a mathematical expression for the stress tensor. The fluid element shown in Fig. 3.5 is considered with this purpose. This element moves within the control volume, and because of this, the stress tensor will undergo deformations. Stokes' law of viscosity for an isotropic fluid establishes the relations presented in Eqs. (3.34a) and (3.34b) [43].

$$\sigma_{ii} = -p + 2\mu\dot{\varepsilon}_{ii} + \left(\mu' - \frac{2}{3}\mu\right)\left(\dot{\varepsilon}_{ii} + \dot{\varepsilon}_{jj} + \dot{\varepsilon}_{kk}\right) \tag{3.34a}$$

$$\sigma_{ij} = \sigma_{ji} = 2\mu\dot{\varepsilon}_{ij} \tag{3.34b}$$

The term $\dot{\varepsilon}_{ij}$ represents the deformation rates, p is the pressure, μ is the viscosity coefficient, and μ' is the second viscosity coefficient. When $i = j$ the stresses are normal to the surface and cause linear deformations. On the other hand, when $i \neq j$ the stresses act on the surface, causing angular deformations. A common practice in combustion studies is to adopt a value of zero for the second viscosity coefficient ($\mu' = 0$) because there are no theoretical bases for its calculation or reliable experimental procedures for its measurement [9]. It is important to recognize that the normal components of the stress tensor include pressure. Since the components point away from the surface, pressure appears with a negative sign in Stokes' law of viscosity. The deformation rates can be related to the velocity of the fluid element. For this purpose, points A, B, C, and D are considered in Fig. 3.5. These points move with respect to the origin. Thus, if the velocity of point A ($\vec{V}_A$) is taken as reference, the velocities of points B, C, and D are given by the following expressions.

$$\vec{V}_B = \vec{V}_A + \left(\partial \vec{V}_A/\partial r\right)dr \quad \vec{V}_C = \vec{V}_A + \left(\partial \vec{V}_A/\partial \theta\right)d\theta$$
$$\vec{V}_D = \vec{V}_A + \left(\partial \vec{V}_A/\partial \varphi\right)d\varphi$$

The above velocities should be determined by applying the chain rule to derive the vector $\vec{V}_A$. The results obtained are presented below. Initially, the velocity of point A is written in terms of its components.

$$\vec{V}_A = u_r \vec{e}_r + u_\theta \vec{e}_\theta + u_\varphi \vec{e}_\varphi \tag{3.35a}$$

Deriving the above expression in terms of the coordinates in the spherical system and grouping the terms with the same orthogonal unit vectors yield

$$\vec{V}_B = \left(u_r + \frac{\partial u_r}{\partial r}dr\right)\vec{e}_r + \left(u_\theta + \frac{\partial u_\theta}{\partial r}dr\right)\vec{e}_\theta + \left(u_\varphi + \frac{\partial u_\varphi}{\partial r}dr\right)\vec{e}_\varphi \tag{3.35b}$$

$$\vec{V}_C = \left(u_r - u_\theta d\theta + \frac{\partial u_r}{\partial \theta} d\theta\right)\vec{e}_r + \left(u_\theta + u_r d\theta + \frac{\partial u_\theta}{\partial \theta} d\theta\right)\vec{e}_\theta + \left(u_\varphi + \frac{\partial u_\varphi}{\partial \theta} d\theta\right)\vec{e}_\varphi \tag{3.35c}$$

$$\vec{V}_D = \left(u_r - \sin\theta u_\varphi d\varphi + \frac{\partial u_r}{\partial \varphi} d\varphi\right)\vec{e}_r + \left(u_\theta - \cos\theta u_\varphi d\varphi + \frac{\partial u_\theta}{\partial \varphi} d\varphi\right)\vec{e}_\theta + \left(u_\varphi + \sin\theta u_r d\varphi + \cos\theta u_\theta d\varphi + \frac{\partial u_\varphi}{\partial \varphi} d\varphi\right)\vec{e}_\varphi \tag{3.35d}$$

The expressions presented in Eqs. (3.35a)–(3.35d) will be used in the analysis of angular deformations.

3.5.1 Linear Deformations

The linear deformation of the lengths $\overline{AB}$, $\overset{\frown}{AC}$ e $\overset{\frown}{AD}$ shown in Fig. 3.6 is considered. The term $\overline{AB}$ is first evaluated. At time t_0 points A and B are found at the locations on the solid lines.

After a time interval dt, point A is displaced by a distance equal to $u_r dt$, and point B is displaced by a distance equal to $(u_r + [\partial u_r/\partial r]dr)dt$. It is observed that the displacement occurs in the radial coordinate, and because of this, the distance traveled by points A and B is determined using the components of the respective velocities in r. The difference between the lengths $L_{2,\,r}$ and $L_{1,\,r}$ is equal to the difference between the displacements of points A and B in the coordinate r.

$$L_{2,r} - L_{1,r} = \left(u_r + \frac{\partial u_r}{\partial r} dr\right)dt - u_r dt = \frac{\partial u_r}{\partial r} dr dt$$

Referring back to Fig. 3.6, the characteristic lengths of the fluid element are now expressed in terms of differentials. Thus, the initial distances are as follows: between A and B, it is dr; between A and C, it is $rd\theta$; and between A and D, it is $r \sin\theta\, d\theta$. This change was adopted because the deformation being analyzed is that of a fluid element and not of the control volume, since the latter does not undergo deformations. The rate of change of length per unit of original length is known as the rate of linear deformation $\dot{\varepsilon}_{rr}$.

$$\dot{\varepsilon}_{rr} = \frac{1}{dr}\frac{L_{2,r} - L_{1,r}}{dt} = \frac{\partial u_r}{\partial r} \tag{3.36}$$

In the case of length $\overset{\frown}{AC}$, for the time interval dt, point A is displaced by a distance equal to $u_r dt$. At the same time interval, the angle θ between A and C changes by an

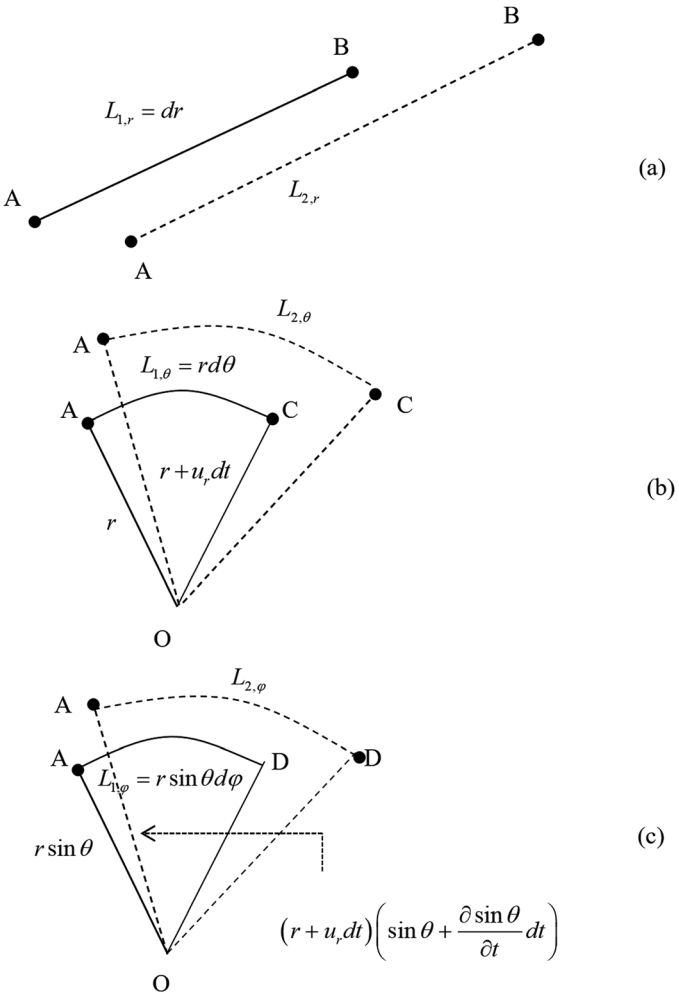


Fig. 3.6 Linear deformation: (**a**) coordinate r; (**b**) coordinate θ; (**c**) coordinate φ

amount equal to $(1/r)(\partial u_\theta/\partial\theta)d\theta dt$. Based on these observations, the lengths $L_{1,\,\theta}$ and $L_{2,\,\theta}$ are determined as follows.

$$L_{1,\theta} = rd\theta$$

$$L_{2,\theta} = (r + u_r dt)\left(d\theta + \frac{1}{r}\frac{\partial u_\theta}{\partial\theta} d\theta dt\right) = rd\theta + u_r d\theta dt + \frac{\partial u_\theta}{\partial\theta} d\theta dt$$

The term multiplied by $(dt)^2$ was considered negligible. Now the strain rate can be determined.

$$\dot{\varepsilon}_{\theta\theta} = \frac{1}{rd\theta}\frac{L_{2,\theta} - L_{1,\theta}}{dt} = \frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r} \tag{3.37}$$

In the case of length $\widehat{AD}$, the distance between A and D is calculated by multiplying the projection of r on the plane of φ, that is, $r\ \sin\theta$ times the angle between A and D, which is $d\varphi$. After the time interval dt, and considering that there is displacement in the other two coordinates, the variation of the projection takes into account the change in r, introduced as $u_r dt$, and the change in $\sin\theta$, introduced as $(\partial \sin\theta/\partial t)dt$. In the same time interval, the angle φ between A and D changes by an amount equal to $(1/rsin\theta)(\partial u_\varphi/\partial\varphi)d\varphi dt$. Thus, the initial and final lengths will be given by the expressions below.

$$L_{1,\varphi} = r\sin\theta d\varphi$$

$$L_{2,\varphi} = = (r + u_r dt)\left(\sin\theta + \cos\theta\frac{u_\theta}{r}dt\right)\left(d\varphi + \frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}d\varphi dt\right)$$

Assuming as negligible all terms where the dt appears raised to a power greater than or equal to 2,

$$L_{2,\varphi} = r\sin\theta d\varphi + u_r\sin\theta d\varphi dt + u_\theta\cos\theta d\varphi dt + \frac{\partial u_\varphi}{\partial\varphi}d\varphi dt$$

Therefore, the strain rate is as shown in Eq. (3.38).

$$\dot{\varepsilon}_{\varphi\varphi} = \frac{1}{r\sin\theta d\varphi}\frac{L_{2,\varphi} - L_{1,\varphi}}{dt} = \frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi} + \frac{u_r}{r} + \frac{u_\theta}{r}\cot\theta \tag{3.38}$$

Thus, the expressions given in Eqs. (3.36)–(3.38) determine the linear deformations as a function of the velocity components and the coordinates in the spherical system.

3.5.2 Angular Deformation

Now the angular deformations must be determined. Observing Fig. 3.7 it is possible to establish the relationship between the angular deformations $d\gamma_a$ and $d\gamma_b$ with the velocities and the coordinates r and θ. Therefore, it is necessary to use Eqs. (3.35a)–(3.35d) which relate the variations in velocity at points B, C, and D with the variation in velocity at point A.

$$d\gamma_a \cong \mathrm{tag}(d\gamma_a) = \frac{\left(u_r - u_\theta d\theta + \frac{\partial u_r}{\partial\theta}d\theta\right)dt - u_r dt}{rd\theta} = \left(\frac{1}{r}\frac{\partial u_r}{\partial\theta} - \frac{u_\theta}{r}\right)dt$$

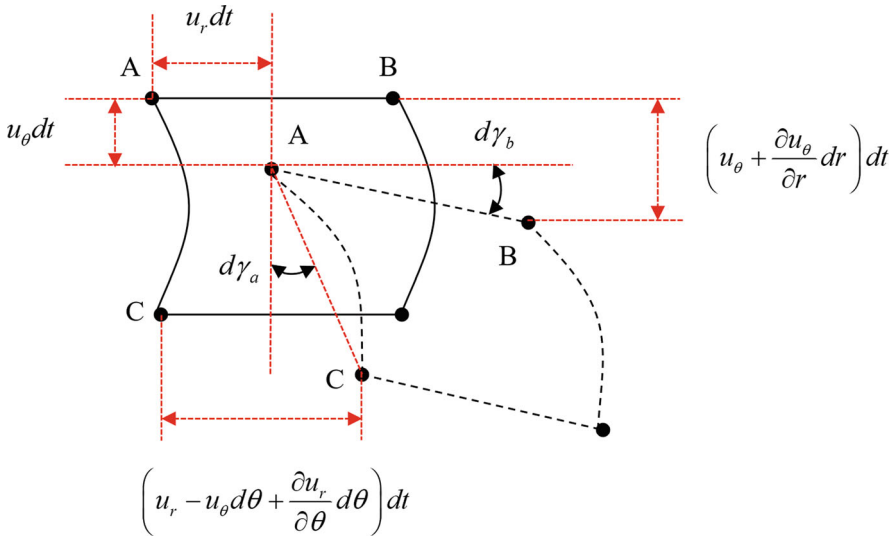


Fig. 3.7 Angular deformation due to the action of stress tensor components $\sigma_{r\theta} = \sigma_{\theta r}$

$$d\gamma_b \cong \text{tag}(d\gamma_b) = \frac{\left(u_\theta + \frac{\partial u_\theta}{\partial r} dr\right) dt - u_\theta dt}{dr} = \frac{\partial u_\theta}{\partial r} dt$$

Note that for an infinitesimally small angle, its tangent can be approximated as equal to the angle itself in radians [$d\gamma \cong \text{tag}(d\gamma)$]. This is easily seen when the tangent function is written as a Taylor series. From now on it will always be considered that the infinitesimal angle represents its own tangent. Dividing the two previous expressions by dt and adding them together, the total angular deformation rate is obtained.

$$\frac{d\gamma_{ab}}{dt} = \frac{d\gamma_a}{dt} + \frac{d\gamma_b}{dt} = \frac{1}{r}\frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r}$$

The rate of angular deformation $\dot{\varepsilon}_{r\theta}$ is equal to half the total angular strain rate, as shown in Eq. (3.39).

$$\dot{\varepsilon}_{r\theta} = \frac{1}{2}\left(\frac{1}{r}\frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r}\right) \tag{3.39}$$

Observing Fig. 3.8, it is possible to establish the relation between the angular deformations $d\gamma_c$ and $d\gamma_d$ with the velocities and the coordinates r and φ. Therefore, writing the angular deformations in terms of the lengths shown in Fig. 3.8, the following expressions are obtained,

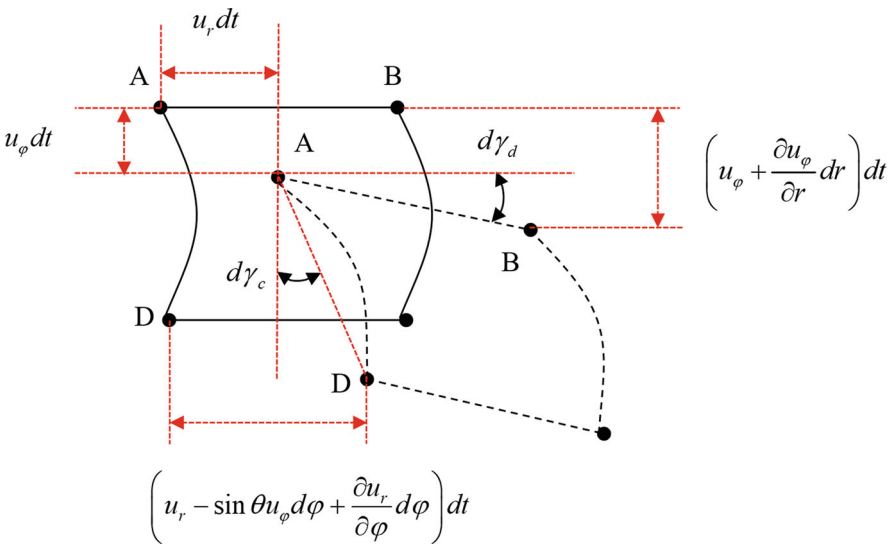


Fig. 3.8 Angular deformation due to the action of stress tensor components $\sigma_{r\varphi} = \sigma_{\varphi r}$

$$d\gamma_c = \frac{\left(u_r - \sin\theta u_\varphi d\varphi + \frac{\partial u_r}{\partial \varphi} d\varphi\right) dt - u_r dt}{r \sin\theta d\varphi} = \left(\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial \varphi} - \frac{u_\varphi}{r}\right) dt$$

$$d\gamma_d = \frac{\left(u_\varphi + \frac{\partial u_\varphi}{\partial r} dr\right) dt - u_\varphi dt}{dr} = \frac{\partial u_\varphi}{\partial r} dt$$

Dividing the two previous expressions by dt and adding them together, the total angular deformation is obtained.

$$\frac{d\gamma_{cd}}{dt} = \frac{d\gamma_c}{dt} + \frac{d\gamma_d}{dt} = \frac{1}{r\sin\theta}\frac{\partial u_r}{\partial \varphi} + \frac{\partial u_\varphi}{\partial r} - \frac{u_\varphi}{r}$$

The rate of angular deformation $\dot{\varepsilon}_{r\theta}$ is equal to half the total angular deformation rate as given in Eq. (3.40).

$$\dot{\varepsilon}_{r\varphi} = \frac{1}{2}\left(\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial \varphi} + \frac{\partial u_\varphi}{\partial r} - \frac{u_\varphi}{r}\right) \tag{3.40}$$

Finally, from Fig. 3.9, it is possible to establish the relation between the angular deformations $d\gamma_e$ and $d\gamma_f$ with the velocities and coordinates θ and φ. Therefore, the angular deformations can be written as shown below.

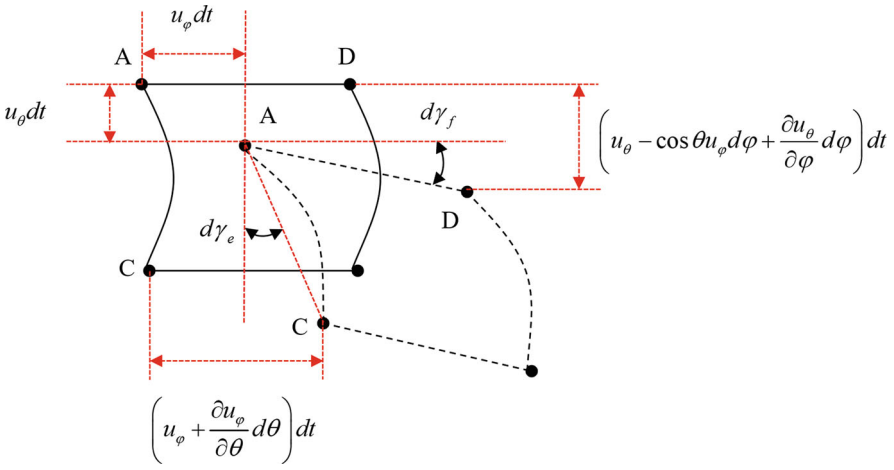


Fig. 3.9 Angular deformation due to the action of stress tensor components $\sigma_{\theta\varphi} = \sigma_{\varphi\theta}$

$$d\gamma_e = \frac{\left(u_\varphi + \frac{\partial u_\varphi}{\partial \theta} d\theta\right) dt - u_\varphi dt}{r d\theta} = \frac{1}{r} \frac{\partial u_\varphi}{\partial \theta} dt$$

$$d\gamma_f = \frac{\left(u_\theta - \cos\theta u_\varphi d\varphi + \frac{\partial u_\theta}{\partial \varphi} d\varphi\right) dt - u_\theta dt}{r \sin\theta d\varphi} = \left(\frac{1}{r\sin\theta} \frac{\partial u_\theta}{\partial \varphi} - \frac{\cot\theta}{r} u_\varphi\right) dt$$

Dividing the two previous expressions by dt and adding them together, the total angular deformation is obtained.

$$\frac{d\gamma_{ef}}{dt} = \frac{d\gamma_e}{dt} + \frac{d\gamma_f}{dt} = \frac{1}{r} \frac{\partial u_\varphi}{\partial \theta} + \frac{1}{r\sin\theta} \frac{\partial u_\theta}{\partial \varphi} - \frac{\cot\theta}{r} u_\varphi$$

The rate of angular deformation $\dot{\varepsilon}_{r\theta}$ is equal to half the total angular deformation rate, as shown in Eq. (3.41).

$$\dot{\varepsilon}_{\theta\varphi} = \frac{1}{2}\left(\frac{1}{r} \frac{\partial u_\varphi}{\partial \theta} + \frac{1}{r\sin\theta} \frac{\partial u_\theta}{\partial \varphi} - \frac{u_\varphi}{r} \cot\theta\right) \tag{3.41}$$

Equations (3.36)–(3.41) express the linear and angular deformations as functions of velocity. When these equations are substituted into Eqs. (3.34a) and (3.34b), the scalar components of the stress tensor as a function of velocity, pressure, and viscosity will be obtained. Thus, substituting Eqs. (3.36)–(3.41) into Eqs. (3.34a) and (3.34b), applying the consideration which states that the second coefficient of viscosity is zero ($\mu' = 0$), and recognizing the divergence of velocity ($\nabla \cdot \vec{V}$) in the

expressions for the normal components of the stress tensor, yield the relations shown in Eqs. (3.42a)–(3.42f).

Scalar components of the stress tensor in spherical coordinates

$$\sigma_{rr} = -p + 2\mu \frac{\partial u_r}{\partial r} - \frac{2}{3}\mu \nabla \cdot \vec{V} \tag{3.42a}$$

$$\sigma_{\theta\theta} = -p + 2\mu \left(\frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r} \right) - \frac{2}{3}\mu \nabla \cdot \vec{V} \tag{3.42b}$$

$$\sigma_{\varphi\varphi} = -p + 2\mu \left(\frac{1}{r \sin\theta} \frac{\partial u_\varphi}{\partial \varphi} + \frac{u_r}{r} + \frac{u_\theta}{r} \cot\theta \right) - \frac{2}{3}\mu \nabla \cdot \vec{V} \tag{3.42c}$$

$$\sigma_{r\theta} = \sigma_{\theta r} = \mu \left(\frac{1}{r} \frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r} \right) \tag{3.42d}$$

$$\sigma_{r\varphi} = \sigma_{\varphi r} = \mu \left(\frac{1}{r \sin\theta} \frac{\partial u_r}{\partial \varphi} + \frac{\partial u_\varphi}{\partial r} - \frac{u_\varphi}{r} \right) \tag{3.42e}$$

$$\sigma_{\theta\varphi} = \sigma_{\varphi\theta} = \mu \left(\frac{1}{r \sin\theta} \frac{\partial u_\theta}{\partial \varphi} + \frac{1}{r} \frac{\partial u_\varphi}{\partial \theta} - \frac{u_\varphi}{r} \cot\theta \right) \tag{3.42f}$$

In Eqs. (3.42a)–(3.42f), it can be recognized that in each of the scalar components of the stress tensor, the respective scalar component of the second-order tensor appears, generated by the sum of the velocity gradient tensor and the transposed tensor of the velocity gradient [see Eq. (A.22)]. Taking advantage of this relation, a compact expression for the scalar components of the stress tensor can be obtained as shown in Eq. (3.43).

Stress tensor in compact form

$$\overleftrightarrow{\sigma}_{ij} = \mu \left[\nabla \vec{V} + \left(\nabla \vec{V} \right)^T \right]_{ij} + \left[-p - \frac{2}{3}\mu \nabla \cdot \vec{V} \right] \vec{e}_i \, \vec{e}_j \delta_{ij} \tag{3.43}$$

where

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

The momentum equation in spherical coordinates is given by Eqs. (3.32a)–(3.32c), in which the stress tensor components are those given in Eqs. (3.42a)–(3.42f). This set of expressions is called the Navier–Stokes equations. Likewise, the momentum equation in compact form is given in Eq. (3.33), where the stress components are those given by Eq. (3.43). It is possible to derive Eqs. (3.42a)–(3.42f) and equate the result with Eqs. (3.32a)–(3.32c) and thus obtain a single

equation. However, in most problems certain assumptions are made, which simplify the equations, so it is useful to apply the simplifications separately.

A set of equations commonly used in the study of flame propagation are Euler equations. These equations can be reasonably applied to fluids at high pressures, for which body forces, viscous stresses, and heat transfer are neglected [44, 45]. The compact form of the Euler equation is obtained by neglecting body forces and assuming an inviscid fluid. Therefore, from Eqs. (3.33), (3.34a), and (3.34b).

Compact form of Euler's equation

$$\rho \frac{D\vec{V}}{Dt} = -\nabla \cdot p \tag{3.44}$$

The Navier–Stokes equations have been successfully applied in numerical simulations of flame propagation in closed ducts in two dimensions [14]. Euler equations have been applied in numerical simulations involving compressible flow [46, 47] as well as in rotating detonation engines (RDE) [48]. The energy conservation equation will be considered in the next section.

3.6 Energy Conservation Equation

The last of the conservation equations to be derived in this chapter is the energy conservation equation. In the context of combustion processes, energy conservation includes a term related to chemical reactions. Therefore, the law of energy conservation establishes the following balance.

Rate of energy accumulation in the control volume	=	Rate of energy entering the control volume	−	Rate of energy leaving the control volume
	+	Net heat exchange rate	+	Heat generation rate
	+	Net rate of work done on the control volume		

The rates of energy entering and leaving the control volume are due to the flow of the species in the mixture. In these terms there are two components: one due to the flow of the species at the average velocity of the mixture and another one due to the diffusion of the chemical species. The net rate of heat exchange includes heat exchanged due to the temperature gradient (conduction) and the concentration gradient (Dufour effect). Heat exchange related to the Dufour effect will be ignored in this analysis. The rate of heat generation is related to processes caused by the

action of electric fields, sparks, or other heat fluxes. The net rate of work is related to the forces acting on the control volume (surface and body forces) which perform work on the control volume. In the most general case, part of all this energy can be accumulated within the control volume. This is shown by the rate of energy accumulation. The derivation of the energy conservation equation will be separated into several parts to provide a better overview.

3.6.1 The Rates of Energy Entering and Leaving the Control Volume

The mathematical expression of the law of energy conservation for 1 kg of mixture has the form shown in Eq. (3.45). Note that the equation shown below is also a form of the first law of thermodynamics applied to a control volume [49].

$$\frac{\partial e}{\partial t} = \dot{e}_{\text{in}} - \dot{e}_{\text{out}} + \dot{q}_{\text{cond},vc} + \dot{q}_{ge} + \dot{w} \tag{3.45}$$

The total specific energy (e) includes internal energy and kinetic energy; therefore, the total specific energy can be written in terms of enthalpy.

$$e = h - \frac{p}{\rho} + \frac{V^2}{2} \tag{3.46}$$

The total energy of the flow per unit surface is equal to the sum of the products of the mass flux of each species with the respective energy of the species.

$$E/A = \sum \dot{m}_k e_k$$

Using Eq. (3.46) it will be possible to determine an expression for the product of the species' mass and its specific enthalpy.

$$\sum \dot{m}_k e_k = \sum \dot{m}_k h_k + \left(-\frac{p}{\rho} + \frac{V^2}{2} \right) \sum \dot{m}_k$$

The mass flux of the species, appearing as the first term on the right-hand side of the above expression, is equal to the sum of the flux due to the bulk flow velocity ($\rho Y_k \vec{V}$) and the flux due to the diffusion of the species ($\vec{\dot{m}}_{k,\text{dif}}$).

$$\sum \dot{m}_k h_k = \rho \vec{V} \sum Y_k h_k + \sum \vec{\dot{m}}_{k,\text{diff}} h_k$$

From Eq. (3.13) it is known that $\sum \dot{m}_k = \rho \vec{V}$. With this information and using the two previous equations, Eq. (3.47) is obtained.

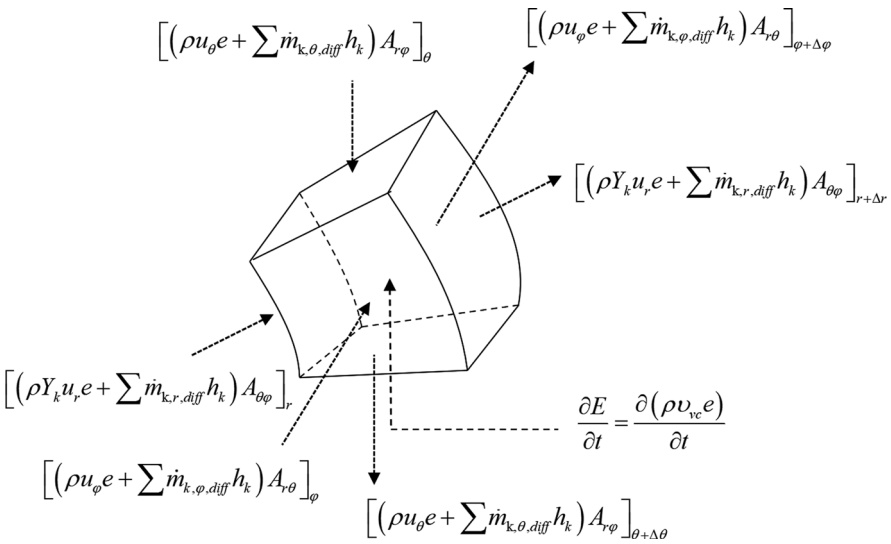


Fig. 3.10 Control volume in spherical coordinates showing energy rates

$$\sum \dot{m}_k e_k = \rho \vec{V} e + \sum \vec{\dot{m}}_{k,\mathrm{diff}} h_k \tag{3.47}$$

The expression shown in Eq. (3.47) will be necessary to evaluate the first two terms on the right-hand side of Eq. (3.45). In order to do so, the rates of energy shown in Fig. 3.10 are considered. The remaining terms will be evaluated separately. It is important to mention that the heat generation term ($\dot{q}_{ge}$) also includes the heat exchanged by thermal radiation and by other heat sources within the control volume [50].

The analysis is analogous to that presented in the case of mass conservation. Thus, writing the balance, dividing by $\Delta\varphi\Delta\theta\Delta r$, taking the limit for Δr, $\Delta\theta$, $\Delta\varphi \to 0$, and rearranging, the following result is obtained.

$$\frac{\partial(\rho e)}{\partial t} = -\frac{1}{r^2}\frac{\partial(r^2[\rho u_r e])}{\partial r} - \frac{1}{r\sin\theta}\frac{\partial([\rho u_\theta e]\sin\theta)}{\partial\theta} - \frac{1}{r\sin\theta}\frac{\partial(\rho u_\theta e)}{\partial\varphi}$$

$$-\frac{1}{r^2}\frac{\partial(r^2[\sum \dot{m}_{k,r,\mathrm{dif}} h_k])}{\partial r} - \frac{1}{r\sin\theta}\frac{\partial([\sum \dot{m}_{k,\theta,\mathrm{diff}} h_k]\sin\theta)}{\partial\theta} - \frac{1}{r\sin\theta}\frac{\partial(\sum \dot{m}_{k,\varphi,\mathrm{diff}} h_k)}{\partial\varphi}$$

$$+\dot{q}_{\mathrm{cond},vc} + \dot{q}_{ge,vc} + \dot{w}_{vc}$$

By applying the derivatives to the first terms on the left- and right-hand sides and rearranging, an expression that includes the total mass conservation equation (which is zero) is obtained. Thus, the above expression simplifies to,

$$\rho\frac{\partial e}{\partial t}+\rho u_r\frac{\partial e}{\partial r}+\rho u_\theta\frac{1}{r}\frac{\partial e}{\partial \theta}+\rho u_\varphi\frac{1}{r\sin\theta}\frac{\partial e}{\partial \varphi}=-\frac{1}{r^2}\frac{\partial(r^2[\sum \dot{m}_{k,r,\mathrm{diff}}h_k])}{\partial r}$$
$$-\frac{1}{r\sin\theta}\frac{\partial([\sum \dot{m}_{k,\theta,\mathrm{diff}}h_k]\sin\theta)}{\partial\theta}-\frac{1}{r\sin\theta}\frac{\partial(\sum \dot{m}_{k,\varphi,\mathrm{diff}}h_k)}{\partial\varphi}$$
$$+\dot{q}_{\mathrm{cond},vc}+\dot{q}_{ge,vc}+\dot{w}_{vc} \tag{3.48}$$

The material derivative of the total specific energy and the divergence of the sum of the products of the mass diffusion fluxes with the species enthalpies are both recognized in Eq. (3.48). Therefore, Eq. (3.49) is the compact form of Eq. (3.48).

$$\rho\frac{De}{Dt}=-\nabla\cdot\left(\sum\vec{\dot{m}}_{k,\mathrm{dif}}h_k\right)+\dot{q}_{\mathrm{cond},vc}+\dot{q}_{ge,vc}+\dot{w}_{vc} \tag{3.49}$$

The rates of energy entering and leaving the control volume could also be called heat exchanged by convection [40].

3.6.2 Heat Transfer by Conduction

The term involving heat transfer by conduction can be written in terms of temperature by applying Fourier's law. Thus, Eq. (3.50) is obtained for the spherical coordinate system.

$$\vec{q}=-k\nabla T=-k\frac{\partial T}{\partial r}\vec{e}_r-k\frac{1}{r}\frac{\partial T}{\partial\theta}\vec{e}_\theta-k\frac{1}{r\sin\theta}\frac{\partial T}{\partial\varphi}\vec{e}_\varphi \tag{3.50}$$

Equation (3.50) gives the heat transfer by conduction per unit area in vector form. Therefore, it is possible to apply an analysis similar to that presented for the energy rates using the scalar components of the vector. Therefore, the net rate of heat transfer by conduction will be obtained ($\dot{q}_{\mathrm{cond},vc}$).

$$\dot{q}_{\mathrm{cond},vc}=-\frac{1}{r^2}\frac{\partial(q_r r^2)}{\partial r}-\frac{1}{r\sin\theta}\frac{\partial(q_\theta\sin\theta)}{\partial\theta}-\frac{1}{r\sin\theta}\frac{\partial q_\varphi}{\partial\varphi}=\nabla\cdot(k\nabla T)$$

Using the expression for Fourier's law given in Eq. (3.50), the heat transfer by conduction is obtained in spherical coordinates and as a function of temperature.

$$\dot{q}_{\mathrm{cond},vc}=\frac{1}{r^2}\frac{\partial}{\partial r}\left(kr^2\frac{\partial T}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(k\frac{\partial T}{\partial\theta}\sin\theta\right)\frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial\varphi}\left(k\frac{\partial T}{\partial\varphi}\right) \tag{3.51}$$

In compact form, the expression shown in Eq. (3.51) is equal to the divergence of k∇T.

$$\dot{q}_{\text{cond},vc} = \nabla \cdot (k \nabla T) \tag{3.52}$$

3.6.3 Work Performed by Body Forces

The work term ($\dot{W}$) is divided into two parts: (a) the work performed by surface forces ($\dot{W}_s$) and (b) the work performed by body forces ($\dot{W}_c$). The work done by body forces will be determined in this section. It is considered that the sum of the body forces acting on each species, in each coordinate, represents the body forces acting on the control volume.

$$\rho \vec{B}_{vc} = \rho \left(\sum_{k=1}^{N} Y_k f_{k,\text{r}} \right) \vec{e}_r + \rho \left(\sum_{k=1}^{N} Y_k f_{k,\theta} \right) \vec{e}_\theta + \rho \left(\sum_{k=1}^{N} Y_k f_{k,\varphi} \right) \vec{e}_\varphi$$

The work done by the body forces on each species will be equal to the product of the body force acting on the species with the velocity of the species ($\vec{V}_k$). Therefore, the total work of the body forces will be equal to the sum of the work performed on each species. Using the expression for the velocity of the species given in Eq. (3.14), the expression given in Eq. (3.53) is obtained for the work of the body forces per unit volume, where $u_{k,\text{ dif},\,i}$ is the scalar component of the species k diffusion velocity in the coordinate i. Thus, the work performed by the body forces is given by Eq. (3.53).

$$\begin{aligned} \dot{w}_c = {} & \rho \left(\sum_{k=1}^{N} Y_k f_{k,\text{r}} [u_r + u_{k,\text{diff},r}] \right) + \rho \left(\sum_{k=1}^{N} Y_k f_{k,\theta} [u_\theta + u_{\text{k,diff},\theta}] \right) \\ & + \rho \left(\sum_{k=1}^{N} Y_k f_{k,\varphi} [u_\varphi + u_{\text{k,diff},\varphi}] \right) \end{aligned} \tag{3.53}$$

3.6.4 Work Performed by Surface Forces

As explained earlier, surface forces are produced by stress on the volume element. To determine an expression for the work done by surface forces, the scalar components of the stress tensor acting in the radial direction are considered, as shown in Fig. 3.11. The work done along the radial component can be represented by the following expression.

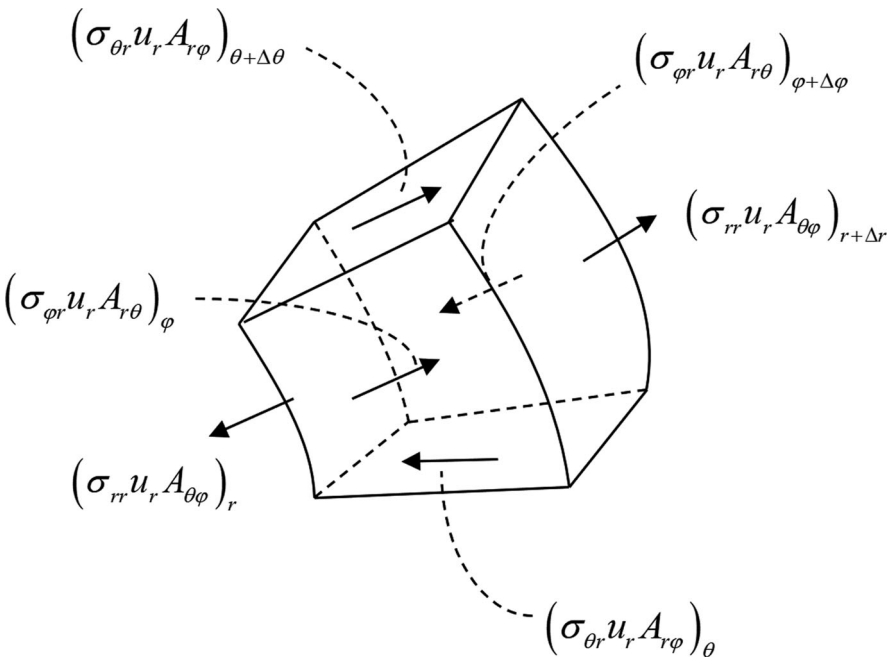


Fig. 3.11 Control volume in spherical coordinates, showing the work done by stress in the radial direction

$$\dot{w}_{s-r,vc}r^2\sin\theta\Delta\varphi\Delta\theta\Delta r = \left(\sigma_{rr}u_r r^2\sin\theta\Delta\varphi\Delta\theta\right)_{r+\Delta r} - \left(\sigma_{rr}u_r r^2\sin\theta\Delta\varphi\Delta\theta\right)_r$$
$$+\left(\sigma_{\theta r}u_r r\sin\theta\Delta\varphi\Delta r\right)_{\theta+\Delta\theta} - \left(\sigma_{\theta r}u_r r\sin\theta\Delta\varphi\Delta r\right)_{\theta} + \left(\sigma_{\varphi r}u_r r\Delta\theta\Delta r\right)_{\varphi+\Delta\varphi}$$
$$-\left(\sigma_{\varphi r}u_r r\Delta\theta\Delta r\right)_{\varphi}$$

Dividing by $\Delta\varphi\Delta\theta\Delta r$, taking the limit for Δr, $\Delta\theta$, $\Delta\varphi \to 0$, and rearranging, Eq. (3.54) is obtained.

$$\dot{w}_{s-r,vc} = \frac{1}{r^2}\frac{\partial\left(\sigma_{rr}u_r r^2\right)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial\left(\sigma_{\theta r}u_r\sin\theta\right)}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial\left(\sigma_{\varphi r}u_r\right)}{\partial\varphi} \tag{3.54}$$

Similarly, the work done along the coordinates θ and φ has the form shown in Eqs. (3.55) and (3.56), respectively.

$$\dot{w}_{s-\theta,vc} = \frac{1}{r^2}\frac{\partial\left(\sigma_{r\theta}u_\theta r^2\right)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial\left(\sigma_{\theta\theta}u_\theta\sin\theta\right)}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial\left(\sigma_{\varphi\theta}u_\theta\right)}{\partial\varphi} \tag{3.55}$$

$$\dot{w}_{s-\varphi,vc} = \frac{1}{r^2}\frac{\partial\left(\sigma_{r\varphi}u_\varphi r^2\right)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial\left(\sigma_{\theta\varphi}u_\varphi \sin\theta\right)}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial\left(\sigma_{\varphi\varphi}u_\varphi\right)}{\partial\varphi} \tag{3.56}$$

Adding Eqs. (3.54)–(3.56), an expression for the work done by surface forces in spherical coordinates is obtained.

$$\begin{aligned}\dot{w}_s = {} & \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2\left(\sigma_{rr}u_r + \sigma_{r\theta}u_\theta + \sigma_{r\varphi}u_\varphi\right)\right] + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta} \\ & \times\left[\sin\theta\left(\sigma_{\theta r}u_r + \sigma_{\theta\theta}u_\theta + \sigma_{\theta\varphi}u_\varphi\right)\right] + \frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi} \\ & \times\left[\sigma_{\varphi r}u_r + \sigma_{\varphi\theta}u_\theta + \sigma_{\varphi\varphi}u_\varphi\right]\end{aligned} \tag{3.57}$$

Equation (3.57) can be expressed in compact form as the divergence of the scalar product of the stress tensor and velocity.

$$\dot{w}_s = \nabla\cdot\left(\overleftrightarrow{\sigma}\cdot\vec{V}\right) \tag{3.58}$$

3.6.5 Energy Conservation Equation as a Function of Total Specific Energy

The energy conservation equation can be written in terms of the total specific energy, which was given in Eq. (3.49); by substituting the expressions for the heat transfer by conduction, Eq. (3.52); the work done by body forces, Eq. (3.53); and the work done by surface forces, Eq. (3.58). The result is the expression given in Eq. (3.59).

Energy conservation equation in general form and as a function of total specific energy

$$\rho\frac{De}{Dt} = -\nabla\cdot\left(\sum\overrightarrow{\dot{m}}_{k,\mathrm{diff}}h_k\right) + \nabla\cdot(k\nabla T) + \nabla\cdot(\overleftrightarrow{\sigma}\cdot\vec{V}) + \dot{w}_c + \dot{q}_{\mathrm{gen},vc} \tag{3.59}$$

As aforementioned, the heat generation (or dissipation) term, $\dot{q}_{\mathrm{gen},vc}$ includes heat transfer by thermal radiation from the flame.

3.6.6 Mechanical Energy Equation

Although Eq. (3.59) represents the energy conservation equation, it is still possible to make its mathematical expression more detailed. To this end, the equation for mechanical energy will first be obtained. Returning to the momentum equation (Eqs. 3.32a–3.32c), the equations that define the surface force (Eq. 3.25), and the

body force (Eq. 3.21), the scalar product of these variables with the velocity ($\vec{V}$) will be determined. Such operations will be performed in parts to show the procedure.

First, the momentum side of Eqs. (3.32a)–(3.32c) will be rearranged. When multiplying scalarly by the velocity and performing the sum, some terms cancel each other out, thus obtaining the expression shown below.

$$\left(\vec{f}_{vc}+\rho\vec{B}_{vc}\right)\cdot\vec{V}=\frac{\rho}{2}\left(\frac{\partial u_r^2}{\partial t}+u_r\frac{\partial u_r^2}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_r^2}{\partial\theta}+\frac{u_\varphi}{r\sin\theta}\frac{\partial u_r^2}{\partial\varphi}\right)$$
$$+\frac{\rho}{2}\left(\frac{\partial u_\theta^2}{\partial t}+u_r\frac{\partial u_\theta^2}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_\theta^2}{\partial\theta}+\frac{u_\varphi}{r\sin\theta}\frac{\partial u_\theta^2}{\partial\varphi}\right)$$
$$+\frac{\rho}{2}\left(\frac{\partial u_\varphi^2}{\partial t}+u_r\frac{\partial u_\varphi^2}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_\varphi^2}{\partial\theta}+\frac{u_\varphi}{r\sin\theta}\frac{\partial u_\varphi^2}{\partial\varphi}\right) \tag{3.60}$$

Applying the scalar product of the velocity with the surface force, as well as the body force, and recognizing that the material derivative of kinetic energy appears in Eq. (3.60), the mechanical energy conservation equation is obtained.

$$\rho\frac{D}{Dt}\left(\frac{V^2}{2}\right)=\left[\frac{u_r}{r^2}\frac{\partial(r^2\sigma_{rr})}{\partial r}+\frac{u_r}{r\sin\theta}\frac{\partial}{\partial\theta}(\sigma_{\theta r}\sin\theta)+\frac{u_r}{r\sin\theta}\frac{\partial\sigma_{\varphi r}}{\partial\varphi}-u_r\frac{\sigma_{\theta\theta}+\sigma_{\varphi\varphi}}{r}\right]$$
$$+\left[\frac{u_\theta}{r^2}\frac{\partial}{\partial r}\left(r^2\sigma_{r\theta}\right)+\frac{u_\theta}{r\sin\theta}\frac{\partial}{\partial\theta}(\sigma_{\theta\theta}\sin\theta)+\frac{u_\theta}{r\sin\theta}\frac{\partial\sigma_{\varphi\theta}}{\partial\varphi}+u_\theta\frac{\sigma_{\theta r}}{r}-u_\theta\frac{\sigma_{\varphi\varphi}}{r}\cot\theta\right]$$
$$+\left[\frac{u_\varphi}{r^2}\frac{\partial\left(r^2\sigma_{r\varphi}\right)}{\partial r}+\frac{u_\varphi}{r\sin\theta}\frac{\partial}{\partial\theta}\left(\sigma_{\theta\varphi}\sin\theta\right)+\frac{u_\varphi}{r\sin\theta}\frac{\partial\sigma_{\varphi\varphi}}{\partial\varphi}+u_\varphi\frac{\sigma_{\varphi r}}{r}+u_\varphi\frac{\sigma_{\varphi\theta}}{r}\cot\theta\right]$$
$$+\rho\left(\left[\sum_{k=1}^{N}Y_k f_{k,\mathrm{r}}\right]u_r+\left[\sum_{k=1}^{N}Y_k f_{k,\theta}\right]u_\theta+\left[\sum_{k=1}^{N}Y_k f_{k,\varphi}\right]u_\varphi\right) \tag{3.61}$$

3.6.7 Energy Conservation Equation as a Function of Enthalpy

In this section, the energy conservation equation will be obtained as a function of the specific enthalpy. For this purpose, the mechanical energy conservation equation will be used, and the viscous dissipation function will appear (Φv) on the conservation equation. The energy conservation equation is written using Eqs. (3.49), (3.51), (3.53), and (3.57). From the expression thus obtained, the mechanical energy equation, Eq. (3.61), is subtracted, thus yielding the expression shown in Eq. (3.62).

$$\rho\frac{De}{Dt}-\rho\frac{D}{Dt}\left(\frac{V^2}{2}\right)+\nabla\cdot\left(\sum\vec{m}_{k,\mathrm{diff}}h_k\right)=$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(kr^2\frac{\partial T}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(k\frac{\partial T}{\partial\theta}\sin\theta\right)\frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial\varphi}\left(k\frac{\partial T}{\partial\varphi}\right)+\sigma_{rr}\left(\frac{\partial u_r}{\partial r}\right)$$

$$+\sigma_{\theta\theta}\left(\frac{1}{r}\frac{\partial u_\theta}{\partial\theta}+\frac{u_r}{r}\right)+\sigma_{\varphi\varphi}\left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}+\frac{u_r}{r}+\frac{u_\theta}{r}\cot\theta\right)+\sigma_{r\theta}\left(\frac{\partial u_\theta}{\partial r}+\frac{1}{r}\frac{\partial u_r}{\partial\theta}-\frac{u_\theta}{r}\right)$$

$$+\sigma_{r\varphi}\left(\frac{\partial u_\varphi}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial\varphi}-\frac{u_\varphi}{r}\right)+\sigma_{\theta\varphi}\left(\frac{1}{r}\frac{\partial u_\varphi}{\partial\theta}+\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial\varphi}-\frac{u_\varphi}{r}\cot\theta\right)$$

$$+\rho\left(\sum_{k=1}^{N}Y_k f_{k,\mathrm{r}}u_{k,\mathrm{diff},r}\right)+\rho\left(\sum_{k=1}^{N}Y_k f_{k,\theta}u_{\mathrm{k,diff},\theta}\right)+\rho\left(\sum_{k=1}^{N}Y_k f_{k,\varphi}u_{\mathrm{k,diff},\varphi}\right)$$

$$+\dot{q}_{ge,vc} \tag{3.62}$$

On the right-hand side of Eq. (3.62), it is possible to recognize the double dot product of the stress (a second-order symmetric tensor) with the gradient of the velocity vector. Therefore, the compact form of this conservation equation is as shown in Eq. (3.63).

$$\rho\frac{De}{Dt}-\rho\frac{D}{Dt}\left(\frac{V^2}{2}\right)+\nabla\cdot\left(\sum\vec{m}_{k,\mathrm{diff}}h_k\right)=\nabla\cdot(k\nabla T)+\overleftrightarrow{\sigma}$$

$$:\nabla\vec{V}+\rho\sum_{k=1}^{N}Y_k\vec{f}_k\cdot\vec{V}_{k,\mathrm{diff}}+\dot{q}_{\mathrm{gen},vc} \tag{3.63}$$

The double dot product of the stress tensor and the velocity gradient can be written in terms of the viscous dissipation function (Φ_v).

$$\vec{\sigma}:\nabla\vec{V}=-p\left(\nabla\cdot\vec{V}\right)+\mu\Phi_v \tag{3.64}$$

where

Viscous dissipation function in spherical coordinates

$$\Phi_v=+2\left(\frac{\partial u_r}{\partial r}\right)^2+2\left(\frac{1}{r}\frac{\partial u_\theta}{\partial\theta}+\frac{u_r}{r}\right)^2+2\left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}+\frac{u_r}{r}+\frac{u_\theta}{r}\cot\theta\right)^2$$

$$+\left(\frac{1}{r}\frac{\partial u_r}{\partial\theta}+\frac{\partial u_\theta}{\partial r}-\frac{u_\theta}{r}\right)^2+\left(\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial\varphi}+\frac{\partial u_\varphi}{\partial r}-\frac{u_\varphi}{r}\right)^2$$

$$+\left(\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial\varphi}+\frac{1}{r}\frac{\partial u_\varphi}{\partial\theta}-\frac{u_\varphi}{r}\cot\theta\right)^2$$

$$-\frac{2}{3}\left(\frac{1}{r^2}\frac{\partial(r^2u_r)}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial(u_\theta\sin\theta)}{\partial\theta}+\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}\right)^2 \tag{3.65}$$

The viscous dissipation function represents the conversion of mechanical energy into heat, which occurs in flows due to the viscosity of the fluids. The Appendix of the book shows the derivation of Eqs. (3.64) and (3.65) on Section A.15. The specific total energy can be expressed in terms of the specific enthalpy using Eq. (3.46). Therefore, applying the material derivative,

$$\rho\frac{De}{Dt}=\rho\frac{Dh}{Dt}+\rho\frac{D}{Dt}\left(\frac{V^2}{2}\right)-\frac{Dp}{Dt}+\frac{p}{\rho}\frac{D\rho}{Dt}$$

Note that the following relation can be obtained from the total mass conservation equation.

$$\frac{\partial\rho}{\partial t}+\nabla\cdot\left(\rho\vec{V}\right)=\frac{D\rho}{Dt}+\rho\nabla\cdot\vec{V}=0$$

Therefore, from the two previous expressions, it is possible to write Eq. (3.66), which relates the material derivative of the specific total energy to the material derivative of the specific enthalpy.

$$\rho\frac{De}{Dt}=\rho\frac{Dh}{Dt}+\rho\frac{D}{Dt}\left(\frac{V^2}{2}\right)-\frac{Dp}{Dt}-p\left(\nabla\cdot\vec{V}\right) \tag{3.66}$$

Using Eqs. (3.64) and (3.66) into Eq. (3.63), the general form of the energy conservation equation as a function of specific enthalpy is obtained.

Energy conservation equation in general form as a function of specific enthalpy

$$\rho\frac{Dh}{Dt}+\nabla\cdot\left(\sum\vec{m}_{k,\text{diff}}h_k\right)=\nabla\cdot(k\nabla T)+\frac{Dp}{Dt}+\mu\Phi_v$$
$$+\rho\sum_{k=1}^{N}Y_k\vec{f}_k\cdot\vec{V}_{k,\text{diff}}+\dot{q}_{\text{gen},vc} \tag{3.67}$$

3.6.8 Energy Conservation Equation as a Function of Temperature

The most used form of the energy conservation equation contains the temperature in explicit form, so this formulation will also be obtained in this section. The enthalpy of the mixture is given by the sum of the enthalpies of the species ($h = \sum Y_k h_k$). Likewise, the enthalpies of the species are given by the sum of the enthalpies of formation and the sensible enthalpies. Applying the material derivative to the sum of the enthalpies of the species,

$$\frac{Dh}{Dt}=\sum Y_k\frac{Dh_k}{Dt}+\sum h_k\frac{DY_k}{Dt}$$

In the equation for the mass conservation of the chemical species, Eq. (3.11), the term $\nabla\cdot\left(\rho_k\vec{V}\right)$ is developed to obtain the equation of conservation of total mass and

the divergence of the mole fraction. The following relation is obtained after some rearrangements.

$$\frac{DY_k}{Dt} = \frac{1}{\rho}\left(\dot{m}_{k,R} - \nabla\cdot\overrightarrow{\dot{m}}_{k,\text{diff}}\right) \tag{3.68}$$

Therefore, the material derivative of enthalpy can be written as shown in Eq. (3.69).

$$\frac{Dh}{Dt} = \sum Y_k \frac{Dh_k}{Dt} + \frac{1}{\rho}\sum\left(h_k \dot{m}_{k,R}\right) - \frac{1}{\rho}\sum\left(h_k \nabla\cdot\overrightarrow{\dot{m}}_{k,\text{diff}}\right) \tag{3.69}$$

Equation (3.69) will be used to represent the left-hand side of Eq. (3.67).

$$\rho\frac{Dh}{Dt} + \nabla\cdot\left(\sum\overrightarrow{\dot{m}}_{k,\text{diff}}h_k\right) = \rho\sum Y_k\frac{Dh_k}{Dt} + \sum\left(h_k\dot{m}_{k,R}\right) - \sum\left(h_k\nabla\cdot\overrightarrow{\dot{m}}_{k,\text{diff}}\right) + \nabla\cdot\left(\sum\overrightarrow{\dot{m}}_{k,\text{diff}}h_k\right) \tag{3.70}$$

Evaluating the third term on the right-hand side of Eq. (3.70), the following is observed.

$$\sum\left(h_k\nabla\cdot\overrightarrow{\dot{m}}_{k,\text{diff}}\right) = \sum\left[\frac{h_k}{r^2}\frac{\partial\left(r^2\dot{m}_{k,r,\text{diff}}\right)}{\partial r} + \frac{h_k}{r\sin\theta}\frac{\partial\left(\dot{m}_{k,\theta,\text{diff}}\sin\theta\right)}{\partial\theta} + \frac{h_k}{r\sin\theta}\frac{\partial\dot{m}_{k,\varphi,\text{diff}}}{\partial\varphi}\right]$$

Evaluating the last term on the right-hand side of Eq. (3.70), it is observed that the divergence can be applied within the summation, that is, the divergence can be applied to each product $\overrightarrow{\dot{m}}_{k,\text{dif}}h_k$.

$$\nabla\cdot\left(\sum\overrightarrow{\dot{m}}_{k,\text{diff}}h_k\right) = \nabla\cdot\left(\overrightarrow{\dot{m}}_{1,\text{diff}}h_1 + \overrightarrow{\dot{m}}_{2,\text{diff}}h_2 + \overrightarrow{\dot{m}}_{3,\text{diff}}h_3 + \cdots + \overrightarrow{\dot{m}}_{k,\text{diff}}h_k\right) = \sum\left(\nabla\cdot[\overrightarrow{\dot{m}}_{k,\text{diff}}h_k]\right)$$

Then, applying the divergence,

$$\sum\left(\nabla\cdot[\overrightarrow{\dot{m}}_{k,\text{diff}}h_k]\right) = \sum\left[\frac{h_k}{r^2}\frac{\partial\left(r^2\dot{m}_{k,r,\text{diff}}\right)}{\partial r} + \frac{h_k}{r\sin\theta}\frac{\partial\left(\dot{m}_{k,\theta,\text{diff}}\sin\theta\right)}{\partial\theta} + \frac{h_k}{r\sin\theta}\frac{\partial\dot{m}_{k,\varphi,\text{diff}}h_k}{\partial\varphi} + \dot{m}_{k,r,\text{diff}}\frac{\partial h_k}{\partial r} + \frac{\dot{m}_{k,\theta,\text{diff}}}{r}\frac{\partial h_k}{\partial\theta} + \frac{\dot{m}_{k,\varphi,\text{diff}}}{r\sin\theta}\frac{\partial h_k}{\partial\varphi}\right]$$

Afterwards, subtracting the last two terms of Eq. (3.70), it is observed that

$$\nabla\cdot\left(\sum\overrightarrow{\dot{m}}_{k,diff}h_k\right)-\sum\left(h_k\nabla\cdot\overrightarrow{\dot{m}}_{k,diff}\right)=\sum\left(\overrightarrow{\dot{m}}_{k,diff}\cdot\nabla h_k\right)$$

Finally, substituting the above expression into Eq. (3.70), the expression given in Eq. (3.71) is obtained.

$$\rho\frac{Dh}{Dt}+\nabla\cdot\left(\sum\overrightarrow{\dot{m}}_{k,\mathrm{diff}}h_k\right)=\rho\sum Y_k\frac{Dh_k}{Dt}+\sum\left(h_k\dot{m}_{k,R}\right)+\sum\left(\overrightarrow{\dot{m}}_{k,\mathrm{diff}}\cdot\nabla h_k\right)\tag{3.71}$$

Equation (3.71) will now be written in terms of temperature. Then, the following expression for the specific enthalpy of a species is considered.

$$h_k=h_{f,k}^0+\int_{T_{ref}}^{T}c_{p,k}dT\tag{3.72}$$

Substituting Eq. (3.72) into Eq. (3.71), it is possible to write the sensible enthalpy of the species in terms of the temperature.

$$\rho\frac{Dh}{Dt}+\nabla\cdot\left(\sum\overrightarrow{\dot{m}}_{k,\mathrm{diff}}h_k\right)=\rho\frac{D}{Dt}\int_{T_{\mathrm{ref}}}^{T}c_pdT+\sum\left(h_{f,k}^0\dot{m}_{k,R}\right)+\sum\left(\dot{m}_{k,R}\int_{T_{\mathrm{ref}}}^{T}c_{p,k}dT\right)+\sum\overrightarrow{\dot{m}}_{k,\mathrm{diff}}\cdot\nabla\left(\int_{T_{\mathrm{ref}}}^{T}c_{p,k}dT\right)\tag{3.73}$$

Finally, substituting Eq. (3.73) into Eq. (3.67) and using Fick's law, Eq. (3.16), the compact form of the energy conservation equation as a function of temperature is obtained, as shown in Eq. (3.74).

Energy conservation equation in compact form and as a function of temperature

$$\rho\frac{D}{Dt}\left(\int_{T_{\mathrm{ref}}}^{T}c_pdT\right)=\frac{Dp}{Dt}+\mu\Phi_v+\rho\sum_{k=1}^{N}Y_k\vec{f}_k\cdot\vec{V}_{k,\mathrm{dif}}+\dot{q}_{\mathrm{gen},vc}+\nabla\cdot(k\nabla T)-\sum\left(h_{f,k}^0\dot{m}_{k,R}\right)-\sum\left(\dot{m}_{k,R}\int_{T_{\mathrm{ref}}}^{T}c_{p,k}dT\right)+\sum\left[\rho D\nabla Y_k\cdot\nabla\left(\int_{T_{\mathrm{ref}}}^{T}c_{p,k}dT\right)\right]\tag{3.74}$$

The derivation of the conservation equations ends with Eq. (3.74). Summarizing, the following expressions were derived.

- The total mass conservation equation, Eq. (3.8)
- The species mass conservation equation, Eq. (3.11)
- The momentum equation, or Navier–Stokes equation, Eq. (3.33)
- The stress tensor constitutive equations, Eq. (3.43)
- The momentum equation for inviscid flow, or Euler equation, Eq. (3.44)
- The energy conservation equation as a function of temperature and mixture properties, Eq. (3.74)

Compiling the results obtained during the derivation, it is possible to write the energy conservation equation in spherical coordinates, as given in Eq. (3.75).

Energy conservation equation in spherical coordinates and as a function of temperature

$$\rho\left[\frac{\partial}{\partial t}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right)+u_r\frac{\partial}{\partial r}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right)+\frac{u_\theta}{r}\frac{\partial}{\partial \theta}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right)+\frac{u_\varphi}{r\sin\theta}\frac{\partial}{\partial \varphi}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right)\right]=$$

$$+\frac{1}{r^2}\frac{\partial}{\partial r}\left(kr^2\frac{\partial T}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(k\frac{\partial T}{\partial \theta}\sin\theta\right)\frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial \varphi}\left(k\frac{\partial T}{\partial \varphi}\right)$$

$$+\rho\sum D\frac{\partial Y_k}{\partial r}\frac{\partial}{\partial r}\left(\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right)+\frac{\rho}{r^2}\sum D\frac{\partial Y_k}{\partial \theta}\frac{\partial}{\partial \theta}\left(\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right)$$

$$+\frac{\rho}{r^2\sin^2\theta}\sum D\frac{\partial Y_k}{\partial \varphi}\frac{\partial}{\partial \varphi}\left(\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right)-\sum\left(h_{f,k}^{0}\dot{m}_{k,R}\right)$$

$$-\sum\left(\dot{m}_{k,R}\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right)+\frac{Dp}{Dt}+\mu\Phi_v+\rho\sum_{k=1}^{N}Y_k\vec{f}_k\cdot\vec{V}_{k,\text{diff}}+\dot{q}_{\text{gen},vc} \quad (3.75)$$

Equation (3.75) can be further simplified by assuming that c_p and k are constant. In particular, the approach of performing the deductions in spherical coordinates was adopted; therefore, the conservation equations were also obtained in spherical coordinates.

- The total mass conservation equation in spherical coordinates, Eq. (3.5)
- The species mass conservation equation in spherical coordinates, Eq. (3.10)
- The momentum equation, or Navier–Stokes equation, in spherical coordinates, Eqs. (3.32a)–(3.32c)
- The stress tensor constitutive equations in spherical coordinates, Eqs. (3.42a)–(3.42f).
- The energy conservation equation as a function of temperature and mixture properties, in spherical coordinates, Eq. (3.75)

In the following chapters, simplified versions of the conservation equations will be used; however, knowledge of the derivation of such equations is valuable to better understand the mathematical representation of physical phenomena related to flame propagation.

3.7 Summary of Conservation Equations in Different Coordinate Systems

This chapter is closed by writing the conservation equation in explicit form in the Cartesian, cylindrical, and spherical coordinate systems.

3.7.1 Conservation Equations in Rectangular System

The total mass conservation equation in rectangular coordinates is given in Eq. (3.76).

$$\frac{\partial \rho}{\partial t} = -\frac{\partial (\rho u_x)}{\partial x} - \frac{\partial (\rho u_y)}{\partial y} - \frac{\partial (\rho u_z)}{\partial z} \tag{3.76}$$

The mass conservation equation of species A in rectangular coordinates using Fick's law is given in Eq. (3.77).

$$\frac{\partial (\rho Y_A)}{\partial t} = -\frac{\partial}{\partial x}(\rho u_x Y_A) - \frac{\partial}{\partial y}(\rho u_y Y_A) - \frac{\partial}{\partial z}(\rho u_z Y_A)$$
$$+\frac{\partial}{\partial x}\left(\rho D \frac{\partial Y_A}{\partial x}\right) + \frac{\partial}{\partial y}\left(\rho D \frac{\partial Y_A}{\partial y}\right) + \frac{\partial}{\partial z}\left(\rho D \frac{\partial Y_A}{\partial z}\right) + \dot{m}_{A,R} \tag{3.77}$$

The momentum equation in rectangular coordinates in terms of the stress tensor is given in Eqs. (3.78a)–(3.78c).

At coordinate x

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{yx}}{\partial y} + \frac{\partial \sigma_{zx}}{\partial z} + \rho B_x = \rho\left(\frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} + u_z \frac{\partial u_x}{\partial z}\right) \tag{3.78a}$$

At coordinate y

$$\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{zy}}{\partial z} + \rho B_y = \rho\left(\frac{\partial u_y}{\partial t} + u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} + u_z \frac{\partial u_y}{\partial z}\right) \tag{3.78b}$$

At coordinate z

$$\frac{\partial \sigma_{xz}}{\partial x}+\frac{\partial \sigma_{yz}}{\partial y}+\frac{\partial \sigma_{zz}}{\partial z}+\rho B_z=\rho\left(\frac{\partial u_z}{\partial t}+u_x\frac{\partial u_z}{\partial x}+u_y\frac{\partial u_z}{\partial y}+u_z\frac{\partial u_z}{\partial z}\right) \tag{3.78c}$$

The components of the stress tensor in rectangular coordinates are given in Eqs. (3.79a)–(3.79f).

$$\sigma_{xx}=-p+2\mu\frac{\partial u_x}{\partial x}-\frac{2}{3}\mu\nabla\cdot\vec{V} \tag{3.79a}$$

$$\sigma_{yy}=-p+2\mu\frac{\partial u_y}{\partial y}-\frac{2}{3}\mu\nabla\cdot\vec{V} \tag{3.79b}$$

$$\sigma_{zz}=-p+2\mu\frac{\partial u_z}{\partial z}-\frac{2}{3}\mu\nabla\cdot\vec{V} \tag{3.79c}$$

$$\sigma_{xy}=\sigma_{yx}=\mu\left(\frac{\partial u_x}{\partial y}+\frac{\partial u_y}{\partial x}\right) \tag{3.79d}$$

$$\sigma_{xz}=\sigma_{zx}=\mu\left(\frac{\partial u_x}{\partial z}+\frac{\partial u_z}{\partial x}\right) \tag{3.79e}$$

$$\sigma_{yz}=\sigma_{zy}=\mu\left(\frac{\partial u_y}{\partial z}+\frac{\partial u_z}{\partial y}\right) \tag{3.79f}$$

The divergence of the velocity vector is given in Eq. (3.80).

$$\nabla\cdot\vec{V}=\frac{\partial u_x}{\partial x}+\frac{\partial u_y}{\partial y}+\frac{\partial u_z}{\partial z} \tag{3.80}$$

The energy conservation equation in terms of temperature and in rectangular coordinates is given by Eq. (3.81).

$$\rho\left[\frac{\partial}{\partial t}\left(\int_{T_{\text{ref}}}^{T}c_p dT\right)+u_x\frac{\partial}{\partial x}\left(\int_{T_{\text{ref}}}^{T}c_p dT\right)+u_y\frac{\partial}{\partial y}\left(\int_{T_{\text{ref}}}^{T}c_p dT\right)+u_z\frac{\partial}{\partial z}\left(\int_{T_{\text{ref}}}^{T}c_p dT\right)\right]=$$

$$+\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right)+\frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right)+\frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right)-\sum\left(h^0_{f,k}\dot{m}_{k,R}\right)-\sum\left(\dot{m}_{k,R}\int_{T_{\text{ref}}}^{T}c_{p,k}dT\right)$$

$$+\rho\left[\sum D\frac{\partial Y_k}{\partial x}\frac{\partial}{\partial x}\left(\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right)+\sum D\frac{\partial Y_k}{\partial y}\frac{\partial}{\partial y}\left(\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right)\right.$$
$$\left.+\sum D\frac{\partial Y_k}{\partial z}\frac{\partial}{\partial z}\left(\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right)\right]$$
$$+\frac{Dp}{Dt}+\mu\Phi_v+\rho\sum_{k=1}^{N}Y_k\vec{f}_k\cdot\vec{V}_{k,\text{dif}}+\dot{q}_{\text{gen},vc} \tag{3.81}$$

The energy conservation equation in terms of temperature, with c_p and k constant, and in rectangular coordinates is given by Eq. (3.82).

$$\rho c_p\left[\frac{\partial T}{\partial t}+u_x\frac{\partial T}{\partial x}+u_y\frac{\partial T}{\partial y}+u_z\frac{\partial T}{\partial z}\right]=k\left[\frac{\partial^2 T}{\partial x^2}+\frac{\partial^2 T}{\partial y^2}+\frac{\partial^2 T}{\partial z^2}\right]-\sum\left(h^0_{f,k}\dot{m}_{k,R}\right)$$
$$-\sum\left(\dot{m}_{k,R}c_{p,k}T\right)+\rho\left[\sum Dc_{p,k}\frac{\partial Y_k}{\partial x}\frac{\partial T}{\partial x}+\sum Dc_{p,k}\frac{\partial Y_k}{\partial y}\frac{\partial T}{\partial y}+\sum Dc_{p,k}\frac{\partial Y_k}{\partial z}\frac{\partial T}{\partial z}\right]$$
$$+\frac{Dp}{Dt}+\mu\Phi_v+\rho\sum_{k=1}^{N}Y_k\vec{f}_k\cdot\vec{V}_{k,\text{diff}}+\dot{q}_{\text{gen},vc} \tag{3.82}$$

The viscous dissipation function in rectangular coordinates is given by Eq. (3.83) [37].

$$\Phi_v=2\left[\left(\frac{\partial u_x}{\partial x}\right)^2+\left(\frac{\partial u_y}{\partial y}\right)^2+\left(\frac{\partial u_z}{\partial z}\right)^2\right]+\left[\frac{\partial u_y}{\partial x}+\frac{\partial u_x}{\partial y}\right]^2$$
$$+\left[\frac{\partial u_z}{\partial y}+\frac{\partial u_y}{\partial z}\right]^2 \tag{3.83}$$
$$+\left[\frac{\partial u_x}{\partial z}+\frac{\partial u_z}{\partial x}\right]^2-\frac{2}{3}\left[\frac{\partial u_x}{\partial x}+\frac{\partial u_y}{\partial y}+\frac{\partial u_z}{\partial z}\right]^2$$

The Nabla operator (∇) in rectangular coordinates is given in Eq. (3.84).

$$\nabla=\vec{i}\frac{\partial}{\partial x}+\vec{j}\frac{\partial}{\partial y}+\vec{k}\frac{\partial}{\partial z} \tag{3.84}$$

Note that the material derivative of any scalar is defined, as shown in Eq. (3.85), for all coordinate systems.

$$\frac{DT}{Dt}=\frac{\partial T}{\partial t}+\vec{V}\cdot\nabla T \tag{3.85}$$

3.7.2 Conservation Equations in Cylindrical System

The total mass conservation equation in cylindrical coordinates is given in Eq. (3.86).

$$\frac{\partial \rho}{\partial t} = -\frac{1}{r}\frac{\partial}{\partial r}(\rho u_r r) - \frac{1}{r}\frac{\partial}{\partial \theta}(\rho u_\theta) - \frac{\partial}{\partial z}(\rho u_z) \tag{3.86}$$

The mass conservation equation of species A in spherical coordinates using Fick's law is given in Eq. (3.87).

$$\begin{aligned}\frac{\partial(\rho Y_A)}{\partial t} = &-\frac{1}{r}\frac{\partial}{\partial r}(r\rho u_r Y_A) - \frac{1}{r}\frac{\partial}{\partial \varphi}(\rho u_\varphi Y_A) - \frac{\partial}{\partial z}(\rho u_z Y_A) + \frac{1}{r}\frac{\partial}{\partial r}\left(r\rho D\frac{\partial Y_A}{\partial r}\right) \\ &+ \frac{1}{r}\frac{\partial}{\partial \varphi}\left(\rho D\frac{\partial Y_A}{\partial \varphi}\right) + \frac{\partial}{\partial z}\left(\rho D\frac{\partial Y_A}{\partial z}\right) + \dot{m}_{A,R}\end{aligned} \tag{3.87}$$

The momentum equation in rectangular coordinates in terms of the stress tensor is given in Eqs. (3.88a)–(3.88c).

At coordinate r

$$\frac{1}{r}\frac{\partial}{\partial r}(r\sigma_{rr}) + \frac{1}{r}\frac{\partial \sigma_{\varphi r}}{\partial \varphi} + \frac{\partial \sigma_{zr}}{\partial z} - \frac{\sigma_{\varphi\varphi}}{r} + \rho B_r = \rho\left(\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + \frac{u_\varphi}{r}\frac{\partial u_r}{\partial \varphi} + u_z\frac{\partial u_r}{\partial z} - \frac{u_\varphi^2}{r}\right) \tag{3.88a}$$

At coordinate φ

$$\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\sigma_{r\varphi}) + \frac{1}{r}\frac{\partial \sigma_{\varphi\varphi}}{\partial \varphi} + \frac{\partial \sigma_{z\varphi}}{\partial z} + \rho B_\varphi = \rho\left(\frac{\partial u_\theta}{\partial t} + u_r\frac{\partial u_\varphi}{\partial r} + \frac{u_\varphi}{r}\frac{\partial u_\varphi}{\partial \varphi} + u_z\frac{\partial u_\varphi}{\partial z} + \frac{u_r u_\varphi}{r}\right) \tag{3.88b}$$

At coordinate z

$$\frac{1}{r}\frac{\partial}{\partial r}(r\sigma_{rz}) + \frac{1}{r}\frac{\partial \sigma_{\varphi z}}{\partial \varphi} + \frac{\partial \sigma_{zz}}{\partial z} + \rho B_z = \rho\left(\frac{\partial u_z}{\partial t} + u_r\frac{\partial u_z}{\partial r} + \frac{u_\varphi}{r}\frac{\partial u_z}{\partial \varphi} + u_z\frac{\partial u_z}{\partial z}\right) \tag{3.88c}$$

The components of the stress tensor in cylindrical coordinates are given in Eqs. (3.89a)–(3.89f).

$$\sigma_{rr} = -p + 2\mu\frac{\partial u_r}{\partial r} - \frac{2}{3}\mu\nabla\cdot\vec{V} \tag{3.89a}$$

$$\sigma_{\varphi\varphi} = -p + 2\mu\left(\frac{1}{r}\frac{\partial u_\varphi}{\partial \varphi} + \frac{u_r}{r}\right) - \frac{2}{3}\mu\nabla\cdot\vec{V} \tag{3.89b}$$

$$\sigma_{zz} = -p + 2\mu\frac{\partial u_z}{\partial z} - \frac{2}{3}\mu\nabla\cdot\vec{V} \tag{3.89c}$$

$$\sigma_{r\varphi} = \sigma_{\varphi r} = \mu\left(\frac{1}{r}\frac{\partial u_r}{\partial \varphi} + \frac{\partial u_\varphi}{\partial r} - \frac{u_\varphi}{r}\right) \tag{3.89d}$$

$$\sigma_{rz} = \sigma_{zr} = \mu\left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r}\right) \tag{3.89e}$$

$$\sigma_{\varphi z} = \sigma_{z\varphi} = \mu\left(\frac{1}{r}\frac{\partial u_y}{\partial \varphi} + \frac{\partial u_\varphi}{\partial z}\right) \tag{3.89f}$$

The divergence of the velocity vector is given in Eq. (3.90).

$$\nabla\cdot\vec{V} = \frac{1}{r}\frac{\partial}{\partial r}(ru_r) + \frac{1}{r}\frac{\partial u_\varphi}{\partial \varphi} + \frac{\partial u_z}{\partial z} \tag{3.90}$$

The energy conservation equation in terms of temperature and in cylindrical coordinates is given by Eq. (3.91).

$$\rho\left[\frac{\partial}{\partial t}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right) + u_r\frac{\partial}{\partial r}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right) + \frac{u_\varphi}{r}\frac{\partial}{\partial \varphi}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right) + u_z\frac{\partial}{\partial z}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right)\right] =$$

$$+\frac{1}{r}\frac{\partial}{\partial r}\left[rk\frac{\partial}{\partial r}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right)\right] + \frac{1}{r^2}\frac{\partial}{\partial \varphi}\left[k\frac{\partial}{\partial \varphi}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right)\right] + \frac{\partial}{\partial z}\left[k\frac{\partial}{\partial z}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right)\right]$$

$$-\sum\left(h^0_{f,k}\dot{m}_{k,R}\right) - \sum\left(\dot{m}_{k,R}\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right) + \rho\sum D\frac{\partial Y_k}{\partial r}\frac{\partial}{\partial r}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right)$$

$$+\rho\sum\frac{D}{r^2}\frac{\partial Y_k}{\partial \varphi}\frac{\partial}{\partial \varphi}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right) + \rho\sum D\frac{\partial Y_k}{\partial z}\frac{\partial}{\partial z}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right)$$

$$+\frac{Dp}{Dt} + \mu\Phi_v + \rho\sum_{k=1}^{N} Y_k\vec{f}_k\cdot\vec{V}_{k,\text{diff}} + \dot{q}_{\text{gen},vc} \tag{3.91}$$

The energy conservation equation in terms of temperature, with c_p and k constant, and in cylindrical coordinates is given by Eq. (3.92).

$$\rho c_p\left[\frac{\partial T}{\partial t}+u_r\frac{\partial T}{\partial r}+\frac{u_\varphi}{r}\frac{\partial T}{\partial \varphi}+u_z\frac{\partial T}{\partial z}\right]=kc_p\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 T}{\partial \varphi^2}+\frac{\partial^2 T}{\partial z^2}\right]$$

$$+\rho\left[\sum Dc_p\frac{\partial Y_k}{\partial r}\frac{\partial T}{\partial r}+\sum\frac{Dc_p}{r^2}\frac{\partial Y_k}{\partial \varphi}\frac{\partial T}{\partial \varphi}+\sum Dc_p\frac{\partial Y_k}{\partial z}\frac{\partial T}{\partial z}\right]-\sum\left(h^0_{f,k}\dot{m}_{k,R}\right)$$

$$-\sum\left(\dot{m}_{k,R}c_{p,k}T\right)+\frac{Dp}{Dt}+\mu\Phi_v+\rho\sum_{k=1}^{N}Y_k\vec{f}_k\cdot\vec{V}_{k,\text{diff}}+\dot{q}_{\text{gen},vc} \tag{3.92}$$

The viscous dissipation function in cylindrical coordinates is given by Eq. (3.93) [37].

$$\Phi_v=2\left[\left(\frac{\partial u_r}{\partial r}\right)^2+\left(\frac{1}{r}\frac{\partial u_\varphi}{\partial \varphi}+\frac{u_r}{r}\right)^2+\left(\frac{\partial u_z}{\partial z}\right)^2\right]+\left[r\frac{\partial}{\partial r}\left(\frac{u_\varphi}{r}\right)+\frac{1}{r}\frac{\partial u_r}{\partial \varphi}\right]^2$$

$$+\left[\frac{1}{r}\frac{\partial u_z}{\partial \varphi}+\frac{\partial u_\varphi}{\partial z}\right]^2+\left[\frac{\partial u_r}{\partial z}+\frac{\partial u_z}{\partial r}\right]^2-\frac{2}{3}\left[\frac{1}{r}\frac{\partial}{\partial r}(ru_r)+\frac{1}{r}\frac{\partial u_\varphi}{\partial \varphi}+\frac{\partial u_z}{\partial z}\right]^2 \tag{3.93}$$

The Nabla operator (∇) in spherical coordinates is given in Eq. (3.94). See the Appendix of the book for details about how to obtain ($\nabla\cdot\vec{V}$).

$$\nabla=\vec{e}_r\frac{\partial}{\partial r}+\vec{e}_\varphi\frac{1}{r}\frac{\partial}{\partial \varphi}+\vec{e}_z\frac{\partial}{\partial z} \tag{3.94}$$

3.7.3 Conservation Equations in Spherical System

The total mass conservation equation in spherical coordinates is given in Eq. (3.5).

$$\frac{\partial \rho}{\partial t}=-\frac{1}{r^2}\frac{\partial(\rho u_r r^2)}{\partial r}-\frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\rho u_\theta\sin\theta)-\frac{1}{r\sin\theta}\frac{\partial(\rho u_\varphi)}{\partial \varphi} \tag{3.5}$$

The mass conservation equation of species A in spherical coordinates using Fick's law is given in Eq. (3.19).

$$\frac{\partial(\rho Y_A)}{\partial t}=-\frac{1}{r^2}\frac{\partial(\rho Y_A u_r r^2)}{\partial r}-\frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\rho Y_A u_\theta\sin\theta)-\frac{1}{r\sin\theta}\frac{\partial(\rho Y_A u_\varphi)}{\partial \varphi}$$

$$+\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\rho D\frac{\partial Y_A}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\rho D\frac{\partial Y_A}{\partial \theta}\sin\theta\right)$$

$$+\frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial \varphi}\left(\rho D\frac{\partial Y_A}{\partial \varphi}\right)+\dot{m}_{A,R} \tag{3.19}$$

The momentum equation in spherical coordinates in terms of the stress tensor is given in Eqs. (3.32a)–(3.32c).

At coordinate r

$$\frac{1}{r^2}\frac{\partial(r^2\sigma_{rr})}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sigma_{\theta r}\sin\theta)+\frac{1}{r\sin\theta}\frac{\partial\sigma_{\varphi r}}{\partial\varphi}-\frac{\sigma_{\theta\theta}+\sigma_{\varphi\varphi}}{r}+\rho B_r=\rho\left(\frac{\partial u_r}{\partial t}+u_r\frac{\partial u_r}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_r}{\partial\theta}+\frac{u_\varphi}{r\sin\theta}\frac{\partial u_r}{\partial\varphi}-\frac{u_\theta^2+u_\varphi^2}{r}\right) \tag{3.32a}$$

At coordinate θ

$$\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\sigma_{r\theta})+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sigma_{\theta\theta}\sin\theta)+\frac{1}{r\sin\theta}\frac{\partial\sigma_{\varphi\theta}}{\partial\varphi}+\frac{\sigma_{\theta r}}{r}-\frac{\sigma_{\varphi\varphi}}{r}\cot\theta+\rho B_\theta=\rho\left(\frac{\partial u_\theta}{\partial t}+u_r\frac{\partial u_\theta}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_\theta}{\partial\theta}+\frac{u_\varphi}{r\sin\theta}\frac{\partial u_\theta}{\partial\varphi}+\frac{u_r u_\theta}{r}-\frac{u_\varphi^2}{r}\cot\theta\right) \tag{3.32b}$$

At coordinate φ

$$\frac{1}{r^2}\frac{\partial(r^2\sigma_{r\varphi})}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sigma_{\theta\varphi}\sin\theta)+\frac{1}{r\sin\theta}\frac{\partial\sigma_{\varphi\varphi}}{\partial\varphi}+\frac{\sigma_{\varphi r}}{r}+\frac{\sigma_{\varphi\theta}}{r}\cot\theta+\rho B_\varphi=\rho\left(\frac{\partial u_\varphi}{\partial t}+u_r\frac{\partial u_\varphi}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_\varphi}{\partial\theta}+\frac{u_\varphi}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}+\frac{u_r u_\varphi}{r}+\frac{u_\theta u_\varphi}{r}\cot\theta\right) \tag{3.32c}$$

The components of the stress tensor in spherical coordinates are given in Eqs. (3.42a)–(3.42f).

$$\sigma_{rr}=-p+2\mu\frac{\partial u_r}{\partial r}-\frac{2}{3}\mu\nabla\cdot\vec{V} \tag{3.42a}$$

$$\sigma_{\theta\theta}=-p+2\mu\left(\frac{1}{r}\frac{\partial u_\theta}{\partial\theta}+\frac{u_r}{r}\right)-\frac{2}{3}\mu\nabla\cdot\vec{V} \tag{3.42b}$$

$$\sigma_{\varphi\varphi}=-p+2\mu\left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}+\frac{u_r}{r}+\frac{u_\theta}{r}\cot\theta\right)-\frac{2}{3}\mu\nabla\cdot\vec{V} \tag{3.42c}$$

$$\sigma_{r\theta} = \sigma_{\theta r} = \mu\left(\frac{1}{r}\frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r}\right) \tag{3.42d}$$

$$\sigma_{r\varphi} = \sigma_{\varphi r} = \mu\left(\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial \varphi} + \frac{\partial u_\varphi}{\partial r} - \frac{u_\varphi}{r}\right) \tag{3.42e}$$

$$\sigma_{\theta\varphi} = \sigma_{\varphi\theta} = \mu\left(\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial \varphi} + \frac{1}{r}\frac{\partial u_\varphi}{\partial \theta} - \frac{u_\varphi}{r}\cot\theta\right) \tag{3.42f}$$

The divergence of the velocity vector can be obtained from Eq. (3.6).

$$\nabla\cdot\vec{V} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 u_r\right) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\left(u_\theta\sin\theta\right) + \frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi} \tag{3.6}$$

The energy conservation equation in spherical coordinates and as a function of temperature is given in Eq. (3.75).

$$\begin{aligned}
&\rho\left[\frac{\partial}{\partial t}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right) + u_r\frac{\partial}{\partial r}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right) + \frac{u_\theta}{r}\frac{\partial}{\partial\theta}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right) + \frac{u_\varphi}{r\sin\theta}\frac{\partial}{\partial\varphi}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right)\right] = \\
&+\frac{1}{r^2}\frac{\partial}{\partial r}\left(kr^2\frac{\partial T}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(k\frac{\partial T}{\partial\theta}\sin\theta\right)\frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial\varphi}\left(k\frac{\partial T}{\partial\varphi}\right) \\
&+\rho\sum D\frac{\partial Y_k}{\partial r}\frac{\partial}{\partial r}\left(\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right) + \frac{\rho}{r^2}\sum D\frac{\partial Y_k}{\partial\theta}\frac{\partial}{\partial\theta}\left(\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right) \\
&+\frac{\rho}{r^2\sin^2\theta}\sum D\frac{\partial Y_k}{\partial\varphi}\frac{\partial}{\partial\varphi}\left(\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right) - \sum\left(h_{f,k}^0\dot{m}_{k,R}\right) - \sum\left(\dot{m}_{k,R}\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right) \\
&+\frac{Dp}{Dt} + \mu\Phi_v + \rho\sum_{k=1}^{N} Y_k\vec{f}_k\cdot\vec{V}_{k,\text{diff}} + \dot{q}_{\text{gen},vc}
\end{aligned} \tag{3.75}$$

The energy conservation equation in spherical coordinates, with cp and k constant, and as a function of temperature is given in Eq. (3.95).

$$\rho c_p\left[\frac{\partial T}{\partial t}+u_r\frac{\partial T}{\partial r}+\frac{u_\theta}{r}\frac{\partial T}{\partial \theta}+\frac{u_\varphi}{r\sin\theta}\frac{\partial T}{\partial \varphi}\right]=$$
$$+k\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial T}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\frac{\partial T}{\partial \theta}\sin\theta\right)\frac{1}{r^2\sin^2\theta}\frac{\partial^2 T}{\partial \varphi^2}\right]$$
$$+\rho\sum Dc_{p,k}\frac{\partial Y_k}{\partial r}\frac{\partial T}{\partial r}+\frac{\rho}{r^2}\sum Dc_{p,k}\frac{\partial Y_k}{\partial \theta}\frac{\partial T}{\partial \theta}+\frac{\rho}{r^2\sin^2\theta}\sum Dc_{p,k}\frac{\partial Y_k}{\partial \varphi}\frac{\partial T}{\partial \varphi}$$
$$-\sum\left(h^0_{f,k}\dot{m}_{k,R}\right)-\sum\left(\dot{m}_{k,R}c_{p,k}T\right)+\frac{Dp}{Dt}+\mu\Phi_v+\rho\sum_{k=1}^{N}Y_k\vec{f}_k\cdot\vec{V}_{k,\text{dif}}$$
$$+\dot{q}_{\text{gen},vc} \tag{3.95}$$

The viscous dissipation function in spherical coordinates is given in Eq. (3.65).

$$\Phi_v=2\left(\frac{\partial u_r}{\partial r}\right)^2+2\left(\frac{1}{r}\frac{\partial u_\theta}{\partial \theta}+\frac{u_r}{r}\right)^2+2\left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial \varphi}+\frac{u_r}{r}+\frac{u_\theta}{r}\cot\theta\right)^2$$
$$+\left(\frac{1}{r}\frac{\partial u_r}{\partial \theta}+\frac{\partial u_\theta}{\partial r}-\frac{u_\theta}{r}\right)^2+\left(\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial \varphi}+\frac{\partial u_\varphi}{\partial r}-\frac{u_\varphi}{r}\right)^2$$
$$+\left(\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial \varphi}+\frac{1}{r}\frac{\partial u_\varphi}{\partial \theta}-\frac{u_\varphi}{r}\cot\theta\right)^2$$
$$-\frac{2}{3}\left(\frac{1}{r^2}\frac{\partial(r^2u_r)}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial(u_\theta\sin\theta)}{\partial \theta}+\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial \varphi}\right)^2 \tag{3.65}$$

The Nabla operator (∇) in spherical coordinates is given in Eq. (3.7). See the Appendix of the book for details about how to obtain ($\nabla\cdot\vec{V}$).

$$\nabla=\vec{e}_r\frac{\partial}{\partial r}+\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial \theta}+\vec{e}_\varphi\frac{1}{r\sin\theta}\frac{\partial}{\partial \varphi} \tag{3.7}$$

3.8 Closure

This chapter presented the derivation of the total mass conservation equation, species mass conservation equation, momentum equation (Navier–Stokes equation), and energy conservation equation. The derivations were presented by following a very simple method which involves the balance across an infinitesimal control volume, followed by the application of the partial derivative definition. The differential equations were therefore obtained in a straightforward manner. Moreover, the

constitutive relations for the stress tensor were also derived. Tensors relations were used in this derivation, and more details are provided in the Appendix of the book. All the derivations were performed in the spherical coordinate system with the objective of showing the nuances of considering the interactions between unit orthogonal vectors in non-Cartesian (non-rectangular) coordinate systems. The derived equations were also presented in general vector form, which makes it easy to write them in other coordinate systems. Finally, the equations were also presented in cylindrical and Cartesian (rectangular) coordinates form.

Appendix

Supplementary Material to the Derivation of the Momentum Equation

- *Equation prior to obtaining* Eq. (3.24)

$$\begin{aligned}
\vec{f}_{vc}(r^2 sin\theta\Delta\varphi\Delta\theta\Delta r) &= (\sigma_{rr} r^2 sin\theta\Delta\varphi\Delta\theta\,\vec{e}_r)_{r+\Delta r} - (\sigma_{rr} r^2 sin\theta\Delta\varphi\Delta\theta\,\vec{e}_r)_r \\
&+ (\sigma_{\theta r} r sin\theta\Delta\varphi\Delta r\,\vec{e}_r)_{\theta+\Delta\theta} - (\sigma_{\theta r} r sin\theta\Delta\varphi\Delta r\,\vec{e}_r)_\theta \\
&+ \left(\sigma_{\varphi r} r\Delta\theta\Delta r\,\vec{e}_r\right)_{\varphi+\Delta\varphi} - \left(\sigma_{\varphi r} r\Delta\theta\Delta r\,\vec{e}_r\right)_\varphi \\
&+ (\sigma_{r\theta} r^2 sin\theta\Delta\varphi\Delta\theta\,\vec{e}_\theta)_{r+\Delta r} - (\sigma_{r\theta} r^2 sin\theta\Delta\varphi\Delta\theta\,\vec{e}_\theta)_r \\
&+ (\sigma_{\theta\theta} r sin\theta\Delta\varphi\Delta r\,\vec{e}_\theta)_{\theta+\Delta\theta} - (\sigma_{\theta\theta} r sin\theta\Delta\varphi\Delta r\,\vec{e}_\theta)_\theta \\
&+ \left(\sigma_{\varphi\theta} r\Delta\theta\Delta r\,\vec{e}_\theta\right)_{\varphi+\Delta\varphi} - \left(\sigma_{\varphi\theta} r\Delta\theta\Delta r\,\vec{e}_\theta\right)_\varphi \\
&+ (\sigma_{r\varphi} r^2 sin\theta\Delta\varphi\Delta\theta\,\vec{e}_\varphi)_{r+\Delta r} - (\sigma_{r\varphi} r^2 sin\theta\Delta\varphi\Delta\theta\,\vec{e}_\varphi)_r \\
&+ (\sigma_{\theta\varphi} r sin\theta\Delta\varphi\Delta r\,\vec{e}_\varphi)_{\theta+\Delta\theta} - (\sigma_{\theta\varphi} r sin\theta\Delta\varphi\Delta r\,\vec{e}_\varphi)_\theta \\
&+ \left(\sigma_{\varphi\varphi} r\Delta\theta\Delta r\,\vec{e}_\varphi\right)_{\varphi+\Delta\varphi} - \left(\sigma_{\varphi\varphi} r\Delta\theta\Delta r\,\vec{e}_\varphi\right)_\varphi
\end{aligned} \tag{3.96}$$

- *Equations prior to obtaining* Eq. (3.25)

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial r}(\sigma_{rr} r^2 sin\theta\,\vec{e}_r) = \vec{e}_r \frac{1}{r^2}\frac{\partial(r^2\sigma_{rr})}{\partial r} \tag{3.97a}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial\theta}(\sigma_{\theta r} r sin\theta\,\vec{e}_r) = \vec{e}_r \frac{1}{r sin\theta}\frac{\partial}{\partial\theta}(\sigma_{\theta r} sin\theta) + \vec{e}_\theta \frac{\sigma_{\theta r}}{r} \tag{3.97b}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial \varphi}(\sigma_{\varphi r} r \vec{e}_r) = \vec{e}_r \frac{1}{r sin\theta}\frac{\partial \sigma_{\varphi r}}{\partial \varphi} + \vec{e}_\varphi \frac{\sigma_{\varphi r}}{r} \tag{3.97c}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial r}(\sigma_{r\theta} r^2 sin\theta \vec{e}_\theta) = \vec{e}_\theta \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 \sigma_{r\theta}) \tag{3.97d}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial \theta}(\sigma_{\theta\theta} r sin\theta \vec{e}_\theta) = \vec{e}_\theta \frac{1}{r sin\theta}\frac{\partial}{\partial \theta}(\sigma_{\theta\theta} sin\theta) - \vec{e}_r \frac{\sigma_{\theta\theta}}{r} \tag{3.97e}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial \varphi}(\sigma_{\varphi\theta} r \vec{e}_\theta) = \vec{e}_\theta \frac{1}{r sin\theta}\frac{\partial \sigma_{\varphi\theta}}{\partial \varphi} + \vec{e}_\varphi \frac{\sigma_{\varphi\theta}}{r} cot\theta \tag{3.97f}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial r}(\sigma_{r\varphi} r^2 sin\theta \vec{e}_\varphi) = \vec{e}_\varphi \frac{1}{r^2}\frac{\partial (r^2 \sigma_{r\varphi})}{\partial r} \tag{3.97g}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial \theta}(\sigma_{\theta\varphi} r sin\theta \vec{e}_\varphi) = \vec{e}_\varphi \frac{1}{r sin\theta}\frac{\partial}{\partial \theta}(\sigma_{\theta\varphi} sin\theta) \tag{3.97h}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial \varphi}(\sigma_{\varphi\varphi} r \vec{e}_\varphi) = \vec{e}_\varphi \frac{1}{r sin\theta}\frac{\partial \sigma_{\varphi\varphi}}{\partial \varphi} - \vec{e}_r \frac{\sigma_{\varphi\varphi}}{r} - \vec{e}_\theta \frac{\sigma_{\varphi\varphi}}{r} cot\theta \tag{3.97i}$$

- *Equation prior to obtaining* Eq. (3.29)

$$\begin{aligned}
\frac{\partial(\rho \vec{V})}{\partial t}(r^2 sin\theta \Delta\varphi \Delta\theta \Delta r) = {} & \left[\rho u_r u_r r^2 sin\theta \Delta\varphi \Delta\theta \vec{e}_r\right]_r - \left[\rho u_r u_r r^2 sin\theta \Delta\varphi \Delta\theta \vec{e}_r\right]_{r+\Delta r} \\
& + \left[\rho u_\theta u_r r sin\theta \Delta\varphi \Delta r \vec{e}_r\right]_\theta - \left[\rho u_\theta u_r r sin\theta \Delta\varphi \Delta r \vec{e}_r\right]_{\theta+\Delta\theta} \\
& + \left[\rho u_\varphi u_r r \Delta\theta \Delta r \vec{e}_r\right]_\varphi - \left[\rho u_\varphi u_r r \Delta\theta \Delta r \vec{e}_r\right]_{\varphi+\Delta\varphi} \\
& + \left[\rho u_r u_\theta r^2 sin\theta \Delta\varphi \Delta\theta \vec{e}_\theta\right]_r - \left[\rho u_r u_\theta r^2 sin\theta \Delta\varphi \Delta\theta \vec{e}_\theta\right]_{r+\Delta r} \\
& + \left[\rho u_\theta u_\theta r sin\theta \Delta\varphi \Delta r \vec{e}_\theta\right]_\theta - \left[\rho u_\theta u_\theta r sin\theta \Delta\varphi \Delta r \vec{e}_\theta\right]_{\theta+\Delta\theta} \\
& + \left[\rho u_\varphi u_\theta r \Delta\theta \Delta r \vec{e}_\theta\right]_\varphi - \left[\rho u_\varphi u_\theta r \Delta\theta \Delta r \vec{e}_\theta\right]_{\varphi+\Delta\varphi}
\end{aligned}$$

$$+\left[\rho u_r u_\varphi r^2 sin\theta \Delta\varphi \Delta\theta \vec{e}_{\varphi}\right]_r - \left[\rho u_r u_\varphi r^2 sin\theta \Delta\varphi \Delta\theta \vec{e}_{\varphi}\right]_{r+\Delta r}$$

$$+\left[\rho u_\theta u_\varphi r sin\theta \Delta\varphi \Delta r \vec{e}_{\varphi}\right]_\theta - \left[\rho u_\theta u_\varphi r sin\theta \Delta\varphi \Delta r \vec{e}_{\varphi}\right]_{\theta+\Delta\theta}$$

$$+\left[\rho u_\varphi u_\varphi r \Delta\theta \Delta r \vec{e}_{\varphi}\right]_\varphi - \left[\rho u_\varphi u_\varphi r \Delta\theta \Delta r \vec{e}_{\varphi}\right]_{\varphi+\Delta\varphi}$$

$$+\left(\vec{f}_{vc} + \rho \vec{B}_{vc}\right)\left(r^2 sin\theta \Delta\varphi \Delta\theta \Delta r\right) \tag{3.98}$$

- *Equations prior to obtaining* Eq. (3.31)

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial r}\left(\rho u_r u_r r^2 sin\theta \vec{e}_r\right) = \vec{e}_r\left[\rho u_r \frac{\partial u_r}{\partial r} + \frac{u_r}{r^2}\frac{\partial}{\partial r}(r^2 \rho u_r)\right] \tag{3.99a}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial \theta}\left(\rho u_\theta u_r r sin\theta \vec{e}_r\right) = \vec{e}_r\left[\frac{\rho u_\theta}{r}\frac{\partial u_r}{\partial \theta} + \frac{u_r}{r sin\theta}\frac{\partial}{\partial \theta}(\rho u_\theta sin\theta)\right] + \vec{e}_\theta \frac{\rho u_\theta u_r}{r} \tag{3.99b}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial \varphi}\left(\rho u_\varphi u_r r \vec{e}_r\right) = \vec{e}_r\left[\frac{\rho u_\varphi}{r sin\theta}\frac{\partial u_r}{\partial \varphi} + \frac{u_r}{r sin\theta}\frac{\partial(\rho u_\varphi)}{\partial \varphi}\right] + \vec{e}_\varphi \frac{\rho u_\varphi u_r}{r} \tag{3.99c}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial r}\left(\rho u_r u_\theta r^2 sin\theta \vec{e}_\theta\right) = \vec{e}_\theta\left[\rho u_r \frac{\partial u_\theta}{\partial r} + \frac{u_\theta}{r^2}\frac{\partial}{\partial r}(r^2 \rho u_r)\right] \tag{3.99d}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial \theta}\left(\rho u_\theta u_\theta r sin\theta \vec{e}_\theta\right) = \vec{e}_\theta\left[\frac{\rho u_\theta}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{u_\theta}{r sin\theta}\frac{\partial}{\partial \theta}(\rho u_\theta sin\theta)\right] - \vec{e}_r \frac{\rho u_\theta u_\theta}{r} \tag{3.99e}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial \varphi}\left(\rho u_\varphi u_\theta r \vec{e}_\theta\right) = \vec{e}_\theta\left[\frac{\rho u_\varphi}{r sin\theta}\frac{\partial u_\theta}{\partial \varphi} + \frac{u_\theta}{r sin\theta}\frac{\partial(\rho u_\varphi)}{\partial \varphi}\right] + \vec{e}_\varphi \frac{\rho u_\varphi u_\theta}{r}\cot\theta \tag{3.99f}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial r}\left(\rho u_r u_\varphi r^2 sin\theta \vec{e}_\varphi\right) = \vec{e}_\varphi\left[\rho u_r \frac{\partial u_\varphi}{\partial r} + \frac{u_\varphi}{r^2}\frac{\partial}{\partial r}(r^2 \rho u_r)\right] \tag{3.99g}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial \theta}\left(\rho u_\theta u_\varphi r sin\theta \vec{e}_\varphi\right) = \vec{e}_\varphi\left[\frac{\rho u_\theta}{r}\frac{\partial u_\varphi}{\partial \theta} + \frac{u_\varphi}{r sin\theta}\frac{\partial}{\partial \theta}(\rho u_\theta sin\theta)\right] \tag{3.99h}$$

$$\frac{1}{r^2 sin\theta}\frac{\partial}{\partial \varphi}\left(\rho u_\varphi u_\varphi r \vec{e}_\varphi\right) = \vec{e}_\varphi\left[\frac{\rho u_\varphi}{r sin\theta}\frac{\partial u_\varphi}{\partial \varphi} + \frac{u_\varphi}{r sin\theta}\frac{\partial(\rho u_\varphi)}{\partial \varphi}\right] - \frac{\rho u_\varphi u_\varphi}{r}\vec{e}_r - \frac{\rho u_\varphi u_\varphi}{r}\cot\theta \vec{e}_\theta \tag{3.99i}$$

- *Expressions in full for* Eqs. (3.34a) and (3.34b)

$$\sigma_{rr} = -p + 2\mu\dot{\varepsilon}_{rr} + \left(\mu' - \frac{2}{3}\mu\right)\left(\dot{\varepsilon}_{rr} + \dot{\varepsilon}_{\theta\theta} + \dot{\varepsilon}_{\varphi\varphi}\right) \tag{3.100a}$$

$$\sigma_{\theta\theta} = -p + 2\mu\dot{\varepsilon}_{\theta\theta} + \left(\mu' - \frac{2}{3}\mu\right)\left(\dot{\varepsilon}_{rr} + \dot{\varepsilon}_{\theta\theta} + \dot{\varepsilon}_{\varphi\varphi}\right) \tag{3.100b}$$

$$\sigma_{\varphi\varphi} = -p + 2\mu\dot{\varepsilon}_{\varphi\varphi} + \left(\mu' - \frac{2}{3}\mu\right)\left(\dot{\varepsilon}_{rr} + \dot{\varepsilon}_{\theta\theta} + \dot{\varepsilon}_{\varphi\varphi}\right) \tag{3.100c}$$

$$\sigma_{r\theta} = \sigma_{\theta r} = 2\mu\dot{\varepsilon}_{r\theta} \tag{3.100d}$$

$$\sigma_{r\varphi} = \sigma_{\varphi r} = 2\mu\dot{\varepsilon}_{r\varphi} \tag{3.100e}$$

$$\sigma_{\theta\varphi} = \sigma_{\varphi\theta} = 2\mu\dot{\varepsilon}_{\theta\varphi} \tag{3.100f}$$

Premixed Flames

4

Nomenclature

A	Área, m^2
Ar	Arrhenius number
B_C	Collision number, kg/m^3 s
c_p	Heat capacity at constant pressure, J/kg K
c_i	Concentration, $kmol/m^3$
Da	Damköhler number
D_{ij}	Binary diffusivity coefficient, m^2/s
E	Energia, J
E_a	Activation energy, KJ/mol
H	Enthalpy, J/kg
$h^\circ_{f,i}$	Enthalpy of formation of species i, J/kg
Δh	Sensible enthalpy, J/kg
Le	Lewis number
m	Mass, kg
$\dot{m}$	Mass flow, kg/s
m°	Mass flux, kg/m^2 s
p	Pressure, Pa
q_C	Heat liberated by unit mass of fuel, J/kg
$\dot{Q}$	Heat flux, W/m^2
$s_u{}^\circ$	Laminar flame speed, m/s
T	Temperature, K
T_a	Activation temperature, K
t	Time, s
V	Volume, m^3
v_i	Stoichiometric coefficient of species i
x	Coordinate

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Y Mass fraction
y Coordinate
w Reaction rate, kg/m^3 s
$\dot{W}_t$ Shaft power, W
W_i Molecular weight, kg/kmol
z Coordinate
Ze Zel'dovich number

Greek Letters

δ_D Length of the preheating zone, m
δ_R Length of the reaction zone, m
$\sigma_{i,f}$ The mass ratio of chemical species "i" reacting per unit mass of fuel
Φ Weighted mass fraction
Ξ Dimensionless horizontal coordinate
ρ Density, kg/m^3
Δ Variation
θ Dimensionless temperature
λ Thermal conductivity, W/m K

Subscripts

b Burned
d Deficient reactant
u Unburned
f Fuel
fr Frontier
ig Ignition
P Products
R Reactants

4.1 Introduction

The theory of laminar premixed flames is presented in several classical textbooks in the field of combustion. The focus of this chapter is to present the classical developments and the general understanding of laminar premixed flames. Therefore, the chapter starts with a description of the premixed flame characteristics, followed by a simple derivation of the one-dimensional conservation equations applied to the flame. Then, the one-step chemical reaction and the dimensionless

conservation equations are introduced. The chapter then introduces the dimensionless numbers that serve to characterize premixed flames. Finally, a solution for the governing equations is presented. The study of premixed flames is also presented by Law [11], Williams [36], Clavin e Searby [39], Glassman et al. [10], Kuo [9], and Turns [7], among others.

4.2 Description and Flame Characteristics

Consider a laminar and one-dimensional premixed flame. This flame can be divided into two zones—the preheating zone and the reaction zone—as shown schematically in Fig. 4.1. In the preheating zone, the reactant mixture is heated by heat conduction from the reaction zone. In the reaction zone, chemical reactions take place, in which energy is released from the fuel. The characteristic dimensions of the premixed flame are the length of the preheating zone δ_D and the length of the reaction zone δ_R. This representation is a well-known description of the laminar premixed flame [7, 10].

Assuming a coordinate system fixed to the flame, the latter can be considered a control volume, as shown in Fig. 4.2. Therefore, the rate of mass accumulation within the flame is determined by the relationship in Eq. (4.1).

$$\frac{\partial m}{\partial t} = [\dot{m}]_x - [\dot{m}]_{x+\Delta x} \tag{4.1}$$

If the cross-sectional area is constant and represented by A, the velocity by u, and the density by ρ, then, the mass within the control volume is equal to $m = \rho A \Delta x$, and the mass flow rate is equal to $\dot{m} = \rho s A$. Substituting these relationships into Eq. (4.1), dividing by Δx, and taking the limit when $\Delta x \to 0$, the law of mass conservation is obtained.

$$\frac{\partial \rho}{\partial t} = -\frac{\partial (\rho s)}{\partial x} \tag{4.2}$$

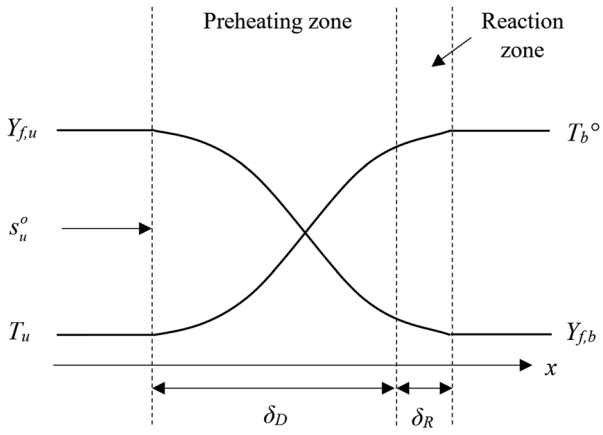


Fig. 4.1 Representation of the unidimensional premixed flame

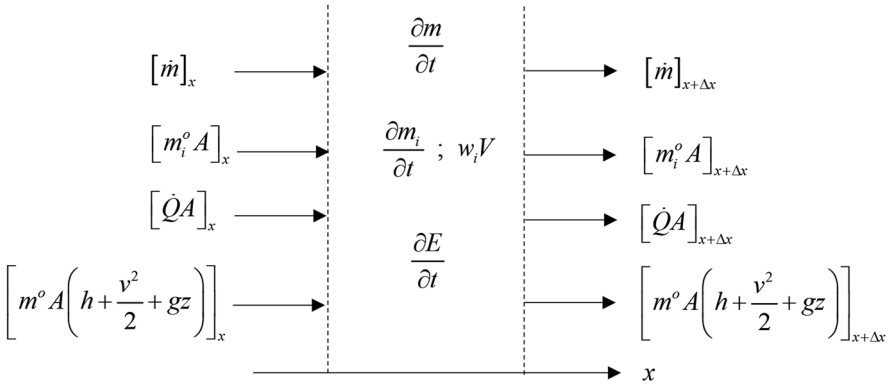


Fig. 4.2 Representation of the flame as a control volume

In steady state, the derivative as a function of time on the left-hand side of Eq. (4.2) is zero. Thus, the mass conservation equation establishes the relationship between the velocity of the reactants ($s_u°$) and the velocity of the products ($s_b°$).

$$m^\circ = \rho_u s_u^\circ = \rho_b s_b^\circ \tag{4.3}$$

The laminar flame speed is defined as the speed at which the reactants approach the flame ($s_u°$). This speed is a property of the fuel–air mixture. However, the application of the mass conservation equation shows that the variable that is conserved is the mass flux ($m°$), which is directly related to $s_u°$ through the reactants' density (ρ_u). Note that, in other chapters of the book the mass flux will be presented as m'', on the present case, the $m°$ is used to emphasize that this mass flux is related to the laminar flame speed ($s_u°$), which is a property of the mixture.

The mass fraction of the fuel in the reactants is represented as $Y_{f,u}$. In a lean mixture, the deficient species is the fuel, whereas in a rich mixture, the deficient species is the oxidant. The heat released per unit mass of the deficient species is denoted by q_C. If the control volume is adiabatic and the heat capacity of the gas mixture is constant (c_p), the conservation of energy between the inlet and outlet flows gives the following results for the adiabatic flame temperature ($T_b°$).

$$T_b^\circ = T_u + \frac{q_C Y_{d,u}}{c_p} \tag{4.4}$$

There are two heat fluxes in the preheating zone. These heat fluxes are due to convection and conduction heat transfer and must balance each other within the preheating zone because the chemical reaction is not yet active. The convection heat flux is given by $m°dh \sim m°c_p(T_b°–T_u)$, while the conduction heat flux is given by $\lambda(dT/dx) \sim \lambda(T_b°–T_u)/\delta_D$. Thus, the following relationship between the mass flow, the thermal conductivity, the heat capacity, and the length of the preheating zone is obtained.

$$m^{\circ} \sim \frac{\lambda}{\delta_D c_p} \tag{4.5}$$

The heat released in the reaction zone per unit area is given by $q_C w_b \delta_R$. Here w_b is the reaction rate in kg/m^3·s. It is assumed that the reaction takes place at a temperature T_b°, which would correspond to a very thin reaction zone. The convective heat flux is comparatively lower than the conductive heat flux in the reaction zone [11]. Therefore, the heat released by chemical reactions is balanced by the conduction heat flux from the reaction zone $\lambda(dT/dx) \sim \lambda(T_b^{\circ}–T_u)/\delta_R$. This produces the following expression.

$$\delta_R \sim \sqrt{\frac{\lambda\left(T_b^{\circ} - T_u\right)}{q_C w_b}} = \sqrt{\frac{\left(\lambda/c_p\right)}{w_b}} \tag{4.6}$$

The mass flux of the deficient species is given by the product $Y_{d,u} m^{\circ}$. According to the mass conservation law, the chemical reaction must consume the mass flux $w_b \delta_R$. Therefore, the following expression relates the mass flux to the reaction rate.

$$m^{\circ} \sim w_b \delta_R \tag{4.7}$$

The relationship between the lengths of the preheating and reaction zones results from the expressions in Eqs. (4.5) and (4.7).

$$\delta_R \delta_D \sim \frac{\lambda/c_p}{w_b} \tag{4.8}$$

The heat transfer between the preheating zone and the reaction zone must be continuous. This means that the temperature derivatives at the boundary separating the two zones must be equal to guarantee this continuity, that is, $(dT/dx)_R = (dT/dx)_D$. If the temperature at the boundary of the zones is T_{fr}, the derivatives can be approximated by $(dT/dx)_D \sim (T_b^{\circ}–T_u)/\delta_D$ and $(dT/dx)_R \sim (T_b^{\circ}–T_{fr})/\delta_R$. Thus, the following expression is obtained.

$$\frac{\delta_D}{\delta_R} \sim \frac{T_b^{\circ} - T_u}{T_b^{\circ} - T_{fr}} \tag{4.9}$$

The denominator on the right-hand side of Eq. (4.9) can be approximated by writing the derivative of the reaction rate as a function of temperature,

$$\left(\frac{dw}{dT}\right)_b \sim \frac{w_b}{T_b^{\circ} - T_{fr}}$$

Then,

$$T_b^\circ - T_{fr} \sim \left[\frac{1}{w_b}\left(\frac{dw}{dT}\right)_b\right]^{-1} \tag{4.10}$$

The reaction rate is related to temperature by an exponential function, $w_b \sim \exp(-T_d/T_b^\circ)$. Here the activation temperature (T_a) is determined by the quotient between the activation energy and the universal gas constant $T_a = E_a/R$. Therefore, the term involving the derivative of the reaction rate is expressed as shown below.

$$\frac{1}{w_b}\left(\frac{dw}{dT}\right)_b = \left[\frac{d}{dT}(\ln w_b)\right] \sim \frac{T_a}{\left(T_b^\circ\right)^2} \tag{4.11}$$

The expression obtained in Eq. (4.11) is used to express the difference $T_b^\circ - T_{fr}$ in Eq. (4.10). Next, the relationship between the characteristic lengths of the two zones of the one-dimensional flame is determined.

$$\frac{\delta_D}{\delta_R} \sim Ze \tag{4.12}$$

Ze is the Zel'dovich number and is given by the following relationship between the flame temperature, activation temperature, and reactants mixture temperature.

$$Ze = \frac{\left(T_b^\circ - T_u\right)T_a}{\left(T_b^\circ\right)^2} \tag{4.13}$$

The Zel'dovich number measures the sensitivity of the reaction with respect to temperature. Moreover, analyzing the terms of Eq. (4.13), it is possible to see that the activation energy of the fuel also plays an important role in determining the Zel'dovich number and that the value of *Ze* affects the reaction rate, as will be shown next. Multiplying Eq. (4.5) by Eq. (4.7) and substituting the relationship obtained in Eq. (4.12), the result in Eq. (4.14) is obtained. In addition, substituting Eq. (4.5) into Eq. (4.14) produces the expression given in Eq. (4.15).

$$m^\circ \sim \sqrt{\frac{\left(\lambda/c_p\right)w_b}{Ze}} \tag{4.14}$$

$$\delta_D \sim \sqrt{\frac{\left(\lambda/c_p\right)Ze}{w_b}} \tag{4.15}$$

The reaction rate (w) can also be given in kmol/m^3·s for an irreversible elementary reaction by using the Arrhenius-type relation. Using the perfect gas equation, it is possible to relate the concentration of the mixture to the pressure, $p = cRT$. Thus, if the reaction order is n, it is possible to relate the reaction rate to the pressure, this results in Eq. (4.16).

Table 4.1 Relevant premixed flames interactions

Variable	Increase of T	Increase of p (with $n > 1$)
w_b	Increases	Increases
m°	Increases	Increases
s_u°	Increases	Increases/decreases
δ_D, δ_R	Decreases	Decreases

$$\omega_b = Be^{-T_a/T_b^\circ} \prod_{i=1}^{N} c_i^{\nu_i'} \sim p^n e^{-T_a/T_b^\circ} \tag{4.16}$$

The above expression implies that for a global order $n > 1$, the increase in mixture pressure increases the reaction rate. This is related to the increase in the frequency of molecular collisions at higher pressures. The influence of pressure on mass flux is obtained by recognizing that $w_b \sim \omega_b$ and substituting Eq. (4.16) into Eq. (4.14).

$$m^\circ \sim p^{n/2} \left(\lambda/c_p\right)^{1/2} e^{-T_a/2T_b^\circ} Ze^{-1/2} \tag{4.17}$$

Expressions for the laminar flame speed can be obtained directly by realizing that $m^\circ = \rho_u s_u^\circ$ and that by the equation of state of the ideal gases $\rho_u \sim p$. The result of this substitution is shown in Eq. (4.18).

$$s_u^\circ \sim p^{n/2-1} \left(\lambda/c_p\right)^{1/2} e^{-T_a/2T_b^\circ} Ze^{-1/2} \tag{4.18}$$

Note that when the reaction rate increases, the mass flux rate also increases. The effect of the Zel'dovich number (Ze) is that the mass flux decreases when it increases. In other words, a higher Ze number means a lower reaction rate. Table 4.1 summarizes the interactions between the different variables related to flame propagation.

In the case of increasing temperature, it is noticed from Eqs. (4.16)–(4.18) that the reaction rate, the mass flux rate, and the laminar flame speed increase. However, the characteristic lengths of the reaction and preheating zones decrease, as shown by the inspection of Eqs. (4.6) and (4.15), respectively. The effect of pressure is analogous on the reaction rate, on the mass flux rate, and on the characteristic lengths. On the other hand, the effect of pressure on the laminar flame speed depends on the reaction order.

In this section, a phenomenological analysis of the propagation of a laminar flame has been presented. In the following section, the conservation equations for chemical species and energy will be established.

4.3 Energy and Chemical Species Conservation Equations

The equations governing the phenomenon of the propagation of a one-dimensional flame are the equations of conservation of mass, conservation of chemical species, and conservation of energy. Although the conservation equations were derived in general form in Chap. 3, it is interesting to perform a more specific derivation to illustrate the interactions taking place in the flame. The equation for the conservation of mass was derived in Sect. 4.2. The species mass conservation equation is obtained by considering Fig. 4.2 again. There, the mass flux of the chemical species (m_i°), the accumulation of mass of the species (dm_i/dt), and the reaction rate of the species ($w_i V$) must be balanced.

$$\frac{dm_i}{dt} = \left[m_i^\circ A\right]_x - \left[m_i^\circ A\right]_{x+\Delta x} + w_i V \tag{4.19}$$

It is possible to express the volume as $V = A\Delta x$ and the mass of the species as $m_i = \rho Y_i A \Delta x$. Therefore, dividing by $A\Delta x$ and taking the limit for $\Delta x \to 0$, the following expression is obtained.

$$\frac{d(\rho Y_i)}{dt} = -\frac{dm_i^\circ}{dx} + w_i \tag{4.20}$$

According to Eq. (4.20), if the process takes place in steady state, it follows that the variation of the mass flux of the chemical species in x is equal to the volumetric reaction rate of the chemical species ($dm_i^\circ/dx = w_i$).

Now it is interesting to consider the mass flux of species (m_i°), which can be represented by two components. The first component refers to the bulk mass flux of the mixture (m°) that transports the chemical species. The second component refers to the mass flux due to mass diffusion of the species from areas where it has a higher concentration to areas where it has a lower concentration. This representation of m_i° is given mathematically in Eq. (4.21), where Fick's law has been used to represent the mass flux of the species due to mass diffusion.

$$m_i^\circ = Y_i m^\circ - \rho D_{ij} \frac{dY_i}{dx} \tag{4.21}$$

Assuming steady state and substituting Eq. (4.21) into Eq. (4.20), the species mass conservation equation is determined, as shown in Eq. (4.22). In addition, it could be assumed that the mass diffusivity (D_{ij}) is constant.

$$m^\circ \frac{dY_i}{dx} - \frac{d}{dx}\left(\rho D_{ij} \frac{dY_i}{dx}\right) = w_i \tag{4.22}$$

The energy conservation equation is obtained by considering the energy flows shown in Fig. 4.2. The term on the left-hand side of Eq. (4.23) represents the energy accumulation in the control volume. The terms on the right-hand side represent enthalpy, kinetic energy, potential energy, and heat fluxes associated with the process, respectively.

$$\frac{dE}{dt} = \left[\dot{Q}A\right]_x - \left[\dot{Q}A\right]_{x+\Delta x} - \dot{W}_t + m^{\circ}A\left[\left(h + \frac{v^2}{2} + gz\right)_x - \left(h + \frac{v^2}{2} + gz\right)_{x+\Delta x}\right] \tag{4.23}$$

Assuming steady state, negligible changes in kinetic and potential energies, and that no work is done, then, several terms on Eq. (4.23) are dropped. Afterwards, the equation thus obtained is divided by $A\Delta x$, and the limit is taken for $\Delta x \to 0$, yielding Eq. (4.24).

$$-\frac{d\dot{Q}}{dx} = m^{\circ}\frac{dh}{dx} \tag{4.24}$$

The heat flux is related to various heat transfer mechanisms such as conduction, mass diffusion, and thermal radiation. Heat transfer by radiation and mass diffusion is considered negligible. Therefore, it is assumed that the heat flux is caused only by conduction.

$$\dot{Q} = -\lambda\frac{dT}{dx} \tag{4.25}$$

Since the mass flux of the mixture is constant, it is possible to write $m^{\circ}dh/dx = d(m^{\circ}h)/dx$. Then, the enthalpy flux of the mixture is represented by the sum of the enthalpy fluxes of the individual species, as shown in Eq. (4.26).

$$\frac{d(m^{\circ}h)}{dx} = \frac{d}{dx}\sum m_i^{\circ}h_i = \sum m_i^{\circ}\frac{dh_i}{dx} + \sum h_i\frac{dm_i^{\circ}}{dx} \tag{4.26}$$

The process takes place in a steady state. Therefore, $dm_i^{\circ}/dx = w_i$. Since the sensible enthalpies are small compared to the formation enthalpies, the second term on the right-hand side of Eq. (4.26) can be represented by Eq. (4.27).

$$\sum h_i\frac{dm_i^{\circ}}{dx} = \sum\left(h_{f,i}^{\circ} + \Delta h_i\right)w_i \sim \sum h_{f,i}^{\circ}w_i \tag{4.27}$$

The specific enthalpy of a species (h_i) results from the sum of its sensible enthalpy (Δh_i) and its enthalpy of formation ($h_{f,i}^{\circ}$). The values of $h_{f,i}^{\circ}$ are constant as soon as the reference temperature is chosen, that is, $h_{f,i}^{\circ}$ does not vary along the x-axis. Replacing the expression $h_i = \Delta h_i + h_{f,i}^{\circ}$ and performing the derivations, the following is obtained for the first term on the right-hand side of Eq. (4.26).

$$\sum m_i^{\circ}\frac{dh_i}{dx} = \sum Y_i m^{\circ}\frac{d\Delta h_i}{dx} + \sum\frac{1}{h_i}\frac{d\Delta h_i}{dx}\left(-\rho D_{ij}\frac{dY_i}{dx}h_i\right) \tag{4.28}$$

The second term on the right-hand side of Eq. (4.28) represents the product of the normalized change in the species' sensible enthalpy and the species' diffusion heat transfer. Originally, it was assumed that the heat transfer by mass diffusion was

negligible, so this term is considered to be equal to zero. Substituting Eqs. (4.25), (4.27), and (4.28) into Eq. (4.24) gives the energy conservation equation in steady state.

$$\sum Y_i m^\circ \frac{d\Delta h_i}{dx} + \frac{d}{dx}\left(-\lambda \frac{dT}{dx}\right) = -\sum h_{f,i}^\circ w_i \tag{4.29}$$

Finally, assuming that the heat capacities at constant pressure of the species ($c_{p,i}$) are constant with respect to temperature, that the mass diffusivities are constant and equal to D, and that the thermal conductivity of the mixture, λ, is also constant, Eqs. (4.22) and (4.29) simplify, yielding the following expressions.

$$m^o \frac{dY_i}{dx} - \rho D \frac{d^2 Y_i}{dx^2} = w_i \tag{4.30}$$

$$m^\circ c_p \frac{dT}{dx} - \lambda \frac{d^2 T}{dx^2} = -\sum h_{f,i}^\circ w_i \tag{4.31}$$

The expressions given in Eqs. (4.3), (4.30), and (4.31) represent a theoretical model for a one-dimensional laminar premixed flame in steady state and with the simplifications assumed in their derivation. For example, it was assumed that the heat capacities are constant and equal and that the heat transfer by radiation and mass diffusion is negligible, among other assumptions.

4.4 The One-Step Reaction Representation

In a one-step chemical kinetics reaction, the fuel and the oxidant are assumed to react irreversibly with each other to form the products. If the fuel is represented by F, the oxidant by O, and the products by P, the one-step reaction is written, as shown by Eq. (4.32).

$$\nu'_F F + \nu'_O O \rightarrow \nu''_P P \tag{4.32}$$

Equation (4.32) assumes that some of the reactants are in deficit, as explained earlier. Following Law [11], the mass ratio of chemical species "i" reacting per unit mass of fuel ($\sigma_{i,f}$) is defined by Eq. (4.33). Note that, in a complete combustion reaction, the air-to-fuel ratio (AF) is a parameter with similar characteristics to $\sigma_{i,f}$. Here, $m_{i,r}$ and $m_{f,r}$ are the mass of the species i and the mass of the fuel reacted in the process, respectively.

$$\sigma_{i,f} = \frac{m_{i,r}}{m_{f,r}} = \frac{W_i\left(\nu''_i - \nu'_i\right)}{W_f\left(\nu''_f - \nu'_f\right)} \tag{4.33}$$

The heat released by the combustion reaction per unit mass of fuel consumed in the reaction ($q_{C,f}$) is the ratio of the sum of the enthalpies of formation of products

and reactants to the mass of fuel reacted. Since the products are formed from several chemical species, the following expression can be written in general terms for N species in the mixture of products and reactants.

$$q_{C,f} = \frac{\sum_{i=1}^{N} h_{f,i}^{\circ} W_i \left(\nu_i'' - \nu_i'\right)}{W_f \left(\nu_f'' - \nu_f'\right)} = \sum_{i=1}^{N} h_{f,i}^{\circ} \sigma_{i,f} \tag{4.34}$$

Equation (4.34) gives the energy released in the formation of combustion products divided by the mass of fuel consumed in the reaction. It is important to emphasize that the form given in Eq. (4.34) is valid in both the first and second cases, when the fuel is the deficient species and when the oxidant is the deficient species in the reactants.

The mass fraction of the fuel in the mixture is represented by Y_f. Therefore, it is possible to relate the mass fraction of any chemical species to Y_f by the following relationship.

$$Y_i = \frac{m_{i,r}}{m_T} = \frac{m_{i,r}}{m_{f,r}/Y_f} = \sigma_{i,f} Y_f \tag{4.35}$$

Recall that the reaction rate of a chemical species (ω_i) is directly related to the reaction rate of the *i*th reaction ($\omega_{R,i}$) via the expression given in Eq. (4.36). Moreover, the reaction rate is given by a modified Arrhenius-type relation as shown in Eq. (4.37).

$$\frac{\omega_i}{\nu_i'' - \nu_i'} = \omega_{R,i} \tag{4.36}$$

$$\omega_{R,i} = BT^{\alpha} e^{-E_a/RT} \prod_{i=1}^{N} (c_i)^{\nu_i'} \tag{4.37}$$

The reaction rate of the species involved in the one-step irreversible reaction is given in Eq. (4.38), which represents the reaction rate of species "*i*" in kg/m^3 s units.

$$w_i = W_i \left(\nu_i'' - \nu_i'\right) BT^{\alpha} (c_O)^{\nu_O'} (c_F)^{\nu_F'} e^{-T_a/T} \tag{4.38}$$

In the case of a one-step irreversible reaction, it is customary to write the reaction rate taking into account the fuel as a deficient species. However, this procedure will be presented in the next section.

4.5 Dimensionless Conservation Equations

The dimensionless form of the conservation equations is very useful for representing the flame propagation process in general. Dimensionless means that the problem variables are normalized with respect to characteristic values of parameters that are relevant to the problem in question. In the case of the propagation of a one-dimensional laminar flame, the dimensionless representation will be performed as described below.

(a) The species mass fraction is already dimensionless. However, it can be weighted by using the initial mass fraction of the chemical species, $Y_{i,u}$, as the characteristic value.

$$\varphi_i = \frac{Y_i}{Y_{i,u}} = \frac{Y_i}{\sigma_{i,f} Y_{f,u}} \tag{4.39}$$

(b) The temperature is made dimensionless by using the temperature rise resulting from the combustion of one unit mass of fuel consumed in the reaction as the characteristic value. Using Eq. (4.4), the following expression is therefore obtained.

$$\theta = \frac{T}{q_{C,f} Y_{f,u} / c_p} \tag{4.40}$$

(c) The distance is made dimensionless by using the flame thickness as the characteristic length, which in turn is approximated by the preheating zone length, according to Eq. (4.5), yielding the following expression.

$$\xi = \frac{m^\circ x}{\lambda / c_p} \tag{4.41}$$

It is now possible to express the temperature derivatives as a function of the coordinate in terms of the new dimensionless variables by applying the chain rule as shown below.

$$\frac{dT}{dx} = \frac{dT}{d\xi}\frac{d\xi}{dx} = \frac{m^\circ}{\lambda / c_p}\frac{dT}{d\xi} \Rightarrow \frac{dT}{dx} = \frac{m^\circ}{\lambda / c_p}\frac{q_{C,f} Y_{i,u}}{c_p}\frac{d\theta}{d\xi} \tag{4.42}$$

$$\frac{d^2T}{dx^2} = \frac{d}{d\xi}\left[\frac{m^\circ}{\lambda / c_p}\frac{q_{C,f} Y_{i,u}}{c_p}\frac{d\theta}{d\xi}\right]\frac{d\xi}{dx} \Rightarrow \frac{d^2T}{dx^2} = \left(\frac{m^\circ}{\lambda / c_p}\right)^2 \frac{q_{C,f} Y_{i,u}}{c_p}\frac{d^2\theta}{d\xi^2} \tag{4.43}$$

Similarly, the derivatives of the mass fraction of a species as a function of the coordinate are obtained by applying the chain rule.

$$\frac{dY}{dx} = \frac{dY}{d\xi}\frac{d\xi}{dx} = \frac{m^{\circ}}{\lambda/c_p}\frac{dY}{d\xi} \Rightarrow \frac{dY}{dx} = \frac{m^{\circ}}{\lambda/c_p} Y_{i,u}\frac{d\varphi_i}{d\xi} \tag{4.44}$$

$$\frac{d^2Y}{dx^2} = \frac{d}{d\xi}\left[\frac{m^{\circ}}{\lambda/c_p} Y_{i,u}\frac{d\varphi_i}{d\xi}\right]\frac{d\xi}{dx} \Rightarrow \frac{d^2Y}{dx^2} = \left(\frac{m^{\circ}}{\lambda/c_p}\right)^2 Y_{i,u}\frac{d^2\varphi_i}{d\xi^2} \tag{4.45}$$

Considering the one-step irreversible reaction shown in Eq. (4.32), the sum of the right-hand side of Eq. (4.31) can be represented as shown in Eq. (4.46). Furthermore, noting that $w_i/w_f = \sigma_{i,f}$, and recognizing that the definition of $q_{C,f}$ appears on the right-hand side, the following is obtained.

$$\sum h_{f,i}^{\circ} w_i = \left(h_{f,f}^{\circ} + h_{f,o}^{\circ}\sigma_{P,o} + h_{f,p}^{\circ}\sigma_{P,f}\right) w_f = q_{C,f} w_f \tag{4.46}$$

Substituting Eqs. (4.42), (4.43), and (4.46) into Eqs. (4.30) and (4.31) and making the necessary simplifications, the expressions given below are obtained.

$$\frac{d\varphi_f}{d\xi} - \left(\frac{\rho D}{\lambda/c_p}\right)\frac{d^2\varphi_f}{d\xi^2} = \frac{\lambda/c_p}{Y_{f,u}(m^{\circ})^2} w_f \tag{4.47}$$

$$\frac{d\theta}{d\xi} - \frac{d^2\theta}{d\xi^2} = -\frac{\lambda/c_p}{Y_{f,u}(m^{\circ})^2} w_f \tag{4.48}$$

The form of the above equations reveals the presence of the Lewis number (Le) in Eq. (4.47), which establishes the balance between thermal diffusivity and mass diffusivity.

$$Le = \frac{\lambda/\rho c_p}{D} = \frac{\alpha}{D} \tag{4.49}$$

An expression for w_f in terms of the mass fraction of the fuel can also be obtained. Note that the concentration of a perfect gas is given by $c_i = pi/RT$, where the partial pressure of the gas, p_i, can be related to the mass fraction and the total pressure by $Y_i = x_i W_i/W$. Thus, remembering that the density of the ideal gas is given by $\rho = pW/RT$, it is possible to write the following expression for the concentration of the species.

$$c_i = \frac{\rho Y_i}{W_i} \tag{4.50}$$

Replacing Eq. (4.50) into Eq. (4.38), multiplying by the appropriate terms to get the dimensionless variables θ and φ, rearranging, and assuming that all the fuel is consumed in the reaction, the expression given in Eq. (4.51) is thus obtained.

$$w_f = -Y_{f,u} B_C \theta^{\alpha - \nu'_F - \nu'_O} \varphi_o{}^{\nu'_O} \varphi_f{}^{\nu'_F} e^{-\theta_a/\theta} \tag{4.51}$$

where

$$B_C = B\left(\frac{W_f \nu'_f}{Y_{f,u}}\right)\left(\frac{q_{C,f} Y_{f,u}}{c_p}\right)^{\alpha}\left(\sigma_{o,f}\frac{pc_p}{Rq_{C,f}}\frac{W}{W_o}\right)^{\nu'_O}\left(\frac{pc_p}{Rq_{C,f}}\frac{W}{W_f}\right)^{\nu'_F} \tag{4.52}$$

The density-weighted collision rate of the reaction, B_C, has units of mass per second and per unit volume, kg/m^3 s. The characteristic diffusion time can be defined as a function of a characteristic length (δ_0) and the thermal diffusivity, as shown below.

$$t_{\text{dif}} = \frac{\delta_0^2}{\alpha} = \frac{\delta_0^2}{\lambda/c_p \rho_o} \tag{4.53}$$

In the above expression, ρ_o is a reference density. Similarly, the characteristic time of collisions can be given as a function ρ_o and B_C as follows.

$$t_{\text{col}} = \frac{\rho_o}{B_C} \tag{4.54}$$

Therefore, the Damköhler number of collisions (Da_C) is defined as the ratio between the characteristic diffusion time and the characteristic collision time, as shown in Eq. (4.54), where Eq. (4.5) was used to express Da_C in terms of the mass flow rate.

$$Da_C = \frac{\delta_0^2/\left(\lambda/\rho_o c_p\right)}{\rho_o/B_C} = \frac{\lambda/c_p}{\left(m^\circ\right)^2} B_C \tag{4.55}$$

Substituting Eqs. (4.49), (4.51), and (4.55) into Eqs. (4.47) and (4.48) yields the following dimensionless conservation equations.

$$\frac{d\varphi_f}{d\xi} - \frac{1}{Le}\frac{d^2\varphi_f}{d\xi^2} = -Da_C \theta^{\alpha - \nu'_F - \nu'_O} \varphi_o{}^{\nu'_O} \varphi_f{}^{\nu'_F} e^{-\theta_a/\theta} \tag{4.56}$$

$$\frac{d\theta}{d\xi} - \frac{d^2\theta}{d\xi^2} = Da_C \theta^{\alpha - \nu'_F - \nu'_O} \varphi_o{}^{\nu'_O} \varphi_f{}^{\nu'_F} e^{-\theta_a/\theta} \tag{4.57}$$

The procedure for determining the collision rate and the Damköhler number requires, in addition to the properties of the mixture at a given temperature, knowledge of the pre-exponential factor B and the experimental laminar flame speed. The Zel'dovich number also requires a value for the activation energy. Before proceeding to discuss the solution of Eqs. (4.56) and (4.57), the dimensionless numbers commonly associated with premixed flames are further discussed.

4.6 Characteristic Dimensionless Numbers of Premixed Flames

In Sect. 4.2, the properties of the premixed flame were discussed, and the Zel'dovich number was defined. Therefore, in Sect. 4.5, the conservation equations were made dimensionless, and the Lewis number and Damköhler number of collisions were defined. In this section, these dimensionless numbers are discussed in more detail. Moreover, some calculation examples will be also presented.

4.6.1 The Zel'dovich Number

The Zel'dovich number is a measure of the influence of temperature and activation energy on the combustion reaction. The Zel'dovich number is rewritten, showing this time the direct dependence on the activation energy of the reaction.

$$Ze = \frac{E_a\left(T_b^\circ - T_u\right)}{R\left(T_b^\circ\right)^2} \tag{4.58}$$

For a one-step reaction, such as that given in Eq. (4.32), the activation energy is directly related to the mass flux, $m^\circ \sim \exp(-E_a/2RT_b^\circ)$. Thus, isolating E_a, Eq. (4.59) is obtained. The Eq. (4.59) allows for the determination of the activation energy for a one-step reaction from data of m° and T_b°. Similar expressions to obtain the activation energy, E_a, can be found in the work by Zel'dovich et al. [51], Law [11], Williams [36], and Liñán and Williams [52], among others.

$$\frac{E_a}{2R} = -\frac{\partial \ln\ m^\circ}{\partial\left(1/T_b^\circ\right)} \tag{4.59}$$

The factor 2 in Eq. (4.59) arises because m° and w_b are connected by a square root, as shown in Eq. (4.14).

Example 4.1
Determine the Zel'dovich number of ethanol–air mixtures with different equivalence ratios.

Solution
In order to determine the Zel'dovich number of a fuel–air mixture, it is necessary to know the pressure and temperature conditions to determine the activation energy in the context of a one-step reaction. The experimental data obtained by Dirrenberger et al. [53] for the laminar flame speed of ethanol–air mixtures at 298 K and 100 kPa are adopted for this problem. It should be noted that both the adiabatic flame temperature, T_b, and the density, ρ_u, were calculated using the Cantera software [54]. The data required for the calculations are listed in Table 4.2.

The value of the activation energy is obtained by applying least squares to the straight line shown in Fig. 4.3(a), from which it follows that $E_a = 105.7$ kJ/mol. It is

Table 4.2 Data necessary to obtain the activation energy (Ea) at an initial pressure and temperature of 100 kPa and 298 K

ϕ	s_u° (cm/s)	ρ_u (kg/m^3)	m° (kg/m^2 s)	T_b (K)
Experimental data by Dirrenberger et al. [53]		Calculated data with Cantera software [54]		
0.60	17.51	1.19	0.21	1705.8
0.66	22.37	1.20	0.27	1811.3
0.71	26.79	1.20	0.32	1894.6
0.77	31.43	1.20	0.38	1988.7
0.82	34.39	1.20	0.41	2061.2
0.88	37.13	1.20	0.45	2138.4
0.93	39.67	1.21	0.48	2192.4
0.99	41.35	1.21	0.50	2238.2
1.04	42.19	1.21	0.51	2252.7
1.10	43.24	1.21	0.52	2237.0
1.16	42.61	1.22	0.52	2199.9
1.21	41.57	1.22	0.51	2163.9

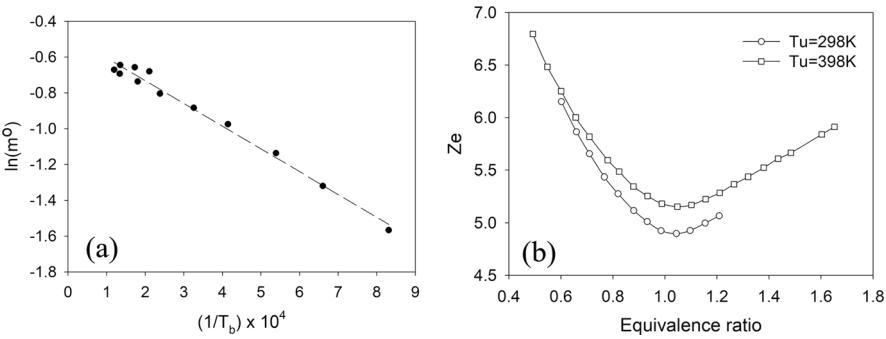


Fig. 4.3 Determination of the activation energy (E_a) by the method of least squares with the data from Table 4.1

therefore possible to calculate the Zel'dovich number for each of the equivalence ratios listed in Table 4.2. The calculation for the stoichiometric mixture is shown below, and the other values can be seen in Fig. 4.3(b).

$$Ze = \frac{105.7(2238.2 - 298)}{8.3144 \times 10^{-3} \times 2238.2^2} = 4.92$$

Figure 4.3(b) also depicts the Zel'dovich numbers obtained for the experimental values of Dirrenberger et al. [53] for ethanol–air mixtures at 100 kPa and 398 K. It can be observed that in this case, the value of the Zel'dovich number increases with the initial temperature, which is due to the fact that the activation energy obtained with the data at 398 K is equal to $E_a = 119.2$ kJ/mol, which is higher than the E_a obtained at 298 K. Note that the E_a depends on the m°, which is increasing with

increasing T_u. However, the increase in the experimental $m°$ does not seem to be enough to produce a decrease in the activation energy.

The Zel'dovich number (Ze) can also be interpreted as a measure of the thickness of the reaction zone relative to the preheating zone, as shown in Eq. (4.12). A very large Zel'dovich number means that the reaction zone is very thin relative to the preheating zone. Considering the Zel'dovich number as a measure of the sensitivity of the reaction with respect to temperature means that a mixture with a high Ze value is very sensitive to temperature fluctuations in the reaction zone, which may be caused by heat losses, for example.

4.6.2 The Lewis Number

The Lewis number determines the relationship between the thermal diffusivity and the mass diffusivity of a mixture. Equivalently, the Lewis number can be interpreted as the ratio between the Schmidt number ($Sc = \nu/D$) and the Prandtl number ($Pr = \nu/\alpha$), as shown below.

$$Le = \frac{\alpha}{D} = \frac{Sc}{Pr} \tag{4.60}$$

In a flame, heat transfer and mass transfer occur simultaneously, which makes the Lewis number important. When the Lewis number is equal to one, heat diffusion and mass diffusion are in balance. On a local scale, this means that the total enthalpy is conserved locally. The three possible cases are illustrated and described in Fig. 4.4.

The Lewis number varies with the temperature and composition of the mixture. However, it is possible to characterize a fuel–air mixture by its initial Lewis number. For a mixture of perfect gases, it is relatively easy to determine the specific mass, the heat capacity at constant pressure, the thermal conductivity, and consequently the thermal diffusivity. The greatest difficulty, however, lies in determining the mass diffusivity.

The Lewis numbers of the fuel (Le_F) and the oxidant (Le_O) are determined from the thermal conductivity of the mixture and the mass diffusivity of the fuel and the oxidant, respectively. Therefore, according to Matalon [55], the effective Lewis number follows from Eq. (4.50) for lean and rich mixtures.

$$Le_{\text{eff}} = \begin{cases} \dfrac{Le_O + (1-\Phi)Le_F}{2-\Phi} & \text{Lean mixture} \\ \dfrac{Le_O + Le_F}{2} & \text{Stoichiometric mixture} \\ \dfrac{Le_F + (1+\Phi)Le_O}{2+\Phi} & \text{Rich mixture} \end{cases} \tag{4.61}$$

where

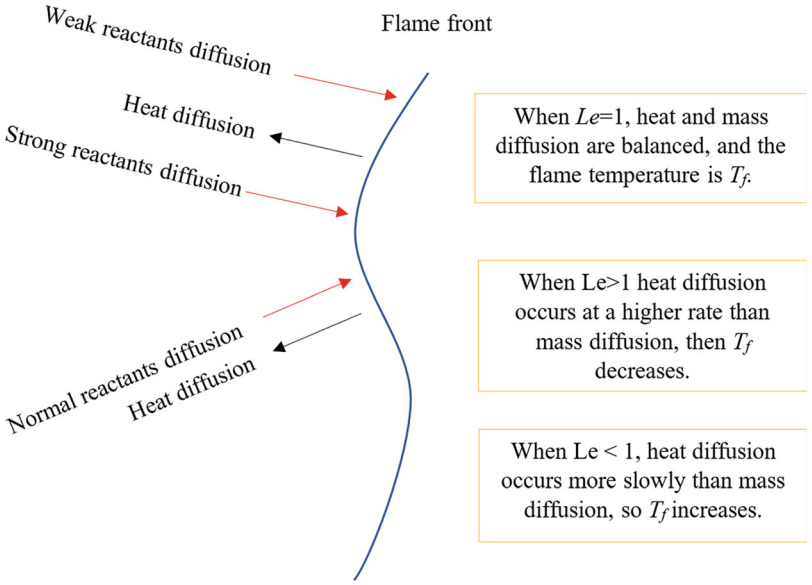


Fig. 4.4 Representation of heat and mass diffusion at the flame front

$$\Phi \sim Ze(\phi - 1) \tag{4.62}$$

For very lean or very rich mixtures, the effective Lewis number can be considered equal to the Lewis number of the deficient reactant. Qiao et al. [56] proposed to determine the Lewis number based on the thermal diffusivity of the mixture and the binary mass diffusivity between the fuel and the oxidant (oxygen).

$$Le_{\text{eff}} = \frac{\alpha_{\text{mis}}}{D_{F-O_2}} \tag{4.63}$$

According to Gárzon et al. [57], the Lewis number can be determined with Eq. (4.63), but taking into account the mass diffusivity of the fuel in the mixture ($D_{F\text{-mix}}$), as shown in Eq. (4.64).

$$D_{F-\text{mix}} = (1 - Y_{F,\text{mix}}) \left(\sum_{k=1}^{N} \frac{x_k}{D_{F,k}} \right)^{-1} \tag{4.64}$$

The effective Lewis number of fuel mixtures can be determined by various approximations as presented by Bouvet et al. [58].

$$Le_{\text{eff},H} = 1 + \frac{q_{F_1}(Le_{F_1} - 1) + q_{F_2}(Le_{F_2} - 1)}{q_{F_1} + q_{F_2}} \quad \text{Based on heat released} \tag{4.65}$$

$$Le_{\text{eff},V} = x_1 Le_{F_1} + x_2 Le_{F_2} \quad \text{Based on volume concentration} \tag{4.66}$$

$$Le_{\text{eff},D} = \frac{\alpha_{\text{mis}}}{x_1 D_{F_1-N_2} + x_2 D_{F_2-N_2}} \quad \text{Based on mass diffusion} \tag{4.67}$$

x_i is the mole fraction of each fuel (F_i) in the mixture, and q_{F_i} is the heat released in the combustion reaction of the fuel. Note that the results obtained with Eqs. (4.65)–(4.67) are not the same. According to the evaluation of Bouvet et al. [58], the Lewis number based on volumetric concentration ($Le_{\text{eff},V}$) gives the best qualitative results. Dinkelacker et al. [59], on the other hand, obtained better results with the diffusion-based Lewis number ($Le_{\text{eff},D}$).

Example 4.2
Determine the effective Lewis number of mixtures involving ethanol and air.

Solution
To determine the effective Lewis number, we need the properties of the mixture, such as the thermal diffusivity and the mass diffusivity of the fuel in the oxidant. These calculations can be done directly, as shown in the literature [7, 60]. However, to avoid very extensive calculations, we will use the properties determined with the software Cantera [54].

Thermodynamic and transport properties are shown in Table 4.3. It can be seen that the mass diffusivities used in the calculations were those of the fuel and oxidant with respect to nitrogen. In view of the fact that combustion takes place with air, this simplification was assumed.

The Lewis numbers given in Table 4.2 and Fig. 4.5 correspond to the initial mixtures before the combustion process. It should be noted that during the combustion process, the temperature and composition of the mixture change. Therefore, the Lewis number also changes during combustion. The Zel'dovich number is also presented in Table 4.3, as determined in Example 4.1.

The value of Le_{eff} for stoichiometric and lean ethanol–air mixtures at 358 K and 100 kPa are all greater than one ($Le_{\text{eff}} \geq 1$). This means that preferential diffusion is present in these mixtures. Moreover, thermal diffusion dominates over mass diffusion. The balance between mass and thermal diffusion, or equidiffusion condition, is approached only for rich mixtures ($\phi \geq 1.3$). These results are in accordance with those presented in the literature [57].

In the experimental work by Nzinga et al. [61, 62], the propagation and extinction of flames involving mixtures of natural gas, hydrogen, helium, and air were studied. The mixtures were characterized by their Lewis and Zel'dovich numbers, which were calculated by using simulated data and by considering methane as representative of natural gas in the calculations. The results of such calculations can be observed in Table 4.4.

Table 4.3 Determination of the effective Lewis number of ethanol–air mixtures at 358 K and 100 kPa according to Eq. (4.61), as recommended by Matalon [55]

ϕ	ρ (kg/m^3)	c_p (J/kg K)	k (W/m K)	α (m^2/s) $\times 10^5$	$D_{F\text{-}N2}$ (m^2/s) $\times 10^5$	$D_{O2\text{-}N2}$ (m^2/s) $\times 10^5$	Le_F	Le_O	Ze	Φ	Le_{eff}
0.60	0.993	1054.7	0.0298	2.845	1.6915	2.8702	1.68	0.99	6.34	−2.536	1.53
0.70	0.996	1060.5	0.0297	2.810	1.6910	2.8679	1.66	0.98	5.87	−1.761	1.48
0.80	1.000	1066.1	0.0296	2.777	1.6906	2.8657	1.64	0.97	5.48	−1.097	1.43
0.90	1.004	1071.7	0.0295	2.744	1.6901	2.8634	1.62	0.96	5.26	−0.526	1.36
1.00	1.007	1077.2	0.0294	2.713	1.6897	2.8613	1.61	0.95	5.18	0.000	1.28
1.10	1.011	1082.5	0.0293	2.682	1.6893	2.8592	1.59	0.94	5.18	0.518	1.20
1.20	1.014	1087.8	0.0293	2.652	1.6889	2.8571	1.57	0.93	5.31	1.061	1.14
1.30	1.018	1092.9	0.0292	2.624	1.6885	2.8551	1.55	0.92	5.47	1.640	1.09
1.40	1.021	1097.9	0.0291	2.596	1.6881	2.8531	1.54	0.91	5.63	2.253	1.06
1.50	1.024	1102.9	0.0290	2.569	1.6877	2.8511	1.52	0.90	5.72	2.862	1.03
1.60	1.028	1107.7	0.0289	2.542	1.6873	2.8492	1.51	0.89	5.89	3.536	1.00

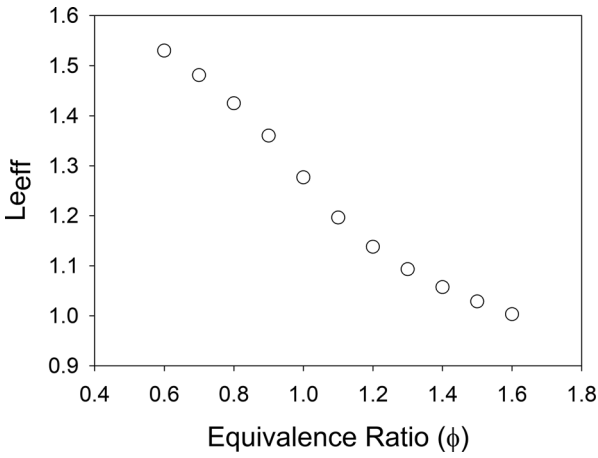


Fig. 4.5 Effective Lewis numbers of ethanol–air mixtures at 358 K and 100 kPa

Table 4.4 Effective Lewis numbers and Zel'dovich numbers for mixtures of methane, hydrogen, and helium at 40, 60, and 80 kPa and at room temperature. Data from Nzinga et al. [61, 62]

Mixtures	Le_{eff}	Ze		
		40 kPa	60 kPa	80 kPa
[50% CH_4 + 50% H_2] + air	0.60	6.09	6.61	6.89
[75% CH_4 + 25% H_2] + air	0.75	7.36	7.85	7.95
[15% CH_4 + 85% H_2] + 20% He + air	0.81	4.24	4.65	4.19
[75% CH_4 + 25% H_2] + 10% He + air	0.93	7.57	8.10	8.20
[15% CH_4 + 85% H_2] + 30% He + air	0.99	4.45	4.91	5.27
[100% CH_4] + air	1.09	8.20	8.58	8.60
[90% CH_4 + 10% H_2] + 10% He + air	1.14	8.17	8.44	8.62
[100% CH_4] + 10% He + air	1.36	8.45	8.66	8.77

Nzinga et al. [61, 62] calculated the Lewis number of the pure fuels by applying the Eq. (4.63) proposed by Qiao et al. [56]. Afterwards, the effective Lewis number of the mixture was calculated by applying the Eq. (4.66) proposed by Bouvet et al. [58]. Note that the Lewis number is not significantly affected by the change in initial pressure of the mixture. On the other hand, the Zel'dovich number is clearly affected by initial pressure.

Taking as reference the Le_{eff} and Ze values of natural gas in air, it is observed that the addition of hydrogen tends to decrease the Le_{eff} and Ze; on the other hand, the addition of helium increases these values. The presence of hydrogen in the mixture will produce preferential diffusion, in which the mass diffusion dominates over the thermal diffusion. The helium increases the Le_{eff} because it has a high thermal conductivity in comparison to the other gases present in the mixtures.

4.6.3 The Damköhler Number

The Damköhler number changes its meaning, depending on the time scale used for its definition. For example, the Damköhler number of collisions is defined as shown below. Note that Eqs. (4.68) and (4.55) are equivalent.

$$Da_C = \frac{\text{Diffusion characteristic time}}{\text{Collisions characteristic time}} = \frac{\lambda/c_p}{(m^\circ)^2} B_C \tag{4.68}$$

If the flow is non-diffusive, the Da_C can be defined as follows.

$$Da_C = \frac{\text{Flow characteristic time}}{\text{Collisions characteristic time}} = \frac{\delta_o/v_o}{\rho_o} B_C \tag{4.69}$$

The Damköhler number of reactions is determined by taking into account the effectiveness of the collisions, which is characterized by the Arrhenius factor. Therefore, the following relationship can be established.

$$Da = \frac{\text{Diffusion characteristic time}}{\text{Reaction characteristic time}} = Da_C \exp\left(-\frac{E_a}{RT_b^\circ}\right) \tag{4.70}$$

Here, the characteristic time scale may be the diffusion or flow time divided by the reaction time. In terms of the combustion phenomenon, the Damköhler number of reactions is more significant than the Damköhler number of collisions. This is because not every collision causes a reaction.

Example 4.3
Determine the collision rate and the Damköhler number of collisions and reactions for a stoichiometric ethanol–air mixture.

Solution
A combustion reaction is assumed in order to visualize the process globally and to obtain some parameters necessary for the calculation.

$$C_2H_5OH + \frac{v_{ar}^s}{\phi}(O_2 + 3.76N_2) \rightarrow \begin{array}{c} 2(1-\beta)CO_2 + 2\beta CO + 3(1-\gamma)H_2O \\ +3\gamma H_2 + \chi\left(\frac{v_{ar}^s}{\phi} - v_{ar}^s\right)O_2 + 3,76\frac{v_{ar}^s}{\phi}N_2 \end{array}$$

The factors β and γ represent the fraction of the number of moles of carbon and hydrogen that form incomplete combustion products such as CO and H_2, respectively. Similarly, the factor χ takes the value of one when the mixture is lean or stoichiometric and zero when the reaction is rich. A stoichiometric mixture and complete combustion are assumed. Ethanol requires 3 moles of oxygen for stoichiometric combustion ($v_{ar}^s = 3$). From Eq. (4.33) the following values are obtained.

$$\sigma_{O_2} = \frac{3 \times 32}{1 \times 46} = 2.087; \quad \sigma_{CO_2} = -\frac{2 \times 44}{1 \times 46} = -1.913; \quad \sigma_{H_2O} = -\frac{3 \times 18}{1 \times 46} = -1.173$$

The value of $q_{C,f}$ is determined from Eq. (4.34). The enthalpy of formation of ethanol is $\overline{h}^{\circ}_{f,C_2H_5OH} = -234$ kJ/mol [4]. The enthalpies of formation of carbon dioxide and water vapor are $\overline{h}^{\circ}_{f,CO_2} = -393.522$ kJ/mol and $\overline{h}^{\circ}_{f,H_2O} = -241.826$ kJ/mol, respectively [2].

The enthalpy of formation of the products must be determined, taking into account the mole fractions of CO_2 and H_2O. For the stoichiometric reaction involving 1 kmol of ethanol, the total number of moles of the products is 16.28 kmol, and the total mass of the products is 457.84 kg. Thus, the formation enthalpies on a mass basis are given below.

$$h^{\circ}_{f,C_2H_5O} = \frac{-234 \times 10^3}{46} = -5086.957 \text{ kJ/kg}$$

$$h^{\circ}_{f,CO_2} = \frac{-393.522 \times 10^3}{44} = -8943.681 \text{ kJ/kg}$$

$$h^{\circ}_{f,H_2O} = \frac{-241.826 \times 10^3}{18} = -13,434.778 \text{ kJ/kg}$$

The heat released in this case is equivalent to the lower heating value of ethanol because the complete combustion of a stoichiometric mixture was assumed in the calculations.

$$q_{C,f} = -5086.957 + 8943.681 \times 1.913 + 13,434.778 \times 1.173 = 27,781.3 \text{ kJ/kg}$$

The initial mass fraction of the fuel is

$$Y_{f,u} = \frac{1 \times 46}{1 \times 46 + 3 \times 32 + 3 \times 3.76 \times 28} = 0.1004$$

The total number of moles of the reactants is 15.28 kmol. Therefore, the molecular mass of the reactants is also determined as shown below.

$$W = \frac{1}{15.28}46 + \frac{3}{15.28}32 + \frac{11,28}{15.28}28 = 29.97 \text{ kg/kmol}$$

The density-weighted collision rate (B_C) of the reaction can now be determined. We have assumed a pressure of 101.3 kPa and the value $R = 8.3144$ kJ/kmol K for the universal gas constant. The heat capacity of air at 1200 K is used for the calculations, $c_p = 1.175$ kJ/kg K [63]. The pre-exponential factor (B) and the exponents m and n were determined by Westbrook and Dryer [35] for various hydrocarbons. The value $B = 1.5 \times 10^{12}$ $(\text{mol/cm}^3)^{1-m-n}$ s^{-1} is used for ethanol.

At this point we are only interested in the order of magnitude, so we assume $m = n = 0$. We also assume that $\alpha = 1$.

$$B_C \sim \left(\frac{1.5 \times 10^{12} \times 10^6}{10^3}\right)\left(\frac{46 \times 1}{0.1004}\right)\left(\frac{27781.3 \times 0.1004}{1.175}\right)^1$$
$$\left(\frac{2.087 \times 101.3 \times 1.175}{8.3144 \times 27781.3}\frac{29.97}{32}\right)^3\left(\frac{101.3 \times 1.175}{8.3144 \times 27781.3}\frac{29.97}{46}\right)^1$$

Solving the above expression yields

$$B_C \sim 5.61 \times 10^8 \text{ kg/m}^3 \text{ s}$$

To determine the Damköhler number, we have assumed the thermal conductivity of air at 1200 K and obtained $\lambda = 0.0763$ W/m K. The mass flow rate is determined from the experimental laminar flame speed for the ethanol–air mixture, $s_u^\circ = 0.4135$ m/s, obtained by Dirrenberger et al. [53] at 298 K and 101 kPa. Under these pressure and temperature conditions, the specific mass is $\rho_u = 1.21$ kg/m^3. Therefore, we can calculate the Damköhler number of collisions from Eq. (4.68).

$$Da_C = \frac{0.0763/1175}{(0.4135 \times 1.21)^2}\left(5.61 \times 10^8\right) = 1.46 \times 10^5$$

The Damköhler number of reactions results from Eq. (4.59), with $E_a/R = 15{,}098$ K [35] and $T_b^\circ = 2354$ K.

$$Da = 1.46 \times 10^5 \exp\left(-\frac{15098}{2354}\right) = 2.39 \times 10^2$$

In the calculations presented in this example, it was necessary to assume a value for the pre-exponential factor and the activation energy of a one-step reaction for the combustion of the ethanol–air mixture. These values were taken from the work of Westbrook and Dryer [19] and are also reproduced in the book by Turns [7].

4.7 Analytical Solution of the Governing Equations

The equations governing the phenomenon can now be solved. However, to solve the equations, the so-called cold boundary condition must be taken into account. This difficulty arises because when $\xi \to -\infty$, the derivatives of θ and φ must physically have a value equal to zero; nevertheless, the right-hand side of Eqs. (4.56) and (4.57) have finite values even for this boundary condition. To avoid this inconsistency, it is therefore assumed that in the preheating zone, the reaction rate is zero until a distance known as ξ_{ig} is reached. This is equivalent to assume the existence of an ignition temperature.

4.7.1 The Energy Conservation Equation in the Preheating Zone

To solve the energy conservation equation in the preheating zone, it is first assumed that the term on the right-hand side of Eq. (4.56) is zero. Therefore, the equation to be solved is as follows.

$$\frac{d\theta_p}{d\xi} - \frac{d^2\theta_p}{d\xi^2} = 0 \tag{4.71}$$

Subject to boundary conditions,

$$\theta_p(\xi \to -\infty) = \theta_u; \quad \theta_p(\xi_{ig}) = \theta_{ig} \tag{4.72}$$

The subscript p in Eq. (4.71) means that the dimensionless temperature corresponds to the preheating zone. Integrating Eq. (4.71) once, the following relationship is obtained.

$$\frac{d\theta_p}{d\xi} = \theta_p + C_1 \tag{4.73}$$

Integrating the above equation,

$$\ln(\theta_p + C_1) = \xi + C_2 \tag{4.74}$$

Applying the boundary conditions given in Eq. (4.72), the integration constants given in Eq. (4.75) are obtained.

$$C_1 = -\theta_u; \quad C_2 = \ln(\theta_{ig} - \theta_u) - \xi_{ig} \tag{4.75}$$

Substituting the above results into Eq. (4.74) and rearranging, the equation of the dimensionless temperature for the preheating zone is obtained.

$$\frac{\theta_p - \theta_u}{\theta_{ig} - \theta_u} = \exp(\xi - \xi_{ig}) \tag{4.76}$$

4.7.2 The Combined Temperature and Concentration Equation

A combined differential equation for temperature and concentration can be obtained by the addition of Eqs. (4.56) and (4.57). The resulting differential equation is shown in Eq. (4.77). Thus, a relationship can be established between the dimensionless mass fraction and the dimensionless temperature by solving Eq. (4.77).

$$\frac{d^2\theta}{d\xi^2} + \frac{1}{Le}\frac{d^2\varphi_f}{d\xi^2} - \frac{d(\theta + \varphi_f)}{d\xi} = 0 \tag{4.77}$$

The boundary conditions for Eq. (4.77) are as follows.

$$\text{For } \xi \to -\infty, \quad \theta = \theta_u; \quad \varphi_f = 1; \quad \frac{d\theta}{d\xi} = \frac{d\varphi_f}{d\xi} = 0 \tag{4.78}$$

$$\text{For } \xi \to +\infty, \quad \theta = \theta_b; \quad \varphi_f = 0; \quad \frac{d\theta}{d\xi} = \frac{d\varphi_f}{d\xi} = 0 \tag{4.79}$$

The Lewis number is considered constant, and Eq. (4.77) is integrated once.

$$\frac{d\theta}{d\xi} + \frac{1}{Le}\frac{d\varphi_f}{d\xi} = \theta + \varphi_f + C_1 \tag{4.80}$$

Note that, in order to proceed with the integration, it is usually assumed that the Lewis number is equal to one [7, 11]. In this case, however, it is only assumed that the Lewis number is constant, which is somehow less restrictive. By rearranging the above expression,

$$\frac{d\left(\theta + \varphi_f\right)}{Le\left(\theta + \varphi_f + C_1\right)} = d\xi$$

Integrating,

$$\theta + \varphi_f + C_1 = C_2 \exp(Le{\cdot}\xi) \tag{4.81}$$

Applying the first boundary condition ($\xi \to -\infty$), it is possible to determine the first constant of integration because the right-hand side of Eq. (4.81) goes to zero. Applying the second boundary condition ($\xi \to +\infty$), it is found that the second constant of integration must be zero to ensure that the solution takes on a finite value.

$$C_1 = -(\theta_u + 1); \quad C_2 = 0 \tag{4.82}$$

Substituting and rearranging the integration constants in Eq. (4.81), the expression given in Eq. (4.83) is obtained. Note that the relationship $\theta_b = 1 + \theta_u$, which was given in Eq. (4.40), has also been used.

$$\varphi_f = \theta_b - \theta \tag{4.83}$$

The expression obtained in Eq. (4.83) is of great importance because it allows for writing the scaled mass fraction in terms of the dimensionless temperature.

4.7.3 The Energy Conservation Equation in the Reaction Zone

Having obtained the expression which relates temperature and concentration in Eq. (4.83), now Eqs. (4.56) and (4.57) can be decoupled. This is because φ_f can now be expressed in terms of θ. On the other hand, from Eqs. (4.35) and (4.39), it is noted that $\varphi_0 = \varphi_f$. Thus, substituting Eq. (4.83) into Eq. (4.57) and using the fact that $\varphi_0 = \varphi_f$, the following is obtained.

$$\frac{d\theta}{d\xi} - \frac{d^2\theta}{d\xi^2} = Da_C \theta^{\alpha - \nu'_F - \nu'_O} (\theta_b - \theta)^{\nu'_F + \nu'_O} e^{-\theta_a/\theta} \tag{4.84}$$

It is assumed that there is an exponent β that satisfies the following condition: $Le \cdot (\theta_b - \theta)^{\beta - \nu'_F - \nu'_O} = \theta^{\alpha - \nu'_F - \nu'_O}$. Therefore, the a simplified form of Eq. (4.84) can be written as shown below.

$$\frac{d\theta}{d\xi} - \frac{d^2\theta}{d\xi^2} = LeDa_C (\theta_b - \theta)^{\beta} e^{-\theta_a/\theta} \tag{4.85}$$

The Lewis number was introduced into Eq. (4.85) because this dimensionless number has an important physical meaning. Moreover, the fact that the exponent β has to be determined was used to introduce the Lewis number. Assuming that the dominant terms in the reaction zone are those related to diffusion and chemical reaction [11], the Eq. (4.85) can be simplified. A demonstration of this consideration is given in the next subsection. It should be accepted without demonstration for now.

$$\frac{d^2\theta_r}{d\xi^2} = -LeDa_C (\theta_b - \theta_r)^{\beta} e^{-\theta_a/\theta_r} \tag{4.86}$$

The subscript "r" in Eq. (4.86) means that the dimensionless temperature corresponds to the reaction zone. The differential equation is subject to the boundary conditions shown below.

$$\theta(\xi_{fr}) = \theta_{fr}; \quad \theta(\xi_b) = \theta_b; \quad \left(\frac{d\theta_r}{d\xi}\right)_{\xi = \xi_b} = 0 \tag{4.87}$$

The first boundary condition specifies that the dimensionless temperature at the boundary between the preheating zone and the reaction zone has a value equal to θ_{fr}. The second boundary condition specifies that the dimensionless temperature derivative at the end of the reaction zone is equal to zero. It can be observed that the second derivative on the left-hand side of Eq. (4.86) can be conveniently written as shown in Eq. (4.88).

$$\frac{d^2\theta_r}{d\xi^2} = \frac{1}{2}\frac{d}{d\theta_r}\left(\frac{d\theta_r}{d\xi}\right)^2 \tag{4.88}$$

Replacing Eq. (4.88) into Eq. (4.86) and applying an integration between θ_b and θ_r on the right-hand side and from 0 to $(\partial\theta_r/\partial\xi)$ on the left-hand side.

$$\left(\frac{d\theta_r}{d\xi}\right)^2 = -2LeDa_C \int_{\theta_b}^{\theta_r} (\theta_b - \theta_r)^{\beta} e^{-\theta_a/\theta_r} d\theta_r \tag{4.89}$$

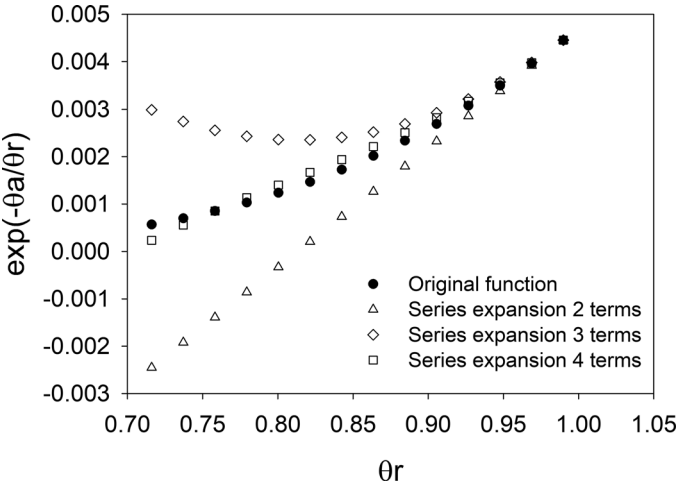


Fig. 4.6 Comparison of the exponential function $e^{-\theta a/\theta r}$ with its series approximation truncated with 2, 3, and 4 terms

The exponential function will be approximated by a series around θ_b, truncated at the fourth term. The result of this approximation is shown in Eq. (4.91), with the signs of the terms $(\theta_b - \theta_r)$ changed to facilitate multiplication by the term $(\theta_b - \theta_r)^\beta$ in Eq. (4.89).

$$e^{-\theta_a/\theta_r} = e^{-\theta_a/\theta_b}\left[1 - (\theta_b - \theta_r)\frac{\theta_a}{\theta_b^2} + \frac{1}{2}(\theta_b - \theta_r)^2\left(\frac{\theta_a}{\theta_b^2}\right)^2 - \frac{1}{6}(\theta_b - \theta_r)^3\left(\frac{\theta_a}{\theta_b^2}\right)^3\right]$$

From the definition of the Zel'dovich number given in Eq. (4.58) and of the Arrhenius number, the following can be deduced.

$$Ze = \frac{E_a\left(T_b^\circ - T_u\right)}{R\left(T_b^\circ\right)^2} = \frac{\theta_a}{\theta_b^2} \quad \text{e} \quad Ar = \frac{E_a}{RT_b^\circ} = \frac{\theta_a}{\theta_b} \tag{4.90}$$

Therefore, it is possible to write the following series expansion.

$$e^{-\theta_a/\theta_r} = e^{-Ar}\left[1 - (\theta_b - \theta_r)Ze + \frac{1}{2}(\theta_b - \theta_r)^2 Ze^2 - \frac{1}{6}(\theta_b - \theta_r)^3 Ze^3\right] \tag{4.91}$$

Data from previous examples were used to test the validity of the series approximation, the results are shown in Fig. 4.6. Note that for an stoichiometric ethanol–air mixture, the adiabatic flame temperature is $T_b^\circ = 2353.7$ K when the initial conditions are 101.3 kPa and 298 K. This value corresponds to a $\theta_b = 1.14$. The value of $\theta_{fr} = 0.84$ corresponds to $T_{fr} = 1726.8$ K.

Figure 4.6 shows that for $0.84 \leq \theta_r \leq \theta_b$, the series approximation with three terms is a good approximation to the exponential function $e^{-\theta a/\theta r}$. Since the function

is being analyzed in the reaction zone, the fact that the approximation is valid from $\theta_r = 0.84$ is compatible with the assumption of the existence of an ignition temperature θ_{fr}. It is also possible to write Eq. (4.91) in general form, as shown in Eq. (4.92), where n is the number of terms in the series.

$$e^{-\theta_a/\theta_r} = e^{-Ar} \sum_{n=1}^{3} \frac{(-1)^{n-1}}{(n-1)!} (\theta_b - \theta_r)^{n-1} Ze^{n-1} \quad \text{for} \quad 0.84 \le \theta_r \le \theta_b \tag{4.92}$$

Returning to the integral given in Eq. (4.89) and adopting the variable $(\theta_b - \theta_r)$, it is seen that $d(\theta_b - \theta_r) = -\mathrm{d}\theta_r$. Therefore, changing the limits of integration, the following expression is obtained.

$$\int_{\theta_b}^{\theta_r} (\theta_b - \theta_r)^{\beta} e^{-\theta_a/\theta_r} d\theta_r = -e^{-Ar} (\theta_b - \theta_r)^{\beta+1} \sum_{n=1}^{3} \times \frac{1}{(n-1)!} \frac{(-1)^{n-1}}{\beta + n} Ze^{n-1} (\theta_b - \theta_r)^{n-1} \tag{4.93}$$

Substituting the result of the integral from Eq. (4.93) into the original expression of Eq. (4.89) and rearranging conveniently,

$$\frac{-d(\theta_b - \theta_r)}{(\theta_b - \theta_r)^{\frac{\beta+1}{2}} \sqrt{a + b(\theta_b - \theta_r) + c(\theta_b - \theta_r)^2}} = \sqrt{2LeDa_C} e^{-Ar/2} d\xi \tag{4.94}$$

where

$$a = \frac{1}{\beta + 1}; \quad b = -\frac{1}{\beta + 2} Ze; \quad c = \frac{1}{2(\beta + 3)} Ze^2 \tag{4.95}$$

The integral on the left-hand side of Eq. (4.94) can be simplified by performing the substitution $u = (\theta_b - \theta_r)$ and assuming that $\beta \sim 1$.

$$\int_{u*}^{u} \frac{-du}{u\sqrt{a + bu + cu^2}} = \sqrt{2LeDa_C} e^{-Ar/2} \int_{\xi_b}^{\xi} d\xi \tag{4.96}$$

The expression on the left-hand side in Eq. (4.96) has a known indefinite integral available in the literature (see Sect. 5.4.13 in Zwillinger et al. [64]). The result is denoted as Ω_r.

$$\Omega_r(u) = \frac{1}{\sqrt{a}} \ln \left(\frac{2\sqrt{a^2 + ab(\theta_b - \theta_r) + ac(\theta_b - \theta_r)^2} + b(\theta_b - \theta_r) + 2a}{(\theta_b - \theta_r)} \right) \tag{4.97}$$

Therefore, substituting Eq. (4.97) into Eq. (4.96), the result is an implicit function of the dimensionless temperature, as shown in Eq. (4.98).

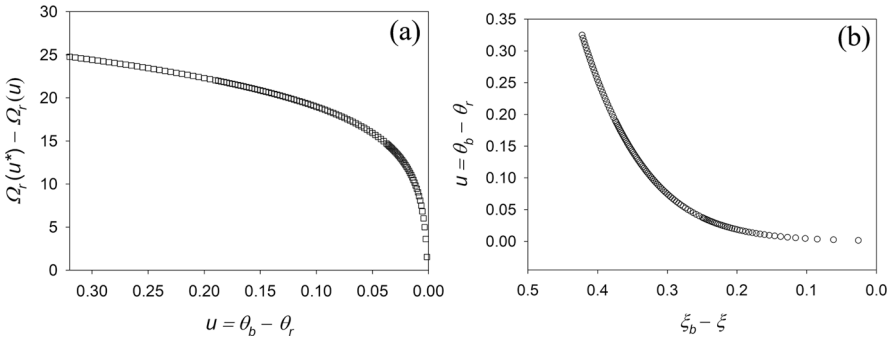


Fig. 4.7 Analysis of the reaction zone. (**a**) Integral $\Omega r(u^*) - \Omega_r(u)$ as a function of the difference $\theta_b - \theta_r$. (**b**) Dimensionless coordinate $\xi_b - \xi$ as a function of the difference $\theta_b - \theta_r$

$$\xi_b - \xi = \frac{\Omega_r(u*) - \Omega_r(u)}{\sqrt{2LeDa_C}e^{-\theta_a/2\theta_b}} \tag{4.98}$$

Note that the function Ω_r is not defined in $u^* = 0$. Since it is expected that $(\theta_b - \theta_r)$ approaches zero when $(\xi_b - \xi)$ also approaches zero, then, the values of u^* must be small in order to obtain a solution. The definition of the dimensionless distance in Eq. (4.41) uses the mass flow (m°) and the ratio (λ/c_p) to make the value of the x-coordinate dimensionless. Note that the value of λ/c_p increases with increasing temperature and that m° is constant. Therefore, the difference $\xi_b - \xi$ must be greater than zero, under the condition that the value of x is bounded within the reaction zone. This last condition is also compatible with the assumption of a very thin reaction zone. On the other hand, given that ξ_b is a constant and ξ must increase with increasing x, it would be expected that the difference, $\xi_b - \xi$, increases with increasing temperature difference.

Assuming that the Da_C and Le values obtained in the previous examples for a stoichiometric mixture of ethanol and air are constant and that $u^* = 0.001$, the graphs shown in Fig. 4.7 were constructed. It can be observed that as the difference $\theta_b - \theta_r$ decreases, the value of the integral, $\Omega_r(u^*) - \Omega_r(u)$, decreases. This was to be expected since the integral Ω includes the variable part of the exponential Arrhenius term expanded in a series. Also note that for the lower values of the dimensionless temperature difference, $\theta_b - \theta_r$, the dimensionless coordinate difference, $\xi_b - \xi$, decreases. Therefore, if the value of θ_r is closer to that of θ_b, the location within the reaction zone is closer to ξ_b.

It is even possible to write an expression for the laminar flame speed. To do this, Eq. (4.89) is considered once again, and it is found that the temperature gradients at the boundary between the reaction zone and the preheating zone must be the same. The ignition temperature is considered to be very close to the temperature at the end of the reaction zone ($\theta_{fr} \sim \theta_b$), and it is found that $\theta_b - \theta_u = 1$. Furthermore, integrating Eq. (4.71) once, between θ_u and θ_{fr}, the following result is obtained.

$$\left(\frac{d\theta_r}{d\xi}\right)_{\xi=\xi_{fr}} = \left(\frac{d\theta_p}{d\xi}\right)_{\xi=\xi_{fr}} = \theta_{fr} - \theta_u \sim \theta_b - \theta_u = 1 \tag{4.99}$$

By substituting Eq. (4.99) into Eq. (4.89) and using the definition of the Damköhler number of collisions, it is possible to obtain an expression for the laminar flame speed.

$$\left(\rho_u s_u^\circ\right)^2 = 2Le\left(\lambda/c_p\right)B_C\left[-\int_{\theta_b}^{\theta_r}(\theta_b - \theta_r)^\beta e^{-\theta_a/\theta_r}d\theta_r\right] \tag{4.100}$$

In this case, it is not desirable to obtain an expression relating dimensionless temperature to distance. Therefore, the integral on the right-hand side of Eq. (4.100) is treated differently. First, the exponent θ_a/θ_r is written, exploiting the definition of $u = (\theta_b - \theta_r)$, and a series expansion is applied, as shown below.

$$\frac{\theta_a}{\theta_r} = \frac{\theta_a}{\theta_b - u} = \frac{Ar}{1 - u/\theta_b} \sim Ar\left[1 + \frac{u}{\theta_b} + \left(\frac{u}{\theta_b}\right)^2 + \cdots\right] \tag{4.101}$$

If only the first two terms of the series are assumed, the integral becomes

$$\int_{\theta_b}^{\theta_r}(\theta_b - \theta_r)^\beta e^{-\theta_a/\theta_r}d\theta_r = -e^{-Ar}\int_0^u u^\beta e^{-Ze\cdot u}du \tag{4.102}$$

Assuming again that $\beta \sim 1$ and integrating by parts, the following is obtained.

$$\int_0^u u^\beta e^{-ze\cdot u}du = \frac{1}{Ze^2}\left[1 - (1 + Ze\cdot u)e^{-Ze\cdot u}\right] \tag{4.103}$$

Substituting into Eq. (4.100) and rearranging, an expression for the laminar flame speed is obtained.

$$s_u^\circ = \frac{\sqrt{2Le\left(\lambda/c_p\right)B_C e^{-Ar}}}{\rho_u Ze}\left[1 - (1 + Ze\cdot u)e^{-Ze\cdot u}\right]^{1/2} \tag{4.104}$$

Since the boundary between the preheating zone and the reaction zone is being analyzed, the term between brackets on the right-hand side of Eq. (4.103) must be considered ~1, because the term involving the chemical reaction must be much less than unity ($e^{-Ze\cdot u} \sim 0$) [11]. This consideration can be evaluated by taking the value $Ze = 5.18$ obtained for a stoichiometric mixture of ethanol and air and assuming a value of $u = 0.35$ at the boundary. Therefore, the term in parentheses is equal to 0.55. The larger the product $Ze \times u$, the closer the value of the term inside the parentheses

is to 1. Thus, applying such consideration, the expression shown in Eq. (4.105) is obtained. Note that, the relationship (2Le ×Da)/$Ze^2 = 1$ can be obtained from the expression below.

$$s_u^\circ = \frac{\sqrt{2Le(\lambda/c_p)B_C e^{-Ar}}}{\rho_u Ze} \tag{4.105}$$

Example 4.4
Determine the laminar flame speed of a stoichiometric mixture of ethanol and air at atmospheric pressure and an initial temperature of 298 K.

Solution
To solve this problem we will use the results from the previous examples. In Example 4.2, it was found that for a stoichiometric mixture of ethanol and air, $Le_{\text{eff}} = 1.28$ and $Ze = 5.18$. In Example 4.3, $B_C = 5.61 \times 10^8$ kg/m^3 s was determined, and a specific mass $\rho_u = 1.21$ kg/m^3 was assumed. In Example 4.1, the activation energy was determined to be $E_a = 105.7$ kJ/mol. The adiabatic flame temperature for the stoichiometric mixture of ethanol and air is $T_b^\circ = 2353.7$ K. The Arrhenius number results from Eq. (4.90).

$$Ar = \frac{105.7}{8.3144 \times 10^{-3} \times 2353.7} = 5.401$$

We assume the value of $\lambda/c_p = 6.494 \times 10^{-5}$, which corresponds to air at 300 K, and substitute the values into Eq. (4.104).

$$s_u^\circ = \frac{\sqrt{2 \times 1.28 \times 6.494 \times 10^{-5} \times 5.61 \times 10^8 \times e^{-5.401}}}{1.21 \times 5.18} = 3.27 \text{ m/s}$$

The result obtained with the simplified theory is eight times larger than the experimental laminar flame speed m/s determined by Dirrenberger et al. [53] at 298 K and 101 kPa. In the derivation of Eq. (4.105), the following term was disregarded.

$$\left[1 - (1 + Ze{\cdot}u)e^{-Ze{\cdot}u}\right] = \left[1 - (1 + 5.18 \times 0.1)e^{-5.18 \times 0.1}\right] = 0.096$$

The value of 0.096 was determined by taking into account $u = 0.1$. Introducing this term into the calculation, as shown by Eq. (4.104), the following result is obtained.

$$s_u^\circ = \frac{\sqrt{2 \times 1.28 \times 6.494 \times 10^{-5} \times 5.61 \times 10^8 \times e^{-5.401}}}{1.21 \times 5.18} 0.096^{0.5} = 1.01 \text{ m/s}$$

Although the result has improved, it is still twice the experimental value. The theory developed makes it possible to evaluate the behavior and interrelation

between different dimensionless numbers, although quantitatively it does not have the necessary accuracy compared to experimental values.

4.7.4 The Energy Conservation Equation in the Reaction Zone

It is important to provide a demonstration regarding neglecting the convective term in the integration of Eq. (4.85) inside the reaction zone. Following Law [11], Eq. (4.85) will be integrated once across the reaction zone, from its left boundary, ξ_{r1}, to its right boundary, ξ_{r2}.

$$\left[\theta(\xi_{r2}) - \theta(\xi_{r1})\right] - \left[\left(\frac{d\theta}{d\xi}\right)\Big|_{\xi_{r2}} - \left(\frac{d\theta}{d\xi}\right)\Big|_{\xi_{r1}}\right] = LeDa_C \int_{\xi_{r1}}^{\xi_{r2}} (\theta_b - \theta)^{\beta} e^{-\theta_a/\theta} d\xi \tag{4.105}$$

Assuming a thin reaction zone, it is possible to note that $\theta(\xi_{r1}) \sim \theta(\xi_{r1}) \sim \theta_b$. This means that the convective term $(d\theta/d\xi)$ is negligible in Eq. (4.85). Moreover, the derivative of temperature at the right-hand side boundary of the reaction zone must be zero for physical consistency.

$$\left(\frac{d\theta}{d\xi}\right)\Big|_{\xi_{r1}} = LeDa_C \int_{\xi_{r1}}^{\xi_{r2}} (\theta_b - \theta)^{\beta} e^{-\theta_a/\theta} d\xi \tag{4.106}$$

Other interesting observations can be obtained. For instance, Eq. (4.71) is integrated once in the preheating zone. Noting that $\theta(\xi \rightarrow -\infty) = \theta_u$ and that $\theta(\xi_{r1}) \sim \theta_b$, the following is obtained.

$$\left(\frac{d\theta_p}{d\xi}\right)\Big|_{\xi_{r1}} - \left(\frac{d\theta_p}{d\xi}\right)\Big|_{\xi \rightarrow -\infty} = \theta_b - \theta_u \tag{4.107}$$

The second term on the left-hand side of the above equation must be zero for physical consistency. Also, note that from Eqs. (4.40) and (4.4), the difference $(\theta_b - \theta_u) = 1$ is obtained. Thus, the derivative of temperature at the left boundary of the reaction zone is equal to unity.

$$\left(\frac{d\theta_p}{d\xi}\right)\Big|_{\xi_{r1}} = \theta_b - \theta_u = 1 \tag{4.108}$$

Substituting Eq. (4.108) into Eq. (4.106) and rearranging,

$$LeDa_C \delta_R \frac{1}{\delta_R} \int_{\xi_{r1}}^{\xi_{r2}} (\theta_b - \theta)^\beta e^{-\theta_a/\theta} d\xi = 1 \tag{4.109}$$

In the above equation, the definition of an average function appeared. Therefore,

$$LeDa_C \left(e^{-\theta_a/\theta} \right)_{av} \left[(\theta_b - \theta)^\beta \right]_{av} \delta_R = LeDa \left[(\theta_b - \theta)^\beta \right]_{av} \delta_R = 1 \tag{4.111}$$

Note that the relationship between the Damköhler numbers of collisions and reactions, Eq. (4.70), has been used. It is now assumed that $\beta \sim 1$ and that $\theta_{av} = (\theta_b + \theta_{fr})/2$. Then, the following is observed by considering Eq. (4.9).

$$\theta_b - \theta_{av} = \frac{1}{2} \frac{T_b - T_{fr}}{T_b - T_u} \sim \frac{1}{2} \frac{\delta_R}{\delta_D} \sim \frac{1}{2} Ze^{-1} \tag{4.112}$$

Introducing a factor b such that $\delta_R/2b \sim Ze^{-1}$, Eq. (4.111) becomes

$$b \frac{LeDa}{Ze^2} = 1 \tag{4.113}$$

In the theoretical development presented by Law [11], the factor in Eq. (4.113) is $b = 2$ for both the Frank–Kamenetskii solution and the asymptotic analysis solution. On the other hand, the factor $b = 1$ was obtained for the integral analysis presented by Law. A factor b = 2 would be obtained from the Eq. (4.105) developed in the previous sub-section. In the present discussion, it is noticed that b is of order $\delta_D/2$.

4.8 A More Detailed Model for the Laminar Premixed Flame

The previous section shows that, although the one-dimensional model with one-step chemical kinetics provides insights into the behavior of the laminar premixed flame propagation, the quantitative results are not well approximated with the experimental ones. Therefore, in this section a one-dimensional model of the laminar flame propagation, considering detailed chemical kinetics and steady-state conditions, will be presented. In order to do so, the conservation equations in rectangular coordinates, developed in Chap. 3, will be invoked. The total mass conservation equation, Eq. (3.76), in steady state and one dimension simplifies to the expression shown below, with the x-direction taken as the dimension of interest.

$$\frac{\partial(\rho u_x)}{\partial x} = 0 \Rightarrow m_x^\circ = \text{constant} \tag{4.114}$$

Note that the only information provided by the total mass conservation equation is that the mass flux in the x-direction is constant. The momentum equation in the x-direction is given in Eqs. (3.78a)–(3.78c). Assuming an inviscid one-dimensional flow, the stress tensor's component of interest will be the σ_{xx}, and this component

will only provide a pressure gradient in the x-direction. Additionally, imposing the steady-state condition, the following expression is obtained.

$$-\frac{dp}{dx}+\rho B_x=\rho u_x\frac{du_x}{dx} \tag{4.115}$$

Further considering that the effect of the body forces, for example, gravity force, and that the variation of kinetic energy are both negligible, the above expression simplifies to Eq. (4.116), which simply states that in such conditions the flame propagates at constant pressure.

$$\frac{dp}{dx}=0\Rightarrow p=\text{constant} \tag{4.116}$$

Therefore, the mass conservation equation and the momentum equation are just providing the information about two constant physical quantities. Now, the energy conservation equation, given in Eq. (3.81), is simplified by considering steady-state conditions, one-dimensional flame propagation, negligible viscous dissipation, no heat generation, no work being done on the control volume, and constant pressure, as suggested by the result obtained in Eq. (4.116).

$$\begin{aligned}\rho u_x\frac{d}{dx}\left(\int_{T_{\text{ref}}}^{T}c_p dT\right)&=\frac{d}{dx}\left(k\frac{dT}{dx}\right)-\sum\left(h_{f,k}^{\circ}\dot{m}_{k,R}\right)\\&-\sum\left(\dot{m}_{k,R}\int_{T_{\text{ref}}}^{T}c_{p,k}dT\right)+\sum\rho D\frac{dY_k}{dx}\frac{d}{dx}\left(\int_{T_{\text{ref}}}^{T}c_{p,k}dT\right)\end{aligned} \tag{4.117}$$

The net production rate of the species k per unit volume ($\dot{m}_{k,R}$) is obtained for each chemical species in the system using the developments from Chap. 2, Eq. (2.73), for a detailed chemical kinetic mechanism. The reaction rate constants, k_{fi} and k_{bi}, are obtained by the methods already discussed in Chap. 2.

$$\dot{m}_{k,R}=W_k\widehat{\omega}_k=W_k\sum_{i=1}^{L}\left(\nu_{ji,P}-\nu_{ji,R}\right)\left[k_{fi}\prod_{j=1}^{N}\left(C_j\right)^{\nu_{ji,R}}-k_{bi}\prod_{j=1}^{N}\left(C_j\right)^{\nu_{ji,P}}\right] \tag{4.118}$$

The species mass conservation equation for the kth species in the mechanism is given in Eq. (3.77). Applying the steady-state and one-dimensional conditions, the following is obtained.

$$\frac{d}{dx}\left(\rho u_x Y_k\right)=+\frac{d}{dx}\left(\rho D\frac{dY_k}{dx}\right)+\dot{m}_{k,R} \tag{4.119}$$

Equation (4.119) must be written for $N-1$ species in the mechanism. The closing is provided by the total mass conservation as shown below.

$$\sum_{j=1}^{N} Y_j = 1 \tag{4.120}$$

The model is now complete. Observe that the mass flux is an eigenvalue for this model. The numerical solution of models similar to this one is discussed in the work by Kee et al. [65] and is also provided in the open-source software Cantera [54]. Therefore, it will not be further discussed here.

Example 4.5
Use the Cantera software to determine the laminar flame velocity of ethanol, hydrogen, and ammonia in air at different equivalence ratios, with an initial temperature and pressure of 298 K and 101.3 kPa, respectively.

Solution
This example will be solved using the Cantera software [54] and its routine "adiabatic_flame" which determines the laminar flame velocity of a one-dimensional steady-state premixed flame in adiabatic conditions and at constant pressure. Note that the equations used by the software are based on the book by Kee et al. [66]. The "adiabatic_flame" routine requires the specification of a chemical kinetic mechanisms. Therefore, the mechanism by Marinov [28] was used for ethanol, the San Diego Mechanism [67] was applied to hydrogen, and the mechanism by Otomo et al. [68] was applied for ammonia. The simulation results will be compared to the experimental data by Dirrenberger et al. [53] in the case of ethanol, by Grosseuvres et al. [69] in the case of hydrogen, and by Lhuillier et al. [70] in the case of ammonia. The simulation results are presented in Fig. 4.8.

It is observed that the simulated and experimental data present a very good agreement, which reflects the results of years of development of the chemical kinetic mechanism and the numerical methods involved in this calculation. The laminar flame speeds of ethanol, hydrogen, and ammonia are in a different scale, as can be seen by the comparison shown in Fig. 4.8, where the vertical scale is logarithmic.

4.9 Closure

This chapter presented the fundamental theory of steady one-dimensional premixed flames. The developments of the previous chapters were integrated here to understand the classical laminar premixed flame theory. The chapter started with a phenomenological discussion, followed by the derivation of the one-dimensional conservation equations applied to the control volume of the flame. Proceeding in this manner helps the reader to understand the nuances of the analysis. Nevertheless, the one-dimensional equations could also have been obtained from the more general expressions derived in Chap. 3. Afterwards, the dimensionless forms of the one-dimensional equations were obtained by introducing characteristic physical quantities. The dimensionless numbers of Lewis, Zel'dovich, and Damköhler were

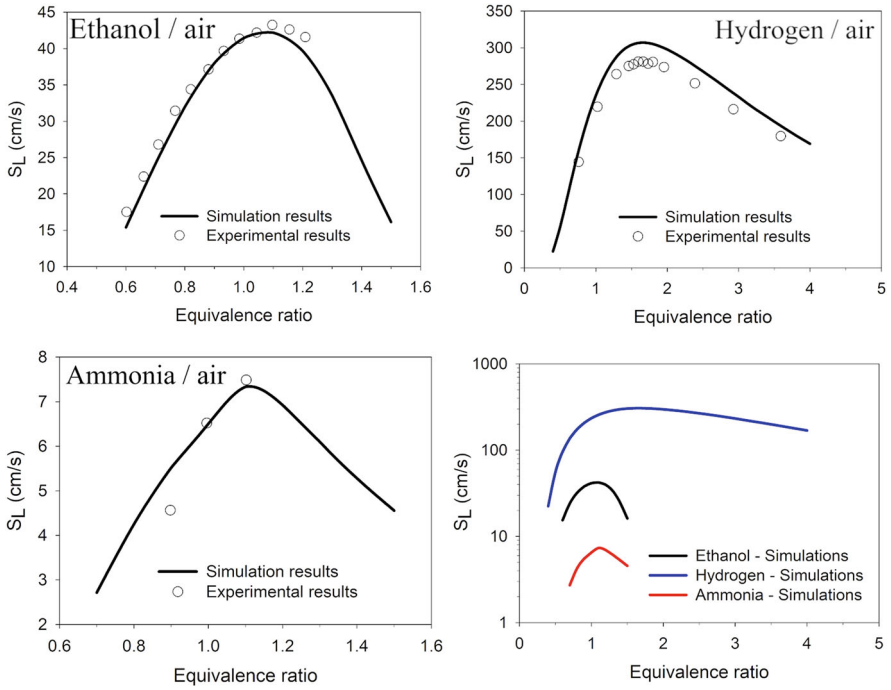


Fig. 4.8 Laminar flame speed of ethanol/air, hydrogen/air, and ammonia/air at 298 K and 100 kPa for different equivalence ratios. Experimental data by Dirrenberger et al. [53] for ethanol, by Grosseuvres et al. [69] for hydrogen, and by Lhuillier et al. [70] for ammonia

introduced and discussed through examples. The chapter also discussed the analytical solution of the dimensionless conservation equations using a Taylor series approximation to simplify the integration of the reaction rate term. Moreover, the chapter also introduced a more sophisticated model that accounts for detailed chemical kinetics and must be solved through numerical simulations.

Flame Dynamics

5

Nomenclature

A	Flame area, m^2
c	Speed of sound, m/s
c_p	Heat capacity at constant pressure, J/kg K
D_i	Mass diffusivity of species i, m^2/s
D	Diameter, m
D_h	Hydraulic diameter, m
E_a	Activation energy, kJ/mol
Ka	Karlovitz number
$Ka°$	Karlovitz number for $s_f°$
l_d	Diffusion length, m
Le	Lewis number
L_M	Markstein length, m
$Ma°$	Markstein number
M_a	Flame Mach number
M_u	Reactants Mach number
$\dot{m}_R$	Reaction rate, kg/m^3 s
$\vec{n}$	Normal unit vector
p	Pressure, Pa
$\tilde{p}$	Dimensionless pressure
q	Heat released per unit mass of reactant, J/kg
$\tilde{q}$	Dimensionless heat released per unit mass of reactant
r	Radius
s_f	Laminar flame speed of a stretched flame, m/s
$s_f°$	Laminar flame speed of an unstretched flame, m/s
$\vec{s}$	Flame surface velocity vector, m/s

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s_n Normal flame surface velocity, m/s
$\tilde{s}_t$ Tangential velocity vector, m/s
T Temperature, K
T_b Adiabatic flame temperature, K
T_f Stretched flame temperature, K
$\tilde{T}$ Dimensionless temperature
t Time, s
u Flow field velocity component in x-coordinate, m/s
V Velocity, m/s
$\vec{V}$ Flow velocity vector, m/s
v Dimensionless velocity component in x-coordinate, m/s
$\vec{w}$ Flow field transversal velocity vector, m/s
w Dimensionless velocity component in y-coordinate or radial direction, m/s
x Coordinate
y Coordinate
Y_A Mass fraction of species A
z Coordinate
Ze Zel'dovich number
z Coordinate
Ze Zel'dovich number

Greek Letters

γ Ratio of heat capacities
δ Small variation
∇ Nabla operator
ξ Dimensionless position in x-coordinate
Θ Expansion ratio
η Dimensionless position in y-coordinate or r-coordinate
ρ Density, kg/m^3
κ Stretch rate, s^{-1}
λ Thermal conductivity, W/m K
$\tilde{\lambda}$ Dimensionless thermal conductivity
τ Dimensionless time
σ Factor accounting for compressibility effects

Subscripts

b Burned gas
u Unburned gas

F	Fuel
f	Flame front
O	Oxidizer
tip	Flame tip
wall	Side wall of the duct or tube

5.1 Introduction

This chapter discusses theoretical and experimental observations of flame dynamics. The discussion starts by considering the flame as a discontinuity. Then, the flame stretch theory is introduced, followed by a discussion on the governing equations that would be applied to the flame. The phenomenon of flame front inversion, also known as tulip flame formation, is also discussed in this chapter by presenting selected experimental and theoretical developments. The flame acceleration inside a duct without obstacles is also presented by considering experimental and theoretical developments. The chapter closes by considering the dimensionless form of the conservation equations applied to the propagating flame.

5.2 The Flame as a Discontinuity

Before starting a more general discussion, consider a planar flame front propagating with a velocity V_f with respect to a coordinate system fixed to the laboratory frame. The first situation shown in Fig. 5.1(a) shows the planar flame propagating at the speed V_f inside a tube with both ends opened. Since the burned gas is at a higher temperature than the unburned gas, its density must be lower. This means that, in order to satisfy the mass conservation law, the burned gas velocity (V_b) must be higher than the unburned gas velocity (V_u) and that the flame propagation velocity realtive to the unburned gas must be $s_f = V_f - V_u$. In the case of a flame propagating from the closed end of a tube, the burned gas is at rest behind the flame because the ignition end is closed. Thus, the flame propagates at a velocity denoted by V_f. This situation is illustrated in Fig. 5.1(b). The unburned gas has a velocity V_u, which must be lower than V_b. Additionally, the relative velocity of the flame with respect to the unburned gas is still represented by $s_f = V_f - V_u$. The third situation involves a flame propagating from the open end of a tube towards the closed end, as illustrated in Fig. 5.1(c). In this case the unburned gas is at rest, and the burned gas presents a velocity equal to V_b the relative velocity of the flame with respect to the unburned gas is still represented by s_f. Note that the burned gas and the flame front move in opposite directions. Closing both ends of the duct would produce an increase in total pressure and density, the following must hold: $V_b > V_u$ and $s_f = V_f - V_u$. The relative velocity of the flame with respect to the burned gases is determined as the diference $V_b - V_f$. Applying the mass conservation across the flame front, it can be observed that $V_b = V_u \Theta_0$, where the expansion ratio is represented by $\Theta_0 = \rho_u/\rho_b$.

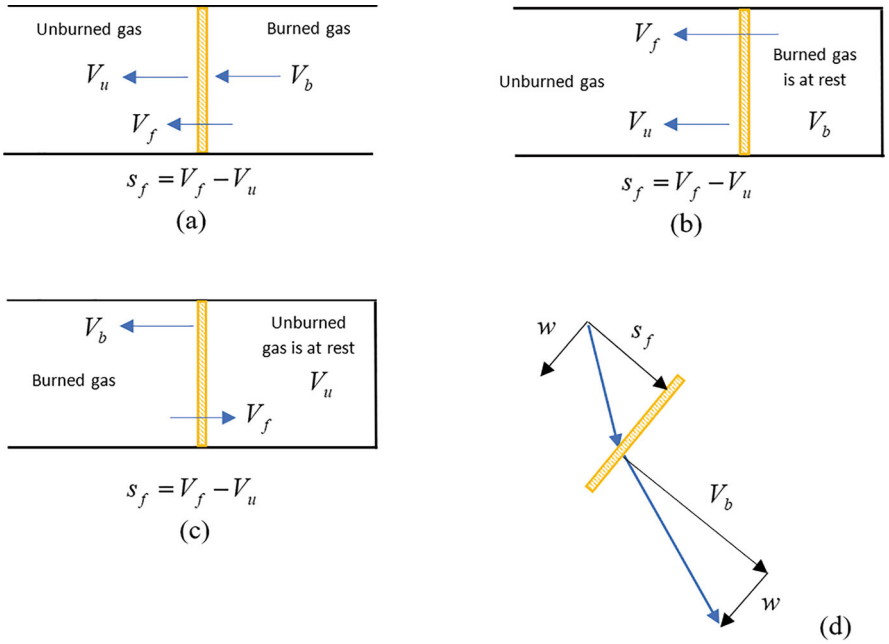


Fig. 5.1 Schematic representation of velocities across a planar flame front. (**a**) The piston effect. (**b**) The deflection

Considering an inclined planar flame with the coordinated system attached to the flame, as shown in Fig. 5.1(d), the velocities of burned and unburned gases are decomposed into normal and tangential components. Furthermore, due to conservation of momentum, the tangential velocity components must be equal. This causes the streamlines to be deflected in the direction normal to the flame.

The four situations considered so far are restricted to planar flames. In the first three cases, it was interesting to consider a coordinate system attached to the laboratory frame, while in the fourth case, it was more adequate to consider the coordinate system attached to the flame. The theory of normal and oblique shock waves is also developed by considering a coordinate system attached to the flame, as will be seen in Chap. 6.

A more general flame is introduced, this flame is treated as a discontinuity that separates the unburned and burned gases, as shown in Fig. 5.2. Following the developments by Markstein [71], the flame structure is neglected. That means the chemical reactions, heat, and species diffusion are not considered in the analysis. The flame speed is taken as s_f, and it is defined as the normal component of the unburned gas velocity relative to the flame surface.

Imagine that the flame front is described by the function $x = f(y, z, t)$. This surface can also be represented in implicit form, as shown in Eq. (5.1).

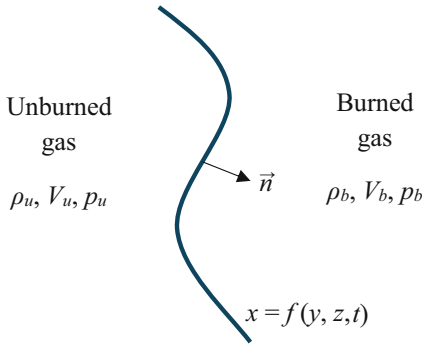


Fig. 5.2 Representation of a flame as a discontinuity

$$F\left(\vec{X}, t\right) = x - f(y, z, t) = 0 \tag{5.1}$$

In the above equation, the vector $\vec{X}$ defines the surface position in terms of the rectangular coordinates, which means that the velocity of this surface can be determined as $\vec{s} = d\vec{X}/dt$. Applying the material derivative to the function F, Eq. (5.2) is obtained.

$$\frac{DF}{Dt} = \frac{\partial F}{\partial t} + \frac{d\vec{X}}{dt} \cdot \nabla F = 0 \tag{5.2}$$

Additionally, the expression for the normal unit vector of the flame surface is given in Eq. (5.3) [42].

$$\vec{n} = \frac{\nabla F}{|\nabla F|} = \frac{\vec{i} - \frac{\partial f}{\partial y}\vec{j} - \frac{\partial f}{\partial z}\vec{k}}{\left[1 + \left(\frac{\partial f}{\partial y}\right)^2 + \left(\frac{\partial f}{\partial z}\right)^2\right]^{1/2}} \tag{5.3}$$

Dividing both sides of Eq. (5.2) by the gradient modulus, $|\nabla F|$, and observing the appearance of the normal unit vector's definition, the expression for the normal velocity of a surface element is obtained.

$$s_n = \vec{s} \cdot \vec{n} = \frac{-\left(\frac{\partial F}{\partial t}\right)}{|\nabla F|} = \frac{\frac{\partial f}{\partial t}}{\left[1 + \left(\frac{\partial f}{\partial y}\right)^2 + \left(\frac{\partial f}{\partial z}\right)^2\right]^{1/2}} \tag{5.4}$$

The velocity field of the flow is represented by $\vec{V} = u\,\vec{i} + \vec{w}$, where $\vec{w}$ is the bidimensional transversal component that includes the velocity components in the y and z coordinates. The flame speed (s_f) is obtained as the relative velocity between the normal surface velocity (s_n) and the flow velocity component normal to the surface ($\vec{V} \cdot \vec{n}$). Therefore, using Eqs. (5.3) and (5.4), the following is obtained.

$$s_f = \vec{V}\cdot\vec{n} - s_n = \frac{u - \vec{w}\cdot\left(f_y\,\vec{j} + f_z\,\vec{k}\right) - f_t}{\left(1 + f_y^2 + f_z^2\right)^{1/2}} \tag{5.5}$$

In the above expression, the velocity $\vec{V}$ is determined in the vicinity of the flame surface from the unburned gas side. In order to simplify the notation, the subscripts y, z, and t were used to represent the partial derivatives of the function f. The subscripts u and b denote the unburned and burned gases, respectively. Other authors, for example, Markstein [71], initiate the analysis by assuming that s_f is prescribed. In the present case, the development of Matalon [72] is preferred. Then, the velocity s_f is given by Eq. (5.5), and it is not necessarily a prescribed value. Note again that Eq. (5.2) provides the relationship between the flame surface function, F, and the flame surface velocity, $d\vec{X}/dt$. That expression must be satisfied in both the normal and tangential directions. Therefore, it is possible to write a new expression involving the relative flame speed, s_f. Multiplying Eq. (5.5) by the gradient of F and rearranging, yield the following expression.

$$\frac{d\vec{X}}{dt}\cdot\nabla F = \vec{V}\cdot\nabla F - s_f|\nabla F| \tag{5.6}$$

Then, substituting Eq. (5.6) into Eq. (5.2) gives

$$\frac{\partial F}{\partial t} + \vec{V}\cdot\nabla F - s_f|\nabla F| = 0 \tag{5.7}$$

The flame speed used in Eq. (5.7) can be written for the side of the burned or unburned gases, that is, $s_{f,b}$ or $s_{f,u}$ can be used. The tangential velocity is obtained from the cross product between the velocity and the normal unit vector. As previously discussed, mass conservation imposes the equality of the tangential velocities of the burned and unburned gases. This relationship is given in Eq. (5.8).

$$\left[\vec{V}\times\vec{n}\right]_b = \left[\vec{V}\times\vec{n}\right]_u \tag{5.8}$$

The mass conservation equation across the flame front is written in terms of the flame speed, as shown by Eq. (5.9). Similarly, the momentum equation is given by Eq. (5.10).

$$\rho_b\left(\vec{V}\cdot\vec{n} - s_n\right)_b = \rho_u\left(\vec{V}\cdot\vec{n} - s_n\right)_u \tag{5.9}$$

$$p_b + \rho_b\left(\vec{V}\cdot\vec{n}\right)_b\left(\vec{V}\cdot\vec{n} - s_n\right)_b = p_u + \rho_u\left(\vec{V}\cdot\vec{n}\right)_u\left(\vec{V}\cdot\vec{n} - s_n\right)_u \tag{5.10}$$

A full description of the velocity field can be obtained by solving the Euler equations with the restrictions given in Eqs. (5.8)–(5.10).

Example 5.1
A flame propagates spherically in a stoichiometric mixture. Determine the unit normal vector and the normal velocity of the flame by applying the theoretical developments presented in this section. Assume two cases: $r(t) = v_0 t$ and r (t) $= v_0 t^3$, where v_0 is a constant coefficient.

Solution
Since the flame is propagating spherically, the equation of the sphere is used to represent the coordinate x. The analysis will be restricted to positive values of x ($x \geq 0$), which represents the upper hemisphere.

$$x = f(y, z, t) = \sqrt{r^2 - y^2 - z^2}$$

The surface function according to Eq. (5.1) would be

$$F\left(\vec{X}, t\right) = x - \sqrt{r^2 - y^2 - z^2} = 0$$

The gradient of the above expression is determined as shown below.

$$\nabla F = \vec{i} - \frac{y}{\sqrt{r^2 - y^2 - z^2}}\vec{j} - \frac{z}{\sqrt{r^2 - y^2 - z^2}}\vec{k}$$

The modulus of the gradient is obtained by taking the square root of the sum of the squared components.

$$|\nabla F| = \left(1 + \frac{y^2 + z^2}{r^2 - y^2 - z^2}\right)^{1/2} = \frac{r}{\sqrt{r^2 - y^2 - z^2}}$$

The normal unit vector will be determined by Eq. (5.3).

$$\vec{n} = \frac{\nabla F}{|\nabla F|} = \frac{\sqrt{r^2 - y^2 - z^2}}{r}\vec{i} - \frac{y}{r}\vec{j} - \frac{z}{r}\vec{k}$$

Equivalently,

$$\vec{n} = \frac{\nabla F}{|\nabla F|} = \frac{x}{r}\vec{i} - \frac{y}{r}\vec{j} - \frac{z}{r}\vec{k}$$

Note that this unit vector is representing only the upper hemisphere of the flame. Therefore, it is important to keep the representation of the x-component of the normal unit vector in terms of r, y, and z in order to determine the stretch rate correctly in Example 5.2. The normal velocity of the flame can be obtained by using Eq. (5.4). However, the partial derivative $\partial f / \partial t$ is needed, and it will present different values depending on the dependence of the radius with time.

$$\frac{\partial f}{\partial t} = \frac{r(dr/dt)}{\sqrt{r^2 - y^2 - z^2}}$$

Then,

$$s_n = \frac{r(dr/dt)}{\sqrt{r^2 - y^2 - z^2}} \left(\frac{r}{\sqrt{r^2 - y^2 - z^2}} \right)^{-1} = \frac{dr}{dt}$$

The above result is in accordance with what is expected from an spherically propagating flame. In the first case, $dr/dt = v_{0,}$ and in the second case, $dr/dt = 3v_0t^2$. Therefore, the normal velocity of the flame would present two expressions, depending on dr/dt.

For the first case,

$$s_n = v_0$$

For the second case,

$$s_n = 3v_0t^2$$

It is interesting to consider now the lower hemisphere ($x < 0$). In this case, the following expressions would have been obtained.

$$x = f(y, z, t) = -\sqrt{r^2 - y^2 - z^2}$$

$$\nabla F = \vec{i} + \frac{y}{\sqrt{r^2 - y^2 - z^2}} \vec{j} + \frac{z}{\sqrt{r^2 - y^2 - z^2}} \vec{k}$$

$$\vec{n} = \frac{\nabla F}{|\nabla F|} = -\frac{x}{r} \vec{i} + \frac{y}{r} \vec{j} + \frac{z}{r} \vec{k}$$

$$\frac{\partial f}{\partial t} = -\frac{r(dr/dt)}{\sqrt{r^2 - y^2 - z^2}}$$

The influence of area change in the burning velocity can also be assessed by using the expression given in Eq. (5.5). Assume that the flow velocity is given by a single component $\vec{V} = u \vec{i}$. Moreover, assume that $f_t = 0$ and that only two dimensions are relevant, such that $f_z = 0$. Applying these considerations into Eq. (5.6) yields the expression shown below.

$$\frac{u}{s_f} = 1 + \frac{s_n}{s_f} = \left(1 + f_y^2\right)^{1/2} \tag{5.11}$$

Note that the relationship $\vec{V}\cdot\vec{n} = s_f + s_n$ was used in Eq. (5.11) to express the component u in terms of $s_f + s_n$. Assuming that, $\vec{V} = u\,\vec{i}$ and $f_t = 0$, without the restriction of f_z being equal to zero, the following is obtained from Eq. (5.5).

$$\frac{u}{s_f} = 1 + \frac{s_n}{s_f} = \left(1 + f_y^2 + f_z^2\right)^{1/2} = \left(1 + |\nabla_t f|^2\right)^{1/2} \tag{5.12}$$

In the above equation, the tangential Nabla operator (∇_t) takes the partial derivatives in the coordinates y and z; thus, $\nabla_t f = f_y\,\vec{j} + f_z\,\vec{k}$. In Eqs. (5.11) and (5.12), it is observed that the flame velocity is influenced by the change in flame surface. Moreover, it is clear that $u/s_f > 1$. A higher flow field velocity means that the relative flame velocity will also be higher. In more general terms, it can be assumed that the flow field velocity has all its components different than zero and also that $f_t \neq 0$. Therefore, from Eqs. (5.4) and (5.5), the following is observed.

$$\frac{\vec{V}\cdot\nabla F}{s_f + s_n} = \frac{u - \vec{w}\cdot\left(f_y\,\vec{j} + f_z\,\vec{k}\right)}{s_f + s_n} = \left(1 + |\nabla_t f|^2\right)^{1/2} \tag{5.13}$$

The left-hand side of Eq. (5.13) is the directional velocity of the flame surface in the direction of the velocity field of the flow, normalized by the sum $s_f + s_n$, which gives a scalar greater than unity. It is also known that the gradient vector will have a greater module in locations where the level curves are closer to each other. Therefore, the relative flame velocity will be higher in the regions where the flame surface is steeper.

5.3 Flame Stretch

The flame stretch is related to the variation of the flame front area. The development by Matalon [73] is followed in this section. The deformation of the area element can be measured by the parameter given in Eq. (5.14). That parameter is called the flame stretch rate (κ) and is measured in 1/s units.

$$\kappa = \frac{1}{A}\frac{dA}{dt} \tag{5.14}$$

In the original study, Matalon defined the flame stretch as $\kappa^* = (l_d/s_f^\circ)\,\kappa$. That is, in the work by Matalon, what is today known as the stretch rate [11] was multiplied by the ratio of the diffusion length [see Eq. (5.15)] and the adiabatic flame speed (s_f°) of the planar flame.

$$l_d = \frac{\lambda/\rho_u c_p}{s_f^\circ} = \frac{\alpha}{s_f^\circ} \tag{5.15}$$

The strategy to obtain an expression for κ consists of writing the derivative of the flame front area in terms of the limit of the variation $A(t + \delta t) - A(t)$ when $\delta t \rightarrow$

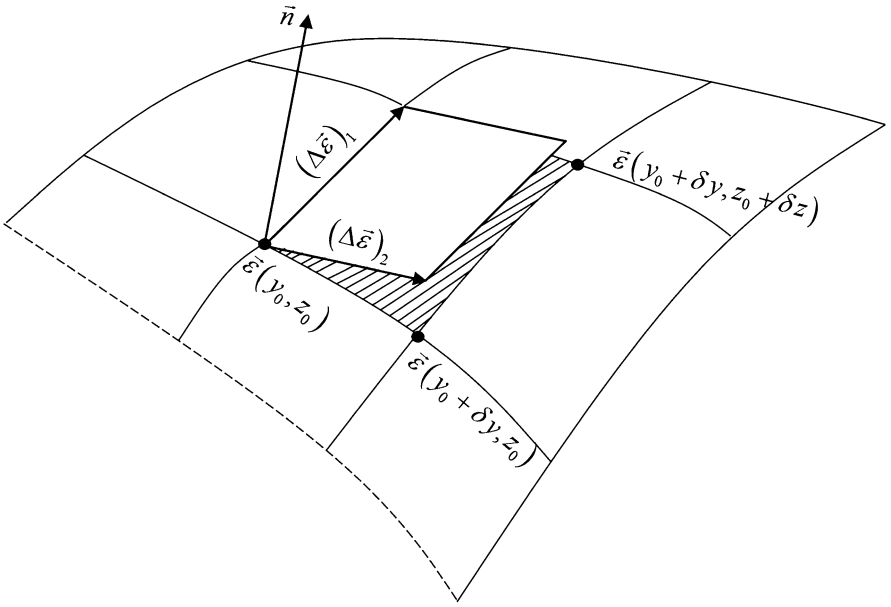


Fig. 5.3 Schematic representation of the area element used to analyze flame stretch. (Adapted from Matalin [73])

0. For the points where $\kappa > 0$, the flame is stretched. On the other hand, for points where $\kappa < 0$, the flame is compressed. Using Eq. (5.1) it is possible to write an expression for the x coordinate of the flame front. Then, a parametric representation is given by Eq. (5.16).

$$\vec{\varepsilon}\,(y,z,t) = f(y,z,t)\,\vec{i} + y\,\vec{j} + z\,\vec{k} \tag{5.16}$$

The point $\varepsilon\ (y_0, z_0)$ is located on the flame surface at the time t, and it belongs to the area element bounded by the four curves identified as $\varepsilon\ (y_0, z)$, $\varepsilon\ (y_0 + \delta y, z)$, $\varepsilon\ (y, z_0)$, and $\varepsilon\ (y, z_0 + \delta z)$. This situation is shown in Fig. 5.3, where the curves are easily identified by locating the points y_0, $y_0 + \delta y$, z_0 and $z_0 + \delta z$.

This geometric image forms a parallelepiped, with its sides determined by the expressions shown in Eqs. (5.17) and (5.18).

$$\left(\Delta\vec{\varepsilon}\right)_1 = \frac{\partial\vec{\varepsilon}}{\partial y}\delta y = \left(f_y\delta y\right)\vec{i} + (\delta y)\,\vec{j} \tag{5.17}$$

$$\left(\Delta\vec{\varepsilon}\right)_2 = \frac{\partial\vec{\varepsilon}}{\partial z}\delta z = \left(f_z\delta z\right)\vec{i} + (\delta z)\,\vec{k} \tag{5.18}$$

The area of the parallelepiped is obtained by taking the module of the cross product of the vectors given in Eqs. (5.17) and (5.18), this provides the result shown in Eq. (5.19).

$$A(t) = \left| \left(\Delta \vec{\varepsilon} \right)_1 \times \left(\Delta \vec{\varepsilon} \right)_2 \right| = \sqrt{\left(1 + f_y^2 + f_z^2 \right)} (\delta y \delta z) \tag{5.19}$$

The flame moves with a velocity represented by $\vec{s} = s_1 \vec{i} + s_2 \vec{j} + s_3 \vec{k}$; therefore, after a time interval δt, the flame surface can be represented by Eq. (5.20), where $f = f(y, z, t)$.

$$\vec{\varepsilon}(y, z, t + \delta t) = (f + s_1 \delta t) \vec{i} + (y + s_2 \delta t) \vec{j} + (z + s_3 \delta t) \vec{k} \tag{5.20}$$

The change on the sides of the parallelepiped as function of time is obtained by derivation of Eq. (5.20), following the definitions given in Eqs. (5.17) and (5.18).

$$\left(\Delta \vec{\varepsilon} \right)_1^* = \left(f_y \delta y + \frac{\partial s_1}{\partial y} \delta t \delta y \right) \vec{i} + \left(\delta y + \frac{\partial s_2}{\partial y} \delta t \delta y \right) \vec{j} + \left(\frac{\partial s_3}{\partial y} \delta t \delta y \right) \vec{k} \tag{5.21}$$

$$\left(\Delta \vec{\varepsilon} \right)_2^* = \left(f_z \delta z + \frac{\partial s_1}{\partial z} \delta t \delta z \right) \vec{i} + \left(\frac{\partial s_2}{\partial z} \delta t \delta z \right) \vec{j} + \left(\delta z + \frac{\partial s_3}{\partial z} \delta t \delta z \right) \vec{k} \tag{5.22}$$

The area at the time $t + \delta t$ is obtained by taking the cross product of the vectors given in Eqs. (5.21) and (5.22).

$$\vec{A}(t + \delta t) = \left(\Delta \vec{\varepsilon} \right)_1^* \times \left(\Delta \vec{\varepsilon} \right)_2^* \tag{5.23}$$

Assuming the function in Eq. (5.1) and taking the derivatives and gradient shown in Eq. (5.2), it is possible to obtain the component s_1 of the flame velocity as shown below.

$$s_1 = f_t + s_2 f_y + s_3 f_z \tag{5.24}$$

It is interesting to define the operator called the tangential material derivative, as given in Eq. (5.25).

$$\frac{D''}{Dt} = \frac{\partial}{\partial t} + s_2 \frac{\partial}{\partial y} + s_3 \frac{\partial}{\partial z} \tag{5.25}$$

Taking the partial derivative of the velocity component s_1, with respect to y, and noting that the tangential material derivative of f_y appears during that operation, the expression given in Eq. (5.26) can be used to represent $\partial s_1 / \partial y$. An analogous expression can be obtained for the partial derivative of s_1 with respect to z.

$$\frac{\partial s_1}{\partial y} = \frac{D'' f_y}{Dt} + f_y \frac{\partial s_2}{\partial y} + f_z \frac{\partial s_3}{\partial y} \tag{5.26}$$

Therefore, the derivatives of s_1 and s_2 can be obtained and used to determine the cross product shown in Eq. (5.23). Afterwards, the module is taken to obtain $A\ (t + \delta t)$ as shown in Eq. (5.27). Note that the terms multiplied by $(\delta t)^2$ were neglected. Further details can be found in the Appendix at the end of this chapter.

$$A(t+\delta t) = \sqrt{1+f_y^2+f_z^2}\delta y \delta z \left[1 + \delta t \left\{ \frac{\partial s_2}{\partial y} + \frac{\partial s_3}{\partial z} + \frac{\frac{D''}{Dt}\sqrt{1+f_y^2+f_z^2}}{\sqrt{1+f_y^2+f_z^2}} \right\}\right] \quad (5.27)$$

Equations (5.19) and (5.27) will be simplified by making $N = \sqrt{1+f_y^2+f_z^2}$. Then, the limit of the difference of areas $[A(t + \delta t) - A(t)]$ is taken as $\delta t \to 0$. Note that the first term on the right-hand side of Eq. (5.27) is equal to A(t). The result of this procedure is shown in Eq. (5.28).

$$\frac{1}{A}\lim_{\delta t \to 0}\frac{A(t+\delta t)-A(t)}{\delta t} = \frac{1}{N}\left(\frac{\partial N}{\partial t} + \nabla_t \cdot N\vec{s}\right) \quad (5.28)$$

where the tangential Nabla operator is given by the expression below.

$$\nabla_t = (0)\,\vec{i} + \frac{\partial}{\partial y}\vec{j} + \frac{\partial}{\partial z}\vec{k} \quad (5.29)$$

Therefore, the flame stretch is obtained by substituting Eq. (5.28) into Eq. (5.14).

$$\kappa = \frac{1}{N}\left(\frac{\partial N}{\partial t} + \nabla_t \cdot N\vec{s}\right) \quad (5.30)$$

The above expression was obtained for a particular solution, which characterizes the flame front as given by Eq. (5.1). It is possible to imagine that a point on the flame surface element changes its position due to the variation of the surface element, as given by Eq. (5.30), and also due to changes on the velocity field of the fluid. However, that point remains on the flame surface. The rate of change of this point's position is equal to the tangential velocity of the fluid [73].

Recalling that $\vec{s}$ and $\vec{V}$ are the velocities of the flame surface and the fluid, respectively, it is noticed that the mass conservation imposes that the tangential velocities of the flame and the fluid must be equal in the vicinity of the flame.

$$\vec{s}_t = \vec{w} = \vec{n} \times \left(\vec{V} \times \vec{n}\right) \quad (5.31)$$

Nevertheless, the expression given in Eq. (5.30) was derived for the flame surface velocity, $\vec{s}$. Hence, taking a point in the vicinity of the flame, the following is obtained.

$$\vec{s} = \vec{n} \times \left(\vec{V} \times \vec{n}\right) + \left(\vec{s} \cdot \vec{n}\right)\vec{n} \quad (5.32)$$

Using the expression for N and taking the time derivative, the following is obtained.

$$\frac{1}{N}\frac{\partial N}{\partial t}=\frac{1}{N}\frac{\partial}{\partial t}\sqrt{1+f_y^2+f_z^2}=\frac{f_y f_{ty}}{N^2}+\frac{f_z f_{tz}}{N^2} \tag{5.33}$$

The substitution of Eq. (5.32) into the second term on the right-hand side of Eq. (5.30), the application of the identities for the cross and dot products, the substitution of the relations for $\vec{n}$ and $\vec{s}\cdot\vec{n}$ given in Eqs. (5.3) and (5.4), and some rearrangements, yield the expressions shown in Eqs. (5.34) and (5.35).

$$\frac{1}{N}\left\{\nabla_t\cdot\left[N\vec{n}\times\left(\vec{V}\times\vec{n}\right)\right]\right\}=-\left[\nabla_t\times\left(\vec{V}\times\vec{n}\right)\right]\cdot\vec{n} \tag{5.34}$$

$$\frac{1}{N}\nabla_t\cdot\left[N\left(\vec{s}\cdot\vec{n}\right)\vec{n}\right]=-\frac{f_y f_{ty}}{N^2}-\frac{f_z f_{tz}}{N^2}+\frac{f_t}{N}\nabla_t\cdot\vec{n} \tag{5.35}$$

Note that the Nabla operator can be written as the sum of a normal and tangential components ($\nabla=\nabla_n+\nabla_t$), which in the present case implies that $\nabla_n=(\partial/\partial x)\,\vec{i}$, which this yields $\nabla_n\cdot\vec{n}=0$ and $\nabla_n\times\left(\vec{V}\times\vec{n}\right)\cdot\vec{n}=-\nabla_n\cdot\left[\vec{n}\times\left(\vec{V}\times\vec{n}\right)\right]=0$. Then, substituting Eqs. (5.33)–(5.35) into Eq. (5.30) provides the result shown below for the stretch rate.

$$\kappa=-\left\{\left[\nabla\times\left(\vec{V}\times\vec{n}\right)\right]\cdot\vec{n}+\frac{F_t}{|\nabla F|}\nabla\cdot\vec{n}\right\} \tag{5.36}$$

The expression given in Eq. (5.33) was obtained by Matalon [73] through a slightly different procedure. Another expression for the flame stretch is presented by Law [11]. However, note that Law [11] defined the unit vector with opposite sign than that defined in Eq. (5.3). It can be observed that the flame stretch is influenced by two terms: the first one is related to the flow velocity in the vicinity of the flame, and the second one is related to the nonstationary behavior of the flame. Also note that, for an irrotational flow the first term inside the brackets on the right-hand side of Eq. (5.36) is zero $\nabla\times\left(\vec{V}\times\vec{n}\right)=0$ [74].

Example 5.2
Determine the stretch rate for a spherically propagating flame which propagates in stationary unburned gas.

Solution
Since the unburned gas is stationary, the flow velocity vector will be equal to zero. Therefore, Eq. (5.36) becomes

$$\kappa=-\frac{F_t}{|\nabla F|}\nabla\cdot\vec{n}$$

The normal unit vector for a spherically propagating flame was obtained in Example 5.1. Therefore, the divergence of the unit normal vector will be directly obtained.

$$\nabla \cdot \vec{n} = \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) \cdot \left(\frac{\sqrt{r^2 - y^2 - z^2}}{r} \vec{i} - \frac{y}{r} \vec{j} - \frac{z}{r} \vec{k} \right) = -\frac{2}{r}$$

The expressions for F_t and $|\nabla F|$ were also obtained in Example 5.1.

$$\frac{F_t}{|\nabla F|} = \frac{r(dr/dt)\left(\sqrt{r^2 - y^2 - z^2}\right)^{-1}}{r\left(\sqrt{r^2 - y^2 - z^2}\right)^{-1}} = \frac{dr}{dt}$$

Finally, the stretch rate of a spherically propagating flame is obtained.

$$\kappa = \frac{2}{r} \frac{dr}{dt}$$

The above response is the correct one for the complete spherical flame. However, note that the normal unit vector could have been represented as shown below for the upper hemisphere.

$$\vec{n} = \left(\frac{x}{r} \vec{i} - \frac{y}{r} \vec{j} - \frac{z}{r} \vec{k} \right)$$

This representation of the normal unit vector leads to the following divergence.

$$\nabla \cdot \vec{n} = \left(\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) \cdot \left(\frac{\sqrt{r^2 - y^2 - z^2}}{r} \vec{i} - \frac{y}{r} \vec{j} - \frac{z}{r} \vec{k} \right) = -\frac{1}{r}$$

Then, the stretch rate would be

$$\kappa_{\text{upper}} = \frac{1}{r} \frac{dr}{dt}$$

This second result is not incorrect; however, in this case, only the stretch rate of the upper hemisphere is being measured. If the lower hemisphere is considered in the calculations, the four last expressions given in Example 5.1 would need to be used. This leads to $\kappa_{\text{lower}} = (1/r)(dr/dt)$, which means that the total stretch rate should be the sum of the stretch rates for both hemispheres.

Example 5.3

Determine an expression for the flame stretch, κ, of a flame with a hemispheric format, as shown in Fig. 5.4. The flame is propagating horizontally inside a tube.

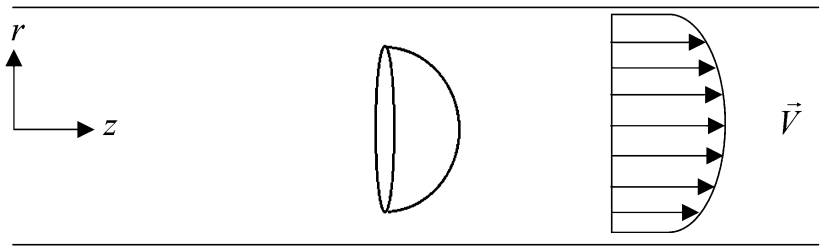


Fig. 5.4 Hemispheric flame propagating inside a tube

Solution

Since the geometry is axisymmetric, it is useful to adopt cylindrical coordinates for this problem. The flame surface equation is given by the function below.

$$F(r,\theta,z,t)=\sqrt{r^2+z^2}-R(t)=0 \tag{5.37}$$

Note that $r^2 + z^2 = R^2$ is the equation of a sphere in cylindrical coordinates and $R(t)$ represents the flame radius evolution. On the other hand, the flow velocity is represented by $\vec{V}=u(r,z)\,\vec{e}_z$, where the unit vector is obtained from

$$\vec{n}=\frac{\nabla F}{|\nabla F|}=\frac{r}{R}\,\vec{e}_r+\frac{z}{R}\,\vec{e}_z \tag{5.38}$$

In the above equation, $R=\sqrt{r^2+z^2}$, $|\nabla F| = 1$, and the Nabla operator is in cylindrical coordinates. Now it is possible to calculate the different terms necessary for Eq. (5.36).

$$\nabla\times\left(\vec{V}\times\vec{n}\right)=-\frac{r}{R}\frac{\partial u}{\partial z}\vec{e}_r+\left(\frac{r}{R}\frac{\partial u}{\partial r}+\frac{u}{R}\right)\vec{e}_z$$

$$\left[\nabla\times\left(\vec{V}\times\vec{n}\right)\right]\cdot\vec{n}=\frac{r}{R^2}\left(z\frac{\partial u}{\partial r}-r\frac{\partial u}{\partial z}+z\frac{u}{r}\right)$$

$$\nabla\cdot\vec{n}=\frac{2}{R}$$

$$F_t=-\frac{dR}{dt}$$

Then, substitution into Eq. (5.36) provides the following result for the stretch rate.

$$\kappa=-\left[\frac{r}{R^2}\left(z\frac{\partial u}{\partial r}-r\frac{\partial u}{\partial z}+z\frac{u}{r}\right)-\frac{2}{R}\frac{dR}{dt}\right] \tag{5.39}$$

The Eq. (5.39) determines the stretch rate for a flame with hemispheric format propagating horizontally inside a tube. In order to further develop the above expression, it is necessary to know the velocity distribution of the unburned gas and also the rate of change of the radius with time. Assume that the radius changes linearly with time as $R(t) = v_0 t$, where v_0 is a constant coefficient. In addition, consider that the velocity of the unburned gas ahead of the flame is given by the expression for a fully developed incompressible flow inside a pipe [74].

$$\vec{V} = u(r)\,\vec{e}_z = \frac{B}{4\mu}\left(\frac{D^2}{4} - r^2\right)\vec{e}_z$$

The incompressible flow consideration is reasonable for low Mach numbers. The coefficient $B = 128(\dot{m}/\rho)\mu/(\pi D^4)$ is a constant which depends on the fluid properties (μ, ρ), the mass flow rate ($\dot{m}$), and the tube diameter (D) [74]. Therefore, Eq. (5.39) becomes

$$\kappa = -\left\{\frac{z}{v_0^2 t^2}\left[\frac{B}{4\mu}\left(\frac{D^2}{4} - r^2\right) - \frac{Br^2}{2\mu}\right] - \frac{2}{t}\right\}$$

More examples on the application of the flame stretch equation can be found in the original article by Matalon [73].

Finally, the influence of flame stretch in the laminar flame velocity and temperature is discussed. In their work, Matalon and Matkowsky [72] obtained the following relationship between the laminar flame speeds of an stretched flame (s_f) and that of an unstretched flame (s_f°). In the work by Matalon [73], a relationship between the flame temperatures of stretched and unstretched flames was also presented. Those expressions are given in Eqs. (5.40) and (5.41).

$$s_f = s_f^{\,z}\left[1 - \beta\left(l_d/s_f{}^\circ\right)\kappa\right] \tag{5.40}$$

$$T_f = T_b\left[1 - \beta_T\left(l_d/s_f{}^\circ\right)\kappa\right] \tag{5.41}$$

The expressions for β and β_T were presented by Matalon [73] and were slightly modified by Quines et al. [75], as shown in Eqs. (5.42) and (5.43). In the development by Matalon [73], the dimensionless heat release rate is $\tilde{q} = q/c_p T_u$, and q is the heat of combustion per unit mass of the reactant. The integral shown in Eqs. (5.42) and (5.43) was taken from $-\infty$ to 0, because the coordinate system is attached to the flame front in the analysis [72]. Therefore, that term can be interpreted as the heat released up to the flame front.

$$\beta = \frac{Ze}{2}(Le - 1)\frac{T_u}{T_b - T_u}\left[\int_{-\infty}^{0} \ln(1 + \tilde{q}e^x)dx\right] + \frac{T_b}{T_b - T_u}\ln\left(\frac{T_b}{T_u}\right) \tag{5.42}$$

$$\beta_T = Ze(Le-1)\left[\int_{-\infty}^{0} \ln(1+\widetilde{q}e^{x})dx\right]\left(\frac{T_b}{E_a/R}\,\frac{T_u}{T_b-T_u}\right) \tag{5.43}$$

The derivation of Eqs. (5.40)–(5.42) will not be presented here. The interested reader should refer to the work by Matalon and Matkowsky [72]. A flame is defined as positively stretched when it is convex towards the reactants. Therefore, it is observed from Eq. (5.40) that for positive stretch, the flame can accelerate depending on the value of the Lewis number. It is possible to identify a critical value of the Lewis (Le^*) number below which a positively stretched flame will accelerate. On the other hand, if $Le > Le^*$, the positively stretched flame will decelerate. In order to obtain the critical value of the Lewis number, it is enough to set $\beta = 0$ in Eq. (5.42).

$$Le^* = 1 - 2Ze^{-1}\frac{T_b}{T_u}\left[\ln(T_b/T_u)/\int_{-\infty}^{0} \ln(1+\widetilde{q}e^{x})dx\right] \tag{5.44}$$

Note that for a positively stretched flame, the flame temperature will decrease for Lewis numbers greater than one ($Le > 1$), as can be inferred from Eq. (5.43). On the other hand, for a flame with positive stretch and $Le < 1$, the flame temperature increases. In addition to the effect of the Lewis number, Eqs. (5.42) and (5.44) reveal the effect of the Zel'dovich number.

5.4 Dimensionless Numbers Related to the Change of Area in the Flame

The flame stretch defined in Eq. (5.14) has units of s^{-1}. Taking into consideration the ratio of the diffusion zone length and the unstretched laminar flame speed (l_d/s_f°), it is possible to obtain the dimensionless number, known as the Karlovic number (Ka°).

$$Ka^\circ = \frac{l_d}{s_f{}^\circ}\kappa = \frac{k/c_p}{\left(\rho_u s_f{}^\circ\right)^2}\rho_u\kappa \tag{5.45}$$

The Karlovic number has been defined above by using the mass flux for unstretched conditions. Nevertheless, a more general Karlovic number would be defined by allowing for different values of the mass flux.

$$Ka = \frac{k/c_p}{\left(\rho_u s_f\right)^2}\rho_u\kappa = Ka^\circ\,\frac{\left(\rho_u s_f{}^\circ\right)^2}{\left(\rho_u s_f\right)^2} \tag{5.46}$$

The right-hand side of Eq. (5.42) shows a dependence on the Zel'dovich and Lewis numbers, That term can be easily rearranged as shown below.

$$\beta = -\frac{Ze}{2}\left(\frac{1}{Le}-1\right)Le\frac{T_u}{T_b-T_u}\left[\int_{-\infty}^{0}\ln(1+\widetilde{q}e^{x})dx\right]+\frac{T_b}{T_b-T_u}\ln\left(\frac{T_b}{T_u}\right) \quad (5.47)$$

The first product involving the Zel'dovich and Lewis numbers appearing on the right-hand side of Eq. (5.47) is known as the Markstein number.

$$Ma^{\circ} = \frac{Ze}{2}\left(\frac{1}{Le}-1\right) \quad (5.48)$$

Equation (5.40) provides another interesting observation. It is first rearranged as shown below.

$$s_f = s_f{}^{\circ} + s_f{}^{\circ}\left[-\beta\left(l_d/s_f{}^{\circ}\right)\right]\kappa \quad (5.49)$$

Then, the product $-\beta(l_d/s_f^{\circ})\kappa$ would be given by the following expression.

$$-\beta\left(\frac{l_d}{s_f{}^{\circ}}\kappa\right) = (Ma^{\circ}Ka^{\circ})Le\frac{T_u}{T_b-T_u}\left[\int_{-\infty}^{0}\ln(1+\widetilde{q}e^{x})dx\right]+Ka^{\circ}\frac{T_b}{T_b-T_u}\ln\left(\frac{T_b}{T_u}\right) \quad (5.50)$$

The product $\sigma^{\circ} = Ma^{\circ}Ka^{\circ}$ accounts for the contribution of the Zel'dovich number, Lewis number, and stretch rate. In his book, Law introduces the Markstein length (L_M) as the length that satisfies the relationship given in Eq. (5.51). It is interesting to compare Eq. (5.51) with Eq. (5.49) to notice that the term ($-\beta l_d\,\kappa$) is performing the same function as the L_M.

$$s_f = s_f{}^{\circ} + L_M\kappa \quad (5.51)$$

Equation (5.51) is generally used in fundamental experiments to determine the laminar flame velocity. In a latter article, Kelly and Law [76] proposed the following nonlinear relationship for the determination of the laminar flame velocity.

$$\left(\frac{s_f}{s_f{}^{\circ}}\right)^2\ln\left(\frac{s_f}{s_f{}^{\circ}}\right)^2 = -2\frac{L_u\kappa}{s_f{}^{\circ}} \quad (5.52)$$

In his work on flame dynamics, Matalon [55] provided an expression for the dependence of the flame velocity on the Markstein length and the stretch rate, as shown in Eq. (5.52), which can be considered an alternative to Eq. (5.40).

$$s_f = s_f{}^{\circ} - L_M^{*}\kappa \quad (5.53)$$

The equation for the Markstein length is given as follows by Matalon [55].

$$L_M^* = -L_M = l_d \left[\frac{\Theta}{\Theta - 1} \int_1^{\Theta} \frac{\tilde{\lambda}(\tilde{T})}{\tilde{T}} d\tilde{T} + \frac{Ze(Le_{\text{eff}} - 1)}{2(\Theta - 1)} \int_1^{\Theta} \frac{\tilde{\lambda}(\tilde{T})}{\tilde{T}} \ln\left(\frac{\Theta - 1}{\tilde{T} - 1} \right) d\tilde{T} \right] \tag{5.54}$$

In the expression above, the following definitions were adopted, $\Theta = \rho_u/\rho_b$, $\tilde{T} = T/T_u$ and $\tilde{\lambda} = \lambda/\lambda_u$ is the dimensionless thermal conductivity of the mixture which is assumed to depend on temperature only. Moreover, the effective Lewis number is calculated for lean and rich conditions in accordance with Eqs. (5.55) and (5.56), respectively.

$$Le_{\text{eff-lean}} = \frac{Le_O + (1 - \Phi) Le_F}{2 - \Phi} \tag{5.55}$$

$$Le_{\text{eff-rich}} = \frac{Le_F + (1 + \Phi) Le_O}{2 + \Phi} \tag{5.56}$$

where the Φ is a function of the Zel'dovich number and the equivalence ratio, as shown in Eq. (5.57). Additionally, the Lewis number of the fuel (Le_F) is calculated as the ratio of the thermal diffusivity of the mixture (α_{mix}) to the mass diffusivity of the fuel in the mixture ($D_{\text{F-mix}}$). Analogously, the Lewis number of the oxidizer (Le_O) is calculated as shown in Eq. (5.58).

$$\Phi \sim Ze(\phi - 1) \tag{5.57}$$

$$Le_F = \frac{\alpha_{\text{mix}}}{D_{F\text{-mix}}}; \quad Le_O = \frac{\alpha_{\text{mix}}}{D_{O\text{-mix}}} \tag{5.58}$$

Finally, Matalon [55] also provided an expression for the variation of the flame temperature with the stretch rate for a lean mixture, which can be considered as an alternative to Eq. (5.41) when a lean mixture is being considered.

$$T_f = T_b - T_u (Le_F - 1) \frac{l_d}{s_f^\circ} \kappa \left[\int_1^{\Theta} \frac{\widetilde{\lambda}}{\widetilde{T}} d\widetilde{T} \ln\left(\frac{\Theta - 1}{\widetilde{T} - 1} \right) d\widetilde{T} \right] \tag{5.59}$$

It is interesting to compare the right-hand side of Eq. (5.59) with the expression for β_T given in Eq. (5.43) to note the difference between both formulations.

5.5 Flame Front Inversion Phenomenon

An interesting phenomenon related to flame dynamics is the inversion of the flame front, which takes place when a flame propagates inside a closed or half-open duct (or tube). In this case, half-open means that the duct is closed at the ignition end but open at the opposite end [77]. The phenomenon of flame front inversion was called

tulip flame by Salamandra in an article published in 1958 [78]. However, the first published observation dates back to Ellis in 1928 [79]. There is a rule attributed to Ellis which states that tulip flames form in closed ducts with an aspect ratio ($AR = L/D_h$) greater than 2. The closed and half-open duct configurations are shown in Fig. 5.5.

There are several experimental articles that show flame propagation inside a closing duct [62, 75, 77, 80, 81]. The formation of the tulip flame is divided into four main stages:

(a) The spherical flame propagation
(b) The finger flame propagation
(c) The flat flame formation
(d) The tulip flame formation

Figure 5.6 shows the propagation of a stoichiometric H_2/C_3H_8/air flame observed by Mendiburu et al. [81]. The fuel mixture was formed by 75% H_2 and 25% C_3H_8, and the equivalence ratio (ϕ) was 1.0. The experiments were conducted at the Combustion Laboratory of Federal University of Rio Grande do Sul. Three of the four stages were observed; the spherical flame propagation stage was not observed because the experimental configuration did not allow for visualization in the vicinity of the ignition source. The theoretical work by Valiev et al. [82] provides interesting results by applying a relatively simple analysis. The dimensionless variables adopted for the analysis are presented in Eqs. (5.60) and (5.61).

$$\tau = \frac{s_f{}^\circ t}{r}; \quad \xi = \frac{x}{r}; \quad \eta = \frac{y}{r} \tag{5.60}$$

$$v = \frac{u_x}{s_f{}^\circ}; \quad w = \frac{u_y}{s_f{}^\circ}; \quad \Theta_0 = \frac{\rho_u}{\rho_b{}^\circ} \tag{5.61}$$

In the above definitions, $\rho_b{}^\circ$ is the density of the product gas at the adiabatic flame temperature. The theory by Valiev et al. [82] will be presented here for ducts and tubes. The following considerations are introduced.

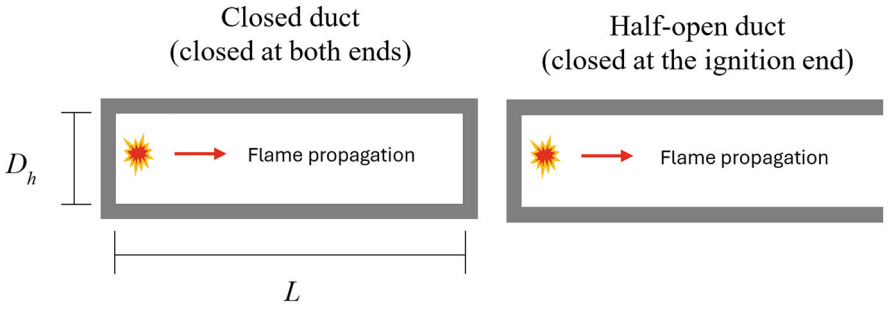


Fig. 5.5 Schematic representation of the closed and half-open duct configurations

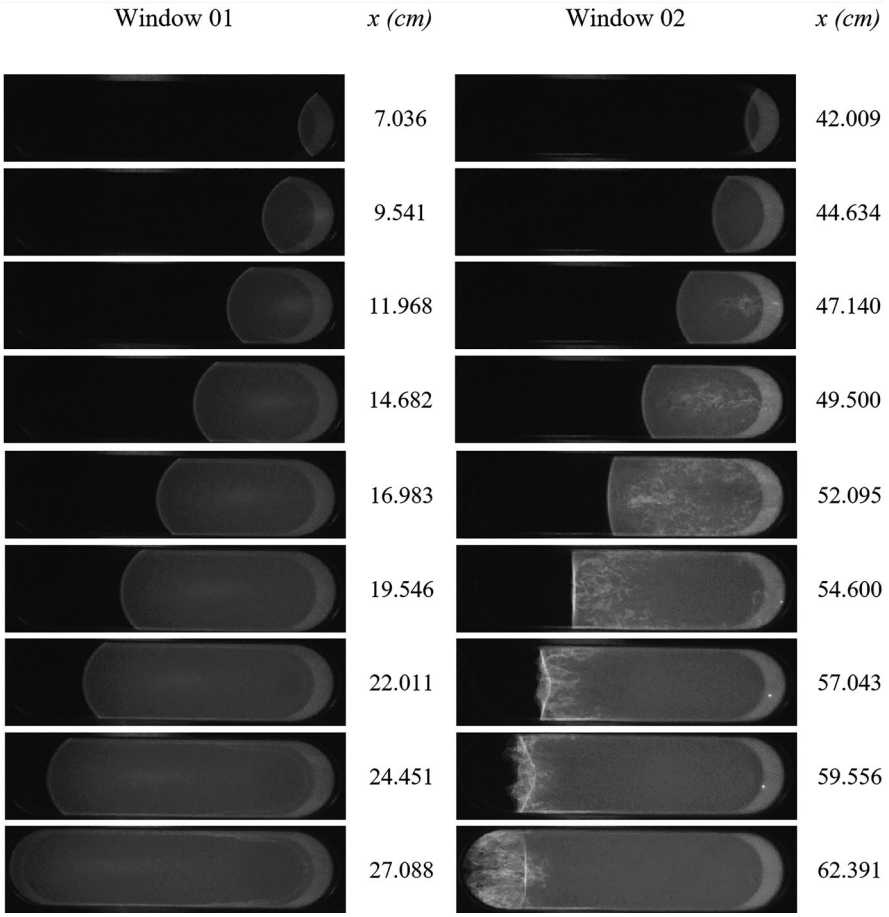


Fig. 5.6 Flame propagation inside a closed duct for a fuel mixture—a stoichiometric mixture of [75%H_2 + 25%C_3H_8] + 5% He + Air. The distance from the ignition source is 4.5 cm, and the flame propagates from right to left [81]. (Reprinted from International Journal of Hydrogen Energy, Vol 78, Andrés Z. Mendiburu, Leonardo P. Norabuena, Vanessa A. Martinez, Rafael Quines da Silva, German R. Chumpitaz, Flame propagation behavior inside a closed duct upstream of a flame arrester housing, 294–302, Copyright (2024), with permission from Elsevier)

(i) Duct walls are adiabatic with no-slip condition.
(ii) Two-dimensional propagation analysis.
(iii) The duct is closed at the ignition end.
(iv) Incompressible flow in the first stage, moderate compression in the second stage.
(v) Rectangular or axysimmetric geometry
(vi) Potential flow, meaning that the flow is incompressible, irrotational, and in steady-state conditions.

5.5.1 Flame Front Inversion Inside a Duct

A rectangular geometry is considered first. An incompressible two-dimensional flow in steady state is assumed. Then, the mass conservation equation [Eq. (3.8)] is written in dimensionless form as shown in Eq. (5.62).

$$\frac{\partial v}{\partial \xi} + \frac{\partial w}{\partial \eta} = 0 \tag{5.62}$$

The boundary conditions for the unburned gas require that the vertical velocity is w_u for η, but it must become $w_u = 0$ for $\eta = 1$. Assuming potential flow for the unburned gases and a horizontal velocity component equal to $v_u = C_1\xi$, then, a solution that satisfies the differential equation and the boundary conditions is $w_u = C_1(1 - \eta)$. Similarly, potential flow is assumed for the burned gas, and the horizontal velocity component is adopted as $v_b = C_2\xi$. The boundary conditions require that the vertical velocity is w_b for η and $w_b = 0$ for $\eta = 0$. This implies that a solution for the vertical velocity component for the burned gas is $w_b = -C_2\eta$.

Assuming a coordinate system fixed to the laboratory frame, the flame velocity is V_f, and the unburned and burned gases have velocity components $u_{i,u}$ and $u_{i,b}$ in the ith coordinate. Then, the mass conservation across the flame requires that $\rho_u(V_f - u_{i,u}) = \rho_b(V_f - u_{i,b})$. Now, the relative velocity between the flame and the unburned gas is $s_f^\circ = V_f - u_{i,u}$, which allows for the substitutions, $V_f = s_f^\circ + u_{i,u}$ and $V_f - u_{i,b} = s_f^\circ + u_{i,u} - u_{i,b}$. Following this reasoning, it can be shown that $u_{i,u} - u_{i,b} = (\rho_u/\rho_b - 1)s_f^\circ$. This idea will be applied to continue the analysis.

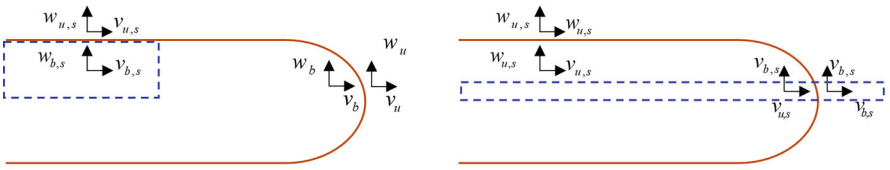
Assuming a planar flame propagating in the vertical direction, the mass conservation equation applied in the vertical direction yields $u_{y,u} - u_{y,b} = (\rho_u/\rho_b - 1)s_f^\circ$. As was discussed in Sect. 5.2, the mass conservation equation requires the equality of the tangential velocity components. Therefore, the dimensionless vertical velocities are related through $w_u - w_b = \Theta_0 - 1$, while the dimensionless horizontal velocity components are related through $v_u = v_b$. Solving Eq. (5.62) and using the relationships between the velocity components, the expressions shown in Eq. (5.63) are obtained. The flame is divided into two regions: the flame skirt region [Fig. 5.7(a)] and the flame tip region [Fig. 5.7(b)].

$$v_u = v_b = (\Theta_0 - 1)\xi; \quad w_u = (\Theta_0 - 1)(1 - \eta); \quad w_b = -(\Theta_0 - 1)\eta \tag{5.63}$$

Valiev et al. applied evolution equations to the control volumes shown in Fig. 5.8. These equations are presented in dimensional form in Eq. (5.64) and in dimensionless form in Eq. (5.65).

$$\frac{dy_f}{dt} - u_{y,u} = s_f^\circ; \quad \frac{dx_f}{dt} - u_{x,u} = s_f^\circ \tag{5.64}$$

$$\frac{d\eta_f}{d\tau} - w_u = 1; \quad \frac{d\xi_{\text{tip}}}{d\tau} - v_b = \Theta_0 \tag{5.65}$$



Flame skirt region. Flame front region.

Fig. 5.7 Control volume for the integration of the equations of flame evolution. (Adapted from Fuel, Vol 373, Andrés Z. Mendiburu, A simple theoretical model for the onset of flame front inversion for flames propagating in closed or half-open ducts, 132304, Copyright (2024), with permission from Elsevier)

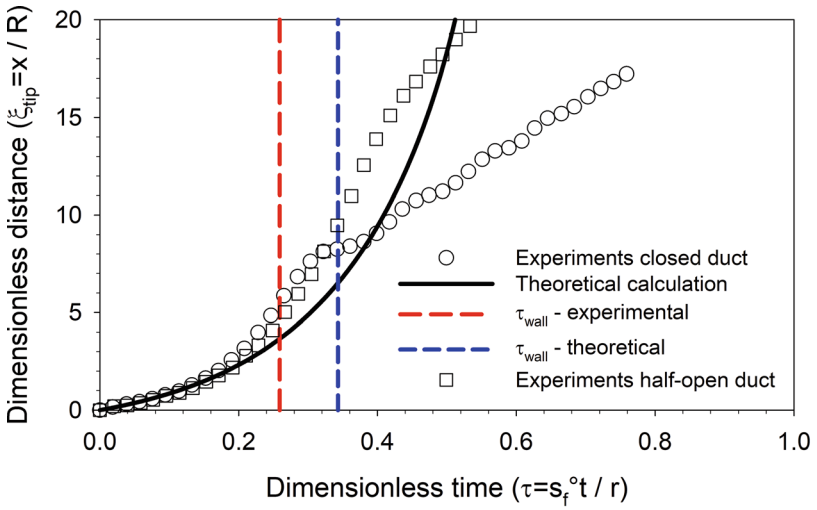


Fig. 5.8 Comparison of theoretical calculations with the experimental results obtained by Yang et al. [84] for a flame propagating in a syngas–air mixture with $\phi = 0.80$. The syngas is formed by 90% H_2 and 10% CO

The vertical evolution equation is integrated from $\eta_f = 0$ to $\eta_f = 1$ and from $\tau = 0$ to τ_{wall}, where τ_{wall} is the time at which the flame skirt touches the side walls of the duct. Thus, the following expression is obtained.

$$\tau_{\text{wall}} = \frac{\ln \Theta_0}{\Theta_0 - 1} \tag{5.66}$$

Integrating for the flame tip position, from $\xi_{\text{tip}} = 0$ to ξ_{tip} and from $\tau = 0$ to τ,

$$\xi_{\text{tip}} = \frac{\Theta_0}{\Theta_0 - 1}\{\exp[(\Theta_0 - 1)\tau] - 1\} \tag{5.67}$$

Substituting Eq. (5.66) into Eq. (5.67), it is observed that $\xi_{\text{tip}}(\tau_{\text{wall}}) = \Theta_0$. This is the flame tip position at the instant when the flame skirt touches the lateral walls of the duct. Compressibility effects were also introduced by Valiev et al. and through the modifications proposed by Mendiburu [83]. The dimensionless variables in Eq. (5.68) are considered to introduce the compressibility effects. In this case, the scaling variables are the stagnation properties [45]. In addition, the Mach numbers relevant to this analysis are also given in Eq. (5.68).

$$\tilde{T} = \frac{T}{T_{u,0}}; \quad \tilde{p} = \frac{p}{p_{u,0}}; \quad \tilde{\rho} = \frac{\rho}{\rho_{u,0}}; \quad M_a = \frac{s_f^{\circ}}{c_u}; \quad M_u = \frac{v_u}{c_u} \tag{5.68}$$

Assuming that the flow is isentropic, the dimensionless variables can be related to the Mach numbers.

$$\widetilde{T}_u = 1 - \frac{\gamma - 1}{2} M_a^2 v_u^2; \quad \widetilde{p}_u = \left(1 - \frac{\gamma - 1}{2} M_a^2 v_u^2\right)^{\frac{\gamma}{\gamma - 1}}; \quad \widetilde{\rho}_u = \left(1 - \frac{\gamma - 1}{2} M_a^2 v_u^2\right)^{\frac{1}{\gamma - 1}} \tag{5.69}$$

Note that $\widetilde{q} = q/c_p T_{u,0}$, $c_p = \gamma R/(\gamma - 1)$, $p_{u,\,0} = \rho_{u,\,0} R T_{u,\,0}$, and $c_{u,0}^2 = \gamma R T_{u,0}$. The dimensionless forms of the conservation equations across the flame are obtained. Note that the conservation equations obtained here are a little bit different than those obtained by Valiev et al. and were presented by Mendiburu [83].

Mass conservation

$$\widetilde{\rho}_u \left(\frac{d\xi_{\text{tip}}}{d\tau} - v_u\right) = \widetilde{\rho}_b \left(\frac{d\xi_{\text{tip}}}{d\tau} - v_b\right) \tag{5.70}$$

Momentum conservation

$$\widetilde{p}_u + \gamma M_a^2 \widetilde{\rho}_u \left(\frac{d\xi_{\text{tip}}}{d\tau} - v_u\right)^2 = \widetilde{p}_b + \gamma M_a^2 \widetilde{\rho}_b \left(\frac{d\xi_{\text{tip}}}{d\tau} - v_b\right)^2 \tag{5.71}$$

Energy conservation

$$\widetilde{T}_u + \widetilde{q} + \frac{\gamma - 1}{2} M_a^2 \left(\frac{d\xi_{\text{tip}}}{d\tau} - v_u\right)^2 = \widetilde{T}_b + \frac{\gamma - 1}{2} M_a^2 \left(\frac{d\xi_{\text{tip}}}{d\tau} - v_b\right)^2 \tag{5.72}$$

The expansion ratio for a burned gas at a temperature different than the adiabatic flame temperature ($\Theta = \rho_u/\rho_b$) can be written as shown in Eq. (5.73).

$$\Theta \sim 1 + \frac{\Theta_0 - 1}{\widetilde{T}_u} \tag{5.73}$$

For a flame propagation process with low values of the Mach number, it is possible to neglect the terms multiplied by M_a^2. Therefore, Eq. (5.74) is obtained.

$$\widetilde{p}_u \sim \widetilde{p}_b; \quad \widetilde{T}_u + (\Theta_0 - 1) = \widetilde{T}_b \tag{5.74}$$

The mass conservation equation applied to the control volume enclosing the vicinity of the flame front is given by Eq. (5.75). Note that, since the flow is isentropic, there is a simple relationship between the density and the pressure derivatives in terms of τ.

$$\frac{\partial w_b}{\partial \eta} + \frac{\partial v_b}{\partial \xi} = -\frac{1}{\gamma \widetilde{p}_u} \frac{d\widetilde{p}_u}{d\tau} \tag{5.75}$$

Following Valiev et al., the functions presented in Eq. (5.76) satisfy the differential equation presented in Eq. (5.75). Therefore, the evolution equation can be written as shown in Eq. (5.77).

$$v_b = \left[(\Theta - 1) - \frac{1}{\gamma \widetilde{p}_u} \frac{d\widetilde{p}_u}{d\tau} \right] \xi; \quad w_b = -(\Theta - 1)\eta \tag{5.76}$$

$$\frac{d\xi_{\text{tip}}}{d\tau} = \left[(\Theta - 1) - \frac{1}{\gamma \widetilde{p}_u} \frac{d\widetilde{p}_u}{d\tau} \right] \xi_{\text{tip}} + \Theta \tag{5.77}$$

Adopting the definition for $M_{a,1}^2$ given in Eq. (5.78), the expression for $\widetilde{T}_u$ can be rewritten as shown also in Eq. (5.78). Additionally, rearranging the expression for the dimensionless pressure given in Eq. (5.69) and deriving it as a function of τ, provide the two expressions shown in Eq. (5.79).

$$M_{a,1}^2 = \frac{\gamma - 1}{2} M_a^2; \quad \widetilde{T}_u = 1 - M_{a,1}^2 v_u^2 \tag{5.78}$$

$$\frac{1}{\widetilde{p}_u} = \left(1 - M_{a,1}^2 v_u^2\right)^{-\frac{\gamma}{\gamma - 1}}; \quad -\frac{1}{\gamma} \frac{d\widetilde{p}_u}{d\tau} = M_a^2 \left(1 - M_{a,1}^2 v_u^2\right)^{\frac{1}{\gamma - 1}} v_u \frac{dv_u}{d\tau} \tag{5.79}$$

For the velocity of the unburned gas, it is known from mass conservation across the flame that, $v_u - v_b = \Theta_0 - 1$. On the other hand, for the velocity of the burned gas, it is known that $v_b = (\Theta_0 - 1)\xi$, so the first expression in Eq. (5.80) can be used to represent v_u. Moreover, the derivative of this expression with respect to τ gives $dv_u/d\tau$, which is also shown in Eq. (5.80).

$$v_u = \sigma(\Theta_0 - 1)\left(\xi_{\text{tip}} + 1\right); \quad \frac{dv_u}{d\tau} = \sigma(\Theta_0 - 1)\frac{d\xi_{\text{tip}}}{d\tau} \tag{5.80}$$

Substituting Eqs. (5.73), (5.78), (5.79), and (5.80) into Eq. (5.77) and rearranging, yields the evolution equation, which has the form shown in Eq. (5.81) with the coefficients given in Eq. (5.82).

$$\frac{d}{d\tau}\left(\xi_{tip}+1\right)=\frac{a_2\left(\xi_{tip}+1\right)^2-a_3\left(\xi_{tip}+1\right)-1}{\left(a_1+a_2\right)\left(\xi_{tip}+1\right)^2-a_1\left(\xi_{tip}+1\right)-1} \tag{5.81}$$

$$a_1=\sigma^2M_a^2(\Theta_0-1)^2;\quad a_2=\sigma^2M_{a,1}^2(\Theta_0-1)^2;\quad a_3=(\Theta_0-1) \tag{5.82}$$

Note that the factor σ introduced here accounts for the effects of compressibility. The integration of Eq. (5.81) is now performed. The details of this integration are discussed on the Appendix of this chapter. Note that the variable for the integration is $z = \xi_{tip} + 1$. Thus, the integration was performed from $\tau = 0$ and $z_0 = 1$ to τ and $z = \xi_{tip} + 1$. The result of the integration is shown in Eq. (5.83) and the new coefficients obtained are listed in Eq. (5.84).

$$\tau=(b_1+1)\xi_{tip}+\frac{b_2}{2a_2}\ln\left(\frac{a_2\left(\xi_{tip}+1\right)^2-a_3\left(\xi_{tip}+1\right)-1}{a_2-a_3-1}\right)$$

$$-\frac{2b_1a_2+b_2a_3}{2a_2}\frac{2}{\sqrt{4a_2+a_3^2}}\left[arc\tan h\left(\frac{2a_2\left(\xi_{tip}+1\right)-a_3}{\sqrt{4a_2+a_3^2}}\right)-arc\tan h\left(\frac{2a_2-a_3}{\sqrt{4a_2+a_3^2}}\right)\right] \tag{5.83}$$

$$b_1=\frac{2}{\gamma-1};\quad b_2=(\Theta_0-1)\left(\frac{\gamma+1}{\gamma-1}\right)-\sigma^2M_a^2(\Theta_0-1)^2 \tag{5.84}$$

The result shown in Eq. (5.83) was obtained by Mendiburu [83]. The article by Valiev et al. provides two other analytical solutions which were not reproduced here. Another important characteristic of Eq. (5.81) is that, since $d(\xi_{tip} + 1) = d\xi_{tip}$, it can be rearranged to obtain the flame tip propagation velocity as shown below.

$$\frac{d\xi_{tip}}{d\tau}=\frac{d\left(x_{tip}/r\right)}{d\left(ts_f{}^\circ/r\right)}=\frac{1}{s_f{}^\circ}\frac{dx_{tip}}{dt}=\frac{V_{tip}}{s_f{}^\circ} \tag{5.85}$$

Example 5.4

Determine the flame tip position as a function of time for a syngas–air mixture with $\phi = 0.8$. The syngas is formed by 90% H_2 and 10% CO, and the flame is propagating inside a closed duct. The initial temperature and pressure conditions are 298 K and 101.3 kPa, respectively. Compare the calculated results with the experimental data obtained by Yang et al. [84].

Solution

The software Cantera [54] is used to obtain the properties necessary to perform the calculations. A chemical kinetic mechanism is needed in order for Cantera to have the species and their properties; therefore, the San Diego Mechanism [67] was adopted. The burned gas properties were obtained by assuming equilibrium after a constant pressure combustion process. The software has a specific function for obtaining results at equilibrium.

$$W_u = 22.75 \text{ kg/kmol}; \quad \rho_u = 0.918 \text{ kg/m}^3; \quad \rho_b = 0.143 \text{ kg/m}^3$$

$$c_p = 1277.50 \text{ J/(kg K)}; \quad c_v = 912.05 \text{ J/(kg K)}; \quad s_f^\circ = 1.531 \text{ m/s}$$

The ratio of heat capacities (γ) and the gas constant are determined.

$$\gamma = \frac{1277.50}{912.05} = 1.40; \quad R = \frac{8314}{22.75} = 365.45 \text{ J/(kg K)}$$

The expansion ratio (Θ_0), the speed of the sound at the initial conditions (c_0), the Mach number at the initial conditions (M_a), and the $M_{a,1}$ are calculated.

$$\Theta_0 = \frac{\rho_u}{\rho_b} = \frac{0.918}{0.143} = 6.42; \quad c_0 = \sqrt{\gamma RT} = \sqrt{1.4 \times 365.4 \times 298} = 390.44 \text{ m/s}$$

$$M_a = \frac{s_f^\circ}{c_0} = \frac{1.531}{390.54} = 3.92 \times 10^{-3}; \quad M_{a,1}^2 = \frac{1.40 - 1}{2}(3.92 \times 10^{-3})^2 = 3.08 \times 10^{-6}$$

The dimensionless time at which the flame skirt touches the side wall of the duct is calculated from Eq. (5.66).

$$\tau_{\text{wall}} = \frac{\ln 6.42}{6.42 - 1} = 0.343$$

The hydraulic diameter of the duct used by Yang et al. [84] in their experiments is $D_h = 100$ mm. Therefore, the time at which the flame skirt touches the side wall of the duct is calculated from Eq. (5.60).

$$t_{\text{wall}} = \frac{\tau_{\text{wall}} D_h}{2 s_f^\circ} = \frac{0.343 \times 0.1}{2 \times 1.531} \times 10^3 = 11.20 \text{ ms}$$

The coefficients a_1, a_2, and a_3 are calculated from Eq. (5.82). In the calculations the value of $\sigma = 1$ is adopted. Therefore, the influence of this factor is not being considered in this example.

$$a_1 = (3.92 \times 10^{-3})^2 (6.42 - 1)^2 = 4.514 \times 10^{-4}$$

$$a_2 = 3.08 \times 10^{-6} (6.42 - 1)^2 = 9.048 \times 10^{-5}$$

$$a_3 = 6.42 - 1 = 5.42$$

The coefficients b_1 and b_2 are determined from Eq. (5.84).

$$b_1 = \frac{2}{1.4-1} = 5.0$$

$$b_2 = (6.42-1)\frac{2.4}{0.4} - \left(3.92\times 10^{-3}\right)^2 (6.42-1)^2 = 32.519$$

With the values of the a_1, a_2, a_3, b_1, and b_2 coefficients, it is possible to use spreadsheet software to calculate the value of τ as function of the values of ξ_{tip}. The comparison between the theoretical calculations and the experimental results is presented in Fig. 5.8.

The theoretical calculations show good agreement with the experiments until the experimental τ_{wall}, represented by the vertical dashed red line, is reached. It is observed that the theory does not predict the change in flame propagation behavior that takes place once the flame skirt touches the side walls of the duct. In the work by Mendiburu [83] an attempt was made to reproduce that behavior by assuming a value of σ different from unity. Another important feature of this theoretical development is that it does not differentiate between closed and half-open ducts.

5.5.2 Flame Front Inversion Inside a Tube

In this section the theoretical development by Valiev et al. [82] is followed. To begin the analysis, consider the mass conservation equation for an incompressible flow in cylindrical coordinates and in terms of dimensionless variables, as shown in Eq. (5.86).

$$\frac{1}{\eta}\frac{\partial}{\partial \eta}(\eta w) + \frac{\partial v}{\partial \xi} = 0 \tag{5.86}$$

The potential flow assumption is adopted for the unburned gas, with the axial velocity component expressed as $v_u = C_1\xi$. The boundary conditions require that the vertical velocity is w_u at η and $w_u = 0$ at $\eta = 1$. Then, a solution for the velocity component that satisfies the differential equation and the boundary conditions is $w_u = (C_1/2)[(1/\eta) - \eta]$. Potential flow is also assumed for the burned gas near the axis, and the axial velocity component is expressed as $v_b = C_2\xi$. The boundary conditions require that the vertical velocity is w_b at η and $w_b = 0$ at $\eta = 0$. Then, a solution that satisfies the differential equation and the boundary conditions is $w_b = -(C_2/2)\eta$.

Following the same reasoning used to integrate Eq. (5.62), it is observed that the dimensionless radial velocities components are related through $w_u - w_b = \Theta_0 - 1$, while the dimensionless horizontal velocity components are related through $v_u = v_b$. Then, the integration constants appearing in the velocity components are $C_1 = C_2 = 2(\Theta_0 - 1)\eta_f$, and the velocities are given by Eq. (5.87). It should be noted that in this case, in contrast to the case of rectangular geometry, the value of C_1 depends on the coordinate of the radially propagating front (η_f). Note that the flame

front is located at η_f in the radial direction, and the relationship $w_u - w_b = \Theta_0 - 1$ was applied at the flame front.

$$v_u = v_b = 2(\Theta - 1)\eta_f \xi; \quad w_u = (\Theta - 1)\eta_f \left(\frac{1}{\eta} - \eta\right); \quad w_b = -(\Theta - 1)\eta_f \eta \tag{5.87}$$

It is straightforward to verify that Eq. (5.65) is still valid if the radius represents the position of the flame front in the vertical direction. Therefore, substituting w_u from Eq. (5.87), the following differential equation is obtained.

$$\frac{d\eta_f}{d\tau} - (\Theta_0 - 1)\left(1 - \eta_f^2\right) = 1 \tag{5.88}$$

By separating the variables, it is observed that

$$\frac{d\eta_f}{1 - [(\Theta_0 - 1)/\Theta_0]\eta_f^2} = \frac{(1/2)d\eta_f}{1 + \left[\sqrt{(\Theta_0 - 1/)\Theta_0}\right]\eta_f} + \frac{(1/2)d\eta_f}{1 - \left[\sqrt{(\Theta_0 - 1/)\Theta_0}\right]\eta_f} = \Theta_0 d\tau$$

It is possible to see that $(\alpha/\Theta_0)d\eta_f = d[1 + (\alpha/\Theta_0)\eta_f]$, in the case of the first term on the rigth-hand side, and that $-(\alpha/\Theta_0)d\eta_f = d[1 - (\alpha/\Theta_0)\,\eta_f]$, in the case of the second term on the right-hand side, where α is defined in Eq. (5.89).

$$\alpha = \sqrt{\Theta_0(\Theta_0 - 1)} \tag{5.89}$$

Therefore, integrating from $\eta = 0$ when $\tau = 0$ to $\eta = \eta_f$ when the elapsed time is τ, the following result is obtained.

$$\tau = \frac{1}{2\alpha} \ln\left(\frac{\Theta_0 + \alpha n_f}{\Theta_0 - \alpha n_f}\right) \tag{5.90}$$

It is possible to isolate the dimensionless radial coordinate η_f by applying an exponential function such that $e^{\alpha\tau} = [(\Theta_0 + \alpha\eta_f)/(\Theta_0 - \alpha\eta_f)]^{1/2}$ and $e^{-\alpha\tau} = [(\Theta_0 - \alpha\eta_f)/(\Theta_0 + \alpha\eta_f)]^{1/2}$. Therefore, relating the left-hand side of Eq. (5.89) to hyperbolic trigonometric functions, the Eq. (5.91) for η_f is obtained.

$$\eta_f = \frac{\Theta_0}{\alpha} \tanh(\alpha\tau) \tag{5.91}$$

Eq. (5.90) relates the dimensionless position of the radially propagating flame front (η_f) to the dimensionless time (τ). When the flame front touches the side walls of the tube, $\eta_f = 1$, and the dimensionless time, τ_{wall}, will be given by Eq. (5.92).

$$\tau_{\text{wall}} = \frac{1}{2\alpha} \ln\left(\frac{\Theta_0 + \alpha}{\Theta_0 - \alpha}\right) \tag{5.92}$$

Equation (5.65) provides also a relation between the most advanced point of the flame front in the horizontal direction (ξ_{tip}) and the velocity of the burned gases.

Substituting the velocity v_b given in Eq. (5.87) and using the expression for η_f given in Eq. (5.91), the following differential equation is obtained.

$$\frac{d\xi_{\text{tip}}}{d\tau} - 2\alpha\tanh(\alpha\tau)\xi_{\text{tip}} = \Theta_0 \tag{5.93}$$

It is possible to rewrite Eq. (5.93) by multiplying all terms by $\cosh^2(\alpha\tau)$; thus, the following relationship is observed.

$$\frac{\cosh^2(\alpha\tau)\frac{d\xi_{\text{tip}}}{d\tau} - 2\alpha\sinh(\alpha\tau)\cosh(\alpha\tau)\xi_{\text{tip}}}{\cosh^4(\alpha\tau)} = \frac{\Theta_0}{\cosh^2(\alpha\tau)}$$

It is evident that the term on the left-hand side of the expression above is the derivative of the quotient $\xi_{\text{tip}}/\cosh^2(\alpha\tau)$ as a function of τ. Furthermore, after separating variables, the right-hand side will have the form $d\tau/\cosh^2(\alpha\tau)$, which has a known solution. The result of the integration is presented in Eq. (5.94), where the identity $sinh(2x) = 2sinh(x)cosh(h)$ was used.

$$\xi_{\text{tip}} = \frac{\Theta_0}{2\alpha}\sinh(2\alpha\tau) \tag{5.94}$$

Substituting Eq. (5.92) into Eq. (5.94), it is observed that $\xi_{\text{tip}}(\tau_{\text{wall}}) = \Theta_0$. The same result was also observed in the case of the rectangular geometry. Considering the case where the time is sufficiently long, it can be assumed $tanh(\alpha\tau) \sim 1$ and the Eq. (5.93) becomes

$$\frac{d\xi_{\text{tip}}}{d\tau} = 2\alpha\xi_{\text{tip}} + \Theta_0 \tag{5.95}$$

Integrating the expression given in Eq. (5.95) from $\tau = 0$ where $\xi_{\text{tip}} = 0$ up to time τ where the dimensionless distance traveled by the most advanced point of the flame front is ξ_{tip}, the following result is obtained.

$$\xi_{\text{tip}} = \frac{\Theta_0}{2\alpha}[\exp(2\alpha\tau) - 1] \tag{5.96}$$

The expressions for $\widetilde{T}_u$, $\widetilde{p}_u$ e $\widetilde{\rho}_u$, given in Eq. (5.69) are also valid for this case. The expression for v_u is obtained using Eqs. (5.65) and (5.95) while observing that $v_b = v_u - (\Theta_0 - 1)$.

$$v_u = 2\alpha\xi_{\text{tip}} + (\Theta_0 - 1)\,; \quad dv_u = 2\alpha d\xi_{\text{tip}} \tag{5.97}$$

Applying the same strategy that was adopted in the analysis of the rectangular geometry, based on the mass conservation equation, the following is written.

$$\frac{1}{\eta}\frac{\partial}{\partial\eta}(\eta w_b) + \frac{\partial v_b}{\partial\xi} = -\frac{1}{\gamma\widetilde{p}_u}\frac{d\widetilde{p}_u}{d\tau} \tag{5.98}$$

The integration is performed in the vicinity of the axis. Therefore, the following functions are adopted, which satisfy Eq. (5.98).

$$v_b = 2(\Theta - 1)\eta_f \xi - \frac{1}{\gamma \tilde{p}_u}\frac{d\tilde{p}_u}{d\tau}\xi\,; \quad w_b = -(\Theta - 1)\eta_f \eta \tag{5.99}$$

Substituting v_b from Eq. (5.99) into Eq. (5.65), noting that Θ_0 is now replaced by Θ, gives the equation for the evolution of the flame front in the axial direction.

$$\frac{d\xi_{\text{tip}}}{d\tau} = \left[2(\Theta - 1)\eta_f - \frac{1}{\gamma \tilde{p}_u}\frac{d\tilde{p}_u}{d\tau}\right]\xi_{\text{tip}} + \Theta \tag{5.100}$$

Similarly to Eq. (5.88), the equation for the evolution of the flame front in the radial direction, accounting for compressibility effects, is given by Eq. (5.101).

$$\frac{d\eta_f}{d\tau} - (\Theta - 1)\left(1 - \eta_f^2\right) = 1 \tag{5.101}$$

The solution to Eq. (5.101) is the same as that presented previously in Eq. (5.94), but with the term $\widehat{\alpha} = \sqrt{\Theta(\Theta - 1)}$ instead of α.

$$\eta_f = \frac{\Theta}{\hat{\alpha}}\tan h(\hat{\alpha}\tau) \tag{5.102}$$

It will be necessary to express $\widehat{\alpha}$ in terms of v_u. Substituting into the equation for $\widehat{\alpha}$ the expressions for Θ and $\tilde{T}_u$ given in Eqs. (5.73) and (5.69), respectively, and neglecting the terms with M_a^4 give the result shown in Eq. (5.103).

$$\widehat{\alpha} = \alpha\sqrt{1 + M_{a,1}^2 \alpha_1 v_u^2} \tag{5.103}$$

where

$$\alpha_1 = \frac{2\Theta_0 - 1}{\alpha^2} \tag{5.104}$$

In the present case it will be easier to solve the differential equation for the velocity of the unburned gas, v_u, and relate it to ξ_{tip} by using Eq. (5.97). Then, Eqs. (5.73), (5.78), (5.79), (5.97), and (5.102) are substituted into Eq. (5.100).

$$\frac{dv_u}{d\tau} = \frac{2\hat{\alpha}\tanh(\hat{\alpha}\tau)[v_u - (\Theta_0 - 1)] + 2\alpha(\Theta_0 - 1)(1 - M_{a,1}^2 v_u^2)^{-1} + 2\alpha}{1 - M_a^2 v_u[v_u - (\Theta_0 - 1)](1 - M_{a,1}^2 v_u^2)^{-1}} \tag{5.105}$$

The above equation should be solved by numerical methods. However, the validity of Eq. (5.93) will be evaluated instead. From the calculations performed in Example 5.3, it was observed that $M_{a,1}^2$ is of order 10^{-6}. Thus, it can be assumed that $M_{a,1}^2 \alpha_1 v_u^2 << 1$, and from Eq. (5.103) it is observed that $\widehat{\alpha} \sim \alpha$. Some other terms were also estimated, and the results are shown in Table 5.1. Additionally,

Table 5.1 Estimation of some terms of Eqs. (5.105) and (5.106) with $\Theta_0 = 6.422$, $\gamma = 1.4$ taken from Example 5.4. The following functions were calculated, $f(v_u) = 1 - M_{a,1}^2 v_u^2$ and $g(\xi_{tip}) = 1 - 4\alpha^2 M_a^2 \xi_{tip}^2 - 2\alpha M_a^2(\Theta_0 - 1)\xi_{tip}$

ξ_{tip}	v_u	$\widehat{\alpha}$	$f(v_u)$	$g(\xi_{tip})$
0.1	6.60	5.90	1.000	1.000
0.2	7.78	5.90	1.000	1.000
0.5	11.32	5.90	1.000	0.999
1.0	17.22	5.90	0.999	0.997
1.5	23.12	5.90	0.998	0.994
2.0	29.03	5.90	0.997	0.989
5.0	64.43	5.91	0.987	0.942
7.0	88.03	5.92	0.976	0.888
10.0	123.44	5.95	0.953	0.776
12.0	147.04	5.97	0.933	0.680
15.0	182.45	6.00	0.897	0.504
20.0	241.46	6.08	0.820	0.124

substituting the expressions in Eq. (5.97) into the above differential equation and rearranging yield the following evolution equation.

$$\frac{d\xi_{tip}}{d\tau} = \frac{2\alpha \tanh(\alpha\tau)\xi_{tip} + \Theta_0}{1 - 2\alpha M_a^2(\Theta_0 - 1)\xi_{tip} - 4\alpha^2 M_a^2 \xi_{tip}^2} \tag{5.106}$$

It is important to notice that Eq. (5.106) provides a relationship between the flame tip velocity and the flame tip position, as was shown in Eq. (5.85).

Therefore, assuming that, $f(v_u)$ and $g(\xi_{tip})$ are both approximately equal to unity. This allows for the simplification of Eq. (5.105) to obtain Eq. (5.106) and also for the simplification of Eq. (5.106) to obtain Eq. (5.93). Therefore, the solution without compressible effects (Eq. 5.93) can be reasonably applied up to $\xi_{tip} < 7$.

Example 5.5

Determine the flame tip position as a function of time for a syngas–air mixture with $\phi = 0.8$. The syngas is formed by 90% H_2 and 10% CO, and the flame is propagating inside a closed tube. The initial temperature and pressure conditions are 298 K and 101.3 kPa, respectively. Compare the results with those obtained in Example 5.4.

Solution

The solution will be presented without considering the compressibility effects. Moreover, the parameters which are necessary for this calculation are only Θ_0 and α. The former was determined as $\Theta_0 = 6.42$ in Example 5.4, and the latter will be determined below.

$$\alpha = \sqrt{6.42(6.42 - 1)} = 5.90$$

The time at which the flame skirt touches the side wall of the tube is determined by Eq. (5.91). The diameter of the tube will be considered equal to the hydraulic diameter of the duct used in Example 5.4, that is, $D = 0.1$ m.

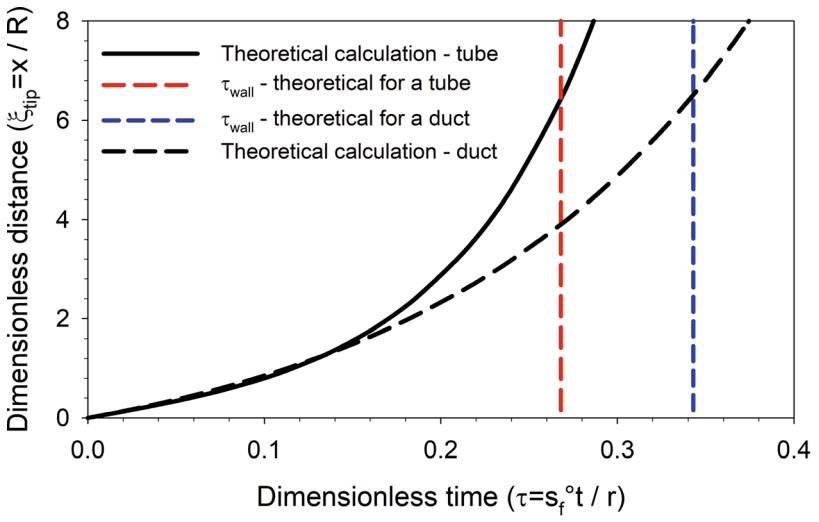


Fig. 5.9 Comparison of theoretical calculations for a flame propagating in a syngas–air mixture with $\phi = 0.80$, considering a closed tube and a closed duct. The syngas is formed by 90% H_2 and 10% CO

$$\tau_{\text{wall}} = \frac{1}{2 \times 5.90} \ln\left(\frac{6.42 + 5.90}{6.42 - 5.90}\right) = 0.269$$

$$t_{\text{wall}} = \frac{\tau_{\text{wall}} D_h}{2 s_f^{\circ}} = \frac{0.269 \times 0.1}{2 \times 1.531} \times 10^3 = 8.78 \text{ ms}$$

Note that, for the case of the closed duct, the $\tau_{\text{wall}} = 0.343$ which representes a time $t_{\text{wall}} = 11.20$ ms. Therefore, for a tube and a duct with equal hydraulic diameters, a mixture propagating inside the tube will reach the side walls faster than the same mixture propagating inside the duct. The flame tip position is calculated by Eq. (5.93), and the results are shown in Fig. 5.9. The results are presented up to $\xi_{\text{tip}} = 8$, because it would not be accurate to use the solution without compressibility effects at further distances. It is observed that at the early stages, the flame tip position is similar regardless of the flame geometry. However, the flame propagating inside the tube shows a steeper slope, which means a higher velocity, starting around $\tau = 0.16$.

The flame tip velocity can also be obtained by calculating the derivative $d\xi_{\text{tip}}//d\tau$ given in Eqs. (5.81) and (5.106) for a duct and a tube, respectively. The results of this calculations are presented in Fig. 5.10. It is observed that the flame tip velocity for the mixture propagating inside the tube is higher than the velocity inside the duct for the early stages of flame propagation.

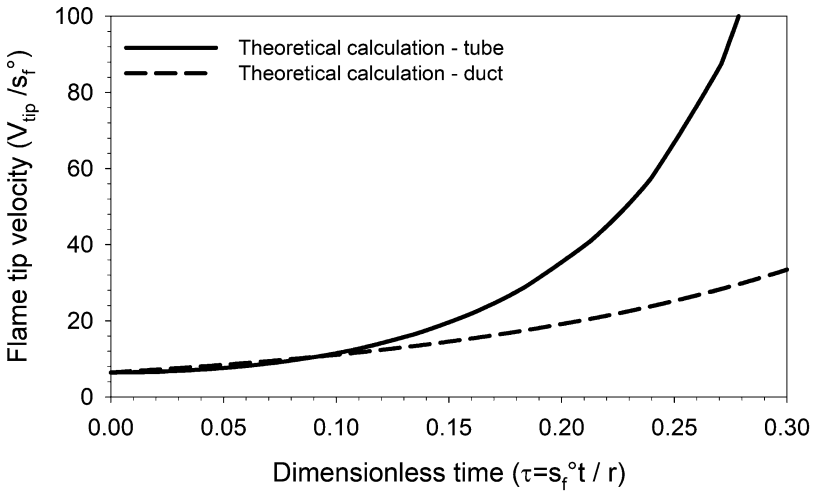


Fig. 5.10 Comparison of the theoretical flame tip velocity for a syngas–air mixture with $\phi = 0.80$ considering a closed tube and a closed duct. The syngas is formed by 90% H_2 and 10% CO

As a final remark, note that Eqs. (5.81) and (5.106) each present a maximum velocity. Moreover, the velocities become negative for sufficiently large values of ξ_{tip}. In the case of Eq. (5.81), the maximum could be found by setting its numerator equal to zero. However, in the case of Eq. (5.106), the maximum is reached for small values of the denominator which becomes negative for sufficiently large values of ξ_{tip}. Therefore, the following equations are good measures of the distance at which the maximum velocity is reached.

$$1 + (\Theta_0 - 1)(\xi_{tip} + 1) - \kappa^2 M_{a,1}^2 (\Theta_0 - 1)^2 (\xi_{tip} + 1)^2 = 0 \tag{5.107}$$

$$1 - 2\alpha M_a^2 (\Theta_0 - 1)\xi_{tip} - 4\alpha^2 M_a^2 \xi_{tip}^2 = 0 \tag{5.108}$$

The order of magnitude of the maximum flame tip propagation velocity, obtained from Eqs. (5.81) and (5.106), is far from what is observed in the experiments. Further developments are needed in order to obtain models that can predict the experimental behavior more accurately.

5.6 The Governing Equations in Dimensionless Form

In this section a classical representation of the conservation equations is discussed. This formulation was used by Matalon and Matkowsky [72] to study the flame as a gas dynamic discontinuity. The flame is represented by a thin control volume, inside which a thinner reaction zone can be found, as shown in Fig. 5.11. The following considerations are applied to the phenomenon [85]:

(i) The propagation velocity is much lower than the speed of sound. Thus, the acoustic perturbations propagate relatively fast, and the pressure is almost instantaneously equalized within the system.
(ii) The reactants mixture is treated as a mixture of ideal gases with constant heat capacities, c_p and c_v, with a constant $\gamma = c_p/c_v$, and a representative mean molecular weight, W_m. Moreover, c_p is equal for all the species.
(iii) The chemical process is modelled as a single-step Arrhenius-type reaction.
(iv) The combustion process is adiabatic.
(v) The products of properties λ/c_p and ρD are assumed constant.

It is considered that the chemical reaction is dominated by the species in deficit, which has a mass fraction Y_A. The reaction rate is then given by Eq. (5.109).

$$\dot{m}_R = -B\rho Y_A e^{-E/RT} \tag{5.109}$$

Note that, since $\dot{m}_R$ must be in kg/m^3 s, the units of B must be 1/s. The conservation equations applied to this phenomenon are the mass conservation equation, Eq. (3.8); the species mass conservation equation, Eq. (3.17); the momentum equation, Eq. (3.33); and the energy conservation equation, Eq. (3.74). The effect of body forces is neglected. Furthermore, the terms involving viscous dissipation, heat loss, and work done by the surface forces are neglected. Since all c_p are equal, the following is obtained.

$$\sum\left(\dot{m}_{k,R}\int_{T_{\text{ref}}}^{T} c_p dT\right) = 0; \quad \sum\left[\rho D\nabla Y_k \cdot \nabla\left(\int_{T_{\text{ref}}}^{T} c_{p,k} dT\right)\right] = 0$$

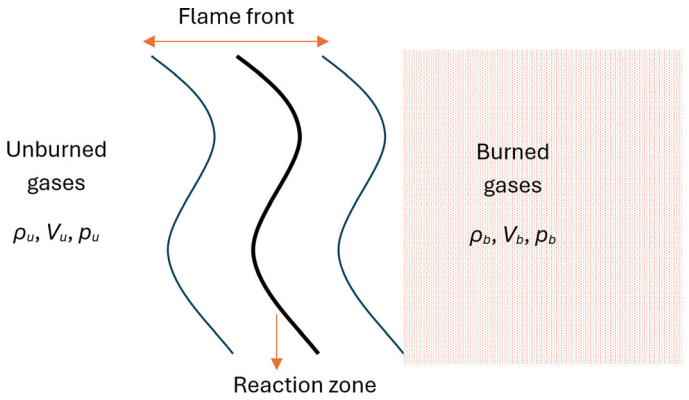


Fig. 5.11 Schematics of flame thin control volume and the even thinner reaction zone. (Adapted from Matalon and Matkowsky [72])

The above expressions mean that the sensible enthalpy lost by a particular species is gained by other species and that the summation of the mass fraction gradients for all the species is equal to zero.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho \vec{V} \right) = 0 \tag{5.110}$$

$$\frac{\partial (\rho Y_A)}{\partial t} + \nabla \cdot \left(\rho \vec{V} Y_A - \rho D \nabla Y_A \right) = \dot{m}_R \tag{5.111}$$

$$\rho \frac{D \vec{V}}{Dt} = \nabla \cdot \overleftrightarrow{\sigma} \tag{5.112}$$

$$\rho c_p \frac{DT}{Dt} = \frac{Dp}{Dt} + \nabla \cdot (k \nabla T) - \sum \left(h^0_{f,k} \dot{m}_{k,R} \right) \tag{5.113}$$

Note that the stress tensor is given by Eq. (3.43), and it is rewritten below for completeness.

$$\overleftrightarrow{\sigma}_{ij} = \mu \left[\nabla \vec{V} + \left(\nabla \vec{V} \right)^T \right]_{ij} + \left[-p - \frac{2}{3} \mu \nabla \cdot \vec{V} \right] \vec{e}_i \vec{e}_j \delta_{ij} \tag{5.114}$$

The conservation equations, given in Eqs. (5.110)–(5.114), will be made dimensionless using the following scaling parameters:

- The adiabatic laminar flame velocity of a plane flame (s_f°)
- Initial temperature (T_0)
- Initial pressure [$p_0 = \rho_0 (s_f^\circ)^2$]
- Initial density (ρ_0)
- Characteristic length of the problem (L)
- Characteristic time ($t_0 = L/s_f^\circ$)

Therefore, the following dimensionless variables are defined:

- Dimensionless velocity ($\vec{v} = \vec{V}/s_f{}^\circ$)
- Dimensionless temperature ($\widetilde{T} = T/T_0$)
- Dimensionless pressure ($\widetilde{p} = p/p_0$)
- Dimensionless density ($\widetilde{\rho} = \rho/\rho_0$)
- The dimensionless coordinates ($\widetilde{x}_i = x_i/L$)
- The dimensionless time ($\tau = t s_f{}^\circ /L$)

Then, taking ϕ as any scalar quantity, the following relationships between the derivatives and the differential operators can be easily verified.

$$\frac{\partial \phi}{\partial x_i} = \frac{\phi_0}{L}\frac{\partial \widetilde{\phi}}{\partial \widetilde{x}_i} \quad \frac{\partial^2 \phi}{\partial x_i^2} = \frac{\phi_0}{L^2}\frac{\partial^2 \widetilde{\phi}}{\partial \widetilde{x}_i^2} \quad \frac{\partial \phi}{\partial t} = \phi_0 \frac{{s_f}^\circ}{L}\frac{\partial \widetilde{\phi}}{\partial \tau} \tag{5.115}$$

$$\nabla \phi = \frac{\phi_0}{L}\widetilde{\nabla}\widetilde{\phi} \quad \nabla \cdot \nabla \phi = \Delta \phi = \frac{\phi_0}{L^2}\widetilde{\Delta}\widetilde{\phi} \quad \frac{D\phi}{Dt} = \phi_0 \frac{{s_f}^\circ}{L}\frac{\widetilde{D}\widetilde{\phi}}{\widetilde{D}\tau} \tag{5.116}$$

The operators are given below for completeness.

$$\widetilde{\nabla}\widetilde{\phi} = \vec{e}_1 \frac{\partial \widetilde{\phi}}{\partial \widetilde{x}_1^2} + \vec{e}_2 \frac{\partial \widetilde{\phi}}{\partial \widetilde{x}_2^2} + \vec{e}_3 \frac{\partial \widetilde{\phi}}{\partial \widetilde{x}_3^2} \tag{5.117}$$

$$\widetilde{\Delta}\widetilde{\phi} = \frac{\partial^2 \widetilde{\phi}}{\partial \widetilde{x}_1^2} + \frac{\partial^2 \widetilde{\phi}}{\partial \widetilde{x}_2^2} + \frac{\partial^2 \widetilde{\phi}}{\partial \widetilde{x}_3^2} \tag{5.118}$$

$$\frac{\widetilde{D}\widetilde{\phi}}{\widetilde{D}\tau} = \frac{\partial \widetilde{\phi}}{\partial \tau} + \vec{v} \cdot \widetilde{\nabla}\widetilde{\phi} \tag{5.119}$$

Applying the above relations to the mass conservation equation, the term $\rho_0 {s_f}^\circ / L$ will appear multiplying each term on the left-hand side; therefore, the following is obtained.

$$\frac{\partial \widetilde{\rho}}{\partial \tau} + \widetilde{\nabla} \cdot \left(\widetilde{\rho} \vec{v} \right) = 0 \tag{5.120}$$

The species conservation equation is made dimensionless by applying the dimensionless derivatives, then using the mass conservation (Eq. 5.110), and finally rearranging the terms, making use of the definitions of δ and l_d. More details are presented in the Appendix of this chapter.

$$\widetilde{\rho}\left(\frac{\partial \widetilde{Y}_A}{\partial \tau} + \vec{v}\,\widetilde{\nabla}\widetilde{Y}_A \right) - \delta Le^{-1}\widetilde{\Delta}\widetilde{Y}_A = -\delta \widetilde{m}_R \tag{5.121}$$

where the dimensionless reaction rate is shown in Eq. (5.122). Moreover, the pre-exponential factor is chosen in such a way that the dimensionless flame speed of the plane adiabatic flame is unity.

$$\widetilde{m}_R = \Lambda \delta^{-2} \widetilde{\rho} \widetilde{Y}_A \exp\left(-\frac{E}{RT} \right) \tag{5.122}$$

Applying the derivatives in dimensionless form to the momentum equation, shown in Eq. (5.123) is obtained. More details on this procedure are given in the Appendix of this chapter.

$$\widetilde{\rho}\left(\frac{\partial \vec{\widetilde{v}}}{\partial \tau}+\vec{\widetilde{v}}\cdot\widetilde{\nabla}\vec{\widetilde{v}}\right)=-\widetilde{\nabla}\widetilde{p}+\delta Pr\left[\widetilde{\Delta}\widetilde{\vec{v}}+\frac{1}{3}\widetilde{\nabla}\left(\widetilde{\nabla}\cdot\vec{\widetilde{v}}\right)\right] \tag{5.123}$$

Finally, the energy conservation equation is made dimensionless by a procedure analogous to the previous ones.

$$\widetilde{\rho}\left(\frac{\partial \widetilde{T}}{\partial \tau}+\widetilde{v}\cdot\nabla\widetilde{T}\right)-\delta\widetilde{\Delta}\widetilde{T}=\delta q\Omega \tag{5.124}$$

The dimensionless heat release rate, in terms of the species in deficit, A, is given by Eq. (5.125). More details on this procedure are given in the Appendix of this chapter.

$$q=\frac{Y_{A,0}}{c_pT_0}\sum\left[h^0_{f,F}+AF\cdot h^0_{f,Ox}-(1+AF)h^0_{f,P}\right] \tag{5.125}$$

The conservation equations can also be written in a form in which the coordinate system is attached to the flame front as shown by Matalon and Matkowsky [72]. The new coordinate system is presented as reference below. However, the interested reader should refer to the work of Matalon and Matkowsky [72] for further details on how the conservation equations are written in this new coordinate system.

$$\xi=\widetilde{x}-f(\widetilde{y},\widetilde{z},\tau);\quad x=x;\quad y=y;\quad z=z;\quad t=t \tag{5.126}$$

5.7 Closure

This chapter discussed the interesting topic of flame dynamics. It started by presenting the flame as a discontinuity, introducing the vector equations of the flame surface, the definitions of the normal unity vector, the normal flame propagation velocity, and the relative flame speed. These definitions are given in detailed form. It should be pointed out that these definitions are related to lectures on fluid dynamics. Afterwards, the concept of flame deformation, or stretch, is introduced. The flame stretch theory was presented by deriving the stretch rate equation and analyzing the effect of flame stretch on the flame speed and temperature. The dimensionless numbers of Karlovic and Markstein were introduced together with the Markstein length, which is important to fundamental experiments on laminar flame velocity. Afterwards, the chapter presented some theoretical and experimental aspects of a flame propagating inside a closed (or half-open) duct (or tube). Finally, the chapter was closed by introducing the dimensionless form of the governing equations.

Appendix

Some Steps for the Derivation of the Stretch Rate Relationship

The area at time $t + \delta t$ is given by Eq. (5.127).

$$\vec{A}(t+\delta t) = \vec{i}\left[\left(\delta y + \frac{\partial s_2}{\partial y}\delta t\delta y\right)\left(\delta z + \frac{\partial s_3}{\partial z}\delta t\delta z\right) - \left(\frac{\partial s_3}{\partial y}\delta t\delta y\right)\left(\frac{\partial s_2}{\partial z}\delta t\delta z\right)\right]$$
$$- \vec{j}\left[\left(f_y\delta y + \frac{\partial s_1}{\partial y}\delta t\delta y\right)\left(\delta z + \frac{\partial s_3}{\partial z}\delta t\delta z\right) - \left(\frac{\partial s_3}{\partial y}\delta t\delta y\right)\left(f_z\delta z + \frac{\partial s_1}{\partial z}\delta t\delta z\right)\right]$$
$$+ \vec{k}\left[\left(f_y\delta y + \frac{\partial s_1}{\partial y}\delta t\delta y\right)\left(\frac{\partial s_2}{\partial z}\delta t\delta z\right) - \left(\delta y + \frac{\partial s_2}{\partial y}\delta t\delta y\right)\left(f_z\delta z + \frac{\partial s_1}{\partial z}\delta t\delta z\right)\right] \tag{5.127}$$

Effecting the internal products, neglecting terms multiplied by $(\delta t)^2$, and using the expressions for $\partial s_1/\partial y$ and $\partial s_1/\partial z$,

$$\vec{A}(t+\delta t) = \vec{i}\left(\frac{1}{\delta t} + \frac{\partial s_3}{\partial z} + \frac{\partial s_2}{\partial y}\right)\delta t\delta y\delta z - \vec{j}\left(\frac{f_y}{\delta t} + \frac{D''f_y}{Dt} + f_y\frac{\partial s_3}{\partial z} + f_y\frac{\partial s_2}{\partial y}\right)\delta t\delta y\delta z$$
$$- \vec{k}\left(\frac{f_z}{\delta t} + \frac{D''f_z}{Dt} + f_z\frac{\partial s_3}{\partial z} + f_z\frac{\partial s_2}{\partial y}\right)\delta t\delta y\delta z \tag{5.128}$$

Now the module of the above vector is obtained.

$$\frac{|A(t+\delta t)|^2}{(\delta t\delta y\delta z)^2} = \left(1+f_y^2+f_z^2\right)\frac{1}{\delta t^2} + \left(1+f_y^2+f_z^2\right)\frac{2}{\delta t}\frac{\partial s_2}{\partial y} + \left(1+f_y^2+f_z^2\right)\frac{2}{\delta t}\frac{\partial s_3}{\partial z}$$
$$+\left(1+f_y^2+f_z^2\right)\left(\frac{\partial s_2}{\partial y}\right)^2 + \left(1+f_y^2+f_z^2\right)\left(\frac{\partial s_3}{\partial z}\right)^2 + \left(1+f_y^2+f_z^2\right)\left(2\frac{\partial s_2}{\partial y}\frac{\partial s_3}{\partial z}\right)$$
$$+\frac{1}{\delta t}\frac{D''}{Dt}\left(1+f_y^2+f_z^2\right) + \left(\frac{\partial s_2}{\partial y}+\frac{\partial s_3}{\partial z}\right)\frac{D''}{Dt}\left(1+f_y^2+f_z^2\right) + \left(\frac{D''f_z}{Dt}\right)^2 + \left(\frac{D''f_y}{Dt}\right)^2 \tag{5.129}$$

For $N^2 = \left(1+f_y^2+f_z^2\right)$, we have $D''(N^2)/Dt = 2N(D''N/Dt)$. This allows simplifying the expression for the area. Again, neglecting the terms multiplied by $(\delta t)^2$ and rearranging,

$$\frac{|A(t+\delta t)|^2}{(\delta y\delta z)^2} = \left(1+f_y^2+f_z^2\right)\left[1 + \frac{\partial s_2}{\partial y}\delta t + \frac{\partial s_3}{\partial z}\delta t + \delta t\frac{\frac{D''}{Dt}\left(1+f_y^2+f_z^2\right)^{1/2}}{\left(1+f_y^2+f_z^2\right)^{1/2}}\right]^2 \tag{5.130}$$

Obtaining the Dimensionless Form of the Conservation Equations

Applying Eqs. (5.115) and (5.116) into Eq. (5.110), the following is obtained.

$$\rho_0 \frac{s_L}{L} \frac{\partial \widetilde{\rho}}{\partial t} + \rho_0 \frac{s_L}{L} \widetilde{\nabla} \cdot \left(\widetilde{\rho} \, \vec{v} \right) = 0 \tag{5.131}$$

Then,

$$\frac{\partial \widetilde{\rho}}{\partial t} + \widetilde{\nabla} \cdot \left(\widetilde{\rho} \, \vec{v} \right) = 0 \tag{5.132}$$

Applying Eqs. (5.115) and (5.116) into Eq. (5.111) and rearranging by using the properties of the divergence and derivatives,

$$Y_A \left[\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \left(\rho \vec{V} \right) \right] + \rho \frac{\partial Y_A}{\partial t} + \rho \vec{V} \vec{\nabla} Y_A - \nabla Y_A [\nabla \cdot (\rho D)] - \rho D (\Delta Y_A) = \dot{m}_R \tag{5.133}$$

Comparison of the expression above with Eq. (5.110) and the assumption which established that ρD is a constant, show that the first and fourth terms on the left-hand side must be equal to zero, then,

$$\rho \left(\frac{\partial Y_A}{\partial t} + \vec{V} \vec{\nabla} Y_A \right) - \rho D (\Delta Y_A) = \dot{m}_R \tag{5.134}$$

Applying dimensionless derivatives and rearranging,

$$\rho \left(\frac{\partial \widetilde{Y}_A}{\partial \tau} + \vec{v} \, \widetilde{\nabla} Y_A \right) - \frac{\rho D}{s_f{}^\circ L} \left(\widetilde{\Delta} \widetilde{Y}_A \right) = \frac{L}{s_f{}^\circ} \frac{1}{Y_{A,0}} \dot{m}_R \tag{5.135}$$

It is possible now to multiply and divide by thermal diffusivity in the second term of the left-hand side and to divide all terms by the initial density. Also, by using the definition of the reaction rate given in Eq. (5.109) and rearranging,

$$\widetilde{\rho} \left(\frac{\partial \widetilde{Y}_A}{\partial \tau} + \vec{v} \, \widetilde{\nabla} Y_A \right) - Le^{-1} \frac{k/\rho_0 c_p}{s_f{}^\circ L} \widetilde{\Delta} \widetilde{Y}_A = - \frac{LB}{s_f{}^\circ} \widetilde{\rho} \widetilde{Y}_A e^{-E/RT} \tag{5.136}$$

It is possible to make some more simplifications to the second term on the left-hand side and to the term on the right-hand side of Eq. (5.136) to obtain exactly the desired form of this conservation equation.

$$\widetilde{\rho} \left(\frac{\partial \widetilde{Y}_A}{\partial \tau} + \vec{v} \, \widetilde{\nabla} Y_A \right) - \left(\frac{l_d}{L} \right) Le^{-1} \widetilde{\Delta} \widetilde{Y}_A = - \left(\frac{l_d}{L} \right) \left(\frac{L}{l_d} \right)^2 \left(\frac{l_d B}{s_f{}^\circ} \right) \widetilde{\rho} \widetilde{Y}_A e^{-E/RT} \tag{5.137}$$

Now the attention is turned to the momentum equation, which must be made dimensionless by applying the dimensionless derivatives and operators. However, the first step is to apply the operators.

$$\rho\left(\frac{\partial \vec{V}}{\partial t}+\vec{V}\cdot\nabla\vec{V}\right)=-\nabla p+\mu\Delta\vec{V}+\frac{1}{3}\mu\nabla\left(\nabla\cdot\vec{V}\right) \tag{5.138}$$

Note that by comparing the present formulation of the stress tensor and the momentum equation presented by Matalon, it is inferred that the following should be obtained.

$$\mu\left[\nabla\cdot\left(\nabla\vec{V}\right)^{T}\right]-\frac{2}{3}\mu\nabla\left(\nabla\cdot\vec{V}\right)=\frac{1}{3}\mu\nabla\left(\nabla\cdot\vec{V}\right) \tag{5.139}$$

The demonstration of the above equation will not be presented here. Now it is possible to apply the dimensionless derivatives and operators.

$$\rho\frac{\left(s_f{}^{\circ}\right)^2}{L}\left(\frac{\partial \vec{v}}{\partial\tau}+\vec{v}\cdot\widetilde{\nabla}\vec{v}\right)=-\frac{p_0}{L}\widetilde{\nabla}\widetilde{p}+\mu\frac{s_f{}^{\circ}}{L^2}\left[\widetilde{\Delta}\widetilde{v}+\frac{1}{3}\widetilde{\nabla}\left(\widetilde{\nabla}\cdot\vec{v}\right)\right] \tag{5.140}$$

The Prandtl number is defined as $\mathrm{Pr}=\mu c_p/k$, and knowing that $\delta=(k/\rho_0 c_p)/(s_f{}^{\circ}L)$, the following can be written,

$$\widetilde{\rho}\left(\frac{\partial \vec{v}}{\partial\tau}+\vec{v}\cdot\widetilde{\nabla}\vec{v}\right)=-\frac{p_0}{\left(s_f{}^{\circ}\right)^2\rho_0}\widetilde{\nabla}\widetilde{p}+\delta\mathrm{Pr}\left[\widetilde{\Delta}\widetilde{v}+\frac{1}{3}\widetilde{\nabla}\left(\widetilde{\nabla}\cdot\vec{v}\right)\right] \tag{5.141}$$

In the above expression, it is noticed that $p_0/[\rho_0(s_f^{\circ})^2]=1$. For the energy conservation equation, dropping the term Dp/Dt and applying the dimensionless derivatives and operators, the following is obtained.

$$\widetilde{\rho}c_pT_0\frac{s_f{}^{\circ}}{L}\left(\frac{\partial\widetilde{T}}{\partial\tau}+\widetilde{v}\cdot\nabla\widetilde{T}\right)=\frac{k}{\rho_0}\frac{T_0}{L^2}\widetilde{\Delta}\widetilde{T}-\frac{Y_{A,0}}{Y_{A,0}}\sum\left[h_{f,F}^{\circ}+\nu h_{f,Ox}^{\circ}-(1+\nu)h_{f,P}^{\circ}\right]\frac{\dot{m}_{k,R}}{\rho_0} \tag{5.142}$$

Rearranging and using the definitions,

$$\widetilde{\rho}\left(\frac{\partial\widetilde{T}}{\partial\tau}+\widetilde{v}\cdot\nabla\widetilde{T}\right)=\delta\widetilde{\Delta}\widetilde{T}-q\frac{L}{s_f{}^{\circ}}\frac{1}{\rho_0Y_{A,0}}\dot{m}_{k,R} \tag{5.143}$$

Applying the same procedure applied for the reaction rate in Eq. (5.135),

$$\widetilde{\rho}\left(\frac{\partial\widetilde{T}}{\partial\tau}+\widetilde{v}\cdot\nabla\widetilde{T}\right)=\delta\widetilde{\Delta}\widetilde{T}+q\left(\frac{l_d}{L}\right)\left(\frac{L}{l_d}\right)^2\left(\frac{l_dB}{s_f{}^{\circ}}\right)\widetilde{\rho}\widetilde{Y}_Ae^{-E/RT} \tag{5.144}$$

where the dimensionless heat release is

$$q=\frac{Y_{A,0}}{c_pT_0}\sum\left[h_{f,F}^{\circ}+AF\cdot h_{f,Ox}^{\circ}-(1+AF)h_{f,P}^{\circ}\right] \tag{5.145}$$

Integration of the Differential Equation of Flame Propagation Inside a Duct

The following rational function appears during the integration of Eq. (5.81). Therefore, the division is shown first.

$$\frac{(a_1+a_2)z^2-a_1z-1}{a_2z^2-a_3z-1} \tag{5.146}$$

$a_2z^2-a_3z-1\overline{)(a_1+a_2)z^2-a_1z-1}$ $-(a_1+a_2)z^2+\left(\frac{a_1a_3}{a_2}+a_3\right)z+\left(\frac{a_1}{a_2}+1\right)$ $\left(\frac{a_1a_3}{a_2}+a_3-a_1\right)z+\frac{a_1}{a_2}$	Dividing $(a_1+a_2)z^2$ by a_2z^2, we have $\left(\frac{a_1}{a_2}+1\right)$ We multiply the denominator by the above term and subtract from the numerator

$$\frac{(a_1+a_2)z^2-a_1z-1}{a_2z^2-a_3z-1}=\left(\frac{a_1}{a_2}+1\right)+\frac{\left(\frac{a_1a_3}{a_2}+a_3-a_1\right)z+\frac{a_1}{a_2}}{a_2z^2-a_3z-1} \tag{5.147}$$

We can check that the above result is correct.

$$\frac{(a_1+a_2)z^2-a_1z-1}{a_2z^2-a_3z-1}=\frac{\left(\frac{a_1}{a_2}+1\right)(a_2z^2-a_3z-1)+\left(\frac{a_1a_3}{a_2}+a_3-a_1\right)z+\frac{a_1}{a_2}}{a_2z^2-a_3z-1} \tag{5.148}$$

$$\frac{(a_1+a_2)z^2-a_1z-1}{a_2z^2-a_3z-1}=\frac{(a_1+a_2)z^2+\left(\frac{a_1a_3}{a_2}-\frac{a_1a_3}{a_2}+a_3-a_3-a_1\right)z-\left(\frac{a_1}{a_2}-\frac{a_1}{a_2}+1\right)}{a_2z^2-a_3z-1} \tag{5.149}$$

We have made $z=(\xi_{tip}+1)$ for simplicity, and the coefficients are defined below.

$$\frac{(a_1+a_2)z^2-a_1z-1}{a_2z^2-a_3z-1}=(b_1+1)+\frac{b_2z}{a_2z^2-a_3z-1}+\frac{b_1}{a_2z^2-a_3z-1} \tag{5.150}$$

where

$$a_1=\sigma^2M_a^2(\Theta_0-1)^2;\quad a_2=\sigma^2M_{a,1}^2(\Theta_0-1)^2;\quad a_3=(\Theta_0-1) \tag{5.151}$$

$$b_1=\frac{a_1}{a_2}=\frac{2}{\gamma-1} \tag{5.152}$$

$$b_2=\frac{a_1a_3+a_3a_2-a_1a_2}{a_2}=(\Theta_0-1)\left(\frac{\gamma+1}{\gamma-1}\right)-\sigma^2M_a^2(\Theta_0-1)^2 \tag{5.153}$$

Using the result obtained in Eq. (5.147) and rearranging, the following can be written.

$$\frac{(a_1+a_2)z^2-a_1z-1}{a_2z^2-a_3z-1}=(b_1+1)+\frac{b_2}{2a_2}\,\frac{2a_2z-a_3}{a_2z^2-a_3z-1}+\frac{1}{2a_2}\,\frac{2b_1a_2+b_2a_3}{a_2z^2-a_3z-1} \tag{5.154}$$

Then, integrating

$$\int\frac{(a_1+a_2)z^2-a_1z-1}{a_2z^2-a_3z-1}dz=\begin{array}{l}(b_1+1)\displaystyle\int dz+\frac{b_2}{2a_2}\int\frac{2a_2z-a_3}{a_2z^2-a_3z-1}dz\\ +\dfrac{2b_1a_2+b_2a_3}{2a_2}\displaystyle\int\frac{1}{a_2z^2-a_3z-1}dz\end{array} \tag{5.155}$$

Term by term, we have the following.

$$\int_{z_0}^{z}dz=z-z_0 \tag{5.156}$$

$$\int_{z_0}^{z}\frac{2a_2z-a_3}{a_2z^2-a_3z-1}dz=\ln\left(\frac{a_2z^2-a_3z-1}{a_2z_0^2-a_3z_0-1}\right) \tag{5.157}$$

$$\int_{z_0}^{z}\frac{1}{a_2z^2-a_3z-1}dz=-\frac{2}{\sqrt{4a_2+a_3^2}}\left[arc\tanh\left(\frac{2a_2z-a_3}{\sqrt{4a_2+a_3^2}}\right)-arc\tanh\left(\frac{2a_2z_0-a_3}{\sqrt{4a_2+a_3^2}}\right)\right] \tag{5.158}$$

or

$$\int_{z_0}^{z}\frac{1}{a_2z^2-a_3z-1}dz=\frac{1}{\sqrt{4a_2+a_3^2}}\left[\ln\frac{2a_2z-a_3-\sqrt{4a_2+a_3^2}}{2a_2z-a_3+\sqrt{4a_2+a_3^2}}-\ln\frac{2a_2z_0-a_3-\sqrt{4a_2+a_3^2}}{2a_2z_0-a_3+\sqrt{4a_2+a_3^2}}\right] \tag{5.159}$$

The above results can be verified by taking the derivatives of the functions on the right-hand side of each equation. Therefore, it is possible to write the following solution.

$$\begin{array}{l}\displaystyle\int_{z_0}^{z}\frac{(a_1+a_2)z^2-a_1z-1}{a_2z^2-a_3z-1}dz=(b_1+1)(z-z_0)+\frac{b_2}{2a_2}\ln\left(\frac{a_2z^2-a_3z-1}{a_2z_0^2-a_3z_0-1}\right)\\ -\dfrac{2b_1a_2+b_2a_3}{2a_2}\,\dfrac{2}{\sqrt{4a_2+a_3^2}}\left[arc\tan h\left(\dfrac{2a_2z-a_3}{\sqrt{4a_2+a_3^2}}\right)-arc\tan h\left(\dfrac{2a_2z_0-a_3}{\sqrt{4a_2+a_3^2}}\right)\right]\end{array} \tag{5.160}$$

Compressible Flow and Shock Waves

6

Nomenclature

A	Cross-sectional area, m^2
A^*	Critical area, m^2
c	Speed of sound, m/s
c^*	Speed of sound at the critical conditions, m/s
c_p	Heat capacity at constant pressure, J/kg K
D	Diameter, m
D^*	Critical diameter, m
F	Force, N
h	Enthalpy, J/kg
h_0	Stagnation enthalpy, J/kg
J	Impulse function, N
$\dot{m}$	Mass flow rate, kg/s
$\dot{m}''$	Mass flux rate, kg/m^2 s
M	Mach number
M^*	Dimensionless velocity comparing ($M^* = V/V^*$)
p	Pressure, Pa
p_0	Stagnation pressure, Pa
p^*	Critical pressure ($M = 1$), Pa
R	Universal gas constant, J/mol K
R_g	Gas constant, J/kg K
s	Entropy, J/kg K
s_0	Stagnation entropy, J/kg K
T	Temperature, K
T_0	Stagnation temperature, K
T^*	Critical temperature ($M = 1$), K
u	Horizontal velocity component in shock polar representation, m/s

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V Velocity, m/s
V^* Critical velocity ($M = 1$), m/s
v Vertical velocity component in shock polar representation, m/s

Greek Letters

α Deflection angle, °
β Wave angle, °
γ Ratio of heat capacities
ρ Density, kg/m^3
ρ_0 Stagnation density, kg/m^3

Subscripts

1 Conditions before a shock wave
2 Conditions after a shock wave
n Normal component
t Tangent component

6.1 Introduction

The present chapter is dedicated to the theory of compressible flow, with a focus on phenomena that are relevant to the objective of this book. The classical theory of compressible flow can be found in well-known textbooks like those by Shapiro [45, 86], Zucrow and Hoffman [87, 88], and Anderson [89]. However, the focus here is to gather the theoretical background that is more useful for the study of flame propagation and detonations. Moreover, the present chapter is limited to ideal gases. Therefore, the chapter starts with the presentation of the isentropic flow of an ideal gas, followed by the theory of normal shock waves, and closes with the study of oblique shock waves.

6.2 Isentropic Flow of an Ideal Gas

A flow in a pipe or tube of short length can be considered adiabatic. Thus, if frictional effects can be neglected, the reversible condition may be assumed, and the flow is isentropic. This consideration is useful to study nozzles and diffusers. Furthermore, the flow can be considered one-dimensional if the variation of the fluid properties along the cross-sections can be neglected compared to their variation along the streamlines. This section will apply these considerations to obtain a simple

representation of the compressible flow of an ideal gas, known as isentropic flow of an ideal gas.

6.2.1 General Description

An enthalpy–entropy diagram can be plotted for an isentropic flow, as shown in Fig. 6.1. The vertical line representing constant entropy is expected to intersect the horizontal lines representing constant enthalpy at different values. The maximum possible enthalpy for the given entropy is reached when all the energy is in the form of enthalpy, that is, the kinetic energy and hence the velocity are both zero. The stagnation enthalpy (h_0) is therefore defined as the value of the enthalpy for which the velocity is zero for a given entropy. Writting the energy conservation equation between the stagnation state and another arbitrary state, the expression shown in Eq. (6.1) gives h_0.

$$h_0 = h + \frac{V^2}{2} \tag{6.1}$$

Since h_0 and s_0 are available, it is possible to obtain other stagnation properties, such as the stagnation temperature (T_0), the stagnation pressure (p_0), and the stagnation density (ρ_0). The Mach number is defined as the ratio between the flow velocity (V) and the speed of sound (c). For $M < 1$ the flow is subsonic, for $M = 1$ the flow is sonic, and for $M > 1$ the flow is supersonic.

$$M = \frac{V}{c} \tag{6.2}$$

The speed of sound is defined as the square root of the partial derivative of pressure as a function of density for an isentropic process [$c = \sqrt{(\partial p/\partial \rho)_s}$]. For an ideal gas, Eq. (6.3) thus gives the speed of sound.

$$c = \sqrt{\gamma R_g T} \tag{6.3}$$

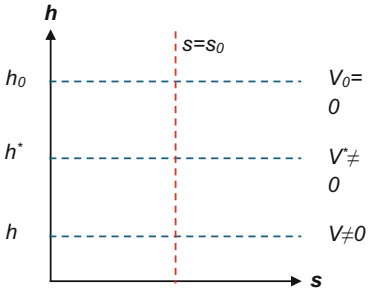


Fig. 6.1 Representation of the stagnation enthalpy in the h vs. s diagram

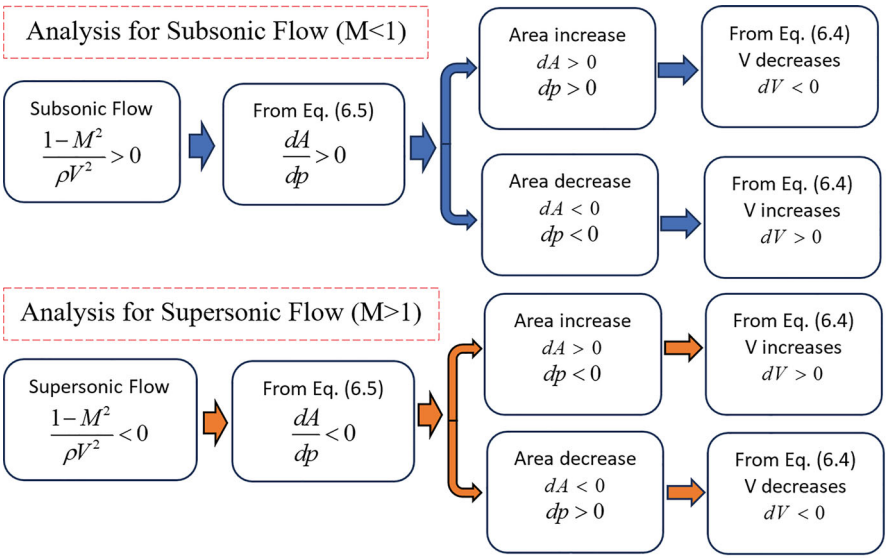


Fig. 6.2 Behavior of the compressible flow for area change

where $R_g = R/W$ is the universal gas constant divided by the molecular weight of the gas. It is known that as the velocity of the one-dimensional flow increases, its pressure decreases. This observation is derived by considering an infinitesimal control volume, applying the first law of thermodynamics to both sides ($dh + VdV = 0$), then, combining that expression with the Tds relationship ($Tds = dh - dp/\rho$), for which $s =$ constant, and rearranging. Applying that procedure, the following is obtained.

$$\frac{dp}{\rho} = -VdV \tag{6.4}$$

Analyzing Eq. (6.4) it is clear the if $dp > 0$, then $dV < 0$. Another important result is the behavior of the flow when the area changes. Depending on whether it is subsonic or supersonic, the flow responds differently to the change in area. Consider the mass conservation equation, which can be written in logarithmic form [$ln(\rho AV) =$ constant]. Then, taking the differential [$d\ ln(\rho AV) = 0$], using Eq. (6.4), and noting that the definitions of the speed of sound and Mach number can be applied, the expression shown in Eq. (6.5) is obtained. The behavior resulting from the analysis of Eq. (6.5) together with Eq. (6.4) is shown in Fig. 6.2.

$$\frac{1}{A}\frac{dA}{dp} = \frac{1-M^2}{\rho V^2} \tag{6.5}$$

Note that the behavior shown in Fig. 6.2 has implications for the operation of nozzles and diffusers. The former leads to a reduction in cross-sectional area, while

the latter leads to an increase in cross-sectional area. Additionally, the analysis of Eqs. (6.5) and (6.4) can give insights into the flow characteristics inside in-line flame arresters, since these generally present an area increase. For sonic flow, the right-hand side of Eq. (6.5) is zero, which means that $dA/dp = 0$ and $dA/dV = 0$. Therefore, at $M = 1$, the area is an extremum. Moreover, it can be shown that at $M = 1$, the area is a minimum.

As shown in Fig. 6.2, at subsonic conditions ($M^2 < 1$), an increase in area produces an increase in pressure and a decrease in velocity. Analogously, at subsonic conditions, decreasing the area implies decreasing the pressure and increasing the velocity. Conversely, at supersonic conditions ($M^2 > 1$), the increase in area produces a decrease in pressure and an increase in velocity. Similarly, at supersonic conditions, the decrease in area produces an increase in pressure and a decrease in velocity.

The fact that the area is a minimum at sonic conditions can be demonstrated by writing Eq. (6.5) as $dlnA/dp$ and taking the second derivative.

$$\frac{d^2}{dp^2}(\ln A) = \frac{d}{dp}\left(\frac{1-M^2}{\rho V^2}\right) = \frac{d}{dp}\left(\frac{1}{\rho V^2}\right) - \frac{d}{dp}\left(\frac{1}{\rho c^2}\right)$$

Note that the following relationships are valid: $(d\rho/dp)_s = 1/c^2$, $dV/dp = -1/(\rho V)$, and, since $c^2 = \gamma p/\rho$, $dc^2/dp = (1/\rho)(\gamma - 1/c)$. The derivatives on the right-hand side of the above equation are now taken.

$$\frac{d}{dp}\left(\frac{1}{\rho V^2}\right) = -\frac{V^2(d\rho/dp)_s + 2\rho V(dV/dp)}{\rho^2 V^4} = \frac{2-M^2}{\rho^2 V^4}$$

$$\frac{d}{dp}\left(\frac{1}{\rho c^2}\right) = -\frac{c^2(d\rho/dp)_s + \rho\left(dc^2/dp\right)}{\rho^2 c^4} = \frac{1-c(1+\gamma)}{\rho^2 c^5}$$

Substituting the above derivatives into the expression for the d^2 (ln A)/dp^2 and noting that $M^2 = 1$, the following is observed.

$$\frac{d^2}{dp^2}(\ln A) = \frac{d}{dp}\left(\frac{1-M^2}{\rho V^2}\right) = \frac{2-M^2}{\rho^2 V^4} + \frac{c(1+\gamma)-1}{\rho^2 c^5} > 0$$

Therefore, for positive values of A, it is clear that the ln$A$ must have its minimum value for the minimum value of A. It is then demonstrated that the area is a minimum at the sonic conditions.

6.2.2 Adiabatic Flow Conditions

For an ideal gas with constant heat capacity, the enthalpy change between any state and the stagnation state can be written as follows: $h_0 - h = c_p\,(T_0 - T)$. Since h_0 can

be related to h and V through Eq. (6.1), the following expression for the velocity is obtained.

$$V^2 = 2c_p(T_0 - T) = \frac{2\gamma}{\gamma - 1} R_g (T_0 - T) \tag{6.6}$$

Note that the c_p value for an ideal gas can be written as shown in Eq. (6.7).

$$c_p = \frac{\gamma}{\gamma - 1} R_g \tag{6.7}$$

At the sonic conditions, the temperature is T^*, and the velocity is $V^* = c^* = \sqrt{\gamma R_g T^*}$. Therefore, it is possible to substitute $T = T^*$ and $V = V^*$ in Eq. (6.6) and rearrange to obtain Eq. (6.8).

$$\frac{T^*}{T_0} = \frac{2}{\gamma + 1} \tag{6.8}$$

For an ideal gas at an arbitrary temperature, Eq. (6.6) can be rearranged to give the ratio $T_0/T = 1 + V^2/(2c_pT)$. After some more rearrangements, and using the expressions for c_p, c, and M, The Eq. (6.9) is obtained.

$$\frac{T_0}{T} = 1 + \frac{\gamma - 1}{2} M^2 \tag{6.9}$$

It is interesting to relate the flow velocity to the critical velocity. This can be done by introducing a dimensionless velocity, which is shown in Eq. (6.10).

$$M^* = \frac{V}{V^*} = \frac{V}{c^*} \tag{6.10}$$

This dimensionless velocity (M^*) can be related to the Mach number (M) of the flow by the following equations.

$$M*^2 = \frac{(\gamma + 1)M^2}{2 + (\gamma - 1)M^2}; \quad M^2 = \frac{2M*^2}{(\gamma + 1) - (\gamma - 1)M*^2} \tag{6.11}$$

The flow per unit area can also be obtained by making $\dot{m}/A = \rho V$. This expression can be modified by introducing the term $(T_0/T_0)^{1/2}$, using the ideal gas law to express $\rho = p/R_gT$, rearranging to obtain the expression for c, and using Eqs. (6.7) and (6.9). The expression shown below is obtained.

$$\dot{m}'' = \frac{\dot{m}}{A} = \left(\frac{\gamma p^2}{T_0 R_g}\right)^{1/2} M \left(1 + \frac{\gamma - 1}{2} M^2\right)^{1/2} \tag{6.12}$$

This expression allows the local Mach number to be determined if the mass flow rate, area, pressure, and stagnation temperature are known.

Example 6.1
Air is flowing in a tube with 10 cm diameter at a speed of 1000 m/s. If the temperature of the gas is 800 K and its pressure is 200 kPa, calculate the other properties. Also, explain what happens if the area increases or decreases. Assume a ratio of heat capacities equal to $\gamma = 1.4$ and a molecular weight for air equal to $W = 28.96$ kg/kmol.

Solution
The first step may be to calculate the speed of sound under the given conditions and obtain the Mach number. The universal gas constant is $R = 8314.4$ J/(kmol K).

$$R_g = \frac{8314.4}{28.96} = 287.1 \text{ J}/(\text{kg K})$$

$$c = \sqrt{1.4 \times 287.1 \times 800} = 567.06$$

$$M = \frac{1000}{567.06} = 1.763$$

We can also determine the mass flux and the mass flow rate inside the tube by determining the stagnation temperature and then applying Eq. (6.12).

$$T_0 = 800 \times \left(1 + \frac{1.4-1}{2} 1.763^2\right) = 1297.3 \text{ K}$$

$$\dot{m}'' = \left(\frac{1.4 \times 200,000^2}{1297.3 \times 287.1}\right) \times 1.763 \left(1 + \frac{1.4-1}{2} 1.763^2\right)^{1/2} = 870.53 \text{ kg}/\left(\text{m}^2 \text{ s}\right)$$

$$\dot{m} = 870.53 \times \left(\frac{\pi \times 0.1^2}{4}\right) = 6.837 \text{ kg/s}$$

Finally, it is interesting to find some properties for the sonic condition ($M = 1$). These are the critical temperature and velocity.

$$T^* = 1297.3 \times \frac{2}{1.4+1} = 1081.1 \text{ K}$$

$$M*^2 = \frac{(1.4+1) \times 1.763^2}{2 + (1.4-1) \times 1.763^2} = 1.517$$

$$V^* = c^* = \frac{1000}{1.517} = 659.2 \text{ m/s}$$

These simple calculations provide some insight into the flow conditions inside the duct.

6.2.3 Isentropic Flow Conditions

The consideration of isentropic flow implies that the stagnation temperature is constant due to the adiabatic conditions. This can be deduced from Fig. 6.1 and from the energy conservation equation. For an isentropic flow, the stagnation properties can be related to the flow properties by the following relationships.

$$\frac{T}{T_0} = \left(\frac{p}{p_0}\right)^{\frac{\gamma-1}{\gamma}}; \quad \frac{p}{p_0} = \left(\frac{\rho}{\rho_0}\right)^{\gamma} \tag{6.13}$$

The relationship between the stagnation temperature and the flow temperature was given in Eq. (6.9). Substituting Eq. (6.13) into this expression gives Eqs. (6.14) and (6.15), which yield the relations between stagnation pressure and stagnation density with flow pressure and density.

$$\frac{p_0}{p} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{\frac{\gamma}{\gamma-1}} \tag{6.14}$$

$$\frac{\rho_0}{\rho} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{\frac{1}{\gamma-1}} \tag{6.15}$$

The critical state for an isentropic flow is an important condition for the design of nozzles and diffusers, as it is related to the manifestation of shock waves. The properties of the critical state are obtained by substituting $M = 1$ in Eqs. (6.9), (6.14), and (6.15).

$$\frac{T^*}{T_0} = \frac{2}{\gamma+1}; \quad \frac{p^*}{p_0} = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}}; \quad \frac{\rho^*}{\rho_0} = \left(\frac{2}{\gamma+1}\right)^{\frac{1}{\gamma-1}} \tag{6.16}$$

The stagnation pressure is introduced into the calculation of the mass flux by substituting Eq. (6.14) into Eq. (6.12). With this new expression, the mass flux can be determined as a function of the stagnation properties and the Mach number.

$$\dot{m}'' = \left(\frac{\dot{m}}{A}\right) = \left(\frac{\gamma}{R_g}\right)^{1/2} \frac{p_0}{T_0^{1/2}} M \left(1 + \frac{\gamma-1}{2}M^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}} \tag{6.17}$$

Since the area at sonic conditions (critical state) is a minimum, the maximum mass flux can be determined by substituting $M = 1$ into the above expression. The result of this procedure is shown in Eq. (6.18).

$$\left(\frac{\dot{m}}{A}\right)_{\max} = \frac{p_0}{T_0^{1/2}} \left(\frac{\gamma}{R_g}\right)^{1/2} \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \tag{6.18}$$

Finally, a useful relationship is obtained by dividing Eq. (6.18) by Eq. (6.17).

$$\frac{A}{A^*} = \frac{1}{M}\left[\left(\frac{2}{\gamma+1}\right)\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}} \tag{6.19}$$

Another interesting quantity in compressible flow is the impulse function, which is the sum of forces related to pressure and momentum.

$$F = pA + \rho A V^2 \tag{6.20}$$

Using the ideal gas law, and after some rearrangements, the impulse function can be written in terms of the Mach number.

$$F = pA\left(1+\gamma M^2\right) \tag{6.21}$$

The net thrust (J) in the internal solid surfaces in contact with the flow is defined as the change in the impulse function, that is, $J = F_2 - F_1$.

Example 6.2
Given the flow described in Example 6.1, assume an isentropic flow to determine the remaining stagnation properties, the critical properties, the maximum mass flux, and the critical area.

Solution
The ideal gas equation is first used to determine the density of the flow under the given conditions of pressure and temperature.

$$\rho = \frac{p}{R_g T} = \frac{200,000}{287.1 \times 800} = 0.8708 \text{ kg/m}^3$$

The Mach number of the flow was determined in the previous example as $M = 1.763$. The stagnation pressure and density are obtained from Eqs. (6.14) and (6.15), respectively.

$$p_0 = 200 \times 10^3 \times \left(1+\frac{1.4-1}{2}1.763^2\right)^{\frac{1.4}{1.4-1}} \text{ Pa}$$

$$\rho_0 = 0.8708 \times \left(1+\frac{1.4-1}{2}1.763^2\right) = 2.917 \text{ kg/m}^3$$

The critical properties are then calculated with Eq. (6.16),

$$T^* = 1297.3 \times \frac{2}{1.4+1} = 1081.1 \text{ K}$$

$$p^* = 1086.1 \times 10^3 \times \left(\frac{2}{1.4+1}\right)^{\frac{1.4}{1.4-1}} = 573.8 \times 10^3 \text{ Pa}$$

$$\rho^* = 2.917 \times \left(\frac{2}{1.4+1}\right)^{\frac{1}{1.4-1}} = 1.850 \text{ kg/m}^3$$

The maximum mass flow rate and critical area are determined by Eqs. (6.18) and (6.19), respectively.

$$\left(\frac{\dot{m}}{A}\right)_{\max} = \frac{1086.1 \times 10^3}{1297.3^{1/2}} \left(\frac{1.4}{287.1}\right)^{1/2} \left(\frac{2}{1.4+1}\right)^{\frac{1.4+1}{2(1.4-1)}} = 1218.6 \text{ kg/}\left(\text{m}^2 \text{ s}\right)$$

$$\frac{A}{A^*} = \frac{1}{1.763}\left[\left(\frac{2}{1.4+1}\right)\left(1 + \frac{1.4-1}{2} 1.763^2\right)\right]^{\frac{1.4+1}{2(1.4-1)}} = 1.4$$

Then, the critical area and critical diameter can also be obtained.

$$A^* = \frac{\left(\pi \times 0.1^2\right)/4}{1.4} = 5.61 \times 10^{-3} \text{ m}^2$$

$$D^* = \sqrt{\frac{4A^*}{\pi}} \times 100 = \sqrt{\frac{4 \times 5.61 \times 10^{-3}}{\pi}} \times 100 = 8.46 \text{ cm}$$

6.3 Normal Shock Waves

6.3.1 General Description

It is assumed that P_1, T_1, and M_1 are known in section A_1 of a stream tube. On the other hand, the value of M_2 is desired in section A_2 of the same stream tube. Furthermore, considering that the flow is isentropic, the stagnation properties are the same, and it can be stated that $A_2/A_1 = (A_2/A^*)/(A_1/A^*)$. Therefore, from Eq. (6.19) it can be shown that the following relationship must hold.

$$aM_2^2 + bM_2 + c = 0 \tag{6.22}$$

$$a = \frac{\gamma - 1}{\gamma + 1}; \quad b = -\left(\frac{A_2}{A_1}\frac{1}{M_1}\right)^{\frac{2(\gamma-1)}{\gamma+1}} \left(\frac{2}{\gamma+1}\right)\left(1 + \frac{\gamma-1}{2}M_1^2\right); \quad c = \frac{2}{\gamma+1} \tag{6.23}$$

Thus, assuming a value for A_2/A_1 and M_1, the discriminant of Eq. (6.22) establishes the conditions under which real solutions for M_2 can be found, that is, $b^2 - 4\,ac \geq 0$. Moreover, in the case where $b^2 - 4\,ac < 0$, there are no real solutions for M_2, which means that isentropic flow cannot be maintained under such

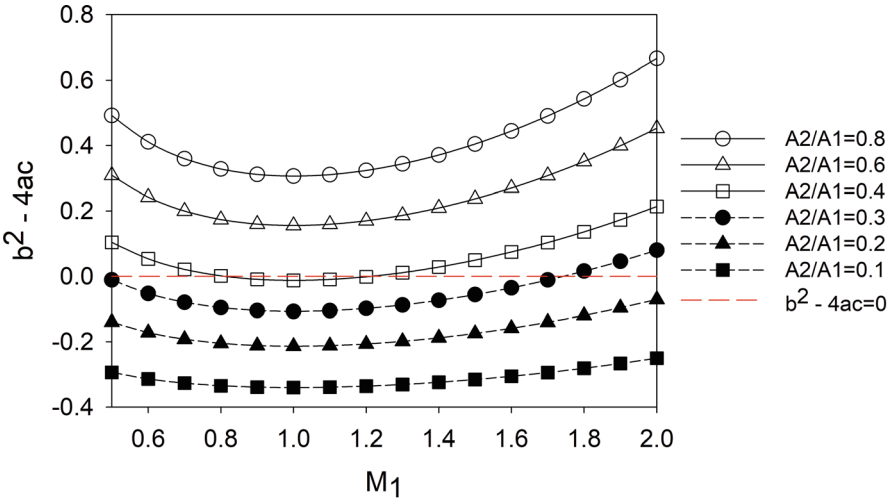


Fig. 6.3 Behavior of the discriminant for fixed values of A_2/A_1

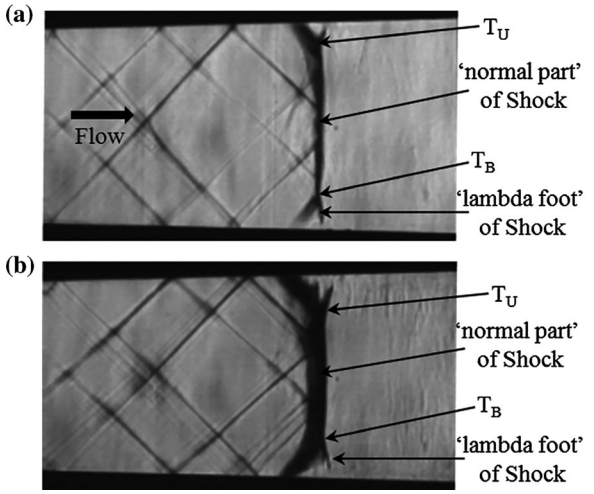


Fig. 6.4 Schlieren photograph of normal shock waves in ducts. (**a**) Aspect ratio 1.1 and (**b**) aspect ratio 2.2 [91]. (Reprinted from Shock Waves, Vol 30, S. Vaisakh, P. R. Namratha, T. M. Muruganandam, Effect of duct aspect ratio on normal shock wave/boundary layer interaction, 215–219, Copyright (2019), with permission from Springer Nature)

conditions. A plot of the discriminant values for fixed A_2/A_1 ratios and different M_1 values can be found in Fig. 6.3.

Observing that, for supersonic values of M_1 it is sometimes obtained that $b^2 - 4\,ac < 0$, the question arises: What happens if isentropic flows are not possible under the given conditions? The answer is that there is a shock wave that acts as a "discontinuity" on the flow. Experimentally observed normal shock waves can be found in the work by Davidson and Babinsky [90]. Also, Fig. 6.4 shows normal

shock waves formed in supersonic flow inside a duct. The T_U and T_B represent the upper and bottom triple points.

Although the changes that take place within the shock wave are not discontinuous, their thickness is of the same order of magnitude as the length of the molecular free path. Therefore, the theory of the continuum treats shock waves as discontinuities and examines the changes they cause in the flow properties. In particular, the shock wave increases the pressure and thus also changes the velocity.

A normal shock wave in steady-state conditions can be analyzed by applying mass, momentum, and energy conservation equations in one dimension. For example, take the shock wave in Fig. 6.4b. Assume that it is perfectly normal to the flow and that it represents a control volume whose coordinate system is bound to itself. Then, the conservation equations can be written.

$$\dot{m}'' = \rho_1 V_1 = \rho_2 V_2 \tag{6.24}$$

$$p_1 + \dot{m}'' V_1 = p_2 + \dot{m}'' V_2 \tag{6.25}$$

$$h_0 = h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \tag{6.26}$$

The above equations must be solved in combination with an equation of state and thermodynamic fundamental equations (tables or correlations) to obtain the specific enthalpy and the specific entropy from, for example, the pressure and density data. Since the pressure, density, and velocity must change across the shock wave, the enthalpy and entropy must also change. The expressions below represent equations of state in implicit form.

$$s = s(p, \rho); \quad h = h(p, \rho); \quad p = p(s, \rho) \tag{6.27}$$

Assume that the parameters at the initial state are known, that is, h_1, p_1, ρ_1, s_1, and V_1 are all known. To determine the parameters at the final state (h_2, p_2, ρ_2, s_2, and V_2), there are six equations for five unknowns. Therefore, there are two meaningful procedures to obtain the desired parameters. The first one defines the **Fanno line** and consists of solving together Eqs. (6.24), (6.26), and (6.27). The second one defines the **Rayleigh line** and consists of solving Eqs. (6.24), (6.25), and (6.27).

It is known from classical thermodynamics that, $Tds = dh - dp/\rho$, and any process must satisfy $ds \geq 0$. It is also known from experiments that there is an increase in pressure due to the shock. Therefore, it is found that for a normal shock, the velocity must decrease (Eq. 6.25), the density must increase (Eq. 6.24), the enthalpy must increase (Eq. 6.26), and the entropy must also increase. Since the shock is considered a discontinuity, the process could be represented as a jump in an $h - s$ diagram. It is interesting to imagine an infinitesimal jump or, in other words, a shock wave that leaves the state in the $h - s$ diagram almost unchanged. Since entropy can only increase (or ideally be constant), this process can only occur at a

point of maximum entropy for which $ds = 0$, and the thermodynamic relationship is then the one shown below.

$$dh = \frac{dp}{\rho} \tag{6.28}$$

To investigate what happens in this state of maximum entropy, note that both sides of Eqs. (6.24)–(6.26) are constant. They can therefore be differentiated to obtain the following expressions.

$$Vd\rho + \rho dV \tag{6.29}$$

$$dp + pVdV \tag{6.30}$$

$$dh + VdV = 0 \tag{6.31}$$

Since the Fanno and Rayleigh lines are obtained by certain combinations of the conservation equations, the differential forms of these equations should be combined in the same order. Thus, the substitution of Eq. (6.29) into Eq. (6.31) and the subsequent substitution of Eq. (6.28) under the assumption of isentropic conditions (because s is a maximum) shows that the velocity of the flow is equal to the speed of sound in the state of maximum entropy. The same condition results by substitution of Eq. (6.29) into Eq. (6.30). Therefore, there is a point of maximum entropy for which $M = 1$. There are two states with this property, one on the Fanno line and another one on the Rayleigh line.

$$V = \sqrt{(\partial p/\partial \rho)_s} \tag{6.32}$$

Restricting the analysis to the Fanno line, shown in Fig. 6.5 in the $h - s$ plane, designating the state of maximum entropy as state a_F, and taking the state 1 as the initial state, then, a sequence of states can be achieved by following a continuous thermodynamic process. The final state of that process would be the state a_F. It can be inferred that the state a_F can be reached by increasing or decreasing the enthalpy, depending on whether the initial state was chosen with $h_1 < h_{aF}$ or $h_1 > h_{aF}$. In both cases the entropy must increase. This procedure generates two branches of the Fanno line: an upper branch and a lower branch.

Since the velocity also decreases with increasing enthalpy, the lower branch must represent supersonic conditions ($M > 1$). On the other hand, since decreasing enthalpy means increasing velocity, the upper branch must represent subsonic conditions ($M < 1$). A normal shock way cannot be represented by a continuous thermodynamic process along the Fanno line. However, the lower and upper branches may be connected by a discontinuous process (jump), which represents a normal shock wave. This means that normal shock waves take place by going from a state with $M > 1$ to a state with $M < 1$. Similar reasoning can be applied to the

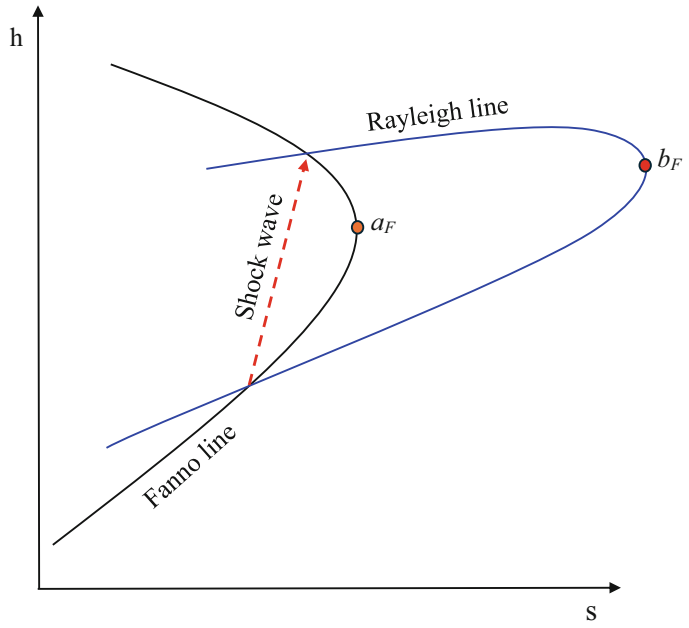


Fig. 6.5 Representation of Fanno line and Rayleigh line on the $h - s$ plane

construction of the Rayleigh line, and the same conclusions would be reached. The representation of the Fanno line and the Rayleigh line can be found in Fig. 6.5.

The point of intersection between the Fanno line and the Rayleigh line is the point at which the conservation equations and the equations of state are simultaneously satisfied. Starting from these intersections, enthalpy can be changed by a continuous thermodynamic process to reach the state of maximum entropy on the Rayleigh line (b_F).

6.3.2 Governing Equations for Ideal Gases

This section assumes ideal gas behavior and constant heat capacity. Then, the following equations of state are applicable.

$$p = \rho R_g T \; ; \quad h = c_p T \tag{6.33}$$

The energy conservation equation (Eq. 6.26) shows that the states before and after the shock have the same stagnation temperature. Thus, employing Eq. (6.9) to relate the flow temperatures to the stagnation temperature yields the expression shown below.

$$\frac{T_2}{T_1} = \frac{T_0/T_1}{T_0/T_2} = \frac{2 + (\gamma - 1)M_1^2}{2 + (\gamma - 1)M_2^2} \tag{6.34}$$

The mass conservation equation (Eq. 6.24) can be rearranged by dividing both sides by $\rho_1 V_1$. Then the pressure from the equation of state can be introduced to replace the density. Furthermore, making $V = M(\gamma R_g T)^{1/2}$, the following relation can be obtained.

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^2 \left(\frac{M_2}{M_1}\right)^2 \tag{6.35}$$

Since Eqs. (6.34) and (6.35) result from the application of Eqs. (6.24) and (6.26), respectively, and from the application of the equations of state, then, their combination must yield an expression for the Fanno line of an ideal gas.

$$\frac{p_2}{p_1} = \frac{M_1}{M_2}\left(\frac{2 + (\gamma - 1)M_1^2}{2 + (\gamma - 1)M_2^2}\right)^{1/2} \tag{6.36}$$

Similarly, the momentum conservation equation (Eq. 6.25) can be combined with the mass conservation equation (Eq. 6.24) to obtain the expression of the Rayleigh line for a perfect gas. Note that the relationship $V = M(\gamma p/\rho)^{1/2}$ was also used to derive Eq. (6.37).

$$\frac{p_2}{p_1} = \frac{1 + \gamma M_1^2}{1 + \gamma M_2^2} \tag{6.37}$$

Equating the right-hand sides of Eqs. (6.36) and (6.37), squaring both sides, multiplying term by term, and rearranging, the following is obtained.

$$(\gamma - 1)\left(M_2^4 - M_1^4\right) - 2\gamma M_1^2 M_2^2\left(M_2^2 - M_1^2\right) + 2\left(M_2^2 - M_1^2\right) = 0 \tag{6.38}$$

Equation (6.38) immediately reveals a trivial solution, $M_2 = M_1$. However, a nontrivial solution, shown in Eq. (6.39), is obtained by dividing Eq. (6.38) by $({M_2}^2 - {M_1}^2)$ and rearranging.

$$M_2^2 = \frac{(\gamma - 1)M_1^2 + 2}{2\gamma M_1^2 - (\gamma - 1)} \tag{6.39}$$

The critical properties are given in Eq. (6.16). Since they depend only on the stagnation properties (γ = constant), it is clear that $c^* = {c_1}^* = {c_2}^*$. Using Eq. (6.11) to write M_1 and M_2 in terms of ${M^*}_1$ and ${M^*}_2$, replacing those expressions into Eq. (6.39) and rearranging give the Prandtl relationship shown below.

$$\left(M_1^*\right)^2\left(M_2^*\right)^2 = \left(\frac{V_1}{c^*}\right)^2\left(\frac{V_2}{c^*}\right)^2 = 1 \tag{6.40}$$

Equation (6.40) implies that if $M^*_1 > 1$, then it must be that $M^*_2 < 1$. This can also be interpreted as saying that if V_1 is supersonic, then V_2 must be subsonic. The expressions given in Eqs. (6.34) and (6.36) relate T_2/T_1 and p_2/p_1 with M_1 and M_2. However, M_1 and M_2 are now related through Eq. (6.39). Hence, it is possible to express T_2/T_1 and p_2/p_1 as functions of M_1 only. Those expressions are presented in the classical book by Shapiro [45]. As must be known from an undergraduate course on thermodynamics, for a perfect gas, the entropy variation is given by the expression shown in Eq. (6.41).

$$s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} = c_p \ln \left[\frac{T_2/T_1}{(p_2/p_1)^{\frac{\gamma-1}{\gamma}}} \right] \tag{6.41}$$

The fact that the stagnation temperatures are equal ($T_{01} = T_{02}$) does not mean that the stagnation pressures are also equal. In fact, $p_{01} \neq p_{02}$. Using Eq. (6.13), which gives a relationship between T/T_0 and p/p_0, into Eq. (6.41) gives the following relationship.

$$\frac{s_2 - s_1}{R_g} = -\ln \frac{p_{02}}{p_{01}} \tag{6.42}$$

Since entropy must increase, stagnation pressure must decrease across the shock wave. This is observed in Eq. (6.42) because if $p_{02}/p_{01} < 1$, the natural logarithm is negative, and the minus sign outside makes the term on the right-hand side positive. The following can be derived from Eqs. (6.14) and (6.37).

$$\frac{p_{02}}{p_{01}} = \frac{p_{02}}{p_2}\frac{p_1}{p_{01}}\frac{p_2}{p_1} = \left[\frac{2 + (\gamma - 1)M_2^2}{2 + (\gamma - 1)M_1^2} \right]^{\frac{\gamma}{\gamma-1}} \left(\frac{1 + \gamma M_1^2}{1 + \gamma M_2^2} \right) \tag{6.43}$$

Therefore, the value of the left-hand side of Eq. (6.42) can be calculated by taking the values of M_1, calculating M_2 from Eq. (6.39), and then calculating p_{02}/p_{01} from Eq. (6.43). The results of these calculations are shown in Figs. 6.6 and 6.7.

It can be seen from Fig. 6.6 that, for $M_1 < 1$ the entropy would decrease, which is impossible according to the second law of thermodynamics. Therefore, the only possible process for a shock wave is that $M_1 > 1$ and $M_2 < 1$. That is, a supersonic flow becomes subsonic after passing through a normal shock wave. Figure 6.7 shows some solutions that are possible for different values of M_1.

Example 6.3

Air at 300 K and 100 kPa enters a normal shock wave at 500 m/s. Determine the Mach number, temperature, pressure, stagnation temperature, and stagnation pressure of the air before and after the shock wave. Assume that $\gamma = 1.4$.

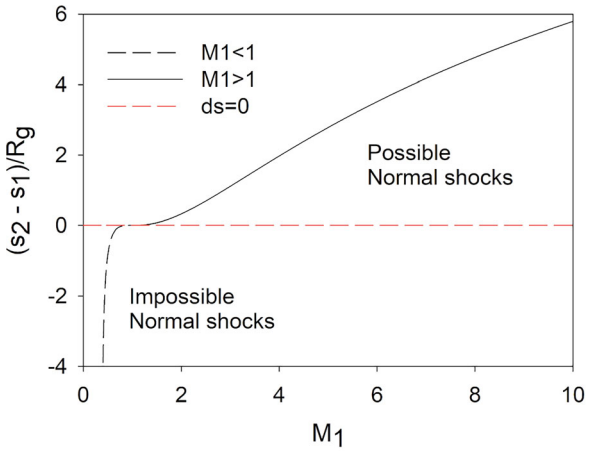


Fig. 6.6 Possible and impossible normal shocks

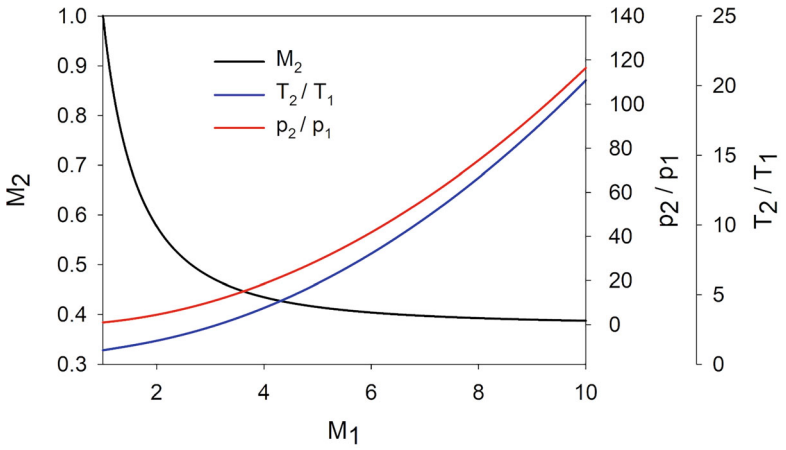


Fig. 6.7 Solutions obtained for M_2, T_2/T_1, and p_2/p_1 for different M_1

Solution

The universal gas constant is $R = 8314.4$ J/(kmol K), and as shown in Example 6.1, for air $R_g = 287.1$ J/kg K. Therefore, the speed of sound and the Mach number before the shock can be calculated.

$$c_1 = \sqrt{1.4 \times 287.1 \times 300} = 347.2; \quad M_1 = \frac{500}{347.2} = 1.44$$

The stagnation temperature and pressure at the initial conditions are determined from Eqs. (6.9) and (6.14). Also, observe that the stagnation temperature before and after the shock is equal $T_{02} = T_{01}$.

$$T_{02} = T_{01} = 300 \times \left(1 + \frac{1.4-1}{2} 1.44^2\right) = 424.4 \text{ K}$$

$$p_{01} = 100\left(1 + \frac{1.4-1}{2} 1.44^2\right)^{\frac{1.4}{1.4-1}} = 336.8 \text{ kPa}$$

The Mach number after the shock (M_2) is determined by Eq. (6.39).

$$M_2 = \sqrt{\frac{(1.4-1) \times 1.44^2 + 2}{2 \times 1.4 \times 1.44^2 - (1.4-1)}} = 0.723$$

The stagnation pressure after the shock wave is determined by Eq. (6.43).

$$p_{02} = 336.8 \left[\frac{2 + (1.4-1) \times 0.723^2}{2 + (1.4-1) \times 1.44^2}\right]^{\frac{1.4}{1.4-1}} \left(\frac{1 + 1.4 \times 1.44^2}{1 + 1.4 \times 0.723^2}\right) = 319.2 \text{ kPa}$$

The temperature and pressure after the shock are determined by Eqs. (6.9) and (6.14).

$$T_2 = 424.4 \times \left(1 + \frac{1.4-1}{2} 0.723^2\right)^{-1} = 384.2 \text{ K}$$

$$p_2 = 319.2\left(1 + \frac{1.4-1}{2} 0.723^2\right)^{-\frac{1.4}{1.4-1}} = 225.38 \text{ kPa}$$

The air velocity after the shock is finally determined.

$$c_2 = \sqrt{1.4 \times 287.1 \times 384.2} = 392.9$$

$$V_2 = c_2 \times M_2 = 392.9 \times 0.723 = 284.1 \text{ m/s}$$

6.4 Interpretation of the Formation of Shock Waves

The formation of a normal shock wave can be explained by considering a gas and a piston which are initially at rest. At a certain point in time, the piston begins to move from left to right, pushing the gas closer to its wall, as shown in Fig. 6.8.

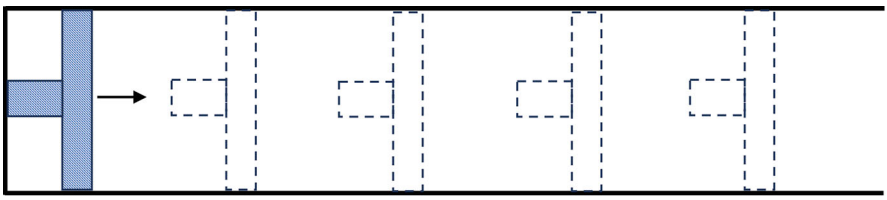


Fig. 6.8 Simple representation for the formation of shock waves

The piston accelerates instantaneously by a certain amount at constant time intervals. Therefore, for a given time interval, when the piston presses on the nearest (infinitesimal) mass of the fluid, a pressure pulse is generated and transmitted through the fluid. The mass closest to the piston moves at the speed of the piston and has a slightly increased pressure due to the passage of the original pressure wave, which continues to propagate to increase the pressure and imprints velocity on the next (infinitesimal) mass of the fluid. The second acceleration of the piston results in another pressure wave, which further increases the pressure of the next mass of fluid that was also accelerated with the piston. This process is repeated with successive accelerations until finally a shock wave is created.

6.5 Oblique Shock Waves

6.5.1 Oblique Shock Wave Theory

An oblique shock wave can be formed, for example, by the interaction of a supersonic flow with a change of direction on a wall. In that case the oblique shock will be attached to the surface. However, this is not mandatory, since detached oblique shocks also occur. A shadowgraph of oblique shocks formed over the surface of an aircraft model inside a supersonic pressure tunnel is shown in Fig. 6.9a. In addition, the schematic representation of the oblique shock is shown in Fig. 6.9b. Since there is an angle between the streamlines and the shock, the flow velocity (V) is conveniently divided into two components: a velocity component normal to the shock (V_n) and a velocity component tangential to the shock (V_t). Therefore, Eq. (6.44) gives the relationship between these velocities.

$$V^2 = V_t^2 + V_n^2 \tag{6.44}$$

In reference to Fig. 6.9, the angle between the shock wave and the horizontal is called the wave angle (β). On the other hand, the deflection angle (α) represents the change of direction of the supersonic flow and is related to a variation on the physical

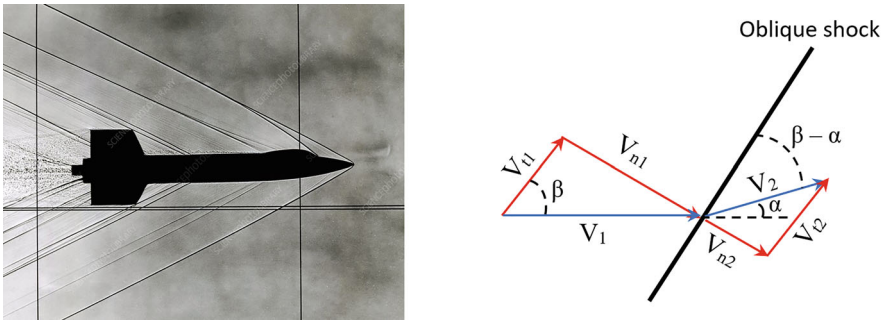


Fig. 6.9 Oblique shock wave. Left: image by NASA [92]. Right: schematic representation

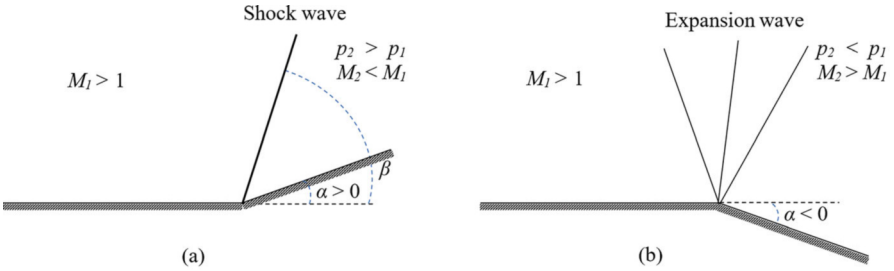


Fig. 6.10 Representation of the effect of slopes on the way of a uniform supersonic flow. (**a**) Attached straight oblique shock and (**b**) expansion wave

boundaries of the flow. A simple representation of the effect of deflection is obtained by considering a uniform supersonic flow which arrives to a boundary with slope α, as shown in Fig. 6.10.

Measuring the angle counterclockwise with respect to the horizontal, for $\alpha > 0$, a shock wave is formed. Therefore, the flow increases its pressure, density, and temperature while decreasing its Mach number. On the other hand, for $\alpha < 0$, an expansion wave is formed, and the flow decreases its pressure, density, and temperature while increasing its Mach number. An important difference between a shock wave and an expansion wave is that the former is a discontinuity, that is, there is a "jump" in the properties' values across it, while in the latter, the properties change continuously.

In this case the momentum equation must be written for the tangential and normal directions, identified by the subscripts t and n, respectively. The conservation equations applied to an ideal gas that crosses the shock give the relationships between the conditions before and after the discontinuity.

Mass conservation equation in normal direction

$$\dot{m}'' = \rho_1 V_{n1} = \rho_2 V_{n2} \tag{6.45}$$

Momentum equation in tangential direction

$$\dot{m}'' V_{t1} = \dot{m}'' V_{t2} \tag{6.46}$$

Momentum equation in normal direction

$$p_1 + \rho_1 V_{n1}^2 = p_2 + \rho_2 V_{n2}^2 \tag{6.47}$$

Energy conservation equation

$$c_p T_1 + \frac{V_1^2}{2} = c_p T_2 + \frac{V_2^2}{2} = c_p T_0 \tag{6.48}$$

The momentum equation in the tangential direction (Eq. 6.46) immediately reveals that $V_{t1} = V_{t2}$. The Rankine–Hugoniot equation can now be derived by

rearranging Eq. (6.47) and using Eq. (6.45) to express the normal velocities in terms of the pressures and densities.

$$V_{n1}^2 = \frac{p_2 - p_1}{\rho_2 - \rho_1}\frac{\rho_2}{\rho_1}; \quad V_{n2}^2 = \frac{p_2 - p_1}{\rho_2 - \rho_1}\frac{\rho_1}{\rho_2} \tag{6.49}$$

Note that Eq. (6.7) can be used to express the c_p in terms of γ and R_g. Eq. (6.44) gives V in terms of V_t and V_n, and the temperature can be related to pressure and density by $T = p/\rho R_g$. The energy conservation equation can now be transformed into the Rankine–Hugoniot equation by substitution of Eqs. (6.46) and (6.49) into Eq. (6.48). An intermediary step for the derivation of the Rankine–Hugoniot equation is shown in Eq. (6.50).

$$\frac{\gamma+1}{\gamma-1}\frac{p_2}{p_1} - \frac{p_2}{p_1}\frac{\rho_2}{\rho_1} = \frac{\gamma+1}{\gamma-1}\frac{\rho_2}{\rho_1} - 1 \tag{6.50}$$

Then, rearranging Eq. (6.50), the Rankine–Hugoniot equation is obtained. The equation provides a relationship between the ratios p_2/p_1 and ρ_2/ρ_1, as shown in the expressions given in Eq. (6.51).

$$\frac{p_2}{p_1} = \frac{\frac{\gamma+1}{\gamma-1}\frac{\rho_2}{\rho_1} - 1}{\frac{\gamma+1}{\gamma-1} - \frac{\rho_2}{\rho_1}}; \quad \frac{\rho_2}{\rho_1} = \frac{\frac{\gamma+1}{\gamma-1}\frac{p_2}{p_1} + 1}{\frac{\gamma+1}{\gamma-1} + \frac{p_2}{p_1}} \tag{6.51}$$

Note that the Rankine–Hugoniot equations depend only on the γ parameter and on thermodynamic properties. Therefore, it can be used for oblique and normal shocks. Further examination of Fig. 6.9b shows that there is an angle β between the projection line of V_1 and the oblique shock. The angle between the projection of V_1 and V_2 is α. Therefore, the angle between the V_2 and the oblique shock is $\beta - \alpha$. It is interesting to obtain mathematical expressions relating thermodynamic properties, the initial Mach number, and these angles. For instance, for the velocity before the shock,

$$V_{t1} = V_1 \cos\beta \quad V_{n1} = V_1 \sin\beta \tag{6.52}$$

On the other hand, for the velocity after the shock,

$$V_{t2} = V_2 \cos(\beta - \alpha) \quad V_{n2} = V_2 \sin(\beta - \alpha) \tag{6.53}$$

These relationships can be used together with Eqs. (6.45) and (6.46) to give the result shown in Eq. (6.54).

$$\frac{\rho_2}{\rho_1} = \frac{\tan\beta}{\tan(\beta - \alpha)} \tag{6.54}$$

Equation (6.54) can also be rearranged by noting that $V_{2n}/V_{1n} = \rho_1/\rho_2$, that $\rho_1 {V_1}^2 = \gamma p_1 {M_1}^2$ and by using Eqs. (6.45) and (6.54), together with the velocity relationships, to get the expressions relating the pressure ratio to the angles.

$$\frac{p_2}{p_1} = 1 + \gamma M_1^2 \sin^2\beta\left(1 - \frac{\rho_1}{\rho_2}\right) \tag{6.55}$$

$$\frac{p_1}{p_2} = 1 + \gamma M_2^2 \sin^2(\beta - \alpha)\left(1 - \frac{\rho_2}{\rho_1}\right) \tag{6.56}$$

The developments to this point allowed to obtain mathematical relationships between six variables, that is, β, α, M_1, M_2, p_2/p_1, and ρ_1/ρ_2. Equations (6.51), (6.54), (6.55), and (6.56) provide four expressions; thus, two independent variables can be adopted to obtain the other four. However, these equations must be solved simultaneously. In order to get a first step towards the solution, it is interesting to obtain a relationship between ρ_2/ρ_1, β, and M_1. This is accomplished by substitution of Eq. (6.56) into Eq. (6.51) and rearranging to obtain Eq. (6.57), which is a quadratic equation.

$$\frac{\gamma+1}{\gamma-1}M_1^2\sin^2\beta\left(\frac{\rho_1}{\rho_2}\right)^2 - \left(\frac{2}{\gamma-1} + \frac{2\gamma}{\gamma-1}M_1^2\sin^2\beta\right)\left(\frac{\rho_1}{\rho_2}\right) + \frac{2}{\gamma-1} + M_1^2\sin^2\beta = 0 \tag{6.57}$$

Once the ratio between the densities has been determined from the assumed values of M_1 and β, it is possible to determine the deflection angle α by rearranging Eq. (6.54). Applying trigonometric identities, the expression shown below is thus obtained.

$$\tan\alpha = \frac{\tan\beta\left(\frac{\rho_2}{\rho_1} - 1\right)}{\tan^2\beta + \frac{\rho_2}{\rho_1}} \tag{6.58}$$

Example 6.4
Determine the variations across a shock wave for a supersonic flow with $M_1 = 3$ and different values of the wave angle β.

Solution
The first step of the calculation procedure is to assume values of the initial Mach number (M_1) and the wave angle (β) to determine the value of the density ratio (ρ_2/ρ_1) from Eq. (6.57). Suppose that $\beta = 30°$ and $M_1 = 3$. Then, from Eq. (6.57),

$$13.50\left(\frac{\rho_1}{\rho_2}\right)^2 - 20.75\left(\frac{\rho_1}{\rho_2}\right) + 7.25 = 0$$

Solving the quadratic equation,

Table 6.1 Determination of properties variations across an oblique shock wave with $M_1 = 3$ and different wave angles

β (°)	M_{n1}	ρ_2/ρ_1	α (°)	p_2/p_1	T_2/T_1	M_2
20	1.026	1.044	0.773	1.062	1.017	2.960
30	1.500	1.862	12.774	2.458	1.320	2.367
40	1.928	2.559	21.846	4.172	1.630	1.894
50	2.298	3.082	28.860	5.995	1.945	1.482
60	2.598	3.447	33.320	7.708	2.236	1.123
70	2.819	3.683	33.277	9.105	2.472	0.814
80	2.954	3.815	23.927	10.017	2.626	0.576
90	3.000	3.857	0.000	10.333	2.679	0.475

$$\frac{\rho_1}{\rho_2} = 0.537; \quad \frac{\rho_2}{\rho_1} = 1.862$$

Then, the values of ρ_2/ρ_1, M_1, and β are used to determine the value of the deflection angle (α) from Eq. (6.58).

$$\tan\alpha = \frac{\tan 30(1.862 - 1)}{\tan^2 30 + 1.862} = 12.774\,^\circ$$

The value of ρ_2/ρ_1 used in Eq. (6.51) allows for the determination of p_2/p_1.

$$\frac{p_2}{p_1} = \frac{\frac{1.4+1}{1.4-1} \times 1.862 - 1}{\frac{1.4+1}{1.4-1} - 1.862} = 2.458$$

Then, the temperature ratio T_2/T_1 can be obtained by applying the equation of state of the ideal gas.

$$\frac{T_2}{T_1} = \frac{\rho_1}{\rho_2}\frac{p_2}{p_1} = 0.537 \times 2.458 = 1.320$$

Finally, the value of p_1/p_2 can be used in Eq. (6.56) to determine the final Mach number (M_2). First, Eq. (6.56) is rearranged to calculate M_2.

$$M_2 = \sqrt{\frac{(1 - \rho_2/\rho_1)^{-1}(p_1/p_2 - 1)}{\gamma \sin^2(\beta - \alpha)}}$$

Substituting the values,

$$M_2 = \sqrt{\frac{(1 - 1.862)^{-1}\left(2.458^{-1} - 1\right)}{1.4 \times \sin^2(30 - 12.774)}} = 2.367$$

Some sample results are presented in Table 6.1 for $M_1 = 3$ and different values of β. Note that the initial Mach number in the normal direction is given by $M_{n1} = M_1 \sin\beta$.

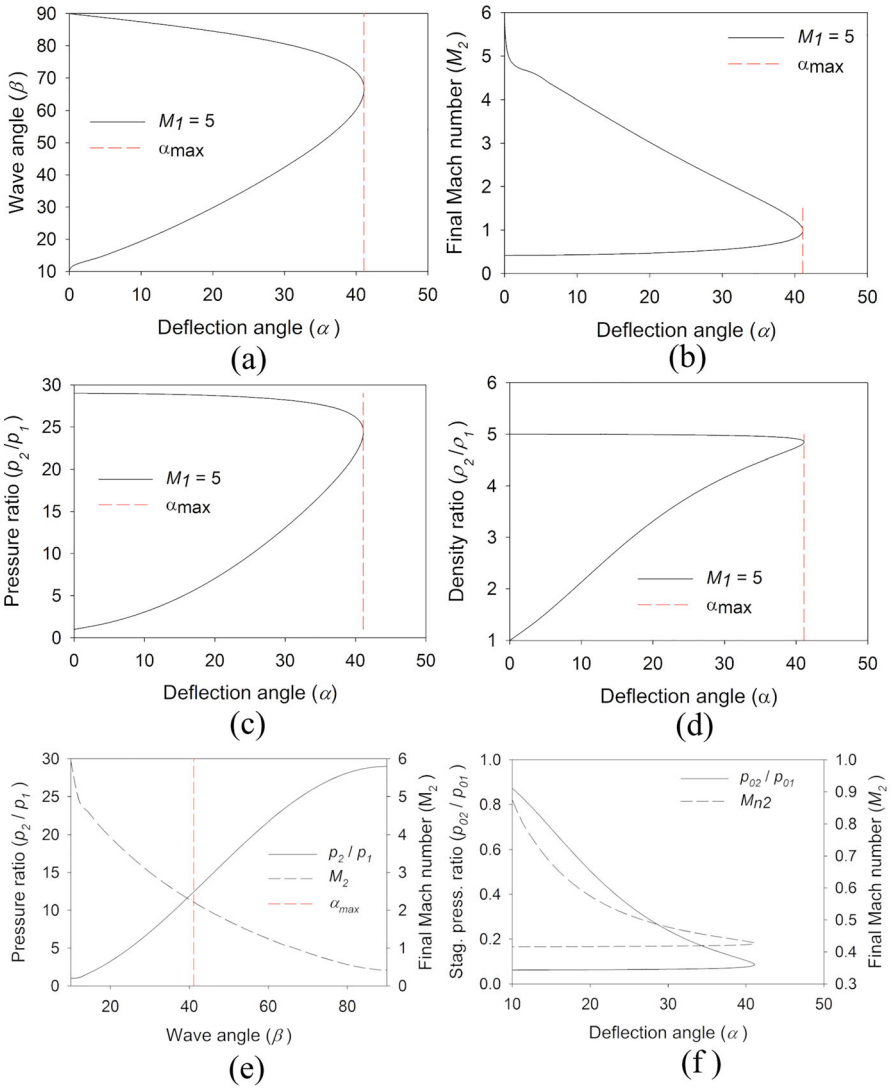


Fig. 6.11 Results obtained for a supersonic flow considering different deflection angles (α) with $M_1 = 5$ and $\gamma = 1.4$

Adopting an initial Mach number and solving for different values of the deflection angle reveals several important characteristics of the solution. The graphs in Fig. 6.11 were plotted for different values of α for $M_1 = 5$ and $\gamma = 1.4$. The Fig. 6.11 shows that for each deflection angle, there are essentially two possible solutions for each parameter. However, at the inflection point where the deflection angle is maximum (α_{max}), there is a single solution.

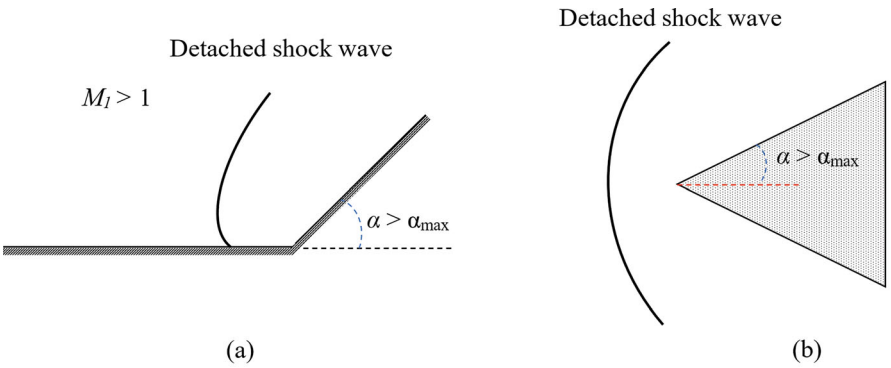


Fig. 6.12 Representation of detached shock waves. (**a**) Shock wave formed due to a wall deflection. (**b**) Shock waved formed over a wedge

Consider first the two possible solutions for each parameter. In the case of the wave angle (Fig. 6.11a), waves with two possible slopes are possible. Fig. 6.11e reveals that the higher β produce higher pressure ratios and lower final Mach numbers. It is evident that small β can even produce supersonic flow after the shock ($M_2 > 1$). This result might seem like a paradox when compared to the theory for the normal shock wave, where it has been demonstrated that the flow must be subsonic after the shock. Nevertheless, there is no contradiction as will be shown below. Analyzing Fig. 6.11f together with Eq. (6.42), it is evident that the stagnation pressure ratio is equal or lower than one, and with this the entropy change is always equal or greater than zero. On the same graph, the final Mach number of the normal velocity component (M_{n2}) is presented, and it is always lower than one. Note that $M_{n2} = M_2 \sin(\beta - \alpha)$. Therefore, the restriction that the flow must be subsonic after the shock is satisfied in the normal direction.

The existence of two solutions gives rise to two types of shock waves, namely, strong shock waves and weak shock waves. The first ones are characterized by higher pressure ratios, which as have been discussed, are related to higher wave angles and lower final Mach numbers. On the other hand, weak shock waves produce lower pressure increase and are characterized by higher Mach numbers after the shock.

Note that for any given initial Mach number, M_1, there is a maximum deflection angle, α_{max}, for which a solution for the straight oblique shock exists. Analogously, for a given α, there exists a minimum M_1 for which a solution exists. In a case where the deflection angle is greater than α_{max}, the shock will be curved and detached from the surface. Additionally, supersonic flows interacting with blunt bodies also produce detached shock waves. An example of an attached shock wave was shown in Fig. 6.10a. Detached shock waves are presented in Fig. 6.12.

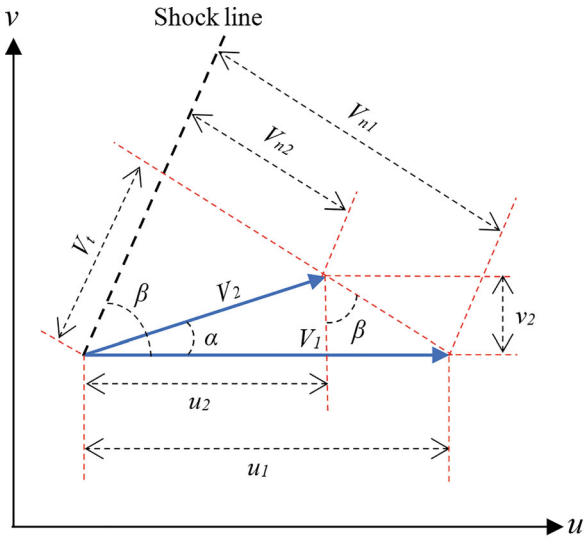


Fig. 6.13 Representation of velocity components in the plane $u - v$

6.5.2 Representation on the Hodograph Plane

Another interesting representation of oblique shocks is given in terms of the velocity components u and v, which are the horizontal and vertical velocity components in a two-dimensional flow. This representation is called shock polar. Assume that the velocity V_1 is in the horizontal direction, that is, $V_1 = u_1$. Therefore, the relationship between the velocities is shown in Fig. 6.13. Analyzing Fig. 6.13, it is possible to obtain some relationships between the normal velocity components, the angles, and the velocity coordinates (u, v). First, the components V_{n1} and V_t are related to u_1 and β as shown below.

$$V_{n1} = u_1 \sin\beta\,; \quad V_t = u_1 \cos\beta \tag{6.59}$$

Then, the difference between the normal velocity components is obtained as function of the velocity coordinates. Rearranging, Eq. (6.60) is obtained.

$$V_{n2} = V_{n1} - \sqrt{v_2^2 + (u_1 - u_2)^2} \tag{6.60}$$

The angle β appearing in the small triangle in Fig. 6.13 can be related to the velocity coordinates through the following expressions.

$$\sin\beta = \frac{u_1 - u_2}{\sqrt{v_2^2 + (u_1 - u_2)^2}} \quad \cos\beta = \frac{v_2}{\sqrt{v_2^2 + (u_1 - u_2)^2}} \tag{6.61}$$

In order to move forward and introduce a direct relationship between the angle β and the velocity coordinates, it is necessary to introduce a more general form of the Prandtl relationship. Therefore, the energy conservation equation (Eq. 6.48) is written in the form shown in Eq. (6.62).

$$\frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1}+\frac{V_{n1}^2+V_t^2}{2}=\frac{\gamma}{\gamma-1}\frac{p_2}{\rho_2}+\frac{V_{n2}^2+V_t^2}{2}=\frac{\gamma}{\gamma-1}\frac{p_0}{\rho_0} \tag{6.62}$$

From the definition of the speed of sound, it is noticed that $\gamma p_0/\rho_0 = {c_0}^2$. Also, the speed of sound at the critical conditions can be related to ${c_0}^2$.

$$c_0^2=\gamma R_g T_0=\gamma R_g T^*\left(\frac{T_0}{T^*}\right)=c*^2\frac{\gamma+1}{2} \tag{6.63}$$

Substitution of Eq. (6.63) into Eq. (6.62) and rearranging give the following expressions.

$$p_1=\rho_1\left(\frac{\gamma+1}{\gamma}\frac{c*^2}{2}-\frac{\gamma-1}{\gamma}\frac{V_{n1}^2+V_t^2}{2}\right) \tag{6.64}$$

$$p_2=\rho_2\left(\frac{\gamma+1}{\gamma}\frac{c*^2}{2}-\frac{\gamma-1}{\gamma}\frac{V_{n2}^2+V_t^2}{2}\right) \tag{6.65}$$

Equations (6.64) and (6.65) are used in the momentum equation in the normal direction (Eq. 6.47), and Eqs. (6.45) and (6.46) are used to simplify the expression. Finally, the Prandtl relation is obtained.

$$V_{n2}V_{n1}=c*^2-\frac{\gamma-1}{\gamma+1}V_t^2 \tag{6.66}$$

Substituting Eqs. (6.59), (6.60), and (6.61) into the Prandtl relation and rearranging yield an expression which allows to plot $v_2/c*$ vs $u_2/c*$ for a given value of $u_1/c*$. The derivation of the relationship shown in Eq. (6.67) is presented in more detail in the Appendix of this chapter.

$$\left(\frac{v_2}{c^*}\right)^2=\left(\frac{u_1}{c^*}-\frac{u_2}{c^*}\right)^2\frac{\left(\frac{u_1}{c^*}\right)\left(\frac{u_2}{c^*}\right)-1}{\frac{2}{\gamma+1}\left(\frac{u_1}{c^*}\right)^2-\left(\frac{u_1}{c^*}\right)\left(\frac{u_2}{c^*}\right)+1} \tag{6.67}$$

The relationship given in Eq. (6.67) provides the conditions in the hodograph plane, that is, in the plane formed by the velocity components u and v. However, direct relationships can be obtained for the dimensionless velocities $u_1/c*$, $u_2/c*$, and, $v_2/c*$ using Eqs. (6.9) and (6.10), together with the expressions $u_1 = V_1$, $u_2 = V_2\cos\alpha$, and $v_2 = V_2\sin\alpha$. The results of the algebraic manipulations are presented in Eqs. (6.68)–(6.70).

$$\frac{u_1}{c^*} = M_1 \sqrt{\frac{\gamma + 1}{2 + (\gamma - 1)M_1^2}} \tag{6.68}$$

$$\frac{u_2}{c^*} = M_2 \cos\alpha \sqrt{\frac{\gamma + 1}{2 + (\gamma - 1)M_2^2}} \tag{6.69}$$

$$\frac{v_2}{c^*} = M_2 \sin\alpha \sqrt{\frac{\gamma + 1}{2 + (\gamma - 1)M_2^2}} \tag{6.70}$$

Note that the value of u_1/c^* does not depend on the angles. Therefore, it has a unique value for a given M_1. On the other hand, the values of u_2/c^* and v_2/c^* will change.

Example 6.5
Using the results obtained in Example 6.4 for $\beta = 30°$, $M_1 = 3$, $T_0 = 600$ K, and $\gamma = 1.4$, determine the values of u_1, u_2, v_2, V_{n1}, V_{t1}, V_{n2}, and V_{t2}.

Solution
The solution to this example is obtained by applying Eqs. (6.68)–(6.70).

$$\frac{u_1}{c^*} = 3 \times \sqrt{\frac{1.4 + 1}{2 + (1.4 - 1) \times 3^2}} = 1.963$$

$$\frac{u_2}{c^*} = 2.367 \times \cos(12.774) \times \sqrt{\frac{1.4 + 1}{2 + (1.4 - 1) \times 2.367^2}} = 1.737$$

$$\frac{v_2}{c^*} = 2.367 \times \sin(12.774) \times \sqrt{\frac{1.4 + 1}{2 + (1.4 - 1) \times 2.367^2}} = 0.393$$

The value of c^* can be determined from Eq. (6.63).

$$c^* = \sqrt{\frac{2\gamma R_g T_0}{\gamma + 1}} = \sqrt{\frac{2 \times 1.4 \times 287.1 \times 600}{1.4 + 1}} = 448.3 \text{ m/s}$$

The previous results are used to obtain the desired velocities.

$$u_1 = 1.963 \times 448.3 = 880.0 \text{ m/s}$$

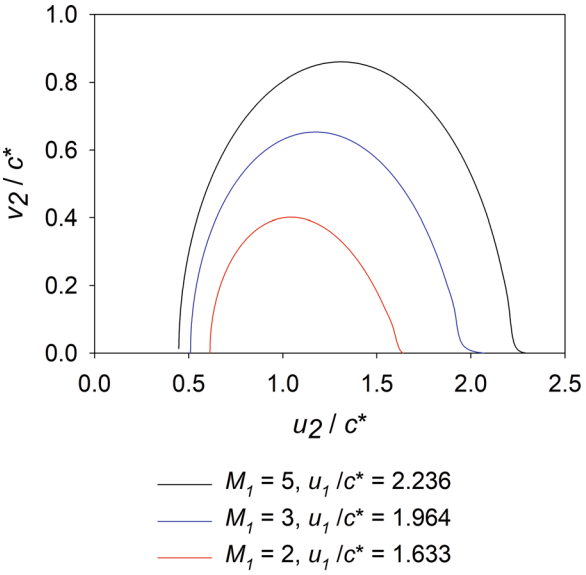


Fig. 6.14 Representation of the velocity components on the hodograph plane

$$u_2 = 1.737 \times 448.3 = 778.7 \text{ m/s}$$

$$v_2 = 0.393 \times 448.3 = 176.1 \text{ m/s}$$

Equation (6.59) is used to determine V_{n1} and $V_{t1} = V_{t2}$. Additionally, Eq. (6.60) is used to determine V_{n2}.

$$V_{n1} = 880 \times \sin 30 = 440 \text{ m/s}$$

$$V_{t1} = V_{t2} = 880 \times \cos 30 = 762.1 \text{ m/s}$$

$$V_{n2} = 440 - \sqrt{176.1^2 + (880 - 778.7)^2} = 236.9 \text{ m/s}$$

This example illustrates how the shock polar development can be used to determine the velocity components involved in the phenomenon.

Using the same results plotted in Fig. 6.11, the curve shown in Fig. 6.14 was obtained by applying Eqs. (6.68)–(6.70). Further exploration of the behavior of u_2/c^* and v_2/c^* can be performed by assuming different values of M_1, calculating u_1/c^*, and applying Eq. (6.67). Moreover, note that the curves in Fig. 6.14 exhibit a maximum for the quotient v_2/c^* and u_2/c^*.

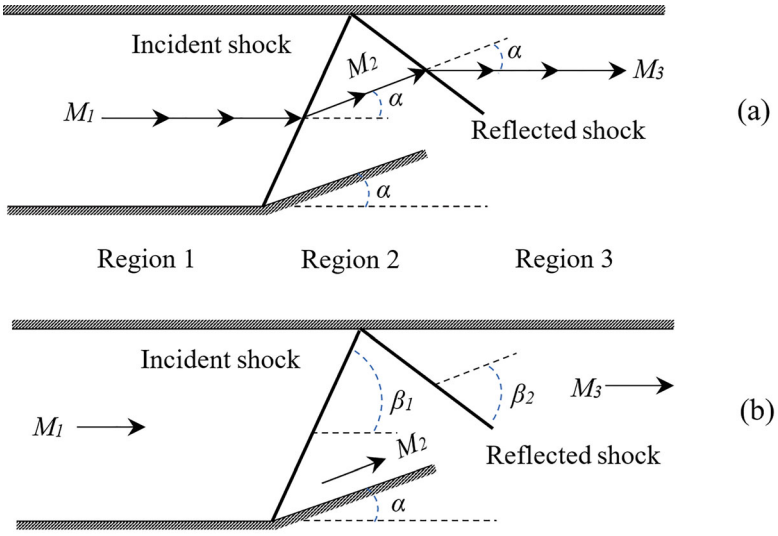


Fig. 6.15 Reflection of shock wave on a solid wall

6.5.3 Phenomenology of Shock Waves Reflection and Refraction

6.5.3.1 Shock Reflection on a Solid Boundary

Consider now the situation of a shock wave generated in the lower boundary of a wall which arrives at the upper boundary. The uniform supersonic flow arriving at the shock will be deflected by angle α once it has crossed the shock. Since the flow must be parallel to the wall near the upper boundary, then there must be a shock that corrects its direction by a deflection angle α, as shown in Fig. 6.15. The initial uniform flow arrives at the slope on the lower boundary coming from region 1. That flow has an initial Mach number M_1, and the deflection angle is $\alpha < \alpha_{\max}$. Once the flow reaches the slope, an incident shock wave is formed, as shown in Fig. 6.15a. The theory developed in this section makes it possible to determine M_2, p_2/p_1, ρ_2/ρ_1, and β_1. Once in region 2, the flow Mach number is M_2, and since the flow near the upper boundary must be parallel to the wall, there should exist a reflected shock that causes a deflection angle α. The properties M_3, p_3/p_2, ρ_3/ρ_2, and β_2 are then determined by applying the theory. Note that $\beta_1 \neq \beta_2$. This type of reflection is called regular reflection.

Another possible situation is that there is an incident shock with a Mach number for the upstream flow M_1, such that it satisfies the condition $\alpha < \alpha_{\max}$, allowing for the straight, attached oblique shock. However, at the same time, the reflected shock cannot be a straight, attached oblique shock if, for the Mach number M_2, the deflection angle is $\alpha > \alpha_{\max}$. Therefore, in this situation a normal shock forms near the upper wall and it transits into a curved shock, as shown in Fig. 6.16. This type of reflection is called Mach reflection.

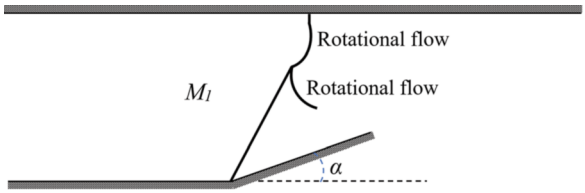


Fig. 6.16 Representation of Mach reflection

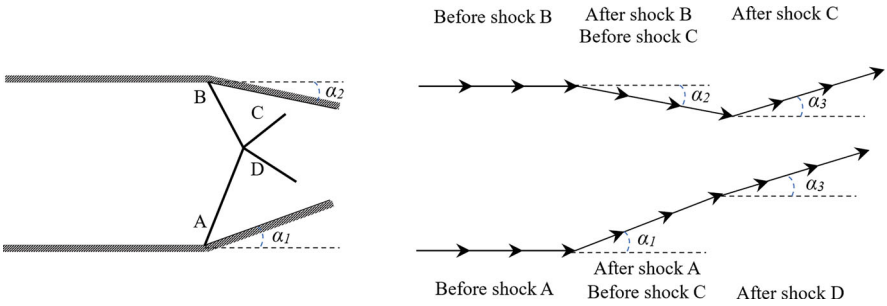


Fig. 6.17 Intersection of left-running and right-running shock waves

6.5.3.2 Shocks Intersection

The intersection of shock waves depends on the family of the shocks. The concept of family is related to the linearized theory of supersonic flows. However, for simplicity, suppose the configuration shown in Fig. 6.17. Shock A is called a left-running shock wave, while shock B is called a right-running shock wave. Shock A will be refracted and continue as shock C, while shock B will be refracted and continue as shock D. In the region after shocks C and D, the pressure is uniform, and the velocity direction is the same for all streamlines. However, note that shocks A and B could have different strengths, and this means that the magnitudes of the flow velocities in the region after shocks C and D could be different. Moreover, the entropies are also different for the flows coming across different shocks.

When the wall at the lower boundary presents two different slopes, two shock waves of different strengths will be formed. Shock wave A will overtake shock wave B, because its wave angle is lower than that of shock B. Therefore, the velocity of the flow after crossing A will be higher than the velocity after crossing B. Since these shocks intersect and they are both left-running shocks, they will form a third shock wave, named D, as shown in Fig. 6.18.

In the region after shocks B and D, the flow must have the same pressure and the same direction of the velocity. Since shock B deflects the flow by the angle α_2, shock D must do the same. However, this situation is extremely restricting because the Mach number of the flow arriving at D is M_1, which means that for given M_1 and α_2, all the other properties are determined. This situation implies that another wave must be formed at the intersection of shocks A and B. This wave is actually an expansion

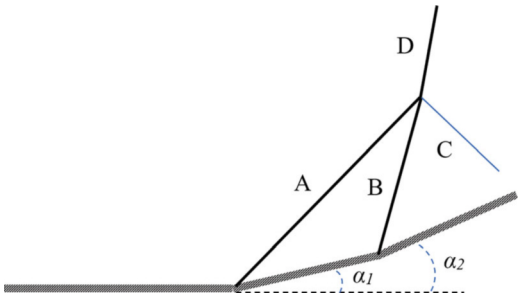


Fig. 6.18 Intersection of two left-running shock waves

wave (or weak shock wave), and it is called the Prandtl–Meyer expansion or weak shock wave. It is depicted as wave C in Fig. 6.18, and it is this wave that allows for the pressure and velocity direction to be equal for flows coming through shocks B and D.

6.6 Closure

This chapter discussed selected aspects of compressible flow theory, which are of key importance to the study of detonation phenomena. Following classical developments on compressible flow theory, the chapter started with the isentropic flow theory, presenting the relationships between stagnation properties and flow properties. Moreover, the different qualitative behaviors of subsonic and supersonic flows in geometries with area increase (or decrease) were discussed. Through this approach it was made possible to understand the general characteristics of steady one-dimensional compressible flow. Afterwards, the discussion was focused on the normal shock wave theory and oblique shock wave theory. The shock waves are treated as discontinuities that cause changes in thermodynamic properties, such as temperature, pressure, and density, among others, as well as in the flow Mach number. Important aspects, such as the Rankine–Hugoniot relation, were presented. The chapter is closed with a brief discussion on shock reflection. Several examples were provided to showcase the application of the theory.

Appendix

Derivation of the Eq. (6.65)

The derivation begins with the substitution of Eqs. (6.59) and (6.60) into the Prandtl relation, given in Eq. (6.66)

$$V_{n1}^2 - V_{n1}\sqrt{v_2^2+(u_1-u_2)^2} = u_1^2\sin^2\beta - u_1\sin\beta\sqrt{v_2^2+(u_1-u_2)^2} = c*^2 - \frac{\gamma-1}{\gamma+1}u_1^2\cos^2\beta$$

The angle β is then eliminated by introducing the sine and cosine formulas given in Eq. (6.59).

$$u_1^2\frac{(u_1-u_2)^2}{v_2^2+(u_1-u_2)^2} - \left(u_1^2-u_1u_2\right) = c*^2 - \frac{\gamma-1}{\gamma+1}u_1^2\frac{v_2^2}{v_2^2+(u_1-u_2)^2}$$

Dividing through by c^{*2} and performing some rearrangement,

$$\left(\frac{u_1}{c^*}\right)^2\frac{\left(\frac{u_1}{c^*}-\frac{u_2}{c^*}\right)^2}{\left(\frac{v_2}{c^*}\right)^2+\left(\frac{u_1}{c^*}-\frac{u_2}{c^*}\right)^2} - \left[\left(\frac{u_1}{c^*}\right)^2 - \left(\frac{u_1}{c^*}\right)\left(\frac{u_2}{c^*}\right)\right] = 1 - \frac{\gamma-1}{\gamma+1}\frac{\left(\frac{u_1}{c^*}\right)^2\left(\frac{v_2}{c^*}\right)^2}{\left(\frac{v_2}{c^*}\right)^2+\left(\frac{u_1}{c^*}-\frac{u_2}{c^*}\right)^2}$$

The procedure must bring the term $v^2/c*$ as the dependent variable. Then, performing the algebraic multiplication and simplifications between the terms of the above equation, the expression given in Eq. (6.65) is obtained.



Detonation Theory

7

Nomenclature

AF	Air-to-fuel ratio, kg_{air}/kg_F
A	Pre-exponential factor, $m^3/kg\ s$
c	Speed of sound, m/s
c_p	Heat capacity at constant pressure, J/kg K
E_a	Energia de ativação
H	Enthalpy, J
h	Specific enthalpy, J/kg
h_f°	Formation enthalpy, J/kg
Δh_c	Enthalpy of reaction per 1 kg of fuel, J/kg_F
m	Mass, kg
$\dot{m}$	Mass flow rate, kg/s
$\dot{m}''$	Mass flux rate, $kg/m^2\ s$
$\dot{m}_{k,R}$	Reaction rate in mass basis, $kg/m^3\ s$
M	Mach number
N	Number of moles, mol
p	Pressure, Pa
$\tilde{p}$	Dimensionless pressure
q	Dimensionless heat release
R	Universal gas constant, J/mol K
R_g	Gas constant, J/kg K
S	Entropy, J/K
s	Specific entropy, J/kg K
$\tilde{s}$	Dimensionless specific entropy
T	Temperature, K
$\tilde{T}$	Dimensionless temperature

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t	Time, s
u	Velocity, m/s
$\vec{V}$	Velocity vector, m/s
V	Volume, m^3
v	Specific volume, m^3/kg
$\tilde{v}$	Dimensionless specific volume
W	Molecular weight, kg/kmol
x	Horizontal coordinate
Y	Mass fraction

Greek Letters

β	Coefficient of thermal expansion, K^{-1}
γ	Ratio of heat capacities
$\dot{\sigma}$	Thermicity, s^{-1}
σ	Thermicity coefficient
Δ_i	Induction length, m
Δ_e	Energy release pulse width, m
∇	Nabla operator
$\dot{\omega}$	Reaction rate in mole basis, $kmol/m^3$ s
ρ	Density, kg/m^3
μ	Chemical potential, J/mol

Subscripts

0	Initial conditions of a process
1	Conditions of unburned mixture
2	Conditions of burned mixture
CJ	Chapman–Jouguet
F	Fuel
H	Hugoniot curve
LCJ	Lower Chapman–Jouguet point
m	Mixture
Ox	Oxidants
P	Products
UCL	Upper Chapman–Jouguet point
VN	von Neumann state

7.1 Introduction

Detonation is a form of combustion in which the propagating wave attains supersonic velocities and produces considerable pressure and temperature increase together with the chemical reaction. Therefore, under controlled conditions, the detonations present great potential for propulsion and energy conversion applications. However, they are also the source of concerns in industry, where safety measures are considered so they could not develop under uncontrolled circumstances. The classical detonation theory involves the representation of the detonation wave as a discontinuity, the theory of Chapman–Jouguet, and the ZND theory developed simultaneously by Zel'dovich, von Neumann, and Döring. Other theoretical aspects such as the Hugoniot curve and its analysis will also be considered. The interested reader should refer to the textbooks by Lee [93], Nettleton [94], Fickett and Davis [95], Dremin [96], and Zhang [97] for further developments on detonations.

7.2 The Detonation Wave as a Discontinuity

The discussion is started by considering a planar detonation (or deflagration) wave, as shown in Fig. 7.1. The coordinate system is fixed to the detonation wave so that the unburned mixture enters the wave with a velocity u_1 and the burned mixture leaves the wave with a velocity u_2. The thermodynamic properties that characterize these mixtures are pressure (p), temperature (T), and density (ρ).

The detonation wave is regarded here as a discontinuity and is taken as a control volume in which the process takes place in steady state. The states at the inlet and exit of the control volume are defined by the abovementioned thermodynamic properties (p, T, ρ). The conservation of mass establishes the relationship shown in Eq. (7.1).

$$\dot{m}'' = \rho_1 u_1 = \rho_2 u_2 \tag{7.1}$$

The conservation of momentum gives the expression in Eq. (7.2), and the conservation of energy gives the expression in Eq. (7.3). The total enthalpy (h) is the sum of the sensible enthalpy (Δh) and the enthalpy of formation (h_f°), as known from previous discussions. Although sensible enthalpy could be simplified by assuming a constant heat capacity, this simplification is not yet applied. The equation of state for ideal gases establishes a relationship between the pressure, temperature, and density, as shown in Eq. (7.4).

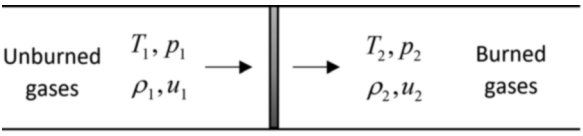


Fig. 7.1 A detonation wave as a discontinuity

$$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2 \tag{7.2}$$

$$h_1 + \frac{1}{2}u_1^2 = h_2 + \frac{1}{2}u_2^2 \tag{7.3}$$

$$p = \rho R_g T \tag{7.4}$$

As already discussed in Chap. 6, the gas constant is $R_g = R/W$. There are five unknowns which appear in Eqs. (7.1)–(7.4), namely u_1, u_2, T_2, p_2, and ρ_2. It is clear that the system of equations is not complete, because there are more unknowns than equations. Before proceeding to analyze the solution to the system of equations, it is possible to obtain the equation of the Rayleigh line by using the momentum and mass conservation equations. To do this, Eq. (7.2) is rearranged, and Eq. (7.1) is used to obtain the expression shown in Eq. (7.5).

Rayleigh line equation

$$\frac{p_2}{p_1} = \left[1 + \frac{(\dot{m}'')^2}{\rho_1 p_1}\right] - \frac{(\dot{m}'')^2}{\rho_1 p_1}\frac{\rho_1}{\rho_2} \tag{7.5}$$

Rearranging the expression in Eq. (7.5), it is possible to observe that the Rayleigh line defines situations in which the solution is impossible within the universe of the real numbers. These situations can be analyzed by applying Eq. (7.6) or (7.7).

$$\dot{m}'' = \sqrt{\frac{p_2 - p_1}{1/\rho_1 - 1/\rho_2}} = \sqrt{p_1\rho_1\frac{p_2/p_1 - 1}{1 - \rho_1/\rho_2}} \tag{7.6}$$

$$\rho_2^2 u_2^2 = \rho_1^2 u_1^2 = p_1\rho_1\left(\frac{p_2/p_1 - 1}{1 - \rho_1/\rho_2}\right) \tag{7.7}$$

It is observed that when the final pressure is greater than the initial pressure ($p_2 > p_1$) and the final density is greater than the initial density ($\rho_2 > \rho_1$), both the numerator and the denominator are positive numbers, and there is a positive real root for Eq. (7.6). Likewise, if the final pressure is smaller than the initial pressure ($p_2 < p_1$) and the final density is smaller than the initial density ($\rho_2 < \rho_1$), both the numerator and the denominator are negative numbers, and there is also a positive real root. Any other combination would not result in positive real roots in Eq. (7.6) and is, therefore, physically impossible, as the expression represents a mass flux, which must be a positive real number.

Among the two cases described above that produce physically possible solutions, the first one corresponds to a detonation, while the second one corresponds to a deflagration. Detonations are characterized by the propagation of a supersonic wave, whereas deflagrations are characterized by the propagation of a subsonic wave. The

speed of sound in a perfect gas (c) and the Mach number (M) were defined in Chap. 6, but for the sake of completeness, they are given in Eqs. (7.8) and (7.9).

$$c^2 = \gamma R_g T = \gamma \frac{p}{\rho} \tag{7.8}$$

$$M^2 = \frac{u^2}{c^2} \tag{7.9}$$

Therefore, the Rayleigh line (Eq. 7.5) can be written as a function of the reactants' Mach number as shown in Eq. (7.10).

$$\frac{p_2}{p_1} = \left(1 + \gamma_1 M_1^2\right) - \left(\gamma_1 M_1^2\right) \frac{\rho_1}{\rho_2} \tag{7.10}$$

Another important relationship is the Rankine–Hugoniot curve, which results from the energy conservation equation, Eq. (7.3). The derivation begins with the momentum conservation equation (Eq. 7.2), from which the following expression can be obtained for the velocities of the unburned gases.

$$u_1^2 = \left(\frac{p_2}{p_1} - 1\right) \frac{p_1}{\rho_1} + \frac{\rho_2}{\rho_1} u_2^2 \tag{7.11}$$

A similar expression can be obtained for the velocity of the burned gases, as shown in Eq. (7.12).

$$u_2^2 = -\left(\frac{p_2}{p_1} - 1\right) \frac{p_1}{\rho_2} + \frac{\rho_1}{\rho_2} u_1^2 \tag{7.12}$$

Substituting Eqs. (7.11) and (7.12) in Eq. (7.3) and rearranging, it will be observed that the definition of the squared mass flux rate appears and cancels out. An intermediary step on this algebraic rearrangement is shown below.

$$2(h_2 - h_1) = \left(\frac{p_2}{p_1} - 1\right) \frac{p_1}{\rho_1} + \left(\frac{p_2}{p_1} - 1\right) \frac{p_1}{\rho_2} + \frac{\rho_2^2 u_2^2}{\rho_1 \rho_2} - \frac{\rho_1^2 u_1^2}{\rho_2 \rho_1}$$

After some more rearrangement, the result shown in Eq. (7.13) is obtained for the enthalpy variation. This equation is known as the Rankine–Hugoniot curve, which is obtained as a function of the total enthalpies.

Rankine–Hugoniot curve

$$h_2 - h_1 = \frac{1}{2} \frac{p_1}{\rho_1} \left(\frac{p_2}{p_1} - 1\right) \left(1 + \frac{\rho_1}{\rho_2}\right) \tag{7.13}$$

This relationship can be developed for a perfect gas, since the ratio $\gamma = c_p/c_v$ and $R_g = c_p - c_v$ can be used to express the heat capacity of a perfect gas at constant pressure.

$$c_p = \frac{\gamma}{\gamma - 1} R_g \tag{7.14}$$

Therefore, the total enthalpy of a perfect gas can be calculated by taking into account the enthalpy of formation and the sensible enthalpy with constant heat capacity as shown below.

$$h = c_p T + h_f^\circ = \frac{\gamma}{\gamma - 1} \frac{p}{\rho} + h_f^\circ \tag{7.15}$$

The enthalpy difference for perfect gases in Eq. (7.13) can be replaced by Eq. (7.15) to obtain the expression shown in Eq. (7.16).

$$h_{f,1}^\circ - h_{f,2}^\circ = \left(\frac{\gamma_2}{\gamma_2 - 1} \frac{p_2}{\rho_2} - \frac{\gamma_1}{\gamma_1 - 1} \frac{p_1}{\rho_1} \right) - \frac{1}{2}(p_2 - p_1)\left(\frac{1}{\rho_1} + \frac{1}{\rho_2} \right) \tag{7.16}$$

By algebraically manipulating Eq. (7.16), the following result can be obtained for the Rankine–Hugoniot curve of a perfect gas.

$$\frac{p_2}{p_1} = \left(\alpha_1 - \frac{\rho_1}{\rho_2} + 2q \right) \Big/ \left(\alpha_2 \frac{\rho_1}{\rho_2} - 1 \right) \tag{7.17}$$

where

$$\alpha = \frac{\gamma + 1}{\gamma - 1} \tag{7.18}$$

$$q = \frac{h_{f,1}^\circ - h_{f,2}^\circ}{p_1 / \rho_1} \tag{7.19}$$

Recalling that the equation of a hyperbola would be given by $(p_2/p_1 + A)(\rho_2/\rho_1 + B) = D$, if this expression is expanded, the result shown in Eq. (7.20) is obtained.

$$\frac{p_2}{p_1} \frac{\rho_1}{\rho_2} + B \frac{p_2}{p_1} + A \frac{\rho_1}{\rho_2} = D - AB \tag{7.20}$$

The alternative relationship for the Rankine–Hugoniot curve, shown in Eq. (7.21), results from a comparison between Eq. (7.20) and the extended form of Eq. (7.17). Another representation of the Rankine–Hugoniot curve is shown in Eq. (7.22).

$$\left(\frac{p_2}{p_1} + \frac{1}{\alpha_2} \right) \left(\frac{\rho_1}{\rho_2} - \frac{1}{\alpha_2} \right) = \beta \tag{7.21}$$

$$\frac{p_2}{p_1} = \frac{\beta}{\rho_1/\rho_2 - \alpha_2^{-1}} - \frac{1}{\alpha_2} \tag{7.22}$$

where

$$\beta = 2\frac{q}{\alpha_2} + \frac{\alpha_1}{\alpha_2} - \frac{1}{\alpha_2^2} \tag{7.23}$$

Example 7.1
Determine the value of β for the Rankine–Hugoniot equation given in Eq. (7.21) for a stoichiometric mixture of methane and air.

Solution
In order to determine the magnitude of the term on the right-hand side of Eq. (7.21), the propagation of a stoichiometric methane–air flame is considered. The complete combustion condition is imposed, and the initial pressure and temperature conditions are 101.325 kPa and 298.15 K, respectively. The complete combustion reaction is shown below, where it can be observed that there are 9.52 mol of air for every mole of methane.

$$CH_4 + 2(O_2 + 3.76N_2) \rightarrow CO_2 + 2H_2O + 7.52N_2$$

The molecular mass of the initial mixture of methane and air is obtained by taking into account the molar fractions of methane and air. Note that the molecular weights of methane and air are taken as 16 kg/kmol and 28.84 kg/kmol, respectively.

$$W_1 = 0.095 \times 16 + 0.905 \times 28.84 = 27.620 \text{ kg/kmol}$$

Therefore, the initial density of the mixture is determined by using Eq. (7.4).

$$\rho_1 = \frac{101.325 \times 10^3}{(8314.4/27.62) \times 298.15} = 1.129 \text{ kg/m}^3$$

The quotient of the universal gas constant and the molecular weight of the gas is $R_g = 0.301$ kJ/kg·K. The mass of the gas was calculated by considering the complete combustion reaction, and it was determined as $m = 0.29$ kg. The difference in enthalpies of formation is obtained by using the enthalpies of formation of the reactants and products at the reference temperature of 298.15 K and the atmospheric pressure of 101.325 kPa.

$$h_{f,1}^{\circ} - h_{f,2}^{\circ} = \frac{1}{0.29}[-74.873 - (-393.522) - 2(-241.826)] = 2766.556 \text{ kJ/kg}$$

Thus, the dimensionless heat release term is given by

$$q = \frac{2766.556 \times 10^3}{101.325 \times 10^3 / 1.129} = 30.826$$

For simplicity, the ratios of heat capacities are considered constant and equal to $\gamma_1 = 1.41$ and $\gamma_2 = 1.26$, respectively. Assuming these values, $\alpha_1 = 5.88$ and $\alpha_2 = 8.69$ are obtained. Therefore, it can be verified that the right-hand side of Eq. (7.17) has the following value.

$$\beta = 2\frac{30.826}{8.69} + \frac{5.88}{8.69} - \frac{1}{8.69^2} = 7.75$$

Strong detonations: $\rho_1/\rho_{UCJ} > \rho_1/\rho_2$ Weak detonations: $\rho_1/\rho_{UCJ} < \rho_1/\rho_2 < 1$	Weak deflagrations: $\rho_1/\rho_A \le \rho_1/\rho_2 < \rho_1/\rho_{LCJ}$ Strong deflagrations: $\rho_1/\rho_{LCJ} < \rho_1/\rho_2$	No solution: $1 \le \rho_1/\rho_2 < \rho_1/\rho_A$

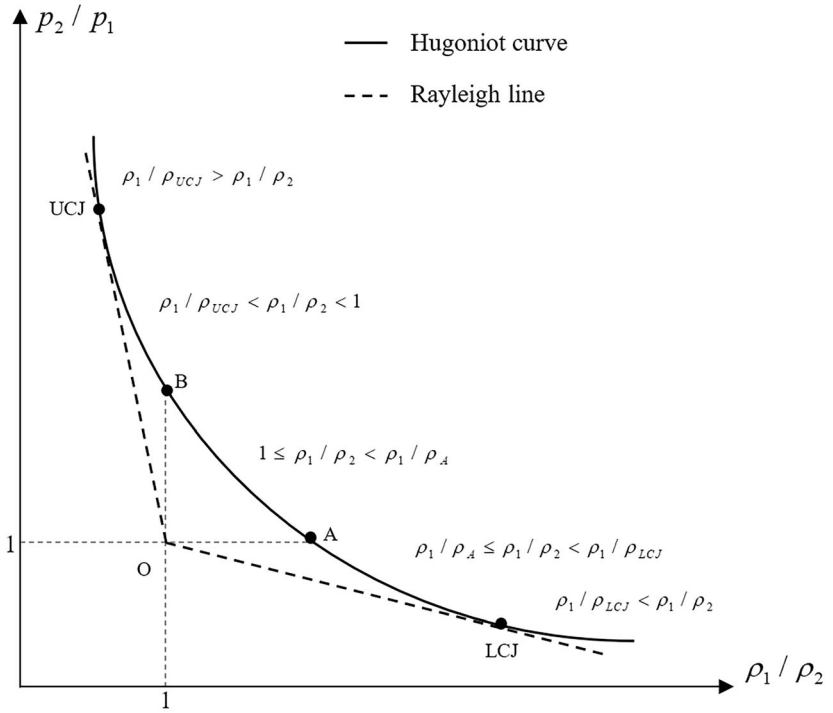


Fig. 7.2 Hugoniot curve and Rayleigh line

7.3 The Hugoniot Curve

The Hugoniot curve is the set of all possible thermodynamic states, represented by the dimensionless variables p_2/p_1 and ρ_1/ρ_2, starting from the point (1, 1), as shown in Fig. 7.2. The Hugoniot curve is intersected by the vertical line representing a constant volume process and by the horizontal line representing a constant pressure process. Additionally, there are two Rayleigh lines that pass through the point (1, 1) and are tangent to the Hugoniot curve. The first Rayleigh line is oriented in the direction of decrease of the ρ_1/ρ_2 value, intercepting the Hugoniot curve at the upper Chapman–Jouguet point (UCJ). The second Rayleigh line is oriented in the direction of increase of the ρ_1/ρ_2 value and intercepts the Hugoniot curve at the lower Chapman–Jouguet point (LCJ). The attention will be focused on these points in the next section.

These four intersections divide the Hugoniot curve into five regions. The first region is formed by thermodynamics states in the range $1 \leq \rho_1/\rho_2 < \rho_1/\rho_A$, where $p_2/p_1 - 1 > 0$. For region one, Eq. (7.5) of the Rayleigh line states that the mass flux rate would be an imaginary number. Therefore, there are no physically possible solutions in this region. In other words, a gas cannot simultaneously increase its pressure and decrease its density in a process such as the one described in this chapter.

The weak detonation region is the one in which the domain and the range are, respectively, $\rho_1/\rho_{UCJ} < \rho_1/\rho_2 < 1$ and $p_2/p_1 < p_{UCJ}/p_1$. The strong detonation region is the one in which the domain and the range are $\rho_1/\rho_{UCJ} > \rho_1/\rho_2$ and $p_2/p_1 > p_{UCJ}/p_1$, respectively. Weak and strong detonations are generally not observed in experiments.

The region of weak deflagrations is the one in which the domain and the range are $\rho_1/\rho_A \leq \rho_1/\rho_2 < \rho_1/\rho_{LCJ}$ and $p_2/p_1 > p_{LCJ}/p_1$, respectively. Weak deflagrations are frequently observed experimentally. The region of strong deflagrations is that in which the domain and the range are $\rho_1/\rho_{LCJ} < \rho_1/\rho_2$ and $p_2/p_1 < p_{LCJ}/p_1$, respectively. Strong deflagrations are generally not observed experimentally.

In an exothermic reaction, the heat released is $q > 0$, and due to the difference in magnitude between $2q/\alpha_2$ and $-1/{\alpha_2}^2$, it is noted that $\beta > 0$. Therefore, for Eq. (7.22) to be physically possible, the inequality in Eq. (7.24) must be fulfilled.

$$\frac{\rho_1}{\rho_2} - \frac{1}{\alpha_2} > 0 \tag{7.24}$$

Considering γ to be constant, taking the first and second derivatives of Eq. (7.22) with respect to ρ_1/ρ_2, and using the relationship given in Eq. (7.24), it is observed that the first derivative of the pressure ratio as a function of the density ratio is always negative. On the other hand, the second derivative is always positive.

$$\frac{d(p_2/p_1)}{d(\rho_1/\rho_2)} = -\frac{\beta}{\left(\rho_1/\rho_2 - \alpha_2^{-1}\right)^2} < 0 \tag{7.25}$$

$$\frac{d^2(p_2/p_1)}{d(\rho_1/\rho_2)^2} = \frac{2\beta}{\left(\rho_1/\rho_2 - \alpha_2^{-1}\right)^3} > 0 \tag{7.26}$$

This confirms that the Hugoniot curve is a concave function that decreases with increasing ρ_1/ρ_2, as shown in Eq. (7.27).

$$\frac{d^2(p_2/p_1)}{d(\rho_1/\rho_2)^2} > 0 \tag{7.27}$$

Example 7.2
Determine the temperature change along the Hugoniot curve for a stoichiometric methane air mixture undergoing a combustion reaction. Afterwards, determine the corresponding mass flux of the intersection with the Rayleigh line.

Solution
Consider that the initial conditions are known, assuming $T_1 = 298$ K, and $p_1 = 101.325$ kPa. Therefore, the initial density can be determined as $\rho_1 = 1.129$ kg/m^3, as was shown in Example 7.1. Moreover, the values $\alpha_1 = \alpha_2 = 6$ and $\beta = 7.75$ were obtained for this mixture in Example 7.1 by assuming that $\gamma_1 = 1.41$ and $\gamma_2 = 1.26$. The value of p_2/p_1 can be obtained for different values of ρ_2/ρ_1. As an exercise consider $\rho_2/\rho_1 = 1.67$. Note that there are 10.52 mol in the products; thus, it is possible to obtain the molecular weight $W_2 = 27.57$ g/mol.

$$\frac{p_2}{p_1} = \frac{7.75}{1.67^{-1} - 8.69^{-1}} - \frac{1}{8.69} = 15.86$$

The relationship between the temperatures is obtained by dividing Eq. (7.4) written for two conditions, being these before and after the process.

$$\frac{T_2}{T_1} = \frac{p_2}{p_1}\frac{\rho_1}{\rho_2}\frac{W_2}{W_1} = 15.86 \times \frac{1}{1.67} \times \frac{27.57}{27.62} = 9.48$$

Therefore, for the adopted initial conditions, the final pressure would be $p_2 = 1607$ kPa, and the final temperature would be $T_2 = 2825$ K. These orders of magnitude give a good understanding of the typical pressure and temperature increase across a detonation wave. The Rayleigh line equation can be written as shown in Eq. (7.6). Thus, the mas flux can be readily obtained.

$$\dot{m}'' = \sqrt{101{,}325 \times 1.129 \times \frac{15.86 - 1}{1 - 1.67^{-1}}} = 2058.4 \text{ kg/m}^2 \text{ s}$$

Then, the velocity of the reactants would be

$$u_1 = \frac{\dot{m}''}{\rho_1} = \frac{2058.4}{1.129} = 1823.2 \text{ m/s}$$

Again, the above result gives a good understanding in terms of order of magnitude of the velocity of the reactants relative to the detonation wave.

Now that the Hugoniot curve and the Rayleigh line have been studied and analyzed in order to establish physically possible processes, it is time to introduce the Chapman–Jouguet theory.

7.4 The Chapman–Jouguet Theory

The Chapman–Jouguet points are those common to the Rayleigh line and the Hugoniot curve when the former is tangent to the latter. As these are common points, it is possible to determine the velocity of the product gases (u_2) by looking at the slopes of the tangents and equating them. The slope of the tangent to the Hugoniot curve can be obtained by derivation of Eq. (7.21), where β and α_2 are taken as constants.

$$\left[\frac{d(p_2/p_1)}{d(\rho_1/\rho_2)}\right]_H = -\frac{p_2/p_1 + \alpha_2^{-1}}{\rho_1/\rho_2 - \alpha_2^{-1}} \tag{7.28}$$

The slope of the Rayleigh line is obtained directly from the graph in Fig. 7.2, from which it is evident that the slope is negative.

$$\left[\frac{d(p_2/p_1)}{d(\rho_1/\rho_2)}\right]_R = -\frac{p_2/p_1 - 1}{1 - \rho_1/\rho_2} \tag{7.29}$$

The slopes must be the same, since the Rayleigh line is the tangent to the Hugoniot curve at the Chapman–Jouguet points; therefore, from Eqs. (7.28) and (7.29), the following is obtained.

$$(p_2/p_1 - 1)\left(\rho_1/\rho_2 - \alpha_2^{-1}\right) = \left(p_2/p_1 + \alpha_2^{-1}\right)(1 - \rho_1/\rho_2)$$

Replacing the expression for α_2 in Eq. (7.29) and rearranging,

$$\gamma_2 p_2 \rho_2 = \rho_1 p_1 \frac{(p_2/p_1 - 1)}{(1 - \rho_1/\rho_2)}$$

Applying Eq. (7.7) to the right-hand side of the previous expression, it is found that the velocity of the products at the Chapman–Jouguet points is equal to the speed of sound, as shown in Eq. (7.30). This result is generally known as the Chapman–Jouguet criterion.

$$u_2^2 = \gamma_2 p_2/\rho_2 = c_2^2 \quad \text{or} \quad M_2 = u_2/c_2 = 1 \tag{7.30}$$

Example 7.3
Determine in a simplified way the velocity of a detonation wave, at the upper Chapman–Jouguet point (UCJ), which propagates in a stoichiometric mixture of ethanol and air.

Solution
The unknowns in this problem are the final pressure, the density of the product gases, and the velocity of the reactants mixture. The data for the initial pressure and the initial temperature must be known; therefore, the initial conditions of 101.235 kPa and 298.15 K are assumed.

The system of equations consists of the equations for the conservation of mass (Eq. 7.1), momentum (Eq. 7.2), and the conservation of energy (Eq. 7.3). These equations were algebraically manipulated using the equation of state of the ideal gas (Eq. 7.4), the enthalpies were written in simplified form, (Eq. 7.14), and the velocity of the product gases was replaced by the speed of sound of the products, (Eq. 7.30). The results of these procedures are the three functions shown in Eqs. (7.31)–(7.33).

$$f_1(p_2, T_2, u_1) = \frac{u_1}{u_2} - \frac{\rho_2}{\rho_1} = 0 \tag{7.31}$$

$$f_2(p_2, T_2, u_1) = \frac{\rho_2}{\rho_1}\left(\frac{\gamma_2 + 1}{\gamma_2} - \frac{1}{\gamma_2}\left[\frac{p_2}{p_1}\right]^{-1}\right) - \left[\frac{u_1}{u_2}\right]^2 = 0 \tag{7.32}$$

$$f_3(p_2, T_2, u_1) = \frac{\gamma_2 + 1}{\gamma_2 - 1} - \frac{2}{\gamma_2}\left(\frac{\gamma_1}{\gamma_1 - 1} + \frac{\Delta h_R}{p_1/\rho_1}\right)\left(\frac{p_2}{p_1}\right)^{-1}\frac{\rho_2}{\rho_1} - \left(\frac{u_1}{u_2}\right)^2 = 0 \tag{7.33}$$

Normally, the solution would be determined directly by using a programming platform such as Python or another programming language. However, this example is about describing the solution procedure, and Python will be used only to perform some operations. The Newton–Raphson method is suitable for solving this type of nonlinear system of algebraic equations. First, however, the three functions are written by adopting the variables and parameters defined in Eqs. (7.34) and (7.35).

$$x_1 = p_1/p_2 \quad x_2 = \rho_2/\rho_1 \quad x_3 = u_1/u_2 \tag{7.34}$$

$$a = \frac{\gamma_2 + 1}{\gamma_2}; \quad b = \frac{1}{\gamma_2}; \quad d = \frac{\gamma_1}{\gamma_1 - 1} + \frac{\Delta h_R}{p_1/\rho_1} \tag{7.35}$$

By applying these variables and parameters, it is possible to simplify the system, as shown in Eqs. (7.36)–(7.38).

$$f_1(p_2, T_2, u_1) = x_3 - x_2 = 0 \tag{7.36}$$

$$f_2(p_2, T_2, u_1) = ax_2 - bx_2x_1 - x_3^2 = 0 \tag{7.37}$$

$$f_3(p_2, T_2, u_1) = \alpha_2 - 2bdx_2x_1 - x_3^2 = 0 \tag{7.38}$$

Then, the Jacobian matrix is written as shown below.

$$J = \begin{bmatrix} \partial f_1/\partial x_1 & \partial f_1/\partial x_2 & \partial f_1/\partial x_3 \\ \partial f_2/\partial x_1 & \partial f_2/\partial x_2 & \partial f_2/\partial x_3 \\ \partial f_3/\partial x_1 & \partial f_3/\partial x_2 & \partial f_3/\partial x_3 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 1 \\ -bx_2 & a - bx_1 & -2x_3 \\ -2bdx_2 & -2bdx_1 & -2x_3 \end{bmatrix} \tag{7.39}$$

The partial derivatives of the Jacobian matrix were determined analytically, but, in most applications they are determined numerically by the central finite difference method [98]. In the Newton–Raphson method, the vector with the values of the unknowns in the nth iteration, $x^{[n]}$; the vector with the values of the functions in the nth iteration, $f\,(x^{[n]})$; and the inverse Jacobian matrix $J^{-1}(x^{[n]})$ are used to obtain a new estimate of the unknowns, $x^{[n+1]}$, according to the matrix operation shown in Eq. (7.40).

$$x^{[n+1]} = x^{[n]} - J^{-1}\left(x^{[n]}\right).f\left(x^{[n]}\right) \tag{7.40}$$

The criteria for the termination of the iterations are shown in Eqs. (7.41) and (7.42). The values on the right-hand side were taken from [98]. The first criterion refers to the value of the sum of the squares of the functions f_i. Since these functions must all be equal to zero, the first criterion specifies a maximum value for this sum of squares. The second criterion refers to the change in the value of the unknowns between two iterations. For this purpose, the value of the norm $\|dx\|$ is determined, which can be interpreted as the distance between two vectors. These convergence criteria do not have to be fulfilled simultaneously.

$$\sum_{i=1}^{N} \left[f_i\left(x_i^{[n]}\right)\right]^2 \leq 2.22 \times 10^{-16} \tag{7.41}$$

$$\|dx\| = \left\| -J^{-1}\left(x^{[n]}\right) f\left(x^{[n]}\right) \right\| \leq 10^{-6} \tag{7.42}$$

The numerical calculations of the problem are now carried out. The global reaction shown below is taken into account.

$$C_2H_5OH + 3(O_2 + 3.76N_2) \rightarrow 2CO_2 + 3H_2O + 11.28N_2$$

On the reactants side, the total mass of the mixture, the molecular mass of the reactants, and the density of the reactants are determined. The sensible enthalpy of the reactants is zero ($\Delta h_1 = 0$), as they are at the reference temperature of 298.15 K.

$$m = (1 \times 46 + 3 \times 4.76 \times 28.84) \times 10^{-3} = 0.458 \text{ kg}$$

$$M_1 = 0.065 \times 46 + 0.935 \times 28.84 = 29.956 \text{ kg/kmol}$$

$$\rho_1 = \frac{101.325}{(8.3144/29.956) \times 298.15} = 1.224 \text{ kg/m}^3$$

The ratio of the specific heat capacities γ_1 is determined by taking into account the temperature of the reactants. The heat capacities at constant pressure of ethanol and air at 298.15 K are taken from thermodynamic tables available online [2, 4].

$$\overline{c}_{p,C_2H_5OH} = 65.21 \text{ J/(mol K)}, \overline{c}_{p,ar} = 29.177 \text{ J/mol K}$$

$$\overline{c}_{p,\text{Re}} = 0.065 \times 65.210 + 0.935 \times 29.177 = 31.520 \text{ J/(mol K)}$$

$$\overline{c}_{v,\text{Re}} = 31.520 - 8.3144 = 23.206 \text{ J/(mol K)}$$

$$\gamma_1 = 31.52/23.206 = 1.359$$

The molecular mass and the gas constant can be determined for the product gases.

$$W_2 = 0.122 \times 44 + 0.184 \times 18 + 0.694 \times 28 = 28.112 \text{ kg/kmol}$$

$$R_2 = 8.3144/28.112 = 0.296 \text{ kJ/kg K}$$

$$h_R = h_{f,1}^{\circ} - h_{f,2}^{\circ} = \frac{1}{0.458}[-234.95 - 2(-393.522) - 3(-241.826)]$$

$$h_R = 2789.46 \text{ kJ/kg}$$

The ratio of the specific heat capacities of the products, γ_2, cannot be determined without knowing the value of the final temperature of the process. This temperature is also an unknown, although it is directly linked to the values of p_2 and ρ_2 via the perfect gas equation. Considering the real value of γ_2 in the iterative solution involves calculating the temperature at each step of the process and recalculating the value of γ_2.

Table 7.1 Ratio of the specific heat capacities of products at different temperatures

T_2 (K)	2400	2600	2800	3000
$\overline{c}_{p,CO_2}$ (J/mol K)	61.287	61.647	61.952	62.229
$\overline{c}_{p,H_2O}$ (J/mol K)	53.444	54.329	55.089	55.748
$\overline{c}_{p,N_2}$ (J/mol K)	36.511	36.713	36.883	37.030
$\overline{c}_{p,\text{Pr}}$ (J/mol K)	42.650	42.997	43.291	43.549
$\overline{c}_{v,\text{Pr}}$ (J/mol K)	34.334	34.682	34.977	35.234
γ_2	1.242	1.240	1.238	1.236

Table 7.2 Results of iterations

	$x^{[0]}$	$x^{[1]}$	$x^{[2]}$	$x^{[3]}$
x_1	0.066667	0.05874489	0.05879574	0.05879574
x_2	1.75000	1.76055907	1.76049304	1.76049304
x_3	1.75000	1.76055907	1.76049304	1.76049304
$\sum_{i=1}^{N} \left[f_i\left(x_i^{[n]}\right)\right]^2$	5.78×10^{-1}	2.46×10^{-5}	3.21×10^{-16}	3.21×10^{-16}
$\|dx\|$		1.69×10^{-2}	1.13×10^{-8}	$<10^{-16}$

For the sake of simplicity, the value of γ_2 is determined for a temperature range $T_2 = [2600, 3000]$ K, as shown in Table 7.1. It can be observed that the value of γ_2 presents relatively small changes in the range under consideration. Therefore, an average value of $\gamma_2 = 1.239$ is assumed. The results presented in Table 7.1 were calculated according to the following equation.

$$\overline{c}_{p,\mathrm{Pr}} = 0.122 \times \overline{c}_{p,\mathrm{CO_2}} + 0.184 \times \overline{c}_{p,\mathrm{H_2O}} + 0.694 \times \overline{c}_{p,\mathrm{N_2}}$$

The constant values that appear in Eqs. (7.36)–(7.38) are determined below.

$$a = \frac{1.239 + 1}{1.239} = 1.808; \quad b = \frac{1}{1.239} = 0.808$$

$$\alpha_2 = \frac{1.239 + 1}{1.239 - 1} = 9.369; \quad d = \frac{1.359}{1.359 - 1} + \frac{2789.46}{101.325/1.224} = 37.482$$

As a starting point for the iterations, it is assumed that the final pressure is 15 times the initial pressure, that the final density is 1.75 times the initial density, and that the velocity of the reactants is 1.75 times the velocity of the products. It is important to recall that the speed of the products is equal to the speed of sound at the product's temperature. The procedure was carried out in the order shown below. The results of the iterations are shown in Table 7.2.

1. Determine the initial value of the unknowns $x^{[0]}$.
2. Determine the value of the functions f_i ($x^{[n]}$) from Eqs. (7.36)–(7.38).
3. Determine the Jacobian matrix from Eq. (7.39).
4. Determine the new value of the unknowns by applying Eq. (7.40).
5. Check the first convergence criterion from Eq. (7.41).
6. Check the second convergence criterion from Eq. (7.42).
7. Repeat steps 2–6 until one or both convergence criteria are fulfilled.

As can be observed, three iterations were necessary to fulfil the second convergence criterion. Since the vector of unknowns does not change significantly between the second and third interactions, the process is terminated. The physical variables are now retrieved.

$$p_2 = p_1 x_1^{-1} = 1723.3 \text{ kPa}$$

$$\rho_2 = \rho_1 x_2 = 2.154 \text{ kg/m}^2$$

$$u_2 = \gamma_2 p_2 / \rho_2 = 995.6 \text{ m/s}$$

$$u_1 = u_2 x_3 = 1752.8 \text{ m/s}$$

$$T_2 = p_2 / (\rho_2 R_2) = 2702.9 \text{ K}$$

A detonation wave from a stoichiometric ethanol–air mixture thus travels at 1752.8 m/s (the speed of the reactants relative to the wave). After the passage of the wave, the product gases have a pressure that is 17 times greater than the initial pressure, a density that is 1.76 times greater than the initial density, and a temperature of 2702.9 K. For comparison, the temperature of the adiabatic flame of a stoichiometric mixture of ethanol–and air at constant pressure is 2358.8 and it is 2850.7 K at constant volume.

Up to Sect. 7.4, the Hugoniot curve, the Rayleigh line, its derivation from the conservation equations, and some application examples were presented. Now an analysis will be carried out, which considers a process that connects the state at one of the Chapman–Jouguet points with another state on the Hugoniot curve.

7.5 Analysis of the Hugoniot Curve

A thermodynamic process that takes place along the Hugoniot curve can be represented by the curved line on the $\tilde{p} - \tilde{v}$ plane. Similarly, an isentropic process that occurs between the same states is represented on this plane. The comparison between these processes is shown in Fig. 7.3. From this comparison, conclusions can

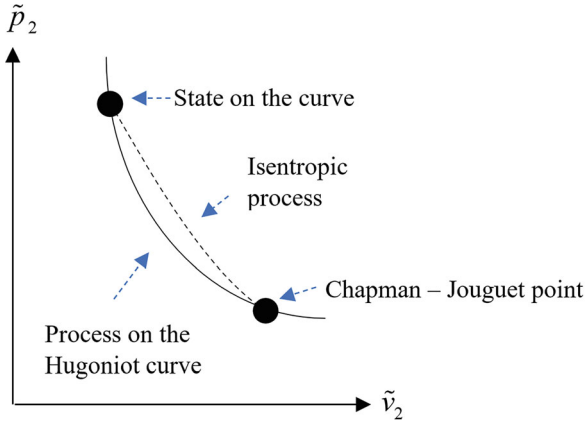


Fig. 7.3 Comparison between isentropic process and a process on the Hugoniot curve

be drawn about the velocity of the product gases compared to the speed of sound in the different regions of the Hugoniot curve.

The analysis is generalized by introducing dimensionless variables to represent the process, since the Hugoniot curve provides all possible states starting from an initial state determined by the state variables p_1 and ρ_1. The dimensionless variables on the side of the products are defined in Eqs. (7.43) and (7.44). Similar definitions also apply to the reactants.

$$\tilde{p}_2 = \frac{p_2}{p_1} \; ; \quad \tilde{v}_2 = \frac{\rho_1}{\rho_2} \; ; \quad \tilde{T}_2 = \frac{T_2}{T_1} \tag{7.43}$$

$$\tilde{h}_2 = \frac{h_2}{p_1/\rho_1} \; ; \quad \tilde{h}_1 = \frac{h_1}{p_1/\rho_1} \; ; \quad \tilde{s}_2 = \frac{s_2 T_1}{p_1/\rho_1} \tag{7.44}$$

The analysis begins with the consideration of Eq. (7.13) for the Hugoniot curve and its representation as a function of dimensionless variables.

$$\widetilde{h}_2 = \frac{1}{2}(\widetilde{p}_2 - 1)(1 + \widetilde{v}_2) + \widetilde{h}_1 \tag{7.45}$$

An infinitesimal change in the enthalpy of the products can be represented by the differential of $\tilde{h}_2$.

$$d\widetilde{h}_2 = \frac{1}{2}(1 + \widetilde{v}_2)d\widetilde{p}_2 + \frac{1}{2}(\widetilde{p}_2 - 1)d\widetilde{v}_2 \tag{7.46}$$

One of the differential relationships between thermodynamic properties is given by $T_2 ds_2 = dh_2 - v_2 dp_2$. This expression, written as a function of dimensionless variables, is shown in Eq. (7.47).

$$\widetilde{T}_2 d\widetilde{s}_2 = d\widetilde{h}_2 - \widetilde{v}_2 d\widetilde{p}_2 \tag{7.47}$$

Substitution of Eq. (7.47) into Eq. (7.46) and some rearrangement give the following expression.

$$\widetilde{T}_2 d\widetilde{s}_2 = \frac{1}{2}(1 - \widetilde{v}_2)d\widetilde{p}_2 + \frac{1}{2}(\widetilde{p}_2 - 1)d\widetilde{v}_2 \tag{7.48}$$

Dividing and multiplying the second term on the right-hand side of the above equation by $(1 - \tilde{v}_2)$ and rearranging it, Eq. (7.49) is obtained.

$$\widetilde{T}_2 \frac{d\widetilde{s}_2}{d\widetilde{v}_2} = \frac{1}{2}(1 - \widetilde{v}_2)\left(\frac{d\widetilde{p}_2}{d\widetilde{v}_2} - \frac{\widetilde{p}_2 - 1}{\widetilde{v}_2 - 1}\right) \tag{7.49}$$

Equation (7.49) shows that $d\tilde{s}_2/d\tilde{v}_2$ is a function of the tangent to the Hugoniot curve and to the tangent of the Rayleigh line. The former is represented by the derivative of the pressure as a function of the specific volume ($d\tilde{p}_2/d\tilde{v}_2$), as can be shown from Eq. (7.28). On the other hand, the latter is represented by the ratio $(\tilde{p}_2 - 1)/(\tilde{v}_2 - 1)$ as can be shown from Eq. (7.29). Therefore, when Eq. (7.49) is

written at the Chapman–Jouguet points, the derivative of entropy as a function of specific volume is zero. In other words, the entropy has a maximum or a minimum at the Chapman–Jouguet points.

$$\left(\frac{d\tilde{s}_2}{d\tilde{v}_2}\right)_{CJ} = 0 \tag{7.50}$$

Deriving Eq. (7.49) a second time in terms of specific volume, still using the Chapman–Jouguet points, with the first derivative of entropy equal to zero (Eq. 7.50), the following result is obtained.

$$\left(\frac{d^2\tilde{s}_2}{d\tilde{v}_2^2}\right)_{CJ} = \frac{1}{2\tilde{T}_2}(1-\tilde{v}_2)\left(\frac{d^2\tilde{p}_2}{d\tilde{v}_2^2}\right)_{CJ} \tag{7.51}$$

The mathematical expression in Eq. (7.27) states that the second derivative of pressure as a function of specific volume is always positive $d^2\tilde{p}_2/d\tilde{v}_2^2 > 0$. Therefore, the sign of the second derivative of entropy is determined by the sign of the term $(1-\tilde{v}_2)$. At the upper Chapman–Jouguet point, it is observed that $(1-\tilde{v}_2) > 0$. Therefore, this point is a minimum in terms of entropy. On the other hand, at the lower Chapman–Jouguet point, it is noticed that $(1-\tilde{v}_2) < 0$, and the entropy is a maximum at this point.

$$\left(\frac{d^2\tilde{s}_2}{d\tilde{v}_2^2}\right)_{\text{UCJ}} > 0 \tag{7.52}$$

$$\left(\frac{d^2\tilde{s}_2}{d\tilde{v}_2^2}\right)_{\text{LCJ}} < 0 \tag{7.53}$$

In general, the entropy differential can be represented as in Eq. (7.54).

$$d\tilde{s}_2 = \left(\frac{\partial\tilde{s}_2}{\partial\tilde{p}_2}\right)_{\tilde{v}_2} d\tilde{p}_2 + \left(\frac{\partial\tilde{s}_2}{\partial\tilde{v}_2}\right)_{\tilde{p}_2} d\tilde{v}_2 \tag{7.54}$$

Then, for an isentropic process, the following is obtained.

$$\left(\frac{\partial\tilde{p}_2}{\partial\tilde{v}_2}\right)_{\tilde{s}_s} = -\left(\frac{\partial\tilde{s}_2}{\partial\tilde{v}_2}\right)_{\tilde{p}_2} \Big/ \left(\frac{\partial\tilde{s}_2}{\partial\tilde{p}_2}\right)_{\tilde{v}_2} \tag{7.55}$$

The derivative of the entropy on the Hugoniot curve is given by Eq. (7.56).

$$\left(\frac{\partial\tilde{s}_2}{\partial\tilde{v}_2}\right)_H = \left(\frac{\partial\tilde{s}_2}{\partial\tilde{p}_2}\right)_{\tilde{v}_2}\left(\frac{\partial\tilde{p}_2}{\partial\tilde{v}_2}\right)_H + \left(\frac{\partial\tilde{s}_2}{\partial\tilde{v}_2}\right)_{\tilde{p}_2} \tag{7.56}$$

After some rearrangement,

$$\left(\frac{\partial \widetilde{p}_2}{\partial \widetilde{v}_2}\right)_H = \left[\left(\frac{\partial \widetilde{s}_2}{\partial \widetilde{v}_2}\right)_H - \left(\frac{\partial \widetilde{s}_2}{\partial \widetilde{v}_2}\right)_{\widetilde{p}_2}\right] \Big/ \left(\frac{\partial \widetilde{s}_2}{\partial \widetilde{p}_2}\right)_{\widetilde{v}_2} \tag{7.57}$$

Subtracting Eq. (7.55) from Eq. (7.57),

$$\left(\frac{\partial \widetilde{p}_2}{\partial \widetilde{v}_2}\right)_H - \left(\frac{\partial \widetilde{p}_2}{\partial \widetilde{v}_2}\right)_{\widetilde{s}_s} = \left(\frac{\partial \widetilde{s}_2}{\partial \widetilde{v}_2}\right)_H \Big/ \left(\frac{\partial \widetilde{s}_2}{\partial \widetilde{p}_2}\right)_{\widetilde{v}_2} \tag{7.58}$$

Rewriting Eq. (7.49) by taking the derivatives on the Hugoniot curve, Eq. (7.59) is obtained.

$$\frac{2\widetilde{T}_2}{1-\widetilde{v}_2}\left(\frac{d\widetilde{s}_2}{d\widetilde{v}_2}\right)_H = \left(\frac{d\widetilde{p}_2}{d\widetilde{v}_2}\right)_H - \frac{\widetilde{p}_2 - 1}{\widetilde{v}_2 - 1} \tag{7.59}$$

Then, substituting Eq. (7.58) in the expression above gives,

$$\frac{\widetilde{p}_2 - 1}{\widetilde{v}_2 - 1} - \left(\frac{\partial \widetilde{p}_2}{\partial \widetilde{v}_2}\right)_{\widetilde{s}_s} = \left(\frac{d\widetilde{s}_2}{d\widetilde{v}_2}\right)_H \left(\frac{1}{(\partial \widetilde{s}_2/\partial \widetilde{p}_2)_{\widetilde{v}_2}} - \frac{2\widetilde{T}_2}{1-\widetilde{v}_2}\right) \tag{7.60}$$

The definition of the speed of sound establishes that it is equal to the derivative of pressure as a function of density at isentropic conditions, $c_2^2 = (\partial p_2/\partial \rho_2)_{s_2}$. Using the chain rule, it can be seen that $\rho_2^2 c_2^2 = -(\partial p_2/\partial v_2)_{s_2}$. Applying the chain rule again and taking dimensionless variables into account, the expression shown in Eq. (7.61) can be verified.

$$\left(\frac{\partial \widetilde{p}_2}{\partial \widetilde{v}_2}\right)_{s_2} = -\frac{\rho_2^2 c_2^2}{p_1 \rho_1} \tag{7.61}$$

From the equation of the Rayleigh line, which is shown in Eq. (7.7), the following is obtained.

$$\frac{\widetilde{p}_2 - 1}{\widetilde{v}_2 - 1} = -\frac{\rho_2^2 u_2^2}{p_1 \rho_1} \tag{7.62}$$

Substituting Eqs. (7.61) and (7.62) into Eq. (7.60),

$$\frac{\rho_2^2}{p_1 \rho_1}\left(c_2^2 - u_2^2\right) = \left(\frac{d\widetilde{s}_2}{d\widetilde{v}_2}\right)_H \left[\frac{1}{(\partial \widetilde{s}_2/\partial \widetilde{p}_2)_{\widetilde{v}_2}} - \frac{2\widetilde{T}_2}{1-\widetilde{v}_2}\right] \tag{7.63}$$

Therefore, Eq. (7.63) can be used to determine whether the velocity of the product gases will be greater or less than the speed of sound. For this purpose, the terms on the right-hand side of Eq. (7.63) must be analyzed. Taking into account Eq. (7.15) for the enthalpy, the enthalpy differential can be written as follows.

$$d\tilde{h}_2 = \frac{\gamma_2}{\gamma_2 - 1}\tilde{v}_2 d\tilde{p}_2 + \frac{\gamma_2}{\gamma_2 - 1}\tilde{p}_2 d\tilde{v}_2 \tag{7.64}$$

Replacing Eq. (7.64) into Eq. (7.47) and rearranging,

$$d\tilde{s}_2 = \frac{\tilde{v}_2}{\gamma_2 - 1}\frac{1}{\tilde{T}_2}d\tilde{p}_2 + \frac{\tilde{v}_2}{\gamma_2 - 1}\frac{1}{\tilde{T}_2}d\tilde{v}_2 \tag{7.65}$$

Comparing Eq. (7.65) with Eq. (7.54), the following is observed.

$$\left(\frac{\partial \tilde{s}_2}{\partial \tilde{p}_2}\right)_{\tilde{v}_2} = \frac{\tilde{v}_2}{\gamma_2 - 1}\frac{1}{\tilde{T}_2} \tag{7.66}$$

Replacing Eq. (7.66) into Eq. (7.63) gives

$$\frac{\rho_2^2}{p_1\rho_1}\left(c_2^2 - u_2^2\right) = \frac{\tilde{T}_2}{\tilde{v}_2}\left(\frac{d\tilde{s}_2}{d\tilde{v}_2}\right)_H \left(\gamma_2 - \frac{1+\tilde{v}_2}{1-\tilde{v}_2}\right) \tag{7.67}$$

As already mentioned, it is necessary to determine the sign of the term on the right-hand side of Eq. (7.67), in order to know whether the velocity of the product gases is greater or less than the speed of sound. The sign of the first term on the right-hand side of Eq. (7.67), $\tilde{T}_2/\tilde{v}_2$, is always positive, whereas the sign of the second term can be determined by taking into account that the entropy has a minimum at the UCJ point and a maximum at the LCJ point.

Strong detonations and strong deflagrations

$$\left(\frac{d\tilde{s}_2}{d\tilde{v}_2}\right)_H < 0 \tag{7.68}$$

Weak detonations and weak deflagrations

$$\left(\frac{d\tilde{s}_2}{d\tilde{v}_2}\right)_H > 0 \tag{7.69}$$

All that remains is to determine the sign of the third term on the right-hand side of Eq. (7.67). For this purpose, the above term is written as shown below.

$$A = \gamma_2 - \frac{1+\tilde{v}_2}{1-\tilde{v}_2} = \frac{(\gamma_2 - 1) - \tilde{v}_2(\gamma_2 + 1)}{1 - \tilde{v}_2} \tag{7.70}$$

Figure 7.2 shows that the dimensionless specific volume in the deflagration region is always greater than one, $(1 - \tilde{v}_2) < 0$. Therefore, the third term is always positive.

In the regions of strong and weak deflagrations

$$A = \gamma_2 - \frac{1+\widetilde{v}_2}{1-\widetilde{v}_2} > 0 \tag{7.71}$$

In the case of detonations, it is noted that $(1-\tilde{v}_2) > 0$. Therefore, the term whose sign must be determined is the numerator on the right-hand side in Eq. (7.70).

$$A_1 = A(1-\widetilde{v}_2) = (\gamma_2 - 1) - \widetilde{v}_2(\gamma_2 + 1) \tag{7.72}$$

The sign of Eq. (7.72) does not change if both terms on the right-hand side are divided by $\gamma_2 - 1$. Therefore, a third term is defined.

$$A_2 = \frac{A_1}{\gamma_2 - 1} = 1 - \alpha_2 \widetilde{v}_2 \tag{7.73}$$

In Eq. (7.17) the numerator is always positive; therefore, the denominator is always positive. The following inequality results from analyzing Eq. (7.17).

$$\alpha_2 \widetilde{v}_2 - 1 > 0 \tag{7.74}$$

Comparing Eqs. (7.74) and (7.73), it is observed that $(1 - \alpha_2 \tilde{v}_2) < 0$. Applying this result to Eq. (7.70), the following expression is obtained.

In the regions of strong and weak detonations

$$A = \gamma_2 - \frac{1+\widetilde{v}_2}{1-\widetilde{v}_2} < 0 \tag{7.75}$$

Strong detonations and weak deflagrations

$$c_2^2 - u_2^2 > 0 \tag{7.76}$$

Weak detonations and strong deflagrations

$$c_2^2 - u_2^2 < 0 \tag{7.77}$$

Therefore, the flow on the products' side is subsonic in the case of strong detonations and weak deflagrations, whereas the flow on the products' side is supersonic in the case of weak detonations and strong deflagrations.

Example 7.4
Determine the velocity of a detonation wave at the upper Chapman–Jouguet point (UCJ) propagating in a stoichiometric mixture of ethanol and air.

Solution
In this case, the problem is solved with the help of the Python programming language [99], which is used via IPython [100]. A very useful tool for solving combustion problems and implementing new solutions is the software Cantera

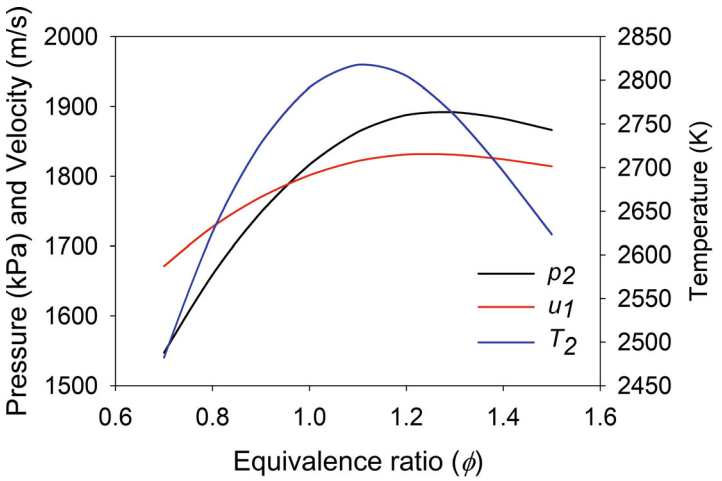


Fig. 7.4 Results obtained for different equivalence ratios with $T_1 = 298$ K and $p_1 = 101.325$ kPa

[54]. Two basic libraries for working with scientific programming in Python are NumPy [101] and SciPy [102]. There are several books dedicated to scientific programming with Python [103]. Finally, it is interesting to note that there are computational routines developed by Shepherd [104] for analyzing detonations.

The solution applies the equations for the conservation of mass (Eq. 7.1), the conservation of momentum (Eq. 7.2), and the conservation of energy (Eq. 7.3). The solution routine implemented in Python is presented below. It was necessary to use a file with information on thermodynamic properties, transport properties, and chemical kinetic mechanisms (although the chemical kinetic mechanism was not used in this example). There are several published mechanisms for the combustion of ethanol. In this example, the mechanism developed by Marinov et al. [28, 105] was used.

The results are $T_2 = 2791.9$ K, $p_2 = 1816.8$ kPa, and $u_1 = 1801.7$ m/s. Logically, there is a small difference between these results and those obtained with the simplified solution (1752.8 m/s) in Example 7.2. The main reason for this divergence is the consideration of the constant γ_2 that was made in Example 7.2. The results obtained for different equivalence ratios are presented in Fig. 7.4. It is possible to observe that for $\phi < 1$, the parameters T_2, p_2, and u_1 increase with increasing equivalence ratio. Once the mixtures become rich, that is, $\phi > 1$, then the parameters T_2, p_2, and u_1 start to decrease.

It is clear that solving this problem with the tools described is very practical and faster compared to the previously implemented manual solution. The solution routine is presented in the Appendix of this chapter.

7.6 The ZND Theory

The unidimensional theory of detonations was developed independently by Zel'dovich, von Neumann, and Döring. This theory proposes that the detonation wave is preceded by a shock wave. Therefore, the reactants are first taken to a state of higher pressure and temperature by the shock wave, and after this the detonation wave will produce the final variation of pressure and temperature. There is an induction zone and a reaction zone, which are placed between the normal shock wave and the Chapman–Jouguet detonation wave, the latter representing the detonation wave. This idea is depicted in Fig. 7.5.

The conservation equations were derived in Chap. 3, the total mass conservation and the species mass conservation equations are presented in Eqs. (7.78) and (7.79). The momentum equation is written for an inviscid flow, as shown in Eq. (7.80), and the energy conservation equation in terms of temperature is written in Eq. (7.81).

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho \vec{V} \right) = 0 \tag{7.78}$$

$$\frac{\partial (\rho Y_k)}{\partial t} + \nabla \cdot \left(\rho \vec{V} Y_k - \rho D \vec{\nabla} Y_k \right) = \dot{m}_{k,R} \tag{7.79}$$

$$\rho \frac{D\vec{V}}{Dt} = -\nabla \cdot p \tag{7.80}$$

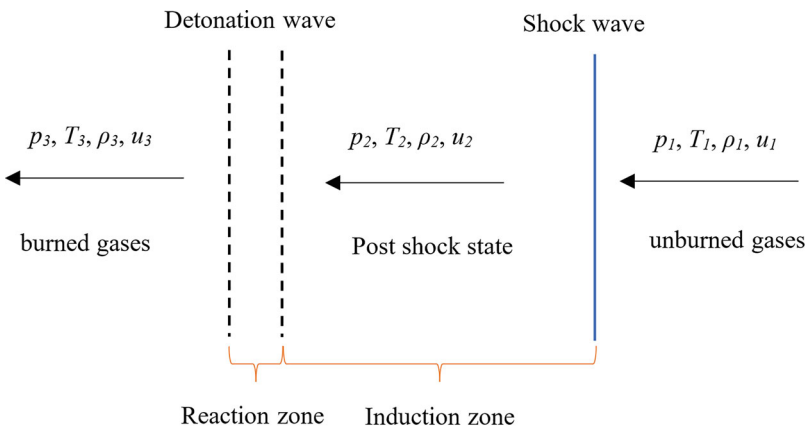


Fig. 7.5 Representation of the ZND theory

$$\rho\frac{D}{Dt}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right)=\frac{Dp}{Dt}+\mu\Phi_v+\rho\sum_{k=1}^{N}Y_k\vec{f}_k\cdot\vec{V}_{k,\text{dif}}+\dot{q}_{\text{gen},vc}+\nabla\cdot(k\nabla T)-\sum\left(h_{f,k}^0\dot{m}_{k,R}\right)$$
$$-\sum\left(\dot{m}_{k,R}\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right)+\sum\left[\rho D\nabla Y_k\cdot\nabla\left(\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right)\right] \tag{7.81}$$

The process is considered in steady state and in one dimension, that is, the derivatives as function of t, y, and z are zero. In addition, the velocity has only one component. The energy conservation equation is simplified by dropping the volumetric heat generation, the viscous dissipation, and the surface work, before applying the other assumptions. Moreover, the heat conductivity is assumed to be constant, the heat capacities of the species are assumed to be constant and equal, and the diffusion processes are considered not important. Therefore, the expressions given above are simplified as shown in the expressions below.

$$\rho\frac{du}{dx}+u\frac{d\rho}{dx}=0 \tag{7.82}$$

$$\rho u\frac{dY_k}{dx}=\dot{m}_{k,R} \tag{7.83}$$

$$\frac{dp}{dx}+\rho u\frac{du}{dx}=0 \tag{7.84}$$

$$\rho uc_p\frac{dT}{dx}=u\frac{dp}{dx}-\sum\left(h_{f,k}^0\dot{m}_{k,R}\right) \tag{7.85}$$

The energy conservation equation can be further simplified by making $c_p/R_g=\gamma/(\gamma-1)$ and $T=p/\rho R_g$. Thus, replacing these expressions in Eq. (7.85) gives,

$$u\frac{\gamma}{\gamma-1}\left(\frac{dp}{dx}-\frac{p}{\rho}\frac{d\rho}{dx}\right)=u\frac{dp}{dx}-\sum\left(h_{f,k}^0\dot{m}_{k,R}\right) \tag{7.86}$$

Further simplification of this expression produces the relation shown in Eq. (7.87),

$$u\frac{dp}{dx}-uc^2\frac{d\rho}{dx}=-(\gamma-1)\sum\left(h_{f,k}^0\dot{m}_{k,R}\right) \tag{7.87}$$

Equations (7.82)–(7.84) and (7.87) form the system of equations for the ZND theory. Moreover, note that there is a species conservation equation for each chemical species participating in the process; therefore, a chemical kinetic mechanism would be necessary to solve the system. Another important expression is

obtained by noting that $d\rho/dx = -(\rho/u)(du/dx)$ and $dp/dx = -(\rho u)(du/dx)$. Therefore, replacing these expressions in Eq. (7.86) gives the expression shown in Eq. (7.88).

$$\frac{du}{dx} = -\frac{(\gamma - 1)\sum\left(h_{f,k}^{0}\dot{m}_{k,R}\right)}{\rho c^2\left(1 - M^2\right)} \tag{7.88}$$

Example 7.5
Determine the post-shock state (or von Neumann state) for the stoichiometric mixture of the ethanol–air mixture considered in Example 7.3.

Solution
The ZND model considers that there is a precursor shock wave that preheats and compresses the reactants. Moreover, this shock is normal and has the same velocity as the detonation wave because both waves are attached. The von Neumann state (post-shock state) is, therefore, determined by using the Chapman–Jouguet velocity as the velocity of the normal shock wave. The pre-shock conditions are given as the state 1 in Example 7.3.

The equations for the normal shock wave are given in Chap. 6 for the case of an ideal gas with constant heat capacity. Considering the conditions after the normal shock wave, the Mach number is given by Eq. (6.39), the pressure by Eq. (6.37), the temperature by Eq. (6.35), and the density by the equation of state of the ideal gases. These expressions are reproduced here for simplicity.

$$M_{\mathrm{VN}}^2 = \frac{(\gamma - 1)M_1^2 + 2}{2\gamma M_1^2 - (\gamma - 1)} \tag{7.95}$$

$$\frac{p_{\mathrm{VN}}}{p_1} = \frac{1 + \gamma M_1^2}{1 + \gamma M_{\mathrm{VN}}^2} \tag{7.96}$$

$$\frac{T_{\mathrm{VN}}}{T_1} = \left(\frac{p_{\mathrm{VN}}}{p_1}\right)^2\left(\frac{M_{\mathrm{VN}}}{M_1}\right)^2 \tag{7.97}$$

At the initial conditions, the temperature is 298.15 K, the pressure is 101.325 kPa, the ratio of heat capacities is $\gamma_1 = 1.359$, and the wave velocity is u_1 from Example 7.3. Then, the following is obtained.

$$c_1 = 335.35\ \mathrm{m/s}, \quad u_1 = 1752.80\ \mathrm{m/s}, \quad M_1 = 5.23$$

Since the normal shock wave theory assumes a constant ratio of heat capacities, the value $\gamma = 1.299$ will be adopted. Then, using Eqs. (7.95)–(7.97),

$$M_{\mathrm{VN}} = \left[\frac{(1.299-1)\times 5.23^2 + 2}{2\times 1.299\times 5.23^2 - (1.299-1)}\right]^{1/2} = 0.38$$

$$p_{\mathrm{VN}} = 101.325 \times \frac{1 + 1.299\times 5.23^2}{1 + 1.299\times 0.38^2} = 3116.9 \text{ kPa}$$

$$T_{\mathrm{VN}} = 298.15 \times \left(\frac{3116.9}{101.325}\right)^2 \left(\frac{0.38}{5.23}\right)^2 = 1489.4 \text{ K}$$

Since there is no chemical reaction across the normal shock, the molecular weight will be the same, and $R_g = 0.2776$ kJ/kg K. Therefore, the speed of sound of the flow after the shock is determined in order to obtain the velocity of the flow relative to the normal shock.

$$\rho_{\mathrm{VN}} = \frac{3116.9}{0.2776\times 1489.4} = 7.539 \text{ kg/m}^3$$

$$c_{\mathrm{VN}} = \left(\frac{1.299\times 3116.9\times 10^3}{7.539}\right)^{1/2} = 732.84 \text{ m/s}$$

$$u_{\mathrm{VN}} = 0.38\times 732.84 = 278.48 \text{ m/s}$$

Note that the pressure after the detonation wave was determined in Example 7.3 as 1723.3 kPa, while the pressure after the shock wave was determined as 3116.9 kPa. The difference in pressure is evident; however, this is compensated by the increase in velocity after the detonation, $u_2 = 995.6$ m/s as obtained in Example 7.3, compared to the velocity after the normal shock, 278.48 m/s.

Now that Example 7.5 has been solved, it is interesting to notice that the pressure is much higher at the von Neumann state (VNS) than at the UCJ. Note that $p_{\mathrm{VN}} = 3116.9$ kPa, while $p_{\mathrm{UCJ}} = 1816.8$ kPa. Although the value of p_{VN} is only a first approximation, it is clear that the reactants reach a much higher pressure after the shock, and they attain a subsonic velocity (u_{VN}). Therefore, the chemical reaction process will produce a decrease in pressure accompanied by a corresponding increase in velocity, thus making the products reach a sonic velocity at the UCJ state. This process is schematically represented in Fig. 7.6. In that figure there are two Hugoniot curves; the first one corresponds to the normal shock process, while the second one corresponds to the chemical equilibrium process. In the first one, the reactants composition is not changing, while in order to reach the second one, it is necessary to change the composition through chemical reactions. Moreover, since this representation must be consistent with the Chapman–Jouguet theory, it is necessary that the Rayleigh line intercepts both Hugoniot curves as shown in Fig. 7.6.

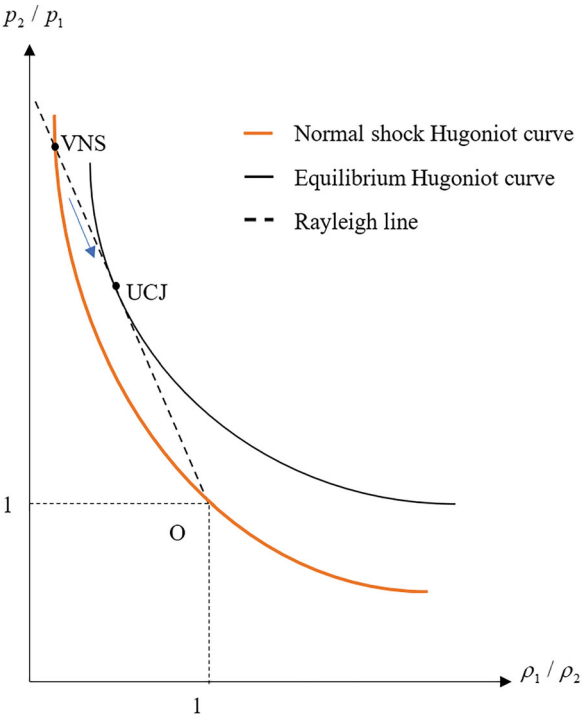


Fig. 7.6 Schematic representation of the detonation process according to ZND theory which is consistent with Chapman–Jouguet theory

Example 7.6
Evaluate the effect of temperature, pressure, and equivalence ratio on the von Neumann state (or post-shock state) for hydrogen–air mixtures.

Solution
This problem will be solved by implementing a Python routine. The software Cantera [54] together with a chemical kinetic mechanism for hydrogen–air [106] will be used to obtain the thermodynamic properties. The system of equations to be solved is formed by Eqs. (7.1)–(7.3); however, the state 2 will be the von Neumann state. The results of these calculations are presented in Fig. 7.7. The Python code is provided in the Appendix of this chapter.

The results depicted in Fig. 7.7 show that the post-shock temperature increases with the equivalence ratio until a maximum value is reached around $\phi = 1.1$, as observed in Fig. 7.7a. A similar behavior is observed for the post-shock pressure in Fig. 7.7b. However, the post-shock velocity keeps increasing with equivalence ratio for the range of ϕ considered in this example, as shown in Fig. 7.7e and f. Increasing the initial temperature results in an increase in post-shock temperature (Fig. 7.7c) and in a decrease in post-shock pressure (Fig. 7.7d). On the other hand, increasing the initial pressure results in a small increase in post-shock temperature (Fig. 7.7b) and in an increase in post-shock pressure (Fig. 7.7d). Finally, increasing the initial

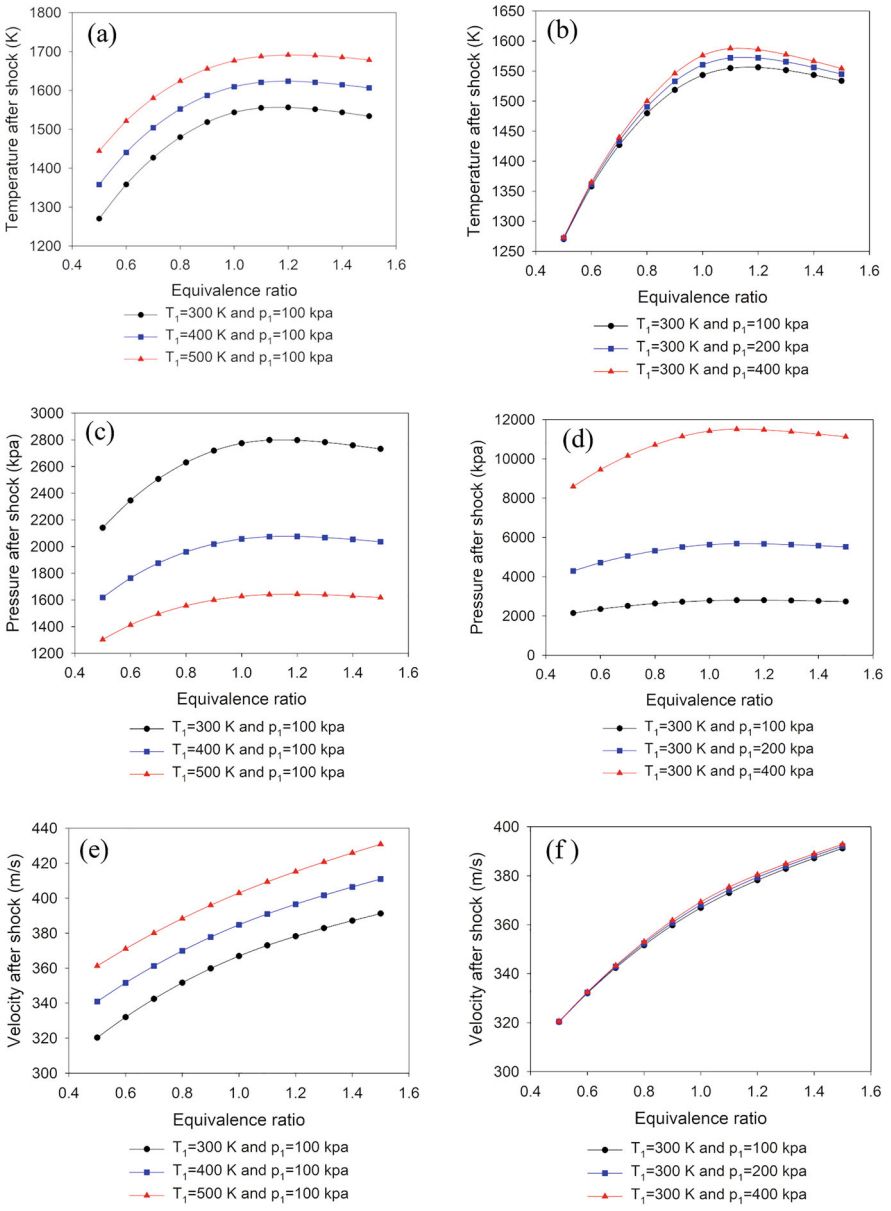


Fig. 7.7 Effect of equivalence ratio, initial temperature, and initial pressure on the post-shock conditions for hydrogen–air mixtures

temperature causes an increase in post-shock velocity (Fig. 7.7e), while increasing the initial pressure causes almost no effect in post-shock velocity (Fig. 7.7f).

7.6.1 The One-Step Chemical Kinetic Approximation

Consider the case of a one-step chemical reaction, as shown in Eq. (7.89), which considers that fuel and oxidant react to form products.

$$F + Ox \rightarrow P \tag{7.89}$$

For 1 kg of fuel, there will be AF kg of oxidant and $(1 + AF)$ kg of products. Thus, the reaction rates of the oxidant and products can be written in terms of the reaction rate of the fuel, that is, $\dot{m}_{Ox,R} = AF\dot{m}_{F,R}$ and $\dot{m}_{P,R} = -(1 + AF)\dot{m}_{F,R}$. Applying this relations, the summation term in Eq. (7.88) can be written as shown in Eq. (7.90) [7]. Note that the fuel's reaction rate is negative because it is being consumed by the reaction.

$$\sum\left(h^{\circ}_{f,k}\dot{m}_{k,R}\right) = \left[h^{\circ}_{f,F} + AFh^{\circ}_{f,Ox} - (1 + AF)h^{\circ}_{f,P}\right]\dot{m}_{F,R} = \Delta h_c \dot{m}_{F,R} \tag{7.90}$$

The fuel reaction rate for a one-step chemical reaction can be expressed in terms of the chemical–diffusive models developed and optimized in the last decade [34, 107, 108]. In terms of a weighted mass fraction ($\varphi_F = Y_F/Y_{F,\,0}$), the fuel's reaction rate is given by Eq. (7.91).

$$\frac{d\varphi_F}{dt} = -A\rho\varphi_F \exp\left(-\frac{E_a}{RT}\right) \tag{7.91}$$

In the above equation, φ_F ranges from 1 to 0 from the beginning to the end of the reaction. However, another form for the one-step reaction is suggested in the literature on detonations [93, 109]. Several works in the literature use a progress variable (λ). The formulation in terms of φ_F can be easily converted into a formulation in terms of the progress variable by making $\varphi_F = 1 - \lambda$. Then, $d\varphi_F/dt = -d\lambda/dt$. Thus, using these relationships into Eq. (7.91) yields the expression shown below.

$$\frac{d\lambda}{dt} = A\rho(1 - \lambda)\exp\left(-\frac{E_a}{RT}\right) \tag{7.92}$$

The reaction rate of the fuel in mass basis is $\dot{m}_{F,R} = \rho dY_F/dt = -Y_{F,0}\rho d\lambda/dt$. Substitution of Eq. (7.90) and (7.92) into Eq. (7.88) gives

$$\frac{du}{dx} = \frac{(\gamma - 1)Y_{F,0}\Delta h_c}{c^2\left(1 - M^2\right)}u\frac{d\lambda}{dx} \tag{7.93}$$

Equation (7.92) was obtained after recognizing that the derivative of Y_F is given in terms of t in Eq. (7.91) and that the derivative of u is given in terms of x in Eq. (7.92). Therefore, the connection is obtained by noting that $dx = udt$. then,

$$\frac{d\lambda}{dx} = \frac{1}{u}\frac{d\lambda}{dt} = A\frac{\rho}{u}(1-\lambda)\exp\left(-\frac{E_a}{RT}\right) \tag{7.94}$$

The integration of Eq. (7.92) requires that the pressure, density, and temperature be solved simultaneously. Therefore, the differential equations given in Eqs. (7.82), (7.84), and (7.93) are solved together with Eq. (7.92). In addition, the algebraic expressions given in Eqs. (7.8) and (7.9) and the ideal gas law, $T = p/\rho R_g$, must be solved at the same time to connect the different variables. It is important to point out that the value of R_g is changing with the reaction because it depends on the molecular weight of the reaction. The integration of the ZND model is performed starting from the von Neumann state, which is the state after the passage of the normal shock attached to the detonation wave. In the following two examples, some interesting aspects of the integration process will be evaluated.

Consider now the mass conservation equation given in Eq. (7.82), and note that for any function, $d\ln f/dx = (1/f)df/dx$. Therefore, Eq. (7.82) can be written in terms of $y_2 = \ln u$ and $y_4 = \ln\rho$, as shown below.

$$\frac{dy_4}{dx} = -\frac{dy_2}{dx} \tag{7.98}$$

The above equation can be integrated from the von Neumann state to an arbitrary state, yielding the expression shown in Eq. (7.99). Therefore, the differential equation representing the density is solved in terms of the new variable $y_2 = \ln u$.

$$y_4 = y_{4,\mathrm{VN}} - \left(y_2 - y_{2,\mathrm{VN}}\right) \tag{7.99}$$

New variables are defined as $y_1 = \ln(1-\lambda)$, or $y_1 = \ln\varphi_F$, and $y_3 = \ln p$. The single step-chemical reaction (Eq. 7.94) can then be written in terms of the new variables, as shown in Eq. (7.100). The momentum conservation equation (Eq. 7.84) is also written as shown in Eq. (7.101) by algebraic manipulation and variable substitution.

$$\frac{dy_1}{dx} = -A\exp\left[(y_4 - y_2) - \frac{E_a}{W_m R_g T}\right] \tag{7.100}$$

$$\frac{dy_3}{dx} = -\gamma M^2\frac{dy_2}{dx} \tag{7.101}$$

The equation for the velocity is also modified by using the variable substitution, obtaining the expression shown in Eq. (7.102).

$$\frac{dy_2}{dx} = -\frac{(\gamma-1)Y_{F,0}\Delta h_c}{c^2\left(1-M^2\right)}e^{y_1}\frac{dy_1}{dx} \tag{7.102}$$

Then, by using the natural logarithm and exponential functions properties, the temperature, the speed of sound, and the Mach number can be written as shown below.

$$R_g T = \exp(y_3 - y_4) \tag{7.103}$$

$$c^2 = \exp(\ln\gamma + y_3 - y_4) \tag{7.104}$$

$$\gamma M^2 = \exp(2y_1 - y_3 + y_4) \tag{7.105}$$

Therefore, the ZND theory with one-step chemistry can be evaluated by solving the three differential equations given in Eqs. (7.100)–(7.102). The density is determined in each step through Eq. (7.99), and the other necessary parameters (or coefficients) are calculated by Eqs. (7.103)–(7.105).

Example 7.7
Implement the solution for the ZND theory with one-step chemistry described by Eqs. (7.100)–(7.102) and closed by Eqs. (7.99) and (7.103)–(7.105). Consider a stoichiometric hydrogen–air mixture.

Solution
This problem will be solved by implementing the solution in Python and by using Cantera. The simulation initiates at the von Neumann state (or post-shock state); therefore, the routine for the post-shock state developed in Example 7.6 must be used. As a first step, the initial mole fractions and initial molecular weights are determined. It has been shown in Chap. 1 that for this mixture, $v_{ar}^s = 0.5$, and the molecular weights of hydrogen and air are 2 kg/kmol and 28.84 kg/kmol, respectively.

$$Y_{F,0} = \frac{1\times 2}{1\times 2 + 0.5\times 4.76\times 28.84} = 0.0283$$

$$Y_{\text{air}} = 1 - Y_F = 0.971$$

$$W_m = 0.0283\times 2 + 0.971\times 28.84 = 28.07 \text{ kg/kmol}$$

$$R_g = \frac{8314.4}{28.07} = 296.2 \text{ J/kg K}$$

The air-to-fuel ratio can be determined as

$$AF = \frac{0.5 \times 4.76 \times 28.84}{2} = 34.32 \text{ kg}_{\text{air}}/\text{kg}_{\text{F}}$$

The heat of combustion is determined by noting that the formation enthalpy of hydrogen and air is zero, while the formation enthalpy of the products is related only to water vapor ($\overline{h}^{\circ}_{f,\text{H}_2\text{O}} = -241.826$ kJ/mol). However, observe that there is 1 mol of H_2O and 1.88 mol of N_2 in the products. This means that the mass fraction of water in the products must be taken into account to determine the enthalpy of formation of the products.

$$Y_{\text{H}_2\text{O}} = \frac{1 \times 18}{1 \times 18 + 1.88 \times 28} = 0.254$$

$$h^{\circ}_{f,\text{H}_2\text{O}} = \frac{-241,826}{18} = -13,434 \text{ J/kg}$$

$$\overline{h}^{\circ}_{f,\text{Pr}} = 0.254 \times (-13,434) = -3412.23 \text{ kJ/kg}_{\text{Pr}}$$

Then, the enthalpy of combustion can be determined from the expression in Eq. (7.90).

$$\Delta h_c = 0 + 34.32 \times 0 - (1 + 34.32) \times 3412.23 = 120{,}519.97 \text{ kJ/kg}_{\text{F}}$$

It is interesting to express the above result as a dimensionless number in order to use it in the simulations.

$$\widetilde{q} = \frac{\Delta h_c}{R_g T_u} = \frac{120{,}519.97 \times 10^3}{296.2 \times 298} = 1365.4$$

The activation energy and the pre-exponential factor are taken from Lu et al. [108], as shown below. Also, the value of $\gamma = 1.23$ was adopted for this example.

$$\frac{E_a}{RT_u} = 54.95; \quad A = 1.064 \times 10^{10} \text{ m}^3$$

The following routine was implemented in Python.

- Define the parameters: R, ϕ, v^s_{ar}, $Y_{F,0}$, T_u, p_u.
- Define the initial state as the von Neumann state using a separate routine.
- Define the parameters: $\tilde{q}$, E_a, W_m, R_g, A, γ.
- Define the distance to be considered in the integration ($x_f - x_0$).
- Define the variables: $y_1 = \ln\varphi_F$, $y_2 = \ln u$, $y_3 = \ln p$ and $y_4 = \ln\rho$.
- y_4 is solved by Eq. (7.99) in each step.
- The R_gT, c^2, and M^2 are obtained, respectively, from Eqs. (7.103)–(7.105) in each step.
- The three differential equations to be solved are given for y_1, y_2, and y_3 in Eqs. (7.100)–(7.102).

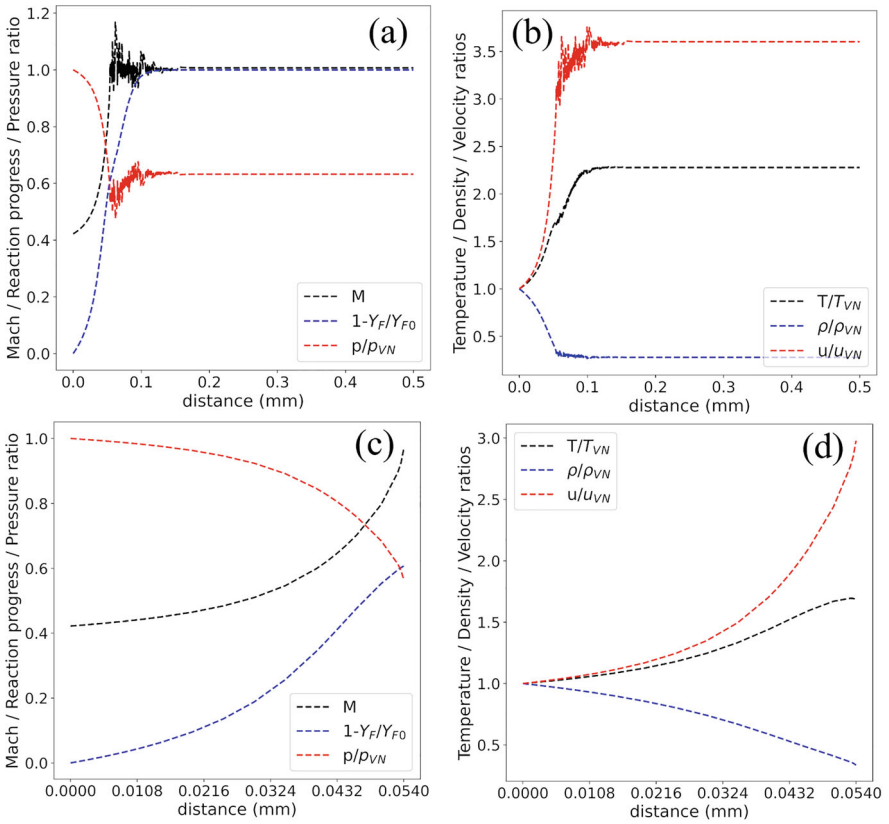


Fig. 7.8 Solution obtained for the ZND model with one-step chemical kinetic

- The numerical solution was obtained by using the method LSODA available in Python libraries.

The results of this solution are depicted in Fig. 7.8 in terms of M, T/T_{VN}, $1 - Y_F/Y_{F,0}$, p/p_{VN}, ρ/ρ_{VN}, and u/u_{VN}. Fig. 7.8a and b demonstrate that the solution oscillates once the condition $M \sim 1$ is reached. Afterwards, the solution tends to be a constant value of all the unknowns. The ZND model with one-step chemistry can also be solved by starting from an unburned gas velocity equal to the Chapman–Jouguet velocity, as was shown in the literature [110]. However, the solution presented here considers the change of the variables with the distance, being the starting point just after the normal shock wave. The oscillation in the solution is related to the form of Eq. (7.102) in which the term $(1 - M^2)$ appears in the denominator, thus creating a singularity when $M = 1$. The results in Fig. 7.8c and d show the integration up to the condition $M = 1$. There are no singularities in this region.



Fig. 7.9 Shadowgraph showing the onset of detonation wave [113]. (Reprinted from Progress in Energy and Combustion Science, Vol 34, G. Ciccarelli, S. Dorofeev, Flame acceleration and transition to detonation in ducts, 499 – 550, Copyright (2008), with permission from Elsevier)

For the initial conditions, the post-shock conditions are $p_{\mathrm{VN}} = 2832.3$ kPa, $T_{\mathrm{VN}} = 1533.7$ K, $\rho_{\mathrm{VN}} = 4.6$ kg/m^3, and $u_{\mathrm{VN}} = 365.9$ m/s. At the upper Chapman–Jouguet point, for this mixture, and the same initial conditions, $p_{\mathrm{CJ}} = 1527.4$ kPa, $T_{\mathrm{CJ}} = 2935.2$ K, and $u_{\mathrm{CJ}} = 1972.2$ m/s. As expected, the pressure decreases from p_{VN} to a value around 0.632 p_{VN}, the temperature increases from T_{VN} to around 2.299 T_{VN}, the density decreases from ρ_{VN} to 0.287 ρ_{VN}, and the velocity increases from u_{VN} to 3.486 u_{VN}. Therefore, the pressure, temperature, density, and velocity stabilize to $p = 1790$ kPa, $T = 3525.9$ K, $\rho = 1.32$ kg/m^3, and $u = 1275.5$. The temperature and pressure are higher than those obtained at the upper Chapman–Jouguet point; therefore, it can be inferred that this situation represents a strong detonation (see Fig. 7.2) because $p_2/p_1 > p_{\mathrm{CJ}}/p_1$. Moreover, it is observed that the sonic conditions ($M = 1$) are achieved before the chemical reaction has been completed ($Y_F/Y_{F,0} < 1$), and this means that the flow is chocked due to heat transfer [45, 110].

Note that the values chosen for W_m, $\tilde{q}$, E_a/RT_u, A, and γ are used to exemplify the solution of the ZND model with one-step chemistry. However, they should not be taken as representative of the hydrogen–air mixture in different situations, since they

have not been validated against different combustion processes and experimental data. The interested reader should refer to the work by Lu et al. [108] for more details about the obtention and validation of one-step chemical kinetic models.

To understand the limitations of ZND theory, it is interesting to consider Fig. 7.9, which shows the onset of a detonation wave. As can be observed, the detonation wave is not a planar wave as assumed by the ZND theory. Therefore, curvature effects have been considered by several researchers [111, 112]. The discussion of curvature effects is out of the scope of the present book. However, it should be considered for further study by the interested reader.

7.6.2 The Thermicity

In their formulation of the ZND theory, Kao et al. [114] introduced a parameter called thermicity (σ), which was defined for a mixture by the expression shown in Eq. (7.106). This parameter allows to relate the flow conditions and the chemical reactions.

$$\dot{\sigma} = \sum_{i=1}^{N} \sigma_i \frac{DY_i}{Dt} \tag{7.106}$$

In the above equation, σ_i is called the thermicity coefficient. In order to understand the meaning of thermicity from a thermodynamics point of view, it is necessary to understand its derivation, which will be obtained by following Kao et al. [114]. The literature on classic thermodynamics will be useful for this derivation [5]. The starting point will be the definition of pressure in terms of other thermodynamic parameters.

$$p = p(S, V, N_1, N_2, \ldots, N_r) \tag{7.107}$$

A small change in pressure can be related to small changes in the other thermodynamic parameters by the expression shown below.

$$dp = \left(\frac{\partial p}{\partial V}\right)_{S,N_i} dV + \left(\frac{\partial p}{\partial S}\right)_{V,N_i} dS + \sum_{i=1}^{r} \left(\frac{\partial p}{\partial N_i}\right)_{S,V,N_{j\neq i}} dN_i \tag{7.108}$$

Each partial derivative on the right-hand side of Eq. (7.108) will be analyzed separately. The **first partial derivative** can be analyzed by introducing the density, $\rho = m/V$. It is noted that $d\rho/dV = -m/V^2$.

$$\left(\frac{\partial p}{\partial V}\right)_{S,N_i} = \left(\frac{\partial p}{\partial \rho}\right)_{S,N_i} \left(\frac{d\rho}{dV}\right) = \left(\frac{\partial p}{\partial \rho}\right)_{S,N_i} \left(-\frac{m}{V^2}\right) \tag{7.109}$$

Since the square of the speed of sound of a nonreacting (frozen) mixture is $c^2 = (\partial p/\partial \rho)_{S,N_i}$, the following expression is obtained.

$$\left(\frac{\partial p}{\partial V}\right)_{S,N_i} = \left(\frac{\partial p}{\partial \rho}\right)_{S,N_i}\left(\frac{d\rho}{dV}\right) = -\frac{m}{V^2}c^2 \tag{7.110}$$

Note that in their work, Kao et al. [114] emphasize the use of the speed of sound of the nonreactive mixture. That speed was designated as c^2 in Eq. (7.110) for simplicity. Then, multiplying by dV on the left-hand side and by $dV = -d\rho(\mathrm{V}^2/\mathrm{m})$ on the right-hand side of the above expression,

$$\left(\frac{\partial p}{\partial V}\right)_{S,N_i} dV = c^2 d\rho \tag{7.111}$$

The **second partial derivative** on the right-hand side of Eq. (7.108) must be treated by looking for the appearance of thermophysical properties of the substance. Particularly, the coefficient of thermal expansion (β) and the heat capacity at constant pressure (c_p) will appear on this derivation.

$$\left(\frac{\partial p}{\partial S}\right)_{V,N_i} = -\frac{(\partial V/\partial S)_{p,N_i}}{(\partial V/\partial p)_{S,N_i}} = -\left(\frac{\partial p}{\partial V}\right)_{S,N_i}\frac{(\partial V/\partial T)_{p,N_i}}{(\partial S/\partial T)_{p,N_i}} \tag{7.112}$$

Remember that the coefficient of thermal expansion and the heat capacity at constant pressure are defined as shown in Eqs. (7.113) and (7.114) [5].

$$\beta = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p \tag{7.113}$$

$$c_p = \frac{T}{m}\left(\frac{\partial S}{\partial T}\right)_p \tag{7.114}$$

Going back to Eq. (7.113) and using the expressions shown in Eqs. (7.110), (7.113), and (7.114), the following relationship is obtained.

$$\left(\frac{\partial p}{\partial S}\right)_{V,N_i} = c^2\rho\frac{T\beta}{c_p}\frac{1}{m} \tag{7.115}$$

Also, note that $ds = (1/\mathrm{m})dS$. Therefore,

$$\left(\frac{\partial p}{\partial S}\right)_{V,N_i} dS = c^2\rho\frac{T\beta}{c_p}ds \tag{7.116}$$

Now the **third partial derivative** on the right-hand side of Eq. (7.108) is considered. In this case it is desired to have the derivatives in terms of constant temperature and pressure.

$$\left(\frac{\partial p}{\partial N_i}\right)_{S,V,N_{j\neq i}} = -\left(\frac{\partial p}{\partial V}\right)_{S,N_i}\left(\frac{\partial V}{\partial N_i}\right)_{p,S,N_{j\neq i}} = \frac{m}{V^2}c^2\left(\frac{\partial V}{\partial N_i}\right)_{p,S,N_{j\neq i}} \tag{7.117}$$

It is easier to analyze the derivative on the last term of Eq. (7.117) separately.

$$\left(\frac{\partial V}{\partial N_i}\right)_{p,S,N_{j\neq i}} = -\left(\frac{\partial V}{\partial S}\right)_{p,N_i}\left(\frac{\partial S}{\partial N_i}\right)_{p,V,N_{j\neq i}} = -\left(\frac{\partial T}{\partial p}\right)_{S,N_i}\left(\frac{\partial S}{\partial N_i}\right)_{p,V,N_{j\neq i}} \tag{7.118}$$

In the above equation, the Maxwell relation, $(\partial V/\partial S)_{p,N_i} = (\partial T/\partial p)_{S,N_i}$, has been used. Moreover, applying the procedures for reducing the partial derivatives of thermodynamic parameters [5] and observing that another Maxwell relation, $(\partial S/\partial p)_{T,N_i} = -(\partial V/\partial T)_{p,N_i}$, can be used, the following can be demonstrated.

$$\left(\frac{\partial T}{\partial p}\right)_{S,N_i} = -\frac{(\partial S/\partial p)_{T,N_i}}{(\partial S/\partial T)_{p,N_i}} = \frac{(\partial V/\partial T)_{p,N_i}}{(\partial S/\partial T)_{p,N_i}} = \frac{TV\beta}{mc_p} \tag{7.119}$$

Then, substituting Eqs. (7.118) and (7.119) into Eq. (7.117),

$$\left(\frac{\partial p}{\partial N_i}\right)_{S,V,N_{j\neq i}} = \frac{m}{V^2}c^2\left(\frac{\partial V}{\partial N_i}\right)_{p,S,N_{j\neq i}} = -\frac{1}{V}c^2\frac{T\beta}{c_p}\left(\frac{\partial S}{\partial N_i}\right)_{p,V,N_{j\neq i}} \tag{7.120}$$

The differential equation for enthalpy as a function of its natural thermodynamic parameters is given by Eq. (7.121).

$$dH = TdS + Vdp + \sum_{i=1}^{r}\mu_i dN_i \tag{7.121}$$

Therefore, it is possible to introduce the entropy and the chemical potential into Eq. (7.120) by making

$$dS = \frac{1}{T}dH - \frac{V}{T}dp - \frac{1}{T}\sum_{i=1}^{r}\mu_i dN_i \tag{7.122}$$

Then, by using Eq. (7.122) into the derivative involving entropy in Eq. (7.120) and dropping the term involving dp because the process is happening at constant pressure,

$$\left(\frac{\partial S}{\partial N_i}\right)_{p,V,N_{j\neq i}} = \frac{1}{T}\left(\frac{\partial H}{\partial N_i}\right)_{p,V,N_{j\neq i}} - \frac{1}{T}\mu_i \tag{7.123}$$

The derivative on the right-hand side of Eq. (7.123) must have T and p as the constant thermodynamic parameters. Therefore, taking the differential of H as function of p, V, and N_i, the expression shown in Eq. (7.124) is obtained.

$$dH = \left(\frac{\partial H}{\partial p}\right)_{V,N_i} dp + \left(\frac{\partial H}{\partial V}\right)_{p,N_i} dV + \sum_{i=1}^{r} \left(\frac{\partial H}{\partial N_i}\right)_{p,V,N_{j\neq i}} dN_i \tag{7.124}$$

Dividing by dN_i and taking a process with constant T, p, and $N_{j\neq i}$,

$$\left(\frac{\partial H}{\partial N_i}\right)_{T,p,N_{j\neq i}} = \left(\frac{\partial H}{\partial V}\right)_{p,N_i} \left(\frac{\partial V}{\partial N_i}\right)_{T,p,N_{j\neq i}} + \left(\frac{\partial H}{\partial N_i}\right)_{p,V,N_{j\neq i}} \tag{7.125}$$

Using Eq. (7.121) to introduce the dH into the first term on the right-hand side of Eq. (7.124); then using again the Maxwell relationship, $(\partial V/\partial S)_{p,N_i} = (\partial T/\partial p)_{S,N_i}$ and applying the results shown in Eq. (7.119), the following is obtained after rearranging.

$$\left(\frac{\partial H}{\partial N_i}\right)_{p,V,N_{j\neq i}} = \left(\frac{\partial H}{\partial N_i}\right)_{T,p,N_{j\neq i}} - \rho \frac{c_p}{\beta} \left(\frac{\partial V}{\partial N_i}\right)_{T,p,N_{j\neq i}} \tag{7.126}$$

Therefore, substitution of Eq. (7.126) into Eq. (7.123) yields

$$\left(\frac{\partial S}{\partial N_i}\right)_{p,V,N_{j\neq i}} = \frac{1}{T}\left(\frac{\partial H}{\partial N_i}\right)_{T,p,N_{j\neq i}} - \rho \frac{1}{T} \frac{c_p}{\beta} \left(\frac{\partial V}{\partial N_i}\right)_{T,p,N_{j\neq i}} - \frac{1}{T}\mu_i \tag{7.127}$$

It is easy to demonstrate that the above equation is equivalent to the mathematical relationship shown in Eq. (7.128).

$$\left(\frac{\partial S}{\partial N_i}\right)_{p,V,N_{j\neq i}} = W_i \left[\frac{1}{T}\left(\frac{\partial h}{\partial Y_i}\right)_{T,p,Y_{j\neq i}} + \frac{1}{\rho} \frac{c_p}{T\beta} \left(\frac{\partial \rho}{\partial Y_i}\right)_{T,p,Y_{j\neq i}} - \frac{1}{T} g_i\right] \tag{7.128}$$

Substitution of Eq. (7.128) into Eq. (7.120) produces the expression shown below.

$$\left(\frac{\partial p}{\partial N_i}\right)_{S,V,N_{j\neq i}} = W_i \left[-\frac{1}{V} c^2 \frac{\beta}{c_p} \left(\frac{\partial h}{\partial Y_i}\right)_{T,p,Y_{j\neq i}} - \frac{1}{V} c^2 \frac{1}{\rho} \left(\frac{\partial \rho}{\partial Y_i}\right)_{T,p,Y_{j\neq i}} + \frac{1}{V} c^2 \frac{\beta}{c_p} g_i\right] \tag{7.129}$$

Then, a differential is introduced in both sides of the above equation by noting that $dN_i = (m/W_i)dY_i$. Also, note that when N_j is constant, it is equivalent to say that Y_j is constant.

$$\left(\frac{\partial p}{\partial N_i}\right)_{S,V,N_{j\neq i}} dN_i = \left[-\rho c^2 \frac{\beta}{c_p} \left(\frac{\partial h}{\partial Y_i}\right)_{T,p,N_{j\neq i}} - c^2 \left(\frac{\partial \rho}{\partial Y_i}\right)_{T,p,Y_{j\neq i}} + \rho c^2 \frac{\beta}{c_p} g_i\right] dY_i \tag{7.130}$$

It is time to go back to Eq. (7.108) in order to conclude the derivation. Substituting the three partial derivatives, shown in Eqs. (7.111), (7.116), and

(7.130) into the partial derivative of pressure given in Eq. (7.108), the relationship shown in Eq. (7.131) is obtained.

$$dp=c^2d\rho+c^2\rho\frac{\beta}{c_p}\left(Tds+\sum_{i=1}^{r}g_idY_i\right)+\rho c^2\sum_{i=1}^{r}\left[-\frac{1}{\rho}\left(\frac{\partial\rho}{\partial Y_i}\right)_{T,p,Y_{j\neq i}}-\frac{\beta}{c_p}\left(\frac{\partial h}{\partial Y_i}\right)_{T,p,Y_{j\neq i}}\right]dY_i \tag{7.131}$$

The thermicity coefficients (σ_i) are defined as the terms inside the summation in Eq. (7.131). These coefficients are thermodynamic parameters because they are a combination of the thermodynamic parameters p, h, ρ, and Y and of the thermophysical properties β and c_p.

$$\sigma_i=-\frac{1}{\rho}\left(\frac{\partial\rho}{\partial Y_i}\right)_{T,p,Y_{j\neq i}}-\frac{\beta}{c_p}\left(\frac{\partial h}{\partial Y_i}\right)_{T,p,Y_{j\neq i}} \tag{7.132}$$

Then, the thermicity accounts for the variation of the thermicity coefficients of all species, as shown in Eq. (7.133).

$$\dot{\sigma}=\sum_{i=1}^{r}\left[-\frac{1}{\rho}\left(\frac{\partial\rho}{\partial Y_i}\right)_{T,p,Y_{j\neq i}}-\frac{\beta}{c_p}\left(\frac{\partial h}{\partial Y_i}\right)_{T,p,Y_{j\neq i}}\right]\frac{DY_i}{Dt} \tag{7.133}$$

Considering now an ideal gas, the derivatives can be readily obtained. Note that for an ideal gas, the volumetric expansion coefficient is simply $\beta = 1/T$, while the expression for the mixture's heat capacity at constant pressure is $c_p = \sum Y_i c_{p,\,i}$. The derivative for enthalpy in terms of mass fractions yields the specific enthalpy of the species in mass basis, as shown in Eq. (7.134).

$$\left(\frac{\partial h}{\partial Y_i}\right)_{T,p,Y_{j\neq i}}=h_i \tag{7.134}$$

In order to get the derivative of density, note that $1/W = N/m = \sum Y_i/W_i$. Moreover, for the ideal gas,

$$\rho=\frac{Wp}{RT}=\left(\sum_{i=1}^{r}\frac{Y_i}{W_i}\right)\frac{p}{RT} \tag{7.135}$$

Therefore, the derivative involving density leads to

$$-\frac{1}{\rho}\left(\frac{\partial\rho}{\partial Y_i}\right)_{T,p,Y_{j\neq i}}=\frac{W}{W_i} \tag{7.136}$$

Substitution of Eqs. (7.134) and (7.136) into Eq. (7.132) gives the thermicity coefficient for an ideal gas, as shown in Eq. (7.137). The thermicity for an ideal gas mixture is then given by the expression shown in Eq. (7.138).

$$\sigma_i = \frac{W}{W_i} - \frac{h_i}{c_p T} \tag{7.137}$$

$$\dot{\sigma} = \sum_{i=1}^{r} \left(\frac{W}{W_i} - \frac{h_i}{c_p T} \right) \frac{DY_i}{Dt} \tag{7.138}$$

Example 7.8
Determine the thermicity coefficients for a mixture of ideal gases formed by 0.8 mol of H_2, 0.4 mol of O_2, 1.88 mol of N_2, 0.1 mol of H_2O, 0.075 mol of OH, 0.025 mol of H, and 0.025 mol of O.

Solution
The values of the thermicity coefficients will be more easily found by using spreadsheet software and implementing Eq. (7.137). The necessary thermodynamic properties are the enthalpy and heat capacity at constant pressure, which were obtained from the thermochemical tables available online [2]. The results of these calculations at different temperatures are presented in Table 7.3.

It is observed that the values of σ_i are dominated by the ratio h_i/c_pT which in turn is dominated by the enthalpy of formation of the species. A negative value of σ_i means that the species is formed by endothermic reactions.

A final remark about thermicity is necessary. Although the thermicity coefficient (σ_i) is required to calculate the thermicity ($\dot{\sigma}$) of the reacting system, the dominating term in that calculation is the material derivative of the mass fractions (DY_i/Dt). These derivatives will be positive or negative, depending on the global reaction rate of the species. For example, imagine that a radical species is being produced such that $DY_i/Dt > 0$ and $\sigma_i < 0$, in this case the term corresponding to this species in Eq. (7.138) will be negative. Therefore, when the dominating reactions are endothermic, the sign of thermicity will be negative ($\dot{\sigma} < 0$). On the other hand, the sign of thermicity will be positive ($\dot{\sigma} > 0$) when the exothermic reactions dominate.

7.6.3 Alternative Formulation of the ZND Theory

The ZND theory is formulated by Kao et al. [114] in terms of the differential equations shown below, where the total mass conservation equation is given by Eq. (7.139), the species conservation equation is given by Eq. (7.140), the momentum equation is represented by Eq. (7.141) and the energy conservation equation is represented by Eq. (7.142).

Table 7.3 Thermicity coefficients for the components of an ideal gas mixture

	Input parameters					Thermicity coefficient (σ_i)		
Species	N_i (mol)	x_i	W_i (g/mol)	W/W_i	h_f/c_pT	298 K	1000 K	2000 K
H_2	0.800	0.242	2	10.672	0.000	10.672	3.858	2.818
O_2	0.400	0.121	32	0.667	0.000	0.667	0.199	0.118
N_2	1.880	0.569	28	0.762	0.000	0.762	0.257	0.168
H_2O	0.100	0.030	18	1.186	−33.041	34.227	9.087	3.971
OH	0.075	0.023	17	1.255	5.640	−4.385	−1.067	−0.363
H	0.025	0.008	1	21.343	536.146	−514.803	−131.916	−53.814
O	0.025	0.008	16	1.334	38.301	−36.967	−9.540	−3.948

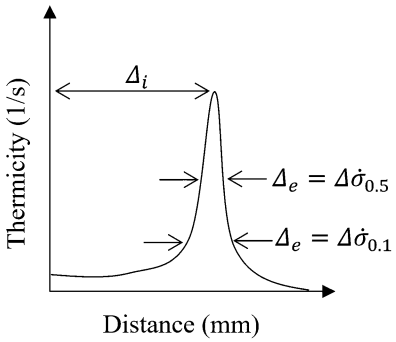


Fig. 7.10 Typical profile of thermicity and definition of induction length

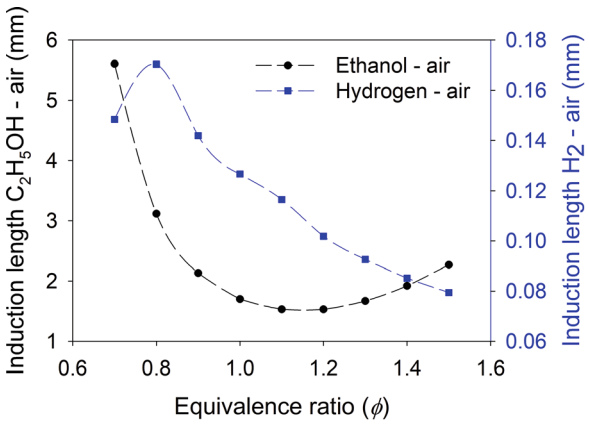


Fig. 7.11 Comparison of induction length of ethanol–air and hydrogen–air detonations at the initial conditions of 298 K and 101.325 kPa for different equivalence ratios

$$u\frac{d\rho}{dx} = -\rho\frac{\dot{\sigma}_i}{\left(1-M^2\right)} \tag{7.139}$$

$$u\frac{dY_i}{dx} = \frac{1}{\rho}W_i\dot{\omega}_i \tag{7.140}$$

$$u\frac{dp}{dx} = -\rho u^2\frac{\dot{\sigma}_i}{1-M^2} \tag{7.141}$$

$$\frac{du}{dx} = \frac{\dot{\sigma}_i}{1-M^2} \tag{7.142}$$

The advantage of this formulation of the ZND theory is that the induction length (Δ_i) and the energy release pulse width (Δ_e) can be determined by analyzing the

evolution of the thermicity. After the mixture has been compressed and heated by the leading shock, the thermicity starts increasing until it reaches a peak. The distance between the shock and the location of this peak is defined as the induction length. This situation is shown in Fig. 7.10. Also, the energy release pulse width can be defined as the width of the thermicity pulse measured at a fraction of its peak. Two examples are shown in Fig. 7.10, for example, it is possible to measure Δ_e at half the thermicity peak ($\Delta\dot{\sigma}_{0.5}$) or at 10% of that peak ($\Delta\dot{\sigma}_{0.1}$).

Example 7.9
Compare the induction lengths of detonations of ethanol–air and hydrogen–air mixtures for different equivalence ratios.

Solution
The Explosion Dynamics Laboratory of the California Institute of Technology developed the Shock and Detonation Toolbox [115], an open-source computational library integrated with Cantera [54], which allows for the calculation of the ZND detonation structure. This example was solved by using this Toolbox. The comparison between the induction length of ethanol–air and hydrogen–air mixtures is shown in Fig. 7.11.

It is observed that the induction length for hydrogen is one order of magnitude lower than the induction length for ethanol. Therefore, the induction length demonstrates the higher reactivity of hydrogen in comparison to ethanol.

7.7 Closure

This chapter discussed the classical detonation theory. The chapter started with the presentation of the detonation wave as a discontinuity through which the mass conservation, momentum, and energy conservation equations must be satisfied. The theoretical discussion is followed by the presentation of the Rayleigh line and the Hugoniot curve. Afterwards, the Chapman–Jouguet theory is discussed. This theoretical approach is applied successfully to determine the detonation wave velocity by considering that the products are at the speed of sound corresponding to their thermodynamic state. Afterwards, the chapter presented the ZND theory, named after Zel'dovich, von Neumann, and Döring. The ZND theory introduces the idea that a detonation wave is preceded by a normal shock wave, which increases the temperature and pressure of the mixture. The distance between the shock wave and the reaction wave is called induction length, and it is a characteristic length of a detonation. Moreover, the concept of thermicity and its derivation from classical thermodynamics were also presented. The chapter was closed with a formulation of the ZND theory, which accounts for detailed chemical kinetic mechanisms.

Appendix

Solution routine for example 7.3

```
import cantera as ct
import numpy as np
from scipy.optimize import fsolve
# Initial conditions
p1 = 101325  # [Pa]
T1 = 298.15  # [K]
# Composition of reactants
ER = 1.5
vars = 3 #Stoichiometric number of mols of O2 for ethanol
a = vars/ER
b = 3.76*a
reactants = 'C2H5OH:{}, O2:{}, N2:{}'.format(1, a, b,)
#Defining the gaseous mixture with the Chemical kinetic mechanism
gas = ct.Solution('ethanol_mech.cti')
gas.TPX = T1, p1, reactants
# Thermodynamic properties
h1 = gas.h
ro1 = gas.density
# The unknowns are: T2 [K], p2 [kPa], u1 [m/s]
# Defining the system of equations
def equations(p):
  T2, p2, u1 = p
# Calculating the equilibrium composition at T2, p2
# Estas funções devem ficar alinhadas para à direita para que o #  # Python
reconheça elas como parte da iteração.
  gas2 = ct.Solution('ethanol_mech.cti')
  gas2.TPX = T2, p2, reactants
  gas2.equilibrate('TP')
  # Thermodynamic properties of products and u2
  h2 = gas2.h
  ro2 = gas2.density
  cp2 = gas2.cp_mass
  cv2 = gas2.cv_mass
  gamma2 = cp2 / cv2
  u2 = (gamma2 * p2 / ro2) ** 0.5
  # Conservation equations
  return (ro1 * u1 - ro2 * u2, p1 + ro1 * u1 ** 2 - p2 - ro2 *
      u2 ** 2, h1 + 0.5 * u1 ** 2 - h2 - 0.5 * u2 ** 2)
# Solution of the system of equation with "fsolve",
# Initial points for iterations 2700 K, 1,5e6 Pa e 1600 m/s
T2, p2, u1 = fsolve(equations, (2700, 1.5e6, 1600), maxfev=10000)
# The pressure p2 is divided by 1000 to be presented in kPa
print(T2, p2 / 1000, u1)
```

Solution routine for example 7.6

```
import cantera as ct
from scipy.optimize import fsolve
from ChapmanJouguetState import CJV

def VNS(T1,p1,ER,vars):
  T2, p2, u1 = CJV(T1,p1,ER,vars)
  a = vars/ER
  b = 3.76*a
  reactants = 'H2:{}, O2:{}, N2:{}'.format(1, a, b,)
  #Defining the gaseous mixture with the Chemical kinetic mechanism
  gas = ct.Solution('Mevel2018.yaml')
  gas.TPX = T1, p1, reactants
  # Thermodynamic properties
  h1 = gas.h
  ro1 = gas.density

  def VNEq(p):
    Tvn, pvn, uvn = p
    # Calculating the equilibrium composition at T2, p2
    # Estas funções devem ficar alinhadas para à direita para que o # #
Python reconheça elas como parte da iteração.
    gas2 = ct.Solution('Mevel2018.yaml')
    gas2.TPX = Tvn, pvn, reactants
    # Thermodynamic properties of gas after shock and uvn
    hvn  = gas2.h
    rovn = gas2.density
    # Conservation equations
    Eqmass = ro1*u1 - rovn*uvn
    Eqmome = p1 + ro1*u1**2 - pvn - rovn*uvn**2
    Eqener = h1 + 0.5*u1**2 - hvn - 0.5*uvn**2

    return Eqmass, Eqmome, Eqener
    # Solution of the system of equation with "fsolve",
    # Initial points for iterations

  Tvn, pvn, uvn = fsolve(VNEq, (1200, 2.0e6, 400), maxfev=10000)
  # apresentação da solução, a divisão de p2 é para converter em kPa

  return (Tvn, pvn, uvn)
```

Flammability Limits

8

Nomenclature

E_a	Activation energy, kJ/kmol
c_p	Heat capacity at constant pressure, kJ/kg K
T	Temperature, K
d	Characteristic length, m
D	Ordinary diffusion coefficients, m^2/s
G	Combined function of stretch rate and the stream function
$h^{\circ}_{f,i}$	Enthalpy of formation of the species i, kJ/kg
$\overline{h}^{\circ}_{f,i}$	Enthalpy of formation of the species i, kJ/mol
$\overline{h}_i$	Absolute enthalpy of species i, kJ/mol
$\Delta\overline{h}_i$	Sensible enthalpy of species i, kJ/mol
k	Thermal conductivity coefficient, W/(m K)
l	Normalized Lewis number deviation from unity
k_{FL}	Coefficient that accounts for the diluent effect
L	Heat losses, kJ
$\dot{m}''$	Mass flux, kg/m^2 s
$\dot{m}_{f,R}$	Reaction rate of fuel, $kg/(m^3$ s)
n_F	Number of moles of fuel, mol
$n_F{}^*$	Number of moles of gaseous fuel which occupies 12 L at T_0, mol
p	Pressure, Pa
p_i	Partial pressure of the ith species in a gas mixture, Pa
Pr	Prandtl number
P_d	Concentration of diluent in the mixture, %
$\dot{q}_{\mathrm{rad}}$	Heat lost by radiation, W/m^3
$\overline{\dot{q}}_{\mathrm{rad}}$	Mean value of heat lost by radiation, W/m^3
Δh_C	Heat of combustion of fuel on mass basis, kJ/kg

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$\Delta\overline{h}_C$ Heat of combustion of fuel on mole basis, kJ
R^2 Squared correlation coefficient
R Radial coordinate, m
s_L Laminar flame speed, m/s
T_∞ Ambient temperature, K
T_b Adiabatic flame temperature, K
T_a Arrhenius temperature, K
v_{air} Number of moles of oxygen in the oxidant, mol
v^s_{air} Number of moles of oxygen in the oxidant for the stoichiometric reaction, mol
W_i Molecular weight of species i, kg/kmol
w Net rate of a reaction, kmol/m^3 s
x Axial coordinate
x_{C} Number of moles of carbon (C) in the fuel, mol
x_{H} Number of moles of hydrogen (H) in the fuel, mol
x_{O} Number of moles of oxygen (O) in the fuel, mol
$_{xi}$ Mole fraction of the species i

Greek Letters

K_p Planck mean absorption coefficient for species i
γ Specific heat ratio
σ_b Stefan–Boltzmann constant (5.670367×10^{-8} W m^{-2} K^{-4})
ξ Dimensionless coordinate
$\alpha(\phi)$ Flammability limit exponent
α_t Thermal diffusivity, m^2
ϕ Equivalence ratio (in the context of mixture composition)
φ Ratio of mass fraction
θ Dimensionless temperature
ρ Density, kg/m^3
δ Flame thickness, m
κ Stretch rate, 1/s
μ Viscosity, Pa s
$\dot{\omega}_i$ Rate of production of species i, kmol/m^3 s
$\overline{\dot{\omega}}_f$ Mean value of the fuel's rate of destruction, kmol/m^3 s

Subscripts

0 Initial or previous condition
B Chain branching reaction
b Burned

d Diluent
F Fuel
T Chain termination reaction
u Unburned

Abbreviations

ARE Absolute value of the relative error, %
AARE Average of the relative errors, %
CAFT Calculated adiabatic flame temperature
FL Flammability limits
LFL Lower flammability limit
UFL Upper flammability limit

8.1 Introduction

The flammability limits are the concentrations limits of a fuel in an oxidant, generally air, for which it is possible to sustain the propagation of a premixed flame. Therefore, they are an important safety parameter. The introduction of alternative fuels such as ethanol, hydrogen, or ammonia in mixtures with other available base fuels will change the combustion characteristics of the base fuel. For example, mixing hydrogen with natural gas will change the adiabatic flame temperature, the laminar flame speed, and the flammability limits, to name a few premixed flame combustion properties. The objective of this chapter is to present the available theoretical developments regarding flammability limits, their experimental measurement, the influence of the initial conditions on the flammability limits, the influence of oxidant composition on these limits, the effect of mixing more than one fuel, and the application of practical methods for the estimation of the flammability limits.

8.2 Flammability Limits Theory

8.2.1 The Existence of Flammability Limits

The flammability limits (FLs) can be considered a fundamental property of a mixture if it can be demonstrated that they depend only on the particular properties of the mixture. Moreover, the flammability limits must be differentiated from the extinction limits of a mixture. The latter are limits that can exist for any mixture composition and depend upon the losses imposed by the system. As pointed out by Law [11], the

system-dependent losses can be heat conduction, heat convection, aerodynamic strain, and buoyancy. The flammability limits of ethanol in air at room temperature and atmospheric pressure are represented schematically in Fig. 8.1, where the experimental values come from Zabetakis [116].

According to Fig. 8.1, the lower flammability limit (LFL) and the upper flammability limit (UFL) of ethanol in air, at room temperature and atmospheric pressure, are 3.3% and 19.0%, respectively. This means that adding more air on the lean mixture or adding more fuel on the rich one would turn the mixture nonflammable. Following the theoretical developments by Mendiburu et al. [117], a one-dimensional flame in steady state, unity Lewis number, one-step chemical reaction, and radiation heat losses represented by ($\dot{q}_{\text{rad}}$) are assumed. This leads to the differential equation shown below.

$$\dot{m}'' \frac{dT}{dx} - \frac{k}{c_p} \frac{d^2T}{dx^2} = -\frac{1}{c_p} \sum h_{f,i}^{\circ} \dot{\omega}_i W_i + \frac{\dot{q}_{\text{rad}}}{c_p} \tag{8.1}$$

Moreover, assuming a linear temperature profile, the above equation will have the following boundary conditions.

$$x \to -\infty; \quad T = T_u; \quad dT/dx = 0 \tag{8.2}$$

$$x = \delta/2; \quad T = (T_b + T_u)/2; \quad dT/dx = (T_b - T_u)/\delta \tag{8.3}$$

The first boundary condition was imposed far from the flame, on the unburned gas side, and the second was imposed at the flame's half-thickness. Integrating Eq. (8.1) gives Eq. (8.4).

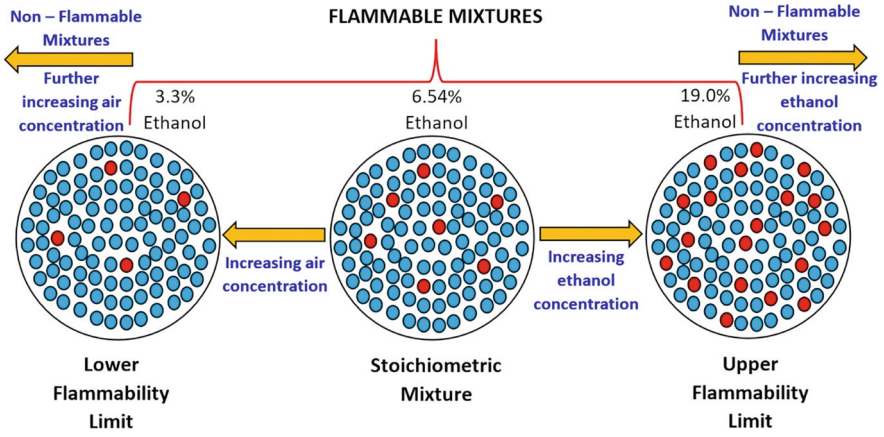


Fig. 8.1 Schematic representation of the flammability limits of ethanol in air at room temperature and atmospheric pressure [117]. (Reprinted from Energy & Fuels, Vol 37/6, Andrés Z. Mendiburu, João A. Carvalho Jr., Yiguang Ju, Flammability Limits: A Comprehensive Review of Theory, Experiments, and Estimation Methods, 4151–4197, Copyright (2023), with permission from American Chemical Society)

$$\frac{1}{2}\left(\frac{\overline{\dot{q}}_{\text{rad}}}{c_p}-\frac{\Delta h_C}{2c_p}\overline{\dot{\omega}}_f W_f\right)\delta^2-\frac{\dot{m}''}{2}(T_b-T_u)\delta+\frac{k}{c_p}(T_b-T_u)=0 \tag{8.4}$$

where the bar over the heat loss and reaction rate terms represents the mean value of these functions. Equation (8.4) is a quadratic equation in which the flame thickness is the unknown. Therefore, in order to obtain real solutions, the discriminant should be equal or greater than zero, which leads to the following expression.

$$c_p(T_b-T_u)(\dot{m}'')^2-8k_t\left(\frac{\overline{\dot{q}}_{\text{rad}}}{c_p}-\frac{\Delta h_C}{2c_p}\overline{\dot{\omega}}_f W_f\right)\geq 0 \tag{8.5}$$

Using the relationship $\dot{m}''=\rho_u s_L$, the following limit is obtained for the flame propagation velocity.

$$s_L\geq\frac{1}{\rho_u}\sqrt{\frac{8k_t}{c_p(T_b-T_u)}\left(\frac{\overline{\dot{q}}_{\text{rad}}}{c_p}-\frac{\Delta h_C}{2c_p}\overline{\dot{\omega}}_f W_f\right)} \tag{8.6}$$

Note that $\overline{\dot{\omega}}_f$ and $\overline{\dot{q}}_{\text{rad}}$ are both negative. Therefore, there is a limit value for the laminar flame propagation velocity, which is determined mainly by the balance between the radiation heat losses and the heat released by chemical reactions. In Eq. (8.6) the radiation heat losses can be made quite complicated just as the reaction rate term. Following the definition of the flammability limit as the last mixture composition which sustains flame propagation, it is noted that $s_L > 0$ at the flammability limits.

8.2.2 The Theory of Flammability Limits for One-Step Reaction

The theoretical model for the flammability limits assuming a one-step chemical reaction was developed by Spalding [118]. Consider the energy conservation equation, Eq. (3.81), and the species mass conservation equation, Eq. (3.77), with the following simplifying assumptions.

- Steady state.
- One-dimensional flame.
- The heat capacities for all species are equal and constant.
- Viscous dissipation is negligible.
- No work is being done on the control volume.
- Heat losses are due to radiation only.
- Unit Lewis number ($Le = \alpha/D = 1$).
- Equal binary diffusion coefficients.
- One-step chemical reaction.

$$\dot{m}'' c_p \frac{dT}{dx} - \frac{d}{dx}\left(k\frac{dT}{dx}\right) = -\Delta h_c \dot{m}_{f,R} + \dot{q}_{\text{rad}} \tag{8.7}$$

$$\dot{m}'' c_p \frac{dY_f}{dx} - \frac{d}{dx}\left(k\frac{dY_f}{dx}\right) = c_p \dot{m}_{f,R} \tag{8.8}$$

The following dimensionless variables are introduced.

$$\theta = \frac{T - T_u}{T_b - T_u}; \quad \xi = \exp\left(\frac{c_p \dot{m}''}{k} x\right); \quad \varphi = \frac{Y_f}{Y_{f,u}} \tag{8.9}$$

Introducing the above variables into the derivatives by considering any scalar function g, the expressions shown below are obtained.

$$\frac{dg}{dx} = \frac{dg}{d\xi}\frac{d\xi}{dx} = \frac{c_p \dot{m}''}{k}\xi\frac{dg}{d\xi} \tag{8.10}$$

$$\frac{d}{dx}\left(k\frac{dg}{dx}\right) = \frac{\left(c_p \dot{m}''\right)^2}{k}\xi\frac{dg}{d\xi} + \frac{\left(c_p \dot{m}''\right)^2}{k}\xi^2\frac{d^2 g}{d\xi^2} \tag{8.11}$$

$$\frac{d^2 T}{d\xi^2} = (T_b - T_u)\frac{d^2\theta}{d\xi^2}; \quad \frac{d^2 Y_f}{d\xi^2} = Y_{f,u}\frac{d^2\varphi}{d\xi^2} \tag{8.12}$$

It is observed that, the variables substitution will cancel the first derivatives in Eqs. (8.7) and (8.8), leaving only the second derivatives. Therefore, substitution of Eqs. (8.10)–(8.12) into Eqs. (8.7) and (8.8) and some rearrangement lead to the expressions shown in Eqs. (8.13) and (8.14).

$$\frac{d^2\theta}{d\xi^2} = \frac{\lambda}{\xi^2}(\phi - K\psi) \tag{8.13}$$

$$\frac{d^2\varphi}{d\xi^2} = -\frac{\lambda}{\xi^2}\phi \tag{8.14}$$

In order to obtain Eq. (8.14) it should be observed that, $\Delta h_c = c_p(T_b - T_u)/Y_{f,\,u}$. The new variables that appear in Eqs. (8.13) and (8.14) are shown in more detail in the equations below. The $\dot{m}_{f,R,b}$ is the reaction rate of the fuel calculated at T_b and $Y_{f,\,u}$. The k_b and $\dot{q}_{\text{rad},b}$ are the heat conductivity and the heat lost by radiation, respectively, both calculated at the temperature T_b.

$$\lambda = \frac{\Delta h_c k_b \dot{m}_{f,R,b}}{\left(c_p \dot{m}''\right)^2 (T_b - T_u)}; \quad \phi = \frac{k}{k_b}\frac{\dot{m}_{f,R}}{\dot{m}_{f,R,b}} \tag{8.15}$$

$$K = \frac{\dot{q}_{\text{rad},b}}{\dot{m}_{f,R,b}\Delta h_c}; \quad \psi = \frac{k}{k_b}\frac{\dot{q}_{\text{rad}}}{\dot{q}_{\text{rad},b}} \tag{8.16}$$

z

Note that since the fuel is being consumed ($\dot{m}_{f,R} < 0$) and heat is being lost ($\dot{q}_{\text{rad}} < 0$), λ must be negative, and K, ϕ, and ψ are positive according to this formulation. Eqs. (8.13) and (8.14) are subject to the following boundary conditions, where the fuel was assumed to be in deficit with respect to the oxidant.

$$\text{At } \xi = 0 \rightarrow \theta = 0 \text{ and } \varphi = 1 \tag{8.17}$$

$$\text{At } \xi = +\infty \rightarrow \frac{d\theta}{d\xi} = 0 \text{ and } \varphi = 0 \tag{8.18}$$

An analytical solution is possible only when the functions ϕ and ψ are simplified. Following Spalding [118], the simplifications shown below are adopted.

$$\phi = \varphi\theta^n \tag{8.19}$$

$$\psi = \theta^m \quad \text{for} \quad \xi > 1 \tag{8.20}$$

$$\psi = 0 \quad \text{for} \quad \xi \leq 1 \tag{8.21}$$

The value of n is between 6 and 15, while the value of m is between 1 and 5, according to Spalding [118]. Note that $\psi = 0$ for $\xi \leq 1$means that the heat transfer is negligible in comparison to chemical reaction upstream of the plane located at $\xi = 1$. This allows for solving the equations in two parts that must match at $\xi = 1$. The plane $\xi = 1$ is such that the fuel is totally consumed once this plane is reached. Equations (8.13) and (8.14) in the region limited to $0 \leq \xi \leq 1$ simplify to the expressions shown below.

$$\frac{d^2\theta}{d\xi^2} = \lambda\frac{\varphi\theta^n}{\xi^2} \tag{8.22}$$

$$\frac{d^2\varphi}{d\xi^2} = -\lambda\frac{\varphi\theta^n}{\xi^2} \tag{8.23}$$

Equations (8.22) and (8.23) are subject to the boundary conditions shown in Eqs. (8.24)–(8.26).

$$\text{At } \xi = 0 \rightarrow \theta = 0 \text{ and } \varphi = 1 \tag{8.24}$$

$$\text{At } \xi = 1 \rightarrow \theta = \theta_1 \text{ and } \frac{d\theta}{d\xi} = \left(\frac{d\theta}{d\xi}\right)_1 \tag{8.25}$$

$$\text{At } \xi = 1 \rightarrow \varphi = 0 \text{ and } \frac{d\varphi}{d\xi} = 0 \tag{8.26}$$

Since Eqs. (8.22) and (8.23) are similar differential equations with different boundary conditions, it can be inferred that they are linearly dependent. Therefore, they must satisfy a relationship as the one shown in Eq. (8.27).

$$\varphi = -\theta + a\xi + b \tag{8.27}$$

From the expression given in Eq. (8.24) it is noticed that $b = 1$. Also, from Eqs. (8.25) and (8.26), it is found that $a = (\theta_1 - 1)$ and $(d\theta/d\xi)_1 = (\theta_1 - 1)$. Then, the following is obtained.

$$\varphi = 1 - \theta - (1 - \theta_1)\xi \tag{8.28}$$

$$\frac{d\theta}{d\xi} = -\left[\frac{d\varphi}{d\xi} + (1 - \theta_1)\right] \tag{8.29}$$

Using the above expressions, it can be shown that

$$\left(\frac{d^2\varphi}{d\xi^2}\right)\left(\frac{d\theta}{d\xi}\right) = -\frac{1}{2}\frac{d}{d\xi}\left(\frac{d\varphi}{d\xi}\right)^2 - (1 - \theta_1)\frac{d}{d\xi}\left(\frac{d\varphi}{d\xi}\right) \tag{8.30}$$

Then, Eq. (8.23) is multiplied by $(d\theta/d\xi)d\xi$, Eq. (8.30) is used on the left-hand side, and the integration is performed in both sides. This leads to the relation in Eq. (8.31). Note that λ is an eigenvalue of the problem.

$$\lambda = \frac{\frac{1}{2}\left[\left(\frac{d\varphi}{d\xi}\right)^2 - \left(\frac{d\varphi}{d\xi}\right)^2_{\xi=0}\right] + (1 - \theta_1)\left[\left(\frac{d\varphi}{d\xi}\right) - \left(\frac{d\varphi}{d\xi}\right)^2_{\xi=0}\right]}{\int_0^{\theta_1} \frac{1 - \theta - (1 - \theta_1)\xi}{\xi^2}\theta^n d\theta} \tag{8.31}$$

Furthermore, it is assumed that the reaction is confined to high temperatures. Then, $(d\varphi/d\xi) = 0$ and $\theta_1 \sim 1$. Since the temperature is not changing at $\xi = 0$, then $(d\theta/d\xi)_{\xi=0} = 0$, and from Eq. (8.28), it is found that $(d\varphi/d\xi)_{\xi=0} = 1$. Therefore, Eq. (8.31) will be solved for $\xi = 1$.

$$\lambda = -\left[2\int_0^{\theta_1} (\theta_1 - \theta)\theta^n d\theta\right]^{-1} \tag{8.32}$$

Integration of Eq. (8.32) leads to the expression shown in Eq. (8.33), being this result valid in the region limited to $0 \leq \xi \leq 1$.

$$\lambda = -\frac{(n+1)(n+2)}{2\theta_1^{n+2}} \tag{8.33}$$

Note that in a case without heat losses, $\theta_1 = 0$. Now it is necessary to evaluate the region characterized by $\xi \geq 1$. In this region the chemical reaction has finished; therefore, Eq. (8.13) reduces to the following expression.

$$\frac{d^2\theta}{d\xi^2} = -\lambda \frac{K\theta^m}{\xi^2} \tag{8.34}$$

Subject to the boundary conditions given in Eqs. (8.35) and (8.36),

$$\text{At } \xi = 1 \rightarrow \theta = \theta_1 \text{ and } \frac{d\theta}{d\xi} = \left(\frac{d\theta}{d\xi}\right)_1 \tag{8.35}$$

$$\text{At } \xi = +\infty \rightarrow \frac{d\theta}{d\xi} = 0 \tag{8.36}$$

Equation (8.34) cannot be integrated easily for an exponent $m \neq 1$. The strategy proposed by Spalding was to assume that the product λK is small, which allows for θ_1^m to be treated as a constant near $\xi = 1$. This allows for the first integration of Eq. (8.34).

$$\frac{d\theta}{d\xi} = \lambda K \frac{\theta_1^m}{\xi} \tag{8.37}$$

The derivative of the dimensionless temperature is postulated to be equal from both sides, that is, coming from regions 1 and $0 \leq \xi \leq 1$. As was previously obtained, adopting $\xi = 1$, the derivative in the region $0 \leq \xi \leq 1$ is $(d\theta/d\xi)_1 = (\theta_1 - 1)$.

$$\lambda K = (\theta_1 - 1)\theta_1^{-m} \tag{8.38}$$

Substitution of Eq. (8.33) into Eq. (8.38) gives

$$K = -\frac{2(\theta_1 - 1)\theta_1^{n+2-m}}{(n+2)(n+1)} \tag{8.39}$$

For positive values of the difference $(n + 2 - m) > 0$ and for a given value of K, there are two values of θ_1 that satisfy Eq. (8.39). However, there will be a critical value K_c which makes the only possible values of θ_1 to be imaginary numbers. At the critical value K_c, the function will have an extremum. Then, the critical value of θ_{1c} is found by differentiating Eq. (8.39) and making $dK_c/d\theta_1 = 0$.

$$\theta_{1c} = \frac{n+2-m}{n+3-m} \tag{8.40}$$

Substitution of Eq. (8.40) into (8.39) gives the critical value K_c.

$$K_c = \frac{2}{(n+1)(n+2)(n+3-m)} \left(\frac{n+2-m}{n+3-m}\right)^{n+2-m} \tag{8.41}$$

A critical value λ_c can also be obtained by substitution of Eqs. (8.40) and (8.41) into Eq. (8.38). The same results would be obtained by substituting Eq. (8.40) into Eq. (8.33).

$$\lambda_c = -\frac{(n+1)(n+2)}{2} \left(\frac{n+3-m}{n+2-m}\right)^{n+2} \tag{8.42}$$

The critical values *Kc* and *λc* are determined by means of Eqs. (8.41) and (8.42), and these are assumed to correspond to the flammability limit. Furthermore, the value of the laminar flame velocity at the critical condition is found by multiplying the expressions for K and λ given in Eqs. (8.15) and (8.16) and rearranging.

$$s_{L,c} = \frac{1}{c_p \rho_u} \sqrt{\frac{\dot{q}_{\text{rad},b} k_b}{(T_b - T_u)\lambda_c K_c}} \tag{8.43}$$

The value of the critical laminar flame velocity could also be determined by rearranging Eq. (8.15).

$$s_{L,c} = \frac{1}{c_p \rho_u} \sqrt{\frac{\Delta h_c k_b \dot{m}_{f,R,b}}{(T_b - T_u)\lambda_c}} \tag{8.44}$$

Since $\dot{q}_{\text{rad},b} < 0$, $\dot{m}_{f,R,b} < 0$, and $\lambda_c < 0$, the terms in the square root of Eqs. (8.43) and (8.44) are positive.

Example 8.1
Analyze the variations of $|\lambda|$ and K as functions of θ and also the variation of $|\lambda|$ as a function of K. Afterwards, determine the critical values λ_c, K_c, and $s_{L,c}$.

Solution
According to the theoretical developments by Spalding, the values of m and n range between 1 to 5 and 6 to 15, respectively. Therefore, three combinations will be tested in this example. Moreover, the values of θ_1 will be varied from 0 to 1. Equations (8.33) and (8.39) are used to obtain the variations of $|\lambda|$ and K as functions of θ_1. Afterwards, the values of $|\lambda|$ are plotted vs K. The results are shown in Figs. 8.2 and 8.3. The dimensionless heat released by chemical reaction $|\lambda|$ decreases as θ_1 approaches unity. This makes sense because the chemical reaction term also depends on the fuel concentration which is decreasing in the reaction. The ratio of heat losses to heat release K achieves a maximum value for θ_1 close to unity. As aforementioned, the reaction rate decreases as the fuel concentration decreases. Figure 8.3 shows the extremum point which should correspond to the critical value of K_c.

One calculation will be demonstrated for $n = 10$ and $m = 3$.

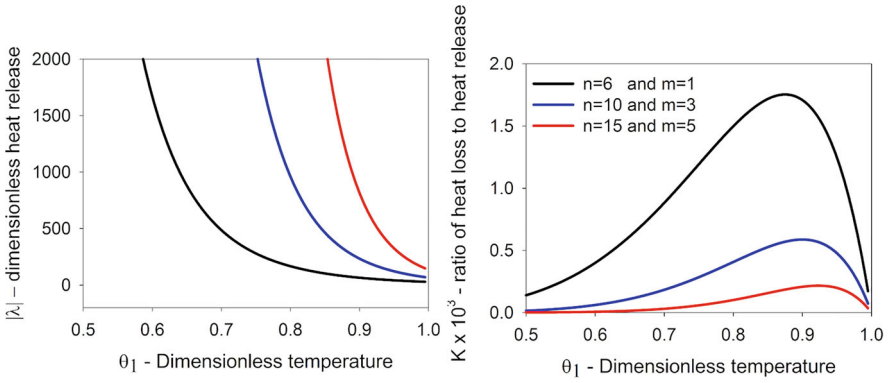


Fig. 8.2 Variation of K and $|\lambda|$ as functions of θ_1

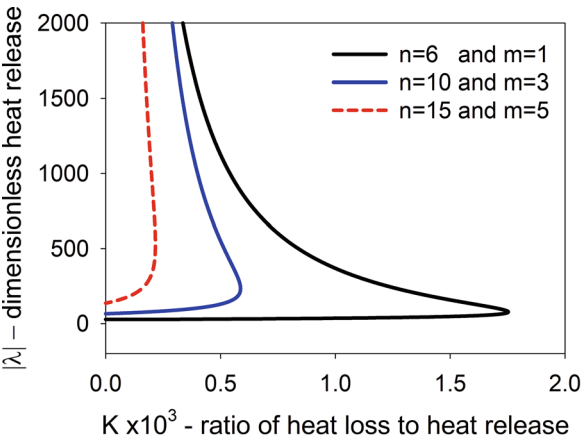


Fig. 8.3 Variation of $|\lambda|$ as a function of K

$$K_c = \frac{2}{(10+1)(10+2)(10+3-3)}\left(\frac{10+2-3}{10+3-3}\right)^{10+2-3} = 5.870\times 10^{-4}$$

$$\lambda_c = -\frac{(10+1)(10+2)}{2}\left(\frac{10+3-3}{10+2-3}\right)^{10+2} = -233.7$$

Following the numerical example presented by Mendiburu et al. [117], at the LFL the properties of the mixture can be approximated by those of air ($c_p = 1004$ J/kg K, $\rho_u = 1.169$ kg/m^3, $k_b = 0.100$ W/m K). Also, $\dot{q}_{\text{rad},b} = -0.4186$ J/cm^3 s, assuming $T_b = 1500$ K and $T_u = 298$, the following is obtained.

$$s_{L,c} = \frac{100}{1004 \times 1.169} \sqrt{\frac{-0.4186 \times 10^6 \times 0.1}{(1500 - 298) \times (-233.7) \times 5.87 \times 10^{-4}}} = 1.36 \text{ cm/s}$$

The above result confirms that, at the flammability limits, the laminar flame velocity is different than zero.

8.2.3 The Theory of Flammability Limits with Detailed Chemistry

A more complete theoretical model for the flammability limits must include detailed chemical kinetic mechanisms and heat losses by radiation. Therefore, assuming a one-dimensional flame, steady-state conditions, no work, negligible pressure variation, and negligible viscous dissipation, the conservation equations are those presented in Eqs. (8.45)–(8.48). Note that, the Eqs. (8.46)–(8.48) are the same equations presented in Sect. 4.8 of Chap. 4.

$$\rho u_x \frac{d}{dx}\left(\int_{T_{\text{ref}}}^{T} c_p dT\right) = \frac{d}{dx}\left(k\frac{dT}{dx}\right) - \sum\left(h_{f,k}^{\circ}\dot{m}_{k,R}\right) - \sum\left(\dot{m}_{k,R}\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right) + \sum \rho D \frac{dY_k}{dx}\frac{d}{dx}\left(\int_{T_{\text{ref}}}^{T} c_{p,k}dT\right) + \dot{q}_{\text{rad}} \tag{8.45}$$

$$\dot{m}_{k,R} = W_k \widehat{\omega}_k = W_k \sum_{i=1}^{L}\left(v_{ji,P} - v_{ji,R}\right)\left[k_{fi}\prod_{j=1}^{N}\left(C_j\right)^{v_{ji,R}} - k_{bi}\prod_{j=1}^{N}\left(C_j\right)^{v_{ji,P}}\right] \tag{8.46}$$

$$\frac{d}{dx}\left(\rho u_x Y_k\right) = +\frac{d}{dx}\left(\rho D\frac{dY_k}{dx}\right) + \dot{m}_{k,R} \tag{8.47}$$

$$\sum_{j=1}^{N} Y_j = 1 \tag{8.48}$$

According to Bertolino et al. [119], the radiation heat losses can be calculated by Eq. (8.49). A similar approach was also adopted by Law and Egolfopoulos [120].

$$\dot{q}_{\text{red}} = 4\sigma\left[\sum_{k=1}^{N} p_k \Phi_k + \beta\right]\left(T^4 - T_{\text{amb}}^4\right) \tag{8.49}$$

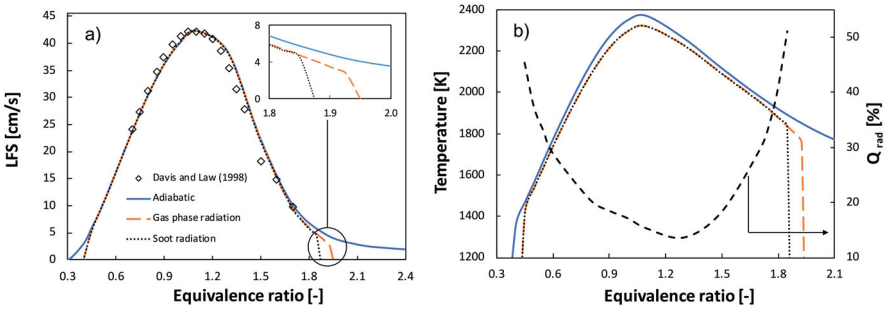


Fig. 8.4 Upper flammability limit of benzene as determined by Bertolino et al. [119]. (**a**) Laminar flame velocity (LFS) drop criterion and (**b**) flame temperature drop criterion

where p_k is the partial pressure of species k, Φ_k is the average Planck's absorptivity coefficient for species k, and β is the emissivity of soot. The value of β is determined as $\beta = Cf_vT$, where C is a coefficient and f_v is the volumetric fraction of the soot. The presence of soot is essential for the correct representation of the upper flammability limits. The criteria adopted by Bertolino et al. [119] to identify the flammability limits was a sudden drop in laminar flame velocity, or in flame temperature, when the system of equations (Eqs. 8.45–8.49) is solved. An example of the determination of the UFL is shown in Fig. 8.4.

The theoretical UFL of benzene can be identified in Fig. 8.4 for $s_L \sim 4$ cm/s and $\phi \sim 1.9$, which corresponds to a theoretical UFL of 5.06%, while the experimental UFL is 6.3% [119]. Therefore, this approach showed good accuracy. A theoretical model based on detailed chemical kinetics, however, without radiation heat losses had been previously proposed by Law and Egolfopoulos [121] in 1990 and was modified by Liang and Law [122] in 2017. The original approach establishes that at the FLs, there is a competition between a chain branching and a chain termination reaction which allows for the determination of a flammability limit coefficient (α), shown in Eq. (8.50), that should be close to unity at the FLs.

$$\alpha_a(\phi) = \frac{\partial(\ln \omega_T)/\partial(\ln \phi)}{\partial(\ln \omega_B)/\partial(\ln \phi)} = \frac{\partial(\ln \omega_T)}{\partial(\ln \omega_B)} \tag{8.50}$$

The subscripts T and B stand for termination and branching reactions, respectively. Note that Eq. (8.50) compares the variation of the reaction rates with the equivalence ratio. The modified version by Liand and Law introduces two other forms for the determination of the flammability coefficient, as shown in Eqs. (8.51) and (8.52). The modified forms of α account for two or three chain branching and chain termination reactions, respectively.

$$\alpha_b(\phi) = \frac{\partial \ln\left[\omega_{T2} + \omega_{B2}\left(\frac{\omega_T}{\omega_T+\omega_B}\right)\right]}{\partial \ln\left[\omega_{B2}\left(\frac{\omega_B}{\omega_T+\omega_B}\right)\right]} \tag{8.51}$$

$$\alpha_c(\phi) = \frac{\partial \ln\left[\omega_{T1} + \omega_{B1}\left(\frac{\omega_{T2}}{\omega_{T2}+\omega_{B2}}\right) + \omega_{B1}\left(\frac{\omega_{B2}}{\omega_{T2}+\omega_{B2}}\right)\left(\frac{\omega_T}{\omega_T+\omega_B}\right)\right]}{\partial \ln\left[\omega_{B1}\left(\frac{\omega_{B2}}{\omega_{T2}+\omega_{B2}}\right)\left(\frac{\omega_B}{\omega_T+\omega_B}\right)\right]} \tag{8.52}$$

The flammability limit coefficients must be near unity at the FLs. Therefore, by varying the equivalence ratio, it is possible to find a value $\alpha \sim 1$ and determine the FLs. Wang et al. [123] determined the FLs of CH_4/SG/air mixtures using the method of flammability coefficients and compared the results with experimental values, as shown in Fig. 8.5. The SG gas was formed by C_2H_6, C_2H_4, CO, and H_2 in the proportion 5:1:1:1.

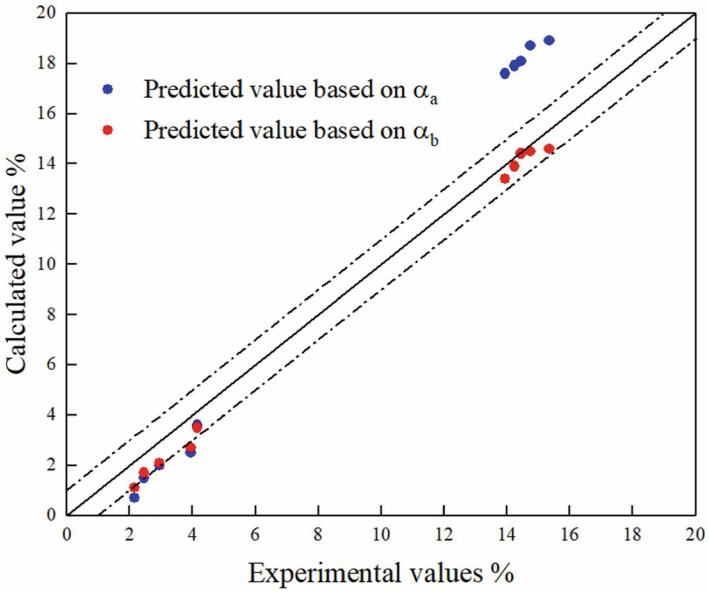


Fig. 8.5 Flammability limits of CH_4/SG/air mixtures determined with the flammability limit coefficients α_a and α_b [123]. The SG gas is formed by C_2H_6, C_2H_4, CO, and H_2 in the proportion 5:1:1:1. (Reprinted from Fuel, Vol 294, Tao Wang, He Liang, Zhenmin Luo, Bin Su, Litao Liu, Yang Su, Xuqing Wang, Fangming Cheng, Jun Deng, Near flammability limits behavior of methane-air mixtures with influence of flammable gases and nitrogen: An experimental and numerical research, 120550, Copyright (2021), with permission from Elsevier)

As shown in Fig. 8.5, the upper flammability limits present a greater deviation when determined with the α_a coefficient. Regardless of the good results obtained with the α_b coefficient, it is important to note that the most complete theoretical approach should include radiation heat losses.

8.2.4 Experimental Determination of Flammability Limits

The flammability limits are generally determined by the ASTM E681 [124] standard, among other standards, as has been discussed in the review article by Mendiburu et al. [117]. A variation of the experimental setup described in ASTM E681 is depicted in Fig. 8.6. This setup was applied to determine the FLs of ethanol and farnesane [125–127]. The spherical vessel is evacuated by using a vacuum pump, and then a fuel/oxidant mixture is prepared by the partial pressures' method. In each experimental test, the fuel concentration is changed to the lean or to the rich sides, depending on whether the LFL or UFL is being determined. The FLs are found as the

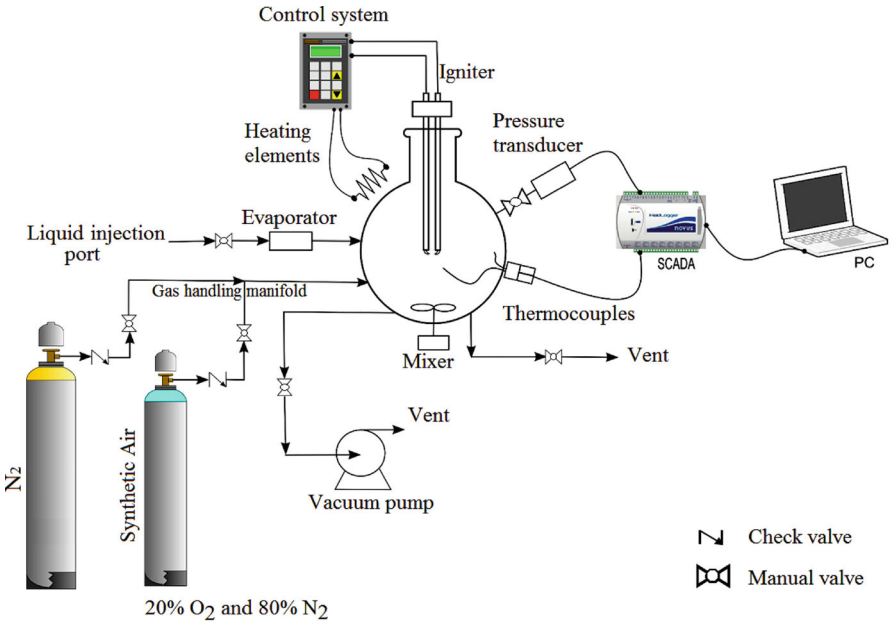


Fig. 8.6 Schematic representation of the experimental equipment used by Coronado et al. [125] to determine the flammability limits. (Reprinted from Journal of Hazardous Materials, Vol 280, Christian J.R. Coronado, João A. Carvalho Jr, José C. Andrade, Andrés Z. Mendiburu, Ely V. Cortez, Felipe S. Carvalho, Beatriz Gonçalves, Juan C. Quintero, Elkin I. Gutiérrez Velásquez, Marcos H. Silva, José C. Santos, Marco. A.R. Nascimento, Flammability limits of hydrated and anhydrous ethanol at reduced pressures in aeronautical applications, 174–184, Copyright (2014), with permission from Elsevier)

intermediate fuel concentrations between two tested mixtures: one that was able to sustain flame propagation and another one that was not able to do so.

The experimental determination depends on a flammability criterion, which in case of the ASTM E681 renders a mixture as flammable if the flame spreads upward and towards the wall, forming an arc greater than 90°. In the experiments performed by Coronado et al. [125], the ignition was provided by two electrodes (6.44 mm apart), with a current of 30 mA and a voltage of 15 kV. There are other flammability criteria, for example, the 5 or 7% pressure rise criteria [128]. Mendiburu et al. [117] provide a more detailed discussion on the experimental determination of the FLs. Considering 1 mol of fuel, the FL is determined as shown below.

$$\mathrm{FL} = \left(\frac{1}{1 + 4.76 v_{ar}} \right) \times 100\% \tag{8.53}$$

The above equation means that the FLs are the fuel mole fractions written in percentage of the total mixture. The mole fraction can also be written in terms of the partial pressure, as would be determined from the experiments. Figure 8.7 depicts the propagation of ethanol/air flames near the flammability limits.

The flames shown in Fig. 8.7 form arcs greater than 90°; thus, both flames comply with the visual flammability criterion. Figure 8.7a depicts an ethanol/air mixture slightly above the LFL, propagating and forming a spherical flame. On the other hand, Fig. 8.7b shows an ethanol/air mixture slightly below the UFL, propagating as a hemispherical flame. This qualitative difference is of great importance for the theoretical analysis at the FLs. Recalling the results of Example 5.2, Eq. (8.54)

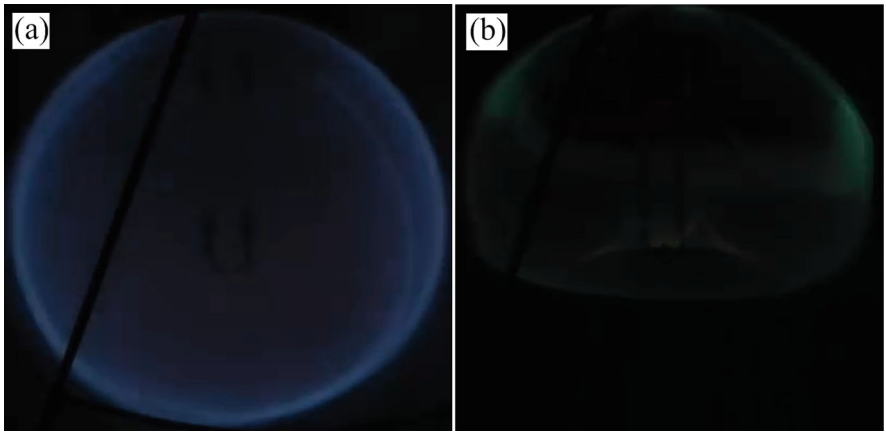


Fig. 8.7 Propagation of ethanol/air flames near the flammability limits. (**a**) Slightly above the LFL of anhydrous ethanol in air at 20 kPa and 50 C. (**b**) Slightly below the UFL of hydrated ethanol at 40 kPa and 280 C. Experiments performed by Mendiburu [129]

represents the stretch rate for a spherically propagating flame, while Eq. (8.55) represents the stretch rate for the upper hemisphere.

Stretch rate near the LFL for Fig. 8.7a

$$\kappa = \frac{2}{r}\frac{dr}{dt} \tag{8.54}$$

Stretch rate near the UFL for Fig. 8.7b

$$\kappa = \frac{1}{r}\frac{dr}{dt} \tag{8.55}$$

As discussed in Chap. 5, the stretch rate affects the flame propagation velocity and the flame temperature.

8.3 Effect of Initial Conditions, Fuel, and Oxidant Compositions

8.3.1 Effect of Initial Temperature

It is expected that the flammability limits will change with the initial temperature because the reaction rates will be affected. However, a simple approach was proposed by Burgess–Wheeler: "For a flammable gas the sum of the heat released at the LFL and the sensible enthalpies of the reactants is constant" [116]. This rule allows for writing the balance shown below.

$$\frac{\mathrm{FL}_T}{100}\left(\Delta\overline{h}_{C,T} + \Delta\overline{h}_{F,T}\right) + \left(1 - \frac{\mathrm{FL}_T}{100}\right)\Delta\overline{h}_{\mathrm{air},T} = \frac{\mathrm{FL}_{T_0}}{100}\Delta\overline{h}_{C,T_0} \tag{8.56}$$

The subscripts T_0 and T represent quantities at a reference initial temperature, T_0, and at an arbitrary initial temperature, T. Note that at the reference temperature, the sensible enthalpies are zero. In the experiments it is generally observed that the LFL decreases and the UFL increases with increasing temperature. Moreover, the FLs vary linearly with the initial temperature, as shown in Eq. (8.57), where the negative sign is applied to the LFL and the positive sign to the UFL.

$$\frac{\mathrm{FL}_T}{\mathrm{FL}_{T_0}} = 1 \mp m(T - T_0) \tag{8.57}$$

The slope, m, can be obtained from Eq. (8.56) by assuming that the sensible enthalpies are very small compared to the heat release and using an average heat capacity at constant pressure for air. The result is shown in Eq. (8.58).

$$m = \pm\left(\frac{\Delta\overline{h}_{C,T_0} - \Delta\overline{h}_{C,T}}{\overline{c}_{p,\mathrm{air}}\Delta T}\frac{\mathrm{FL}_{T_0}}{100} - 1\right)\frac{\overline{c}_{p,\mathrm{air}}}{(\mathrm{FL}_{T_0}/100)\Delta\overline{h}_{C,T}} \tag{8.58}$$

Assuming $T > T_0$, the difference $(\Delta\overline{h}_{C,T_0} - \Delta\overline{h}_{C,T})$ is positive at the LFL, because complete combustion can be assumed, and the heat release decreases with increasing

temperature. For example, in the case of methane, the heat released per mole of fuel at 298 and 500 K is 802.3 kJ/mol and 800.5 kJ/mol, respectively. On the other hand, the difference $(\Delta\overline{h}_{C,T_0} - \Delta\overline{h}_{C,T})$ should be positive at the UFL, because higher concentrations of fuel are expected, which in turn increases the concentration of incomplete combustion products. Note that the first term inside the brackets on the right-hand side of Eq. (8.58) should be smaller than unity. Therefore, the negative sign would be produced at the LFL and a positive sign at the UFL.

8.3.2 Effect of Initial Pressure

As discussed in Chap. 4, increasing the initial pressure will increase the number of collisions between molecules, and consequently the global reaction rate (with order $n > 1$) will also increase, as shown in Eq. (8.59).

$$\omega_b \sim p^n e^{-T_a/T_b^\circ} \tag{8.59}$$

Therefore, it is expected that increasing the initial pressure will decrease the LFL and increase the UFL. This becomes clearer if the equation for the mass flux rate ($\dot{m}''$) is written in terms of the flame thickness and the pressure [117], as shown in Eqs. (8.60) and (8.61), respectively.

$$\dot{m}'' = s_L \rho_u \sim \frac{k/c_p}{\delta} \tag{8.60}$$

$$\dot{m}'' \sim \left[p^n \frac{k}{c_p} e^{-T_a/T_b^\circ} \right]^{1/2} \tag{8.61}$$

As stated by Law [11], the ratio k/c_p presents low sensitivity to pressure variations. Therefore, assuming $n > 0$ and $n \neq 2$, it is observed that increasing the pressure will increase the mass flux rate and decrease the flame thickness. The volumetric heat losses from the flame will therefore decrease for increasing pressure due to lower flame thickness. However, increasing the initial pressure will also lead to an increase in the mean absorption coefficients, which results in increased radiation heat losses. Thus, two mechanisms compete at the FLs when the initial pressure is increased.

The theoretical model for the FLs developed with detailed chemical kinetics and radiation heat losses was able to predict the pressure effect on the UFL of benzene/air and methanol/air mixtures between 1 and 2 atm [119]. As expected, the model depends on the accuracy of the chemical kinetic mechanisms at high and low pressures. Therefore, in some cases the application of the detailed theoretical model is limited by the validity of the detailed chemical kinetics mechanisms at different pressures. The variation of the FLs with pressure is generally represented by a correlation of the form shown in Eq. (8.62).

$$\mathrm{FL}_p = \mathrm{FL}_{p_0} + a \ln \frac{p}{p_0} \tag{8.62}$$

The coefficients for natural gas are presented in the work by Zabetakis [116], while those for ethane and propane are presented by Huang et al. [130]. A list of experimental work in which the FLs have been determined for different fuels and under different initial conditions, is presented in the supplementary material of the review article by Mendiburu et al. [117].

8.3.3 Effect of Diluents and Oxidant Concentrations

In the context of FLs, a diluent is a chemical species that can be considered inert in the combustion reaction. For example, nitrogen (N_2), carbon dioxide (CO_2), water vapor (H_2O), carbon tetrachloride (CCL_4), methyl bromide (CH_3Br), and helium (He) are considered diluents. The effect of diluent addition is to reduce significantly the UFL and to increase the LFL. Moreover, increasing the diluent's concentration eventually leads to the equality of both FLs at the fuel inertization point (FIP). Figure 8.8 shows the change of the FLs of methane in air with the addition of different diluents.

Figure 8.8 shows that the methane's FLs vary more sharply with the addition of CO_2 than with the addition of H_2O, N_2, or He. Moreover, the effects of CCL_4 and CH_3Br are even stronger. The FIP is reached for lower concentrations of CH_3Br, followed by CCL_4, CO_2, H_2O, N_2, and He, in that order. The effect of some diluents can be explained through their capacity to remove heat from the reaction zone which casuses an increase their own enthalpy. Therefore, in some cases, the heat capacity and the heat conductivity can explain the differences between the effects of diluents. Table 8.1 compares the heat capacities and heat conductivities of some diluents, and air, at ambient temperature and atmospheric pressure. The Prandtl number, given in Eq. (8.63), is also shown in that table.

$$Pr = \frac{c_p \mu}{k} \tag{8.63}$$

Since the concentrations are measured by volume, the heat capacities should be considered on a mole basis to compare between different diluents. Comparison of the effects of CO_2 with N_2 reveals that the former has a higher heat capacity and lower heat conductivity than the latter. An analogous observation can be derived by comparing the effects of CO_2 and He. Therefore, in this case the effect of heat capacity seems to be more important than the effect of heat conductivity. It should be noticed that halogenated compounds also modify the chemical reaction by recombination of H atoms to form more stable compounds instead of more reactive radicals [117].

The FLs of different fuels diluted with CO_2 and N_2 were studied by Mendiburu et al. [133]. It was observed that the FIP corresponds to an equivalence ratio greater

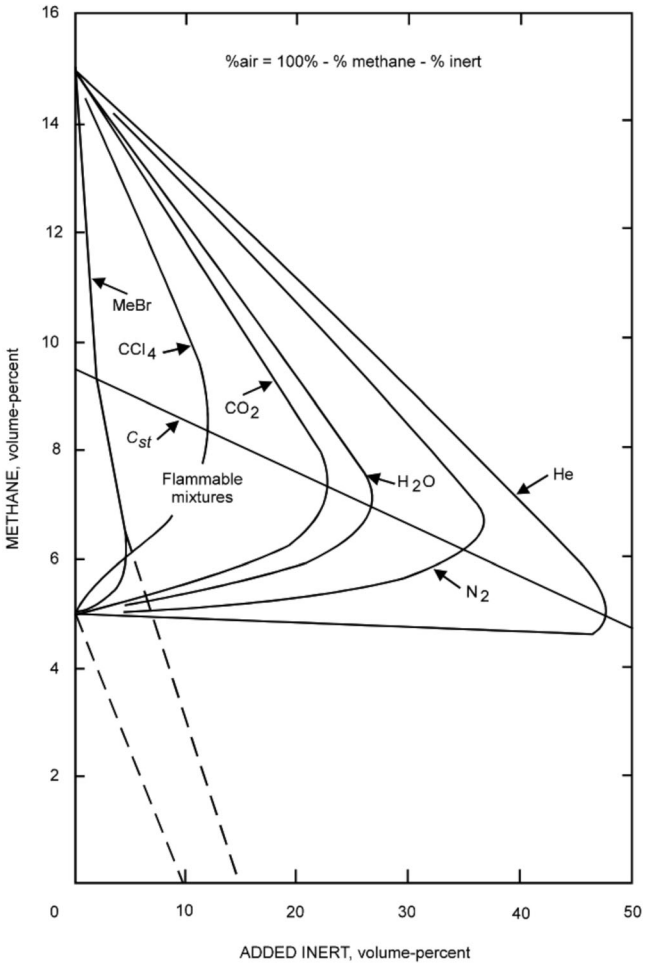


Fig. 8.8 Effect of diluent concentration on the FLs of methane in air [116]. Note that MeBr is CH_3Br

than one ($\phi > 1$) for most fuels, with the exception being H_2, for which $\phi < 1$. The fuel is considered mixed with the diluent, in such a way that the sum of the mole fractions is $x_F + x_d = 1$. Therefore, the number of moles of air at the FL_0 without diluent ($\nu^{FL}_{air,0}$) is obtained from Eq. (8.64), while the number of moles of air at the FL_d with diluent ($\nu^{FL}_{air,d}$) is obtained from Eq. (8.65).

$$\nu^{FL}_{air,0} = \frac{1}{4.76}\left(\frac{100}{FL_0} - 1\right) \tag{8.64}$$

Table 8.1 Heat capacities and heat conductivities of some species at 300 K and 1 atm. Properties of water vapor at 380 K and 1 atm [131]. Properties of CCL_4 and CH_3Br from [132]

Species	W (kg/kmol)	c_p [J/(kg K)]	c_p [J/(mol K)]	$k \times 10^3$ [W(m K)]	Pr
Air	28.97	1007	29.18	26.3	0.707
CH_3Br	94.9	449	42.64	7.8	0.797
CCL_4	153.8	544	83.60	6.1	0.882
N_2	28.01	1041	29.16	25.9	0.716
CO_2	44.01	851	37.46	16.5	0.766
H_2O	18.02	2060	37.12	24.6	1.060
He	4.003	5193	20.79	152.0	0.680

$$v_{\text{air},d}^{\text{FL}} = \frac{1}{4.76}\left(\frac{100}{\text{FL}_d + P_d} - 1\right) \tag{8.65}$$

The P_d represents the percentage of diluent in the mixture such that $x_F = \text{FL}/(\text{FL} + P_d)$. Then, it is possible to find a factor k_{FL} which allows to write,

$$v_{\text{air},d}^{\text{FL}} = x_F v_{\text{air},0}^{\text{FL}} + (1 - x_F)k_{\text{FL}} \tag{8.66}$$

After calculating the values of k_{FL} by using the experimental data of several mixtures, the average values can be obtained with Eq. (8.67). The results of this procedure are presented elsewhere [133].

$$k_{\text{FL}}^{av} = \frac{1}{N}\sum_{i=1}^{N-1} k_{\text{FL}} \tag{8.67}$$

An expression for k_{FL} can be obtained by assuming thermodynamic equilibrium and that all the heat lost from the reaction is absorbed by the diluents or by a term designated as L. Considering given temperature and pressure conditions, writing the global combustion reactions for the cases with and without diluent, applying energy conservation, and isolating the number of moles of air, the following expressions are obtained.

$$v_{\text{air},0}^{\text{FL}} = \frac{1}{\overline{h}_{\text{air}}}\left(\sum_{j=1}^{N} n_j \overline{h}_j + L_0 - \overline{h}_F\right) \tag{8.68}$$

$$v_{\text{air},d}^{\text{FL}} = \frac{1}{\overline{h}_{\text{air}}}\left(\sum_{j=1}^{M} n_k \overline{h}_k + L_d - x_F \overline{h}_F - (1 - x_F)\overline{h}_d\right) \tag{8.69}$$

Note that the products might be different when a diluent is added because the chemistry of the reaction can change. Therefore, the subscripts j and k were used to differentiate between the two cases being considered. Substitution of Eqs. (8.68) and (8.69) into Eq. (8.66) gives

$$k_{\mathrm{FL}} = \frac{1}{(1-x_F)\overline{h}_{\mathrm{air}}}\left(\sum_{j=1}^{N} n_k \overline{h}_k - x_F \sum_{j=1}^{M} n_j \overline{h}_j + (L_d - x_F L_0) - (1-x_F)\overline{h}_d\right) \quad (8.70)$$

A correlation for obtaining the values of k_{FL} was provided by Mendiburu et al. [133]. Also, a semi-empirical method for determining the FIP was proposed in the same article.

8.3.4 Effect of Fuel Composition

Another important aspect of the study of flammability limits is the effect of mixing two or more fuels in the same flammable mixture. For example, natural gas (NG) and liquefied petroleum gas (LPG) are mixtures of several hydrocarbons. Moreover, hydrogen could be added to NG of LPG, further complicating the mixture. In the studies by Barbosa et al. [126, 127], mixtures involving farnesane and jet fuel were studied. Figure 8.9 shows the flame propagation behavior during the experimental determination of the UFLs in air for farnesane and a mixture of jet fuel and farnesane, the mixture had 10% (F10) of farnesane on a mass basis.

The qualitative behavior of flame propagation is evidently different, and the FLs will change accordingly. Consider a flammable mixture near the flammability limit involving two fuels; thus, forming a fuel mixture characterized by, $x_{F,\,\mathrm{A}} + x_{F,\,\mathrm{B}} = 1$. Since the FL for 1 mol of fuel would be given by Eq. (8.53), then, for $x_{F,\mathrm{A}}$ moles of fuel, the following is obtained.

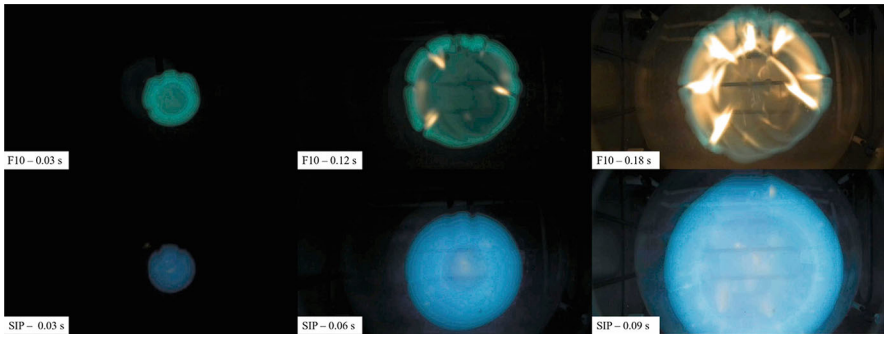


Fig. 8.9 Flame propagation for farnesane and jet fuel/farnesane mixture. The flame propagation at the top is for a jet fuel/farnesane/air mixture (F10) near the UFL at 424 K and 20 kPa. The flame propagation at the bottom is for a farnesane/air mixture near the UFL at 417 K and 20 kPa [127]. (Reprinted from Process Safety and Environmental Protection, Vol 160, Jean A. Barbosa, Christian J.R. Coronado, José C. de Andrade, Celso E. Tuna, Marcos H. Silva, João A. Carvalho Junior, Andrés Z. Mendiburu, Experimental determination of upper flammability limits of synthesized iso-paraffins (SIP), Jet fuel and their mixtures with air at atmospheric and sub-atmospheric pressures, 102–115, Copyright (2022), with permission from Elsevier)

$$\frac{x_{F,\mathrm{A}}}{\mathrm{FL_A}} = \frac{1}{100\%}\left(x_{F,\mathrm{A}} + 4.76 x_{F,\mathrm{A}} \nu_{\mathrm{air,A}}^{\mathrm{FL}}\right) \tag{8.71}$$

Writing a similar equation for fuel B and adding this to the previous expression,

$$\frac{x_{F,\mathrm{A}}}{\mathrm{FL_A}} + \frac{x_{F,\mathrm{B}}}{\mathrm{FL_B}} = \frac{1}{100\%}\left[\left(x_{F,\mathrm{A}} + x_{F,\mathrm{B}}\right) + 4.76\left(x_{F,\mathrm{A}} \nu_{\mathrm{air,A}}^{\mathrm{FL}} + x_{F,\mathrm{B}} \nu_{\mathrm{air,B}}^{\mathrm{FL}}\right)\right] \tag{8.72}$$

The mixture of A and B is now treated as a third fuel, denoted by C. Considering 1 mol of C, the following is obtained.

$$\frac{1}{\mathrm{FL_C}} = \frac{1}{100\%}\left(1 + 4.76 \nu_{\mathrm{air,C}}^{\mathrm{FL}}\right) \tag{8.73}$$

The law of Le Chatelier [134] is an empirical mixing rule which establishes the equality shown in Eq. (8.74). Recalling that $x_{F,\,\mathrm{A}} + x_{F,\,\mathrm{B}} = 1$, allows to make a connection between the number of moles of air required for each fuel.

$$\frac{1}{\mathrm{FL_C}} = \frac{x_{F,\mathrm{A}}}{\mathrm{FL_A}} + \frac{x_{F,\mathrm{B}}}{\mathrm{FL_B}} \tag{8.74}$$

The above expression is equivalent to saying that the number of moles of air required for the fuel mixture C is equal to the weighted sum of the number of moles of air required for each individual fuel in the mixture, with the mole fractions of each fuel in the fuel mixture serving as the weighting factors.

$$\nu_{\mathrm{air,C}}^{\mathrm{FL}} = x_{F,\mathrm{A}} \nu_{\mathrm{air,A}}^{\mathrm{FL}} + x_{F,\mathrm{B}} \nu_{\mathrm{air,B}}^{\mathrm{FL}} \tag{8.75}$$

Using the expression shown in Eq. (8.68) to write the number of moles of air in terms of energy conservation, noting that $\overline{h}_{F,\mathrm{C}} = x_{F,\mathrm{A}} \overline{h}_{F,\mathrm{A}} + x_{F,\mathrm{B}} \overline{h}_{F,\mathrm{B}}$, and rearranging give,

$$\sum_{j=1}^{N} n_i \overline{h}_i + L_{\mathrm{C}} = x_{F,\mathrm{A}}\left(\sum_{j=1}^{M} n_j \overline{h}_j + L_{\mathrm{A}}\right) + x_{F,\mathrm{B}}\left(\sum_{k=1}^{R} n_k \overline{h}_k + L_{\mathrm{B}}\right) \tag{8.76}$$

Therefore, the heat released in the reaction of the fuel mixture plus the heat loss must be equal to the corresponding weighted sum of these parameters for each individual fuel, with the mole fractions of the individual fuels in the fuel mixture being the weighting parameters. The equality given in Eq. (8.76) can be true, when the reaction path and the heat losses are not significantly affected by the addition of another fuel to the flammable mixture. At the LFL the combustion can be assumed as complete; therefore, it is expected that the chemical reaction path is not greatly affected by adding another fuel. Conversely, at the UFL the combustion is incomplete; thus, the presence of another fuel is more likely to affect the reaction path. The above equations can be extended to the case of several fuels. However, as more fuels are added, the chemical reaction path should be more susceptible to changes.

$$\frac{1}{\mathrm{FL_{mix}}} = \sum_{i=1}^{N} \frac{x_{F,i}}{\mathrm{FL}_i} \tag{8.77}$$

As discussed by Mendiburu et al. [134], the law of Le Chatelier works better for estimating the LFL than for the UFL. A complete discussion on the applicability of this law and on the data available for validation can be found in the aforementioned article.

8.4 Methods for Estimating the Flammability Limits

In the context of energy transition, it is important to know the flammability limits of different fuels under different conditions. Therefore, in this section simple methods are presented to have a first approximation of the FLs.

8.4.1 Method to Estimate the FLs at Ambient Temperature and Atmospheric Pressure

The method presented here is called the calculated adiabatic flame temperature (CAFT) method. Several authors [135, 136] have developed alternative versions of this method to estimate the FLs at ambient temperature and atmospheric pressure. A robust version of this method was presented by Mendiburu et al. [117], based on previous work by Mendiburu et al. [137–139]. The CAFT method assumes a complete combustion reaction for the LFL, as shown in Eq. (8.78), or an incomplete combustion reaction, as shown in Eqs. (8.79) and (8.80).

Complete combustion reaction at the LFL

$$\mathrm{C}_{x_\mathrm{C}}\mathrm{H}_{x_\mathrm{H}}\mathrm{O}_{x_\mathrm{O}} + v_{\mathrm{air}}^{\mathrm{LFL}}(\mathrm{O}_2 + 3.76\mathrm{N}_2) \rightarrow x_\mathrm{C}\mathrm{CO}_2 + \frac{x_\mathrm{H}}{2}\mathrm{H}_2\mathrm{O} + \left(v_{\mathrm{air}}^{\mathrm{LFL}} - v_{\mathrm{air}}^{s}\right)\mathrm{O}_2 + 3.76 v_{\mathrm{air}}^{\mathrm{LFL}}\mathrm{N}_2 \tag{8.78}$$

Incomplete combustion reaction at the UFL without condensate

$$\mathrm{C}_{x_\mathrm{C}}\mathrm{H}_{x_\mathrm{H}}\mathrm{O}_{x_\mathrm{O}} + v_{\mathrm{air}}^{\mathrm{UFL}}(\mathrm{O}_2 + 3.76\mathrm{N}_2) \rightarrow n_{\mathrm{CO}_2}\mathrm{CO}_2 + n_{\mathrm{CO}}\mathrm{CO} + n_{\mathrm{H}_2\mathrm{O}}\mathrm{H}_2\mathrm{O} + n_{\mathrm{H}_2}\mathrm{H}_2 + 3.76 v_{\mathrm{air}}^{\mathrm{UFL}}\mathrm{N}_2 \tag{8.79}$$

Incomplete combustion reaction at the UFL with condensate

$$\mathrm{C}_{x_\mathrm{C}}\mathrm{H}_{x_\mathrm{H}}\mathrm{O}_{x_\mathrm{O}} + v_{\mathrm{air}}^{\mathrm{UFL}}(\mathrm{O}_2 + 3.76\mathrm{N}_2) \rightarrow n_{\mathrm{CO}}\mathrm{CO} + n_\mathrm{C}\mathrm{C} + n_{\mathrm{H}_2\mathrm{O}}\mathrm{H}_2\mathrm{O} + n_{\mathrm{H}_2}\mathrm{H}_2 + 3.76 v_{\mathrm{air}}^{\mathrm{UFL}}\mathrm{N}_2 \tag{8.80}$$

In the case of the UFL there are two alternative combustion reactions. This is related to the fact that some fuels present moderate values of the UFL, which means

that there is enough oxygen to form CO_2 and CO. On the other hand, there are also fuels with high UFL values, for which it is necessary to assume that only CO and C are formed in the products due to the low availability of oxygen. Assuming chemical equilibrium for the reactions given in Eqs. (8.79) and (8.80), one auxiliary reaction must be assumed in each case to close the system.

Water–gas homogenous reaction for UFL reaction without condensate

$$H_2O + CO \leftrightarrow CO_2 + H_2 \tag{8.81}$$

Water–gas heterogenous reaction for UFL reaction with condensate

$$C + H_2O \leftrightarrow CO + H_2 \tag{8.82}$$

Assuming that the adiabatic flame temperature at the LFL can be known (T_{LFL}), it is possible to write the energy conservation equation for the reaction given in Eq. (8.78) and calculate the number of moles of oxygen at the LFL ($v_{\mathrm{air}}^{\mathrm{LFL}}$), as shown in Eq. (8.83).

$$v_{\mathrm{air}}^{\mathrm{LFL}} = \frac{\overline{h}_{F,R} + v_{\mathrm{air}}^{s}\overline{h}_{\mathrm{O_2}} - x_{\mathrm{C}}\overline{h}_{\mathrm{CO_2}} - \frac{x_{\mathrm{H}}}{2}\overline{h}_{\mathrm{H_2O}} - \frac{x_{\mathrm{N}}}{2}\overline{h}_{\mathrm{N_2}}}{\overline{h}_{\mathrm{air}} - \overline{h}_{\mathrm{air},R}} \tag{8.83}$$

Note that in this formulation, $\overline{h}_{\mathrm{air}} = \overline{h}_{\mathrm{O_2}} + 3.76\overline{h}_{\mathrm{N_2}}$. The subscript R refers to reactants, which means that the enthalpy should be calculated at the initial temperature. All other enthalpies are calculated at the T_{LFL}. Similarly, assuming that the T_{UFL} is known, writing the energy conservation equation and the expression for the equilibrium constant, the following expression is obtained for the number of moles of oxygen at the UFL ($v_{\mathrm{air}}^{\mathrm{UFL}}$).

$$a_v\left(v_{\mathrm{air}}^{\mathrm{UFL}}\right)^2 + b_v v_{\mathrm{air}}^{\mathrm{UFL}} + c_v = 0 \tag{8.84}$$

The coefficients to be used together with Eq. (8.84) depend on which reaction is being used, that is, without or with condensate.

Coefficients for the combustion reaction at the UFL without condensate

$$a_v = am^2 + 2am; \quad a = 1 - K_{sc} \tag{8.85}$$

$$b_v = 2amn + amx_{\mathrm{O}} + 2an - (2a+1)mx_{\mathrm{C}} - m\frac{x_{\mathrm{H}}}{2} - 2x_{\mathrm{C}} \tag{8.86}$$

$$c_v = an^2 - (2a+1)nx_{\mathrm{C}} + (an - x_{\mathrm{C}})x_{\mathrm{O}} - (n - x_{\mathrm{C}})\frac{x_{\mathrm{H}}}{2} + 2x_{\mathrm{C}}^2 \tag{8.87}$$

$$m = \frac{\overline{h}_{\mathrm{air},R} - 2\overline{h}_{\mathrm{H_2}} - 2\overline{h}_{\mathrm{H_2O}} - 3.76\overline{h}_{\mathrm{N_2}}}{\overline{h}_{\mathrm{CO}} - \overline{h}_{\mathrm{CO_2}} + \overline{h}_{\mathrm{H_2O}} - \overline{h}_{\mathrm{H_2}}} \tag{8.88}$$

$$n = \frac{\overline{h}_{F,R} - x_C\overline{h}_{CO_2} + (2x_C - x_O)\overline{h}_{H_2O} + \left(x_O - \frac{x_H}{2} - 2x_C\right)\overline{h}_{H_2}}{\overline{h}_{CO} - \overline{h}_{CO_2} + \overline{h}_{H_2O} - \overline{h}_{H_2}} \tag{8.89}$$

Coefficients for the combustion reaction at the UFL with condensate

$$a_v = am^2 - 2am + 3.76(a-1)m - 7.52(a-1)\ ; \quad a = 1 + K_{cc}(p_0/p) \tag{8.90}$$

$$b_v = 2amn + am\left(\frac{x_H}{2} - x_O\right) - 2an + (a-1)(3.76[n - x_O] - x_H) \tag{8.91}$$

$$c_v = an^2 + an\left(\frac{x_H}{2} - x_O\right) - (a-1)\frac{x_H}{2}x_O \tag{8.92}$$

$$m = \frac{\overline{h}_{air,R} - 2\overline{h}_{H_2} - 2\overline{h}_{H_2O} - 3.76\overline{h}_{N_2}}{\overline{h}_{CO} - \overline{h}_{CO_2} + \overline{h}_{H_2O} - \overline{h}_{H_2}} \tag{8.93}$$

$$n = \frac{\overline{h}_{F,R} - x_C\overline{h}_C - x_O\overline{h}_{H_2O} + \left(x_O - \frac{x_H}{2}\right)\overline{h}_{H_2}}{\overline{h}_{CO} - \overline{h}_{H_2} - \overline{h}_{H_2O} - \overline{h}_C} \tag{8.94}$$

The correlations to obtain the T_{LFL} and T_{UFL} depend on prior knowledge of the adiabatic flame temperature at stoichiometric composition (T_{stq}).

$$\frac{T_{stq}}{T_{LFL}} = a_0 + a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 \tag{8.95}$$

$$\begin{aligned}\frac{T_{stq}}{T_{UFL}} = {} & a_0 + a_1x_1 + a_2x_1^2 + a_3x_1^3 + a_4x_2 + a_5x_2^2 + a_6x_2^3 + a_7x_3 + a_8x_3^2 \\ & + a_9x_3^3 + a_{10}x_4 + a_{11}x_4^{1/2}\end{aligned} \tag{8.96}$$

where

$$x_1 = \frac{\bar{h}^\circ_{f,F}}{W_F}\ ; \quad x_2 = \frac{\bar{h}^\circ_{f,F}}{x_Cx_Hx_m}\ ; \quad x_3 = \frac{W_F}{x_Cx_Hx_m}\ ; \quad x_4 = \frac{x_Cx_m}{x_H} \tag{8.97}$$

The x_m is introduced to account for compounds without oxygen in their molecules. For the LFL, $x_m = x_O + 1$, and for the UFL, $x_m = \max\{x_O, 1\}$. The units of the parameters x_i will be disregarded because the coefficients a_i have the exact units to make Eqs. (8.95) and (8.96) dimensionless. The coefficients a_i are presented in Table 8.2.

The coefficients in Table 8.2 were obtained for hydrocarbons (C-H) and for oxygenated compounds (C-H-O). The number of compounds used for validation is N. The average of the absolute values of the relative errors (AARE) is lower than 6% for the LFL and lower than 10% for the UFL. The squared correlation coefficient (R^2) is higher than 0.96 for the LFL and higher than 0.93 for the UFL. Therefore, the method presents good accuracy.

Table 8.2 Values of the coefficients a_i to use in Eqs. (8.95) and (8.96) and the validation parameters of the method [117]

	LFL		UFL	
Coefficients	C-H	C-H-O	C-H	C-H-O
a_0	1.4774	1.5101	2.3737	2.5526
a_1	1.0186×10^{-2}	-6.7195×10^{-3}	-2.3822×10^{-1}	-1.7112×10^{-1}
a_2	-2.0090×10^{-3}	1.3081×10^{-2}	1.2076×10^{-2}	-2.9203×10^{-2}
a_3	6.4563×10^{-2}	8.4976×10^{-2}	2.8293×10^{-2}	-5.8587×10^{-4}
a_4	1.1814×10^{-2}	6.4285×10^{-3}	4.6652×10^{-2}	-6.1402×10^{-2}
a_5	–	–	9.8788×10^{-4}	-4.1708×10^{-3}
a_6	–	–	-1.8129×10^{-4}	-4.9859×10^{-5}
a_7	–	–	5.2138×10^{-2}	-4.9785×10^{-1}
a_8	–	–	-1.9116×10^{-1}	1.1146×10^{-1}
a_9	–	–	4.3725×10^{-2}	-4.3060×10^{-3}
a_{10}	–	–	-2.0102×10^{-2}	2.1873×10^{-2}
a_{11}	–	–	-1.3389×10^{-1}	-1.6556×10^{-1}
N	324	374	247	208
AARE (%)	5.53	5.57	8.46	9.24
R^2	0.9608	0.9756	0.9312	0.9318

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Example 8.2
Estimate the flammability limits of ethanol in air at 1 atm and 298 K for the LFL and 330.15 K for the UFL.

Solution
Although the estimation methods are more easily applied by creating a routine in a programming platform such as Python or MATLAB, hand calculations will be presented in this example. The input parameters for ethanol are shown below.

$$\overline{h}^0_{f,F} = -234.95 \text{ kJ/mol}; \quad W_F = 46 \text{ g/mol}; \quad T_{\text{stq}} = 2353.8 \text{ K};$$

$$x_C = 2; \quad x_H = 6; \quad x_O = 1; \quad \nu^s_{\text{air}} = 3$$

(a) *Estimation of the LFL*
The first step is to calculate the values of the parameters x_i from Eq. (8.97) with $x_m = 2$.

$$x_1 = \frac{-234.95}{46} = -5.1077; \quad x_2 = \frac{-234.95}{2 \times 6 \times 2} = -9.7896$$

$$x_3 = \frac{46}{2 \times 6 \times 2} = 1.9167; \quad x_4 = \frac{2 \times 2}{6} = 0.6667$$

The second step is to calculate the T_{LFL} from Eq. (8.95).

$$\frac{T_{\text{stq}}}{T_{\text{LFL}}} = \begin{array}{c} 1.5101 - 6.7195 \times 10^{-3}(-5.1077) + 1.3081 \times 10^{-2}(-9.7896) \\ +8.4976 \times 10^{-2}(1.9167) + 6.4285 \times 10^{-3}(0.6667) \end{array} = 1.5836$$

$$T_{\text{LFL}} = \frac{2353.8}{1.5836} = 1486.3 \text{ K}$$

The third step is to obtain the enthalpies of the products. The enthalpies will be obtained from thermodynamic tables available online [2]. However, it is worth to mention that, in the original work the enthalpies were obtained using the NASA Glenn Polynomials [3].

$$\overline{h}_{\text{CO}_2} = -393.522 + 60.91 = -332.612 \text{ kJ/mol}$$

$$\overline{h}_{\text{H}_2\text{O}} = -241.826 + 47.512 = -194.314 \text{ kJ/mol}$$

$$\overline{h}_{\text{O}_2} = 40.1 \text{ kJ/mol}; \quad \overline{h}_{\text{N}_2} = 37.93 \text{ kJ/mol}$$

$$\overline{h}_{ar} = 40.1 + 3.76 \times 37.93 = 182.717 \text{ kJ/mol}$$

Since the initial temperature is 298 K, the sensible enthalpies of the reactants are zero. The fourth step is to calculate the number of moles of air at the LFL with Eq. (8.83).

$$\nu_{\text{air}}^{\text{LFL}} = \frac{-234.95 + 3(40.1) - 2(-332.612) - 3(-194.314)}{182.717} = 6.203$$

Finally,

$$\text{LFL} = \frac{100\%}{1 + 4.76 \times 6.203} = 3.28\%$$

(b) *Estimation of the UFL*
Since this method is more involving, the calculations were done with the help of spreadsheet software. The first step is to calculate the values of the parameters x_i from Eq. (8.97) with $x_m = 1$.

$$x_1 = \frac{-234.95}{46} = -5.1077; \quad x_2 = \frac{-234.95}{2 \times 6 \times 1} = -19.5791$$

$$x_3 = \frac{46}{2 \times 6 \times 1} = 3.8333; \quad x_4 = \frac{2 \times 1}{6} = 0.3333$$

The second step is to calculate the T_{UFL} from Eq. (8.96).

$$\frac{T_{\text{stq}}}{T_{\text{UFL}}} = 2.1190$$

$$T_{\mathrm{UFL}} = \frac{2353.8}{2.1023} = 1110.8 \ \mathrm{K}$$

The third step is to obtain the enthalpies of the products, reactants, and the equilibrium constant. The enthalpy of ethanol is estimated by considering the c_p at 330.15 K.

$$\overline{h}_{\mathrm{CO_2}} = -393.522 + 39.487 = -354.035 \ \mathrm{kJ/mol}$$

$$\overline{h}_{\mathrm{CO}} = -110.527 + 25.401 = -85.126 \ \mathrm{kJ/mol}$$

$$\overline{h}_{\mathrm{H_2O}} = -241.826 + 30.657 = -211.169 \ \mathrm{kJ/mol}$$

$$\overline{h}_{\mathrm{H_2}} = 24.051 \ \mathrm{kJ/mol}\ ; \quad \overline{h}_{\mathrm{O_2}} = 26.595 \ \mathrm{kJ/mol}\ ; \quad \overline{h}_{\mathrm{N_2}} = 25.121 \ \mathrm{kJ/mol}$$

$$\overline{h}_{ar} = 26.595 + 3.76 \times 25.121 = 121.05 \ \mathrm{kJ/mol}$$

$$\overline{h}_{\mathrm{O_2},R} = 0.9447 \ \mathrm{kJ/mol}; \quad \overline{h}_{\mathrm{N_2},R} = 0.9326 \ \mathrm{kJ/mol}$$

$$\overline{h}_{ar,R} = 0.9447 + 3.76 \times 0.9326 = 4.4512 \ \mathrm{kJ/mol}$$

$$\overline{h}_{F,R} = -234.95 + 0.07023(330.15 - 298.15) = -232.70 \ \mathrm{kJ/mol}$$

The equilibrium constant is calculated with the procedure explained in Chap. 1.

$$K_{sc} = \exp\left[- \frac{(-396.011 + 0.0) - (-186.427 - 210.02)}{8.3144 \times 10^{-3} \times 1110.8} \right] = 0.0472$$

The fifth step is to calculate the coefficients given in Eqs. (8.85)–(8.89).

$$a = 0.9528; \quad m = 8.4369; \quad n = -9.0442$$

$$a_v = 83.8996; \quad b_v = -231.744; \quad c_v = 159.3402$$

The sixth step is to solve the quadratic equation given in Eq. (8.84).

$$83.8996\left(v_{\mathrm{air}}^{\mathrm{UFL}}\right)^2 - 231.744 v_{\mathrm{air}}^{\mathrm{UFL}} + 159.3402 = 0$$

There are two roots for the above quadratic equation.

$$v_{\mathrm{air},1}^{\mathrm{UFL}} = 1.2905 \quad \text{and} \quad v_{\mathrm{air},2}^{\mathrm{UFL}} = 1.4716$$

Therefore,

$$\mathrm{UFL}_1 = \frac{100\%}{1 + 4.76 \times 1.2905} = 14.00\%$$

$$\mathrm{UFL}_2 = \frac{100\%}{1 + 4.76 \times 1.4716} = 12.49\%$$

The highest value of the UFL will be adopted from a safety point of view. The same criterion would be applied if the reactions without and with condensate would both provide real numbers for $v_{\mathrm{air}}^{\mathrm{UFL}}$. In the case of ethanol, the solution with condensate provides an imaginary number. The experimental value of the LFL of ethanol at 101.3 kPa and 23.4 °C was determined by Coronado et al. as 3.79%. In the same article, the UFL of ethanol at 101.3 kPa and 57 °C was determined as 19.65%. Therefore, the hand calculations presented in this example provide a good approximation in the case of the LFL and an acceptable estimation in the case of the UFL. The use of the NASA Glenn coefficients and a programming platform increases the accuracy at the UFL, since it is sensitive to the T_{UFL}, which is estimated through the correlation. For example, using the original routine, the UFL was estimated as 18.08%, which is a considerable change compared to the 14.00% obtained in this example.

8.4.2 Method to Estimate the FLs at Different Initial Temperatures

The variation of the FLs with temperature can be estimated using Eq. (8.57), with an adequate value of the slope m in Eq. (8.58). Zabetakis [116] assumed that the heat released at the FLs was constant and equal to the heat of combustion of 1 mol of fuel. Zabetakis [116] also assumed that $\Delta\overline{h}_{C,T_0} = \Delta\overline{h}_{C,T}$, and adopted the values $\overline{c}_{p,\mathrm{air}} = 0.0075$ kcal/(mol K) and $(\mathrm{FL}_{T_0}/100)\Delta\overline{h}_{C,T_0} = 10.4$ kcal/mol. Therefore, the slope became $m = \pm 0.000721$.

$$\frac{\mathrm{FL}_T}{\mathrm{FL}_{T_0}} = 1 \mp 0.000721(T - T_0) \tag{8.98}$$

It has been observed that the simple method proposed by Zabetakis presents good accuracy with AAREs of 4.82 and 5.01% for the LFL and UFL, respectively [139]. Mendiburu et al. [139] improved the method by considering a parameter I, as shown in Eq. (8.99).

$$I = \frac{W_F}{n_F^* \Delta\overline{h}_C} \tag{8.99}$$

The n_F* are the number of moles of fuel in a gaseous fuel–air mixture occupying a volume of 12 L at a specific pressure and T_0. Then, a correlation was obtained for the slope at the LFL by using this parameter.

$$m_{\mathrm{LFL}} = 8.3959 \times 10^{-4} + 9.6643 \times 10^{-5} I + 2.6402 \times 10^{-6} I^2 + 8.3413 \times 10^{-10} \left(\Delta\overline{h}_C \times I\right) \tag{8.100}$$

In the case of the UFL, a more complex approach was adopted and will not be presented here. The method developed by Mendiburu et al. presented AAREs of 3.25 and 3.60% for the LFL and UFL, respectively. It should be observed that the validation of both methods [139] was performed with 180 data points for the LFL and 129 for the UFL. A collection of experimental slopes to be used with Eq. (8.57) to determine the LFL and UFL of several fuels, including ethanol, hydrogen, and ammonia, at different initial temperatures is presented in the work by Mendiburu et al. [117, 139].

Example 8.3
Estimate the variation of the FLs of ethanol at 1 atm and different initial temperatures. Consider as starting values the experimental data by Coronado et al. [125], where the LFL is 3.79% at 23.4 °C and the UFL is 19.65% at 57 °C.

Solution
The methods by Mendiburu et al. and Zabetakis will be applied. In the case of the method by Mendiburu, only the LFL will be determined explicitly. The heat of combustion of 1 mol of ethanol is determined first.

$$\Delta\overline{h}_C = -234.95 - 2\times(-393.522) - 3(-241.826) = 1277.572 \text{ kJ/mol}$$

The number of moles of the mixture at $T_0 = 296.55$ K inside a 12 L vessel will be

$$n_F^* = \frac{3.79}{100}\frac{101.325\times 12\times 10^{-3}}{8.3144\times 10^{-3}\times 296.55} = 0.01869 \text{ mol}$$

Then, the parameter I will be

$$I = \frac{46}{0.01869\times 1277.572} = 1.9264$$

The slope at the LFL is obtained using Eq. (8.100).

$$m_{\text{LFL}} = 0.001038$$

One calculation will be shown. For example, for $T = 360$ K, Eq. (8.57) gives

$$\text{LFL}_T = 3.79[1 - 0.001038(360 - 296.55)] = 3.54\%$$

Or, by applying the slope proposed by Zabetakis,

$$\text{LFL}_T = 3.79[1 - 0.000721(360 - 296.55)] = 3.61\%$$

The results for the LFL and UFL at different initial temperatures are shown in Fig. 8.10. Note that the UFL calculated by the method developed by Mendiburu et al. [139] was also included in the figure.

As can be observed in Fig. 8.9, the method by Mendiburu et al. estimates correctly the variation of the FLs of ethanol. The method to estimate the UFL was

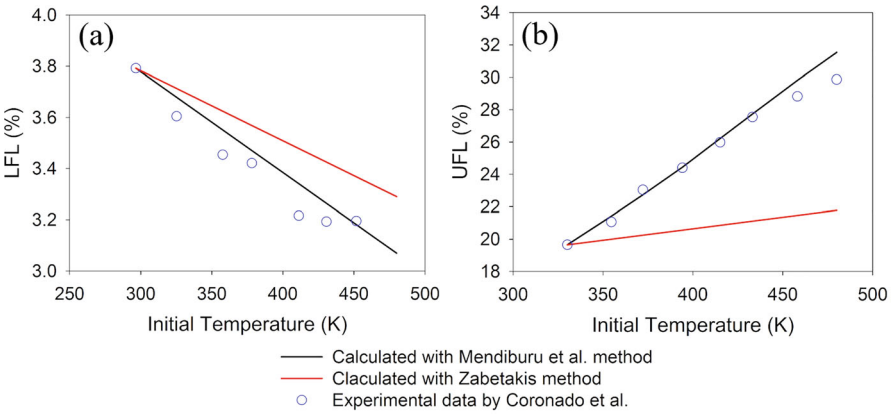


Fig. 8.10 Flammability limits of anhydrous ethanol in air at atmospheric pressure and different initial temperatures. Experimental data by Coronado et al. [125]. (**a**) LFL and (**b**) UFL

not presented here. However, the interested reader can find the method in the work by Mendiburu et al. [117, 139].

8.4.3 Method to Estimate the Effect of Diluents

The effect of diluents can also be estimated using a correlation to determine k_{FL}^{av} and using it in Eq. (8.66) to estimate $v_{\mathrm{air},d}^{\mathrm{FL}}$. This was done in the work by Mendiburu et al. [133] for N_2 and CO_2 dilution. Also, the FIP can be estimated using a correlation to determine the ratio of the flame temperature at FIP (T_{FIP}) to the flame temperature at the LFL (T_{LFL}).

$$\theta_d = \frac{T_{\mathrm{FIP}}}{T_{\mathrm{LFL}}} \tag{8.101}$$

It is possible to relate the equivalence ratio at the FIP (ϕ_{FIP}), the fuel molar fraction at the FIP (x_F^{FIP}), and the number of moles of oxygen at the FIP ($v_{\mathrm{air}}^{\mathrm{FIP}}$) through the expression shown in Eq. (8.102).

$$v_{\mathrm{air}}^{\mathrm{FIP}} = \frac{x_F^{\mathrm{FIP}}}{\phi_{\mathrm{FIP}}} v_{\mathrm{air}}^{s} \tag{8.102}$$

In the aforementioned study, must compounds presented a ϕ_{FIP} between 0.841 and 1.847. However, hydrogen presented a ϕ_{FIP} of 0.463 and 0.468 for dilution with N_2 and CO_2, respectively.

The correlation proposed in Eq. (8.103) is validated for N_2 and CO_2 dilution and can be applied to estimate θ_d, ϕ_{FIP}, k_{LFL}^{av}, and k_{UFL}^{av}.

Table 8.3 Coefficients to be used with the correlation shown in Eq. (8.103) for dilution with N_2. Adapted from Mendiburu et al. [133]

Coefficients	Dilution with N_2			
	k_{LFL}^{av}	k_{UFL}^{av}	θ_d	ϕ_{FIP}
a_0	-2.3808×10^{-1}	1.9958×10^{-1}	9.8694×10^{-1}	1.1825
a_1	0.0	-6.1268×10^{-2}	5.4050×10^{-2}	-1.0775×10^{-1}
a_2	3.3183×10^{-2}	5.6992×10^{-2}	3.6254×10^{-2}	-1.6605×10^{-1}
a_3	0.0	-4.2119×10^{-2}	0.0	0.0
a_4	0.0	-2.7750×10^{-2}	0.0	0.0
a_5	-1.8869×10^{-2}	0.0	0.0	2.4997×10^{-2}
a_6	0.0	-8.2907×10^{-3}	0.0	0.0
LHV_{avg}	–	43.711	42.057	42.057
FL_{avg}	2.904	15.144	3.149	3.149
S_{LHV}	–	7.818	10.159	10.159
S_{FL}	1.392	8.394	1.889	1.889

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Table 8.4 Coefficients to be used with the correlation shown in Eq. (8.103) for dilution with CO_2. Adapted from Mendiburu et al. [133]

Coefficients	Dilution with CO_2			
	k_{LFL}^{av}	k_{UFL}^{av}	θ_d	ϕ_{FIP}
a_0	-4.6388×10^{-1}	4.0034×10^{-1}	9.8435×10^{-1}	1.2354
a_1	0.0	-6.2961×10^{-2}	6.1818×10^{-2}	-1.5933×10^{-1}
a_2	0.0	4.7020×10^{-2}	5.3154×10^{-2}	-3.6088×10^{-1}
a_3	0.0	0.0	0.0	0.0
a_4	1.4387×10^{-3}	-9.4863×10^{-2}	0.0	0.0
a_5	0.0	0.0	0.0	7.7318×10^{-2}
a_6	0.0	0.0	0.0	0.0
LHV_{avg}	39.569	37.539	37.292	37.292
FL_{avg}	–	16.144	2.990	2.990
S_{LHV}	10.062	11.502	11.768	11.768
S_{FL}	–	10.319	2.372	2.372

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$$L = a_0 + a_1 z_1 + a_2 z_2 + a_3 (z_1 z_2) + a_4 z_1^2 + a_5 z_2^2 + a_6 (z_1 z_2)^2 \tag{8.103}$$

where the parameters z_i are given in Eq. (8.104) and the coefficients a_i are presented in Table 8.3 and 8.4.

$$z_1 = \frac{\mathrm{LHV}_i - \mathrm{LHV}_{\mathrm{avg}}}{S_{\mathrm{LHV}}}; \quad z_2 = \frac{\mathrm{FL}_i - \mathrm{FL}_{\mathrm{avg}}}{S_{\mathrm{FL}}} \tag{8.104}$$

The abbreviation LHV_i stands for the lower heating value of the particular fuel, while $\mathrm{LHV}_{\mathrm{avg}}$ is the average lower heating value of the fuels considered to obtain the correlation, and S_{LHV} is the standard deviation. Similar definitions apply to FL_i, $\mathrm{FL}_{\mathrm{avg}}$, and S_{FL}.

To determine the FIP, it is necessary to apply a method similar to the one used to estimate the UFL. The following expression is assumed as the global combustion reaction.

$$\begin{aligned} & x_F^{\mathrm{FIP}}\mathrm{C}_{x_\mathrm{C}}\mathrm{H}_{x_\mathrm{H}}\mathrm{O}_{x_\mathrm{O}} + x_d^{\mathrm{FIP}} M_d + v_{\mathrm{air}}^{\mathrm{FIP}}(\mathrm{O}_2 + 3.76\mathrm{N}_2) \\ & \rightarrow \begin{matrix} n_{\mathrm{CO}_2}\mathrm{CO}_2 + n_{\mathrm{CO}}\mathrm{CO} + n_{\mathrm{H}_2\mathrm{O}}\mathrm{H}_2\mathrm{O} \\ + n_{\mathrm{H}_2}\mathrm{H}_2 + 3.76 v_{\mathrm{air}}^{\mathrm{FIP}}\mathrm{N}_2 + x_d^{\mathrm{FIP}} M_d \end{matrix} \end{aligned} \tag{8.105}$$

The value of x_F^{FIP} can be determined by solving the quadratic equation given in Eq. (8.106), which was obtained after applying species mass conservation, energy conservation, and chemical equilibrium. The water–gas homogeneous reaction, given in Eq. (8.81), was used to apply the equilibrium condition.

$$a_v \left(x_F^{\mathrm{FIP}}\right)^2 + b_v x_F^{\mathrm{FIP}} + c_v = 0 \tag{8.106}$$

$$a = 1 - K_{sc} \tag{8.107}$$

$$a_v = am^2 + 2\frac{v_{ar,1}^s}{\phi_{\mathrm{FIP}}}(am - x_\mathrm{C}) + x_\mathrm{C}[2x_\mathrm{C} - (2a+1)m] + x_\mathrm{O}(am - x_\mathrm{C}) + \frac{x_\mathrm{H}}{2}(x_\mathrm{C} - m) \tag{8.108}$$

$$b_v = 2amn + 2an\frac{v_{ar,1}^s}{\phi_{\mathrm{FIP}}} - (2a+1)nx_\mathrm{C} + anx_\mathrm{O} - n\frac{x_\mathrm{H}}{2} \tag{8.109}$$

$$c_v = an^2 \tag{8.110}$$

$$m = \frac{\left(\begin{matrix} h_F + \frac{v_{\mathrm{air},1}^s}{\phi_{\mathrm{FIP}}}(h_{ar,R} - 3.76h_{\mathrm{N}_2} + 2h_{\mathrm{H}_2} - 2h_{\mathrm{H}_2\mathrm{O}}) - \frac{x_\mathrm{H}}{2}h_{\mathrm{H}_2} \\ + x_C(2h_{\mathrm{H}_2\mathrm{O}} - 2h_{\mathrm{H}_2} - h_{\mathrm{CO}_2}) + x_\mathrm{O}(h_{\mathrm{H}_2} - h_{\mathrm{H}_2\mathrm{O}}) + (h_{d,\mathrm{P}} - h_{d,R}) \end{matrix}\right)}{h_{\mathrm{CO}} - h_{\mathrm{CO}_2} + h_{\mathrm{H}_2\mathrm{O}} - h_{\mathrm{H}_2}} \tag{8.111}$$

$$n = \frac{h_{d,R} - h_{d,\mathrm{P}}}{h_{\mathrm{CO}} - h_{\mathrm{CO}_2} + h_{\mathrm{H}_2\mathrm{O}} - h_{\mathrm{H}_2}} \tag{8.112}$$

Example 8.4
Determine the variation of the FLs of ethanol in air for dilution with N_2 at atmospheric pressure and ambient temperature. Consider the experimental data by Zabetakis [116], for which the LFL and UFL without dilution are 3.40% and 18.70%, respectively.

Solution
The LHV_i for ethanol is 27.79 kJ/g. The other data for this calculation are taken from Table 8.3. The values of k_{LFL}^{av} and k_{UFL}^{av} are determined using the correlation given in Eq. (8.103). Taking out the $a_i = 0$ coefficients,

$$k_{av}^{\text{LFL}} = a_0 + a_2 z_2 + a_5 z_2^2$$

$$k_{av}^{\text{UFL}} = a_0 + a_1 z_1 + a_2 z_2 + a_3 (z_1 z_2) + a_4 z_1^2 + a_6 (z_1 z_2)^2$$

Note that z_1 is not necessary for the determination of k_{LFL}^{av}, while z_1 and z_2 are both necessary for the determination of k_{UFL}^{av}. The z_i parameters for this calculation are determined.

$$z_{2,\text{LFL}} = \frac{3.40 - 2.904}{1.392} = 0.3563$$

$$z_{1,\text{UFL}} = \frac{27.79 - 43.711}{7.818} = -2.0364; \quad z_{2,\text{UFL}} = \frac{18.70 - 15.144}{8.394} = 0.4236$$

The calculations were performed with spreadsheet software.

$$k_{av}^{\text{LFL}} = -0.2287 \quad \text{and} \quad k_{av}^{\text{UFL}} = 0.2636$$

Consider a sample calculation. Assume $x_F = 0.194$ for the LFL in the fuel–diluent mixture, such that $x_d = 0.806$. Then, from Eq. (8.66),

$$v_{\text{air},0}^{\text{LFL}} = \frac{1}{4.76}\left(\frac{100}{3.40} - 1\right) = 5.9689$$

$$v_{\text{air},d}^{\text{LFL}} = 0.194 \times 5.9689 + 0.806 \times (-0.2287) = 0.9736$$

In this case, the FLs are not calculated for 1 mol of fuel; they should be calculated for x_F moles of fuel.

$$\text{LFL}_d = \frac{1 \times 0.194}{1 + 4.76 \times 0.9736} = 3.44\%$$

The above FLs can be used to determine the diluent concentration in the fuel–air–diluent mixture. Note that $x_F = \text{FL}/(\text{FL} + P_d)$.

$$P_d = \left(\frac{1}{x_F} - 1\right)\text{LFL} = \left(\frac{1}{0.194} - 1\right)3.44 = 14.3\%$$

Similarly, consider $x_F = 0.46$ for the UFL, and repeat the same procedure.

$$v_{\text{air},0}^{\text{UFL}} = \frac{1}{4.76}\left(\frac{100}{18.70} - 1\right) = 0.9133$$

$$v_{\text{air},d}^{\text{UFL}} = 0.460 \times 0.9133 + 0.540 \times 0.2636 = 0.5624$$

$$\text{UFL}_d = \frac{1 \times 0.460}{1 + 4.76 \times 0.5624} = 12.51\%$$

$$P_d = \left(\frac{1}{x_F} - 1\right)\text{UFL} = \left(\frac{1}{0.460} - 1\right)12.51 = 14.69\%$$

The LFL and UFL calculated are close to those obtained by Zabetakis for 15% N_2 dilution. Note that the method needs to assume a value of x_F, and the diluent concentration is calculated afterwards. The procedure for determining the FIP is similar to that demonstrated for the UFL without dilution in Example 8.2. In addition, Mendiburu et al. [133] provided a sample calculation in the supplementary material of their work. Figure 8.11 shows results for different concentrations of N_2. In Fig. 8.11 the diluent mole fraction (x_d) corresponds to the fuel–diluent mixture, while the diluent concentration (P_d) corresponds to the fuel–air–diluent mixture. It is noticed that the estimation method works accurately for ethanol.

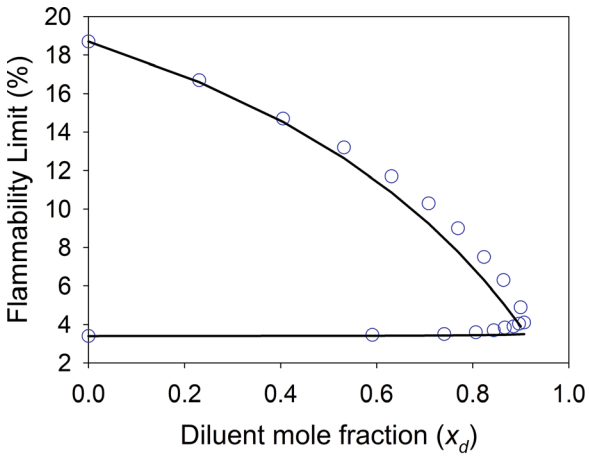


Fig. 8.11 Effect of N_2 dilution on the flammability limits of ethanol in air, where $x_F = (1 - x_d)$ and $P_d = \text{FL}(1/x_F - 1)$. Experimental data by Zabetakis [116] shown as blue circles

The accuracy of the estimation methods presented in this section was tested in the original work, showing AAREs of 4.02% and 3.91% for the LFL and UFL, respectively, with N_2 dilution. In the case of CO_2 dilution, the method showed AAREs of 7.01% and 5.57% for the LFL and UFL, respectively.

8.5 Closure

The flammability limits were discussed in this chapter. These limits are of fundamental importance for premixed flame theory and of practical importance for industrial applications. The presentation started by analyzing the conditions for the existence of the flammability limits, which are related to a minimum laminar flame velocity. Moreover, the chapter presented the derivation of the steady-state one-dimensional theory with one-step chemical kinetics reaction, developed by Spalding, for flammability limits and flame quenching. Spalding's theory introduces a critical condition that defines the flammability limits. The theoretical approach that identifies the flammability limits as a direct competition between chain branching and chain termination reactions was also presented. Afterwards, the chapter discussed a more complex theoretical approach based on detailed chemical kinetic mechanisms and heat losses by thermal radiation. Moreover, an analysis was provided to account for the effects of initial parameters, such as temperature and pressure, and the effects of adding a diluent or another fuel to the flammable mixture. Finally, some estimation methods were presented.

Appendix to the Book: Operations with Vectors and Second-Order Tensors

The material presented in this appendix is intended to complement the derivations of the conservation equations given in Chap. 3. Some of the key operations needed for the derivation of the conservation equations are developed in Cartesian, cylindrical, and spherical coordinate systems, with operations with more extensive procedures being performed only for spherical coordinates. The material was prepared by following the textbooks of Bird et al. [37], Schobeiri [142], Papanastasiou et al. [42], and Tang [143].

Orthogonal Unit Vectors

The unit vectors, denoted as $\vec{e}_1$, $\vec{e}_2$, $\vec{e}_3$, are called an orthogonal set when they fulfill the following relations with respect to the scalar and vector products.

$$\begin{aligned}
&\vec{e}_1 \cdot \vec{e}_1 = \vec{e}_2 \cdot \vec{e}_2 = \vec{e}_3 \cdot \vec{e}_3 = 1; \quad && \vec{e}_1 \cdot \vec{e}_2 = \vec{e}_1 \cdot \vec{e}_3 = \vec{e}_2 \cdot \vec{e}_3 = 0 \\
&\vec{e}_1 \times \vec{e}_2 = \vec{e}_3 && \vec{e}_2 \times \vec{e}_1 = -\vec{e}_3 \\
&\vec{e}_2 \times \vec{e}_3 = \vec{e}_1 && \vec{e}_3 \times \vec{e}_2 = -\vec{e}_1 \\
&\vec{e}_3 \times \vec{e}_1 = \vec{e}_2 && \vec{e}_1 \times \vec{e}_3 = -\vec{e}_2
\end{aligned} \tag{A.1}$$

The relations shown in Eq. (A.1) are true for any orthogonal set of unit vectors, in particular for Cartesian, cylindrical, and spherical coordinate systems.

Vectors and Second-Order Tensors

Any vector $\vec{V}$ is also a first-order tensor and can be represented in terms of its components, as shown below.

$$\vec{V} = u_1 \vec{e}_1 + u_2 \vec{e}_2 + u_3 \vec{e}_3 \tag{A.2}$$

The scalar components u_i define the magnitude of the vector at the respective coordinate, and the orthogonal unit vectors $\vec{e}_i$ specify the direction of the respective

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coordinate. The unit dyadic product is defined as the multiplication of two-unit vectors without using the scalar or cross product, as shown below.

$$\text{Unitary dyadic product} = \vec{e}_i \vec{e}_j \tag{A.3}$$

It is important to realize that $\vec{e}_i \vec{e}_j \neq \vec{e}_j \vec{e}_i$. A second-order tensor can be generated by multiplying two vectors (without being a scalar or vector product). Although this is possible, not all second-order tensors can be obtained by the product of two vectors. A second-order tensor has nine scalar components (Φ_{ij}), and each of them multiplies to a unitary dyadic product ($\vec{e}_i \vec{e}_j$), as shown below.

$$\begin{aligned} \overleftrightarrow{\Phi} = & \Phi_{11} \vec{e}_1 \vec{e}_1 + \Phi_{12} \vec{e}_1 \vec{e}_2 + \Phi_{13} \vec{e}_1 \vec{e}_3 \\ & + \Phi_{21} \vec{e}_2 \vec{e}_1 + \Phi_{22} \vec{e}_2 \vec{e}_2 + \Phi_{23} \vec{e}_2 \vec{e}_3 \\ & + \Phi_{31} \vec{e}_3 \vec{e}_1 + \Phi_{32} \vec{e}_3 \vec{e}_2 + \Phi_{33} \vec{e}_3 \vec{e}_3 \end{aligned} \tag{A.4}$$

Therefore, a second-order tensor is represented by its components Φ_{ij}, in which the subscript "*i*" represents the direction of the normal to the surface on which the tensor component is being applied and the superscript "*j*" represents the direction of the component. A second-order tensor is symmetric when $\Phi_{ij} = \Phi_{ji}$. Therefore, a second-order tensor is antisymmetric when $\Phi_{ij} \neq \Phi_{ji}$. In the context of this book, the stress tensor on a fluid element is a second-order tensor and it is considered symmetric.

Addition and Subtraction of Two Tensors

The sum of two tensors follows the same logic as the sum of two vectors, that is, the components with the same dyadic products add together.

$$\begin{aligned} \overleftrightarrow{\Phi} + \overleftrightarrow{\Gamma} = & (\Phi_{11} + \Gamma_{11}) \vec{e}_1 \vec{e}_1 + (\Phi_{12} + \Gamma_{12}) \vec{e}_1 \vec{e}_2 + (\Phi_{13} + \Gamma_{13}) \vec{e}_1 \vec{e}_3 \\ & + (\Phi_{21} + \Gamma_{21}) \vec{e}_2 \vec{e}_1 + (\Phi_{22} + \Gamma_{22}) \vec{e}_2 \vec{e}_2 + (\Phi_{23} + \Gamma_{23}) \vec{e}_2 \vec{e}_3 \\ & + (\Phi_{31} + \Gamma_{31}) \vec{e}_3 \vec{e}_1 + (\Phi_{32} + \Gamma_{32}) \vec{e}_3 \vec{e}_2 + (\Phi_{33} + \Gamma_{33}) \vec{e}_3 \vec{e}_3 \end{aligned} \tag{A.5}$$

When it is desired to subtract the tensor $\overleftrightarrow{\Phi}$ from the tensor $\overleftrightarrow{\Gamma}$, it will be sufficient to write $\overleftrightarrow{\Phi} - \overleftrightarrow{\Gamma} = \overleftrightarrow{\Phi} + \left(-\overleftrightarrow{\Gamma}\right)$.

The Product of a Tensor and a Scalar

Let s be a scalar and $\overleftrightarrow{\Phi}$ be a second-order tensor, the product $s\overleftrightarrow{\Phi}$ is obtained by multiplying each component of the tensor by the scalar, as shown below.

$$s\overleftrightarrow{\Phi} = \begin{array}{r} s\Phi_{11}\,\vec{e}_1\,\vec{e}_1 + s\Phi_{12}\,\vec{e}_1\,\vec{e}_2 + s\Phi_{13}\,\vec{e}_1\,\vec{e}_3 \\ +s\Phi_{21}\,\vec{e}_2\,\vec{e}_1 + s\Phi_{22}\,\vec{e}_2\,\vec{e}_2 + s\Phi_{23}\,\vec{e}_2\,\vec{e}_3 \\ +s\Phi_{31}\,\vec{e}_3\,\vec{e}_1 + s\Phi_{32}\,\vec{e}_3\,\vec{e}_2 + s\Phi_{33}\,\vec{e}_3\,\vec{e}_3 \end{array} \tag{A.6}$$

The Double Dot Product or Scalar Product of Two Tensors

The double dot product of two second-order tensors produces a scalar. Equivalently, the dot product of two vectors also produces a scalar. The operation is represented below.

$$\overleftrightarrow{\Phi} : \overleftrightarrow{\Gamma} = \begin{bmatrix} \Phi_{11}\,\vec{e}_1\,\vec{e}_1 + \Phi_{12}\,\vec{e}_1\,\vec{e}_2 + \Phi_{13}\,\vec{e}_1\,\vec{e}_3 \\ +\Phi_{21}\,\vec{e}_2\,\vec{e}_1 + \Phi_{22}\,\vec{e}_2\,\vec{e}_2 + \Phi_{23}\,\vec{e}_2\,\vec{e}_3 \\ +\Phi_{31}\,\vec{e}_3\,\vec{e}_1 + \Phi_{32}\,\vec{e}_3\,\vec{e}_2 + \Phi_{33}\,\vec{e}_3\,\vec{e}_3 \end{bmatrix} : \begin{bmatrix} \Gamma_{11}\,\vec{e}_1\,\vec{e}_1 + \Gamma_{12}\,\vec{e}_1\,\vec{e}_2 + \Gamma_{13}\,\vec{e}_1\,\vec{e}_3 \\ +\Gamma_{21}\,\vec{e}_2\,\vec{e}_1 + \Gamma_{22}\,\vec{e}_2\,\vec{e}_2 + \Gamma_{23}\,\vec{e}_2\,\vec{e}_3 \\ +\Gamma_{31}\,\vec{e}_3\,\vec{e}_1 + \Gamma_{32}\,\vec{e}_3\,\vec{e}_2 + \Gamma_{33}\,\vec{e}_3\,\vec{e}_3 \end{bmatrix}$$

The double dot product rule states that, for $i = k$ and $j = l$, the following is obtained $\vec{e}_i\,\vec{e}_j : \vec{e}_k\,\vec{e}_l = 1$. Otherwise, $\vec{e}_i\,\vec{e}_j : \vec{e}_k\,\vec{e}_l = 0$. Applying this rule, the double dot product of tensors $(\overleftrightarrow{\Phi} : \overleftrightarrow{\Gamma})$ is given by Eq. (A.7).

$$\overleftrightarrow{\Phi} : \overleftrightarrow{\Gamma} = \begin{array}{r} \Phi_{11}\Gamma_{11} + \Phi_{12}\Gamma_{12} + \Phi_{13}\Gamma_{13} \\ +\Phi_{21}\Gamma_{21} + \Phi_{22}\Gamma_{22} + \Phi_{23}\Gamma_{23} \\ +\Phi_{31}\Gamma_{31} + \Phi_{32}\Gamma_{32} + \Phi_{33}\Gamma_{33} \end{array} \tag{A.7}$$

The Dot Product of a Tensor and a Vector

The dot product of a second-order tensor with a vector generates a vector, that is, it decreases the order of the initial tensor from "two" to "one."

$$\overleftrightarrow{\Phi} \cdot \vec{V} = \begin{bmatrix} \Phi_{11}\,\vec{e}_1\,\vec{e}_1 + \Phi_{12}\,\vec{e}_1\,\vec{e}_2 + \Phi_{13}\,\vec{e}_1\,\vec{e}_3 \\ +\Phi_{21}\,\vec{e}_2\,\vec{e}_1 + \Phi_{22}\,\vec{e}_2\,\vec{e}_2 + \Phi_{23}\,\vec{e}_2\,\vec{e}_3 \\ +\Phi_{31}\,\vec{e}_3\,\vec{e}_1 + \Phi_{32}\,\vec{e}_3\,\vec{e}_2 + \Phi_{33}\,\vec{e}_3\,\vec{e}_3 \end{bmatrix} \cdot \left[u_1\,\vec{e}_1 + u_2\,\vec{e}_2 + u_3\,\vec{e}_3 \right]$$

Such operation establishes products of the type $\Phi_{ij}u_k\left(\vec{e}_i\,\vec{e}_j \cdot \vec{e}_k\right) = \Phi_{ij}u_k\,\vec{e}_i$ when $j = k$, or $\Phi_{ij}u_k\left(\vec{e}_i\,\vec{e}_j \cdot \vec{e}_k\right) = 0$, when $j \neq k$.

$$\overleftrightarrow{\Phi}\cdot\vec{V} = \begin{array}{r}(\Phi_{11}u_1 + \Phi_{12}u_2 + \Phi_{13}u_3)\,\vec{e}_1 \\ +(\Phi_{21}u_1 + \Phi_{22}u_2 + \Phi_{23}u_3)\,\vec{e}_2 \\ +(\Phi_{31}u_1 + \Phi_{32}u_2 + \Phi_{33}u_3)\,\vec{e}_3\end{array} \tag{A.8}$$

On the other hand, if the order of the dot product was reversed

$$\vec{V}\cdot\overleftrightarrow{\Phi} = \begin{array}{r}(\Phi_{11}u_1 + \Phi_{21}u_2 + \Phi_{31}u_3)\,\vec{e}_1 \\ +(\Phi_{12}u_1 + \Phi_{22}u_2 + \Phi_{32}u_3)\,\vec{e}_2 \\ +(\Phi_{13}u_1 + \Phi_{23}u_2 + \Phi_{33}u_3)\,\vec{e}_3\end{array} \tag{A.9}$$

Analyzing the Eqs. (A.8) and (A.9), it is evident that if the tensor is antisymmetric, then $\overleftrightarrow{\Phi}\cdot\vec{V} \neq \vec{V}\cdot\overleftrightarrow{\Phi}$. If the tensor is symmetric, then the equality, $\overleftrightarrow{\Phi}\cdot\vec{V} = \vec{V}\cdot\overleftrightarrow{\Phi}$, will be true. The operation of the dot product of a tensor and a vector is of great use in fluid mechanics.

Point Located with Cylindrical and Spherical Coordinates

The cylindrical and spherical coordinate systems are of particular interest. Figure A.1 shows point P, located in space using the cylindrical and spherical coordinate systems. The information shown in Fig. A.1 is enough to intuitively understand how to locate a point in each of these coordinate systems. In the cylindrical coordinate system, it is necessary to know two lengths denoted by ρ and z, and an angle denoted by φ. In the spherical coordinate system, it is necessary to know a length denoted by r and two angles denoted by θ and φ.

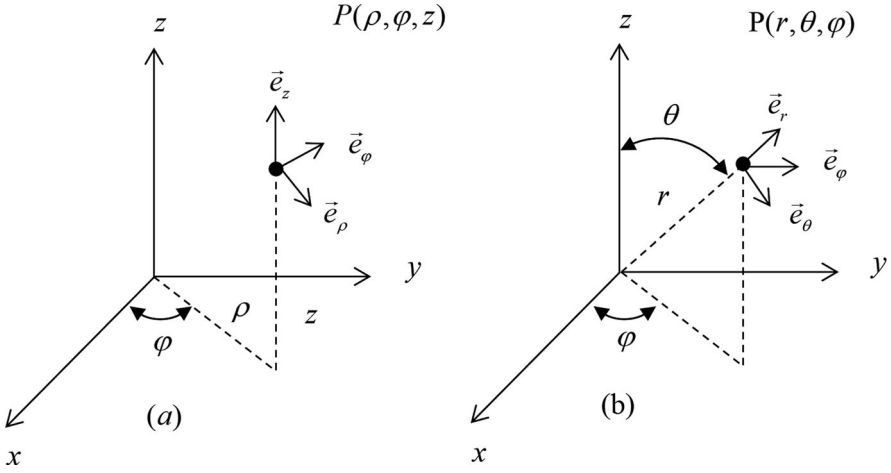


Fig. A.1 Coordinates of point P in: (**a**) cylindrical coordinate system and (**b**) spherical coordinate system

Orthogonal Unit Vectors and Their Derivatives

The orthogonal unit vectors in the cylindrical and spherical coordinate systems were depicted in Fig. A.1. Considering these orthogonal unit vectors as a function of the vectors of the Cartesian coordinate system, the expressions given in Eqs. (A.10) and (A.11) are written.

Orthogonal Unit Vectors for the Cylindrical Coordinate System

$$\begin{aligned}\vec{e}_{\rho} &= \cos\varphi\,\vec{e}_{x} + \sin\varphi\,\vec{e}_{y} \\ \vec{e}_{\varphi} &= -\sin\varphi\,\vec{e}_{x} + \cos\varphi\,\vec{e}_{y} \\ \vec{e}_{z} &= \vec{e}_{z}\end{aligned} \tag{A.10}$$

Orthogonal Unit Vectors for the Spherical Coordinate System

$$\begin{aligned}\vec{e}_{r} &= \sin\theta\cos\varphi\,\vec{e}_{x} + \sin\theta\sin\varphi\,\vec{e}_{y} + \cos\theta\,\vec{e}_{z} \\ \vec{e}_{\theta} &= \cos\theta\cos\varphi\,\vec{e}_{x} + \cos\theta\sin\varphi\,\vec{e}_{y} - \sin\theta\,\vec{e}_{z} \\ \vec{e}_{\varphi} &= -\sin\varphi\,\vec{e}_{x} + \cos\varphi\,\vec{e}_{y}\end{aligned} \tag{A.11}$$

In the Cartesian coordinate system, the orthogonal unit vectors are constant. On the other hand, in the cylindrical coordinate system, only one of the orthogonal unit vectors ($\vec{e}_{z}$) is constant with respect to the system's coordinates. Furthermore, in the spherical coordinate system, none of the orthogonal unit vectors are constant with respect to the system's coordinates. Expressing the unit vectors in terms of the Cartesian coordinates and differentiating by applying the chain rule, the following equations are obtained.

For the Cartesian Coordinate System

$$\begin{aligned}\frac{\partial\vec{e}_{x}}{\partial x} &= \frac{\partial\vec{e}_{x}}{\partial y} = \frac{\partial\vec{e}_{x}}{\partial z} = 0 \\ \frac{\partial\vec{e}_{y}}{\partial x} &= \frac{\partial\vec{e}_{y}}{\partial y} = \frac{\partial\vec{e}_{y}}{\partial z} = 0 \\ \frac{\partial\vec{e}_{z}}{\partial x} &= \frac{\partial\vec{e}_{z}}{\partial y} = \frac{\partial\vec{e}_{z}}{\partial z} = 0\end{aligned} \tag{A.12}$$

For the Cylindrical Coordinate System

$$\begin{aligned}
&\frac{\partial \vec{e}_{\rho}}{\partial \varphi} = \vec{e}_{\varphi}; \quad \frac{\partial \vec{e}_{\varphi}}{\partial \varphi} = -\vec{e}_{\rho} \\
&\frac{\partial \vec{e}_{\rho}}{\partial \rho} = \frac{\partial \vec{e}_{\rho}}{\partial z} = \frac{\partial \vec{e}_{\varphi}}{\partial \rho} = \frac{\partial \vec{e}_{\varphi}}{\partial z} = \frac{\partial \vec{e}_{z}}{\partial \rho} = \frac{\partial \vec{e}_{z}}{\partial \varphi} = \frac{\partial \vec{e}_{z}}{\partial z} = 0
\end{aligned} \tag{A.13}$$

For the Spherical Coordinate System

$$\begin{aligned}
&\frac{\partial \vec{e}_{r}}{\partial \theta} = \vec{e}_{\theta}; \qquad \frac{\partial \vec{e}_{r}}{\partial \varphi} = \sin\theta \vec{e}_{\varphi} \\
&\frac{\partial \vec{e}_{\theta}}{\partial \theta} = -\vec{e}_{r}; \quad \frac{\partial \vec{e}_{\theta}}{\partial \varphi} = \cos\theta \vec{e}_{\varphi} \\
&\frac{\partial \vec{e}_{\varphi}}{\partial \varphi} = -\left(\sin\theta \vec{e}_{r} + \cos\theta \vec{e}_{\theta}\right) \\
&\frac{\partial \vec{e}_{r}}{\partial r} = \frac{\partial \vec{e}_{\theta}}{\partial r} = \frac{\partial \vec{e}_{\varphi}}{\partial r} = \frac{\partial \vec{e}_{\varphi}}{\partial \theta} = 0
\end{aligned} \tag{A.14}$$

The Nabla Operator

The nabla operator $\left(\vec{\nabla}\right)$ is given below in Cartesian, cylindrical, and spherical coordinates.

For the Cartesian Coordinate System

$$\nabla = \vec{e}_{x}\frac{\partial}{\partial x} + \vec{e}_{y}\frac{\partial}{\partial y} + \vec{e}_{z}\frac{\partial}{\partial z} \tag{A.15}$$

For the Cylindrical Coordinate System

$$\nabla = \vec{e}_{\rho}\frac{\partial}{\partial \rho} + \vec{e}_{\varphi}\frac{1}{r}\frac{\partial}{\partial \varphi} + \vec{e}_{z}\frac{\partial}{\partial z} \tag{A.16}$$

For the Spherical Coordinate System

$$\nabla = \vec{e}_r \frac{\partial}{\partial r} + \vec{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \vec{e}_\varphi \frac{1}{r \sin\theta} \frac{\partial}{\partial \varphi} \tag{A.17}$$

The Gradient of a Vector

The gradient of a function can be interpreted as a direct multiplication with the nabla operator and the application of the derivatives imposed by nabla. Moreover, the function can be a scalar, vector, or tensor. Since the nabla operatior is a vector, then, the gradient of a scalar will be a vector, the gradient of a vector will be a second-order tensor, and the gradient of a second-order tensor will be a third-order tensor. In this section, the focus is on determining the gradient of a vector in different coordinate systems.

For the Cartesian Coordinate System

$$\nabla \vec{V} = \vec{e}_x \frac{\partial \vec{V}}{\partial x} + \vec{e}_y \frac{\partial \vec{V}}{\partial y} + \vec{e}_z \frac{\partial \vec{V}}{\partial z}$$

Each of these partial derivatives will be performed separately using the expression for $\vec{V}$ in terms of its components. The partial derivative in x is taken first.

$$\vec{e}_x \frac{\partial \vec{V}}{\partial x} = \vec{e}_x \frac{\partial}{\partial x}\left(u_x \vec{e}_x + u_y \vec{e}_y + u_z \vec{e}_z\right) = \vec{e}_x \vec{e}_x \frac{\partial u_x}{\partial x} + u_x \vec{e}_x \frac{\cancel{\partial \vec{e}_x}}{\cancel{\partial x}} + \vec{e}_x \vec{e}_y \frac{\partial u_y}{\partial x} + u_y \vec{e}_x \frac{\cancel{\partial \vec{e}_y}}{\cancel{\partial x}} + \vec{e}_x \vec{e}_z \frac{\partial u_z}{\partial x} + u_z \vec{e}_x \frac{\cancel{\partial \vec{e}_z}}{\cancel{\partial x}}$$

In the previous expression, it can be seen that the fact that the orthogonal unit vectors are constant in the Cartesian coordinate system simplifies the operation. Performing the same operation for the partial derivatives in y and z, the following is obtained.

$$\nabla \vec{V} = \vec{e}_x \vec{e}_x \frac{\partial u_x}{\partial x} + \vec{e}_y \vec{e}_y \frac{\partial u_y}{\partial y} + \vec{e}_z \vec{e}_z \frac{\partial u_z}{\partial z} + \vec{e}_x \vec{e}_y \frac{\partial u_y}{\partial x} + \vec{e}_x \vec{e}_z \frac{\partial u_z}{\partial x} + \vec{e}_y \vec{e}_x \frac{\partial u_x}{\partial y} + \vec{e}_y \vec{e}_z \frac{\partial u_z}{\partial y} + \vec{e}_z \vec{e}_x \frac{\partial u_x}{\partial z} + \vec{e}_z \vec{e}_y \frac{\partial u_y}{\partial z} \tag{A.18}$$

For the Cylindrical Coordinate System

$$\nabla \vec{V} = \vec{e}_\rho \frac{\partial \vec{V}}{\partial \rho} + \vec{e}_\varphi \frac{1}{r} \frac{\partial \vec{V}}{\partial \varphi} + \vec{e}_z \frac{\partial \vec{V}}{\partial z}$$

Performing the partial derivatives separately,

$$\vec{e}_\rho \frac{\partial \vec{V}}{\partial \rho} = \vec{e}_\rho \frac{\partial}{\partial x}\left(u_\rho \vec{e}_\rho + u_\varphi \vec{e}_\varphi + u_z \vec{e}_z\right) = \vec{e}_\rho \vec{e}_\rho \frac{\partial u_\rho}{\partial \rho} + \vec{e}_\rho u_\rho \cancel{\frac{\partial \vec{e}_\rho}{\partial \rho}}$$
$$+ \vec{e}_\rho \vec{e}_\varphi \frac{\partial u_\varphi}{\partial \rho} + \vec{e}_\rho u_\varphi \cancel{\frac{\partial \vec{e}_\varphi}{\partial \rho}} + \vec{e}_\rho \vec{e}_z \frac{\partial u_z}{\partial \rho} + \vec{e}_\rho u_z \cancel{\frac{\partial \vec{e}_z}{\partial \rho}}$$

Then,

$$\vec{e}_\rho \frac{\partial \vec{V}}{\partial \rho} = \vec{e}_\rho \vec{e}_\rho \frac{\partial u_\rho}{\partial \rho} + \vec{e}_\rho \vec{e}_\varphi \frac{\partial u_\varphi}{\partial \rho} + \vec{e}_\rho \vec{e}_z \frac{\partial u_z}{\partial \rho} \tag{a}$$

$$\vec{e}_\varphi \frac{1}{r} \frac{\partial \vec{V}}{\partial \varphi} = \vec{e}_\varphi \frac{1}{r} \frac{\partial}{\partial \varphi}\left(u_\rho \vec{e}_\rho + u_\varphi \vec{e}_\varphi + u_z \vec{e}_z\right) = \vec{e}_\varphi \vec{e}_\rho \frac{1}{r} \frac{\partial u_\rho}{\partial \varphi}$$
$$+ \vec{e}_\varphi u_\rho \frac{1}{r} \underbrace{\frac{\partial \vec{e}_\rho}{\partial \varphi}}_{\vec{e}_\varphi} + \vec{e}_\varphi \vec{e}_\varphi \frac{1}{r} \frac{\partial u_\varphi}{\partial \varphi} + \vec{e}_\varphi u_\varphi \frac{1}{r} \underbrace{\frac{\partial \vec{e}_\varphi}{\partial \varphi}}_{-\vec{e}_\rho}$$
$$+ \vec{e}_\varphi \vec{e}_z \frac{1}{r} \frac{\partial u_z}{\partial \varphi} + \vec{e}_\varphi u_z \frac{1}{r} \cancel{\frac{\partial \vec{e}_z}{\partial \varphi}}$$

Then,

$$\vec{e}_\varphi \frac{1}{r} \frac{\partial \vec{V}}{\partial \varphi} = \vec{e}_\varphi \vec{e}_\rho \frac{1}{r} \frac{\partial u_\rho}{\partial \varphi} + \vec{e}_\varphi \vec{e}_\varphi \frac{u_\rho}{r} + \vec{e}_\varphi \vec{e}_\varphi \frac{1}{r} \frac{\partial u_\varphi}{\partial \varphi} - \vec{e}_\varphi \vec{e}_\rho \frac{u_\varphi}{r} + \vec{e}_\varphi \vec{e}_z \frac{1}{r} \frac{\partial u_z}{\partial \varphi} \tag{b}$$

$$\vec{e}_z \frac{\partial \vec{V}}{\partial z} = \vec{e}_z \frac{\partial}{\partial z}\left(u_\rho \vec{e}_\rho + u_\varphi \vec{e}_\varphi + u_z \vec{e}_z\right) = \vec{e}_z \vec{e}_\rho \frac{\partial u_\rho}{\partial z} + \vec{e}_z u_\rho \cancel{\frac{\partial \vec{e}_\rho}{\partial z}}$$
$$+ \vec{e}_z \vec{e}_\varphi \frac{\partial u_\varphi}{\partial z} + \vec{e}_z u_\varphi \cancel{\frac{\partial \vec{e}_\varphi}{\partial z}} + \vec{e}_z \vec{e}_z \frac{\partial u_z}{\partial z} + \vec{e}_z u_z \cancel{\frac{\partial \vec{e}_z}{\partial z}}$$

Then,

$$\vec{e}_z \frac{\partial \vec{V}}{\partial z} = \vec{e}_z \vec{e}_\rho \frac{\partial u_\rho}{\partial z} + \vec{e}_z \vec{e}_\varphi \frac{\partial u_\varphi}{\partial z} + \vec{e}_z \vec{e}_z \frac{\partial u_z}{\partial z} \tag{c}$$

Using the previous results,

$$\begin{aligned}\nabla \vec{V} = {} & \vec{e}_\rho \vec{e}_\rho \frac{\partial u_\rho}{\partial \rho} + \vec{e}_\varphi \vec{e}_\varphi \left(\frac{1}{r} \frac{\partial u_\varphi}{\partial \varphi} + \frac{u_\rho}{r} \right) + \vec{e}_z \vec{e}_z \frac{\partial u_z}{\partial z} + \vec{e}_\rho \vec{e}_\varphi \frac{\partial u_\varphi}{\partial \rho} \\ & + \vec{e}_\rho \vec{e}_z \frac{\partial u_z}{\partial \rho} + \vec{e}_\varphi \vec{e}_\rho \left(\frac{1}{r} \frac{\partial u_\rho}{\partial \varphi} - \frac{u_\varphi}{r} \right) + \vec{e}_\varphi \vec{e}_z \frac{1}{r} \frac{\partial u_z}{\partial \varphi} \\ & + \vec{e}_z \vec{e}_\rho \frac{\partial u_\rho}{\partial z} + \vec{e}_z \vec{e}_\varphi \frac{\partial u_\varphi}{\partial z}\end{aligned} \tag{A.19}$$

For the Spherical Coordinate System

$$\nabla \vec{V} = \vec{e}_r \frac{\partial \vec{V}}{\partial r} + \vec{e}_\theta \frac{1}{r} \frac{\partial \vec{V}}{\partial \theta} + \vec{e}_\varphi \frac{1}{r \sin\theta} \frac{\partial \vec{V}}{\partial \varphi}$$

Performing the partial derivatives separately

$$\begin{aligned}\vec{e}_r \frac{\partial \vec{V}}{\partial r} = {} & \vec{e}_r \frac{\partial}{\partial r} \left(u_r \vec{e}_r + u_\theta \vec{e}_\theta + u_\varphi \vec{e}_\varphi \right) = \vec{e}_r \vec{e}_r \frac{\partial u_r}{\partial r} + \vec{e}_r u_r \frac{\partial \cancel{\vec{e}_r}}{\partial r} \\ & + \vec{e}_r \vec{e}_\theta \frac{\partial u_\theta}{\partial r} + \vec{e}_r u_\theta \frac{\partial \cancel{\vec{e}_\theta}}{\partial r} + \vec{e}_r \vec{e}_\varphi \frac{\partial u_\varphi}{\partial r} + \vec{e}_r u_\varphi \frac{\partial \cancel{\vec{e}_\varphi}}{\partial r}\end{aligned}$$

Then,

$$\vec{e}_r \frac{\partial \vec{V}}{\partial r} = \vec{e}_r \vec{e}_r \frac{\partial u_r}{\partial r} + \vec{e}_r \vec{e}_\theta \frac{\partial u_\theta}{\partial r} + \vec{e}_r \vec{e}_\varphi \frac{\partial u_\varphi}{\partial r} \tag{a}$$

$$\begin{aligned}\vec{e}_\theta \frac{1}{r} \frac{\partial \vec{V}}{\partial \theta} = {} & \vec{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} \left(u_r \vec{e}_r + u_\theta \vec{e}_\theta + u_\varphi \vec{e}_\varphi \right) = \vec{e}_\theta \vec{e}_r \frac{1}{r} \frac{\partial u_r}{\partial \theta} \\ & + \vec{e}_\theta u_r \frac{1}{r} \underbrace{\frac{\partial \vec{e}_r}{\partial \theta}}_{\vec{e}_\theta} + \vec{e}_\theta \vec{e}_\theta \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \vec{e}_\theta u_\theta \frac{1}{r} \underbrace{\frac{\partial \vec{e}_\theta}{\partial \theta}}_{-\vec{e}_r} \\ & + \vec{e}_\theta \vec{e}_\varphi \frac{1}{r} \frac{\partial u_\varphi}{\partial \theta} + \vec{e}_\theta u_\varphi \frac{1}{r} \frac{\partial \cancel{\vec{e}_\varphi}}{\partial \theta}\end{aligned}$$

Then,

$$\vec{e}_\theta \frac{1}{r}\frac{\partial \vec{V}}{\partial \theta} = \vec{e}_\theta \vec{e}_r \left(\frac{1}{r}\frac{\partial u_r}{\partial \theta} - \frac{u_\theta}{r}\right) + \vec{e}_\theta \vec{e}_\theta \left(\frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r}\right) + \vec{e}_\theta \vec{e}_\varphi \frac{1}{r}\frac{\partial u_\varphi}{\partial \theta} \quad \text{(b)}$$

$$\vec{e}_\varphi \frac{1}{r\sin\theta}\frac{\partial \vec{V}}{\partial \varphi} = \vec{e}_\varphi \frac{1}{r\sin\theta}\frac{\partial}{\partial \varphi}\left(u_r \vec{e}_r + u_\theta \vec{e}_\theta + u_\varphi \vec{e}_\varphi\right)$$

$$= \frac{1}{r\sin\theta}\left(\vec{e}_\varphi \vec{e}_r \frac{\partial u_r}{\partial \varphi} + \vec{e}_\varphi u_r \underbrace{\frac{\partial \vec{e}_r}{\partial \varphi}}_{=\sin\theta \vec{e}_\varphi} + \vec{e}_\varphi \vec{e}_\theta \frac{\partial u_\theta}{\partial \varphi}\right.$$

$$\left. + \vec{e}_\varphi u_\theta \underbrace{\frac{\partial \vec{e}_\theta}{\partial \varphi}}_{=\cos\theta \vec{e}_\varphi} + \vec{e}_\varphi \vec{e}_\varphi \frac{\partial u_\varphi}{\partial \varphi} + \vec{e}_\varphi u_\varphi \underbrace{\frac{\partial \vec{e}_\varphi}{\partial \varphi}}_{\begin{array}{l}-\sin\theta \vec{e}_r \\ -\cos\theta \vec{e}_\theta\end{array}}\right)$$

Then,

$$\vec{e}_\varphi \frac{1}{r\sin\theta}\frac{\partial \vec{V}}{\partial \varphi} = \vec{e}_\varphi \vec{e}_r \left(\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial \varphi} - \frac{u_\varphi}{r}\right) + \vec{e}_\varphi \vec{e}_\theta \left(\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial \varphi} - \frac{u_\varphi}{r}\cot\theta\right)$$
$$+ \vec{e}_\varphi \vec{e}_\varphi \left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial \varphi} + \frac{u_r}{r} + \frac{u_\theta}{r}\cot\theta\right) \quad \text{(c)}$$

Using the previous results,

$$\nabla \vec{V} = \vec{e}_r \vec{e}_r \frac{\partial u_r}{\partial r} + \vec{e}_\theta \vec{e}_\theta \left(\frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r}\right) + \vec{e}_\varphi \vec{e}_\varphi \left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial \varphi} + \frac{u_r}{r} + \frac{u_\theta}{r}\cot\theta\right)$$
$$+ \vec{e}_r \vec{e}_\theta \frac{\partial u_\theta}{\partial r} + \vec{e}_r \vec{e}_\varphi \frac{\partial u_\varphi}{\partial r} + \vec{e}_\theta \vec{e}_r \left(\frac{1}{r}\frac{\partial u_r}{\partial \theta} - \frac{u_\theta}{r}\right) + \vec{e}_\theta \vec{e}_\varphi \frac{1}{r}\frac{\partial u_\varphi}{\partial \theta}$$
$$+ \vec{e}_\varphi \vec{e}_r \left(\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial \varphi} - \frac{u_\varphi}{r}\right) + \vec{e}_\varphi \vec{e}_\theta \left(\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial \varphi} - \frac{u_\varphi}{r}\cot\theta\right) \quad \text{(A.20)}$$

The Transposed Tensor of a Second-Order Tensor

The previous result will be used to define the transposed tensor as the tensor resulting from the operation in which the rows are exchanged for the columns, in the matrix formed by the scalar components of the tensor, keeping the unitary dyadic products fixed. The transposed tensor of the velocity gradient in spherical coordinates will be obtained. Equation (A.20) is considered, the transposed tensor of the velocity gradient is shown below.

$$\begin{aligned}\left(\nabla\vec{V}\right)^T &= \vec{e}_r\vec{e}_r\frac{\partial u_r}{\partial r}+\vec{e}_r\vec{e}_\theta\left(\frac{1}{r}\frac{\partial u_r}{\partial\theta}-\frac{u_\theta}{r}\right)+\vec{e}_r\vec{e}_\varphi\left(\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial\varphi}-\frac{u_\varphi}{r}\right)\\&+\vec{e}_\theta\vec{e}_r\frac{\partial u_\theta}{\partial r}+\vec{e}_\theta\vec{e}_\theta\left(\frac{1}{r}\frac{\partial u_\theta}{\partial\theta}+\frac{u_r}{r}\right)+\vec{e}_\theta\vec{e}_\varphi\left(\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial\varphi}-\frac{u_\varphi}{r}\cot\theta\right)\\&+\vec{e}_\varphi\vec{e}_r\frac{\partial u_\varphi}{\partial r}+\vec{e}_\varphi\vec{e}_\theta\frac{1}{r}\frac{\partial u_\varphi}{\partial\theta}+\vec{e}_\varphi\vec{e}_\varphi\left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}+\frac{u_r}{r}+\frac{u_\theta}{r}\cot\theta\right)\end{aligned} \tag{A.21}$$

Now the sum of the velocity gradient tensor and the transposed velocity gradient tensor will be determined.

$$\begin{aligned}\nabla\vec{V}+\left(\nabla\vec{V}\right)^T &= \vec{e}_r\vec{e}_r2\left(\frac{\partial u_r}{\partial r}\right)+\vec{e}_r\vec{e}_\theta\left(\frac{1}{r}\frac{\partial u_r}{\partial\theta}+\frac{\partial u_\theta}{\partial r}-\frac{u_\theta}{r}\right)\\&+\vec{e}_r\vec{e}_\varphi\left(\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial\varphi}+\frac{\partial u_\varphi}{\partial r}-\frac{u_\varphi}{r}\right)\\&+\vec{e}_\theta\vec{e}_r\left(\frac{1}{r}\frac{\partial u_r}{\partial\theta}+\frac{\partial u_\theta}{\partial r}-\frac{u_\theta}{r}\right)+\vec{e}_\theta\vec{e}_\theta2\left(\frac{1}{r}\frac{\partial u_\theta}{\partial\theta}+\frac{u_r}{r}\right)\\&+\vec{e}_\theta\vec{e}_\varphi\left(\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial\varphi}+\frac{1}{r}\frac{\partial u_\varphi}{\partial\theta}-\frac{u_\varphi}{r}\cot\theta\right)\\&+\vec{e}_\varphi\vec{e}_r\left(\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial\varphi}+\frac{\partial u_\varphi}{\partial r}-\frac{u_\varphi}{r}\right)+\vec{e}_\varphi\vec{e}_\theta\left(\frac{1}{r}\frac{\partial u_\varphi}{\partial\theta}+\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial\varphi}-\frac{u_\varphi}{r}\cot\theta\right)\\&+\vec{e}_\varphi\vec{e}_\varphi2\left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}+\frac{u_r}{r}+\frac{u_\theta}{r}\cot\theta\right)\end{aligned} \tag{A.22}$$

The result obtained in Eq. (A.22) is of great importance for expressing the scalar components of the tensor in terms of the velocity.

Divergence of a Vector

The divergence is a differentiation operation that involves the dot product of the nabla operator with a vector or tensor. For now, the interest lies in the divergence of a vector. The divergence of a vector in the Cartesian coordinate system is known and therefore will be given directly. However, in the case of cylindrical and spherical coordinate systems, the operations will be shown in more detail.

For the Cartesian Coordinate System

$$\nabla \cdot \vec{V} = \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} \tag{A.23}$$

For the Cylindrical Coordinate System

$$\nabla \cdot \vec{V} = \left(\vec{e}_{\rho} \frac{\partial}{\partial \rho} + \vec{e}_{\varphi} \frac{1}{r} \frac{\partial}{\partial \varphi} + \vec{e}_{z} \frac{\partial}{\partial z} \right) \cdot \left(u_{\rho} \vec{e}_{\rho} + u_{\varphi} \vec{e}_{\varphi} + u_{z} \vec{e}_{z} \right)$$

$$\begin{aligned} \nabla \cdot \vec{V} = & \vec{e}_{\rho} \cdot \frac{\partial}{\partial \rho} \left(u_{\rho} \vec{e}_{\rho} \right) + \vec{e}_{\rho} \cdot \frac{\partial}{\partial \rho} \left(u_{\varphi} \vec{e}_{\varphi} \right) + \vec{e}_{\rho} \cdot \frac{\partial}{\partial \rho} \left(u_{z} \vec{e}_{z} \right) \\ & \vec{e}_{\varphi} \cdot \frac{1}{r} \frac{\partial}{\partial \varphi} \left(u_{\rho} \vec{e}_{\rho} \right) + \vec{e}_{\varphi} \cdot \frac{1}{r} \frac{\partial}{\partial \varphi} \left(u_{\varphi} \vec{e}_{\varphi} \right) + \vec{e}_{\varphi} \cdot \frac{1}{r} \frac{\partial}{\partial \varphi} \left(u_{z} \vec{e}_{z} \right) \\ & \vec{e}_{z} \cdot \frac{\partial}{\partial z} \left(u_{\rho} \vec{e}_{\rho} \right) + \vec{e}_{z} \cdot \frac{\partial}{\partial z} \left(u_{\varphi} \vec{e}_{\varphi} \right) + \vec{e}_{z} \cdot \frac{\partial}{\partial z} \left(u_{z} \vec{e}_{z} \right) \end{aligned}$$

$$\left(\underbrace{\vec{e}_\rho\cdot\vec{e}_\rho}_{1}\right)\frac{\partial u_\rho}{\partial\rho}+u_\rho\left(\vec{e}_\rho\cdot\frac{\partial\vec{e}_\rho}{\partial\rho}\right)+\left(\vec{e}_\rho\cdot\vec{e}_\varphi\right)\frac{\partial u_\varphi}{\partial\rho}+u_\varphi\left(\vec{e}_\rho\cdot\frac{\partial\vec{e}_\varphi}{\partial\rho}\right)+\left(\vec{e}_\rho\cdot\vec{e}_z\right)\frac{\partial u_z}{\partial\rho}+u_z\left(\vec{e}_\rho\cdot\frac{\partial\vec{e}_z}{\partial\rho}\right)$$

$$=\left(\vec{e}_\varphi\cdot\vec{e}_\rho\right)\frac{1}{r}\frac{\partial u_\rho}{\partial\varphi}+\frac{u_\rho}{r}\left(\underbrace{\vec{e}_\varphi\cdot\frac{\partial\overbrace{\vec{e}_\rho}^{\vec{e}_\varphi}}{\partial\varphi}}_{1}\right)+\left(\underbrace{\vec{e}_\varphi\cdot\vec{e}_\varphi}_{1}\right)\frac{1}{r}\frac{\partial u_\varphi}{\partial\varphi}+\frac{u_\varphi}{r}\left(\vec{e}_\varphi\cdot\underbrace{\frac{\partial\vec{e}_\varphi}{\partial\varphi}}_{-\vec{e}_\rho}\right)+\left(\vec{e}_\varphi\cdot\vec{e}_z\right)\frac{1}{r}\frac{\partial u_z}{\partial\varphi}+\frac{u_z}{r}\left(\vec{e}_\varphi\cdot\frac{\partial\vec{e}_z}{\partial\varphi}\right)$$

$$\left(\vec{e}_z\cdot\vec{e}_\rho\right)\frac{\partial u_\rho}{\partial z}+u_\rho\left(\vec{e}_z\cdot\frac{\partial\vec{e}_\rho}{\partial z}\right)+\left(\vec{e}_z\cdot\vec{e}_\varphi\right)\frac{\partial u_\varphi}{\partial z}+u_\varphi\left(\vec{e}_z\cdot\frac{\partial\vec{e}_\varphi}{\partial z}\right)+\left(\underbrace{\vec{e}_z\cdot\vec{e}_z}_{1}\right)\frac{\partial u_z}{\partial z}+u_z\left(\vec{e}_z\cdot\frac{\partial\vec{e}_z}{\partial z}\right)$$

Then,

$$\nabla\cdot\vec{V}=\frac{\partial u_\rho}{\partial\rho}+\frac{u_\rho}{r}+\frac{1}{r}\frac{\partial u_\varphi}{\partial\varphi}+\frac{\partial u_z}{\partial z}=\frac{1}{r}\frac{\partial\left(ru_\rho\right)}{\partial\rho}+\frac{1}{r}\frac{\partial u_\varphi}{\partial\varphi}+\frac{\partial u_z}{\partial z}\tag{A.24}$$

For the Spherical Coordinate System

$$\nabla\cdot\vec{V}=\left(\vec{e}_r\frac{\partial}{\partial r}+\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}+\vec{e}_\varphi\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\right)\cdot\left(u_r\vec{e}_r+u_\theta\vec{e}_\theta+u_\varphi\vec{e}_\varphi\right)$$

$$\nabla\cdot\vec{V}=\begin{array}{c}\vec{e}_r\cdot\frac{\partial}{\partial r}\left(u_r\vec{e}_r\right)+\vec{e}_r\cdot\frac{\partial}{\partial r}\left(u_\theta\vec{e}_\theta\right)+\vec{e}_r\cdot\frac{\partial}{\partial r}\left(u_\varphi\vec{e}_\varphi\right)\\ \vec{e}_\theta\cdot\frac{1}{r}\frac{\partial}{\partial\theta}\left(u_r\vec{e}_r\right)+\vec{e}_\theta\cdot\frac{1}{r}\frac{\partial}{\partial\theta}\left(u_\theta\vec{e}_\theta\right)+\vec{e}_\theta\cdot\frac{1}{r}\frac{\partial}{\partial\theta}\left(u_\varphi\vec{e}_\varphi\right)\\ \vec{e}_\varphi\cdot\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\left(u_r\vec{e}_r\right)+\vec{e}_\varphi\cdot\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\left(u_\theta\vec{e}_\theta\right)+\vec{e}_\varphi\cdot\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\left(u_\varphi\vec{e}_\varphi\right)\end{array}$$

$$\left(\underbrace{\vec{e}_r\cdot\vec{e}_r}_{1}\right)\frac{\partial u_r}{\partial r}+u_r\left(\vec{e}_r\cdot\frac{\partial\cancel{\vec{e}_r}}{\partial r}\right)+\left(\cancel{\vec{e}_r}\cdot\vec{e}_\theta\right)\frac{\partial u_\theta}{\partial r}+u_\theta\left(\vec{e}_r\cdot\frac{\partial\cancel{\vec{e}_\theta}}{\partial r}\right)+\left(\cancel{\vec{e}_r}\cdot\vec{e}_\varphi\right)\frac{\partial u_\varphi}{\partial r}+u_\varphi\left(\vec{e}_r\cdot\frac{\partial\cancel{\vec{e}_\varphi}}{\partial r}\right)$$

$$\left(\cancel{\vec{e}_\theta}\cdot\vec{e}_r\right)\frac{1}{r}\frac{\partial u_r}{\partial\theta}+\frac{u_r}{r}\left(\underbrace{\vec{e}_\theta\cdot\overbrace{\frac{\partial\vec{e}_r}{\partial\theta}}^{\vec{e}_\theta}}_{1}\right)+\left(\underbrace{\vec{e}_\theta\cdot\vec{e}_\theta}_{1}\right)\frac{1}{r}\frac{\partial u_\theta}{\partial\theta}+\frac{u_\theta}{r}\left(\cancel{\vec{e}_\theta}\cdot\overbrace{\frac{\partial\vec{e}_\theta}{\partial\theta}}^{-\vec{e}_r}\right)+\left(\cancel{\vec{e}_\theta}\cdot\vec{e}_\varphi\right)\frac{1}{r}\frac{\partial u_\varphi}{\partial\theta}+\frac{u_\varphi}{r}\left(\vec{e}_\theta\cdot\frac{\partial\cancel{\vec{e}_\varphi}}{\partial\theta}\right)$$

$$=\quad+\frac{1}{r\sin\theta}\left[\left(\cancel{\vec{e}_\varphi}\cdot\vec{e}_r\right)\frac{\partial u_r}{\partial\varphi}+u_r\left(\underbrace{\vec{e}_\varphi\cdot\overbrace{\frac{\partial\vec{e}_r}{\partial\varphi}}^{\sin\theta\,\vec{e}_\varphi}}_{\sin\theta}\right)+\left(\cancel{\vec{e}_\varphi}\cdot\vec{e}_\theta\right)\frac{\partial u_\theta}{\partial\varphi}+u_\theta\left(\underbrace{\vec{e}_\varphi\cdot\overbrace{\frac{\partial\vec{e}_\theta}{\partial\varphi}}^{\cos\theta\,\vec{e}_\varphi}}_{\cos\theta}\right)\right]$$

$$+\frac{1}{r\sin\theta}\left[\left(\underbrace{\vec{e}_\varphi\cdot\vec{e}_\varphi}_{1}\right)\frac{\partial u_\varphi}{\partial\varphi}+u_\varphi\left(\cancel{\vec{e}_\varphi}\cdot\overbrace{\frac{\partial\vec{e}_\varphi}{\partial\varphi}}^{-\sin\theta\,\vec{e}_r\;-\cos\theta\,\vec{e}_\theta}\right)\right]$$

Then,

$$\nabla\cdot\vec{V}=\frac{\partial u_r}{\partial r}+2\frac{u_r}{r}+\frac{1}{r}\frac{\partial u_\theta}{\partial\theta}+\frac{u_\theta}{r}\cot\theta+\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}\tag{A.25a}$$

Or,

$$\nabla\cdot\vec{V}=\frac{1}{r^2}\frac{\partial(r^2u_r)}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(u_\theta\sin\theta)+\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi} \tag{A.25b}$$

Divergence of a Second-Order Tensor

The divergence of a second-order tensor produces a vector. Since this operation is more complex, it will be shown only for spherical coordinate systems.

For the Spherical Coordinate System

$$\nabla\cdot\overleftrightarrow{\Phi}=\left(\vec{e}_r\frac{\partial}{\partial r}+\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}+\vec{e}_\varphi\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\right)\cdot\begin{pmatrix}\Phi_{rr}\vec{e}_r\vec{e}_r+\Phi_{r\theta}\vec{e}_r\vec{e}_\theta+\Phi_{r\varphi}\vec{e}_r\vec{e}_\varphi\\+\Phi_{\theta r}\vec{e}_\theta\vec{e}_r+\Phi_{\theta\theta}\vec{e}_\theta\vec{e}_\theta+\Phi_{\theta\varphi}\vec{e}_\theta\vec{e}_\varphi\\+\Phi_{\varphi r}\vec{e}_\varphi\vec{e}_r+\Phi_{\varphi\theta}\vec{e}_\varphi\vec{e}_\theta+\Phi_{\varphi\varphi}\vec{e}_\varphi\vec{e}_\varphi\end{pmatrix}$$

For the coordinate r

$$\left[\nabla\cdot\overleftrightarrow{\Phi}\right]_r=\vec{e}_r\frac{\partial}{\partial r}\cdot\begin{pmatrix}\Phi_{rr}\vec{e}_r\vec{e}_r+\Phi_{r\theta}\vec{e}_r\vec{e}_\theta+\Phi_{r\varphi}\vec{e}_r\vec{e}_\varphi\\+\Phi_{\theta r}\vec{e}_\theta\vec{e}_r+\Phi_{\theta\theta}\vec{e}_\theta\vec{e}_\theta+\Phi_{\theta\varphi}\vec{e}_\theta\vec{e}_\varphi\\+\Phi_{\varphi r}\vec{e}_\varphi\vec{e}_r+\Phi_{\varphi\theta}\vec{e}_\varphi\vec{e}_\theta+\Phi_{\varphi\varphi}\vec{e}_\varphi\vec{e}_\varphi\end{pmatrix}$$

$$\vec{e}_r\frac{\partial}{\partial r}\cdot\left(\Phi_{rr}\vec{e}_r\vec{e}_r\right)=\underbrace{\vec{e}_r\cdot\left(\vec{e}_r\vec{e}_r\right)}_{\vec{e}_r}\frac{\partial\Phi_{rr}}{\partial r}+\Phi_{rr}\vec{e}_r\cdot\left(\frac{\partial\vec{e}_r}{\partial r}\vec{e}_r\right)+\Phi_{rr}\vec{e}_r\cdot\left(\vec{e}_r\frac{\partial\vec{e}_r}{\partial r}\right)$$

$$\vec{e}_r\frac{\partial}{\partial r}\cdot\left(\Phi_{rr}\vec{e}_r\vec{e}_r\right)=\vec{e}_r\frac{\partial\Phi_{rr}}{\partial r} \tag{r.1}$$

$$\vec{e}_r\frac{\partial}{\partial r}\cdot\left(\Phi_{r\theta}\vec{e}_r\vec{e}_\theta\right)=\vec{e}_r\cdot\left(\vec{e}_r\vec{e}_\theta\right)\frac{\partial\Phi_{r\theta}}{\partial r}+\Phi_{r\theta}\vec{e}_r\cdot\left(\frac{\partial\vec{e}_r}{\partial r}\vec{e}_\theta\right)+\Phi_{r\theta}\vec{e}_r\cdot\left(\vec{e}_r\frac{\partial\vec{e}_\theta}{\partial r}\right)$$

$$\vec{e}_r\frac{\partial}{\partial r}\cdot\left(\Phi_{r\theta}\vec{e}_r\vec{e}_\theta\right)=\vec{e}_\theta\frac{\partial\Phi_{r\theta}}{\partial r} \tag{r.2}$$

$$\vec{e}_r \frac{\partial}{\partial r} \cdot \left(\Phi_{r\varphi} \vec{e}_r \vec{e}_\varphi \right) = \vec{e}_r \cdot \left(\vec{e}_r \vec{e}_\varphi \right) \frac{\partial \Phi_{r\varphi}}{\partial r} + \Phi_{r\varphi} \vec{e}_r \cdot \left(\frac{\partial \cancel{\vec{e}}_r}{\partial r} \vec{e}_\varphi \right) + \Phi_{r\varphi} \vec{e}_r \cdot \left(\vec{e}_r \frac{\partial \cancel{\vec{e}}_\varphi}{\partial r} \right)$$

$$\vec{e}_r \frac{\partial}{\partial r} \cdot \left(\Phi_{r\varphi} \vec{e}_r \vec{e}_\varphi \right) = \vec{e}_\varphi \frac{\partial \Phi_{r\varphi}}{\partial r} \tag{r.3}$$

$$\vec{e}_r \frac{\partial}{\partial r} \cdot \left(\Phi_{\theta r} \vec{e}_\theta \vec{e}_r \right) = \vec{e}_r \cdot \left(\vec{e}_\theta \vec{e}_r \right) \frac{\partial \Phi_{\theta r}}{\partial r} + \Phi_{\theta r} \vec{e}_r \cdot \left(\frac{\partial \cancel{\vec{e}}_\theta}{\partial r} \vec{e}_r \right) + \Phi_{\theta r} \vec{e}_r \cdot \left(\vec{e}_\theta \frac{\partial \cancel{\vec{e}}_r}{\partial r} \right)$$

$$\vec{e}_r \frac{\partial}{\partial r} \cdot \left(\Phi_{\theta r} \vec{e}_\theta \vec{e}_r \right) = 0 \tag{r.4}$$

$$\vec{e}_r \frac{\partial}{\partial r} \cdot \left(\Phi_{\theta\theta} \vec{e}_\theta \vec{e}_\theta \right) = \vec{e}_r \cdot \left(\vec{e}_\theta \vec{e}_\theta \right) \frac{\partial \Phi_{\theta\theta}}{\partial r} + \Phi_{\theta\theta} \vec{e}_r \cdot \left(\frac{\partial \cancel{\vec{e}}_\theta}{\partial r} \vec{e}_\theta \right) + \Phi_{\theta\theta} \vec{e}_r \cdot \left(\vec{e}_\theta \frac{\partial \cancel{\vec{e}}_\theta}{\partial r} \right)$$

$$\vec{e}_r \frac{\partial}{\partial r} \cdot \left(\Phi_{\theta\theta} \vec{e}_\theta \vec{e}_\theta \right) = 0 \tag{r.5}$$

$$\vec{e}_r \frac{\partial}{\partial r} \cdot \left(\Phi_{\theta\varphi} \vec{e}_\theta \vec{e}_\varphi \right) = \vec{e}_r \cdot \left(\vec{e}_\theta \vec{e}_\varphi \right) \frac{\partial \Phi_{\theta\varphi}}{\partial r} + \Phi_{\theta\varphi} \vec{e}_r \cdot \left(\frac{\partial \cancel{\vec{e}}_\theta}{\partial r} \vec{e}_\varphi \right) + \Phi_{\theta\varphi} \vec{e}_r \cdot \left(\vec{e}_\theta \frac{\partial \cancel{\vec{e}}_\varphi}{\partial r} \right)$$

$$\vec{e}_r \frac{\partial}{\partial r} \cdot \left(\Phi_{\theta\varphi} \vec{e}_\theta \vec{e}_\varphi \right) = 0 \tag{r.6}$$

$$\vec{e}_r \frac{\partial}{\partial r} \cdot \left(\Phi_{\varphi r} \vec{e}_\varphi \vec{e}_r \right) = \vec{e}_r \cdot \left(\vec{e}_\varphi \vec{e}_r \right) \frac{\partial \Phi_{\varphi r}}{\partial r} + \Phi_{\varphi r} \vec{e}_r \cdot \left(\frac{\partial \cancel{\vec{e}}_\varphi}{\partial r} \vec{e}_r \right) + \Phi_{\varphi r} \vec{e}_r \cdot \left(\vec{e}_\varphi \frac{\partial \cancel{\vec{e}}_r}{\partial r} \right)$$

$$\vec{e}_r \frac{\partial}{\partial r} \cdot \left(\Phi_{\varphi r} \vec{e}_\varphi \vec{e}_r \right) = 0 \tag{r.7}$$

$$\vec{e}_r \frac{\partial}{\partial r} \cdot \left(\Phi_{\varphi\theta} \vec{e}_\varphi \vec{e}_\theta \right) = \vec{e}_r \cdot \left(\vec{e}_\varphi \vec{e}_\theta \right) \frac{\partial \Phi_{\varphi\theta}}{\partial r} + \Phi_{\varphi\theta} \vec{e}_r \cdot \left(\frac{\partial \cancel{\vec{e}}_\varphi}{\partial r} \vec{e}_\theta \right) + \Phi_{\varphi\theta} \vec{e}_r \cdot \left(\vec{e}_\varphi \frac{\partial \cancel{\vec{e}}_\theta}{\partial r} \right)$$

$$\vec{e}_r \frac{\partial}{\partial r} \cdot \left(\Phi_{\varphi\theta} \vec{e}_\varphi \vec{e}_\theta \right) = 0 \tag{r.8}$$

$$\vec{e}_r \frac{\partial}{\partial r} \cdot \left(\Phi_{\varphi\varphi} \vec{e}_\varphi \vec{e}_\varphi \right) = \vec{e}_r \cdot \left(\vec{e}_\varphi \vec{e}_\varphi \right) \frac{\partial \Phi_{\varphi\varphi}}{\partial r} + \Phi_{\varphi\varphi} \vec{e}_r \cdot \left(\frac{\partial \cancel{\vec{e}}_\varphi}{\partial r} \vec{e}_\varphi \right) + \Phi_{\varphi\varphi} \vec{e}_r \cdot \left(\vec{e}_\varphi \frac{\partial \cancel{\vec{e}}_\varphi}{\partial r} \right)$$

$$\vec{e}_r \frac{\partial}{\partial r} \cdot \left(\Phi_{\varphi\varphi} \vec{e}_\varphi \vec{e}_\varphi \right) = 0 \tag{r.9}$$

Then,

$$\left[\nabla\cdot\overleftrightarrow{\Phi}\right]_r = \vec{e}_r\frac{\partial\Phi_{rr}}{\partial r} + \vec{e}_\theta\frac{\partial\Phi_{r\theta}}{\partial r} + \vec{e}_\varphi\frac{\partial\Phi_{r\varphi}}{\partial r} \tag{A.26}$$

For the coordinate θ

$$\left[\nabla\cdot\overleftrightarrow{\Phi}\right]_\theta = \vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\begin{pmatrix}\Phi_{rr}\vec{e}_r\vec{e}_r + \Phi_{r\theta}\vec{e}_r\vec{e}_\theta + \Phi_{r\varphi}\vec{e}_r\vec{e}_\varphi \\ +\Phi_{\theta r}\vec{e}_\theta\vec{e}_r + \Phi_{\theta\theta}\vec{e}_\theta\vec{e}_\theta + \Phi_{\theta\varphi}\vec{e}_\theta\vec{e}_\varphi \\ +\Phi_{\varphi r}\vec{e}_\varphi\vec{e}_r + \Phi_{\varphi\theta}\vec{e}_\varphi\vec{e}_\theta + \Phi_{\varphi\varphi}\vec{e}_\varphi\vec{e}_\varphi\end{pmatrix}$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{rr}\vec{e}_r\vec{e}_r\right) = \frac{1}{r}\vec{e}_\theta\cdot\left(\vec{e}_r\vec{e}_r\right)\frac{\partial\Phi_{rr}}{\partial\theta} + \frac{\Phi_{rr}}{r}\vec{e}_\theta\cdot\left(\underbrace{\frac{\partial\vec{e}_r}{\partial\theta}}_{\vec{e}_\theta}\vec{e}_r\right) + \frac{\Phi_{rr}}{r}\vec{e}_\theta\cdot\left(\vec{e}_r\underbrace{\frac{\partial\vec{e}_r}{\partial\theta}}_{\vec{e}_\theta}\right)$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{rr}\vec{e}_r\vec{e}_r\right) = \vec{e}_r\frac{\Phi_{rr}}{r} \tag{θ.1}$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{r\theta}\vec{e}_r\vec{e}_\theta\right) = \frac{1}{r}\vec{e}_\theta\cdot\left(\vec{e}_r\vec{e}_\theta\right)\frac{\partial\Phi_{r\theta}}{\partial\theta} + \frac{\Phi_{r\theta}}{r}\vec{e}_\theta\cdot\left(\underbrace{\frac{\partial\vec{e}_r}{\partial\theta}}_{\vec{e}_\theta}\vec{e}_\theta\right) + \frac{\Phi_{r\theta}}{r}\vec{e}_\theta\cdot\left(\vec{e}_r\underbrace{\frac{\partial\vec{e}_\theta}{\partial\theta}}_{-\vec{e}_r}\right)$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{r\theta}\vec{e}_r\vec{e}_\theta\right) = \vec{e}_\theta\frac{\Phi_{r\theta}}{r} \tag{θ.2}$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{r\varphi}\vec{e}_r\vec{e}_\varphi\right) = \frac{1}{r}\vec{e}_\theta\cdot\left(\vec{e}_r\vec{e}_\varphi\right)\frac{\partial\Phi_{r\varphi}}{\partial\theta} + \frac{\Phi_{r\varphi}}{r}\vec{e}_\theta\cdot\left(\underbrace{\frac{\partial\vec{e}_r}{\partial\theta}}_{\vec{e}_\theta}\vec{e}_\varphi\right) + \frac{\Phi_{r\varphi}}{r}\vec{e}_\theta\cdot\left(\vec{e}_r\cancel{\frac{\partial\vec{e}_\varphi}{\partial\theta}}\right)$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{r\varphi}\vec{e}_r\vec{e}_\varphi\right) = \vec{e}_\varphi\frac{\Phi_{r\varphi}}{r} \tag{θ.3}$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{\theta r}\vec{e}_\theta\vec{e}_r\right) = \frac{1}{r}\vec{e}_\theta\cdot\left(\vec{e}_\theta\vec{e}_r\right)\frac{\partial\Phi_{\theta r}}{\partial\theta} + \frac{\Phi_{\theta r}}{r}\vec{e}_\theta\cdot\left(\underbrace{\frac{\partial\vec{e}_\theta}{\partial\theta}}_{-\vec{e}_r}\vec{e}_r\right) + \frac{\Phi_{\theta r}}{r}\vec{e}_\theta\cdot\left(\vec{e}_\theta\underbrace{\frac{\partial\vec{e}_r}{\partial\theta}}_{\vec{e}_\theta}\right)$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{\theta r}\vec{e}_\theta\vec{e}_r\right) = \vec{e}_r\frac{1}{r}\frac{\partial\Phi_{\theta r}}{\partial\theta} + \vec{e}_\theta\frac{\Phi_{\theta r}}{r} \tag{θ.4}$$

$$\vec{e}_\theta \frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{\theta\theta}\vec{e}_\theta\vec{e}_\theta\right)=\frac{1}{r}\vec{e}_\theta\cdot\left(\vec{e}_\theta\vec{e}_\theta\right)\frac{\partial\Phi_{\theta\theta}}{\partial\theta}+\frac{\Phi_{\theta\theta}}{r}\vec{e}_\theta\cdot\left(\underbrace{\frac{\partial\vec{e}_\theta}{\partial\theta}}_{-\vec{e}_r}\vec{e}_\theta\right)+\frac{\Phi_{\theta\theta}}{r}\vec{e}_\theta\cdot\left(\vec{e}_\theta\underbrace{\frac{\partial\vec{e}_\theta}{\partial\theta}}_{-\vec{e}_r}\right)$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{\theta\theta}\vec{e}_\theta\vec{e}_\theta\right)=\vec{e}_\theta\frac{1}{r}\frac{\partial\Phi_{\theta\theta}}{\partial\theta}-\vec{e}_r\frac{\Phi_{\theta\theta}}{r} \qquad (\theta.5)$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{\theta\varphi}\vec{e}_\theta\vec{e}_\varphi\right)=\frac{1}{r}\vec{e}_\theta\cdot\left(\vec{e}_\theta\vec{e}_\varphi\right)\frac{\partial\Phi_{\theta\varphi}}{\partial\theta}+\frac{\Phi_{\theta\varphi}}{r}\vec{e}_\theta\cdot\left(\underbrace{\frac{\partial\vec{e}_\theta}{\partial\theta}}_{-\vec{e}_r}\vec{e}_\varphi\right)+\frac{\Phi_{\theta\varphi}}{r}\vec{e}_\theta\cdot\left(\vec{e}_\theta\cancel{\frac{\partial\vec{e}_\varphi}{\partial\theta}}\right)$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{\theta\varphi}\vec{e}_\theta\vec{e}_\varphi\right)=\vec{e}_\varphi\frac{1}{r}\frac{\partial\Phi_{\theta\varphi}}{\partial\theta} \qquad (\theta.6)$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{\varphi r}\vec{e}_\varphi\vec{e}_r\right)=\frac{1}{r}\vec{e}_\theta\cdot\left(\vec{e}_\varphi\vec{e}_r\right)\frac{\partial\Phi_{\varphi r}}{\partial\theta}+\frac{\Phi_{\varphi r}}{r}\vec{e}_\theta\cdot\left(\cancel{\frac{\partial\vec{e}_\varphi}{\partial\theta}}\vec{e}_r\right)+\frac{\Phi_{\varphi r}}{r}\vec{e}_\theta\cdot\left(\vec{e}_\varphi\underbrace{\frac{\partial\vec{e}_r}{\partial\theta}}_{\vec{e}_\theta}\right)$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{\varphi r}\vec{e}_\varphi\vec{e}_r\right)=0 \qquad (\theta.7)$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{\varphi\theta}\vec{e}_\varphi\vec{e}_\theta\right)=\frac{1}{r}\vec{e}_\theta\cdot\left(\vec{e}_\varphi\vec{e}_\theta\right)\frac{\partial\Phi_{\varphi\theta}}{\partial\theta}+\frac{\Phi_{\varphi\theta}}{r}\vec{e}_\theta\cdot\left(\cancel{\frac{\partial\vec{e}_\varphi}{\partial\theta}}\vec{e}_\theta\right)+\frac{\Phi_{\varphi\theta}}{r}\vec{e}_\theta\cdot\left(\vec{e}_\varphi\underbrace{\frac{\partial\vec{e}_\theta}{\partial\theta}}_{-\vec{e}_r}\right)$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{\varphi\theta}\vec{e}_\varphi\vec{e}_\theta\right)=0 \qquad (\theta.8)$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{\varphi\varphi}\vec{e}_\varphi\vec{e}_\varphi\right)=\frac{1}{r}\vec{e}_\theta\cdot\left(\vec{e}_\varphi\vec{e}_\varphi\right)\frac{\partial\Phi_{\varphi\varphi}}{\partial\theta}+\frac{\Phi_{\varphi\varphi}}{r}\vec{e}_\theta\cdot\left(\cancel{\frac{\partial\vec{e}_\varphi}{\partial\theta}}\vec{e}_\varphi\right)+\frac{\Phi_{\varphi\varphi}}{r}\vec{e}_\theta\cdot\left(\vec{e}_\varphi\cancel{\frac{\partial\vec{e}_\varphi}{\partial\theta}}\right)$$

$$\vec{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}\cdot\left(\Phi_{\varphi\varphi}\vec{e}_\varphi\vec{e}_\varphi\right)=0 \qquad (\theta.9)$$

Then,

$$\left[\nabla\cdot\overleftrightarrow{\Phi}\right]_{\theta}=\begin{array}{c}\vec{e}_{r}\left(\frac{1}{r}\frac{\partial\Phi_{\theta r}}{\partial\theta}+\frac{\Phi_{rr}}{r}-\frac{\Phi_{\theta\theta}}{r}\right)+\vec{e}_{\theta}\left(\frac{1}{r}\frac{\partial\Phi_{\theta\theta}}{\partial\theta}+\frac{\Phi_{r\theta}}{r}+\frac{\Phi_{\theta r}}{r}\right)\\+\vec{e}_{\varphi}\left(\frac{1}{r}\frac{\partial\Phi_{\theta\varphi}}{\partial\theta}+\frac{\Phi_{r\varphi}}{r}\right)\end{array}\tag{A.27}$$

For the coordinate φ

$$\left[\nabla\cdot\overleftrightarrow{\Phi}\right]_{\varphi}=\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\begin{pmatrix}\Phi_{rr}\vec{e}_{r}\vec{e}_{r}+\Phi_{r\theta}\vec{e}_{r}\vec{e}_{\theta}+\Phi_{r\varphi}\vec{e}_{r}\vec{e}_{\varphi}\\+\Phi_{\theta r}\vec{e}_{\theta}\vec{e}_{r}+\Phi_{\theta\theta}\vec{e}_{\theta}\vec{e}_{\theta}+\Phi_{\theta\varphi}\vec{e}_{\theta}\vec{e}_{\varphi}\\+\Phi_{\varphi r}\vec{e}_{\varphi}\vec{e}_{r}+\Phi_{\varphi\theta}\vec{e}_{\varphi}\vec{e}_{\theta}+\Phi_{\varphi\varphi}\vec{e}_{\varphi}\vec{e}_{\varphi}\end{pmatrix}$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{rr}\vec{e}_{r}\vec{e}_{r}\right)$$

$$=\frac{1}{r\sin\theta}\left[\vec{e}_{\varphi}\cdot\left(\vec{e}_{r}\vec{e}_{r}\right)\frac{\partial\Phi_{rr}}{\partial\varphi}+\Phi_{rr}\vec{e}_{\varphi}\cdot\left(\underbrace{\frac{\partial\vec{e}_{r}}{\partial\varphi}}_{\sin\theta\vec{e}_{\varphi}}\vec{e}_{r}\right)+\Phi_{rr}\vec{e}_{\varphi}\cdot\left(\vec{e}_{r}\underbrace{\frac{\partial\vec{e}_{r}}{\partial\varphi}}_{\sin\theta\vec{e}_{\varphi}}\right)\right]$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{rr}\vec{e}_{r}\vec{e}_{r}\right)=\vec{e}_{r}\frac{\Phi_{rr}}{r}\tag{.1}$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{r\theta}\vec{e}_{r}\vec{e}_{\theta}\right)$$

$$=\frac{1}{r\sin\theta}\left[\vec{e}_{\varphi}\cdot\left(\vec{e}_{r}\vec{e}_{\theta}\right)\frac{\partial\Phi_{r\theta}}{\partial\varphi}+\Phi_{r\theta}\vec{e}_{\varphi}\cdot\left(\underbrace{\frac{\partial\vec{e}_{r}}{\partial\varphi}}_{\sin\theta\vec{e}_{\varphi}}\vec{e}_{\theta}\right)+\Phi_{r\theta}\vec{e}_{\varphi}\cdot\left(\vec{e}_{r}\underbrace{\frac{\partial\vec{e}_{\theta}}{\partial\varphi}}_{\cos\theta\vec{e}_{\varphi}}\right)\right]$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{r\theta}\vec{e}_{r}\vec{e}_{\theta}\right)=\vec{e}_{\theta}\frac{\Phi_{r\theta}}{r}\tag{.2}$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{r\varphi}\vec{e}_{r}\vec{e}_{\varphi}\right)=\frac{1}{r\sin\theta}\left[\vec{e}_{\varphi}\cdot\left(\vec{e}_{r}\vec{e}_{\varphi}\right)\frac{\partial\Phi_{r\varphi}}{\partial\varphi}+\Phi_{r\varphi}\vec{e}_{\varphi}\cdot\left(\underbrace{\frac{\partial\vec{e}_{r}}{\partial\varphi}}_{\sin\theta\vec{e}_{\varphi}}\right)+\Phi_{r\varphi}\vec{e}_{\varphi}\cdot\left(\vec{e}_{r}\underbrace{\frac{\partial\vec{e}_{\varphi}}{\partial\varphi}}_{-\left(\begin{matrix}\sin\theta\vec{e}_{r}\\+\cos\theta\vec{e}_{\theta}\end{matrix}\right)}\right)\right]$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{r\varphi}\vec{e}_{r}\vec{e}_{\varphi}\right)=\vec{e}_{\varphi}\frac{\Phi_{r\varphi}}{r}\tag{.3}$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{\theta r}\vec{e}_{\theta}\vec{e}_{r}\right)=\frac{1}{r\sin\theta}\left[\vec{e}_{\varphi}\cdot\left(\vec{e}_{\theta}\vec{e}_{r}\right)\frac{\partial\Phi_{\theta r}}{\partial\varphi}+\Phi_{\theta r}\vec{e}_{\varphi}\cdot\left(\underbrace{\frac{\partial\vec{e}_{\theta}}{\partial\varphi}}_{\cos\theta\vec{e}_{\varphi}}\vec{e}_{r}\right)+\Phi_{\theta r}\vec{e}_{\varphi}\cdot\left(\vec{e}_{\theta}\underbrace{\frac{\partial\vec{e}_{r}}{\partial\varphi}}_{\sin\theta\vec{e}_{\varphi}}\right)\right]$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{\theta r}\vec{e}_{\theta}\vec{e}_{r}\right)=\vec{e}_{r}\frac{\Phi_{\theta r}}{r}\cot\theta\tag{.4}$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{\theta\theta}\vec{e}_{\theta}\vec{e}_{\theta}\right)=\frac{1}{r\sin\theta}\left[\vec{e}_{\varphi}\cdot\left(\vec{e}_{\theta}\vec{e}_{\theta}\right)\frac{\partial\Phi_{\theta\theta}}{\partial\varphi}+\Phi_{\theta\theta}\vec{e}_{\varphi}\cdot\left(\underbrace{\frac{\partial\vec{e}_{\theta}}{\partial\varphi}}_{\cos\theta\vec{e}_{\varphi}}\vec{e}_{\theta}\right)+\Phi_{\theta\theta}\vec{e}_{\varphi}\cdot\left(\vec{e}_{\theta}\underbrace{\frac{\partial\vec{e}_{\theta}}{\partial\varphi}}_{\cos\theta\vec{e}_{\varphi}}\right)\right]$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{\theta\theta}\vec{e}_{\theta}\vec{e}_{\theta}\right)=\vec{e}_{\theta}\frac{\Phi_{\theta\theta}}{r}\cot\theta\tag{.5}$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{\theta\varphi}\vec{e}_{\theta}\vec{e}_{\varphi}\right)=\frac{1}{r\sin\theta}\left[\vec{e}_{\varphi}\cdot\left(\vec{e}_{\theta}\vec{e}_{\varphi}\right)\frac{\partial\Phi_{\theta\varphi}}{\partial\varphi}+\Phi_{\theta\varphi}\vec{e}_{\varphi}\cdot\left(\underbrace{\frac{\partial\vec{e}_{\theta}}{\partial\varphi}}_{\cos\theta\vec{e}_{\varphi}}\vec{e}_{\varphi}\right)+\Phi_{\theta\varphi}\vec{e}_{\varphi}\cdot\left(\vec{e}_{\theta}\quad\underbrace{\frac{\partial\vec{e}_{\varphi}}{\partial\varphi}}_{-\left(\begin{matrix}\sin\theta\vec{e}_{r}\\+\cos\theta\vec{e}_{\theta}\end{matrix}\right)}\right)\right]$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{\theta\varphi}\vec{e}_{\theta}\vec{e}_{\varphi}\right)=\vec{e}_{\varphi}\frac{\Phi_{\theta\varphi}}{r}\cot\theta \tag{.6}$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{\varphi r}\vec{e}_{\varphi}\vec{e}_{r}\right)$$

$$=\frac{1}{r\sin\theta}\left[\vec{e}_{\varphi}\cdot\left(\vec{e}_{\varphi}\vec{e}_{r}\right)\frac{\partial\Phi_{\varphi r}}{\partial\varphi}+\Phi_{\varphi r}\vec{e}_{\varphi}\cdot\left(\begin{matrix}\underbrace{\frac{\partial\vec{e}_{\varphi}}{\partial\varphi}}_{-\sin\theta\vec{e}_{r}}\quad\vec{e}_{r}\\-\cos\theta\vec{e}_{\theta}\end{matrix}\right)+\Phi_{\varphi r}\vec{e}_{\varphi}\cdot\left(\vec{e}_{\varphi}\underbrace{\frac{\partial\vec{e}_{r}}{\partial\varphi}}_{\sin\theta\vec{e}_{\varphi}}\right)\right]$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{\varphi r}\vec{e}_{\varphi}\vec{e}_{r}\right)=\vec{e}_{r}\frac{1}{r\sin\theta}\frac{\partial\Phi_{\varphi r}}{\partial\varphi}+\vec{e}_{\varphi}\frac{\Phi_{\varphi r}}{r} \tag{.7}$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{\varphi\theta}\vec{e}_{\varphi}\vec{e}_{\theta}\right)$$

$$=\frac{1}{r\sin\theta}\left[\vec{e}_{\varphi}\cdot\left(\vec{e}_{\varphi}\vec{e}_{\theta}\right)\frac{\partial\Phi_{\varphi\theta}}{\partial\varphi}+\Phi_{\varphi\theta}\vec{e}_{\varphi}\cdot\begin{pmatrix}\underbrace{\frac{\partial\vec{e}_{\varphi}}{\partial\varphi}}_{-\sin\theta\vec{e}_{r}} & \vec{e}_{\theta}\\ -\cos\theta\vec{e}_{\theta}\end{pmatrix}+\Phi_{\varphi\theta}\vec{e}_{\varphi}\cdot\begin{pmatrix}\vec{e}_{\varphi} & \underbrace{\frac{\partial\vec{e}_{\theta}}{\partial\varphi}}_{\cos\theta\vec{e}_{\varphi}}\end{pmatrix}\right]$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{\varphi\theta}\vec{e}_{\varphi}\vec{e}_{\theta}\right)=\vec{e}_{\theta}\frac{1}{r\sin\theta}\frac{\partial\Phi_{\varphi\theta}}{\partial\varphi}+\vec{e}_{\varphi}\frac{\Phi_{\varphi\theta}}{r}\cot\theta \qquad (.8)$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{\varphi\varphi}\vec{e}_{\varphi}\vec{e}_{\varphi}\right)=\frac{1}{r\sin\theta}\left[\vec{e}_{\varphi}\cdot\left(\vec{e}_{\varphi}\vec{e}_{\varphi}\right)\frac{\partial\Phi_{\varphi\varphi}}{\partial\varphi}+\Phi_{\varphi\varphi}\vec{e}_{\varphi}\right.$$

$$\left.\cdot\begin{pmatrix}\underbrace{\frac{\partial\vec{e}_{\varphi}}{\partial\varphi}}_{-\sin\theta\vec{e}_{r}} & \vec{e}_{\varphi}\\ -\cos\theta\vec{e}_{\theta}\end{pmatrix}+\Phi_{\varphi\varphi}\vec{e}_{\varphi}\cdot\begin{pmatrix}\vec{e}_{\varphi} & \underbrace{\frac{\partial\vec{e}_{\varphi}}{\partial\varphi}}_{-\sin\theta\vec{e}_{r}}\\ & -\cos\theta\vec{e}_{\theta}\end{pmatrix}\right]$$

$$\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\cdot\left(\Phi_{\varphi\varphi}\vec{e}_{\varphi}\vec{e}_{\varphi}\right)=\vec{e}_{\varphi}\frac{1}{r\sin\theta}\frac{\partial\Phi_{\varphi\varphi}}{\partial\varphi}-\vec{e}_{r}\frac{\Phi_{\varphi\varphi}}{r}-\vec{e}_{\theta}\frac{\Phi_{\varphi\varphi}}{r}\cot\theta \quad (.9)$$

Then,

$$\left[\nabla\cdot\overleftrightarrow{\Phi}\right]_{\varphi} = \begin{aligned} &\vec{e}_r\left(\frac{1}{r\sin\theta}\frac{\partial\Phi_{\varphi r}}{\partial\varphi}+\frac{\Phi_{rr}}{r}+\frac{\Phi_{\theta r}}{r}\cot\theta-\frac{\Phi_{\varphi\varphi}}{r}\right)\\ &+\vec{e}_\theta\left(\frac{1}{r\sin\theta}\frac{\partial\Phi_{\varphi\theta}}{\partial\varphi}+\frac{\Phi_{r\theta}}{r}+\frac{\Phi_{\theta\theta}}{r}\cot\theta-\frac{\Phi_{\varphi\varphi}}{r}\cot\theta\right)\\ &+\vec{e}_\varphi\left(\frac{1}{r\sin\theta}\frac{\partial\Phi_{\varphi\varphi}}{\partial\varphi}+\frac{\Phi_{r\varphi}}{r}+\frac{\Phi_{\varphi r}}{r}+\frac{\Phi_{\theta\varphi}}{r}\cot\theta+\frac{\Phi_{\varphi\theta}}{r}\cot\theta\right)\end{aligned} \tag{A.28}$$

Using the results in Eqs. (A.26)–(A.28), the complete expression for the divergence of a second-order tensor will be obtained.

$$\begin{aligned}\nabla\cdot\overleftrightarrow{\Phi} &= \vec{e}_r\left(\frac{\partial\Phi_{rr}}{\partial r}+2\frac{\Phi_{rr}}{r}+\frac{1}{r}\frac{\partial\Phi_{\theta r}}{\partial\theta}+\frac{\Phi_{\theta r}}{r}\cot\theta+\frac{1}{r\sin\theta}\frac{\partial\Phi_{\varphi r}}{\partial\varphi}-\frac{\Phi_{\theta\theta}+\Phi_{\varphi\varphi}}{r}\right)\\ &+\vec{e}_\theta\left(\frac{\partial\Phi_{r\theta}}{\partial r}+2\frac{\Phi_{r\theta}}{r}+\frac{1}{r}\frac{\partial\Phi_{\theta\theta}}{\partial\theta}+\frac{\Phi_{\theta\theta}}{r}\cot\theta+\frac{1}{r\sin\theta}\frac{\partial\Phi_{\varphi\theta}}{\partial\varphi}+\frac{\Phi_{\theta r}}{r}-\frac{\Phi_{\varphi\varphi}}{r}\cot\theta\right)\\ &+\vec{e}_\varphi\left(\frac{\partial\Phi_{r\varphi}}{\partial r}+2\frac{\Phi_{r\varphi}}{r}+\frac{1}{r}\frac{\partial\Phi_{\theta\varphi}}{\partial\theta}+\frac{1}{r\sin\theta}\frac{\partial\Phi_{\varphi\varphi}}{\partial\varphi}+\frac{\Phi_{\varphi r}}{r}+\frac{\Phi_{\theta\varphi}+\Phi_{\varphi\theta}}{r}\cot\theta\right)\end{aligned} \tag{A.29}$$

Expressing the result in the most commonly used form,

$$\begin{aligned}\nabla\cdot\overleftrightarrow{\Phi} &= \vec{e}_r\left(\frac{1}{r^2}\frac{\partial(r^2\Phi_{rr})}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\Phi_{\theta r}\sin\theta)+\frac{1}{r\sin\theta}\frac{\partial\Phi_{\varphi r}}{\partial\varphi}-\frac{\Phi_{\theta\theta}+\Phi_{\varphi\varphi}}{r}\right)\\ &+\vec{e}_\theta\left(\frac{1}{r^2}\frac{\partial(r^2\Phi_{r\theta})}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\Phi_{\theta\theta}\sin\theta)+\frac{1}{r\sin\theta}\frac{\partial\Phi_{\varphi\theta}}{\partial\varphi}+\frac{\Phi_{\theta r}}{r}-\frac{\Phi_{\varphi\varphi}}{r}\cot\theta\right)\\ &+\vec{e}_\varphi\left(\frac{1}{r^2}\frac{\partial\left(r^2\Phi_{r\varphi}\right)}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\left(\Phi_{\theta\varphi}\sin\theta\right)+\frac{1}{r\sin\theta}\frac{\partial\Phi_{\varphi\varphi}}{\partial\varphi}+\frac{\Phi_{\varphi r}}{r}+\frac{\Phi_{\varphi\theta}}{r}\cot\theta\right)\end{aligned} \tag{A.30}$$

The Material (or Substantial) Derivative of a Vector

The material derivative is defined as shown in Eq. (A.31), and this definition is valid for a scalar or a vector.

$$\frac{D}{Dt}() = \frac{\partial}{\partial t}() + \vec{V}\cdot\nabla() \tag{A.31}$$

As can be seen, in the case of a vector, it is first derived with respect to time and subsequently its gradient is multiplied scalarly by the velocity.

For the Spherical Coordinate System

The material derivative of the velocity will be obtained, and the process begins by taking the derivative as a function of time,

$$\frac{\partial \vec{V}}{\partial t} = \frac{\partial}{\partial t}\left(\vec{e}_r u_r + \vec{e}_\theta u_\theta + \vec{e}_\varphi u_\varphi\right)$$
$$= \vec{e}_r \frac{\partial u_r}{\partial t} + \vec{e}_\theta \frac{\partial u_\theta}{\partial t} + \vec{e}_\varphi \frac{\partial u_\varphi}{\partial t} + u_r \frac{\partial \vec{e}_r}{\partial t} + u_\theta \frac{\partial \vec{e}_\theta}{\partial t} + u_\varphi \frac{\partial \vec{e}_\varphi}{\partial t}$$

$$\frac{\partial \vec{V}}{\partial t} = \vec{e}_r \frac{\partial u_r}{\partial t} + \vec{e}_\theta \frac{\partial u_\theta}{\partial t} + \vec{e}_\varphi \frac{\partial u_\varphi}{\partial t} \tag{D.1}$$

The gradient of a vector has already been determined, as shown in Eq. (A.22),

$$\vec{V}\cdot\nabla\vec{V} = \left(\vec{e}_r u_r + \vec{e}_\theta u_\theta + \vec{e}_\varphi u_\varphi\right)\cdot\left(\vec{e}_r\vec{e}_r\frac{\partial u_r}{\partial r} + \vec{e}_r\vec{e}_\theta\frac{\partial u_\theta}{\partial r} + \vec{e}_r\vec{e}_\varphi\frac{\partial u_\varphi}{\partial r}\right)$$
$$\left(\vec{e}_r u_r + \vec{e}_\theta u_\theta + \vec{e}_\varphi u_\varphi\right)\cdot\left(+\vec{e}_\theta\vec{e}_r\left(\frac{1}{r}\frac{\partial u_r}{\partial\theta} - \frac{u_\theta}{r}\right)\right.$$
$$\left. + \vec{e}_\theta\vec{e}_\theta\left(\frac{1}{r}\frac{\partial u_\theta}{\partial\theta} + \frac{u_r}{r}\right) + \vec{e}_\theta\vec{e}_\varphi\frac{1}{r}\frac{\partial u_\varphi}{\partial\theta}\right)$$

$$\left(\vec{e}_r u_r + \vec{e}_\theta u_\theta + \vec{e}_\varphi u_\varphi\right)\cdot\left[\begin{array}{l} +\vec{e}_\varphi\vec{e}_r\left(\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial\varphi} - \frac{u_\varphi}{r}\right) + \vec{e}_\varphi\vec{e}_\theta\left(\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial\varphi} - \frac{u_\varphi}{r}\cot\theta\right) \\ +\vec{e}_\varphi\vec{e}_\varphi\left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi} + \frac{u_r}{r} + \frac{u_\theta}{r}\cot\theta\right) \end{array}\right]$$

The dot product of the previous equation is identical to that determined in Eq. (A.9), which shows the result of the scalar multiplication of a vector and a second-order tensor.

$$\vec{V}\cdot\nabla\vec{V} = \begin{array}{l} \left(u_r\frac{\partial u_r}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_r}{\partial\theta} - \frac{u_\theta^2}{r} + \frac{u_\varphi}{\mathrm{r}\ \sin\ \theta}\frac{\partial u_r}{\partial\varphi} - \frac{u_\varphi^2}{r}\right)\vec{e}_r \\ \left(u_r\frac{\partial u_\theta}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_\theta}{\partial\theta} + \frac{u_r u_\theta}{r} + \frac{u_\varphi}{r\ sin\ \theta}\frac{\partial u_\theta}{\partial\varphi} - \frac{u_\varphi^2}{r}\cot\theta\right)\vec{e}_\theta \\ \left(u_r\frac{\partial u_\varphi}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_\varphi}{\partial\theta} + \frac{u_\varphi}{r\ sin\ \theta}\frac{\partial u_\varphi}{\partial\varphi} + \frac{u_r u_\varphi}{r} + \frac{u_\theta u_\varphi}{r}\cot\theta\right)\vec{e}_\varphi \end{array} \tag{D.2}$$

Finally, the material derivative is obtained using the results shown in Eqs. (D.1) and (D.2).

$$\left(\frac{\partial u_r}{\partial t}+u_r\frac{\partial u_r}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_r}{\partial \theta}+\frac{u_\varphi}{r\ sin\ \theta}\frac{\partial u_r}{\partial \varphi}-\frac{u_\theta^2+u_\varphi^2}{r}\right)\vec{e}_r$$

$$\frac{D\vec{V}}{Dt}=\left(\frac{\partial u_\theta}{\partial t}+u_r\frac{\partial u_\theta}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_\theta}{\partial \theta}+\frac{u_\varphi}{r\ sin\ \theta}\frac{\partial u_\theta}{\partial \varphi}+\frac{u_r u_\theta}{r}-\frac{u_\varphi^2}{r}\cot\theta\right)\vec{e}_\theta$$

$$\left(\frac{\partial u_\varphi}{\partial t}+u_r\frac{\partial u_\varphi}{\partial r}+\frac{u_\theta}{r}\frac{\partial u_\varphi}{\partial \theta}+\frac{u_\varphi}{r\ sin\ \theta}\frac{\partial u_\varphi}{\partial \varphi}+\frac{u_r u_\varphi}{r}+\frac{u_\theta u_\varphi}{r}\cot\theta\right)\vec{e}_\varphi \tag{A.32}$$

The Energy Dissipation Function Due to the Effect of Viscosity

The viscous dissipation function is obtained by performing the product $\vec{\sigma}:\nabla\vec{V}$ using the stress in terms of the velocity. To do this, it will be necessary to observe Eqs. (3.34) and (3.35), in which it has already been assumed that $\mu^{'}=0$; therefore, Eq. (A.33) is considered.

$$\begin{aligned}\vec{\sigma}:\nabla\vec{V}\ =&\sigma_{rr}\left(\frac{\partial u_r}{\partial r}\right)+\sigma_{\theta\theta}\left(\frac{1}{r}\frac{\partial u_\theta}{\partial \theta}+\frac{u_r}{r}\right)+\sigma_{\varphi\varphi}\left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial \varphi}+\frac{u_r}{r}+\frac{u_\theta}{r}\cot\theta\right)\\&+\sigma_{r\theta}\left(\frac{\partial u_\theta}{\partial r}+\frac{1}{r}\frac{\partial u_r}{\partial \theta}-\frac{u_\theta}{r}\right)+\sigma_{r\varphi}\left(\frac{\partial u_\varphi}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial \varphi}-\frac{u_\varphi}{r}\right)\\&+\sigma_{\theta\varphi}\left(\frac{1}{r}\frac{\partial u_\varphi}{\partial \theta}+\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial \varphi}-\frac{u_\varphi}{r}\cot\theta\right)\end{aligned} \tag{A.33}$$

The products are as follows:

$$\sigma_{rr}\left(\frac{\partial u_r}{\partial r}\right)=-p\left(\frac{\partial u_r}{\partial r}\right)+2\mu\left(\frac{\partial u_r}{\partial r}\right)^2-\frac{2}{3}\mu\left(\nabla\cdot\vec{V}\right)\left(\frac{\partial u_r}{\partial r}\right) \tag{a.3}$$

$$\sigma_{\theta\theta}\left(\frac{1}{r}\frac{\partial u_\theta}{\partial \theta}+\frac{u_r}{r}\right)=-p\left(\frac{1}{r}\frac{\partial u_\theta}{\partial \theta}+\frac{u_r}{r}\right)+2\mu\left(\frac{1}{r}\frac{\partial u_\theta}{\partial \theta}+\frac{u_r}{r}\right)^2-\frac{2}{3}\mu\left(\nabla\cdot\vec{V}\right)\left(\frac{1}{r}\frac{\partial u_\theta}{\partial \theta}+\frac{u_r}{r}\right) \tag{b.3}$$

$$\sigma_{\varphi\varphi}\left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}+\frac{u_r}{r}+\frac{u_\theta}{r}\cot\theta\right)=-p\left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}+\frac{u_r}{r}+\frac{u_\theta}{r}\cot\theta\right)$$
$$+2\mu\left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}+\frac{u_r}{r}+\frac{u_\theta}{r}\cot\theta\right)^2-\frac{2}{3}\mu\left(\nabla\cdot\vec{V}\right)\left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}+\frac{u_r}{r}+\frac{u_\theta}{r}\cot\theta\right) \tag{c.3}$$

$$\sigma_{r\theta}\left(\frac{\partial u_\theta}{\partial r}+\frac{1}{r}\frac{\partial u_r}{\partial\theta}-\frac{u_\theta}{r}\right)=\mu\left(\frac{1}{r}\frac{\partial u_r}{\partial\theta}+\frac{\partial u_\theta}{\partial r}-\frac{u_\theta}{r}\right)^2 \tag{d.3}$$

$$\sigma_{r\varphi}\left(\frac{\partial u_\varphi}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial\varphi}-\frac{u_\varphi}{r}\right)=\mu\left(\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial\varphi}+\frac{\partial u_\varphi}{\partial r}-\frac{u_\varphi}{r}\right)^2 \tag{e.3}$$

$$\sigma_{\theta\varphi}\left(\frac{1}{r}\frac{\partial u_\varphi}{\partial\theta}+\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial\varphi}-\frac{u_\varphi}{r}\cot\theta\right)=\mu\left(\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial\varphi}+\frac{1}{r}\frac{\partial u_\varphi}{\partial\theta}-\frac{u_\varphi}{r}\cot\theta\right)^2 \tag{f.3}$$

Adding the terms together, the following is obtained:

$$-p\left(\frac{1}{r^2}\frac{\partial(r^2u_r)}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial(u_\theta sin\,\theta)}{\partial\theta}+\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}\right)$$
$$-\frac{2}{3}\mu\left(\nabla\cdot\vec{V}\right)\left(\frac{1}{r^2}\frac{\partial(r^2u_r)}{\partial r}+\frac{1}{r\sin\theta}\frac{\partial(u_\theta sin\,\theta)}{\partial\theta}+\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}\right)$$
$$\vec{\sigma}:\nabla\vec{V}=+2\mu\left(\frac{\partial u_r}{\partial r}\right)^2+2\mu\left(\frac{1}{r}\frac{\partial u_\theta}{\partial\theta}+\frac{u_r}{r}\right)^2+2\mu\left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial\varphi}+\frac{u_r}{r}+\frac{u_\theta}{r}\cot\theta\right)^2$$
$$\mu\left(\frac{1}{r}\frac{\partial u_r}{\partial\theta}+\frac{\partial u_\theta}{\partial r}-\frac{u_\theta}{r}\right)^2+\mu\left(\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial\varphi}+\frac{\partial u_\varphi}{\partial r}-\frac{u_\varphi}{r}\right)^2$$
$$+\mu\left(\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial\varphi}+\frac{1}{r}\frac{\partial u_\varphi}{\partial\theta}-\frac{u_\varphi}{r}\cot\theta\right)^2$$

Recognizing the definition of the divergence of velocity,

$$\vec{\sigma}:\nabla\vec{V}=-p\left(\nabla\cdot\vec{V}\right)+\mu\Phi_v \tag{A.32}$$

where Φ_v is the viscosity dissipation function and is given in Eq. (A.33),

$$\Phi_v = +2\left(\frac{\partial u_r}{\partial r}\right)^2 + 2\left(\frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r}\right)^2 + 2\left(\frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial \varphi} + \frac{u_r}{r} + \frac{u_\theta}{r}\cot\theta\right)^2 + \left(\frac{1}{r}\frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r}\right)^2 + \left(\frac{1}{r\sin\theta}\frac{\partial u_r}{\partial \varphi} + \frac{\partial u_\varphi}{\partial r} - \frac{u_\varphi}{r}\right)^2 + \left(\frac{1}{r\sin\theta}\frac{\partial u_\theta}{\partial \varphi} + \frac{1}{r}\frac{\partial u_\varphi}{\partial \theta} - \frac{u_\varphi}{r}\cot\theta\right)^2 - \frac{2}{3}\left(\frac{1}{r^2}\frac{\partial (r^2 u_r)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial (u_\theta \sin\theta)}{\partial \theta} + \frac{1}{r\sin\theta}\frac{\partial u_\varphi}{\partial \varphi}\right)^2 \tag{A.33}$$

Some Other Operations in Cartesian Coordinates

$$\nabla \cdot \vec{V} = \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z}$$

$$\nabla\left(\nabla \cdot \vec{V}\right) = \vec{e}_x \frac{\partial}{\partial x}\left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z}\right) + \vec{e}_y \frac{\partial}{\partial y}\left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z}\right) + \vec{e}_z \frac{\partial}{\partial z}\left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z}\right)$$

$$\nabla \vec{V} = \begin{matrix} \vec{e}_x \vec{e}_x \frac{\partial u_x}{\partial x} & + \vec{e}_x \vec{e}_y \frac{\partial u_y}{\partial x} & + \vec{e}_x \vec{e}_z \frac{\partial u_z}{\partial x} \\ + \vec{e}_y \vec{e}_x \frac{\partial u_x}{\partial y} & + \vec{e}_y \vec{e}_y \frac{\partial u_y}{\partial y} & + \vec{e}_y \vec{e}_z \frac{\partial u_z}{\partial y} \\ + \vec{e}_z \vec{e}_x \frac{\partial u_x}{\partial z} & + \vec{e}_z \vec{e}_y \frac{\partial u_y}{\partial z} & + \vec{e}_z \vec{e}_z \frac{\partial u_z}{\partial z} \end{matrix}$$

$$\left(\nabla \vec{V}\right)^T = \begin{matrix} \vec{e}_x \vec{e}_x \frac{\partial u_x}{\partial x} & + \vec{e}_x \vec{e}_y \frac{\partial u_x}{\partial y} & + \vec{e}_x \vec{e}_z \frac{\partial u_x}{\partial z} \\ + \vec{e}_y \vec{e}_x \frac{\partial u_y}{\partial x} & + \vec{e}_y \vec{e}_y \frac{\partial u_y}{\partial y} & + \vec{e}_y \vec{e}_z \frac{\partial u_y}{\partial z} \\ + \vec{e}_z \vec{e}_x \frac{\partial u_z}{\partial x} & + \vec{e}_z \vec{e}_y \frac{\partial u_z}{\partial y} & + \vec{e}_z \vec{e}_z \frac{\partial u_z}{\partial z} \end{matrix}$$

$$\nabla\cdot\left(\nabla\vec{V}\right)^{T}=\vec{e}_{x}\frac{\partial}{\partial x}\left(\frac{\partial u_{x}}{\partial x}+\frac{\partial u_{y}}{\partial x}+\frac{\partial u_{z}}{\partial x}\right)+\vec{e}_{y}\frac{\partial}{\partial y}\left(\frac{\partial u_{x}}{\partial y}+\frac{\partial u_{y}}{\partial y}+\frac{\partial u_{z}}{\partial y}\right)+\vec{e}_{z}\frac{\partial}{\partial z}\left(\frac{\partial u_{x}}{\partial z}+\frac{\partial u_{y}}{\partial z}+\frac{\partial u_{z}}{\partial z}\right)$$

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